

CABBAGE: A common name for a biennial herb, widely cultivated in temperate regions and in the tropics for food and fodder. It belongs to the family Brassicaceae (formerly Cruciferae), which includes brussels sprouts, cauliflower, kale and broccoli. Cabbage is classified as *Brassica oleracea* variety capitata. The wild cabbage is classified as *Brassica oleracea* variety sabauda. It grows in many soil types that are high in nitrogen and rich in organic matter, at temperatures of between 4.4 - 21.1 °C. The head of the cabbage is round and forms on a short thick stem. The leaves are thick and alternating with wavy or lobed edges and the roots are fibrous and shallow. The plant produces large yellow flowers. The densely leaved heads can weigh between 0.5 to 4 kg depending on variety. Cabbage is low in sodium and very low in saturated fat and cholesterol. It is also a good source of thiamin, magnesium and phosphorus, dietary fibre, Vitamin C, Vitamin K, Vitamin B6, folate, calcium, potassium and manganese. It is commonly cooked, boiled, drained and eaten as a vegetable; and also used as fodder for livestock. The leaves are used for medicine to treat stomach pain, excess stomach acid, stomach and intestinal ulcers, asthma and morning sickness. It is also used to prevent weak bones (osteoporosis), as well as cancer.

CABIBBO ANGLE: A measure of the probability that one flavour of quark (either down or strange) will change into another flavour (up) under the action of the weak force. The Cabibbo angle is named after the Italian physicist Nicola Cabibbo, who introduced it in 1963.

CABINET FILE (CAB): In Microsoft Windows system, it is an archival file system which supports lossless data compression, holding a number of compressed files. A related set of cabinet files can be contained in a folder. During installation of a program, the compressed files in a cabinet are decompressed and copied to an appropriate directory for the user.

CABLE: 1. Mechanical cable is a strong thick rope of twisted strands of steel wire, used for lifting, pulling, towing, securing things or for supporting suspension bridges, etc. 2. Electrical cable is a group of conducting metal such as copper, usually several wires stranded together (twisted or braided) and surrounded by an insulating sheath of plastic or rubber. It is used for transmitting electrical signals. Cable for VHF signals, as used in radio, is **coaxial cable**. Cables with a number of cores (multi-core cables) are used for telephone cables, although these are increasingly being replaced by optical cables. 3. A rope or chain attached to an anchor or used for mooring a ship. 4. A long distance communication cable, usually using fibre optics cable.

CABLE MODEM: A device that modulates and demodulates an analogue carrier signal to encode and decode digital information that is transmitted, providing bi-directional data communication. It enables the connection of a PC to a cable TV line for high speed data communication far exceeding what is possible using a standard modem. A 56 K modem can receive data at about 53 Kbps, cable modems support data transfer rates of up to 30 Mbps.

CABLE TELEVISION (CABLE TV; COMMUNITY ANTENNA TELEVISION): The distribution of broadcast signals through cable relay systems. A television distribution system in which station signals, picked up by elevated antennas, are delivered by cable to the receivers of subscribers.

CACAO (COCOA; THEOBROMA CACAO): Also called **cacao bean** or **cocoa bean**, it is the dried partly fermented fatty seeds of the evergreen tree, *Theobroma cacao* of the family Malvaceae, native to tropical America, but now grown mainly in West Africa. The term is also used for the spindly, evergreen tree, which is 5 to 8 metres tall and grows in the shade of giant trees occupying the top layer of the rain forest. It has a taproot, which penetrates far below the soil surface. Its dark green leaves are shiny, leathery, egg-shaped or elliptic in shape and 20 to 35 cm long and 7 to 8 cm wide. The leaf surfaces are hairless or covered in scattered star-shaped hairs. The base of the leaf is rounded or heart-shaped and the tip is long and drawn out allowing water to drip from it. The flowers are small, yellowish white to pale pink and grouped together in clusters arising

directly from the trunk (cauliflory). Flowers are produced throughout the year. The fruit is an egg-shaped ripening yellow to orange and weighs about 500 g, 15 to 25 cm long, with a more or less knobbly surface and lines from top to bottom. The pod contains 30 to 40 seeds, each of which is surrounded by a bitter-sweet white pulp. When the seeds are dried and fermented in the sun they are brownish red and known as cocoa beans. Cocoa has a high food value, containing as much as 20 percent protein, 40 percent carbohydrate and 40 percent fat. The dried and processed cocoa beans are used in making the beverage cocoa drink, chocolate and cocoa butter. The beans also contain up to 10% of phenols and flavonoids which are antioxidants and can inhibit cancer or cardiovascular diseases. They also contain potassium, magnesium, calcium, iron; and theobromine and caffeine, which are alkaloids that stimulate the central nervous system. Cocoa is also used for medicine and cosmetics. Unfermented cocoa seeds and the seed coat are used to treat a variety of ailments, including diabetes, digestive and chest complaints. Cocoa powder, prepared from fermented cocoa beans, is used to prevent heart disease. Cocoa butter is also taken to lower cholesterol levels.

CACHE: A high-speed access area that can be either a reserved section of main memory or a storage device that provides the ability to access data or files at a higher speed. It can be either a reserved section of main memory or a storage device. Cache is implemented both in hardware and software. It serves as an intermediary component between the primary storage appliance and the recipient hardware or software device to reduce the latency in data access. The two main cache types are memory cache and disk cache. Memory cache is a portion on memory of high-speed static RAM (SRAM) and is effective because most programs access the same data or instructions over-and-over.

CACHE CRAMMING: A method of tricking a computer into running Java code that it would not ordinarily run. It consists of placing code in the computer's local disk cache when the computer uses Internet Explorer in certain environments.

CACHE MEMORY: A random access memory (RAM) that a computer microprocessor can access more quickly than it can access regular RAM. As the microprocessor processes data, it looks first in the cache memory and if it finds the data there (from a previous reading of data), does not have to do the more time-consuming reading of data from larger memory.

CACHE SERVER (CACHE ENGINE): A dedicated network server or service acting as a server that saves Web pages or other Internet content locally. By placing previously requested information in temporary storage or cache, a cache server both speeds up access to data and reduces demand on an enterprise's bandwidth. Cache servers also allow users to access content offline, including rich media files or other documents.

CACHING: 1. In computing, it is the process of storing data in a cache. A cache is a temporary storage area. 2. In biology, it is the practice by some organisms, for example, squirrels to store food during the time of plenty to use, when food is not available or is short in supply.

CACODYL (DICACODYL; TETRAMETHYLDIARSINE; ALKARSINE): A poisonous oily flammable arsenic-containing chemical, with the chemical formula $C_4H_{12}As_2$ and a characteristic garlic odour. It undergoes spontaneous combustion in dry air. Cacodyl oxide is prepared by the reaction of potassium acetate with arsenic trioxide.

CACTACEAE: A large family of mostly spiny succulent dicotyledonous xerophytic plants comprising of over 2000 species in about 87 genera. The leaves are alternate, generally extremely reduced and ephemeral or absent or rarely well developed and fleshy and are associated with highly modified axillary buds or shoots called areoles that bear spines. Photosynthesis is carried out by the green expanded stems, which show various advanced characteristics in the arrangement of vascular tissues and many other features. The flowers, however, are borne singly directly on the stem and are comparatively primitive,

having large numbers of stamens, petals, sepals and bracts arranged spirally. The fruit is a berry.

CACTUS: Any of the various types of plants belonging to the family Cactaceae, native to America. The family contains about 1650 species, most of which are adapted to arid climates and hot environments. Cacti are distinctive and unusual plants, showing a wide range of anatomical and physiological features which conserve water. Their stems have expanded into green succulent structures containing the chlorophyll necessary for life and growth, while the leaves have become the spines to reduce transpiration.

CAD: In computing, it is the acronym for **computer-aided design**, which is using computer software in the design and editing of drawings.

CADE LAMB: A lamb reared from a bottle because of the death of its mother.

CADMIUM: A soft silver white, ductile and malleable transition metallic element, with the symbol Cd, atomic number 48, atomic weight 112.41, melting point 320.9 °C (594.05 K), boiling point 765 °C (1 038.15 K), specific gravity 8.65, atomic number 48 and relative atomic mass of 112.40. It is a transition metal in the same group as zinc. It is therefore a common impurity in zinc and it is most often isolated during the production of zinc by vacuum distillation of Zn. Its uses include batteries, pigments, electroplating and as a constituent of alloys. It is a toxic metal known to cause cancer and has become a pollutant because of industrial dumping.

CADMIUM CELL (WESTON CELL): A type of primary voltaic cell or wet-chemical cell with mercury and cadmium electrodes in cadmium sulfate electrolyte, used to supply an accurate voltage for electronic measurements and as a voltage reference for calibration of voltmeters. It was invented by Edward Weston, a British electrical engineer in 1893. It was adopted as the International Standard for EMF between 1911 and 1990. At 20 °C (293.15 K) its voltage is 1.0186 volts. It usually has an H-shaped glass vessel with a mercury anode covered with a paste of cadmium sulfate and mercury(I) sulfate. The cathode is of pure mercury over which a paste of mercurous sulfate and mercury is placed. The electrolyte is a solution of cadmium sulfate octahydrate and the depolariser is a paste of mercurous sulfate.

CADMIUM SULFIDE: A water-insoluble yellow compound, CdS, which occurs naturally as the mineral greenockite, used as a pigment and in semiconductors and fluorescent materials. It forms thermally stable pigments. With the addition of cadmium telluride (CdTe), mercury sulfide (HgS), etc, colours from deep red to yellow are formed. Sandwiched with cadmium sulfide it forms a p-n junction photovoltaic solar cell.

CADUCEUS: In mathematics, it is a pair of curves in space, each of which is a helix and which twists in opposite directions around one another. It is also a representation of a staff with two entwined snakes and two wings at the top, as used by physicians as symbol.

CADUCOUS: A term usually used for floral parts, which fall off very early compared to similar structures in other plants.

CAECUM (CECUM): A pouch in the digestive system of animals in which the large intestine begins. It branches off from the first part of the large intestine and terminates in the ileum.

CAENORHABDITIS ELEGANS: A soil-dwelling transparent, unsegmented, bilaterally symmetrical nematode (roundworm), comprising two sexes (hermaphrodites and males). It is about 1 mm in length and lives in temperate soil environments. It is used experimentally as a model organism in genetics and developmental biology.

CAESAREAN SECTION: A surgical operation for delivering a baby by incising the walls of the mother's abdomen and uterus. It is performed when factors such as an abnormally narrow pelvis, pelvic tumours, haemorrhage, herpes simplex infection, multiple births or an abnormal position of the foetus within the uterus make natural childbirth dangerous.

CAESALPINIA: A genus of flowering plants in the pea family, Fabaceae that are famous for producing large bright flowers with

shades of yellow, fiery red, and orange. It is native to the tropics and subtropics of the Americas. They are deciduous or evergreen shrubs or small trees that grow up to 5 metres tall. Some of the species include *Caesalpinia gilliesii*, commonly called desert or yellow bird of paradise, and *Caesalpinia pulcherrima* with common names like peacock flower, red bird of paradise, Mexican bird of paradise, pride of Barbados, etc. *Caesalpinia pulcherrima* is used as food (the young seeds are eaten), medicine and ornamental plant.

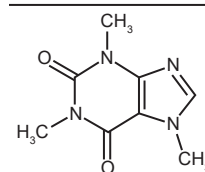
CAESIUM: A soft silvery-white, ductile metallic element, one of the alkali metals, with the symbol Cs, atomic number 55, relative atomic mass 132.905, melting point 28.44 °C (301.59 K) and boiling point 671 °C (944 K). It is the most electro-positive of all the elements. It is one of only five metals that are liquid at or near room temperature. It is used in the manufacture of photoelectric cells and in atomic clocks.

CAESIUM CHLORIDE STRUCTURE: One of the different types of ionic crystal in which the unit cell is cubic with the cation situated at the centre of the cube and the anions at the eight corners of the cube. Each cation or anion has a coordination number of 8. Examples of compounds with this structure are caesium chloride (CsCl), caesium bromide (CsBr), caesium iodide (CsI), caesium cyanide (CsCN) and ammonium chloride (NH₄Cl).

CAESIUM CLOCK: An atomic clock that uses the frequency of radiation absorbed in changing the spin of electrons in caesium-133 atoms when in a magnetic field.

CAESPITOSE: A term used to describe a tufted form of growth, as in many grasses, such as *Deschampsia caespitosa* (tufted hairgrass).

CAFFEINE (1,3,7-TRIMETHYL-1H-PURINE-2,6(3H,7H)-DIONE; 3,7-DIHYDRO-1,3,7-TRIMETHYL-1H-PURINE-2,6-DIONE): A stimulant drug or alkaloid organic substance, with the chemical formula C₈H₁₀O₂N₄. It is found in tea leaves, coffee beans, cola nuts, ilex plants and in small amounts in cocoa. It stimulates the heart and central nervous systems. It can be detrimental to health if drunk excessively (more than 6 average cups per day of tea or coffee).



Chemical structure of caffeine

CAGE: 1. In chemistry, it is an aggregate of molecules, generally in the condensed phase that surrounds the fragments formed, for example, by thermal or photochemical dissociation. Because the cage hinders the separation of the fragments by diffusion, they may preferentially react with one another, called the **cage effect**, but not necessarily to reform the precursor species. 2. A structure made of bars or wires in which birds or animals are kept or carried. 3. An enclosed platform used to raise and lower people and equipment in the shaft of a mine.

CAGE COMPOUND: A polycyclic compound having the shape of a cage. The term is also used for inclusion compounds.

CAGE EFFECT: An effect occurring in certain condensed-phase reactions in which fragments are formed but their diffusion is hindered by the presence of other molecules, which form a "cage" of molecules. Such fragments may recombine.

CAHN-HILLIARD MODEL: An equation of mathematical physics describing the process of phase separation, named after John W. Cahn and J. E. Hilliard, in which the dynamics is dissipative and the order parameter is conserved. The two components of a binary fluid spontaneously separate and form domains pure in each component. If c is the concentration of the fluid, with $c = \pm 1$ indicating domains, then the equation is written as $\frac{\partial c}{\partial t} = D \nabla^2 (c^3 - c - \gamma^2 c)$, where D is a diffusion coefficient with units length square over time and $\sqrt{\gamma}$ gives the length of the transition regions between the domains, $\frac{\partial}{\partial t}$ is the partial time derivative and ∇^2 is the Laplacian in n dimension. Additionally $\mu = c^3 - c - \gamma^2 c$ is identified as a chemical potential.

CAIMAN: A tropical American crocodilian amphibian belonging to the subfamily Caimaninae, of the family Alligatoridae, found in Central and South America in swamps, marshes, mangroves, rivers, and lakes. Caimans have wide, slightly flattened bodies and long vertically flattened, muscular tails that propel them through water. Most caiman species are between 2 m and 6 m long depending on the species, with a maximum weight of 6 to 40 kg depending on species. The black caiman (*Melanosuchus niger*), can grow more than 5 m in length and weigh up to 1,100 kg. The females reproduce by laying eggs in large nests, which can be more than 1.5 metres wide. Their diet consist mostly of fish, but also insects, birds and small mammals and reptiles. Caimans have been hunted by men for food and their skins that are used to make shoes, bags, belts, wallets and some other leather products.

CAISSON: A hollow cylindrical or box-like structure, usually reinforced concrete, sunk into a riverbed to form the foundations of a bridge.

CAISSON DISEASE (BENDS; DECOMPRESSION SICKNESS; DIVERS' DISEASE): A condition that is caused by a sudden and substantial change in atmospheric pressure. It is mostly experienced by deep-sea divers who surface too quickly, causing nitrogen bubbles to be dissolved into the bloodstream and tissues because of a sudden drop in the surrounding pressure. It is characterised by breathing difficulties, joint and muscle pain, cramps, numbness, nausea and paralysis.

CAL: In computing, it is the abbreviation for **computer-assisted learning**, also called **e-learning**, which is the use of all forms of electronically supported learning and teaching.

CALABASH TREE (*Crescentia cujete*): An evergreen, tropical, flowering tree belonging to the family, Bignoniaceae that is native to Central and South America, but now grown throughout the tropics worldwide. It grows to about 6-9 metres in height with a trunk diameter of about 30 centimetres. Its bark is rough and light-grey with pinkish-brown sapwood. Its leaves are simple, obovate and arranged alternately in clusters along the stout twigs. It bears flowers and fruits throughout the year. The flowers are yellow/green with red or purple veins, cup-shaped and borne singly on stout stalks on the trunk and branches. The smooth, large and hard gourd-like fruits are green to brown and poisonous. The tree is grown as an ornamental. The fruits are used to make utensils such as bowls, cups, jugs, water containers and also musical instruments. They are also often decorated as ornaments. The pulp of the fruit has been used traditionally in some areas medicinally to treat various ailments. The seeds are sometimes cooked and eaten. The trees are propagated from seeds or cuttings, but grow slowly. Cultivated varieties may produce larger fruits than the wild trees. The tree is the national tree of St. Lucia, an island country in the eastern Caribbean Sea.

CALAMINE: An ore of zinc, with the chemical formula $ZnCO_3$. It also refers to a pink powder made of a mixture of zinc oxide and 0.5% iron (III) oxide (Fe_2O_3). It is used in lotions and ointments for treating, for example, sunburn, eczema, measles, rash, insect bites and stings.

CALAMUS: 1. In birds, it is the hollow base of a feather shaft, which attaches the feather to the skin. 2. Any of various chiefly tropical Asian climbing palms of the genus *Calamus*, having strong flexible stems used as a source of rattan.

CALCAR: 1. A spur or spur-like appendage such as the base of a petal or on the wing or leg of a bird. 2. A small oven or furnace, used for the calcination of sand and potash and converting them into frit.

CALCARATE (CALCEIFORM): 1. In botany, it refers to a shape in the form of a shoe or slipper, for example, the labellum or the enlarged petal of some orchids. 2. A structure with a calcar; spurred.

CALCAREOUS SOIL: Soil profiles containing both magnesium carbonate ($MgCO_3$) and calcium. Because of the high levels of calcium carbonate ($CaCO_3$), the soils are alkaline ($pH > 7$). Calcareous soils are most often formed from limestone or in dry environments where low rainfall prevents the soils from being leached of carbonates.

Calcareous soils frequently cause nutrient deficiencies for many plants because some nutrients cannot be efficiently absorbed by plant roots if soil pH is too high.

CALCEOLATE (CALCEIFORM): Slipper-shaped or shoe-shaped, an example is the labellum or the enlarged petal of some orchids.

CALCICOLE (CALCIPHILE): A plant which grows well on soils rich in calcium or alkaline soils but not on acidic soil. Examples of calcicole plants are ash trees, honeysuckle, lilac and beet.

CALCIFICATION: A process in which calcium builds up in body tissues or other body parts such as arteries, kidneys, lungs and the brain causing the tissue to harden. This can be a normal or abnormal process. To **calcify** is to harden by the deposition of calcium.

CALCIFUGE: A term used to describe a plant that grows on mainly acidic soil or lime-deficient soils such as loams or clays. Examples include blueberries, camellias, foxgloves, azaleas, magnolias, rhododendrons, and strawberries.

CALCINATION: The oxidation of metals by burning in air or the process of heating a solid to a high temperature but below the melting or fusing point and converting it to a powdery residue by drying, decomposing or oxidising. To **calcine** is to cause something to be reduced to powder by burning or to burn to ashes.

CALCINED COKE: A petroleum coke or coal-derived pitch coke obtained by heat treatment of green coke to about temperatures of 1200 -1350 °C or higher to refine its crystalline structure. Calcined coke is the main raw material for the manufacture of polygranular carbon and polygranular graphite products, for example, carbon and graphite electrodes.

CALCINEURIN (PROTEINPHOSPHATASE 3, PPP3CA): A phosphatase enzyme that plays a crucial role in relaying intracellular signals following activation of T cells of the immune cells, particularly lymphocytes.

CALCINITE: A mineral form of potassium hydrogen carbonate ($KHCO_3$). Potassium hydrogen carbonate is a white crystalline solid, which is soluble in water and insoluble in ethanol. It is prepared by passing carbon dioxide into saturated potassium carbonate solution. It is used in baking, soft-drinks manufacture and in CO_2 fire extinguishers. Because of its buffering capacity, it is added to some detergents and also used as a laboratory reagent.

CALCIPHILIOUS: Thriving or suitable to a calcareous soil; or lime-loving.

CALCISOL (RENDZINA): A type of shallow intrazonal soil rich in lime (calcmorphic) formed from the underlying limestone or chalk rocks. The A horizon is brown or black and rich in humus. The B horizon, if one is distinguishable, is grey or yellowish with limestone fragments. Brown earths develop in similar climatic conditions from other parent rocks.

CALCITE: A colourless or white, (sometimes grey, yellow or green) mineral consisting of calcium carbonate, with the chemical formula $CaCO_3$. It is the most stable polymorph of calcium carbonate. It is present in many types of limestone, chalk and marble.

CALCITONIN (THYROCALCITONIN): A peptide hormone found in fish, reptiles, birds and mammals (vertebrates) that lowers the concentration of calcium in the blood by suppressing the release of calcium from bone and promoting excretion of calcium and phosphate by the kidney.

CALCIUM: A soft grey or silvery metallic element, with the chemical symbol Ca, atomic number 20, relative atomic mass 40.08, melting point 842 °C (1115 K) and boiling point 1484 °C (1757 K). It is in Group 2 (IIA) of the Periodic Table, one of the alkaline earth metals. It is the fifth and third most abundant element and metal respectively in the Earth's crust. Calcium is an essential component of bones, teeth, shells, milk and leaves and it forms 15% of the human body by mass. Calcium is easily leached from soils resulting in an acid soil. This may be countered by adding calcium carbonate, calcium hydroxide (slaked lime) or calcium sulfate (gypsum) to the soil. These compounds also help improve soil structure.

CALCIUM CARBIDE (CALCIUM DICARBIDE; CALCIUM ACETYLIDE): A colourless compound of calcium and carbon, with the chemical formula CaC_2 , molar mass 64.099 g/mol, density 2.22 g/cm³, melting point 2160 °C (2433 K) and boiling point 2300 °C (2573 K). Pure calcium carbide is colourless, but most of the material is produced industrially and is impure, giving it a black or greyish-white colour, depending on the quality. It is produced industrially in an electric arc furnace from a mixture of lime and coke at approximately 2000 °C in the reaction: $\text{CaO} + 3\text{C} \rightarrow \text{CaC}_2 + \text{CO}$. When calcium carbide comes into contact with water, a chemical reaction immediately begins, which yields new compounds, which is used in producing acetylene, a useful flammable gas. It reacts with nitrogen at high temperature to form calcium cyanamide: $\text{CaC}_2 + \text{N}_2 \rightarrow \text{CaCN}_2 + \text{C}$. It is also used in steelmaking to help in desulfurisation of iron and acts as an important fuel in the process of steelmaking. It also acts as a powerful deoxidiser in the work of the process of ladle treatment in steelmaking.

CALCIUM CARBONATE: A white chemical compound, with the chemical formula CaCO_3 that is found in nature as limestone, chalk and marble and produced industrially in an electric arc furnace loaded with a mixture of lime and coke at approximately 2000 °C (2273.15 K). It is insoluble in water and gives carbon dioxide when mixed with an acid. It is a valuable resource used in the making of iron, steel, cement, glass, slaked lime, bleaching powder, sodium carbonate and bicarbonates. Calcium carbonate is the active ingredient in agricultural lime and is usually the principal cause of hard water.

CALCIUM CHLORIDE: A white crystalline compound, with the chemical formula CaCl_2 , which forms several hydrates. It is highly soluble in water and it is deliquescent. Its dehydrated form is used as drying agent. It is used as a de-icing agent, as a refrigerant and in controlling dust.

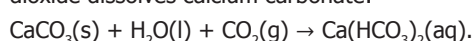
CALCIUM CYANAMIDE: A colourless solid compound, with the chemical formula CaCN_2 , molar mass 80.102 g/mol and density 2.29 g/cm³. It sublimates at 1,300 °C (1,573 K). It is prepared by heating calcium carbide at 800 °C (1073 K) in a stream of nitrogen in the reaction: $\text{CaC}_2 + \text{N}_2 \rightarrow \text{CaCN}_2 + \text{C}$. It is used to produce melanine, urea and as fertiliser because it reacts with water to give ammonia and calcium carbonate.

CALCIUM CYCLAMATE: A calcium salt of cyclamic acid, with the molecular formula $\text{C}_{12}\text{H}_{24}\text{CaN}_2\text{O}_6\text{S}_2$, molar mass 80.102 g/mol that has a very sweet taste and depending on concentration can be 30–50 times sweeter than sugar. It is soluble in water. It was formerly used as an artificial sweetener. Cyclamate was banned in the United States market in 1969 because of possible carcinogenic effects, but current evidence indicates that cyclamate is not carcinogenic. It is still banned in the United States, but approved in much of the world.

CALCIUM DICARBIDE (CALCIUM CARBIDE; CALCIUM ACETYLIDE): A colourless solid compound of calcium and carbon, with the chemical formula CaC_2 , density 2.22 g/cm³, melting point 2160 °C (2433 K) and boiling point 2300 °C (2573 K). It is used in making acetylene gas.

CALCIUM FLUORIDE: A white inorganic crystalline solid, which occurs naturally as the mineral fluorite. Its chemical formula is CaF_2 , molar mass 78.07 g/mol, density 3.18 g/cm³, melting point 1418 °C (1691 K) and boiling point 2533 °C (2806 K). It is the main source of fluorine.

CALCIUM HYDROGEN CARBONATE (CALCIUM BICARBONATE): A substance found in water when rain water passes over limestone rock, with the chemical formula CaC_2 and molar mass 162.1146 g/mol. It is the cause of temporary hardness in water. The compound is stable only in solution. It is formed when water containing carbon dioxide dissolves calcium carbonate:



CALCIUM HYDROGEN PHOSPHATE: A dibasic calcium phosphate made by heating calcium phosphate with 70% sulfuric acid, with the chemical formula $\text{Ca}(\text{H}_2\text{PO}_4)_2$. It is usually found as the dihydrate,

$\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$. It has very low solubility in water (0.02 g per 100 ml at 25 °C, 298 K). Its uses include as fertiliser, as a dietary supplement in breakfast cereals, enriched flour, noodle products and poultry feed.

CALCIUM HYDROXIDE (SLAKED LIME; HYDRATED LIME): A white solid obtained by reacting calcium oxide (lime) with water, a process called slaking. Its chemical formula is $(\text{Ca}(\text{OH})_2)$, density 2.211 g/cm³ (solid) and melting point 580 °C, 853 K (loses water). It is sparingly soluble in water. A solution of the compound in water is called lime water and it is used in the laboratory to test for the presence of carbon dioxide, because it readily reacts with CO_2 , to form calcium carbonate, CaCO_3 . It is also used to reduce soil acidity and as a cheap alkali in many industrial processes. It is used in leather tanning to remove hair from hides and also as whitewash to paint structures or buildings.

CALCIUM MODULATED PROTEIN (CALMODULIN): A calcium-binding protein, consisting of 148 amino-acid residue that is an important regulator of numerous cellular activities, including mediation of processes such as metabolism, apoptosis, muscle contraction, intracellular movement, short-term and long-term memory.

CALCIUM NITRATE: A white deliquescent compound that is very soluble in water, with the chemical formula $\text{Ca}(\text{NO}_3)_2$. It can be prepared by neutralising nitric acid with calcium carbonate and crystallising it from solution as the tetrahydrate $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$. The anhydrous salt can be obtained from the hydrate by heating but it decomposes on strong heating to give the oxide, nitrogen dioxide and oxygen. Calcium nitrate is used as a reagent, a pyrotechnic and a fertiliser.

CALCIUM OCTADECANOATE (CALCIUM STEARATE): An insoluble white salt, with the chemical formula $\text{Ca}(\text{CH}_3(\text{CH}_2)_{16}\text{COO})_2$. It is formed when soap is mixed with water containing calcium ions and the scum produced in hard-water regions. It is used in industry as a metallic soap. It is insoluble in water but slightly soluble in petroleum, benzene and toluene.

CALCIUM OXIDE (QUICK LIME): A white solid compound, with the chemical formula CaO , density 3.35 g/cm³, melting point 2572 °C (2845 K) and boiling point 2850 °C (3123 K). It is formed by thermal decomposition of calcium carbonate or by heating calcium metal in oxygen. The thermal decomposition of CaCO_3 is a completely reversible process, once the quicklime product has cooled, it immediately begins to absorb CO_2 from the air, until, after enough time, it is completely converted back to CaCO_3 . Calcium oxide is used in water and sewage treatment to reduce acidity, to soften or remove Ca ions, as a flocculant and to remove phosphates and other impurities, in paper making to dissolve lignin, in agriculture to improve acidic soils, in gas scrubbers to remove SO_2 from waste gases and to treat many liquid effluents.

CALCIUM PHOSPHATE: A white insoluble powder, with the chemical formula $\text{Ca}_3(\text{PO}_4)_2$. It is found naturally in the mineral apatite and as rock phosphate. It is also the main constituent of animal bones and tooth enamel. It is used in the preparation of phosphate fertilisers.

CALCIUM SILICATES: Compounds having compositions CaSiO_3 , Ca_2SiO_4 , $\text{Ca}_3\text{O} \cdot \text{SiO}_4$ and $\text{Ca}_3\text{Si}_2\text{O}_7$. They are obtained by reacting calcium oxide and silica in various ratios. They are insoluble crystalline compounds, present in various minerals and cements and form part of the slag produced in a blast furnace. Calcium orthosilicate (Ca_2SiO_4) has a low bulk density and high physical water absorption and used as an anti-caking agent and as antacid. It is also used in roads, insulation, bricks and roof tiles.

CALCIUM SULFATE: A white solid compound found in nature as gypsum and anhydrite, with the chemical formula CaSO_4 . It is used in the manufacture of certain paints, ceramics and paper. The naturally occurring forms are used in the manufacture of sulfuric acid. The anhydrous form is used as a drying agent. The hemihydrate ($\text{CaSO}_4 \cdot 0.5\text{H}_2\text{O}$) is known as Plaster of Paris (POP), while the dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) occurs naturally as gypsum.

CALCULUS: 1. Also known as **infinitesimal calculus**, it is a branch of mathematics that permits the manipulation of continuously varying quantities, by two fundamental operations integral and differential calculus. **Differential calculus** is the study of the variation of a function with respect to changes in independent variables, slopes of curves, accelerations, maxima and minima by means of derivatives and differentials. **Integral calculus** is concerned with the study of integrals, especially the methods of ascertaining indefinite integrals and applying them to the solution of differential equations and the determining of areas, volumes and lengths. 2. Any formal theory or set of rules for calculation, as for example, functional calculus, predicate calculus and lambda-calculus. 3. An abnormal solid concretion or stone composed of mineral salts or organic compounds formed in the kidney, bladder or gall bladder. Calculi can also form in gastrointestinal tract called enteroliths, in the salivary duct and also in the crevices of the tonsils. 4. In dentistry, it is a deposit of calcium phosphate, calcium carbonate and organic matter that accumulates on the teeth.

CALCULUS OF VARIATIONS (VARIATIONAL CALCULUS): The branch of calculus concerned with finding the maxima or minima of definite integrals that depends for its values on another function to find functions that maximize or minimize a given function of those functions.

CALCULUX: A small cup shaped structure or a group of small bracts that resembles a calyx, as found in carnation or hibiscus.

CALDERA: A very large basin-shaped crater, often found on top of volcanoes, which may produce a lake. Examples are the Crater Lake in Oregon, USA and Lake Bosomtwe in the Ashanti Region of Ghana.

CALENDAR: A system of measuring time for the needs of civil life, by dividing time into days, weeks, months and years. Calendar divisions are based on the movements of the Earth and the regular appearances of the Sun and the Moon. The measurement of a year is based on one revolution of the Earth around the Sun and is called a seasonal, tropical or solar year. A solar year contains 365 days, 5 hr, 48 min and 45.5 sec.

CALENDAR YEAR (GREGORIAN YEAR): The time of period beginning on January 1 and ending on December 31 with a length of 365 days in an ordinary year and 366 days in a leap year, giving an average length of 365.2425 days.

CALF: 1. The young of cattle, seal, whale and certain other animals. 2. The fleshy part of the back of a human leg below the knee.

CALF MUSCLES: The muscles consisting of gastrocnemius and soleus muscles extending from the back of the knee to the heel. The gastrocnemius starts behind the knee and forms the bulky part of the calf. Under it is the soleus, which is longer and thinner than the gastrocnemius and starts lower down from the back of the tibia (shin). These muscles, together with the plantar muscles, join to form the Achilles tendon, which connects them to the heel.

CALGON: The trade name for water softening agents, sodium hexametaphosphate (amorphous sodium polyphosphate), with the chemical formula $\text{Ca}(\text{C}_{18}\text{H}_{35}\text{O}_{2})_2$, which in water would complex with calcium ion and certain other cations, preventing formation of unwanted salts and interference by those cations with the actions of the soap or the detergents.

CALIBAN PUZZLE: A logic puzzle in which one is asked to infer one or more facts from a set of given facts.

CALIBRATION: 1. Evaluating and adjusting the precision and accuracy of measurement equipment. It is done to determine any deviation and correct errors. 2. It is the process of marking out and interpreting the numerical scale on an instrument. For instance, a voltage scale on a digital (or analogue) voltmeter can be converted to a temperature scale to match the temperature indicated by a particular thermocouple. 3. The set of operations which establish,

under specified conditions, the relationship between values indicated by the analytical instrument and the corresponding known values of an analyte. To **calibrate** is to add a scale to a measuring instrument or to adjust a measuring instrument so that it gives accurate readings.

CALIBRATION COMPONENT: A component of a calibration mixture, present in the gaseous or vapour state, quantitatively and qualitatively defined and used directly for testing and for calibration.

CALIBRATION FUNCTION: The functional (not statistical) relationship for a chemical measurement process, relating the expected value of the observed (gross) signal or response variable $E(y)$ to the analyte amount. The corresponding graphical display for a single analyte is referred to as the **calibration curve**. With additional variables or analytes which occur in multi component analysis, the 'curve' becomes a calibration surface or hyper surface.

CALIBRATION GAS MIXTURE: A gas mixture of known composition, generally comprising one or more calibration components and a complementary gas.

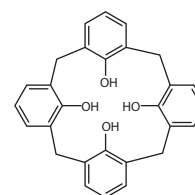
CALIBRATION MATERIAL: A material of known composition or properties which can be presented to the analytical instrument for calibration purposes.

CALIBRATION SAMPLE: The test portion or test solution used for calibration of an analytical procedure. The calibration sample is normally of known weight or volume and is prepared according to specifications.

CALICHE (HARDPAN; CALCRETE; KANKAR; DURICRUST): A sedimentary rock or a hardened deposit of calcium carbonate. Caliche generally forms when minerals leach from the upper layer of the soil and accumulate in the next layer, at depths of approximately 1 to 10 m under the surface. Sometimes, these deposits may contain impurities which determine the colouration. One of the world's largest caliche is in the Makgadikgadi Pans in Botswana, where surface calcretes occur at the location of a now desiccated prehistoric lake.

CALIFORNIUM: A metallic transuranic element, belonging to the actinoids with the chemical symbol Cf, atomic number 98, mass number of most stable isotope 251, melting point about 900 °C (1173 K) and boiling point of about 1470 °C (1743 K). It is produced by bombarding curium atoms with alpha particles. It has several isotopes with half-lives of up to 800 years. Cf-252 with a half-life of 2.64 years is a very strong neutron emitter and is thus extremely radioactive and poisonous. One microgram of Cf-252 spontaneously emits 170 million neutrons per minute.

CALIXARENES: Macrocyclic compounds that have molecules with a cuplike structure. The simplest, calix[4] arene has four phenol molecules joined by four $-\text{CH}_2-$ groups into a ring. Calix[6]arene has six phenol molecules in the ring. Calixarenes have hydrophobic cavities that can play host to smaller molecules or ions in Host-Guest Complexes. Calixarenes are used in commercial applications such as ion selective electrodes, selective membranes and stationary phase in high-performance liquid chromatography (HPLC), which is a technique used to separate, identify, and quantify each component in a mixture



Phenol-derived calixarene

CALL ADMISSION CONTROL (CAC): The practice or process of regulating traffic volume in voice communications, particularly in wireless mobile networks and in VoIP (voice over Internet Protocol, also known as Internet telephony). Call admission control can also be used to ensure or maintain, a certain level of audio quality in voice communications networks or a certain level of performance in Internet nodes and servers where VoIP traffic exists.

CALL CENTRE SUITE: A suite of software products that includes multiple integrated components used in call centres. These software suites provide organisations with everything that is required to perform call centre operations within single platforms. They also include tools for telephone switch functionality, automatic call distribution, intelligent routing, outbound dialing and voice mail.

CALL DETAIL RECORD (CDR): A computer record created by a telephone exchange. It includes the details of a phone call established through the telephone exchange, including an automated record of the length of each telephone call.

CALL FAILURE RATE (CFR): The percentage of calls to an ISP or any network provider that fails to get through. The CFR is commonly used as a statistical measure in assessing Internet service providers. Companies can also measure the CFR for their own employees who dial in for access to the company's network.

CALL FORWARDING: A phone feature that enables users to forward or redirect incoming calls to any alternate number, which may be either a land line or cellular number. Users are also provided with options to divert incoming calls to voice mails. Phones can be set to divert calls without ringing; a diversion can also happen when lines are busy, calls are not answered or phones are switched off. Phones can also be set to divert calls in the absence of network coverage.

CALL MATCHING (VOCAL MATCHING): In zoology, a distinctive vocal interactive process, often displayed by some animal species in which members duplicate each other's call. Examples of animal species that display such traits include some birds, for example, parrots, members of the finch family; and some cetaceans. Call matching may be done for mating, social or warning purposes.

CALL SIGNALLING: A process, set up by the administrator that is used to set up a connection in a telephone network. In VoIP systems, call signaling establishes connections between H.323 or SIP endpoints (or between an endpoint and a gatekeeper).

CALLER ID: A telephone feature that displays a caller's phone number on the recipient's phone device before the call is answered. The phone number, location and associated billing or subscriber name is shown on the handset's display or a separate caller ID box attached to the phone.

CALLER ID SPOOFING: A malicious technique where a caller masquerades as someone else by falsifying the number that appears on the recipient's caller ID. Similar to e-mail spoofing by making a email message appear that it comes from one email address, when in reality it has been spoofed and has come from an entirely different one, caller ID spoofing can make a call appear to come from any phone number the caller chooses.

CALLICLONE: A colony derived from a single plant cell that has been removed from a suspended culture and planted out in semi-solid medium.

CALLIPERS: An instrument used to measure the internal and external diameters of pipes. It is used in many fields such as mechanical engineering, metalworking, woodworking, science and medicine. One type, the vernier calliper is a very precise instrument consisting of an L-shaped frame with a linear scale along its longer arm and an L-shaped sliding attachment with a vernier used to read directly the dimension of an object represented by the separation between the inner or outer edges of the two shorter arms (Fig 1.). A vernier is a short, auxiliary scale which slides along the main instrument scale to permit accurate fractional reading of the least main division of the main scale. Diagram (Fig 2.) shows a vernier scale having 10 intervals equal to 9 intervals, on the main scale. The zero division of the vernier scale is seen reading 101.4 divisions on the main scale. The fractional part of the reading is given with greater certainty by the 4th division of the vernier being exactly in line with a division on the main scale.

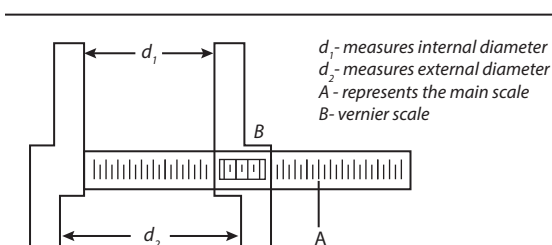


Fig. 1: Vernier (Sliding) callipers

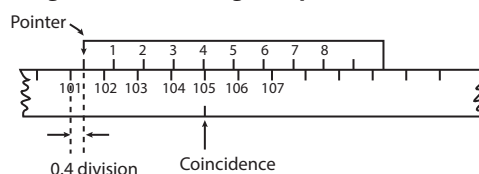


Fig. 2: Vernier scale

CALLISTO: The second largest moon on the planet Jupiter. It is 4 800 km in diameter, making it the third-largest moon in the Solar System and the second largest in the Jovian system, after Ganymede. Callisto has about 99% the diameter of the planet Mercury, but only about a third of its mass. It has an orbital radius of about 1 880 000 km. Callisto is composed of approximately equal amounts of rock and ices. It has the lowest density of the four Galilean moons with a mean density of about 1.83 g/cm³. Scientists have detected water, ice, carbon dioxide, silicates and organic compounds on Callisto. Its highest daytime temperature is around -108 °C (165 K) and night temperatures can reach -193 °C (80 K). It was discovered in 1610 by Galileo Galilei, an Italian astronomer.

CALLOSE: A structural polysaccharide found in the sieve plates of the phloem of higher plants, consisting of glucose residues linked through β (1-3) glycosidic bonds. Callose plays important roles during a variety of processes in plant development and/or in response to multiple biotic and abiotic stresses. It is deposited steadily throughout the growing season eventually causing blockage of the pores of mature sieve tubes. The functions of blocked tubes are taken over by new phloem formed from the cambium. In some species the blockage is reversible, the callose being hydrolysed in the spring. Callose may be laid down very rapidly in response to injury of the phloem.

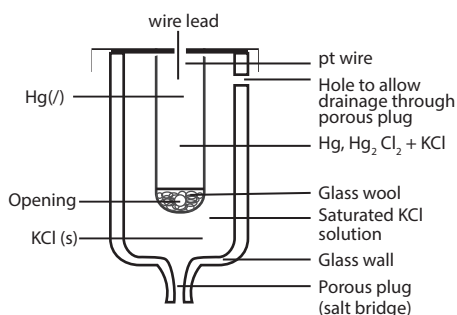
CALLUS: 1. A protective tissue consisting of parenchyma cells that develop over a cut or damaged plant surface, for example, where a branch has been cut off a tree. The callus tissue is produced by the cambium and initially forms a ring of thickening (wound wood) around the wound. Eventually, it may completely cover the exposed wood. Callus tissue is also formed at the base of cuttings before roots are produced. Similarly, callus tissue is important when propagating plants by leaf cuttings. For example, when a Begonia leaf is pegged down on the soil surface and the veins are cut, it is the callus tissue at the wound sites that gives rise to the new plantlets. Grafts also depend on the formation of callus for a successful graft union. The regenerative capacity of callus tissue is utilised considerably in tissue culture. Callus cells are grown in vitro on suitable nutrient media and under correct conditions new plantlets will develop. 2. An accumulation of the carbohydrate callose on phloem sieve plates. 3. Thickened layers of skin that commonly form on the palms of the hands and soles of the feet as a result of continuous pressure or friction such as repeated rubbing against a hard surface or being involved in heavy manual work.

CALMODULIN (CALCIUM MODULATED PROTEIN): Abbreviation for calcium modulated protein, a calcium-binding protein, consisting of 148 amino-acid residue that is an important regulator of numerous cellular activities, including mediation of processes such as metabolism, apoptosis, muscle contraction, intracellular movement, short-term and long-term memory.

CALNEXIN (CNX): A protein, bound to the membrane of the endoplasmic reticulum (ER) of cells. It acts as a molecular chaperone to newly synthesised proteins of the immune-globulin super class. It associates with proteins, including T-cell receptors, MHC class I and II molecules and immunoglobulins that are in a partly folded or part-assembled state and retains them in ER while they undergo further maturation.

CALOMEL (MERCURY(I) CHLORIDE; MERCUROUS CHLORIDE): A white crystalline powder, with the chemical formula Hg_2Cl_2 , density 7.150 g/cm^3 , melting point 525°C , 798 K (triple point), boiling point 383°C (656 K) and sublimes. It is slightly soluble in water and found in nature as the mineral horn quicksilver. It is prepared by sublimation from a mixture of mercury and mercuric chloride or by precipitation from a mercurous chloride solution on adding chloride ion. It is used for fungicide and insecticide and in calomel electrode. Mercurous chloride is less poisonous than mercuric chloride mainly because it is much less soluble.

CALOMEL HALF CELL (CALOMEL ELECTRODE): A half cell type in which the electrode is mercury coated with calomel (Hg_2Cl_2) and the electrolyte is a solution of potassium chloride and saturated calomel. It is used as a reference electrode of known potential.



Calomel half cell (Calomel electrode)

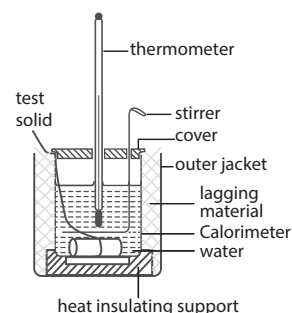
CALOR GAS: A mixture of hydrocarbon compounds, made up mainly of propane (C_3H_8) and butane (C_4H_{10}), which is used as a portable supply of gas.

CALORIC THEORY: An obsolete theory of the nature of heat, which described heat as consisting of a weightless fluid called caloric that flowed from hotter to colder bodies.

CALORIE: The non-SI unit of energy or heat. It is the quantity of heat required to raise the temperature of 1 gram of water by 1°C ; or the centimetre-gram-second (cgs) system unit of heat, now replaced by the joule, which is the SI unit of energy (one calorie is approximately 4.18 joules). There are differently defined calories, which include: (i) **thermochemical calorie** (cal_{th}), which is equal to 4.184 J ; (ii) **international steam table calorie** (cal_{IT}), which is equal to 4.18674 J ; (iii) **15°C calorie** (cal_{15}) = 4.1855 J , the mechanical equivalent of heat, defined as the amount of energy required to warm one gram of air-free water from 14.5°C to 15.5°C at standard atmospheric pressure (101.325 kPa). Some others are **mean calorie** (cal_{mean}) = 4.19002 J , defined as $1/100$ of the amount of energy required to warm one gram of air-free water from 0°C to 100°C at standard atmospheric pressure; **4°C calorie** cal_4 = 4.2045 J (water at its densest), defined as the amount of energy required to warm one gram of air-free water from 3.5°C to 4.5°C at standard atmospheric pressure; and **20°C calorie** cal_{20} = 4.18190 J (20°C = warm indoors), defined as the amount of energy required to warm one gram of air-free water from 19.5°C to 20.5°C at standard atmospheric pressure. The calorie has been replaced by the SI unit of energy, the joule, but is commonly used for energies in the field of chemistry and also used as a unit of food energy measured in kcal (1000 calories) and defined as the heat energy that raises 1 kg one degree by burning in an atmosphere of pure oxygen in a bomb calorimeter. It is often denoted by capitalising the C as **Calorie** and also known as **large calorie** or **grand calorie**. Calorific means relating to heat or calories.

CALORIFIC VALUE: The amount of heat generated by a given mass of fuel when it is completely burned or oxidised by aerobic respiration or the heat per unit mass released by complete combustion of a given substance. It is measured in joules per kilogram (J/kg) and used to express the energy values of fuels.

CALORIMETER: An apparatus consisting of a thin-walled container, used to hold liquids and objects in heat experiments for measuring thermal properties such as calorific value, specific heat capacity, specific latent heat, etc., or for measuring the amount of heat given out or absorbed by a body during processes such as combustion, change in heat energy and change of state. It may be an electromagnetic calorimeter (which is designed to measure energy of particles that interact primarily via electromagnetic interaction) or a hydronic calorimeter (which is designed to measure particles that interact via the strong nuclear force). The various types include the bomb calorimeter, the



A simple Calorimeter for measuring the Specific Heat Capacity of a solid

coffee-cup calorimeter and the Bunsen ice calorimeter (that is used to measure the specific heat capacity of solid in liquid when they are in smaller quantities). A simple calorimeter is made of a heavy copper vessel, polished to reduce heat losses by radiation and covered with insulating material to reduce heat losses by conduction and convection. Sometimes a calorimeter is calibrated to measure the specific heat capacity of a substance. In the diagram below, the deductions or calculations which follow the experiment will be from the fact that: heat lost by the hot solid test material will be equal to the heat gained by the water and the calorimeter, if none of the heat is lost to the surrounding.

CALORIMETRIC TITRATION: A titration performed in a calorimeter which produces a plot of heat change versus volume of titrant. This is the preferred term for experiments in which the main goal is the measurement of thermodynamic parameters. Because such experiments may be performed in any of a variety of types of calorimeters with varying degrees of calorimetric accuracy, it is important that the reporting of such results include an assessment of accuracy and precision. The term stands in contrast to the main emphasis in enthalpimetric analysis.

CALORIMETRY: An experimental method used for the determination of enthalpy change - a science of measuring a quantity of heat, expressed in joules; as distinct from thermometry, the science of measuring the intensity of heat expressed as temperature.

CALOTROPIS: A genus of sub succulent flowering plants in the Dogbane family, Apocynaceae, with the common name milkweed, as all parts exude a milky-white sap which is sticky and hard to wash off. It grows widely in North Africa, Tropical Africa, Western Asia, South Asia and Indochina. It has oval, light green leaves and milky stem, with clusters of waxy flowers that are either white or lavender in colour. The green fruit is globe-like, spongy and hollow but the flesh contains a toxic milky sap that is extremely bitter. Some of the species include *Calotropis procera* with the common names swallow-wort, apple of Sodom, Sodom apple; and *Calotropis gigantea* with the common names gigantic swallow wort, giant milkweed or crown flower. The flowers, bark and the milky latex have high medicinal value and used in the treatment of ailments such as diarrhoea, constipation and stomach ulcers; painful conditions including toothache, cramps and joint pain; and parasitic infections including elephantiasis and worms. Some others include syphilis, boils, inflammation (swelling), epilepsy, hysteria, fever, muscular spasm, warts, leprosy, gout, snakebites and cancer.

CALOTYPE: A paper-based photograph made by using a wax paper negative, allowing multiple prints to be made from a single

photographic image. It is also a photographic process of producing a negative on a plate wetted with silver iodide, now superseded by methods using glass plates.

CALRETICULUM: A protein that acts as a molecular chaperone for certain immunoglobulin-like proteins in the endoplasmic reticulum of cells.

CALUTRON: An electromagnetic apparatus for separating isotopes according to their masses.

CALVE: The process of giving birth by a cow or other similar animals. **Calver** is a cow which has had calves.

CALVIN CYCLE (PHOTOSYNTHETIC CARBON REDUCTION CYCLE; BENSON-CALVIN-BASSHAM CYCLE; CALVIN-BENSON CYCLE): The metabolic pathway of the light-independent stage of photosynthesis, in which carbon dioxide is reduced to carbohydrate using adenosine triphosphate (ATP) and NADPH derived from the light-dependent reactions. The Calvin cycle occurs in the chloroplast stroma and begins with the carboxylation and cleavage of ribulose biphosphate (RUBP) to form two molecules of phosphoglyceric acid. The RUBP is then regenerated by a complex series of reactions involving 3-, 4-, 5-, 6- and 7-carbon sugar phosphates. During this process glucose is formed by reversal of the glycolytic.

CALX: A powdery metal oxide formed by heating an ore in air.

CALYBIUM: A hard, one-loculed dry fruit derived from an inferior ovary. An example is the acorn of oak (*Quercus*), in which the actual nut is in a cup-shaped involucre of imbricate (overlapping) scales.

CALYCLATE: A flower with small bracts around the calyx, which appears as a false outer calyx, as found in *Tridax* species.

CALYPTRA: 1. A layer of cells that cover the developing sporophyte of bryophytes and pteridophytes. In mosses, it forms a hood over the capsule and in liverworts, it forms a sheath at the base of the capsule stalk. 2. Also known as **root cap**, it is a hood-shaped cap of parenchymatous cells at the tip of the root that protects the root meri-stem from abrasion as it grows through the soil. It is derived from a separate meristem, the calyptragen and may be augmented by periclinal divisions of the cells at the surface of the root.

CALYPTROGEN: The region within the root apical meristem that divides to produce the root cap (calyptras). It occurs in grasses and many other plants.

CALYX: A collective term for the sepals of a flower, forming the outermost whorl or a ring of leaves (sepals) enclosing an unopened flower bud that protect them while in bud.

CALYX LIMB (CALYX LOBE; CALYX TOOTH): One of the free portions of a calyx of fused sepals as found in the flowers of *Allamanda*.

CALYX TUBE: The tube-like united portion of a calyx of united sepals.

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CAM: 1. The part of a machine, taking various forms that converts circular motion to linear motion or vice versa or an evenly shaped wheel that rotates and makes an object pressing against its side move back and forth. Cams, for example, open and close the valves in petrol engine. Cams are used in automatic machine tools, printing machines, sewing machines and textile machinery. 2. Abbreviation for **computer aided manufacturing**, which is the use of computers to control production processes.

CAM PLANT: Any of a group of succulent plants that employ crassulacean acid metabolism (CAM) for fixing atmospheric carbon dioxide.

CAMBIUM: A cylindrical tissue layer composed of cells capable of active cell division to form new layers of tissue, producing xylem to the inside of the plant and phloem to the outside for plant growth. It lies between the bark and wood of the stem and forms the annual rings of woods. It originates in all secondary growths in plants and is most active in woody plants. Some types of cambium found in plant stems and roots include the following: **cork cambium**, which is a tissue found in many vascular plants as part of the periderm, creating cells that eventually become bark on the outside and cells that add to the cortex on the inside; **unifacial cambium**, which produces

cells to the interior of its cylinder; and **vascular cambium**, which is a lateral meristem in the vascular tissue of plants that carry water and nutrients throughout the plant.

CAMBRIAN: The earliest geological period of the Palaeozoic era. It is estimated to have begun about 570 million years ago and lasted for some 100 million years. During this period, invertebrate animal life, including trilobites, appeared and marine algae developed.

CAMBRIAN EXPLOSION: A relatively short interval of rapid intense evolution that supposedly occurred in the early to mid Cambrian period. The supposition is based on the sudden appearance in the fossil record from this time of many diverse and novel forms, particularly marine animals, among which can be found representatives of all major groups.

CAMCORDER (CAMERA RECORDER): A portable video camera containing both the image sensor and recording media in a single unit. Camcorders contain three major components, which are: (i) the lens, which gathers and focuses light on the imager; (ii) the imager, usually a charge-coupled-device (CCD) or complimentary metal-oxide semiconductor (CMOS) sensor on modern camcorders used to convert light into electrical signals and record the resulting images; (iii) the recorder, which converts the electric signal into video and encodes it to be stored. Camcorders also have viewfinders, which allow users to see what they are filming.

CAMEL: An animal classified as an ungulate and a member of the order Artiodactyla and family *Camelidae* with a long neck and one or two humps on its back, used in desert countries for transportation. Camels have three-chambered stomachs and digest their food in two steps. The camel can survive about three weeks without drinking, depending on the water content of its food and also a water loss of about 40 percent of its normal body weight.

CAMELID: A member of the family *Camelidae* that includes camels, llamas and their relatives, classified as ungulates.

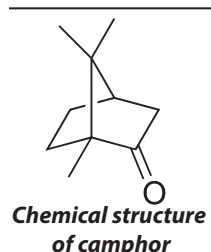
CAMERA: An optical instrument for forming and recording images. It consists of a lens system set in a light proof box inside of which a sensitised film or plate can be placed for photographing. Still video and digital cameras use electronic sensors instead of film and store the image in solid-state memory or on magnetic media or optical disks. Motion picture or cine cameras record movement at regular intervals in a series of frames, which are projected on a screen to create an illusion of movement. Television and video cameras record movement electronically for broadcast and storage on magnetic media or optical disks. Camcorders are video cameras which contain both the image sensor and recording media in a single unit.

CAMOUFLAGE: Colours or structures that allow an animal to blend with its surroundings to avoid detection by other animals. The leaf insect is a notable example. It is flat and green and the female has large leathery forewings with markings that resemble a pattern of vein leaves.

CAMPANULATE: In botany, it refers to parts of plants such as the calyx and corolla that are bell-shaped.

C A M P H O R
(1,7,7-TRIMETHYLBICYCLO[2.2.1] HEPTAN-2-ONE):

A volatile white crystalline bicyclic saturated terpenoid ketone, with the chemical formula $C_{10}H_{16}O$. It is obtained naturally from the wood or leaves of the camphor tree, *Cinnamomum camphora* or synthesised. It is distilled from chips of the root, stem or bark and is used in insect repellents, medicinal inhalants, liniments and in the manufacture of celluloids.



CAMPODACTYLY: A dominant trait in which a muscle is improperly attached to bones in the little finger, causing the finger to be permanently bent.

CAMPTODROMOUS: A term used to describe a form of leaf venation in which the major or secondary veins arising from a single primary

vein come close to the margin, but bend before they get to it. The secondary veins may join together in a series of conspicuous marginal loops (brochidodromous venation, for example, in wild cherry, *Prunus avium*) or may remain separate (eucamptodromous venation, as in downy rose, *Rosa tomentosa*).

CAMPUS AREA NETWORK (CORPORATE AREA NETWORK, CAN): A network of multiple interconnected local area networks (LAN) in a limited geographical area. A CAN is smaller than a wide area network (WAN) or metropolitan area network (MAN). The networking devices will normally be entirely owned by the organisation, college or university.

CAMPYLODROMOUS: A form of leaf venation in which several primary veins originate at the base of the lamina and run in strongly developed, basally recurved arches which converge towards the apex, reaching it or not, as seen in black bryony (*Tamus communis*).

CAMPYLOTROPOUS OVULE: A form of ovule orientation in which the ovule develops horizontally and the funiculus appears to be attached half way between the chalaza and the micropyle. This arrangement is not very common but may be found in the *Malvaceae* and *Caryophyllaceae*.

CAMWOOD (AFRICAN SANDALWOOD; *Baphia nitida*): A shrubby tree, belonging to the family Fabaceae or Leguminosae, growing under bigger trees in rainforests. It is native to West Africa, where it is widespread. The branchlets are smooth and the leaves are simple, alternately arranged, oblong-elliptic, 10-15 centimetres long and strongly tipped. The flowers are bisexual, white with a yellow centre, solitary or up to 4 grouped together on the main branch, faintly fragrant and inconspicuous. The fruits are straight pods, 10-15 centimetres long and 12-16 millimetres wide, sharply pointed at both ends and with 2-4 brown, flat seeds. Its wood is of a very fine colour and is used for making knife handles, violin bows, umbrella handles, walking sticks, table legs, etc. It contains santalin, used as red dye, but is not permanent. Traditionally, the dye is used medicinally to treat sprains, swollen joints and skin diseases. The plants can be used as hedge plants.

CANADA BALSAM (CANADA TURPENTINE; BALSAM OF FIR): A yellowish tinted resin used for mounting specimens in optical microscopy. It is obtained from the resin of the balsam fir tree (*Abies balsamea*). Dissolved in oil it forms a viscous colourless or yellow liquid whose refractive index ($n = 1.55$) is very close to that of glass. It finds use in optics as invisible glue for glass, in oil painting to achieve glow and facilitate fusion and also used to repair scratches in glass.

CANAL: Artificial water way constructed for drainage, irrigation or navigation.

CANAL RAYS: Particles of positive ions produced in a discharge tube by boring holes or canals in the cathode.

CANALICULATE: Having longitudinal channels or grooves usually in relation to petioles or midribs.

CANALICULUS (CANALICULI): A small passageway or channel in the body. Examples include bile canaliculi carrying bile to the bile duct; bone canaliculus, a small channel found in ossified bone; lacrimal canaliculi that are several small ducts in the eye; and dental canaliculi that supply the blood within a tooth.

CANALISATION: A developmental mechanism that limits variation of the phenotype within narrow bounds by repressing underlying genetic variation. It maintains a fairly uniform phenotype over a range of different environments in which the organism might normally occur. There are two types: genetic and environmental canalisations. Genetic canalisation refers to distinct genotypes producing the same phenotype, while environmental canalisation is the same genotype producing the same phenotype in spite of environmental variation.

CANARIUM: A genus of about 75 species of large evergreen tropical and subtropical trees in the family Burseraceae, which grow up to 50 m tall, found in tropical Africa, southern Asia and Australia, India,

southern China, Indonesia and the Philippines. They have alternate, pinnate leaves native. The genus contains 8 species producing edible kernels known commonly as galip nut (*Canarium indicum*). The species, *Canarium schweinfurthii* known commonly as African canarium, African elemi, bush candle tree, gum resin tree, incense tree or purple canary tree is a large deciduous tree reaching up to 50 metres high and up to 2 metres in diameter with spreading crown reaching to the upper canopy of the forest. It is widely distributed in East, Central and West Africa. The bole or trunk is straight and cylindrical. The bark is rough, whitish grey with creases and rough ridges, flaking off in rectangular scales in old trees, exuding a copious fragrant oily resin that solidifies to paraffin-like consistency. The leaves are pinnate, clustered at the end of the branches, with 8-12 pairs of leaflets, mostly opposite, oblong, cordate at base. The creamy white unisexual flowers grow in inflorescences that stand in the axils of the leaves. The fruit is a small drupe, bluish-purple, glabrous. It contains a thin pulp and a hard fluted spindle-shaped stone. The oil contained in the pulp is about 71% palmitic acid and 18% oleic acid. The seed-kernels are also oily and edible and contain several fatty acids including oleic (36%), linoleic (28%), palmitic (26%), stearic (7%). The bark exudes a heavy, sticky oleoresin that smells like turpentine and solidifies to a whitish resin, used as an illuminant, incense and releases a pleasant lavender-like smell. Some other species include *Canarium acutifolium*, *Canarium australasicum*, *Canarium batjanense*, etc.

CANCELLATE: A structure or tissue, latticed with a fine, regular, reticulate pattern as found in the tissue of xylem.

CANCELLATION: The method of simplifying an expression by eliminating terms, for example, by dividing the numerator and the denominator by a common factor. For example, dividing both the numerator and the denominator of $\frac{9}{24}$ by 3 gives $\frac{3}{8}$. Identical terms can be simplified by removing from both sides of an equation by subtraction or division, for example, $3x + y = y + 6$ can be simplified to $3x = 6$ and again to $x = 2$, by cancellation of a common factor on both sides.

CANCELLATION LAWS: The **cancellation law of addition** states that for any real numbers a, b, c , if $a + b = a + c$, then $b = c$. In other words, any real number can be added on both sides of an equation and the equality will still hold. The **cancellation law of multiplication** says that both sides of an equation can be multiplied by any real number and the equation will still hold. For example, in a binary operation $*$ on the set Z , the cancellation law states that

(i) if p, q and r are in z and $p * q = p * r$, then $q = r$; (ii) If $q * p = r * p$, then $q = r$. The law holds for addition, subtraction, multiplication and division of real and complex numbers, except for multiplication by zero and division of zero by another number. The law is not valid for some algebraic structures.

CANCER: A class of diseases caused by mutations in the cells of an organism or a group of diseases characterised by abnormal proliferation of cells; invasion or intrusion on and destruction of adjacent tissues; and sometimes metastasis or spreading to other locations in the body via lymph or blood. Cancer cells are usually degenerate, capable only of reproducing themselves. There are more than 100 types of cancer cells.

CANDELA: The base unit of luminous intensity or the power emitted by a light source in a particular direction. It measures the brightness of a light itself rather than the amount of light falling on an object (**illuminance**). In other words, the candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and has a radiant intensity of $1/683$ per steradian in that direction. Its symbol is cd.

CANDIDIASIS (MONILIA; VAGINAL THRUSH): A mucous membrane infection with the yeast-like fungus *Candida albicans* that inhabits the human digestive tract, the vagina and also the skin

in small amounts. This is a common cause of vaginal infections in women. Its main symptoms include intense itching and creamy white discharge. Also, called **oral candidiasis**, **thrush** or **oral thrush** it is a mucous membrane infection affecting the mouth, tongue and throat of children and adults with reduced immune function, especially those with diabetes or kidney failure. It is characterised by small creamy lesions or eruptions in the affected area.

CANDLE POWER: A unit of measuring light intensity, but now replaced by candela.

CANDLING: 1. The technique of using strong shaded light to detect unfertilised egg or dead embryo in an egg. 2. In ceramics, candling is a method used to slowly heat up the kiln and the ware inside the kiln in order to dry the ware or preheat before firing.

CANE: 1. A hollow jointed stem of certain plants, such as bamboo or sugar cane or such stems. It is used as material for making, for example, furniture and baskets. 2. A thin rod used for supporting plants or as a whip.

CANE RAT (GRASSCUTTER; CUTTING GRASS): Heavily-built burrowing rodent (*Thryonomys swinderianus*) belonging to the family Thryonomyidae with coarse hair, small eyes and ears. Its body length is usually 35 - 60 cm, with a tail length of 65 - 260 mm. An adult's average is about 3.5 - 4.5 kg. It is widely distributed in sub-Saharan Africa, where it is eaten in some African countries. It feeds on many crops and grasses such as sugar cane and is a pest to many crops.

CANE SUGAR (SUCROSE; BEET SUGAR; SACCHAROSE): A nonreducing disaccharide of glucose and fructose linked through a high-energy bond between carbon two of fructose and carbon one of glucose. Sucrose is the major transport sugar in higher plants. It is formed in the chloroplasts and transported through the phloem to other organs and tissues, where it is either metabolised for energy or utilised in the synthesis of reserve and structural polysaccharides.

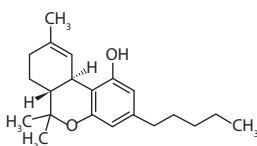
CANESCENT: Grey or white appearance due to a covering of short, fine grey or white hairs.

CANINE: 1. A sharp-pointed tooth found at the front of the mouth, between the incisors and premolars of mammalian carnivores. They are used for catching and killing prey as well as piercing and tearing food. They are absent in herbivores such as rabbit and sheep and are much reduced in humans but are large and highly developed in carnivores for tearing meat. 2. Any domestic or wild dog or doglike mammal such as wolf, jackal and fox in the family Canidae. Canines are characterised by slender and long legs, with a long muzzle, bushy tail, erect pointed ears and well-developed canine and cheek teeth.



CANKER: 1. A general term for a large number of different plant diseases that destroy the woods of plants and trees due to attack by bacteria, fungi mycoplasmas and viruses. 2. A disease that causes ulcerous sores on the ears of animals especially dogs and cats. It may affect horses making their hooves spongy; and also causing an infection in the throats of some birds.

CANNABINOIDS: Structurally related phenolic constituents of the plant *cannabis sativa* or generally referring to compounds not necessarily structurally related but which are known to elicit physiological action through interaction with the cannabinoid receptor found in the brain and the spleen of mammals. Tetrahydrocannabinol (THC) is the most notable cannabinoid with the structure in the diagram.



Chemical structure of THC, a cannabinoid

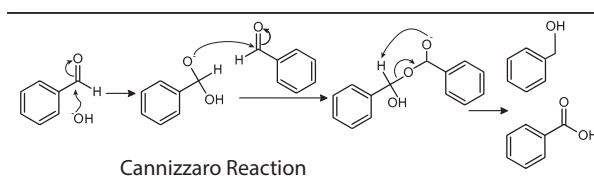
CANNABIS: The hemp plant *Cannabis sativa* belonging to the family Moraceae; or any of the various drugs made from flowers of the hemp plant that are smoked or chewed for their intoxicating effect. India hemp is the only species of *Cannabis*. It is a coarse, tall, hairy annual

herb that provides fibre from its stems, oil from its seeds and drugs from its leaves and flowers.

CANNIBAL: An animal that feeds on its own species; a process of eating one another in a habitat in order to survive.

CANNING: A method of food preservation in which the food is processed and sealed in an airtight container. It was developed by the Frenchman Nicolas Appert in 1810. Canning involves cooking food, sealing it in sterile cans or jars and boiling the containers to kill or weaken any remaining bacteria as a form of sterilisation. Because varieties of food have varying degrees of natural protection against spoilage, it may be necessary that the final step occur in a pressure cooker. High-acid fruits like strawberries require no preservatives to can and only a short boiling cycle, whereas marginal fruits such as tomatoes require longer boiling and addition of other acidic elements. Low acid foods, such as vegetables and meats require pressure canning. Food preserved by canning or bottling is at immediate risk of spoilage once the can or bottle has been opened.

CANNIZZARO REACTION: A disproportionation reaction of aldehyde usually without alpha-hydrogen in presence of a base, named after its discoverer Stanislao Cannizzaro. The products are carboxylic acid and an alcohol. For aldehydes with a hydrogen atom alpha to the carbonyl, the preferred reaction is an aldol condensation originating from deprotonation of this hydrogen.



CANNULA: A tube through which is inserted into a bodily cavity, duct or vessel to drain fluid or administer a substance such as a medication.

CANONICAL: An expression in its standard form such as the simplest form of equation of a type of curve derived from a given equation by a suitable change of variables. For example, $y^2 = 4ax$ and $y^2 = 24ax$ are canonical forms of a parabola.

CANONICAL EQUATIONS OF MOTION (HAMILTON'S EQUATIONS OF MOTION): A set of first-order, highly symmetrical equations describing the motion of a classical dynamical system.

They are given by: $\dot{q} = \frac{\partial H}{\partial p}$, $\dot{p} = -\frac{\partial H}{\partial q}$, where H is the Hamiltonian function $p = \frac{\partial L}{\partial \dot{q}}$.

CANONICAL FORMS (CANONICAL STRUCTURE; CONTRIBUTING STRUCTURE): The different electronic structures which can be drawn to represent a molecule and which contribute to the total wave function of the molecule.

CANONICAL RATE CONSTANT: The rate constant for a system in which the reactants are in thermal equilibrium at a given temperature. In statistical mechanics, the expression canonical ensemble is used to refer to a closed system in thermal equilibrium, the species being present in a statistical distribution. The canonical or thermal rate constant can be obtained from the micro canonical rate constant by summing over the energy, taking into account the statistical distribution.

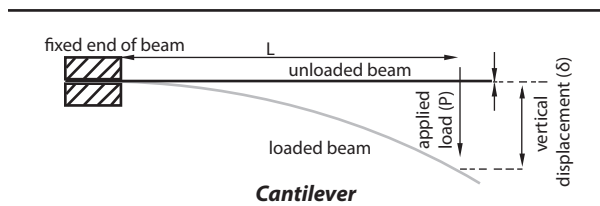
CANONICAL VARIATIONAL TRANSITION-STATE THEORY (CVTST): A development of transition-state theory in which the position of the dividing surface is varied so as to minimise the rate constant at a given temperature.

CANOPUS (ALPHA CARINAE): The second brightest star in the sky after Sirius, located in the constellation Carina, part of the constellation Argo Navis. It is about 120 light years from Earth and thousands of times more luminous than the Sun.

CANOPY: The over-hanging covering, especially of leaves as in the forest or uppermost level of greenery in a forest or woodland.

CANTHAROPHILY: Pollination of plants by beetles. Flowers of such plants tend to be large, greenish or off-white in colour and heavily scented. Scents may be spicy, fruity or similar to decaying organic material. They may also be flattened or dish shaped, with pollen easily accessible, although they may include traps to keep the beetle longer.

CANTILEVER: 1. A beam positioned horizontally supported at one end and carrying a load at the other end or distributed along the unsupported portion. The upper half of the thickness of such a beam is subjected to tensile stress, tending to elongate the fibres; and the lower half to compressive stress, tending to crush them. Cantilevers are employed extensively in building construction and in machines. In building, any beam built into a wall and with the free end projecting forms a cantilever. Longer cantilevers are incorporated in a building when clear space is required below, with the cantilevers carrying structures such as gallery, roof and canopy. 2. A form of wing construction in aircrafts in which no external bracing is used.



CANTOR SET (CANTOR'S TERNARY SET): Consider the closed set $[2, 3]$. When the open interval is removed the middle third obtained is $\left(1\frac{1}{3}, 2\frac{2}{3}\right)$. From each of the remaining intervals remove the open

interval that forms the middle third. The remaining set is the Cantor set when the process is continued indefinitely.

CANTOR SET THEORY (NAIVE SET THEORY): The presentation of the theory of sets in the style of an informal mathematical theory, regarding it as a body of given knowledge rather than the consequences of a set of uninterpreted axioms. In naive set theory, a set is a collection of objects called members or elements that is regarded as being a single object. An object x that is a member of a set A , is written as $x \in A$ and $x \notin A$ indicates that x is not a member of A .

CANTOR-BENDIXSON THEOREM: The theorem stating that the transfinite continuity of derived sets process terminates countably on the real line, in that every closed set in a Polish space is the disjoint union of a countable set and a perfect set.

CANTOR-BERNSTEIN THEOREM (CANTOR-SCHROEDER-BERNSTEIN): The theorem that establishes that two sets are equipollent if there is an injective mapping from each to the other. For example, if there is an injection for two sets $A \subseteq B$ and $B \subseteq A$, then the two sets are of the same cardinality: there is a bijection $A \cong B$. It can also be stated that if $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.

CANTOR, GEORG FERDINAND LUDWIG PHILIPP (1845 – 1918): A mathematician born in St. Petersburg in Russia. He spent most of his life at Halle University in Germany. He invented set theory and established the infinite notation. He was able to show that the set of rational numbers is denumerable. He went on to show that the set of real numbers is not denumerable. Finally, he developed the theory of infinite sets and transfinite numbers. He suffered from chronic mental illness during the later stages of his life.

CANTOR'S PARADOX: In logic, the paradox derived in Cantor's set theory from the supposition of an all-inclusive infinite set that every subset of such a set would be a member of it, but by Cantor's diagonal theorem, every set has strictly more subsets than it has members. There is thus no largest cardinal number. It was named after George Cantor (1845-1918), a German mathematician.

CANTOR'S DIAGONAL THEOREM (CANTOR DIAGONAL ARGUMENT; CANTOR'S DIAGONAL SLASH): The theorem of set theory that there are infinite sets which cannot be put into one-to-one correspondence with the infinite set of natural numbers, which means that the uncountably infinite set of real numbers is more than the countably infinite set of integers. This is proved by a diagonal process.

CANTOR'S INTERSECTION THEOREM: The theorem about compact sets that in a complete metric space, any bounded nonempty closed sets whose diameters decrease to zero contains a unique point of intersection. It was named after George Cantor (1845-1918), German mathematician.

CANYON (GORGE): A deep narrow valley or chasm running through hills or mountains often carved from the landscape by a river or stream. Examples are the Akosombo and Bui gorges in Ghana, where the Volta River runs through and has been used to run plants for hydroelectric power generation; and the Grand Canyon of the Colorado River in Arizona. A gorge may be formed as a waterfall retreats upstream, as a result of the collapse of caves or when the down cutting power of the river is greater than the processes of valley-wall erosion.

CAP: In mathematics, it is the symbol for set intersection, written as $S \cap T$ or S_i .

CAP CELL: A cell, seen in filaments of algae in the order Oedogoniales, that has one to several 'caps' of cell wall material at its anterior end. These result from a distinctive form of cell division that involves the formation of rings of new cell wall material prior to nuclear division.

CAPABILITY CLASS: A soil classification indicating the capability of land usage such as agriculture, forestry, recreation, or wildlife. It is the most intensive, but safe land use, designated by Roman numerals I - VIII and also Arabic numerals are used 1 - 8. In the capability system, all kinds of soil are grouped at three levels: capability class, subclass and unit (5). The soils are classed according to their limitations when they are used for field crops, the risk of damage when they are used and the way they respond to treatment.

CAPABILITY SUBCLASS: A soil group within the capability system designated by adding a small letter, e, w, s or c, to the capability class numeral, for example, IIe. It provides information on the kind of conservation problem or limitation. The letter e shows that the main limitation is risk of erosion unless close-growing plant cover is maintained. **Capability Unit** is a soil group within a subclass.

CAPACITANCE: An electrical property of a capacitor that determines how much charge can be stored in it for a given potential difference between its terminals. In other words, capacitance is the amount of electrical charge that a capacitor can store for each volt of applied potential or the ratio of the magnitude of charge, Q , on either plate to its electric potential V , between the plate. It is expressed as Capacitance (C) = charge stored/potential difference: $= \frac{Q}{V}$ and measured in farads

(the S.I unit of capacitance), with the symbol F . The farad is a very large unit and capacitors are more often rated in terms of microfarads ($1 \mu F = 10^{-6}$ Farad). A capacitor in an ac circuit limits the current flow, similar to a resistor. This applies to an isolated conductor and does not apply to a charged insulator because an insulator cannot have a potential since charge cannot flow across the insulator to equalise the potential.

CAPACITANCE HYGROMETER: A hygrometer, which uses the capacitance variations of a capacitor whose dielectric medium consists of the gas or of a material in contact with this gas.

CAPACITATION: The final stage in the maturation process of a spermatozoon in mammals inside the female reproductive tract as the sperm penetrates the ovum after ejaculation.

CAPACITIVE REACTANCE (X_C): The opposition of capacitance to alternating current, caused by the inductance and capacitance in a circuit rather than by resistance. It is equal to the reciprocal of the product of the angular frequency of the current times the capacitance.

Its symbol is X_C . The equation for calculating the amount of capacitive reactance in an ac circuit is given by: $X_C = \frac{1}{2\pi fC}$, where X_C is the capacitive reactance in ohms, f is the frequency in hertz and C is the capacitance in farads. A capacitor in an ac circuit limits the current flow similar to a resistor, therefore, applying Ohm's Law directly to a capacitor, gives $V_C = I_C \times C$, where V_C is the voltage across the capacitor, I_C the current through the capacitor and X_C the capacitive reactance. When capacitors are connected in parallel, the total charge, Q_t that is stored for the same potential difference, V , equals the sum of the charges stored on the individual capacitors in the circuit:

$Q_t = q_1 + q_2 + \dots + q_n$. For capacitors connected in series, the total charge $\frac{1}{Q_t} = \frac{1}{q_1} + \frac{1}{q_2} + \dots + \frac{1}{q_n}$. The total capacitance, $\frac{1}{Q_t}$, of capacitors in series is always less than the least capacitance in the circuit.

CAPACITOR (CONDENSER): A device for storing electric charge or energy for releasing at a specific time or rate. A capacitor consists of two plates the dielectric. Materials such as air, paper, mica and oil can be used as dielectrics. Most consist of two sheets of metal foil separated by a dielectric or insulator such as glass, paraffin, mica, oil, paper, tantalum or air and rolled into a tube. Capacitors are classified as **non-polarised** for use in alternating current (AC) circuits and **polarised** for use in direct current (DC) circuits. The capacitance of a capacitor depends primarily upon its shape and size and upon the relative permittivity ϵ_r of the dielectric between the plates. Capacitance can be increased by increasing the area of the plates, by decreasing their separation or by changing the dielectric. If an electrical potential (voltage) is applied to the plates of a capacitor, for example, by connecting one plate to the positive and the other to the negative terminal of a direct current source, the plates will become charged, one positively and one negatively. If the externally applied voltage is then removed, the plates of the capacitor remain charged. When power is required, the circuit is switched on to conduct current between the plates and the charge is released. Capacitors are used in electronic circuits for blocking direct current while allowing alternating current to pass; in filter networks, for smoothing the output of power supplies; in the resonant circuits that tune radios to particular frequencies; in electric power transmission systems for stabilising voltage and power flow and for many other purposes.

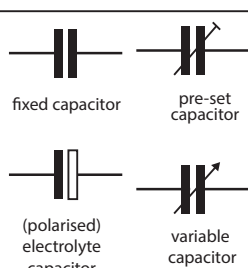
CAPACITOR MICROPHONE: A microphone consisting of a flexible metal diaphragm and a rigid metal plate that together form a two-plate air capacitor. Sound waves set the diaphragm in vibration producing capacitance variations that are converted into audio-frequency signals.

CAPACITY: In mathematics, it is the amount or volume that an object such as a container can hold, usually measured in millilitres (ml), litres (l) or cubic metre (m³) in the SI system.

CAPACITY OF A CUT: When all the maximum flows allowable across all edges coincide, the cuts are added. In graph theory, a cut is a partition of the vertices of a graph into two disjoint subsets.

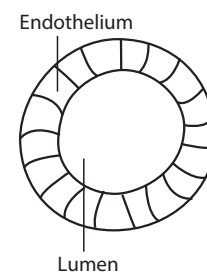
CAPACITY OF AN EDGE: The greatest flow possible along an edge. In graph theory, a cut is a partition of the vertices of a graph into two disjoint subsets.

CAPELLA (ALPHA AURIGAE): The sixth brightest star in the sky, 45 light years from the Earth.



The symbols for various types of capacitor

CAPILLARY: 1. In biology, it refers to tiny, thin walled or microscopic blood vessels which are about 0.008-0.02mm in diameter that form networks in-between cells of tissues and in contact with the walls of the cells, where the arterial and venous circulation meet for exchange of oxygen, nutrients and wastes with body tissues. They receive blood from the artery and pass it on to the veins. They are the narrowest blood vessels in the vertebrate circulatory system. 2. In physics, also called **capillary tube**, it is a narrow thick-walled tube with a small internal diameter, that holds liquid by capillary action.



Vertebrate capillary

CAPILLARY ACTION (CAPILLARITY): The spontaneous movement of water and other liquids inside a tube or capillaries with a small internal diameter; or the phenomenon in which a liquid's surface rises, falls or becomes distorted in shape where it is in contact with a solid. It is caused by the difference between the relative attraction of the molecules of the liquid for each other and for those of the solid. The narrower the tube, the higher the water rises. If a glass tube is inserted into water it rises up the tube but in case of mercury, the level of the liquid in the tube falls. Capillarity is one of the causes of the upward flow of water in the soil and in plants.

CAPILLARY ATTRACTION: A force by which a liquid is drawn along a very narrow tube. It is the force that allows a porous material to soak up a liquid.

CAPILLARY BED: A branching network of capillaries supplied by arterioles and drained by venules in a particular body region.

CAPILLARY COLUMN: A general term for columns having a small diameter. A capillary column may contain a packing or have the stationary phase supported on its inside wall. The former case corresponds to packed capillary column while the latter case corresponds to an open-tubular column. Due to the ambiguity of this term, its use without an adjective is discouraged.

CAPILLARY CONDENSATION: It occurs when, in porous solids, multilayer adsorption from a vapour proceeds to the point at which pore spaces are filled with liquid separated from the gas phase by menisci. The concept loses its sense when the dimensions of the pores are so small that the term meniscus ceases to have a physical significance. Capillary condensation is often accompanied by hysteresis.

CAPILLARY CONSTANT: A measure of surface tension defined, for two immiscible fluids in contact, by $2Tg(D-d)$, where T is the surface tension for the interface, g is the acceleration due to gravity and D and d are the densities of the fluids. Usually one of these is air. The capillary constant is equal to the rise of a liquid in a capillary tube multiplied by the diameter by the radius of the tube.

CAPILLARY ELECTROPHORESIS (CE): A technique for investigating mixtures of charged species in which electrically charged analyte species are separated in a conductive liquid medium under the influence of an electric field.

CAPILLARY FRINGE: A zone of essentially saturated soil just above the water table where groundwater seeps up by capillary action to fill pores. The size distribution of the pores determines the extent and degree of the capillary fringe. If pore size is small and relatively uniform, it is possible that soils can be completely saturated with water for some distance above the water table. When pore size is large, the saturated portion will extend only a few centimetres above the water table. In soils with a wide range in pore size, the unsaturated zone can be several times thicker than the saturated zone.

CAPILLARY POROSITY: The small pores or the bulk volume of small pores that hold water in soils against a tension usually greater than 60 cm of water.

CAPILLARY TUBE: A narrow, thick walled tube with a very small internal diameter so that capillary attraction of a liquid into the tube is significant. It is usually made of glass and used in thermometer and similar pieces of equipment. Properties of fluids such as surface tension and viscosity can be studied using capillary tubes.

CAPILLARY WATER: Water held in the small pores of a soil by particles by capillarity which is available for plant use, usually with a tension greater than 60 cm.

CAPILLARY WAVE (CAPILLARY RIPPLE; RIPPLE): A wave occurring at the interface between two fluids, such as the interface between air and water on oceans and lakes, in which the principal restoring force is controlled by surface tension.

CAPITAL: The upper part of an architectural pillar or column, on top of the shaft and supporting the entablature.

CAPITATE: In botany, it refers to a structure with rounded dead or collected in a head, as a dense terminal cluster of sessile or nearly sessile flowers in the Compositae (Asteraceae).

CAPITELLATE: A flower with small head-like structures or with parts in very small head-shaped clusters.

CAPITULUM: 1. Also known as **head**, it is a flattened or rounded type of inflorescence of numerous, small, stalk-less flowers, which is short and densely clustered to form a single flower-like structure. It is characteristic of plants of the family *Compositae*. Examples are dandelion, sunflowers, broccoli and cauliflower. 2. A rounded enlarged body part such as at the end of a bone or at the tips of an insect's antennae.

CAPON: Castrated male chicken used for meat. The castration improves its growth and the quality of its flesh, which is moister, tender and has more flavour than that of a hen or rooster. The process of turning a rooster into a capon is known as **caponisation**, which is done between six and twenty weeks of age. It may be done by surgically removing the rooster's testes or by injecting oestrogen implants. Due to the lack of sex hormones, capons are not as aggressive as normal roosters; making them easier to handle and allowing them to be kept together with other capons since their reduced aggressiveness prevents them from fighting.

CAPPIE: A disease of sheep, mainly of older lambs and young sheep, associated with thinning of the skull bones. In severe cases the animal cannot eat or close its mouth.

CAPROLACTAM (6-HEXANELACTAM): A cyclic amide of caproic acid with the chemical formula $(\text{CH}_2)_5\text{C}(\text{O})\text{NH}$, density 1.01 g/cm^3 , melting point 69.2°C (342.2 K) and boiling point 270.8°C (543.8 K) at 1013.25 hPa . It is the precursor to Nylon 6, a widely used synthetic polymer. In water, caprolactam hydrolyses to aminocaproic acid, which is used medicinally.

CAPS LOCK: A feature of most keyboards which, when active, capitalises each letter typed on the keyboard. Unlike most keys on the keyboard, the caps lock key is a toggle key, which is always in either an on or off position.

CAPSICUM (CHILLI PEPPER; CHILE; PEPPER): A genus of annual plants in the nightshade family, *Solanaceae*, native of tropical areas consisting of five domesticated species and twenty-six wild species. *Capsicum annum* contains sweet (or mild) peppers and hot Chile peppers and is the most common and extensively cultivated of the five domesticated species. It is a many-branched plant, growing up to 75 centimetres, often shrubby in appearance. It has small white or

occasionally purple flowers borne singly or, rarely, in pairs. The pod-like green fruits are many-seeded berries that vary considerably in size and shape, ripening to yellow orange, red or purple. It is used as flavouring sauces and relishes, as topical analgesic for pain. Red chillies contain high amounts of vitamin C and carotene or provitamin A. Yellow and especially green chillies contain a considerably lower amount of both substances. *Capsicum chinense*, commonly known as goat pepper, habanero, Jamaican hot, etc. originated in the Amazon Basin and is now common throughout the Caribbean, Central and South America and in the tropics. It has rigid, upright stems, large, thin and wrinkled green leaves and 2-6 flowers per knot. The flowers are usually small and white with five petals. The size and shape of the peppers is very varied and can range from small and rounded to elongated, bonnet, bell or lantern shaped. These chillies can be harvested when green or they can be left on the tree and allowed to mature to full colour. The mature fruits have pungent flavour and fruity aroma and have a range of colours from sunshine yellow, fiery orange and bright red to deep purple and dark, almost-black chocolate brown depending upon the selection. *Capsicum frutescens*, also known as tabasco pepper or cayenne pepper has many stems with flowers that have greenish white corollas with no spots and purple anthers. The pods are often small, pointy and grow erect on the plants. *Capsicum pubescens* is the only domesticated *Capsicum* species with no wild form. It has a compact to erect habit, sometimes sprawling and vine like. The leaves are hairy. The flowers have purple corollas, purple and white anthers and stand erect from the leaves. The pods are normally pear- or apple-shaped with black seeds. *Capsicum baccatum* is the least cultivated species, consisting of the South American cultivars known as Aji's, with pods ranging from non-pungent to very hot and yellow or tan spots on the corollas (on the flowers) and by the yellow anthers.

CAPSID: The protein that encloses a virus, consisting of several oligomeric structural subunits made of protein called protomers. The chemical nature of the capsid is important in stimulating the body's immune response against invading virus.

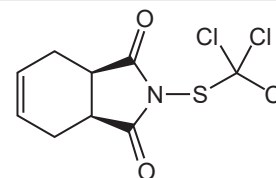
CAPSOMERE (CAPSOMER): Any of the protein subunits that make up the regularly organised outer coat of viruses.

CAPSULAR: Enclosed in or in the form of a capsule.

CAPSULE: 1. A dry, usually many seeded fruit formed from an ovary composed of two or more fused carpels, which splits open when ripe to release the seeds. 2. The spore-containing structures of mosses, ferns and liverworts, borne at the top of a long stalk and may rupture by a variety of mechanisms. 3. A transparent layer of gelatinous or mucilaginous material that envelops some bacterial cells. The polysaccharides of the capsule are often characteristic of a species. Polypeptides may also be present. In some species the capsules hold together a number of bacteria, forming a structure termed a zoogloea. 4. The chamber of a spacecraft containing the astronauts; the only part that returns intact to Earth.

CAPTAN [(3AR,7AS)-2-[(TRICHLOROMETHYL)SULFANYL]-3A,4,7,7A-TETRAHYDRO-1H-ISOINDOLE-1,3(2H)-DIONE]:

A white powder fungicide, with the chemical formula $\text{C}_9\text{H}_8\text{O}_2\text{NSCl}_3$, molar mass $300.59 \text{ g mol}^{-1}$, density 1.74 g/cm^3 and melting point 178°C , 451 K (decomposes). It is used on fruits, flowers and vegetables. It is also used in seed dressings for peas and other vegetables.



Chemical Structure of Captan

CAPTCHA (COMPLETELY AUTOMATED PUBLIC TURING TEST TO TELL COMPUTERS AND HUMANS APART): A program that protects websites against bots by generating and grading tests that humans can pass but current computer programs cannot. It is used to verify that a human, rather than a computer, is entering data. Captchas are commonly seen at the end of filling online forms, which ask the

user to enter text from a distorted image. The text in the image may be wavy, have lines through it or may be highly irregular, making it nearly impossible for an automated program to recognise it.

CAPTODATIVE EFFECT: The effect on the stability of a carbon-centred radical determined by the combined action of a captor (electron withdrawing) and a dative (electron releasing) substituent, both attached to the radical centre. The term is also used for certain unsaturated compounds.

CAPTURE: 1. Any of various processes in which a system of particles absorbs an extra particle or gains an additional particle. 2. The process in which a spacecraft or a natural object, such as an asteroid, comes under the influence of a more massive celestial body's gravity. 3. To record and store data on a computer.

CAPTURE AREA: The total surface area of a telescope or interferometer, upon which radiation falls.

CARAMEL: A complex mixture of substances produced by heating any of a variety of sugars slowly to around 170 °C (443 K), without charring, until they turn brown. As the sugar melts, the molecules break down and re-form into compounds with a characteristic colour and flavour. It is used in colouring and flavouring in foods.

CARAPACE: 1. The dorsal part of the exoskeleton of some crustaceans such as lobsters, shrimps, crayfish and crabs, which extends like a shield over revealed segments of the head and thorax. 2. The dorsal part of the shell of a chelonian reptile (tortoise).

CARAT (KARAT): 1. A unit for measuring the mass of all gemstones except pearls. It is equal to 200 mg and is part of the tray system of weights. 2. A unit of purity or fineness of gold. Pure gold, for example, is 24 carat; 22 karat is the purest used in jewelry, consisting of 22 parts gold and two parts alloy to give greater strength; 18 karat is 75% gold.

CARATHEODORY MEASURABLE: Of a set A with respect to an outer measure, μ^* , having the property that for all B , $\mu^*(B) = (\mu(A) + \mu(B))$.

CARATHEODORY OUTER MEASURE (OUTER MEASURE): A set function constructed preliminary to and sharing many of the properties of a measure. Consider an outer measure μ defined on the class $P(X)$ of subsets of a metric space (X, d) , μ is a Caratheodory outer measure, if $\mu(A \cup B) = \mu(A) + \mu(B)$ for every for every pair of sets $A, B \subset X$ which has positive distance $\inf\{d(x, y) : x \in A, y \in B\} > 0$. A theorem due to Caratheodory shows then that the Borel sets are μ -measurable.

CARATHEODORY, CONSTANTIN (1873 - 1950): Greek-German mathematician who worked as an engineer in Egypt and went on to study mathematics in 1900 at the University of Berlin and at the University of Göttingen, where he obtained his Ph.D. in 1904. He was appointed professor of mathematics at the University of Munich in 1924. His most important work was in the calculus of variations, but he also made important contributions to the theory of functions of several variables, measure theory, thermodynamics and relativity.

CARATHEODORY'S EXTENSION THEOREM: The theorem that if μ is a measure on an algebra A and μ^* is an outer measure of μ , then the collection A^* of μ^* -Caratheodory measurable sets is a sigma-algebra and the restriction of μ^* to A^* is a measure that extends μ . This theorem makes it possible to obtain Lebesgue measure from length measure on the half-open intervals of the real line.

CARATHEODORY'S THEOREM: The theorem that in Euclidean space every point in the convex hull of a given set S lies in some simplex with vertices in S . In other words, the theorem states that if a point x of R^d lies in the convex hull of a set P , there is a subset P' of P consisting of $d + 1$ or fewer points such that x lies in the convex hull of P' . Equivalently, x lies in an r -simplex with vertices in P , where $r \leq d$. For example, consider a set $P = \{(0,0), (0,1), (1,0), (1,1)\}$ which is a subset of R^2 . The convex hull of this set is a square. Consider now a point $x = (1/4, 1/4)$, which is in the convex hull of P . We can then construct a set $\{(0,0), (0,1), (1,0)\} = P'$, the convex hull of which is a

triangle and encloses x and thus the theorem works for this instance, since $|P'| = 3$.

CARAWAY: Seeds of a herb, *Carum carvi* used as a flavouring in bread and pastry.

CARBAMATES: Salts or esters of carbamic acid, H_2NCOOH or their derivatives, including some insecticides, which exert their effect by temporarily inhibiting cholinesterase activity.

CARBANION: 1. A transient negatively charged organic ion that has one more electron than the corresponding free radical. 2. A negatively charged carbon atom usually with three substituents for a total of eight valence electrons. The carbanion exists in a trigonal pyramidal geometry.

CARBAZOLE (DIPHENYLENIMINE; 9H-CARBAZOLE; 9-AZAFLUORENE; DIBENZOPYRROLE; DIBENZO[B,D]PYRROLE; DIPHENYLENEIMINE; DIPHENYLENIMIDE): A white crystalline compound, 9-azafluorene with the chemical formula $C_{12}H_9N$, melting point 244 °C (517 K) and boiling point (628 K). It is found with anthracene and derived from coal tar. Carbazole and its derivatives are widely used as an intermediate in synthesis of pharmaceuticals, agrochemicals, dyes, pigments and other organic compounds.

CARBENE: A highly reactive organic molecule in which the carbon atom has six electrons that do not form bonds, with the general formula $RR'C$. Carbenes are divided into two classes, singlets and triplets. Examples are methylene (H_2C) and dichlorocarbene (Cl_2C). Because of their extreme reactivity, carbenes exist only as transient intermediates and serve as useful intermediates in organic synthesis.

CARBIDE: An inorganic compound made of carbon and one other chemical element usually a metal, silicon or boron. Calcium carbide is useful as a source of acetylene. Carbides of tungsten, silicon and boron are extremely hard and refractory and used as abrasives and in cutting tools.

CARBINES: The neutral species $HC\equiv$ and its derivatives formed by substitution, in which a univalent carbon atom is covalently bonded to one group and also bears three nonbonding electrons.

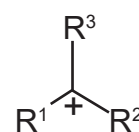
CARBO-REDUCTION: A process in which a metal oxide is reduced in the presence of carbon or a carbon-containing compound.

CARBOCATION: A short-lived positively charged carbon atom in an organic compound. Carbocations may be described as primary, secondary or tertiary depending on how many carbon atoms are directly bonded to the carbocation. A tertiary carbocation has three other carbon atoms directly bonded to it and it is the most stable while the secondary has two other carbon atoms directly attached to it. The primary carbocation which is the least stable has only one carbon directly attached to it. The stability of a carbocation is generally due to inductive effect, resonance effect and hyperconjugation.

CARBOCYCLIC COMPOUNDS: Cyclic compounds such as 1,2-dihydronaphthalene, in which all of the ring members are carbon atoms.

CARBOHYDRATE: A chemical organic compound composed of carbon, hydrogen and oxygen and related compounds with the same basic structure but modified functional group. Many carbohydrates have the general chemical formula $C_n(H_2O)_n$. The carbon (C) atoms are bonded to hydrogen atoms ($-H$), hydroxyl groups ($-OH$) and carbonyl groups ($-C=O$), whose combinations order and geometric arrangement lead to a large number of isomers. Carbohydrates include compounds such as glucose, fructose, cellulose, glycogen, chitin and inulin. Carbohydrates are the main source of energy for living things and the basic compound, hexose, is made by plants through photosynthesis.

CARBOLIC ACID: The former name of **phenol**, also known as **phenolic acid**, with the chemical formula C_6H_5OH , the simplest of



Carbocation structure

the phenols obtained from coal, wood tar or benzene. It is used to manufacture dyes, pharmaceuticals, antiseptics and disinfectants.

CARBON: A non-metallic element, with the chemical symbol C, atomic number 6, atomic weight 12.011, a sublimation point of above 3500 °C (3773 K) and boiling point of 4827 °C (5100 K). It occurs in all organic compounds and is therefore basic to the structure of all living organisms. It occurs in nature in almost pure form in diamond, graphite and fullerenes. It is also the major component of coal, petroleum, asphalt, limestone and most materials made by plants and animals. Carbon can form more compounds than any other element except hydrogen. A carbon atom can chemically combine with atoms of other elements, as well as with other carbon atoms, to form molecules. The uses of carbon and its compounds are varied and many. Some examples of carbon's uses are to form alloys with iron to produce carbon steel, as a lubricant and a pigment, as a moulding material in glass manufacture, in electrodes for dry batteries and in electroplating and electroforming, in brushes for electric motors and as a neutron moderator in nuclear reactors.

CARBON ARTEFACT (ARTIFICIAL CARBON ARTICLE; ARTIFICIAL CARBON BODY): A solid body which consists mainly of carbonaceous material in a distinct shape. Sometimes this term is also used for artificially (in the sense of technically) produced non-shaped carbon materials such as coke, blacks, etc. This application of the term carbon artifact is not recommended.

CARBON ASSIMILATION: The incorporation of carbon from atmospheric carbon dioxide into organic molecules, occurring during photosynthesis.

CARBON BISULFIDE (CARBON DISULFIDE): A colourless highly refractive liquid organic compound with the chemical formula CS₂. It is slightly soluble in water and soluble in ethanol and ether. Its odour when pure is chloroform-like. Its densities are 1.539 g/cm³ (−186 °C), 1.2927 g/cm³ (0 °C) and 1.266 g/cm³ (25 °C) with a melting point of −111.61 °C (161.54 K), and a boiling point of 46.24 °C (319.39 K). It can be prepared by direct reaction of carbon, for example, as charcoal, with sulfur. Pure carbon bisulfide has an ethereal odour but the commercial product is contaminated with a variety of other sulfur compounds and has a most unpleasant smell. Carbon bisulfide is an excellent solvent for oils, waxes, rubber, sulfur and phosphorus, but its use is decreasing because of its high toxicity and its flammability. It is used for the preparation of xanthates in the manufacture of viscose yarns.

CARBON BLACK: A form of carbon obtained by the incomplete combustion or thermal decomposition of hydrocarbons (natural gas or petroleum). It is used in the manufacture of ink, pigment and rubber etc.

CARBON BRUSH: A piece of carbon in electric motors, alternators and electric generators that conveys current between the rotor and the stator.

CARBON CENOSPHERES: Porous or hollow carbonaceous sphere-like particles (frequently in the size range of a few to several hundreds of μm diameter) formed during pyrolysis, also in the course of combustion, of carbonaceous liquid droplets (e.g. heavy fuel) or solid particles (e.g. coal).

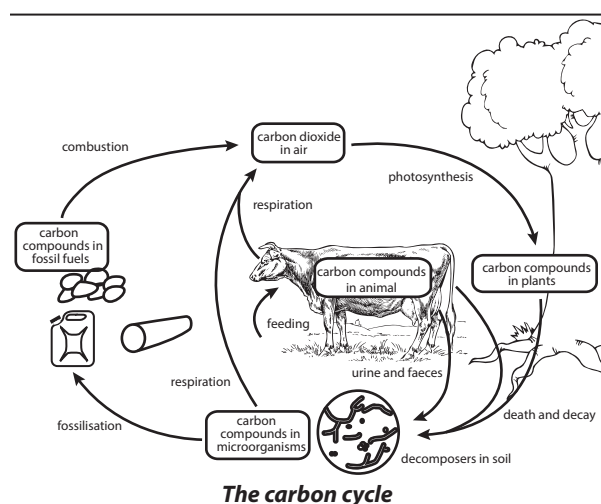
CARBON CLOTH: A textile material consisting of carbon fibres oriented at least in two directions. Carbon cloth is not necessarily woven.

CARBON COPY (CC; FIRST CARBON COPY, FCC): A term used in information technology to refer to an e-mail copy of a note sent to an addressee other than the main addressee. A **blind carbon copy** (BCC) is a copy sent to an addressee that is not visible to the main and carbon copy addressees.

CARBON CREDIT (CARBON TRADING): A system of credits that allows a company or country that reduces its carbon-dioxide emissions below a target level to sell the extra reduction as a credit to a company or country that has not met the target level, ratified in

conjunction with the Kyoto Protocol. Carbon credits and markets are key components of national and international attempts to mitigate the increase in concentrations of greenhouse gases. One carbon credit is equal to one ton of CO₂ or in some markets, carbon dioxide equivalent gases. For example, if an environmentalist group plants enough trees to reduce emissions by one ton, the group will be awarded a credit. If a steel producer has an emissions quota of 10 tons, but is expecting to produce 11 tons, it could purchase this carbon credit from the environmental group.

CARBON CYCLE: The sequence by which carbon circulates and is recycled through the natural world or the movement of carbon between carbon dioxide in the atmosphere, living organisms and fossil fuels, caused mainly by the metabolism of living organisms. The process is as follows: Green plants (producers) use CO₂ in photosynthesis to make carbohydrates, which in turn are used in the processes of metabolism to make all other compounds in their tissues and those of animals that consume them in carbohydrate, protein, and fat by most animals as well as plants that do not carry out photosynthesis and released to the atmosphere through respiration, the decay of animal and plant matter, etc;



CARBON DATING (CARBON-14 DATING): A method of calculating the age of once living or prehistoric objects or materials by measuring the decay of radioactive isotope of carbon (¹⁴C) in them, developed by U.S. physicist Willard Libby in 1947. When an organism dies it stops taking in carbon, so the amount of carbon-14 in its tissues steadily decreases. Because carbon-14 decays at a constant rate, the time since an organism died can be estimated by measuring the amount of radiocarbon in its remains. The method is a useful technique for dating fossils and archaeological specimens from 500 to 50,000 years old.

CARBON DIOXIDE (CO₂): A naturally occurring colourless, odourless, incombustible gas, present in the atmosphere (about 0.03 per cent) and absorbed by plants in photosynthesis. Atmospheric carbon dioxide is formed from multiple natural sources including volcanic outgassing, the combustion of organic matter, and the respiration processes of living aerobic organisms. It is also a by-product of burning fossil fuels and biomass, as well as land-use changes and other industrial processes. CO₂ is also produced from carbohydrates by fermentation, and by reaction of acid with limestone or other carbonates. CO₂ is essential to many biochemical and biological processes. It is the principal anthropogenic greenhouse gas that affects the Earth's radiative balance. It is the reference gas against which other greenhouse gases are measured and therefore has Global Warming Potential of 1.



Chemical structure of CO₂

CARBON DIOXIDE (CO₂) FERTILIZATION (CARBON FERTILIZATION EFFECT, CFE): The increased growth of plants and its corresponding higher yields as a result of increased atmospheric

carbon dioxide concentration due to rising anthropogenic emissions. In other words, the larger the amount of carbon dioxide concentration in the atmosphere, the more some plants flourish, as they use carbon dioxide during photosynthesis to enhance growth and crop yields.

CARBON DIOXIDE LASER (CO₂ LASER): A continuous or pulsed source of coherent radiation normally tunable through the CO₂ vibration-rotation band centred near 10.6 μ m.

CARBON DISULFIDE: A liquid organic compound, with the chemical formula CS₂. It is used as a solvent for oils, fats and rubber and in paint-removers.

CARBON ELECTRODE: An electrode for an electrical application. In its green state it comprises granular carbon material bound with pitch. The granular carbon material may be either needle coke, fine-grained or isotropic coke or reclaimed graphite powder. Electrodes for use in steel production can only be manufactured from needle coke and the green electrodes are fired at temperatures above 2800 K to produce highly graphite electrodes. The other granular carbon materials may be used for aluminium electrodes where the duty is not so severe and the green electrodes are generally fired to lower temperatures.

CARBON FELT: A textile material consisting of, in approximation, randomly oriented and intertwined carbon fibres. Carbon felts are usually fabricated by carbonisation of organic felts but they can also be produced from short carbon fibres.

CARBON FIBRE (GRAPHITE FIBRE; CARBON GRAPHITE): A material consisting of extremely thin fibres about 0.005–0.010 mm in diameter and composed mostly of carbon atoms that are bonded together in microscopic crystals that are more or less aligned parallel to the long axis of the fibre. It is produced by heat treatment from a special grade of courtelle acrylic fibre and it is used for reinforcing plastics. Several thousand carbon fibres are twisted together to form a yarn, which may be used by itself or woven into a fabric.

CARBON FIXATION: The conversion of carbon dioxide into organic compounds during photosynthesis. It represents the first stage of the dark reactions of photosynthesis, in which carbon dioxide from the air is combined with ribulose 1,5-bisphosphate.

CARBON FOOTPRINT: A measure of the environmental impact of an organisation, event, person or product. It measures the total greenhouse gas (GHG) emissions of a given activity or group of activities to provide a measure of the extent to which these activities produce harmful emissions and therefore contribute to global warming.

CARBON MIX: A mixture of filler coke, for example, grains and/or powders of solid carbon materials and a carbonaceous binder and selected additives, prepared in heated mixers at temperatures in the range of 410 – 445 K as a preliminary step for the formation of shaped green bodies.

CARBON MONOXIDE (CARBON(II) OXIDE): An inorganic compound, with the chemical formula CO. It is a highly toxic, colourless, odourless and flammable gas, formed when carbon is oxidised in insufficient oxygen. It is the result of incomplete combustion and it is present in car exhaust fumes and cigarette smoke. It is highly poisonous, forming a stable compound with haemoglobin in the blood and thus preventing haemoglobin from transporting oxygen to the body tissues. In industry, carbon monoxide is used as a reducing agent in metallurgical processes such as iron extraction.

CARBON NANOTUBE (BUCKYTUBE): In nanotechnology, it is a cylindrical carbon molecule that belongs to the fullerene structural family, which also includes buckyballs. Carbon nanotubes are essentially rolled-up sheets of graphite, a few nanometres in diameter and 1–100 microns in length. A nanotube cylinder typically has at least one end capped with a hemisphere of the buckyball structure.

CARBON PAPER: A thin paper, coated with carbon or some other coloured substance, which is placed between sheets of writing paper for producing copies of documents.

CARBON SEQUESTRATION: In ecology, it is the process of removing the atmospheric carbon dioxide (CO₂) content and storing it in a reservoir or carbon sinks (such as oceans, forests or soils) through physical or biological processes. Biological approaches to sequestration include direct removal of carbon dioxide from the atmosphere through land-use change, afforestation, reforestation, and practices that enhance soil carbon in agriculture. Physical approaches include separation and disposal of carbon dioxide from flue gases or from processing fossil fuels to produce hydrogen- and carbon dioxide-rich fractions, and long-term storage in underground in depleted oil and gas reservoirs, coal seams, and saline aquifers.

CARBON SINK: Anything that collects and stores carbon dioxide (CO₂) from the atmosphere. They include trees and other vegetation, forests and oceans. Plants collect carbon dioxide from the atmosphere to use in photosynthesis; some of this carbon is transferred to soil as plants die and decompose. The oceans are a major carbon storage system for carbon dioxide.

CARBON TAX (EMISSIONS TAX): A form of fee charged on the production, distribution or use of fossil fuels such as oil, coal and natural gas, based on how much carbon their combustion emits. A pollution tax on carbon could help cut greenhouse gas emissions.

CARBON TETRACHLORIDE: The former name for tetra chloromethane, with the chemical formula CCl₄. It is a volatile, colourless, poisonous, liquid organic compound that boils at 76.8°C (349.95 K). It is formed by the reaction of chlorine gas with methane with the help of ultra violet light. It is not flammable and is a good solvent for fats, oils and greases. It is often used commercially for dry cleaning and for degreasing metals.

CARBON TRADING (CARBON CREDIT): A system of credits that allows a company or country that reduces its carbon-dioxide emissions below a target level to sell the extra reduction as a credit to a company or country that has not met the target level, ratified in conjunction with the Kyoto Protocol. Carbon credits and markets are key components of national and international attempts to mitigate the increase in concentrations of greenhouse gases. One carbon credit is equal to one ton of CO₂ or in some markets, carbon dioxide equivalent gases. For example, if an environmentalist group plants enough trees to reduce emissions by one ton, the group will be awarded a credit. If a steel producer has an emissions quota of 10 tons, but is expecting to produce 11 tons, it could purchase this carbon credit from the environmental group.

CARBON-12 SCALE: The mass reference scale in which one atom of the ¹²C isotope is taken to have exactly twelve unit of atomic mass (12u), which is relative nuclidic mass unit.

CARBON-BASED ERROR: A term, which is used in computer science, which refers to a problem with a computer or a program that is caused by the user rather than the machine. All life on Earth depends on carbon, which forms the molecular bonds that make life possible. Therefore, to say that an error is carbon-based is another way of saying it is caused by a computer's human user. Carbon-based errors are a common problem for IT professionals, particularly when it comes to inexperienced users.

CARBON-CARBON COMPOSITE: A carbon fibre-reinforced carbon matrix material. The carbon matrix phase is typically formed by solid, liquid or gaseous pyrolysis of an organic precursor material. The matrix is either a graphitisable carbon or non-graphitisable carbon and the carbonaceous reinforcement is fibrous in form. The composite may also contain other components in particulate or fibrous forms.

CARBON-NITROGEN RATIO (CARBON-TO-NITROGEN RATIO; C/N RATIO; C:N RATIO): The ratio of the mass of organic carbon to the mass of total nitrogen in a particular organic substance, used as an indicator for nitrogen limitation of plants and other organisms. For example, a C:N ratio of 24:1, means that there are 24 units of carbon to 1 unit of nitrogen which is the proper carbon and nitrogen that a microorganism must have in order to sustain its

health. Also, for maintaining a compost pile in a good condition, the composting microorganisms, require the correct proportion of carbon for energy and nitrogen for protein production with C:N ratio of around 25-30:1. If the C:N ratio is too high (excess carbon), decomposition slows down. If the C:N ratio is too low (excess nitrogen) the compost pile becomes stinky. It is important that in selecting crops and cover crops rotation, C:N ratio of certain plants that may be included in the rotation should be considered. The following are the C:N ratios of a few crops: rye straw - 82:1, wheat straw - 80:1, oat straw - 70:1, young alfalfa hay - 13:1, mature alfalfa hay - 25:1, and alfalfa - 11.3. The ratios change as the plant grows, so the C:N ratio in a young plant will be much less than that of a matured plant. C:N ratio can identify whether molecules found in the sediment under study come from land-based or algal plants. It has a direct impact on residue decomposition and also nitrogen cycling in soils and can be used in analysing sediments and compost. It also serves as a tool for understanding the sources of sedimentary organic matter, which can lead to information about the ecology, climate, and ocean circulation at different times in Earth's history.

CARBON, NICOLAS LÉONARD SADI (1796-1832): French physicist and military engineer, who in 1824 published "Reflections on the Motive Power of Fire"; an analysis of the efficiency of heat engines, known as Carnot cycle. He died at an early age of cholera. He is often described as the "Father of thermodynamics", being responsible for such concepts as Carnot efficiency, Carnot theorem, Carnot heat engine and others.

CARBON(II) OXIDE: The International Union of Pure and Applied Chemistry's (IUPAC) name for carbon monoxide, with the chemical formula CO. It is an inorganic compound, highly toxic, colourless, odourless, flammable gas, formed when carbon is oxidised in insufficient oxygen. It is the result of incomplete combustion and it is present in car exhaust fumes and cigarette smoke. It is highly poisonous, forming a stable compound with haemoglobin in the blood and thus preventing haemoglobin from transporting oxygen to the body tissues. In industry, carbon monoxide is used as a reducing agent in metallurgical processes such as iron extraction.

CARBON(IV) OXIDE: The International Union of Pure and Applied Chemistry's (IUPAC) name for carbon dioxide with chemical formula CO₂. It is a colourless, odourless, slightly soluble in water, denser than air gas. It makes up 0.03% of air by volume and is the result of combustion. It is formed by the combustion of carbon and its organic compounds, by the reaction of acids on carbonates and as a product of fermentation and respiration. It is a raw material of photosynthesis in plants. It is used in carbonation of drinks, as a coolant in nuclear reactors, as solid carbon dioxide in fire extinguishers and as a refrigerant.

CARBONACEOUS FILM: A type of fossil found in sedimentary rock when organic material is buried in sediment, leaving a thin carbon film of residue.

CARBONACEOUS MESOPHASE: A liquid-crystalline state of pitch which shows the optical birefringence of disc-like (discotic) nematic liquid crystals. It can be formed as an intermediate phase during thermolysis (pyrolysis) of an isotropic molten pitch or by precipitation from pitch fractions prepared by selective extraction.

CARBONATE: An ion and an acid radical with the formula (CO₃)²⁻. It is formed when carbon dioxide dissolves in water to form H₂CO₃ (carbonic acid). The term also refers to any salt formed by this ion and another chemical element, usually a metal. Examples include calcium carbonate, sodium carbonate, etc. Alkali metal carbonates are stable when heated, but other carbonates decompose, releasing carbon dioxide. Carbonates also give off carbon dioxide when treated with dilute acids such as hydrochloric acid.

CARBONATE MINERALS: A group of common rock-forming minerals such as calcite, dolomite, siderite, rhodochrosite and magnesite containing the anion CO₃²⁻ as the fundamental unit in their structure.

CARBONATED: Containing carbon dioxide. Examples are fizzy or carbonated drinks. Carbonated beverages do not require pasteurisation. Carbonated water is water in which carbon dioxide is dissolved under pressure. It forms the basis of many fizzy or soft drinks such as soda water and lemonade.

CARBONATION: 1. A form of chemical weathering caused by weak carbonic acid which is formed when rain water absorbs carbon dioxide from the atmosphere. 2. Addition of carbon dioxide gas to a beverage, imparting sparkle and a tangy taste and preventing spoilage. The liquid is chilled and cascaded down in an enclosure containing carbon dioxide either as dry ice or a liquid under pressure. Increasing pressure and lowering temperature maximise gas absorption.

CARBONIC ACID (CARBON DIOXIDE SOLUTION; DIHYDROGEN CARBONATE; AERIAL ACID; HYDROXYMETHANOIC ACID): A weak acid made by dissolving carbon dioxide in water, with the chemical formula H₂CO₃, molar mass 62.03 g/mol and density 1.0 g/cm³ (dilute solution). It exists only in solution and decomposes readily into water and carbon dioxide. Carbonic acid forms carbonate and bicarbonate salts by reaction with bases. It contributes to the sharp taste of carbonated beverages.

CARBONIC ANHYDRASE: An enzyme, present in red blood cells and kidney cells that catalyses the reaction between carbon dioxide and water to form carbonic acid.

CARBONIFEROUS: A period of geological time 363-290 million years ago. It is the fifth period of the Palaeozoic era; between the Devonian and Permian periods. It lasted for nearly 64 million years during which deposition of plant remains later hardened into coal.

CARBONISATION (CARBONIZATION): A process by which solid residues with increasing content of the element carbon are formed from organic material usually by pyrolysis in an inert atmosphere. As with all pyrolytic reactions, carbonisation is a complex process in which many reactions take place concurrently such as dehydrogenation, condensation, hydrogen transfer and isomerisation. It differs from coalification in that its reaction rate is faster by many orders of magnitude. The final pyrolysis temperature applied controls the degree of carbonisation and the residual content of foreign elements, for example, T ~ 1200 K at the carbon content of the residue exceeds a mass fraction of 90 wt.%, whereas T ~ 1600 K at more than 99 wt.% carbon is found. To **carbonise** is to convert an organic compound into a carbon residue by heating, such as destructive distillation of coal in the absence of air or to coat something with carbon by heating.

CARBONITRILES: In systematic nomenclature, the suffix carbonitrile is used to name compounds RC≡N, where the suffix includes the carbon atom of the -CN. However, carbonitrile is not a class name for nitriles.

CARBONIUM ION: A positively charged fragment that arises from the heterolytic fission of a covalent bond involving carbon. It is a carbocation of the penta- or tetracoordinated nonclassical type such as an ion of the type R₅C⁺.

CARBONYL CHLORIDE (PHOSGENE): A highly toxic colourless gas, with the chemical formula COCl₂. It is produced industrially by passing purified carbon monoxide and chlorine gas through a bed of porous activated carbon, which serves as a catalyst: CO + Cl₂ → COCl₂ (ΔH_{rxn} = -107.6 kJ/mol). It is used in the manufacture of pesticides, plastics and dyes and was used as a chemical weapon in World War I. Inhalation causes severe lung injury several hours after exposure.

CARBONYL COMPOUNDS: Compounds containing the functional group R_nC=O, where R can be any element or group of atoms and n is a positive integer not greater than 2. It may be an inorganic compound such as nickel carbonyl (Ni(CO)₄) or organic compound such as an aldehydes R-CHO or a ketone R₂C=O. It is formed when carbon monoxide combines with a metal.

CARBONYL GROUP: A functional group that consists of an oxygen atom joined by a double bond to a carbon atom. The carbon atom is joined to the remainder of the molecule by two single bonds or one double bond. Typical examples of compounds containing the carbonyl

group are aldehydes $RCHO$, ketones $R_2C=O$, carboxylic acids $RCOOH$, esters $RCOOR$, amides $RCONH_2$ and acid anhydrides $(RCO)_2CO$.

CARBORANE: A compound derived from a polyhedral borane by substitution of carbon for boron atoms. Carboranes are relatively unreactive and thermally stable and form many derivatives, including highly stable organo-inorganic polymers.

CARBORUNDUM: A tradename for the hard substance silicon carbide (SiC), used as an abrasive for cutting, grinding and polishing. Carborundum is almost as hard as diamond.

CARBOXAMIDES: Amides of carboxylic acids, having the structure $RC(=O)NR_2$. The term is used as a suffix in systematic name formation to denote the $-C(=O)NH_2$ group including its carbon atom.

CARBOXAMIDINES: Compounds having the structure $RC(=NR)NR_2$. The term is used as a suffix in systematic nomenclature to denote the $-C(=NH)NH_2$ group including its carbon atom, for example, acetamidine ($CH_3C(=NH)NH_2$) and 2-butyl-4,5-dihydroimidazole.

CARBOXYDISMUTASE (RIBULOSE BIPHOSPHATE CARBOXYLASE; RUBP CARBOXYLASE): An enzyme that catalyses the carboxylation of ribulose biphosphate to two molecules of phosphoglyceric acid. This is the first step in the Calvin cycle. RUBP carboxylase is the fraction 1 protein of chloroplasts. It is very abundant in chloroplasts and may account for up to 15% of the total protein. This enzyme can also act as an oxygenase, forming a molecule of phosphoglyceric acid and a molecule of phosphoglycolic acid from ribulose biphosphate and oxygen. This reaction forms the basis of photorespiration.

CARBOXYHAEMOGLOBIN (CARBONYLHAEMOGLOBIN): A highly stable product formed in the blood when haemoglobin combines with carbon monoxide.

CARBOXYL GROUP: The acidic functional group ($-COOH$) consisting of a carbon atom joined to an oxygen atom by a double bond and to a hydroxyl group that determines the properties of organic acids and amino acids.

CARBOXYLASE: Any enzyme involved in the transfer of carbon dioxide or carboxyl group from a compound. An example is pyruvate carboxylase, which catalyses the formation of oxaloacetate from pyruvic acid. As with many carboxylases, this is an allosteric enzyme, its activity being increased by the binding of acetyl CoA.

CARBOXYLATE: An anion formed from a carboxylic acid.

CARBOXYLIC ACID: The organic acid containing the carboxyl group ($-COOH$) attached to another group which can be hydrogen or a large radical. It has the general formula $C_nH_{2n+1}COOH$, where $n = 0, 1, 2, \dots$. Examples are formic acid (methanoic acid) $HCOOH$, acetic acid (ethanoic acid) CH_3COOH and benzoic acid C_6H_5COOH . Some higher plant cells contain high concentrations of certain carboxylic acids. For example, citrus fruits are rich in citric acid and oxalic acid is present in high concentrations in rhubarb and tobacco leaves.

CARBOXYPEPTIDASE: Any enzyme that catalyses the hydrolysis of amino acid residue from the end containing the free carboxyl group. Pancreatic juice contains a carboxypeptidase that is secreted into the duodenum.

CARBOXYSOME: An organelle found in certain autotrophic bacteria that contains the carbon-fixing enzyme ribulose biphosphate carboxylase. They are made of polyhedral protein shells about 80 to 140 nanometres in diameter.

CARBURATION (CARBURETION): The mixing of a gas such as air with a volatile hydrocarbon fuel such as petrol, kerosene or fuel oil in order to form an explosive mixture. The energy released during combustion is used in internal engines.

CARBURETTOR: A device used in internal-combustion engines to control the metering, delivery and mixing of fuel in the air stream for combustion. It is also used to control the engine power. It may be classified as having either a fixed venturi, in which the diameter of the air opening ahead of the throttle valve remains constant or a variable venturi, which changes area to meet the changing demand.

CARBYLAMINE (ISOCYANIDE): A colourless liquid with a pungent odour that contains the chemical group $-NC$.

CARBONIUM IONS: The cationic species H_2C^+ or substitution derivatives thereof, formally derived by adding a hydron to a carbyne or subtracting an electron from a carbene.

CARCASS: A dead animal's body without the head and inner parts. The dead body of a human is referred to as a **cadaver** or a **corpse**.

CARCASS WEIGHT: The weight of an animal after slaughter and removal of internal organs, head, and the hide.

CARCERULUS: A dry fruit that is a type of schizocarp characteristic of members of the Malvaceae family. It consists of a number of one-seeded portions that adhere to a central axis.

CARCINOGEN: A substance capable of causing cancerous growth in living cells or any agent that increases the chances of a cell becoming cancerous. Exposure to one or more carcinogens, including certain chemicals, radiation and certain viruses, can initiate cancer.

CARCINOMA: A malignant growth arising from the skin, glandular tissues or the mucous membranes that line the gut and lungs, which tends to metastasise to other areas of the body.

CARCINOTRON (BACKWARD WAVE OSCILLATOR, BWO): A vacuum tube that is used to generate microwaves up to the terahertz range. It belongs to the travelling-wave tube family. It is an oscillator with a wide electronic tuning range.

CARD (EXPANSION CARD; BOARD; ADAPTER): In computing, it is a circuitry designed to provide expanded capability to a computer. It is provided on the surface of a standard-size rigid material such as a fibre board and then plugged into one of the computer's expansion slots in its motherboard or backplane. Cards may come in one of two sizes designed to match standard slot dimensions. A card can actually contain the capability within its circuitry such as a video card or it can control through an extended connection a device such as a hard disk drive.

CARD PUNCH: A machine for punching coded set of holes in a card, to be fed through a card reader for inputting data into a computer.

CARD READER: A data input device designed to read card systems such as the punched card readers made of paper tape or cardboard. Modern readers can read from plastic cards with embedded chips.

CARD VERIFICATION VALUE (CVV; CARD VERIFICATION CODE, CVC; CARD SECURITY CODE, CSC): A combination of features used in credit, debit and automated teller machine (ATM) cards for the purpose of establishing the owner's identity and minimising the risk of fraud.

CARDANO'S FORMULA: The name commonly given to the formula for the solution by radicals of the general (normalised) cubic equation,

$$x^3 + ax^2 + bx + c = 0. \text{ On substituting, } x = y - \frac{1}{3}a \text{ the reduced form of}$$

the cubic is obtained, where $y^3 + py + q = 0$, the quadratic form is

$$\text{missing. The discriminant of the cubic is } \Delta^2 = \left[\frac{q^2}{4} + \frac{p^3}{27} \right] \text{ in order to}$$

have repeated roots if and only if $\Delta^2 = 0$, when $\Delta^2 > 0$ the unique

$$\text{real solution to the reduced cubic form is } \left[-\frac{q}{2} + \Delta \right]^{\frac{1}{3}} + \left[-\frac{q}{2} - \Delta \right]^{\frac{1}{3}}.$$

In the case where $\Delta^2 < 0$, there are three real solutions best expressed trigonometrically. A corresponding solution to the general quartic equation exists and is associated with the names of Ludovico Ferrari and Geronimo Cardano, which proceeds by producing a cubic resolvent equation to which the previous formula applies. Although named after Geronimo Cardano, he did not find the solution to the cubic, he published it in his treatise, based on the formula of Niccolò Tartaglia and the solution of the quartic by Ludovico Ferrari.

CARDIAC: A term used to describe activities and parts of the body related to the heart and its functions.

CARDIAC CYCLE: The sequence of events that occur in the heart during one full heartbeat. The cardiac cycle has two phases. They are the **diastole**, when the heart's chambers are relaxed and **systole**, when the chambers contract to move blood from the atria to the ventricles and then into the arteries. During the systolic phase, the atria contract first, followed by contraction of the ventricles. This sequential contraction ensures efficient movement of blood from atria to ventricles and then into the arteries.

CARDIAC GLYCOSIDE: Any of a group of steroid glycosides characterised by their stimulative effects on the heart. Cardiac glycosides resemble saponins in being diterpenes and by foaming in solution. They differ however in having a lactone ring attached at carbon 17 of the steroid nucleus. An important commercial source is the foxglove (*Digitalis purpurea*), which yields digitonin.

CARDIAC MUSCLE: A specialised form of muscle that is peculiar to the vertebrate heart that do not get fatigued. Each cardiac muscle fibre is an individual small cell, tapering at either end and containing a single nucleus.

CARDIAC OUTPUT: The volume of blood pumped out per minute by each ventricle which is also the total blood flow through the pulmonary circuit. For an adult at rest the cardiac output is about 5 litres per minute.

CARDIATAXIA: Extreme irregularity in the action of the heart.

CARDINAL HUMOUR (HUMOUR): Any of the four fluids in the human body, namely the blood, phlegm, yellow bile and black bile that ancient physicians believed when balanced and working well kept a person healthy but if in excess or deficit, caused diseases and disabilities.

CARDINAL NUMBER: 1. A term used in mathematics to describe one of the series of numbers 0, 1, 2, 3, 4, ..., n , where n is the last number. It can also be defined as a measure of the size of a set that does not take into account the order of its members. Cardinal numbers relate to quantity, whereas ordinal numbers such as first, second, third and fourth relate to order. 2. The smallest ordinal number that is equipollent to a given set.

CARDINALITY: The cardinality of a set S , denoted $|S|$, is the number of elements in S . If the set has an infinite number of elements, then its cardinality is ∞ . Consider an infinite set S , the cardinality of S , represented by $n(S)$, is the number of elements in S . The subsets X , Y and Z of the universal set ϵ include the following:

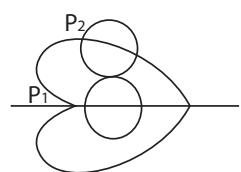
$$(i) n(X \cup Y) = n(X) + n(Y) - n(X \cap Y) \text{ and}$$

$$(ii) n(X \cup Y \cup Z) = n(X) + n(Y) + n(Z) - n(X \cap Y) - n(X \cap Z) - n(Y \cap Z) + n(X \cap Y \cap Z).$$

CARDIO: A medical term used in reference to the heart. It is also used in reference to exercise for cardiovascular fitness or endurance training.

CARDIOID: A heart-shaped curve generated by a fixed point on the circumference of a circle as it rolls around another circle of equal radius, R .

It has an equation of $p = n(1 \pm \cos \theta)$ or $p = a(1 \pm \sin \theta)$ in polar coordinates. Its cartesian equation is $(x^2 + y^2 - 2Rx)^2 = 4R^2(x^2 + y^2)$.



Cardioid

CARDIOLOGY: A branch of medicine concerned with the study of the heart and its diseases and disorders.

CARDIOMYOPATHY: A chronic condition of the heart muscle in which the muscle is abnormally enlarged, thickened and stiffened that makes it harder for the heart to pump and deliver blood to the rest of the body. The main types include **dilated** or **congestive cardiomyopathy** in which the heart cavity is enlarged and stretched, resulting in weak and slow pumping of the blood; **hypertrophic cardiomyopathy** in which the muscles of the

left ventricle become enlarged, obstructing blood flow from the left ventricle; and **restrictive cardiomyopathy** in which the heart muscle of the ventricles becomes rigid, affecting the relaxation period between contractions. The causes may be hereditary, viral infection, long-term severe high blood pressure and alcoholism. Signs and symptoms may include: breathlessness, swelling of the legs, ankles and feet, bloating of the abdomen due to fluid buildup, fatigue, irregular heartbeats, chest pain, dizziness, light-headedness and fainting. It may cause enlargement of the heart as well as heart failure and cardiac arrhythmia.

CARDIOVASCULAR CENTRE: The area of the human brain that is responsible for regulating the rate at which the heart beats. It is found in the medulla and influences the heart rate during exercise or when the body is under trauma.

CARIES: Decay or softening of hard tissues, usually of the teeth, but also of the bone. Dental caries is a bacterial decay of dentine and enamel. Fluoride in small quantities protects against caries.

CARINA: 1. A keel-shaped ridge or structure, such as that on the breastbone of a bird. 2. The fused lower two petals of flowers of many members of the pea family, which form a boat-like structure around the stamens and styles.

CARINAL CANAL (PROTOXYLEM LACUNA): A longitudinal channel in the stem internode of Equisetum (horsetail) and some of its fossil relatives, positioned radially opposite a raised stem ridge (carina). Each carinal canal is arranged on the inner side of a vascular bundle. The carinal canals apparently result from breakdown of protoxylem and are believed to function in water conduction.

CARIUS METHOD: A method for the determination of sculpture, halogens and phosphorus in organic compounds by heating the compound in a sealed tube with fuming nitric acid.

CARLESON'S THEOREM: A fundamental result in mathematical analysis which states that the partial sums of the Fourier series of square-integrable function converge almost everywhere to the given function. It remains true in L_p -space for $p > 1$. For $p = 1$, an example due to Kohnogorov shows that the series may diverge everywhere. It was named after the Swedish mathematician Lennart Carleson (1928-).

CARMICHAEL NUMBER (PSEUDOPRIME): An odd composite number ' n ' which satisfies Fermat's little theorem $b^{n-1} \equiv 1 \pmod{n}$. The first ten Carmichael numbers are 561, 1105, 1729, 2465, 2821, 6601, 8911, 10585, 15841, 29341. Only one composite pseudo-prime, 561 is less than 1000.

CARMINATIVES: Agents or substances containing properties that expel gas from the gastrointestinal tract and prevent it from forming. They are also used to relieve pressure in the oesophagus and heartburn. Examples of carminatives include herbs such as ginger, garlic, pawpaw, cinnamon, coriander, eucalyptus, licorice, mustard and nutmeg oil.

CHOLAGOGUES: Agents or substances containing properties that increase the flow of bile into the intestine, which aids digestion. Cholagogues include herbs such as aloe vera, beet, dandelion root, licorice, ginger, grape and olive oil.

CARNALLITE: A deliquescent mineral consisting of a hydrated mixture of potassium and magnesium chlorides, with the chemical formula $\text{KMgCl}_3 \cdot 6(\text{H}_2\text{O})$. It is mined as a source of potassium and magnesium chloride.

CARNASSIAL TEETH: A powerful scissor-like pair of teeth with sharp edges found in all carnivores except seal and used for shearing flesh. They are found on either side of the jaw and composed of the lower first molar and upper last molar teeth.

CARNATION: A perennial plant, classified as *Dianthus caryophyllus*, a member of the pink family, Caryophyllaceae. It has fragrant white, pink or red flowers with fringed petals, often smelling of cloves.

CARNAUBA WAX: A very hard and brittle natural wax obtained from the leaves of the Brazilian carnauba palm (*Copernicia prunifera*). It is

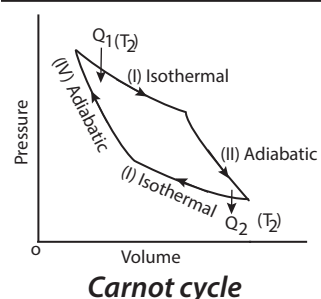
composed mainly of esters of fatty acids (80-85%), fatty alcohols (10-16%), acids (3-6%), hydrocarbons (1-3%) and traces of antioxidants. Its uses include adding sheen to waxes in shoe polishes and in varnishes, cosmetics and for making candles, fibres, etc.

CARNIVORA: An order of class Mammalia consisting of 12 families of primarily predatory mammals, called **carnivores**. Examples are cats, dogs, wolves, foxes, bears, otters, mink, weasels, skunks, seals, walruses and sea lions. They are adapted by their structure to feed upon flesh, though some of them, as the bears, also eat vegetables. They have powerful jaws, with sharp, pointed canine teeth for killing prey and shearing molars for cutting flesh. Carnivores have a keen sense of smell and exhibit complex social interactions. Some such as seals, walruses and sea lions have their limbs modified into flippers for swimming. However, these animals are not completely adapted to life in water, they come ashore to mate and bear their offspring.

CARNIVORE: A heterotrophic flesh-eating mammal of the order Carnivora. They have strong canines called fangs and sharp molars called carnassials adapted for eating flesh. Examples are dogs, cats, bears, leopards and lions. Some carnivores such as dogs, cats and bears eat other foods apart of flesh. There are also several species of carnivorous plants, though most are primarily insectivorous.

CARNIVOROUS PLANT (INSECTIVOROUS PLANT): Any plant that supplements its supply of nitrogen in nitrate-deficient habitat by digesting small animals especially insects. Examples are the Venus fly-trap (*Dionea muscipula*), bladderwort (*Utricularia vulgaris*) and pitcher plant (*Sarracenia flava*).

CARNOT CYCLE: A hypothetical thermodynamic cycle postulated by Nicolas Carnot, a French physicist (1796-1832). According to the Carnot principle, the most efficient cycle of operations for a reversible heat engine consists of four operations on the working substance in the engine as follows: (i) isothermal expansion at thermodynamic temperature T_1 with heat Q_1 taken in; (ii) adiabatic expansion with a fall of temperature to T_2 ; (iii) isothermal compression at temperature T_2 with heat Q_2 given out; (iv) adiabatic compression of the gas with a rise of temperature back to T_1 . The Carnot cycle shows that, even under ideal conditions, a heat engine cannot convert all the heat energy supplied to it into mechanical energy; some of the heat energy must be rejected.



CARNOTITE: A radioactive mineral consisting of hydrated uranium potassium vanadate, with the chemical formula $K_2(UO_2)_2(VO_4)_2 \cdot nH_2O$. It varies in colour from bright yellow to lemon- or greenish-yellow. It is a source of uranium, radium and vanadium.

CARO'S ACID (PEROXOMONOSULFURIC(VI) ACID; PERSULFURIC ACID): A crystalline compound made by the action of hydrogen peroxide on concentrated sulfuric acid, with the chemical formula H_2SO_5 and density 2.239 g/cm^3 . It is a very strong acid as well as a strong oxidising agent.

CAROTENES: Several related unsaturated hydrocarbon substances containing eight isoprene subunits having the formula $C_{40}H_{56}$, which are synthesised by plants but cannot be made by animals. The most commonly occurring carotenes are alpha (α) and beta (β) carotene. Another carotene, lycopene, gives tomato fruits their red colour. Carotenes are orange photosynthetic pigment important for photosynthesis; and produce the orange, yellow and red colours of carrots, tomatoes oranges and crustaceans. They are also found in yolk, butter and in all green leaves. They exist in several isomeric forms and are converted to vitamin A in the liver.

CAROTENOID: Any of a group of yellow orange, red or brown organic pigments found in many living organisms. They are always

present in the chloroplast of plants. They absorb light energy for use in photosynthesis and they also protect chlorophyll from damage by UV rays. There are two kinds of carotenoids: carotenes and xanthophylls. Carotenes contain no oxygen but xanthophylls do.

CAROTID ARTERY: A pair of arteries arising from the aorta on the left and the innominate artery on the right that run up the neck, with each branching into an external and an internal carotid artery. They are the major arteries that supply oxygenated blood to the head.

CAROTID BODY: 1. One of a pair of tissue masses adjacent to the carotid sinus. Each contains receptors that are sensitive to oxygen and carbon dioxide levels in the blood. 2. Relating to either of the two large blood vessels in the neck, which carries blood to the heart.

CAROTID SINUS: An enlarged region of the carotid artery at its major branching point in the neck. Its walls contain many receptors sensitive to changes in pressure and it regulates blood pressure by initiating reflex changes in heart rate and dilation of blood vessels.

CARPAL: 1. Of, relating to or near the carpus. 2. Any of the bones that make up the carpus in human and many other land vertebrates. The carpal bones articulate proximally with the long bones of the radius, indirectly, with the ulna and distally with the five metacarpal bones that make up the palm.

CARPAL TUNNEL SYNDROME (CTS): A medium entrapment neuropathy that causes a condition of paresthesia, pain, numbness, tingling and other symptoms in the distribution of the median nerve due to its compression at the wrist.

CARPEL: The female part of the flower or the ovule and seed producing reproductive organ in the pistil of flowering plants; or one of the leaf-like, seed-bearing structures that constitute the innermost whorl of a flower. One or more carpels make up the pistil. It is the female sporophyll within a flower, a floral organ that bears ovules in angiosperms. A collection of carpels is known as **gynoecium**. Carpellate means having carpels.

CARPET GRASS (AXONOPUS FISSIFOLIUS): A coarse grass belonging to the genus Axonopus that forms a tight matted growth and is widely used in warm humid areas for turf and pasture.

CARPOGONIUM: The female gametangium of the red algae, found at the tips of the female gametophyte. It is a flask-shaped structure from which protrudes a slender elongation called trichogyne.

CARPOPHORE: The receptacle between the carpels as a central axis which is slender and prolonged as found in the fruits of some members of the Umbrelliferae and the Geraniaceae.

CARPOPODIUM (PODOGYNE): A short, thick, pistillate stalk supporting an ovary; a stipe.

CARPOSPORANGIUM: A sporangium that develops directly from the zygote in situ in some of the red algae and which is dependent on the parental tissue for nutrition.

CARPOSPORES: Spores of the red algae, which are produced in the fertilised carpogonium. They develop into a sporophyte or a new gametophyte after release.

CARPOTAXIS: An arrangement of fruits.

CARPUS: 1. The wrist or corresponding part of the forelimb in terrestrial vertebrates, consisting of a number of small bones (carpals). The number of carpal bones varies from species to species; in humans it is eight. 2. The fifth segment from the base of a generalised crustacean appendage.

CARRAGEENAN (CARRAGEENIN): A naturally occurring polysaccharide isolated from red seaweed. The polymer is composed of D-galactose unit, many of which are sulfated. Gelatinous extracts of the seaweed, *Chondrus crispus*, have been used as food additives.

CARRIER: 1. An organism that harbours a particular disease-causing microorganism without ill effects and who can transmit the microorganisms to others; or an individual that has an allele for a defect that is masked by a normal dominant allele. Such

individual therefore, does not suffer from the condition, but may pass on the defective allele to an offspring. 2. In mathematics, also called **multivalued function**, **multifunction**, **point-to-point mapping** or **set-valued function**, it is a mapping that associates a number of different elements of the output with the same element of the input; a mapping from a set into the power set of another set. The inverses of trigonometric, hyperbolic, exponential and integer power functions are multivalued. For example, the function z^2 maps each complex number z to a well-defined number z^2 , while its inverse function $\sqrt{z}$ maps, for example, the value $z = 1$ to $\sqrt{1} = \pm 1$. 3. In chemistry, it is a substance in appreciable amount which, when associated with a trace of a specified substance, will carry the trace with it through a chemical or physical process or prevent the tracer from undergoing nonspecific processes due to its low concentration. 4. In telecommunications, it is the short form of **wireless carrier** also known as **mobile network operator**, **mobile phone operator**, **mobile operator** and **wireless service provider** and refers to a company that provides mobile services.

CARRIER ATOM: A participant atom which is neither core nor peripheral and which carries other groups to and from the molecule containing the core atom(s) but which does not itself become covalently attached to that molecule.

CARRIER GAS (ELUANT GAS; MOBILE PHASE): The gas that carries the sample in gas chromatography. Carrier gases include helium, nitrogen, argon, hydrogen and air. A **gas chromatograph** is a chemical analysis instrument for separating chemicals in a complex sample. A gas chromatograph uses a flow-through narrow tube known as the **column**, through which different chemical constituents of a sample pass in a gas stream known as carrier gas (mobile phase) at different rates depending on their various chemical and physical properties and their interaction with a specific column filling, called the **stationary phase**.

CARRIER MOLECULE: A molecule that plays a role in transporting electrons through the electron transport chain. They are usually proteins bound to a non-protein group. They can undergo oxidation and reduction relatively easily, thus allowing electrons to flow through the system.

CARRIER NETWORK: The proprietary network infrastructure belonging to a telecommunications service provider such as Verizon, AT&T or Sprint. Carrier networks are made up of large, complex configurations of hardware, interconnected to provide communications services to people spread over large geographic areas.

CARRIER PARTICLE: A fundamental boson associated with quantum excitations of the force field corresponding to some fundamental interaction. Gluons are carrier particles for strong interactions (colour force fields), photons are carrier particles of electromagnetic interactions and the W and Z bosons are carrier particles for weak interactions.

CARRIER PROTEIN: 1. A protein to which a specific ligand or hapten is conjugated. 2. Unlabelled protein introduced into an assay at relatively high concentrations, which distributes in a fractionation process in the same manner as labelled protein analyte, present in very low concentrations. 3. Protein added to prevent non-specific interaction of reagents with surfaces, sample components and each other. 4. Protein found in cell membranes, which facilitates transport of a ligand across the membrane.

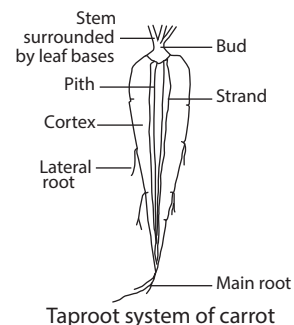
CARRIER SENSE MULTIPLE ACCESS, CSMA (COLLISION DETECT, CD): The protocol for carrier transmission access in Ethernet networks. **Ethernet** is the most widely-installed local area network (LAN) technology. On Ethernet, any device can try to send a frame at any time. Each device senses whether the line is idle and therefore available to be used. If it is, the device begins to transmit its first frame. If another device has tried to send at the same time, a collision is said to occur and the frames are discarded. Each device then waits a random amount of time and retries until it is successful in getting its transmission sent.

CARRIER SENSITIVE ROUTING (CSR): An application that allows Voice over Internet Protocol (VoIP) subscribers to determine and manipulate the routing of individual calls. CSR can maximise cost effectiveness and Quality of Service (QoS) because the routing can be tailored to fit the attributes of the particular carrier used.

CARRIER SIGNAL (CARRIER WAVE): A transmitted electromagnetic pulse or wave at a steady base frequency of alternation on which information can be imposed by increasing signal strength, varying the base frequency, varying the wave phase or other means. This variation is called modulation. With the advent of laser transmission over optical fibre media, a carrier can also be a laser-generated light beam on which information is imposed. Types of analogue modulation of a carrier include amplitude modulation (AM), frequency modulation (FM) and phase modulation. Types of digital modulation include varieties of pulse code modulation (PCM), including pulse amplitude modulation (PAM), pulse duration modulation (PDM) and pulse position modulation (PPM).

CARRIER TO NOISE RATIO (CNR; C/N): In communications, it refers to a measure of the received carrier strength relative to the strength of the received noise. High C/N ratios provide better quality of reception and generally higher communications accuracy and reliability, than low C/N ratios.

CARROT: A biennial herbaceous plant, *Daucus carota*, of the Apiaceae family, which usually grows up to about 1 metre tall with a taproot system. It is grown mainly for its edible tuberous conical taproot, also known as carrot which is used as a vegetable. The root usually has an orange colour, but may also have purple, red, white and yellow colours. The umbels are claret-coloured or pale pink and produce small white, red or purple flowers. It is propagated through the seeds. Carrot has a crisp texture when fresh and can be eaten fresh or cooked. The health benefits of carrot are many, as it is rich in antioxidants



**Longitudinal section
of a carrot**

such as beta carotene, alpha carotene, phytochemicals, calcium and potassium and vitamins A, B1, B2, C and E. It also contains copper, iron, magnesium, manganese, phosphorous and sulfur. Carrot can enhance the quality of breast milk. It can improve the appearance of the skin, hair and nails. When taken daily it can lower cholesterol and blood pressure. Raw carrot contains beta-carotene, which can prevent cancer. Carrot juice, when taken every day prevents bodily infections and is claimed to be valuable for the adrenal glands. Carrot can help improve eye sight; can help increase menstrual flow, regulate blood sugar; and also promote colon health, because it is rich in fibre.

CARROT FLY: A small fly, *Chamaepsila rosae*, with a reddish brown head and yellowish wings. It lays eggs in the surface of the soil near plants and the larvae burrow into the root. The foliage wilts and dies and in severe cases, the crop is lost. It affects mainly carrot but can also attack, for example, parsley and celery.

CARRY: 1. In mathematics, to transfer a digit from a one place-value column to another, for example, from units to tens as in multiplication and addition. 2. In physics, to possess an intrinsic property, such as colour charge or convey (a force) that governs particle interactions. 3. To maintain livestock in good health without increasing their weight or obtaining any products from them.

CARRY-OVER: A process by which materials are carried into a reaction mixture to which they do not belong. These materials can be either parts of a specimen or reagents including the diluent or wash solution. In such cases, carry-over means the transfer of material (specimen or reagents) from one container or from one reaction mixture, to another one. It can be either unidirectional or bidirectional in a series of specimens or assays. The term carry-over effect is used for carry-over from specimen to specimen.

CARRYING CAPACITY: The maximum number of individuals that a particular habitat or environment can support, below which its numbers tend to increase and above which its numbers tend to decrease because of shortages of resources. It is represented by K . The carrying capacity is different for each species in a habitat because of that species' particular food, shelter and social requirements. Species whose populations increase until the populations begin to fluctuate at or near the carrying capacity are referred to as K -selected species or K -strategist species. Species whose populations increase rapidly, often exponentially, quickly filling available environments, are referred to as r -selected species or r -strategist species. For the human population, more complex variables such as sanitation and medical care are sometimes considered as part of the necessary establishment.

CARTESIAN COORDINATES (CARTESIAN COORDINATE SYSTEM; RECTANGULAR COORDINATE SYSTEM; RECTANGULAR COORDINATES): A set of coordinates that gives the position of an object with respect to two or three mutually perpendicular axes (x , y and z) that cross at some specified origin. In the two mutually perpendicular axes (x and y) system, the plane consists of two number lines, one horizontal (x -axis) and the other vertical (y -axis) intersecting at 90° of the origin. Their units are of equal measure. The two axes divide the plane into four quadrants numbered in the anticlockwise direction. The three mutually perpendicular axes (x -, y - and z -axes) system consists of three axes x , y and z that intersect at the origin. In astronomy, heliocentric or geocentric rectangular coordinates are occasionally used to give positions of bodies in the Solar System.

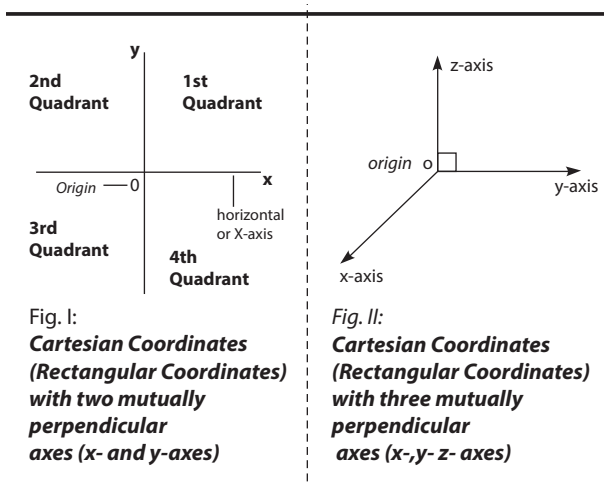


Fig. II:
Cartesian Coordinates (Rectangular Coordinates) with three mutually perpendicular axes (x-, y- and z-axes)

CARTESIAN DISTANCE (EUCLIDEAN DISTANCE): In mathematics, the distance between two points in Euclidean space, defined as the square root of the sum of the squares of the differences between the corresponding coordinates of the points. In a 2-dimensional Euclidean space, the distance $|PQ|$ between

$P = (x_1, y_1)$ and $Q = (x_2, y_2)$, is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. In n dimensions, the Euclidean distance between two points p and q is $\sqrt{\sum_{i=1}^n (p_i - q_i)^2}$,

where p_i or q_i is the coordinate of p or q in dimension i .

CARTESIAN GEOMETRY (ANALYTIC GEOMETRY; ANALYTICAL GEOMETRY; COORDINATE GEOMETRY): The branch of mathematics that studies geometrical relations using algebra with coordinates as references. An example of coordinate geometry is plotting points, lines and curves on an x - and y -axis.

CARTESIAN OVAL: A curve that actually consists of two ovals, one inside the other. It is the locus of a point whose distances s and t from two fixed points S and T satisfy the equation $s + mt = a$.

CARTESIAN PLANE: Cartesian coordinates (x , y) defined by two-dimensional space. It is the Cartesian square of the real line.

CARTESIAN PRODUCT: A set of all pairs of elements (a , b) that can be constructed from given sets, A and B , such that a is a member of A and b is a member of B and is denoted by $A \times B$. The elements (a , b) of $A \times B$ are ordered pairs, where $a \in A$ and $b \in B$. The cross product of two or three dimensional vectors $a = (a_1, a_2, a_3)$ and $b = (b_1, b_2, b_3)$ is $a \times b = [(a_2b_3 - a_3b_2), (a_3b_1 - a_1b_3), (a_1b_2 - a_2b_1)]$; $a \times b$ (read a cross b) is a vector with the following properties: (i) $\|a \times b\| = \|a\| \|b\| \sin \theta_{ab}$, where θ_{ab} is the angle between a and b and $\|a\|$ is the length of vector a . (ii) $a \times b$ is perpendicular to both a and b . (iii) The direction of $a \times b$ is determined by the right hand rule: if you put your right hand so that your finger point in the direction from a to b , then your thumb points in the direction of $a \times b$. (iv) $a \times b = 0$, if a and b are parallel, i.e., if $\theta = 0$. (v) $\|a \times b\| = \|a\| \|b\|$ if a and b are perpendicular. (vi) The cross product is not cumulative since $a \times b = -b \times a$. The Cartesian product of n sets, also known as a n -fold Cartesian product, can be represented by an array of n dimensions, where each element is an n -tuple. An ordered pair is a **2-tuple** or **couple**.

CARTESIAN SPACE (CARTESIAN PLANE; EUCLIDEAN PLANE; EUCLIDEAN SPACE): A two or more dimensional space possessing a scalar product, so that a Euclidean distance may be imposed. In other words, it is a set of points, along with a set of neighbourhoods for each point, that satisfy a set of axioms relating points and neighbourhoods. It is used to denote the position of an element in the system of two or more dimensions. The number of dimensions in Euclidean space is denoted by n . The n -dimensional space denoting real coordinates is represented by $\mathbb{R}^n$ or E^n . A point that is defined in a two-dimensional space, is represented by (x, y) and in three-dimensions by (x, y, z) ; where, x , y , z are the distances from x -, y - and z - axes respectively. One-dimensional Euclidean space is represented by a real number line; while Euclidean space of two dimensions is composed of Cartesian plane with two mutually perpendicular lines.

CARTESIAN SQUARE: The Cartesian product of any set with itself. For example, Cartesian coordinates are the Cartesian square of the real numbers.

CARTESIAN TENSOR: A tensor defined on a vector space with an orthonormal basis.

CARTILAGE (REPLACING BONE; GRISTLE): A firm flexible connective tissue of vertebrates and some invertebrates made up of a protein called collagen. In cartilaginous fish, it forms the skeleton but in other vertebrates it forms the greater part of the embryonic skeleton (endochondral ossification) and is replaced by bones in most areas in the course of development. It also forms structural tissues in the larynx, nose and external ear of mammals. Cartilage comprises a matrix consisting chiefly of a glycosaminoglycan (mucopolysaccharide) called chondroitin sulfate secreted by cells (chondroblasts) that become embedded in the matrix as chondrocytes.

CARTILAGINOUS: A cartilage-like structure, hard and tough but elastic and flexible.

CARTOGRAPHY: The science of producing geographic area graphically, usually by means of maps and charts. Modern cartography uses aerial photography and satellite radar for accuracy. **Cartographer** is a person who draws or produces maps.

CARTRIDGE: 1. In computing, it is a removable unit in a computer, such as an integrated circuit containing software. 2. A toner cartridge holds a physical consumable print substance in a protective plastic shell that is used for a modern laser printer for making actual images on paper.

CARUNCLE: 1. A small outgrowth from the testa of a seed that develops from the funicle or micropyle. Examples include the warty out-growth from the castor-oil (*Ricinus communis*) and cassava (*Manihot esculenta*) seeds. **Caruncular** or **carunculate** means of or pertaining to a caruncle; fibrous with stringy or cord-like seed coat, as mace in Myristica. 2. Any normal or abnormal fleshy outgrowth, such as the comb and wattles of fowl or the mass in the inner canthus of the eye.

CARYOPHYLLIDAE (CENTROSPERMAE): A subclass of the dicotyledons in the class Magnoliopsida containing mostly herbaceous plants usually with bisexual flowers. The seeds often contain perisperm and betalain alkaloids are usually present.

CARYOPSIS: A dry, one seeded fruit in which the testa of the seed becomes fused with the pericarp during its development. It is typical of members of the grass family.

CAS REGISTRY: A database of chemical compounds, certain mixtures and biological sequences maintained by the Chemical Abstracts Service of the American Chemical Society. A CAS registry number has three parts: part 1 has up to six digits; part 2 has two digits and part 3 has one digit, for example, the CAS number for water is 7732-18-5.

CASCADE IMPACTOR: An instrument used for the classification of aerosols according to size and for possible subsequent chemical analysis. Air is drawn through a series of orifices of decreasing size; the air flow is normal to collecting surfaces on which aerosols are collected by inertial impaction. The particles, separated stepwise by their momentum differences into a number of size ranges, are collected simultaneously.

CASCADE-LIQUEFIER: An apparatus for liquefying a gas of low critical temperature, for example, air and oxygen. In the cascade liquefier the critical temperature of the gas is reached by successively steps, using a series of gases having successively lower boiling points.

CASCADE-PROCESS: Any process that takes place in a number of successive steps, because a single step alone cannot produce the desired result. Examples are: (i) the cascade liquefier, in which the critical temperature of the gas is reached step by step, using a series of gases having successively lower boiling points; (ii) the steps involved in the clotting of blood occurring as a cascade; and (iii) the sequence of spontaneous decays by an excited atom or ion.

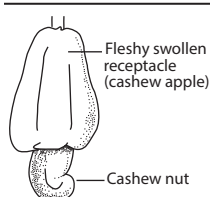
CASCADING STYLE SHEET (CSS): A style sheet language used to format the layout of Web pages. They can be used to define text styles, table sizes and other aspects of Web pages that previously could only be defined in a page's Hypertext Markup Language (HTML).

CASE-HARDENING: 1. The process of hardening of the outer layer of steel or other ferrous alloy with a low carbon content to a depth of about 0.03 mm to make it harder than the core and still maintain properties such as flexibility. The processes of case hardening are carburising, cyaniding, carbonitriding, nitriding, induction hardening and flame hardening. Case hardening is used for tools and certain mechanical components. 2. Formation of a mineral coating, usually iron and magnesium hydroxides and sometimes of amorphous silica on the surface of porous rock by evaporation of a mineral-bearing solution. 3. A condition of stress in timber, in which the outer layers have dried without shrinkage. It is characterised by compressive stress in the outer layers and tensile stress in the centre or core.

CASEIN: The main protein of milk, from which it can be separated by the action of acid, the enzyme rennin or bacteria. It is the main component of cheese. Casein is a very efficient nutrient and contains, in addition to essential amino acids, some carbohydrates as well as calcium and phosphorus.

CASH CROP: A crop grown on a large scale for commercial and industrial purposes. Examples are coffee, cocoa and palm nut.

CASHEW TREE (*Anacardium occidentale*): A small tree or shrub growing up to 6-15 m high, with dense, evergreen deep green foliage, which is red or pale green when young. It is found in north-east of Brazil, West Indies, tropical Africa and Madagascar. The crown is regular and hemispheric with branches sometimes touching the ground. The trunk is short and twisted. The bark is rough, grey, with pinkish slash. Stems are more or less pubescent, grey to brown. The leaves are simple, alternate, smelling strongly of turpentine when crushed. The inflorescence is a terminal cyme, more or less dense. Flowers are greenish, turning



Fruit of cashew tree

purplish with 5 petals. The true fruit of the cashew tree is a kidney or boxing-glove shaped drupe that grows at the end of the yellow or reddish cashew apple. The drupe develops first on the tree and then the pedicel expands to become the cashew apple. Within the true fruit is a single seed, the cashew nut, which is the main commercial part of the tree. It is kidney or bean shaped, with smooth curvy pointed tip. Each nut has two equal halves as occurs in legumes. Cashews have nutty, aromatic, sweet, almond-like taste, when roasted and eaten as a snack, salted or sweetened. They are also widely used in confectionery, as an addition to biscuits, sweets and cakes. Its achene wall oil is toxic and used to produce cashew balm, varnish, indelible ink, insecticide and fix tattoos. The fruit peduncle is rich in vitamins when eaten fresh or dried and used to prepare jams, syrups or alcoholic drinks. All parts of the plant are used for their bactericide, fungicide, vermicide and amoebicide properties. Traditionally, various parts of the plant such as roots, bark and fruit are used to treat ailments such as cough, diabetes, stomatitis, dysentery, measles, haemorrhoids, eczema and skin diseases. The bark tannins are used to produce yellow dyes. The tree is used to produce charcoal and firewood, woodwork and mortars. It also provides shade during the dry season and also for reforestation because of its powerful root system, which prevents soil erosion.

CASIMIR: A conserved quantity which exists because of degeneracy in a Poisson bracket that is singular and noncanonical.

CASIMIR EFFECT: The attractive force between two parallel conducting surfaces in vacuum containing an electromagnetic field.

CASING: 1. A substance or object that covers something to protect it. 2. A steel pipe or tubing, for example, those used during the drilling of a well. 3. A layer of glass that has been fused to an underlying layer of glass of a different colour or of different properties. 4. A thin covering used by meat processors to give sausages their shape. It is usually cylindrical, but other shapes are also available. 5. Top layer of soil used during the fruiting stage in mycological cultivation.

CASORATI-WIERSTRASS THEOREM: A weak form of Picard's theorem, stating that an analytic function comes arbitrarily close to all values in any neighbourhood of an essential singularity (i.e, the image of any ball centred on the singularity is dense in the complex numbers).

CASPARIAN STRIP (CASPARIAN BAND): A band-like wall formation of lignin and suberin occurring in anticlinal, radial and transverse walls of endodermal cells of roots that controls movement of water from the cortex to the stele.

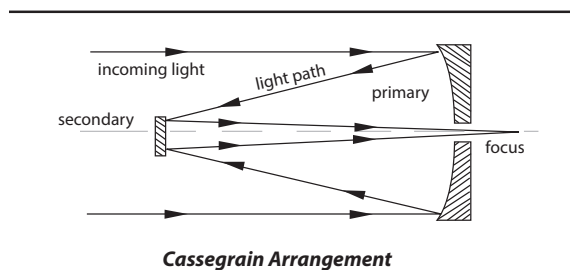
CASPASES (CYSTEINE-ASPARTIC PROTEASES): Any of various enzymes that cleave target proteins during programmed cell death or apoptosis, discovered in the mid 1990s. They are formed from inactive precursors present in the cell in response to certain triggers, such as cell damage.

CASQUE: 1. A helmet-like structure located on the head of a lizard. 2. A helmet-like structure on the head of some birds, for example, a cassowary or on the bill of a hornbill.

CASSAVA (MANIOC; MANDIOC; TAPIOCA): A perennial woody shrub, belonging to the family Euphorbiaceae, which can grow up to about 4 m in height. It is a widely cultivated food crop in tropical and subtropical regions for its edible starchy tuberous root, a major source of carbohydrates and gluten free. Cassava is classified as sweet or bitter. Bitter cassava is classified as *Manihot esculenta*. Sweet cassava is classified as *Manihot dulcis*. Cassava has erect stems and spirally arranged simple lobed leaves with petioles up to 30 cm in length. The plant produces flowers without petals on a raceme. The edible roots of the plant are usually cylindrical and tapered and are white, brown or reddish in colour. Some varieties contain the toxic glycoside called phaseolunatin which is destroyed by boiling. It is usually harvested 9-12 months after planting. It is the source of food products such as garri, fufu, tapioca and starch. Industrially, cassava starch is used for the manufacture of glue, biscuits, pharmaceutical products, noodles, magi cubes, paper-cartons, animal feed, pastries, mosquito coils, confectionaries, ethanol, textile products, dry cell

batteries, toothpaste and biodegradable products. It is also used in the brewery industry as alternative or complementary to sorghum, maize starch and barley. All parts are used as food by most wild and domesticated animals.

CASSEGRAIN TELESCOPE: An astronomical reflecting telescope consisting of a concave primary mirror and a convex secondary mirror, in which the convex secondary mirror reflects the convergent beam from the concave primary mirror through a hole in the primary mirror to an eyepiece in back of the primary mirror. Depending on the curve of the secondary mirror, the light can and most often does, come to focus behind the primary having passed through a hole in the the primary. It was conceived in about 1672 by the Frenchman Guillaume Cassegrain.



CASSIA: 1. A genus of about 100 accepted species of trees, shrubs and herbs in the legume family, Fabaceae, which occurs in tropical and subtropical regions. Some of these species have properties similar to that of the *Senna* species. Many species of genus *Senna* were previously classified as *Cassia*. *Cassia sieberiana*, commonly known as West African laburnum, African laburnum or drumstick tree is a very attractive shrub or small tree of between 10 to 20 m high. It is found in various parts of Africa including the southern part of the Sahel, Senegal, Sudan and Uganda. It is readily recognised by the large pendulous racemes of bright yellow flowers and the persistent long cylindrical very narrow fruit. The leaves are opposite, leaflets vary in shape, ranging from elliptic with an acute apex to broadly ovate or almost obviate with the apex rounded or slightly notched, cuneate or rounded at base. The flowers are very bright yellow during the dry season, arranged either upright or in pendulous racemes. The indehiscent fruits are cylindrical, slightly curved and blackish-brown when ripe. It has numerous seeds, which are hard, glossy light brown, each separated by a transversal membrane. The hard wood is resistant to termites and is used to make railway sleepers, furniture, tools, pestles, mortars and for construction work. It is also an ornamental tree because of its brightly coloured flowers. The parts such as the roots, bark and leaves are variously used as diuretic and vermifuge for treatment of diseases such as elephantiasis, leprosy, diarrhoea, haemorrhoids, dysentery, venereal diseases, dropsy, swelling, gout, arthritis, rheumatism, etc. *Cassia occidentalis* with the common names coffee weed, coffee senna, negro coffee, rubbish cassia or stinking weed is a small shrubby annual with many branches, or a biannual, which grows up to 2 m high. The various parts have been used traditionally for the treatment of many ailments. For example, the seeds are beneficial in treating whooping cough, convulsions, hypertension and some other heart diseases. Some other *Cassia* species include *Cassia alata* also known as *Senna alata*, known commonly as candle bush, candlestick senna or wild senna, which is used to treat ringworm, eczema, athlete's foot, etc; *Cassia Fistula* known commonly as yellow golden shower is a tree from Egypt and India, which is used to treat diseases such as malaria, bronchitis and rheumatism; *Cassia spruceana* Benth.; *Cassia psilocarpa* Welw.; *Cassia manicula* Symon; etc. 2. The English common name of a tropical Asian evergreen tree, *Cinnamomum cassia* in the family Lauraceae, having aromatic bark and used as a substitute for cinnamon.

CASSINI OVALS (OVALS OF CASSINI): The locus of the vertex of a triangle when the sides adjacent to the vertex are held constant and the opposite side is held fixed. When the constant is one quarter of the length of the fixed side, this produces a Lemniscate; for a smaller ratio, two distinct ovals are obtained.

CASSITERITE: A dark-coloured mineral consisting of tin(IV) oxide having the composition SnO_2 , which is the main source of tin.

CAST: 1. A part of an organism that has been shed in a natural recurring process. Examples are an insect casing, a snake skin or worm faeces. 2. Shape solid objects such as glass, engine and its parts, plastics and metal alloys by pouring molten materials into a shaped mould and allowing it to cool to form the shape of the mould. 3. A term used in computer science in the context of C#(C Sharp), which is a method of converting a value from one data type to another. Cast is an explicit conversion by which the compiler is informed about the conversion and the resulting possibility of data loss. C#(C Sharp) is a general object-oriented programming (OOP) language for networking and Web development.

CAST IRON (PIG IRON): A metallic material primarily composed of iron (Fe), carbon (C) and silicon (Si), but may also contain traces of sulfur (S), manganese (Mn) and phosphorus (P). It has a relatively high carbon content of 2% to 5%. It is hard, brittle, non-malleable. Its structure is crystalline and relatively brittle and weak in tension. Cast-iron members fracture under excessive tensile loading with little prior distortion. Cast iron is, however, very good in compression. Because it is produced by casting, it is well suited to the production of intricate shapes, both structural and decorative. It is cheap but invaluable and is used in construction especially for beams and posts that hold up architectural structures and in pipe plumbing; for making cylinder blocks in car engines and other parts; in some type of cooker; and some cooking utensils; etc. It can be cast into different shapes for direct use or can be further treated for special uses.

CASTE (CAST): A division or subset of individuals within a colony of social animals chiefly Hymenoptera (ants, bees, wasps) and the Isoptera (termites), which have become specially adapted to perform a particular function. For example, in honeybees there are queens (fertile females), workers (sterile females) and drones (males).

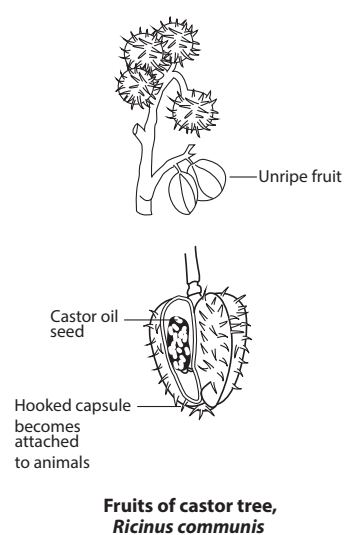
CASTING: The process of making solid objects such as glass, engine and its parts, plastics and metal alloys by pouring molten materials into a shaped mould and allowing it to cool to form the shape of the mould.

CASTING OUT NINES: A method of checking arithmetical calculations that relies on the fact that, modulo 9, the sum of the digits of a sum or product equals the sum or product of the sums of the digits of the numbers added or multiplied, where the process of adding digits is repeated where necessary until the result is less than nine. For example, it can be determined that 365×248 cannot equal 90520, since adding the digits of each of the multiplicands yields 14, whose digits sum to 5, so that their product should have the same nines-complement as $5 \times 5 = 25$, that is 7, whereas similarly casting out nines from the supposed product gives 8. However, since this is merely a necessary and not a sufficient condition, the process can only be used to detect errors, but can neither determine nor verify a result.

CASTNER-KELLENER CELL:

An electrolytic cell used in industry for the production of sodium hydroxide. The iron cell is filled with brine, with liquid mercury as the cathode.

CASTOR OIL: A pale-yellowish oil extracted from the seed of the castor-oil plant (*Ricinus communis*). It contains a mixture of glycerides of fatty acids, the predominant acid being ricinoleic acid, $\text{C}_{17}\text{H}_{32}(\text{OH})\text{COOH}$. It is used as a drying oil in paints and varnishes and medically as a laxative.



CASTRATION (GONADECTOMY): Removal of the testicles of a male animal in order to make reproduction impossible. This may make them docile and also prevent the spread of diseases. Castration methods are generally categorised into two as open and close castration. **Open castration** involves excision of the testes by cutting the scrotal sac and removing the testes by severing the sperm ducts using an emasculator in a manner that minimises bleeding. **Close castration** involves the use of a burdizzo to crush the sperm ducts within the scrotum or an elastic band known as elastrator is placed around the neck of the scrotum to slough off the testes from the sperm ducts within the scrotum.

CASUARINA: A tree with hard, durable wood, a member of the family Casuarinaceae with needle-shaped leaves that form whorls at the end of short branches, valued for its strength and resistance to termites. It is native to Australia and parts of Asia.

CAT: 1. A small furry, domesticated, mainly carnivorous animal, *Felis catus* belonging to the Felidae family. Adults normally weigh between 4 - 5 kg. They have large eye sockets and powerful and specialised jaws. Cats are digitigrades. They walk directly on their toes, with the bones of their feet making up the lower part of the visible leg. Cats have strong, flexible bodies, quick reflexes, sharp retractable claws. They are largely carnivorous and have evolved a simple gut appropriate for raw meat and have teeth adapted for killing prey and tearing meat. They also have the rough tongue to help them clean every piece of meat from an animal bone and also for grooming. They can hear sounds too faint or too high in frequency for human ears, such as those made by mice and other small animals. They can see in near darkness. Like most other mammals, cats have poorer colour vision and a better sense of smell than humans. Cats possess excellent memory and exhibit considerable aptitude for learning by observation and experience. The natural life span of a domestic cat is about 15 years. It is often kept as a pet or for catching mice. 2. A wild animal such as lions, tigers and leopards, which are related to the domestic cat.

CAT SCANNER (COMPUTERIZED AXIAL TOMOGRAPHY SCANNER; CT SCANNER): A ring shaped x-ray machine that rotates through 180° around the horizontal patient, making numerous x-ray measurements every few degrees. It produces cross-sectional views of the internal structure of the body, used for diagnosing diseases.

CAT'S EYE: Any of several types of gemstone which, when cut to form a convex surface, shows a thin line of light like a cat's eye (chatoyancy). The commonest is chalcedony; and the least common is cymophane, a green variety of chrysoberyl, found in Sri Lanka. The effect is due to minute parallel inclusions or cavities.

CATABLASE (CATACASE; CATALASE): An enzyme in the liver of nearly all living organisms that decomposes poisonous hydrogen peroxide to water and oxygen.

CATABOLISM: The destructive metabolic breakdown of large molecules in living cells to smaller ones with the release of energy. It occurs continuously in the body but is accelerated during starvation and many disease processes such as fever. Large polymeric molecules such as polysaccharides, nucleic acids and proteins are first split into their constituent monomeric units, such as amino acids, after which the monomers themselves can be broken down into such simple cellular metabolites as lactic acid, acetic acid, carbon dioxide, ammonia and urea.

CATABOLITE ACTIVATOR PROTEIN (CAP; cAMP RECEPTOR PROTEIN; CRP): In *E. coli*, a helper protein that stimulates gene expression by binding within the promoter region of an operon and enhancing the promoter's ability to associate with RNA polymerase which is also responsible for the dimerisation of the protein and a DNA-binding domain at the carboxyl-terminus (C-terminus) of an amino acid.

CATABOLITE REPRESSION: An intracellular regulatory mechanism in bacteria, which represses transcription of genes associated with catabolism in the presence of glucose.

CATACLASTIC ROCK: A type of metamorphic rock, such as a breccia, containing angular fragments of rock associated with plate tectonic

processes produced by the grinding and crushing action of faults, a process known as cataclasis.

CATALAN NUMBERS: The integers of the form

$$\frac{2n!}{n!9(n+1)!} = \frac{1}{n+1} \binom{2n}{n}. \text{ It is denoted by } C(n) \text{ or } C_n.$$

Catalan numbers form a sequence of natural numbers that occur in various counting problems, often involving recursively-defined objects. The first four are 2, 3, 10 and 420. Catalan number was named after Eugène Charles Catalan (1814–1894), a Belgian mathematician. Catalan numbers arise in a number of problems in combinatorial mathematics and can be used among other things to, for example, describe the following: (i) the number of binary trees with n vertices; (ii) the number of ways to use n rectangles to tile a staircase shape $(1, 2, \dots, n-1, n)$; (iii) the number of ways a polygon with $n+2$ sides can be cut into n triangles; (iv) the number of well formed sequences of $2n$ parentheses; and (v) the number of ways $2n$ ballots can be counted, in order, with n positive and n negative, so that the running sum is never negative.

CATALAN'S CONSTANT: A constant, denoted by K that is the sum of the alternating series

$$\sum_{n=0}^{\infty} (-1)^n (2n+1)^{-2} = 1 - \frac{1}{9} + \frac{1}{25} - \frac{1}{49} \dots$$

It is approximately equal to 0.915965. It commonly appears in estimates of combinatorial functions and in certain classes of sums and definite integrals.

CATALASE: A haemeprotein which catalyses the disproportionation of dihydrogen peroxide to O_2 and water; it also catalyses the oxidation of other compounds, such as ethanol, by dihydrogen peroxide. A non-haemeprotein containing a dinuclear manganese cluster with catalase activity is often called **pseudocatalase**.

CATALOGUE (CATALOG): 1. A directory of files and libraries having reference to their locations. A catalogue may contain other information such as the types of devices in which the files are stored, passwords and blocking factors. 2. To enter information about a file or a library into a catalogue.

CATALYTIC TITRATION: A titration process, which involves a catalyst, where the end-point is detected by the sudden increase or decrease of the rate of a reaction.

CATALYSIS: The process of speeding up a chemical reaction by addition of a catalyst. Catalysis can be classified as **homogeneous catalysis**, in which only one phase is involved or **heterogeneous catalysis**, in which the reaction occurs at or near an interface between phases. Catalysis brought about by one of the products of a reaction is called **autocatalysis**. Catalysis brought about by a group on a reactant molecule itself is called **intramolecular catalysis**.

CATALYST: 1. A substance that increases the rate of a chemical or biochemical reaction without itself changing or becoming part of the end product. Catalysts function by reducing the activation energy between reactants and products. A catalyst is both a reactant and product of the reaction. Catalysts may be inorganic compounds organic compounds or complex combinations. Enzymes are also catalysts. 2. A software suite for Linux and Microsoft Windows included with ATI Radeon video cards that include video drivers and the Catalyst Control Centre (CCC).

CATALYST AGEING: Catalyst deactivation resulting from changes in the structure or in the texture of the catalyst. Changes of this kind are usually irreversible and the catalyst cannot be regenerated.

CATALYST DECAY (CATALYST DEACTIVATION): The conversion in a catalytic reaction performed under constant conditions of reaction, which often decreases with time of run or time on stream.

CATALYTIC ACTIVITY: The increase in the rate of a specified chemical reaction caused by an enzyme under specified assay conditions. It is measured in katals or in moles per second.

CATALYTIC ACTIVITY CONCENTRATION: Catalytic activity of the component divided by the volume of the system.

CATALYTIC ACTIVITY CONTENT: Catalytic activity of the component divided by the mass of the system.

CATALYTIC ACTIVITY FRACTION: Quotient of the catalytic activity of the isozyme and the catalytic activity of all the isozymes of the same enzyme in the system.

CATALYTIC ANTIBODY (ABZYME): An antibody selected for its ability to catalyse a chemical reaction by binding to and stabilising the transition state intermediate.

CATALYTIC CONVERTER: A device fitted to the exhaust system of a motor vehicle in order to reduce toxic emission from the engine by converting carbon monoxide, hydrocarbons and oxides of nitrogen (NO_x) to harmless carbon dioxide, water and nitrogen.

CATALYTIC CRACKING: A process of using catalyst in breaking down long-chain hydrocarbons into more useful shorter-chain ones. It is used in the refinery of crude oil to produce fuels such as petrol and kerosene.

CATALYTIC DEHYDROCYCLIZATION: A reaction in which an alkane is converted into an aromatic hydrocarbon and hydrogen.

CATALYTIC DOMAIN: Any part of a polypeptide chain that possesses a catalytic function. It may contain more than one structural domain.

CATALYTIC GRAPHITISATION/GRAPHITIZATION: The transformation of non-graphitic carbon into graphite by heat treatment in the presence of certain metals or minerals. Catalytic graphitisation gives a fixed degree of graphitisation at lower temperature and/or for a shorter heat treatment time than in the absence of the catalytic additives (or a higher degree of graphitisation at fixed heat treatment conditions). Often it involves dissolution of carbon and precipitation of graphite at the catalyst particles so that non-graphitising carbons can be graphitized by this procedure.

CATALYTIC HYDROCRACKING: A process similar to catalytic cracking in its industrial purpose but effected under hydrogen pressure and on a catalyst containing an ingredient with a hydrogenating function.

CATALYTIC HYDRODESULFURIZATION: A process in which the presence of hydrogen and gas are removed as hydrogen sulfide.

CATALYTIC METHANATION: A process for removing carbon monoxide from gas streams or for producing methane by the reaction.

CATALYTIC REFORMING: A process for increasing the octane number of naphthas. It involves isomerization of alkanes, dehydrogenation of cyclohexanes to aromatic hydrocarbons, isomerization and dehydrogenation of alkylcyclopentanes and dehydrocyclization of alkanes.

CATALYTIC RIBONUCLEIC ACID/RNA (RIBOZYME): A natural or synthetic ribonucleic acid (RNA) molecule that cuts an RNA substrate.

CATALYTIC SITE: The site of an enzyme involved in the catalytic process. It contains the catalytic and binding sites. The structure and chemical properties of the active site allows the recognition and binding of the substrate.

CATAPHYLL: A reduced leaf, such as a bract.

CATAPULT: A mechanism for forcefully propelling stones, spears or other projectiles. A simple example is the y-shaped stick with a piece of elastic attached to it, used especially for shooting stones. Also steam, hydraulic pressure, tension or other force is used to launch gliders, aircraft or missiles from a launching pad such as the deck of a ship.

CATARACT: 1. A medical condition in which the crystalline lens of the eye becomes cloudy, causing partial or total blindness. It is caused by factors such as age, diabetes mellitus, prolonged exposure to ultraviolet rays or trauma. It may also occur in newborns and infants. 2. A large steep waterfall, especially one containing great volumes of water rushing over a precipice.

CATARRH: An inflammation of the mucus membrane of the nose and throat, causing increased flow of mucus. It is a symptom usually associated with the common cold and chesty coughs.

CATASTROPHE THEORY: 1. The mathematical theory that classifies surfaces up to diffeomorphism according to their form. 2. The popular application of the theory to the explanation of abruptly changing phenomena, by, for example, the discontinuity of a line on the topmost fold of a folded surface. It can be represented in two dimensions.

CATASTROPHIC EVENT: In agriculture, it is an unforeseen event caused by human action such as setting uncontrollable fire that can have high impact on agricultural production systems.

CATASTROPHISM: An early belief that events in the history of the earth had occurred in the past as sudden catastrophic events such as floods or earthquakes and by processes unlike those operating today. Periods of catastrophic change were followed by long periods of little change.

CATATONIA: A medical condition that is typically schizophrenia characterised by a range of abnormal movements, muscular rigidity and stupor. The causes include mental disorders, post-traumatic stress disorder and Parkinson's disease.

CATCH CROP: A fast-growing crop planted between the rows of a main slower growing crop. Examples are vegetables such as radishes, onions grown from sets or spinach; quick-growing crops such as rye, millet or buckwheat; or an annual legume, such as soybean, which is valuable as fodder or which increases the soil's fertility. Catch crops can also be defined as crops grown in the time interval between seasons of regular planting. Catch crops are effective in reducing nitrate leaching, because they remove nitrate from the soil at the time the leaching takes place, by absorbing nitrogen in autumn before a spring sown crop.

CATCHMENT AREA: 1. In hydrology, it is also called **catchment basin** and is an area from which water is collected by a river and its tributaries. 2. The area served by an institution, such as a hospital or school.

CATECHOLAMINE: Any one of a class of amines that possesses a catechol ($\text{C}_6\text{H}_4(\text{OH})_2$) group, including dopamine, adrenaline and noradrenaline; and which functions as neurotransmitters or hormones.

CATECHU: Any one of several astringent substances obtained from various tropical plants, especially from the wood of *Acacia catechu* and *Acacia suma* found in East Indian that is rich in tannic acid and used in medicine, dyeing, tanning, etc.

CATEGORICAL DATA: A set of data having values or observations belonging to it that can be sorted according to category. Each value is chosen from a set of non-overlapping categories. For example, shoes in a cupboard can be sorted according to colour: the characteristic 'colour' can have non-overlapping categories 'black', 'brown', 'red' and 'other'. People have the characteristic of 'gender' with categories 'male' and 'female'.

CATEGORICAL VARIABLE (NOMINAL VARIABLE): A variable whose values are not numerical but names or labels that have two or more categories. Examples are gender (e.g., male or female) and colour (e.g., red, green, blue).

CATEGORY: 1. In blogging, it refers to a software feature that allows a blogger to group posts together by topic. Categories can be labelled by short descriptions called tags or elements. 2. A structure consisting of a class of objects, denoted ObC and a class of arrows or morphisms, denoted ArC , such that disjoint classes of morphisms are associated with each pair of objects and the morphisms are associative with an identity under composition. Categories are useful as an abstract model for the study of structures and mappings that preserve these structures. When the categories are small and the morphisms are functions, it is called a **classical category** or **kittygory**. 3. In statistics, category represents a set of people or objects that have common characteristics. To study the people in a population, for example, they can be sorted into "natural" categories, by gender (men and women), for example or into categories defined by other criteria, such as vocation (managers, secretaries, farmers, etc.).

CATEGORY 3 CABLE (CAT 3 CABLE): A type of network cable that consists of four UTP wires, transmits data at 10Mbps and was used in token ring networks, 10BASE-T and is still used in some older phone systems.

CATEGORY 5 CABLE (CAT 5 CABLE): A type of cable that is used extensively in Ethernet connections in local networks, as well as telephony and other data transmissions. Category 5 cable employs a twisted pair design, rather than a coaxial cable or a fibre-optic cable design.

CATEGORY THEORY: The general mathematical theory of structures and systems of structure with many complementary natures, for example, geometric, logical, computational, combinatorial, just as groups are many-faceted algebraic structures. For example, a mapping will be called an identity if and only if the existence of any product ea or be implies that $ea = a$ and $be = b$.

CATENA (SOIL CATENA): A grouping of a sequence of different soil profiles that occur down a slope. These soil profiles are from similar parent materials, and occurring under similar climatic conditions, but have dissimilar characteristics because of variations in relief and changes from one drainage condition to another.

CATENA-: An affix used in inorganic nomenclature to denote a chain structure. It is often used to designate linear polymeric substances.

CATENANE: A supramolecule consisting of two or more large rings that are interlocked like the links in a chain with no chemical bonding between the rings, being bonded together mechanically.

CATENARY: 1. A curve formed when a chain or rope of uniform density hangs from two fixed points. It is represented by the formula $y = \frac{1}{2}a(e^{x/a} + e^{-x/a})$. The value of e is about 2.718. The value of a is the y intercept. The catenary can also be represented by the hyperbolic cosine function $\cosh x$. 2. The measure of the sag at the midpoint, when strands of materials such as fibreglass-reinforced plastics of specified length is suspended at the ends.

CATENATION: 1. The linking of molecules without any direct covalent bonding between them, as when two circular DNA molecules interlock like the links in a chain. 2. Ability of a chemical element to form a long chain by covalent bonds as in carbon compounds. Several of the main group elements are known to catenate. For example, sulfur forms S_n , where $n = 1 - 6$.

CATENOID: The geometrical surface generated by rotating a catenary around its axis of symmetry.

CATERPILLAR: The larva of a butterfly or moth, members of the order Lepidoptera. Caterpillars have distinct heads and are segmented and wormlike. The silkworm, the caterpillar of the domesticated silkmoth, *Bombyx mori* is a primary producer of silk.

CATHEPSINS (CTS): Protein-digesting enzymes (proteases) that are found in lysosomes. There are approximately a dozen members of this family, which are distinguished by their structure, catalytic mechanism and which proteins they breakdown.

CATHETER: A tube used to drain or inject fluid or to apply pressure to a vessel. Urinary catheters are commonly used to drain urine from the bladder. Other types are used to sample blood from the heart or to inject dye into the blood vessels during X-ray screening (see cardiac catheterisation). Catheters are also used to unblock or widen obstructed blood vessels or to control bleeding.

CATHETOMETER: A telescope or microscope fitted with crosswire in the eyepiece and mounted so that it can slide along a graduated scale. Cathetometers are used for accurate measurement of small differences in height, without mechanical contact, for example, between two columns of mercury.

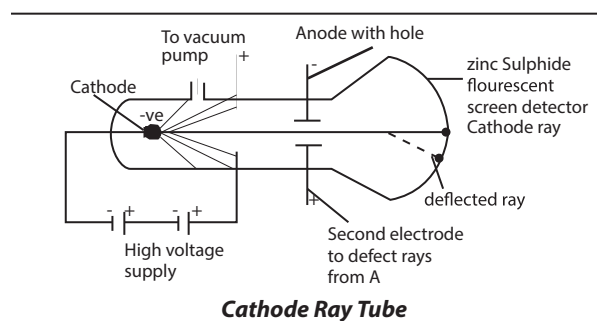
CATHETUS: A line that is perpendicular to another line. Usually, it refers to one of the lines in a right triangle that is not the hypotenuse.

CATHODE: An electrode that has a negative charge or a negative electrode by which an electric current leaves a device such as a

battery. In chemistry, it is the negative electrode of an electrolytic cell toward which positive particles (cations) usually in solutions are attracted. In a galvanic cell it is the electrode at which reduction takes place. In an electron tube, such as a cathode-ray tube, it is the electrode from which electrons stream off and travel through the tube toward the anode.

CATHODE RAY OSCILLOSCOPE (C.R.O.): An electronic measuring instrument that provides a virtual image of electrical signals on a screen corresponding to oscillations of voltage and current by means of the fluorescent screen of a cathode ray tube. It consists of a vacuum tube in which a hot cathode emits electrons that are accelerated as a beam through a relatively high voltage anode, further focused or deflected electrostatically or electromagnetically and allowed to fall on a phosphorescent screen. Oscilloscopes have many applications such as to examine the voltages on circuit boards, radios, TVs and other electronic devices. In medicine, it is used in processes such as the displaying of patients' brain waves and heartbeats for monitoring and diagnostic purposes.

CATHODE RAY TUBE (CRT): An electronic device that uses a beam of electrons to produce spots of light on a fluorescent screen. To form a display on the screen, the electron beam is deflected in the vertical and horizontal directions either by the electrostatic effect of electrodes within the tube or by magnetic fields produced by coils located around the neck of the tube. Cathode-ray tubes are used in television sets, oscilloscopes, computer monitors and radar displays. CRTs have been superseded by more modern display technologies such as liquid crystal display (LCD), Plasma, light-emitting diode (LED) and organic light emitting diode (OLED).



CATHODE RAYS: A stream of electrons emitted from the cathode or the negatively charged electrode of a vacuum tube to the anode or positively charged electrode, across a voltage difference between the electrodes placed at each end. Electrons are repelled by the negative charge; therefore, the cathode is the source of the cathode ray in the vacuum chamber. The cathode ray technology is used in television sets.

CATHODIC PROTECTION: A method of protection of a metal against corrosion by making it the cathode in an electrochemical cell by attaching it to a more easily oxidised metal called the sacrificial anode. Cathodic protection is commonly used to protect, for example, water or fuel pipelines, underground storage tanks, ships, offshore oil platforms or any metallic structure in salty water.

CATION: A positively charged ion in electrolyte attracted to the cathode (negative electrode) during electrolysis. Alternatively, an atom or group of atoms carrying a positive charge, indicated by a superscript plus sign after the chemical symbol. Examples are sodium (Na^+), calcium (Ca^{2+}) and ammonium (NH_4^+).

CATION EXCHANGE: The interchange between cations in a solution and those on the surface of materials such as clay colloid or organic matter.

CATION EXCHANGE CAPACITY (CEC): The quantity of negative charges that exist on the surfaces of clay particles and organic matter in soil that attract positively charged ions or cations through electrostatic forces. In other words, it is the total amount

of exchangeable cations that a soil can adsorb. It is a measure of the soil's ability to hold positively charged ions and its capacity to exchange cations between the soil particles and the soil solution (the water in the soil). Its unit is cmolc/kg (centimol positive charge per kg of soil) or meq/100 g (milli-equivalents per 100 grams of soil). It is a very important soil property influencing soil structure stability, nutrient availability, soil pH and the soil's reaction to fertilisers and other ameliorants. The main ions associated with CEC in soils are the positively charged ions or cations such as potassium (K^+), calcium (Ca^{2+}), magnesium (Mg^{2+}), and ammonium (NH_4^+), which are major plant nutrients. Other cations occupying the CEC in soils are hydrogen (H^+), sodium (Na^+), and aluminum (Al^{3+}). Clay particles and organic matter in the soil are negatively charged. The presence of clay or organic matter in soil means high CEC, hence greater water holding capacity. Sandy soils have low CEC and cannot retain much water. Small increases in organic matter will greatly increase soil CEC. The CEC of a soil is important because it implies the size of the reservoir of nutrients that is available to replenish the nutrients removed by plant uptake. Also, the CEC of a soil can be an indicator of how much leaching of nitrogen and potassium fertilisers will occur; i.e., higher CEC soils will experience less leaching of cationic elements.

CATIONIC DETERGENTS: A type of detergent in which the active part of the molecule is a positive ion (cation or a class of synthetic detergents in which the surface-active part of the molecule is the cation. They usually consist essentially of a quaternary ammonium salt with 12 to 24 carbon atoms in the organic groups attached to the nitrogen atom in the cation. They possess wetting, foaming and emulsifying properties but are not good detergents. It is used mostly in areas in which anionic detergents cannot be used. Cationic-active agents are used as emulsifying agents for asphalt in the surfacing of roads and also possess excellent germicidal properties and are utilised in surgery in dilute form.

CATIONIC POLYMERISATION: An ionic polymerisation in which the kinetic-chain carriers are cations.

CATKIN: A pendulous inflorescence in flowering plants, bearing numerous small, usually unisexual flowers. The calyx and corolla are normally reduced or absent to allow maximum air circulation around the flower and the catkin itself develops in an exposed position on the plant. Male catkins shed vast amounts of light dry pollen. Female catkins usually have long hairy styles and stigmas to enhance pollen interception. Catkins are formed by many tree species, for example, Salicaceae and the male flowers of the Betulaceae (birches, hazels, alders, hornbeams) are always aggregated into catkins.



Catkin

CATTLE: Large domesticated herbivorous mammals such as cows and oxen belonging to the Genus, *Bos*, of the family Bovidae kept for the production of milk, meat and hides and also as draft animals.

CAUCHY, AUGUSTIN LOUIS, BARON (1789 -1857): French mathematician and physicist, who laid the foundations of modern analysis in terms of limits and continuity and developed the theory of functions of complex variables. He became Professor at the École Polytechnique, the Sorbonne and the Collège de France. He published many works, including monographs on definite integrals and on wave propagation and papers on geometry, number theory, elasticity, theory of error, astronomy and optics.

CAUCHY CONDENSATION TEST: The test for the convergence of a series that relies on the fact that if $\{p_n\}$ are decreasing positive terms, then $\sum p_n$ and $\sum 2p_{2n}$ converge or diverge together.

CAUCHY CONDITION: 1. The necessary and sufficient condition for an infinite sequence to converge that the absolute difference between successive terms with sufficiently large indices tends to zero. 2. Any of a number of conditions establishing the convergence

of a sequence, series, product, etc., by verifying that one obtains a Cauchy sequence and then appealing to the completeness of the underlying metric space. For example, if $\{a_i\}$ is a Cauchy sequence of elements of a normed space, then the associated series converges

whenever for every $\epsilon > 0$ there is an N such that $\left\| \sum_{k=m}^j a_k \right\| < \epsilon$ for all $j > m > N$.

3. An initial condition for a second-order partial differential equation in which u and $\partial u / \partial t$ are given at t .

CAUCHY INTEGRAL FORMULA: The fundamental formula

$$f(z) = \frac{1}{2\pi i} \int_C f(w)[w-z]^{-1}, \text{ for a function that is analytic on a region,}$$

where C is any positively oriented simple closed curve surrounding z and lying inside the region.

CAUCHY PRINCIPAL VALUE: The evaluation of an improper integral on the interval $[-\infty, \infty]$ as the symmetric (two-sided limit) of the integral on intervals of the form $[-n, n]$. This may well converge even if the sum of the two ordinary improper integrals over $[-\infty, a]$ and $[a, \infty]$ does not. This is the case for odd functions such as x or $\sin x$.

CAUCHY PRODUCT: The convolution of two sequences given by $C_n = \sum_{k=0}^n a_{n-k} b_k$. If the series $\sum a_n$ sums to A and $\sum b_n$ to B , then the Cauchy product series sums to AB if one of the series is absolutely convergent.

CAUCHY SEQUENCE (FUNDAMENTAL SEQUENCE): An infinite sequence of points or values whose distances between them tend to zero as their indices tend to infinity; $\{a_i\}$ is a Cauchy sequence in a metric space if, for every $\epsilon > 0$, there is an N such that $d(a_i, a_j) < \epsilon$ for all $i, j > N$. An example of Cauchy sequence is $\{1/n\}$.

CAUCHY-GREEN DEFORMATION TENSORS: The tensors $F^T F$ (the right Cauchy-Green deformation tensor) and $F F^T$ (the left Cauchy-Green deformation tensor) for a given deformation gradient F .

CAUCHY-HADAMARD THEOREM: The theorem that the radius of convergence of a Taylor series with coefficients $\{a_n\}$ is $\lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{|a_n|}}$.

CAUCHY-KOWALSKI THEOREM: The theorem that if S_r is a system of r partial differential equations in r unknown functions, $v_1, \dots, v_r$, of $p+1$ real variables, $x_1, \dots, x_{p+1}$ of the form

$$\frac{\partial v_j}{\partial x_{p+1}} = H_j \left(x_1, \dots, x_{p+1}, v_1, \dots, v_r, \frac{\partial v_1}{\partial x_1}, \frac{\partial v_2}{\partial x_2}, \dots, \frac{\partial v_r}{\partial x_p} \right),$$

where there are no derivatives with respect to x_{p+1} on the right-hand side and the H_j are real analytic functions, then there exists a neighbourhood of the origin in $(p+1)$ -dimensional Euclidean space in which S_r may be uniquely solved for analytic functions $v_1, \dots, v_r$.

CAUCHY-RIEMANN EQUATIONS: The equations linking the partial derivatives of the real and imaginary parts of an analytic function of z with real part, x and imaginary part, y . When $f = u + iv$, with u and v real valued, these are $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

When the partial derivatives are continuous, these equations are also sufficient for analyticity, when the domain of analyticity is a region.

CAUCHY-SCHWARZ INEQUALITY (BUNIAKOVSKI'S INEQUALITY): The inequality, valid for any inner product, that

$$\langle x, y \rangle \leq \|x\| \|y\|$$

In a Euclidean space this can be written as Cauchy's inequality.

CAUCHY'S INEQUALITY: The special case of Cauchy-Schwarz

$$\text{inequality in Euclidean space: } \sum_{i=1}^n a_i b_i \leq \sqrt{\left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right)}.$$

This follows from the cosine law.

CAUCHY'S INTEGRAL THEOREM: The theorem of complex analysis stating that the contour integral of an analytic function around a simple closed curve is zero.

CAUCHY'S LEMMA: The result that if G is a finite group and p is a prime number that divides the order of G , then G contains an element of order p .

CAUCHY'S MEAN-VALUE THEOREM (GENERALIZED MEAN-VALUE THEOREM): It states that if f and g are differentiable in an interval (a, b) and continuous on $[a, b]$ then $f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)]$ for some point c in the open interval.

CAUCHY'S RATIO TEST (RATIO TEST): A test for a complex series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent the limit of the ratio of the

absolute values of successive terms is tested. If the term $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$

is less than unity, the series converges absolutely; if it is greater than unity, it diverges.

CAUCHY'S RESIDUE THEOREM (RESIDUE THEOREM OF CAUCHY): A tool used to evaluate line integrals of analytic functions over closed curves. It generalises the Cauchy integral theorem and Cauchy's integral formula that if a function f is analytic in a simply connected domain D except for finitely many isolated singularities, then the contour integral of f over any simple closed curve that misses the singularities is equal to $2\pi i$ times the sum of the residues of f at singularities inside the contour. An equivalent definition is that the integral of a complex function round a Jordan contour is the sum of residues of that function inside the curve multiplied by $2\pi i w$, where w is the winding number of the curve about that point. This is of great use for evaluating curvilinear integrals of meromorphic functions and so certain classes of definite integrals of real functions.

CAUCHY'S ROOT TEST (ROOT TEST): The test for whether a complex series $\sum a_n$ is absolutely convergent by considering $L = \limsup a_n^{1/n}$; if L is less than unity, the series converges absolutely and if it is greater than unity, the series diverges. The radius of convergence of a power series is the reciprocal of L when the $\{a_n\}$ are the coefficients of the series.

CAUCHY'S STRESS PRINCIPLE: The fundamental axiom postulating that the stress vector at a point on a surface of a body depends continuously on the unit outward normal to the surface at that point.

CAUCHY'S STRESS THEOREM: The theorem which states that the stress vector at a point on a given surface of a body is given by σn , where σ is a symmetric second order Cartesian tensor and n is the unit outward normal to the surface at that point.

CAUCHY'S VORTICITY FORMULA: The relation, for a given body executing a circulation preserving motion, of the vorticities ω_r and ω_t respectively in the reference and current configurations; this states that $\omega_t = \frac{F \omega_r}{\det F}$, where F is the deformation gradient.

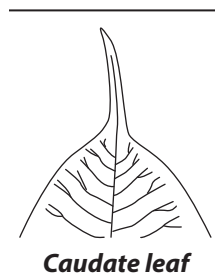
CAUDA EQUINA: A collection of spinal nerves that descends from the lower part (distal end) of the spinal column, beyond the conus medullaris and occupies the lower third of the spinal canal (the central space within the spine). **Cauda equine syndrome** is a rare disorder affecting the cauda equina at the lower (lumbar) end of the spinal cord, in which the nerves at the end of the spinal cord are compressed. It may be caused by a ruptured disk, tumour, infection, fracture or narrowing of the spinal canal. It may also happen because of a violent impact, such as a car crash, a fall from significant height or a penetrating injury. If medical attention is not sought early to relieve the pressure, it can result in permanent paralysis, impaired bladder and/or bowel control, loss of sexual sensation and other problems.

CAUDAL TAIL: A tail-like appendage attached to the pelvic girdle and corresponding to the hind limbs of higher vertebrates; or a bundle

of the spinal cord in vertebrates; or a term used to describe that part of the body of an animal at or near the tail. It includes the caudal fin of a fish and the narrow part of a fish's body just before the tail fin called the caudal peduncle.

CAUDAL VERTEBRAE: The bone of the tail, which articulates with the sacral vertebrae; their number varies with the species.

CAUDATE: Having a tail-like appendage, as shown in the diagram.



Caudate leaf

CAUDEX: 1. The trunk of a palm or tree fern. 2. A short, thick, vertical or branched perennial often with woody base, found mostly in herbaceous perennial. The plural is **caudices** or **caudexes**.

CAULERPALES (SIPHONALES): An order of the Bryopsidophyceae containing algae that lack any dividing walls. They also differ in containing α -carotene in addition to β -carotene. In some classifications these acellular algae are placed in the three orders: Derbesiales, Codiales (including Bryopsis and Codium) and Caulerpaes (including Caulerpa and Halimeda).

CAULESCENT: Having a leafy stem rising above the ground.

CAULICLE: A small stem; a rudimentary stem.

CAULIFLORY: The production of flowers on tissue that has been secondarily thickened, such as the branches or trunks, rather than at apical meristems. In other words flowers and fruits are produced on the older branches or trunks of woody plants rather than from new growth and shoots. This trait is found most frequently in species such as cocoa (*Theobroma cacao*) that make up the lower canopy of tropical forests and the redbud. This may allow the seeds to be dispersed by animals which cannot climb or fly. **Cauliflorous** is the production of flowers and fruits directly from the branches or trunks of woody plants.

CAULIFLOWER (*Brassica oleracea*): A herbaceous annual or biennial plant, grown for its edible head, which is made up of aborted white flower buds. The flowers are unable to produce fruit or seed as they possess only female reproductive organs; the male organs are either underdeveloped or totally lacking. The flowers are diverse in colour such as green, white orange and purple. It is a shallow rooted plant that can grow to 0.8 metres high with a small, thickened stem. It has light green, ribbed leaves that branch off the top of the stem. It is a cool weather vegetable that thrives in organically rich, fertile and consistently moist, well-composted and well-drained loams in full sun. Reproduction is by seeds and can be grown from transplants started in glasshouses or indoors. Its fat and carbohydrates contents are low, but high in roughage, vitamin C and water. It is eaten as a vegetable, either fresh or cooked.

CAULINE: Pertaining to the stem, especially buds and flowers borne directly on the stem, as seen in, for example, cocoa (*Theobroma cacao*).

CAULIS: The main stem of a herbaceous plant.

CAULOCARPIC: Referring to stems, which live for several years.

CAULOID: Stem-like.

CAULOID THEORY: A theory of the origin of the sporophyte of vascular plants that proposes that the plant body has been differentiated from a primeval axis or stem.

CAULOUS: In botany, it refers to branches, which are more or less evenly spaced along the trunk.

CAUSAL AGENT: An organism which causes a disease.

CAUSAL DIAGRAM: A graphical representation of the relationship between variables with arrows from each explanatory variable to each of the dependent variables that it affects.

CAUSAL RELATIONSHIP: A relationship where one variable causes another variable to change. If there is a causal relationship between two variables, they must be correlated. It is cause and effect

relationship in which one event, called the cause makes another event happen, called the effect.

CAUSALITY: Relationship between cause and effect or the principle that future events are caused by past events and that future events cannot be the cause of past events.

CAUSTIC: A term describing chemical substances which attack or corrode organic materials. Caustic substances are strongly alkaline; an example is caustic soda. Concentrated inorganic acids, such as sulfuric acid are also caustic. 2. In optics, the curve or surface formed by the reflection of parallel rays of light in a large-aperture concave mirror.

CAUSTIC LIME (CALCIUM HYDROXIDE; CALCIUM(II) HYDROXIDE; SLAKED LIME; PICKLING LIME; HYDRATED LIME): A soft white powder, with the chemical formula Ca(OH)_2 , density 2.211 g/cm^3 (solid) and melting point 580°C , 853 K (loses water). It is used in making mortar, cements, calcium salts, paints, hard rubber products and petrochemicals.

CAUSTIC POTASH (POTASSIUM HYDROXIDE; POTASH LYE; POTASSIA; POTASSIUM HYDRATE): A caustic white solid, with the chemical formula KOH , density 2.044 g/cm^3 , melting point 406°C (679 K) and boiling point 1327°C (1600 K). It can be found in pure form by reacting sodium hydroxide with impure potassium. It is a strong base and dissolves readily in water, giving off much heat and forming a strongly alkaline, caustic solution. It is used as bleach and in the manufacture of soaps, dyes, alkaline batteries and many potassium compounds. Potassium hydroxide is extremely corrosive to body tissues and especially damaging to the eyes.

CAUSTIC SODA (SODIUM HYDROXIDE; LYE): A strongly alkaline compound, with the chemical formula NaOH , density 2.13 g cm^{-3} , melting point 318°C (591 K) and boiling point 1388°C (1661 K). It is hygroscopic and readily absorbs carbon dioxide from the air and should be stored in an airtight container. It is very soluble in water and is highly exothermic when dissolved in water. It also dissolves in ethanol and methanol. It is used in the manufacture of chemicals and soaps and in petroleum refining.

CAUTERISE (CAUTERIZE): To remove a body tissue by burning or searing by applying heat from electric current, laser, a thermal source, solar radiation; as well as a caustic agent or dry ice. The procedure, called **cauterisation** is usually used to destroy damaged or diseased tissues to prevent infections or to coagulate blood vessels to reduce or stop bleeding. **Cautery** is the process of cauterising. **Cautery**, also known as cauterant, it is any agent or instrument used to destroy abnormal tissue by burning or searing. Some of the processes of cauterisation include: (i) **electrocautery**, which uses electric current to heat an instrument or by directing a high-frequency current through a tissue; (ii) **chemocautery** or **chemical cautery**, using chemical agents such as silver nitrate, trichloroacetic acid, cantharidin and liquid nitrogen.

CAVALIERI'S PRINCIPLE: The principle that solids with the same height and with cross sections of the same area have the same volume. This applies in particular to prisms and cylinders with the equal bases and heights. This is easily established using the integral calculus. It was named after the Italian mathematician and physicist Francesco Cavalieri (1598-1647).

CAVENDISH, HENRY (1731 - 1810): British chemist and physicist, noted for his discovery of hydrogen, which he called "inflammable air". He described the density of inflammable air, which formed water on combustion in 1766. He was distinguished for great accuracy and precision in his researches into the composition of atmospheric air, the properties of different gases, the synthesis of water, the law governing electrical attraction and repulsion and mechanical theory of heat. He used a sensitive torsion balance, called the Cavendish balance to measure the value of the gravitational constant, G , which he used in an experiment called experiment to weigh the Earth.

CAVEOLA (CAVELAE): A small static flask-shaped depression in the plasma membrane of a cell that is lined by transmembrane proteins

called caveolins and contains receptors for various extracellular signaling molecules.

CAVITANDS: Compounds constrained by structure to having or accommodating a cavity large enough to host other molecules.

CAVITATION: The formation of gas or vapour filled cavities in liquids in motion when the pressure is reduced to a critical value while the ambient temperature remains constant or the formation of partial vacuum in fluids at high velocities, produced by propellers or other machine parts in hydraulic engines, in accordance with Bernoulli's principle. Cavitation is undesirable because it produces extensive erosion of the rotating blades and reduces efficiency of flow.

CAVITY: An empty space within something solid. An example is a hole in a tooth.

CAVITY MAGNETRON: A type of vacuum tube or valve in which the electrons emitted by the cathode are subjected to a magnetic as well as an electric field. As the electrons are carried round in the magnetic field they sweep by a series of oscillatory circuits and maintain oscillations in them, which are sources of microwaves. Cavity magnetrons give pulses of hundreds of kilowatts of energy for about a millionth of a second, without destroying them, just as it is possible to get brilliant flashes from an electric light if subjected to over-voltage for a very short interval.

CAVITY RADIATION (BLACKBODY RADIATION): An object or system which absorbs all radiation incident upon it and re-radiates energy which is characteristic of this radiating system only, not dependent upon the type of radiation which is incident upon it.

CAVITY RESONATOR (MICROWAVE RESONANCE CAVITY; MICROWAVE CAVITY; WAVEGUIDE RESONATOR; RHUMBATRON): An enclosure of metallic conductor within which an oscillating electromagnetic field can be maintained. The oscillation takes place at certain frequencies, called its **resonant frequencies**, with greater amplitude than at others. The oscillations in a resonator can be either electromagnetic or mechanical including acoustic. The cavity resonator enclosure has a volume which stores energy oscillating between one form and another. For sound, the oscillation is between displacement and velocity of particles; and for electromagnetic waves, the energy oscillates between the magnetic and the electric fields. Acoustic cavity resonators, in which sound is produced by air vibrating in a cavity with one opening, are known as **Helmholtz resonators**. Resonators are used to either generate waves of specific frequencies or to select specific frequencies from a signal. Resonator pipe organs and flutes use acoustic to increase their sonority.

CAVITY WALL: In architecture, a double wall consisting of two vertical layers separated by air space and joined horizontally by masonry or metal. It is designed to give extra insulation.

CAYLEY ALGEBRA: A non-associative non-commutative division algebra of dimension 8 over the real numbers. It was named after the the English mathematician, Arthur Cayley (1821-1895).

CAYLEY REPRESENTATION THEOREM: The result that every group is isomorphic to a group of permutations; named after the the English mathematician, Arthur Cayley (1821-1895).

CAYLEY-HAMILTON THEOREM: When we consider the characteristics of the polynomial $p(\lambda)$ of an $m \times m$ matrix A , it is defined by $p(\lambda) = \det(A - \lambda I)$. The theorem about the characteristics polynomial is known as Cayley-Hamilton theorem, which states that every square matrix over a commutative ring satisfies its own characteristic equation. According to Cayley-Hamilton's theorem, if the characteristics $p(\lambda)$ of an $m \times m$ matrix A is $p(\lambda) = (-1)^n (\lambda^n + b_{n-1} \lambda^{n-1} + \dots + b_1 \lambda + b_0)$, then $A^n + b_{n-1} A^{n-1} + \dots + b_1 A + b_0 I = 0$.

CAYLEY, ARTHUR (1821-1895): A British mathematician who contributed in the field of pure mathematics. He published a lot of papers on algebra and geometry. He developed the theory of matrices and was one of the main mathematicians who first studied abstract algebra.

CD SPECTRUM (CIRCULAR DICHROISM SPECTRUM): An effect, which is observed when optically active matter absorbs left and right hand circularly polarised light slightly differently. A CD-spectrum is obtained by plotting the variable $I_R - I_L$ against frequency of the incident electromagnetic radiation (190-170nm), where I_R and I_L are the absorption intensities for right and left circularly polarised light. Practical applications include, for example, the determination of configurations of complexes of transitional metals and secondary structures of biomolecules.

CD-i: Abbreviation for **compact disc interactive**, an interactive multimedia CD player that can combine audio, video and text on high-capacity compact discs and a variant of CD-ROM. It also refers to the multimedia Compact Disc standard used by the CD-i console, also known as Green Book, which was developed by Philips and Sony.

CD-QUALITY SOUND: Digitized sound at 44.1 KHz and 16 bits, the standard defined in ISO 10149, known as the Red Book. CD-quality sound was designed to be the minimum standard required to reproduce every sound the human ear can hear. Most audio CDs are recorded to this level.

CD-R: Abbreviation for compact disc-recordable, a type of write once, read many (WORM) compact disc (CD) format that allows one-time recording on a disc or a compact disc on which can be written only once and thereafter is read-only memory. It is used to back up and transfer data to master CD-ROMs, CD-R discs can be written and read by most CD/DVD drives.

CD-ROM: Abbreviation for **compact disc read-only memory**, a device that is based on the audio compact disc and provides read-only access (but not writable) to a large amount of data, up to 640 megabytes for use on computer systems.

CD-ROM EXTENDED ARCHITECTURE (CD-ROM XA): A set of standards for storing multimedia information on CD-ROM. It was introduced by Philips, Sony and Microsoft in 1988 as a partial development of the CD-I standard. It interleaves data (as in CD-I) so that blocks of audio data are sandwiched between blocks of text, graphics or video. This allows parallel streams of data to be handled, so that information can be seen and heard simultaneously.

CD-RW: Abbreviation for **compact disc re-writable**, a compact disc (CD) format that enables repeated audio or other digital data to be written, read, erased and rewritten.

CD4: A glycoprotein predominantly found on the surface of helper T cells consisting of a single polypeptide chain organised into four immunoglobulin-like domains that characterises helper T cells. In humans, it is a receptor for HIV, enabling the virus to gain entry into its host.

CD8: A surface glycoprotein, comprising two distinct chains linked by a disulfide bond that characterises cytotoxic T cells. CD8 binds to any antigen-MHC I complex on the surface of an infected host cell and acts as a coreceptor, helping to activate the T cell and prompting it to kill infected host cell. There are two isoforms of the protein, alpha and beta, each encoded by a different gene. In humans, both genes are located on chromosome 2 in position 2p12.

CDMA (CODE-DIVISION MULTIPLE ACCESS): A coding system, used as a modulation technique, in which multiple channels are independently coded for transmission over a single wideband channel. In some communication systems, CDMA is used as an access

method that permits carriers from different stations to use the same transmission equipment by using a wider bandwidth than the individual carriers. On reception, each carrier can be distinguished from the others by means of a specific modulation code, thereby allowing for the reception of signals that were originally overlapping in frequency and time. Thus, several transmissions can occur simultaneously within the same bandwidth, with the mutual interference reduced by the degree of orthogonality of the unique codes used in each transmission.

CECROPIN: Any of a group of peptides that have potent activity against certain Gram-negative bacteria, such as *E. coli*. Produced by cecropia moths on response to bacterial infection, they disrupt bacteria cell membranes.

Cedrela odorata (BARBADOS CEDAR; BRAZILIAN CEDAR; BRITISH GUIANA CEDAR; BRITISH HONDURAS CEDAR; SPANISH CEDAR): A monoecious, deciduous, medium-sized to large tree in the family Meliaceae that is commonly found in tropical and subtropical forests and is the most commercially important and widely distributed species in the genus Cedrela. It can grow up to 40 metres tall, sometime reaching up to 60 metres in South America. The bole is straight, cylindrical and branchless for up to 25 metres and about 120-300 centimetres in diameter. The buttress is absent or very small. The surface of the bark is rough and fissured, reddish-brown especially near the base of the bole and greyish higher up. The leaves alternate, paripinnate with pairs of leaflets that are opposite to alternate, entire, ovate oblong-lanceolate. It has large and much-branched inflorescences, which bear numerous small, five-part, symmetrical greenish-white flowers. Trees are monoecious; male and female flowers are borne on the same inflorescence but the species is protogynous (female flowers open first). The fruit is a large woody capsule, borne near branch tips. Fruits ripen, split and shed seeds while still attached to the parent tree. The seed is a sharply angled or winged columella. Its aromatic wood is rot resistant and also resistant to termites. It is, therefore, used for making panelling and veneer wood including cigar boxes, light construction, mouldings, cabinets, furniture, for building boats and canoes and for making musical instruments. It is also used as firewood. Medicinally, the root and bark are used to treat several ailments such as fever and pain; and the seeds are believed to have vermifugal properties. It is also planted as an ornamental tree alongside roads and in parks.

CEILING FUNCTION (LEAST INTEGER FUNCTION): In mathematics and computer science, the ceiling function of x is the smallest integer greater than or equal to x . For example, $\text{ceiling}(x) = x$ is the smallest integer not less than x . In the same way that the floor function maps a real number, the smallest following integer: $\text{floor}(x) = x$ is the largest integer not greater than x .

CEILING VALUE (CV): Airborne concentration of a potentially toxic substance which should never be exceeded in a worker's breathing zone.

CEILOMETER: An automatic recording instrument for reading the height of the cloud-base.

CELCIUS: A scale and SI derived unit of measurement of temperature in which water freezes at 0° and boils at 100 °C under standard conditions. Its symbol and abbreviation is C, written as °C (degree Celsius).

CELERIAC: A variety of celery (long-stemmed vegetable) with a thick edible root. The root is used as a vegetable in soups and salads. Unlike other root vegetables, which store a large amount of starch, celery root is only about 5-6% starch by weight.

CELERON: One of the products from a family of microprocessors from Intel that is based on its P6 architecture as that of Pentium II, but is a less powerful version designed for low-cost computers. For example, its cache capacity is 256 kB, whilst that of Pentium 4 is 1 MB; and the fastest bus speed for a Celeron processor is 400 MHz and for Pentium it is 1,066 MHz.

CELESTIAL COORDINATES: Any system of coordinates that can be used to give the position of an object, such as a star, on the celestial sphere. A particular system is defined by the chosen point of observation (the origin), the plane of reference and whether the coordinates are spherical or rectangular.

CELESTIAL EQUATOR: The circle in which the plane of the Earth's equator meets the celestial sphere. As a result of the Earth's axial tilt, the celestial equator is inclined by $\sim 23.5^\circ$ with respect to the ecliptic plane.

CELESTIAL MECHANICS: The study of the motions and forces between celestial bodies moving under their mutual gravitational influence in systems with few bodies. It usually describes and predicts motions in the solar system, both of natural bodies such as planets, satellites, asteroids and comets and of artificial bodies such as space probes. Its subfields are orbital mechanics (astrodynamics), which focuses on the orbits of artificial satellites and lunar theory, which studies the orbit of the Moon.

CELESTIAL POLE: One of the two points in the sky around which the celestial sphere seems to rotate.

CELESTIAL SPHERE: The imaginary sphere of infinite radius within which celestial bodies appear to lie or an imaginary sphere to whose inner surface the heavenly bodies appear to be attached.

CELESTINE (CELESTITE): A white or coloured mineral form of strontium sulfate, SrSO_4 , a source of strontium. It occurs in orthorhombic, tabular crystals and in compact forms.

CELL: 1. Also known as **voltaic cell** or **galvanic cell**, it is a device in which chemical energy is converted into electrical energy. Battery refers to a collection of electrical cells in one unit. A cell consists of a positive electrode (anode) and a negative electrode (cathode) and a third substance, an electrolyte that acts chemically on the electrodes to conduct electricity. The two electrodes are connected to an external circuit by a wire. The electrolyte functions as an ionic conductor between the electrodes, where they deliver a direct current. The voltage of the cell depends on the chemical properties of the substances used. 2. In biology, it is the basic structural and functional unit of a living organism, capable of undergoing all life processes. The average diameter of a cell is 0.01 - 0.1mm and made up of protoplasm enclosed by a cell membrane. Cells can be classified into prokaryotic and eukaryotic cells. Eukaryotic cells contain a nucleus. They comprise protists (single-celled organisms), fungi, plants and animals and are generally 5 - 100 μm in linear dimension. Prokaryotic cells contain no nucleus, are relatively small (1 - 10 μm in diameter) and have a simple internal structure. They include two classes of bacteria: eubacteria including photosynthetic organisms or cyanobacteria, which are common bacteria inhabiting soil, water and larger organisms and archaeobacteria, which grow under unusual conditions. 3. In botany, it is a hollow cavity or compartment within a structure, such as the cavity of the anther containing pollen or the cavity of the ovary containing ovules; a locule. 4. In a spreadsheet, the box where a vertical column and a horizontal row intersect. The address of a cell is the column letter followed by the row number. 5. The box where a column and row in a table intersect.

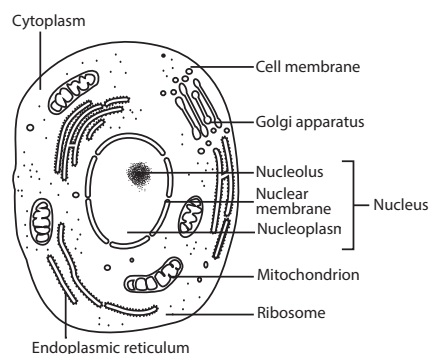


Fig I: A typical animal cell

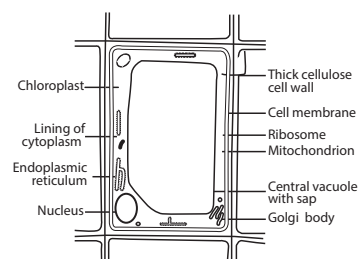


Fig II: A typical plant cell

CELL ADDRESS (CELL NAME): In computer science, it is the location or identity of a particular cell in a spreadsheet. For example, cell D8 is a location on column D row 8.

CELL ADHESIVE MOLECULE (CAM): Any of the proteins in the plasma membrane of animal cells that glue cells to the extra cellular matrix or to each other.

CELL BIOLOGY (CYTOLOGY): The study of the structure organisation and functioning of cellular material.

CELL BODY (PERIKARYON; SOMA; CYTON): The part of a neuron that contains the nucleus. The cell processes that are involved in the transmission and reception of nervous impulses develop as extension from the cell body.

CELL COMMITMENT: The stage in a cell's life when it becomes committed to a certain line of development.

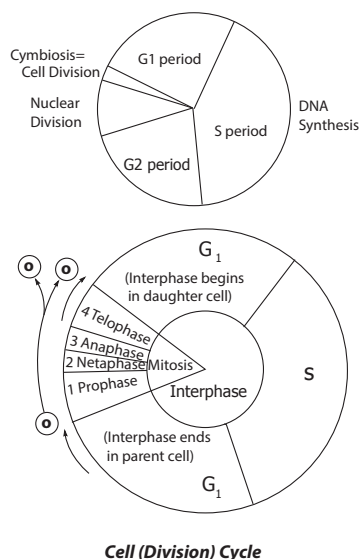
CELL CULTURE: The growing of cells from plants, animals or microorganisms in a culture medium under controlled condition.

CELL CYCLE: The sequence of stages, namely prophase, anaphase, metaphase and telophase that a cell passes through from one cell division to the next.

CELL DIFFERENTIATION (CELLULAR DIFFERENTIATION): 1. The process by which cells acquire their specialisation, such as heart cells, muscle cells and brain cells or the process of change in cells during growth and development, whereby previously undifferentiated cells become specialised for a particular function as a result of structural and functional change. This occurs in developing embryos and occurs numerous times during the development of a multicellular organism as the organism changes from a single zygote to a complex system of tissues and cell types. 2. The structural and functional divergence of cells as they become specialised during a multicellular organism's development. It is dependent on the control of gene expression.

CELL DIVISION: The division of a parent cell into two or more daughter cells, as occurs in amoeba. Cell division in eukaryotes, which creates daughter cells capable of dividing again is called mitosis and

in prokaryotes is called binary fission. In another type of cell division present only in eukaryotes, called meiosis, a cell is permanently transformed into a gamete and cannot divide again until fertilisation.



Cell (Division) Cycle

CELL ERROR: The error which results if the incident beam does not fall perpendicularly to the windows of the cell or if the cell windows are contaminated or have some other imperfection.

CELL EXTENSION: The increase in cell length and volume that occurs a certain distance behind the apex following cell division. The process involves both the uptake of large quantities of water into the cell vacuole by osmosis (vacuolation) and the synthesis of new cytoplasmic and cell wall materials. Cell extension accounts for a far greater proportion of overall plant growth than cell division.

CELL FRACTIONATION: A laboratory technique that uses differential centrifugation to separate the different components of the cell, resulting in nuclear, mitochondrial, microsomal and soluble fractions. The tissue is broken down in a homogenizer and the homogenate is spun in the cell of an ultracentrifuge. Protoplasmic particles settle out in order of size and weight, separation also depending on the speed and duration of centrifugation. After sedimentation is complete the fractions can be identified by their biochemical activity and by electron microscopy.

CELL FUSION (SOMATIC CELL HYBRIDISATION): The cellular process of combining two cells from different tissues of species in a cell culture. The cells are fused and coalesce but their nuclei generally remain separate. The technique is used to determine the control of characteristics exerted by specific chromosomes. Hybrid cells resulting from cell fusion have also been used to produce monoclonal antibodies.

CELL JUNCTION (HOMOPHILIC BINDING; INTERCELLULAR BRIDGE): Any one of various kinds of connection between cells that exists within the tissue of a multicellular organism, consisting of protein complexes. Tight junctions form a seal between adjacent cells, particularly in epithelia, to prevent the passage of material between cells.

CELL LINE: An established clone originally derived from a whole organism through tissue culture.

CELL LINEAGE: The pedigree of cells resulting from binary fission. In other words, it is the developmental pattern of a tissue or part of an organism from particular cells in the fertilised egg or embryo through to their fully differentiated state.

CELL MEMBRANE (PLASMA MEMBRANE): 1. The outer covering of a living cell or a thin semi-permeable layer of protein and fat surrounding cells that controls substances passing between

the cytoplasm and the intercellular space. It controls the flow of substances in and out of the cell by allowing some substances such as water, glucose and amino acids to pass through. 2. Any membrane that is found in a living cell, especially the plasma membrane, which forms the cell boundary.

CELL PHONE JAMMER: A device that is used to block transmission or reception of signals, usually by creating some form of interference at the same frequency ranges that cell phones use. As a result, a cell phone user will either lose the signal or experience a significant loss of signal quality. Police and the military often use them to limit or disrupt communications for security reasons.

CELL PHONE SPAM (SMS SPAM; MESSAGING SERVICE SPAM): Any junk message delivered to a mobile phone as text messaging through the Short Message Service (SMS). The practice can cause major service providers to be overcome by the volume of SMS spam, causing users' screens to freeze and spreading programs that can cause the phones to dial emergency numbers.

CELL PHONE VIRUS: A computer virus specifically adapted for the cellular environment and designed to spread from one vulnerable phone to another.

CELL PLATE: The structure that forms in a dividing plant cell at the end of mitosis; it separates the cytoplasm into two. The cell plate is formed by materials in vesicles from the dictyosomes (Golgi bodies) and arranged by microtubules.

CELL POINTER (ACTIVE CELL; SELECTED CELL): A rectangular box, highlighting the cell in a spreadsheet to identify what cell is being worked with and where data is to be entered.

CELL POTENTIAL: Potential difference, between oxidation and reduction half-cells under non-standard conditions.

CELL PROCESSOR (CELL): A microprocessor chip developed by IBM, Sony and Toshiba that is designed for media processing such as in high-definition TV sets. All phones have a processor of some kind. It may be integrated into the main cell phone chip or be a separate computer chip. The cell processor has 234 million transistors, measures 235 mm² in size, with speeds of more than 4 gigahertz (GHz), a memory bandwidth of 25.6 gigabytes per second (GBps) and has an input/output (I/O) bandwidth of 76.8 GBps. There are nine cores, one of which is the equivalent of a Power PC chip and acts as the controller. The other eight cores are vector processors, each of which is capable of 32 billion (thousand-million) floating point operations per second (GFLOPS).

CELL RELAY: A network technology for data transmission that uses small data packets of a fixed size called cells. Cells are the basic units of data and are widely used in common networks for communication. Just like frames, which are data packets of variable size, cells travel from one computer to another over a network. Asynchronous transfer mode (ATM) is a particularly popular form of cell relay and is based on cell units.

CELL SAP: A dilute fluid found in the large central vacuole of many plant cells. It is made up of water, amino acids, glucose and salts and its functions include storage of useful materials and provision of mechanical support for non-woody plants.

CELL THEORY: The revolutionary idea proposed in 1839 by two Germans, botanist Matthias Schleiden (1804-81) and zoologist Theodor Schwann that plants and animals were made up of cells and that these units were basic to the structure and function of all organisms. This theory laid the foundation of modern biology.

CELL VACUOLE: A membrane-bound sac that is found in many cells, normally acting as a storage organ of various types. A large central vacuole is a particular feature of many plant cells, where it can occupy 80 - 90% of the total cell volume.

CELL VOLTAGE: The difference in electrical potential between the anode and cathode of an electrochemical cell.

CELL WALL: Rigid outer layer that surrounds the plasma membrane of plant, fungal, algal and bacterial but not animal cells. It protects

and gives shape to a cell. Most plant cell walls are composed of the polysaccharide cellulose and may be strengthened by the addition of lignin. The cell walls of fungi consist mainly of chitin. Bacterial cell walls consist of complex polymers of polysaccharides and amino acids called peptidoglycan.

CELL-MEDIATED IMMUNITY: Immune reaction directed against body cells that have been infected by viruses, fungi, protozoans, cancers and bacteria and controlled by T cells, which mature in the thymus. It also plays a major role in transplant rejection.

CELL-PHONE UNLOCKING: The sequence of steps a user must perform in order to unlock a cell phone. This depends on the original communications service provider and on the manufacturer of the phone set. Some cell phone sets can be unlocked by inputting codes to modify the software in the unit. Other sets require the purchase of special cables or modification of the internal hardware as well as reprogramming of the phone's data settings.

CELLOBIOSE: A reducing disaccharide composed of two glucose units linked by a $\beta(1 - 4)$ glycosidic bond. It is formed by the incomplete hydrolysis of cellulose by the enzyme cellulase. Complete hydrolysis to glucose units is achieved by the enzyme cellobiase (β -glycosidase).

CELLOPHANE: A transparent wrapping film made from wood cellulose and widely used for packaging.

CELLS IN FRAMES (CIF): An asynchronous transfer mode (ATM) protocol used to facilitate Ethernet data packet transfers. CIF allows ATM to be implemented using existing Ethernet equipment such as network interface cards and provides advantages such as quality of service without additional hardware expenses.

CELLULAR COMMUNICATION TECHNOLOGY: A wireless communication technology in which several small exchanges called cells are equipped with low-power radio antennas which are strategically located over a wide geographical area and interconnected through a central exchange that enables the use of mobile phones. Each cell is allocated a given number of frequencies (or channels) that allow a large number of subscribers to conduct conversations simultaneously.

CELLULAR NETWORK: Radio network distributed over land through cells, where each cell includes a fixed location transceiver known as base station. These cells together provide radio coverage over larger geographical areas. User equipment (UE), such as mobile phones, is therefore able to communicate even if the equipment is moving through cells during transmission.

CELLULAR PHONE (CELL PHONE; MOBILE PHONE): A telecommunication device that enables simultaneous transmission and reception using radio waves over a networked area and is served through a cell site or base station at a fixed location, enabling calls to transmit wirelessly over a wide range, to a fixed landline or via the Internet. The transmitter's span of coverage is called a cell.

CELLULAR PHONE HACKING: A practice, in which a third party gains access to an individual's cellular phone through a variety of methods. Law enforcement and national governments often use cell phone hacking methods to apprehend criminals and monitor dissidents.

CELLULAR PLASTICS: Solid synthetic materials having an open structure, for example, polystyrene foam used in insulation and packaging.

CELLULAR RESPIRATION: The transfer of energy from various molecules to produce adenosine triphosphate (ATP), which occurs in the mitochondria of eukaryotes and the cytoplasm of prokaryotes. It is the oxidation of glucose or metabolites in a cell to release energy in the form of ATP, with or without the participation of oxygen. In the process, oxygen is consumed and carbon dioxide is generated.

CELLULAR SLIME MOULDS (ACRASIOMYCETES): A class of the Myxomycota containing organisms that grow on dung and decaying vegetation and have a life cycle characterised by a slime-like amoeboid stage and a multicellular reproductive stage. It includes some 25 species divided among 10 genera. Cellular slime moulds are made up of an aggregation of myxamoebae forming a pseudoplasmodium.

These produce uncovered masses of walled spores or pseudospores by a process different from that of other fungi. The cellular slime moulds are often classified as an order (Acrasiales) of the Myxomycetes.

CELLULAR-INTERNET PHONE (CELL INTERNET PHONE; WI-FI CELL PHONE): A cellular telephone that can automatically switch between conventional cellular and Wi-Fi VoIP modes, even during the course of a conversation. A Wi-Fi LAN acts, in effect, as a cellular repeater for such a phone. The phones are designed to prevent a phenomenon called drop-out that occurs when a cell phone set cannot maintain reliable contact with a repeater. An unlocked cell phone is a cellular telephone handset that can be used with more than one service provider, making it easy for a user to switch from one cellular network to another.

CELLULASE: A class of carbohydrate-digesting enzymes (carbohydrase) that hydrolyses cellulose sugars, including cellobiose and glucose. It plays an essential part in abscission in plants by breaking the β -glycosidic links that join the constituent sugar units in cellulose, while microbial cellulases enable the digestion of plant material by herbivores.

CELLULITIS: A bacterial infection underneath the skin surface caused by infection of bacterial strains streptococcus and staphylococcus. It is transmitted through any opening in the body such as a cut, wound, etc. It mainly affects the legs, face and arms. Symptoms include blisters, swelling and redness, pain, fever, etc. It often leads to ulceration and abscess formation.

CELLULOID: A transparent or translucent highly flammable plastic material developed in 1869 and made from cellulose nitrate and plasticised camphor. It was formerly used for toilet articles, novelties and photographic film.

CELLULOLYTIC: The ability to digest cellulose, by means of the enzyme cellulase. For example, cellulolytic bacteria in the stomach of ruminants digest cellulose by means of the enzyme cellulase.

CELLULOSE: Complex carbohydrates composed of long chains of glucose units, with the chemical formula $(C_6H_{10}O_5)_n$. It is the main structural constituent of the cell wall of higher plants. In industry it is obtained from wood pulp and cotton and used in making paper and textiles, ropes, plastics and some explosives. The enzyme cellulase is responsible for the breakdown of cellulose to cellobiose, which is hydrolysed to its constituent glucose units by the enzyme cellobiase. Although cellulose is not acted upon by the digestive juice of the human gut, it is important to the digestive processes. It makes up the roughage that stimulates peristaltic movements of the intestines. The gut of plant-eating animals such as cow, horse and rabbit is especially adapted to the digestion of cellulose. Bacteria in the intestines of these animals act upon the cellulose, converting it into soluble carbohydrates.

CELLULOSE ACETATE: A chemical made by the action of acetic acid on cellulose. It is used in making transparent film, especially photographic film, in the manufacture of adhesives, synthetic fibre and cigarette filters.

CELLULOSE ETHANOATE: A compound prepared by heating cellulose with a mixture of ethanoic anhydride, ethanoic acid and concentrated sulfuric acid. It is used in lacquers, unshatterable glass, varnish and as fibre.

CELLULOSE NITRATE (NITROCELLULOSE; FLASH PAPER): A highly flammable material made by treating cellulose (wood pulp) with concentrated nitric acid. Despite the alternative name *nitrocellulose*, the compound is in fact an ester containing $CONO_2$ groups, not a nitro compound, which would contain $C-NO_2$. It is used in making explosives, plastics and varnishes.

CELLULOSIC ETHANOL: Bioethanol produced from lignocellulose found in wood, grasses or the non-edible parts of plants. Lignocellulose is the structural material that comprises the mass of most plants, composed mainly of cellulose, hemicellulose and lignin and represents much of the mass of plants. Examples of common agricultural sources

of lignocellulose are corn stover, switchgrass, and wheat straw. Unlike alcohol from sources like cane sugar, lignocellulose requires a greater amount of processing to make the sugar monomers available to the microorganisms that are used for the fermentation.

CELLULOSE XANTHATE (VISCOSE): A sticky yellowish or brown syrupy solution made by treating cellulose with sodium hydroxide and carbon disulfide. The solution is then regenerated as continuous filament for making rayon and cellophane.

CELOSIA: Any plant belonging to a genus of herbaceous plants of the family Amaranthaceae. The inflorescence is feathery, yellow to purplish red in colour. Celosia plants are ornamental, leafy vegetables and have medicinal uses.

CELSIUS SCALE: A scale for measuring temperature, named after the Swedish astronomer Anders Celsius (1701–1744). It was formerly called the centigrade. The range from freezing to boiling point of water is divided into 100 degrees, with the freezing point being 0 degrees and the boiling point being 100 degrees.

CELSIUS TEMPERATURE (CENTIGRADE TEMPERATURE): A physical quantity, a temperature, now defined in terms of thermodynamic temperature. It is expressed in the SI unit as degree Celsius ($^{\circ}\text{C}$). Its symbol is $^{\circ}\text{C}$. Celsius temperature scale is defined from the absolute temperature scale through the relation: $\theta = T - 273.15$, where θ is the temperature in degree Celsius and T is the temperature in Kelvin (K).

CEMENT: 1. A substance made by heating chalk or limestone (CaCO_3) and clay together and then powdering the product. When mixed with water, cement would form a paste that sets into a hard material. Portland cement, for example, is a powder obtained from burning together a mixture of lime or chalk and clay and when mixed with water and sand or gravel, turns into mortar or concrete. 2. A bony material that holds the root of a tooth in the jawbone. 3. A substance applied as paste and hardening into a strong consistence for binding together stones and rocks.

CEMENTATION: 1. Any metallurgical process in which the surface of a metal is impregnated by some other substance, especially an obsolete method of making steel by heating bars of wrought iron to red heat for several days in a bed of charcoal. 2. In geology, the hardening and welding of clastic sediments (those formed from preexisting rock fragments) by the precipitation of mineral matter in the pore spaces. It is the last stage in the formation of sedimentary rock. Many minerals may act as cements; the most common is silica (generally quartz), but calcite and other carbonates also occur, as do iron oxides, barite, anhydrite, zeolites and clay minerals.

CEMENTITE (IRON CARBIDE): A hard brittle substance of iron, 93.3% by weight and carbon, 6.67% by weight, with the chemical formula Fe_3C . It forms in some types of cast iron, in carbon steels and in alloys of carbon and iron. It has an orthorhombic crystal structure normally classified as a ceramic.

CENOZOIC (CAINOZOIC; CAENOZOIC): The most recent era of geological time, beginning about 65 million years ago, during which modern plants and animals evolved. It has seen four periods of extensive land elevation separated by periods when the oceans expanded. The Andes, Rockies, Himalayas and Alps were formed in the Cenozoic.

CENSORED DATA: Data items in which the true value is replaced by some other value.

CENSUS: An operation that consists of observing all individuals in a population. A census can refer to a population census, in other words a population count, but it can also refer to inquiries, called **exhaustive inquiries**, where information can be retrieved about population by observing all of the individuals in the population. Exhaustive inquiries are rarely performed, because such inquiries are very expensive for large populations. Sampling, which consists of observing only a portion of the population called the sample, is usually preferred.

CENTESIMAL: In the S.I. system of physical units, the hundredth part of a quantity or value.

CENTI: A prefix with the symbol c , used in the metric system to denote one hundredth ($1/100$). For example, one hundredth ($1/100$) part of a gram is **centigram**; one hundredth ($1/100$) part of a litre is **centilitre**; and one hundredth ($1/100$) part of a metre is **centimetre**.

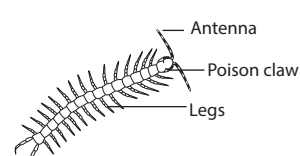
CENTIGRADE SCALE (CENTIGRADE): A scale for temperature measurement, introduced in 1744 until 1948, when the CGPM (Conference General des Poids et Mesures) changed the name to Celsius temperature scale, although the calibration remains the same and both are equal in magnitude. Just as the Celsius temperature scale, the Centigrade temperature scale is divided into 100 degrees. The lower fixed point or ice point is 0°C ; and the its upper fixed point or steam point is 100°C . The difference between these fixed points is called the **fundamental interval**. The Celsius scale is generally used wherever the metric system of units has been adopted and is used in scientific work everywhere.

CENTIMETRE WAVE (CENTIMETRE BAND; SUPER HIGH FREQUENCY, SHF): A radio frequency in the range of 3 and 30 GHz with wavelengths of 10 to 1 centimetres (100 mm to 10 mm). This frequency is used for microwave devices, WLAN, most modern radars, satellite uplinks/downlinks and terrestrial high-speed data links.

CENTIMETRE–GRAM–SECOND SYSTEM OF UNITS (CGS, cgs): An outdated system of units using centimetre as the unit of length, the gram as the unit of mass, and the second as the unit of time that also included electrical and mechanical units. For example, the cgs units of energy and force are the dyne and the erg, abampere (abA) for electric current, abcoulomb (abC) for electric charge, abfarad (abF) for electric capacitance, etc. It was introduced by the British Association for the Advancement of Science in 1874 and was immediately adopted by many working scientists. The cgs system was largely replaced by the MKS (metre, kilogram, second) system with base units of the metre, kilogram, second, ampere, kelvin, candela and mole, which has also been replaced by the current International System of Units (SI). The International System of Units (SI) is an extension of MKS (metre, kilogram, second) system with base units of the metre, kilogram, second, ampere, kelvin, candela and mole.

CENTIMORGAN: A measure of genetic distance that tells how far apart two genes are. Generally one centimorgan equals about 1 million base pairs.

CENTIPEDE: A small crawling arthropod with a long thin body, numerous segments and a pair of legs on each segment. It belongs to the phylum Arthropoda and the class Chilopoda.



External features of a centipede

CENTRAL ANGLE: The angle formed when two radii meet at the centre of a circle with the vertex being the centre of the circle and its rays passing through points on the circumference of the circle.

CENTRAL ATOM: An atom in a molecule of more than two atoms or polyatomic ion that is bonded to more than one surrounding atoms.

CENTRAL DILATION: A dilatation that is not merely a translation because all lines joining corresponding points of a figure and its image are concurrent. For example, two triangles related by a central dilation are said to be perspective triangles because the lines joining corresponding vertices concur.

CENTRAL DOGMA: The basic belief originally held by molecular geneticists, that the flow of genetic information can only occur from DNA to RNA to proteins.

CENTRAL FORCE: A force whose line of action is always directed towards a fixed point; it may attract or repel.

CENTRAL LIMIT THEOREM: The fundamental statistical result that if a sequence of independent identically distributed random variables

each has finite variance, then as their number increases, their sum (or, equivalently, their arithmetic mean) approaches a normally distributed random variable. Hence, in particular, if sufficiently many samples are successively drawn from any population, the sum or mean of the sample values can be thought of, approximately, as an outcome from a normally distributed random variable. Mathematically, if we let $x_1, x_2, \dots, x_n$ be n independent random variables that are identically distributed (with any distribution) with mean μ and a finite variance σ^2 , the sum $S_n = x_1 + x_2 + \dots + x_n$ is defined and the ratio: $\frac{S_n - n\mu}{\sigma\sqrt{n}}$ is

established, where n , μ and $\sigma\sqrt{n}$ represent the mean and the standard deviation of S_n respectively. The central limit theorem establishes that the distribution of the ratio tends to the standard normal distribution when n tends to infinity. This means that $P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x\right) \xrightarrow{n \rightarrow +\infty} \Phi(x)$, where $\Phi(x)$ is the distribution function of the normal distribution expressed by $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt, -\infty < x < \infty$.

CENTRAL MERIDIAN: An imaginary line bisecting the apparent disk of a planet or the Sun and joining the poles of rotation. The passage of a surface feature across the central meridian, due to the axial rotation of a planet, is known as a **central meridian transit**. The timing of central meridian transits is an important means of determining the planetographic longitudes of planetary surface features.

CENTRAL NERVOUS SYSTEM (CNS): The part of the nervous system composed of the brain and spinal cord that coordinates various body functions. The brain and spinal cord are intimately covered by membranes called meninges and bathed in an extracellular fluid called cerebrospinal fluid.

CENTRAL ORBIT: An orbit in which the attractive force between a central mass and test mass is a central force, which is directed towards the central mass.

CENTRAL PROCESSING UNIT (CPU; PROCESSOR): The main unit of a computer, where most of the processing takes place. It includes the control unit (CU) and the arithmetic/logic unit. Its function is to fetch instructions from memory, decode them and execute the program. It also provides timing signals. An integrated circuit that has a complete CPU on a single silicon chip is called a **microprocessor**.

CENTRALIZER: The subgroup, denoted $C_G(x)$, consisting of elements that commute with a given element or subset of a group, G .

CENTRE: 1. The point on a line that is of the same distance from both ends. 2. The interior point that is the same distance from all points on the circumference of a circle, the surface of a sphere or the vertices of a polygon. 3. The point of intersection of the axes of an ellipse or a hyperbola. 4. In affine or Euclidean geometry, the pole of the line of affinity with respect to a conic. 5. The set elements of a group, G , that commute with every member of the group. It is denoted $Z(G)$ and is equal to the intersection of the centralizers of the elements of the group. 6. The point about which a power series is evaluated. It is the centre of the circle of convergence of the given series. 7. A part of the nervous system consisting of a group of nerve cells that coordinate a particular process. An example is the respiratory centre in the vertebrate brainstem, which controls breathing movements.

CENTRE OF CURVATURE: 1. The centre of the circle of curvature of a curve at a given point. 2. The centre of the sphere of which a lens curve or curved mirror forms a part. The radius of curvature is the radius of this sphere.

CENTRE OF DIVERSITY (GENE CENTRE): An area that shows considerable genetic diversity of certain crop plants and their relatives. Some gene centres, the primary centres, are believed to correspond to the region where a particular crop originated. For example, numerous forms of wheat exist in the Middle East and, since many of the wild relatives of wheat are also found in this area, it is believed the crop was first domesticated there. Other centres, the secondary centres, do not contain wild relatives and are believed simply to be areas where

the crops have been cultivated for a considerable period. Ethiopia is an example of a secondary centre. Gene centres are typically found in mountainous regions

CENTRE OF ENLARGEMENT: The point from which an enlargement of a shape takes place. The image created is similar to the object. The size of an enlargement is described by its scale factor. For example, a shape with a scale factor of 3 means that the new shape is three times the size of the original. In the diagram, ABC has been enlarged to A'B'C' with the point O as the centre of enlargement.

CENTRE OF GRAVITY (CENTRE OF INERTIA; MASS CENTROID): The point in an object about which its whole weight is evenly balanced or seems to act if the object is in a uniform gravitational field. It is the point through which any resultant force must act if it is to accelerate the body without causing it to rotate. A terrestrial body would have to be very large for the centre of mass and the centre of gravity to be in two different places. Note that the centre of mass and the centre of gravity coincide because they are at the same point in a uniform gravitational field.

CENTRE OF MASS: 1. The weighted mean position of finite discrete set of particles with position vectors $x_1, x_2, \dots, x_n$ and masses

$m_1, \dots, m_n$. Its position vector is given by $\bar{x} = \left(\sum_{i=1}^n m_i x_i \right) / \left(\sum_{i=1}^n m_i \right)$. 2. For

a continuous body, the position vector given by the corresponding ratio of integrals over the volume of the configuration of the body at

time t , that is, $\bar{x}(t) = \frac{\int \rho x dv}{\int \rho dv}$, where ρ is the density at the point of position vector x .

CENTRE OF ORIGIN: It is usually the location in the world, where the oldest cultivation of a particular crop has been identified.

CENTRE OF PERSPECTIVE: The point where the lines joining corresponding points of two figures that are in perspective meet.

CENTRE OF ROTATION: The point around which a rotation takes place.

CENTRE OF SYMMETRY (CENTRE): A point around which a curve is symmetrical, so that for every point P on the curve there is another P' such that the directed lines joining the centre to P and to P' are of equal length and opposite sense.

CENTRE-OF-MASS ENERGY: In a particle collision, the energy that can go into making new particles. For a collider experiment where two beams of equal energy collide head-on, this is simply the sum of the energy of the two beams. In fixed target experiments, in which a beam of particles strikes a stationary target, the centre-of-mass energy is significantly less than the sum of the energies of the two colliding particles.

CENTRIC BUNDLE (CONCENTRIC BUNDLE): A vascular bundle in which one vascular tissue surrounds the other so they appear to be arranged concentrically when viewed in transverse section.

CENTRIFUGAL: 1. Generally, it is moving or directed away from a centre or axis. 2. In botany, developing from the centre outwards so the oldest structures are in the middle and the youngest at the perimeter. An example is the differentiation of secondary xylem. 3. In human biology, it is transmitting nerve impulses away from the central nervous system; efferent.

CENTRIFUGAL ACCELERATION (CENTRIPETAL ACCELERATION): The acceleration of a component as a result of a uniform rotational motion. Centrifugal acceleration is a vector quantity. Its symbol is a_{rot} .

CENTRIFUGAL BARRIER: The barrier that results, when a reaction without an electronic energy barrier or its reverse, the rotational energy of the transition state gives rise to a reaction barrier.

CENTRIFUGAL FORCE: The apparent force, equal and opposite to the centripetal force, drawing a rotating body away from the centre

of rotation, caused by the inertia of the body. Centrifugal force is a fictitious force because it is a by-product of measuring coordinates with respect to a rotating coordinate system as opposed to an actual push or pull.

CENTRIFUGAL INFLORESCENCE: A flower cluster developing from the centre outward or from the top downward as occurs in a cyme.

CENTRIFUGAL PUMP: A pump that uses a rotating impeller to increase the pressure of a fluid by accelerating it radially outward. It has one or more rotating impellers in a stationary casing, which guides the fluid to and from the impeller or from one impeller to the next in the case of multi stage pumps. Centrifugal pumps are used for large discharge through smaller heads.

CENTRIFUGAL TAIL MOULT: The shedding and replacing of birds' tail feathers. It starts from the innermost pair of feathers and then moves from the centre outward.

CENTRIFUGAL RADIUS: The radius at which a component is spinning at the end of the period of centrifugation. For a component sedimented from a dilute suspension, it can be equated with radius of rotation at the bottom of the centrifuge tube. Its symbol is r .

CENTRIFUGE: 1. An apparatus designed for rapid separation of solid or liquid particles of different densities, for example, milk and cream by rotating them in a container at very high speeds. An ultracentrifuge rotates at much higher speeds (up to a million times per second) and is used to separate colloidal particles such as large protein molecules. 2. A large, motor-driven apparatus with a long arm at the end of which human and animal subjects or equipment can be rotated at various speeds to simulate the prolonged accelerations encountered in high-performance aircraft, rockets and spacecraft.

CENTRINO: An Intel chipset that enables laptops to connect to wireless networks using Wi Fi network card in the laptop's mini-PC1.

CENTRIOLE: A cylindrical structure composed of microtubules, which is commonly found in pairs orientated at right angles to each other in animal cells and have been identified in the cells of some primitive plants, such as brown algae. They are not found in the cells of higher plants. They play a role in the process of cell division by forming spindle fibres.

CENTRIPETAL: 1. In botany, it describes structures that develop from the outside inwards, so the oldest are at the edge and the youngest are in the middle. An example is the differentiation of xylem in club mosses (*Lycopodium*). 2. In physiology, the term refers to transmitting nerve impulses toward the central nervous system.

CENTRIPETAL ACCELERATION: The acceleration of a body which travels along a circular track at constant speed. The acceleration is directed radially toward the centre of the circle and has a magnitude equal to the square of the body's speed along the curve divided by the distance from the centre of the circle to the moving body. The force causing this acceleration is directed also toward the centre of the circle and is named centripetal force. An example of centripetal acceleration is the acceleration toward the centre that holds a satellite in an elliptical orbit. The magnitude of centripetal acceleration, $a_c = v^2/r$, where a_c is centripetal acceleration, v is velocity in metres per second and r is the radius of the circle in metres. The higher the velocity, the greater the centripetal acceleration; and the smaller the curve of the path being travelled, the greater the centripetal acceleration.

CENTRIPETAL FORCE: The force which makes an object move on a circular path or acts radially inward on an object moving in a curved path or acts tangentially to the object's path. It is that part of total gravitational force that keeps a body in its circular track and it is always directed towards the centre of the circle. For example, with a ball whirled in a circle at the end of a length of string, the centripetal force is the tension in the string and the reaction to this force is the centrifugal force. If the string breaks, the ball will move off in a straight line. The straight line motion in the absence of the constraining force is an example of Newton's first law. For a body of mass, m , moving

at constant speed, v , around a circle of radius, r , the magnitude of the centripetal force, F_c is expressed as $F_c = mv^2/r$. The gravitational attraction of the Sun is acentripetal force acting on Earth. This force produces the centripetal acceleration of the orbital motion.

CENTRIPETAL INFLORESCENCE: A flower cluster developing from the edge towards the centre or from bottom upward as occurs in a corymb.

CENTROBLAST: A rapidly proliferating B cell, B lymphocyte, found within germinal centre of a lymph node follicle. They are derived from B cells activated by exposure to a specific antigen in the first phase of humoral immune response; they seed the germinal centres and divide rapidly.

CENTROID: 1. The point within an area or volume at which the centre of mass would be if the surface or body had a uniform density. 2. For a triangle, it is the point of coincidence of the medians. 3. Also called **centre**, a point whose coordinates are the mean values of the coordinates of the points in a given set. 4. Also known as **barycentre**, it is the centre of mass of a system especially of astronomical objects, which moves as though the system's total mass existed at the point and all external forces were applied at the point. For example, Earth-Moon system moves in the Sun's gravitational field as though both masses were located at a centre of mass some 4700 km from the Earth's geometric centre

CENTROLECITHAL: Of an egg, having the yolk concentrated near the centre of the egg, as found in many arthropods. The nucleus is located centrally in a small amount of non-yolky cytoplasm, surrounded by a large mass of yolk; after fertilisation and some nuclear divisions, the nuclei migrate to the periphery to proceed with superficial cleavage, the yolk remaining central.

CENTROMERE: The part of a chromosome that attaches via a plate-like structure called the kinetochore to the spindle during cell division. Centromere has no genetic significance. It usually appears as a constriction when chromosomes duplicate during cell division. The position of the centromere is a distinguishing feature of individual chromosomes.

CENTROSOME (CELL CENTRE; CENTROSPHERES): A specialised region of cytoplasm near the nucleus of a cell of all eukaryote cells except fungi. It organises the polymerisation, position and polar orientation of many of the cell's microtubules throughout the cell cycle.

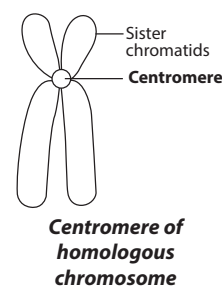
CENTROSPERMAE (CARYOPHYLLIDAE): A subclass of the dicotyledons containing mostly herbaceous plants usually with bisexual flowers. It includes the cacti, carnations, amaranths, ice plants and many carnivorous plants. Many members are succulent, having fleshy stems or leaves. The seeds often contain perisperm and betalain alkaloids are usually present.

CEPHALIC: Pertaining to the head, as in cephalic presentation, the head-first appearance of a baby in the birth canal.

CEPHALISATION/CEPHALIZATION: The tendency among animal groups for the major sense organs, mouth and brain, to be grouped together at the anterior end of the body. This process eventually produces a head region with sensory organs.

CEPHALOCHORDATA: A subphylum of the phylum Chordata that contains only the lancelets, a group of small invertebrate marine fishlike animals.

CEPHALPOD: Any of more than 900 species of relatively intelligent marine predators equipped with a complex and efficient nervous system. They include octopus, squid, cuttlefish, nautilus and extinct ammonites. All cephalopods are capable of swimming by jet propulsion and have mobile tentacles for catching prey. They have large heads, large eyes, prehensile tentacles and, in most species, an ink sac containing a dark fluid used for protection or defence.



CEPHALOSPORIN: Any of a class of broadspectrum antibiotics derived from the fungus *Cephalosporium acremonium*, which are similar in structure and antimicrobial action to penicillins and used to treat infectious diseases such as pneumonia, strep throat, tonsillitis, bronchitis and gonorrhoea. It was first extracted from sewage-contaminated water in 1948 by Italian scientist Giuseppe Brotzu (1895-1976). Other naturally occurring ones have been isolated from moulds taken from soil samples.

CEPHALOTHORAX: The first or anterior major body section of crustaceans (lobsters, crabs, shrimp, crayfish, etc.) and arachnids (spiders, scorpions, etc.). It comprises a fused head and thorax, which is connected to the abdomen.

CEPHEID VARIABLE: A yellow super giant star whose luminosity varies cyclically with a constant period as a result of pulsations. Observation of cepheids can be used to determine the distances to stars and other galaxies.

CERACEOUS: In botany, it refers to structures having a wavy texture or appearance.

CERAMIC: A rigid material that consists of an infinite three-dimensional network of sintered crystalline grains comprising metals bonded to carbon, nitrogen or oxygen. Ceramic generally applies to any class of inorganic, non-metallic product subjected to high temperature during manufacture or use. The classes of materials generally considered to be ceramics are oxides, nitrides, borides, carbides, silicides and sulfides.

CERAMIC FILTER (CERAMIC THIMBLE): A component of a stack sampling system which is suitable for high temperature use.

CERAMIC PRECURSOR: Material that is converted to a ceramic through pyrolysis. Examples are poly(dimethylsilanediyl), poly(carbasilane)s and poly(silazane)s.

CERAMIC YIELD: Mass of ceramic expressed as a percentage of the mass of the ceramic precursor used in the ceramisation process.

CERAMIC-REINFORCED POLYMER: Polymer composite consisting of a polymer continuous phase and disperse phase domains of microscopic ceramic particles.

CERAMISATION/CERAMIZATION: A process in which a ceramic precursor is converted into a ceramic.

CERCARIA: Larva of various parasitic worm flukes that is tadpole-shaped. It becomes the adult fluke after entering the definitive host, a vertebrate.

CERCUS: Any of a pair of many jointed appendages that are present in the last segment of the abdomen in certain insects such as earwigs, cockroaches and mayflies. Cerci often serve as sensory organs, but they may also be used as weapons or copulation aids or they may simply be vestigial structures.

CERE: A raised fleshy or wax-like membranous covering, located at the base of the upper mandible in certain birds, such as parrots, through which the nostrils open.

CEREALS: 1. Plants of the grass family grown for their nutrient rich starchy grains including barley, corn, millet, oats, rice, rye and wheat. In common usage, cereal and grain are the same thing. Cereal grains, a rich source of vitamins, minerals, carbohydrates, fats, oils and protein in their natural form are grown in greater quantities and provide more food energy worldwide than any other type of crop. When refined by the removal of the bran and germ, the remaining endocarp is mostly carbohydrate and lacks the majority of the other nutrients. 2. Breakfast foods prepared from grains and seeds.

CEREBELLUM: The largest and most highly developed part of the brain in vertebrate animals, lying posterior to the pons and medulla oblongata and inferior to the occipital lobes of the cerebral hemispheres. It controls muscular movements, balance and coordination.

CEREBRAL AQUEDUCT (AQUEDUCT OF SYLVIVS): The narrow channel, containing cerebrospinal fluid that connects the third and fourth ventricles of the brain.

CEREBRAL CORTEX: The layer of grey matter that forms the outer layer of the cerebrum in many vertebrates. It is responsible for integrating sensory impulses. It is divided into four lobes: the frontal lobe, the parietal lobe, the temporal lobe and the occipital lobe.

CEREBRAL HAEMORRHAGE (CEREBRAL INFARCTION; STROKE): An interruption of the blood supply to any part of the brain. It can be caused by a blocked blood vessel carrying blood to the brain or by a blood clot. This is called an ischaemic stroke. When a blood vessel breaks open, causing blood to leak into the brain, it is called a **haemorrhagic stroke**. If blood flow is stopped for longer than a few seconds, the brain cannot get blood and oxygen. Brain cells can die, causing permanent damage.

CEREBRAL HEMISPHERE: One of the two symmetrical halves (left and right) of the cerebrum in the front part of the brain.

CEREBROSIDE: Any one of a class of glycolipids in which a single sugar unit, usually glucose or galactose is bound to a sphingolipid. Examples are glucocerebrosides and galactocerebrosides.

CEREBROSPINAL FLUID (CSF): A clear, colourless liquid, similar in composition to lymph that bathes the central nervous system (the brain and spinal cord) of vertebrates. It supports the brain, acts as a lubricant, maintains pressure in the skull and cushions shocks.

CEREBROSPINAL MENINGITIS: Inflammation of the protective membranes surrounding the spinal cord and brains (meninges), caused by infection with viruses, bacteria or other microorganisms and sometimes by drugs, causing high fever and sometimes unconsciousness.

CEREBRUM: The largest and most anterior region of the vertebrate brain, divided into symmetric left and right cerebral hemispheres. It coordinates the senses and is responsible for learning and other high mental faculties. It is the most superior region of the vertebrate central nervous system.

CERENKOV COUNTER: Equipment for detecting and counting high-energy charged particles by observing the Cerenkov radiation produced.

CERENKOV EFFECT: Emission of radiation in the visible and ultraviolet spectrum arising when a charged particle crosses a medium with a velocity greater than that of light in the same medium.

CERENKOV RADIATION: Radiation resulting from the Cerenkov effect. In other words, it is electromagnetic radiation, usually a bluish light radiated by charged particles moving at high speed (greater than speed of light) when the particles pass through a transparent, non conducting, solid material. A typical example is the bluish glow observed in the water of a nuclear reactor, close to the active fuel elements.

CERENKOV, PAVEL ALEKSEYEVICH (1904-90): Soviet physicist and a professor at the Lebedev Institute of Physics in Moscow. In 1934, while observing radioactive radiation underwater, he discovered Cerenkov radiation, which is the emission of a bluish light from a liquid when electrons or other charged atomic particles move through the liquid with a velocity greater than that of light in the same medium. He shared with the Soviet physicists, I. M. Frank and I. Y. Tamm the 1958 Nobel Prize in Physics for this discovery. His research opened the way to new studies of high-energy subatomic particles and of cosmic rays.

CERIUM: A lustrous steel-grey metallic element in the lanthanide series of the Periodic Table with the chemical symbol Ce, atomic number 58, atomic weight 140.12, melting point 795 °C (1068 K) and boiling point 3468 °C (3741 K). It is the most abundant of the rare earth elements. It occurs chiefly in the minerals monazite and bastnaesite, existing in four allotropic states. It is used in lighter flints, tracer bullets, catalytic converters for car exhausts and gas mantles.

CERMET: A composite material containing ceramics and sintered metal. Cermet has properties of both ceramics and metal such as high temperature resistance, corrosion and hardness. It is widely used

in jet engines and nuclear reactors; resistors, capacitors and other electronic components which may experience high temperatures.

CERN (COUNSEIL EUROPÉE POUR LA RECHERCHE NUCLÉAIRE): European laboratory for nuclear and particle physics, headquartered in Geneva, Switzerland. It is the world's largest founded in 1954 by European countries with 20 European member states and the U.S., Japan, Russia and others as observer nations. It invented the World Wide Web in the 1990s to enhance collaboration on research documents pertaining to particle physics.

CERNUOUS: In botany, it refers to drooping or nodding, as occurs in a flower.

CERT: Abbreviation for Computer Emergency Response Team or Computer Emergency Readiness Team; It is an expert group that studies and handles computer security incidents such as network or host activity that potentially threatens the security of computer systems. It includes attempts to gain unauthorized access to a system or its data, unwanted disruption or denial of service, unauthorized use of a system for the processing or storage of data and changes to system hardware, firmware or software characteristics without the owner's knowledge, instruction or consent.

CERTIFIED EMISSION REDUCTION (CER): Carbon credits issued by Clean Development Mechanism (CDM) projects as defined by the Kyoto Protocol; aimed at reducing greenhouse gases in the atmosphere. It is the right to emit 650,000 tonnes of CO₂.

CERTIFIED ETHICAL HACKER (CEH): A professional designation for hackers that perform legitimate services for IT companies and other organisations. A CEH is hired to locate and repair application and system security vulnerabilities to pre-empt exploitations by black hat hackers and others with potentially illegal intentions.

CERTIFIED NOVELL ENGINEER (CNE): A certification for those who install, maintain and support Netware systems and other server or hardware infrastructure from the Novell Company, a leader in networking systems. The CNE certification tests proficiency in aspects of working with local area networks (LANs) and other networking setups.

CERTIFIED SEED: Seed tested successfully for purity, disease and weed contamination and is granted certification for sale to be planted.

CERUMEN (EAR WAX): A yellowish waxy substance secreted by ceruminous glands in the auditory meatus of the outer ear of humans and many other mammals. It protects the skin of the human ear canal, provides cleaning and lubrication and also provides some protection from bacteria, fungi, insects and water.

CERUSSITE: The mineral form of lead carbonate, PbCO₃ usually of secondary origin and formed by weathering of galena.

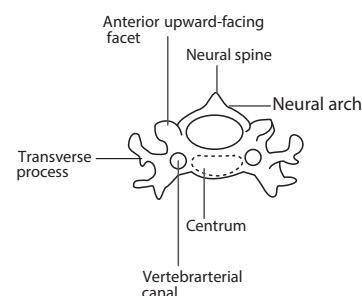
CERVICAL: Describing parts of the body and functions related to the neck and the cervix.

CERVICAL CANCER: Cancer that affects the cervix. It may be caused by the human papillomavirus (HPV) infection, which can be sexually transmitted. Some other examples of causes include taking birth control pills, having many sexual partners, smoking and immunodeficiency. At an early stage it does not indicate any signs or symptoms but can be detected through regular pelvic screening and using cervical smear (Pap test) to detect the disease. At advanced stage, signs and symptoms include vaginal bleeding after intercourse, between periods or after menopause, bloody vaginal discharge and pelvic pain or pain during intercourse. Women who have sexual intercourse before they attain the age 18 and have many sexual partners are at increased risk for cervical cancer. They also stand the risk of getting human papillomavirus (HPV) infection from partners who may be infected by the virus.

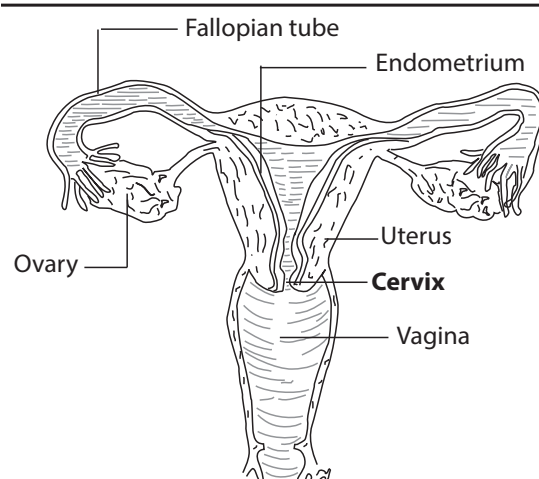
CERVICAL SMEAR TEST (PAP SMEAR TEST; PAPANICOLAOU TEST): A cancer screening test in which cells scraped from the cervix of the uterus (womb) are examined for abnormality under the microscope.

CERVICAL VERTEBRAE: The vertebrae of the neck immediately inferior to the skull. The main functions are to support the head and to provide articulating surfaces against which it moves, relative to the backbone. Most mammals have seven cervical vertebrae.

CERVIX: A narrow neck-like part of the female reproductive system, consisting of a narrow passage between the vagina and the uterus (cervix uteri). Glands in its walls produce mucus, whose viscosity changes according to the oestrous cycle. During labour, the cervix enlarges greatly to allow the passage of the foetus.



Anterior view of cervical vertebra of a mammal



Cervix of the female reproductive system

CESARO SUMMATION: The computation, in summability theory, of the limit of a possibly divergent sequence of numbers as the limit, as n approaches infinity, of the means of the first numbers. The method is regular in that for a convergent sequence this limit exists and agrees with the original limit. This method assigns the limit of $1/2$ to the sequence $1, 0, 1, 0, 1, \dots$ It was named after Italian mathematician Ernesto Cesaro (1859-1906).

CESTODA: A class of flatworms (Platyhelminthes), comprising the tapeworms with about a thousand species. The length of tapeworms varies from 0.6 millimetre to 30 metres, usually found in sperm whales. Tapeworms are parasites that have no head, mouth or digestive system. Members of cestoda live in the digestive tract of vertebrates as adults and often in the bodies of various animals as juveniles. All vertebrate species can be parasitised by at least one species of tapeworm. Some species infest humans after being consumed in under-cooked meat. For example, *Taenia solium*, found in pig can infest humans from eating under-cooked pork. *Taenia saginata* found in cattle can also infest humans from eating under-cooked beef.

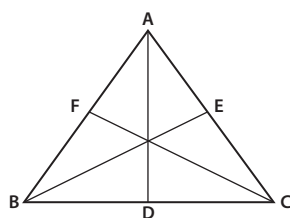
CETACEA: An order of exclusively aquatic placental mammals comprising the whales, dolphins and porpoises, including the largest known animal, the blue whale (*Balaenoptera musculus*), which is over

30m long and over 150 tonnes in weight. The forelimbs of whales are modified as short stabilising flippers and the skin is very thin and almost hairless. They have lungs to breathe atmospheric air and have warm blood. A thick layer of blubber insulates the body against heat loss and is an important food store.

CETANE NUMBER: A number, usually between 30 and 60 that provides a measure of the ignition characteristics of a diesel fuel when it is burnt in standard diesel engine. The higher the cetane number, the more easily the fuel can be ignited.

CETERIS PARIBUS: Latin for "other things being equal"; that is, the assumption that all other things remain constant, for example, in the conditions of a theorem. The term is widely used in mathematics, economics, science and some other fields.

CEVA'S THEOREM: The theorem about triangles in Euclidean plane geometry that three Cevians, one through each vertex of a triangle, are concurrent if and only if the product of the ratios in which they divide the opposite sides is unity. In other words, given a triangle ABC, with the points D, E and F lying on lines BC, CA and AB respectively, the lines AD, BE and CF are concurrent if and only if $(AF/FB)(BD/DC)(CE/EA) = 1$.



Ceva's Theorem

CEVIAN: Any line segment joining a vertex of a triangle to a point on the opposite side or on its extension.

CHADWICK, SIR JAMES (1891-1974): British physicist, who worked at Manchester University under Rutherford. In 1908 intending to study mathematics he was admitted to the physics programs. He studied under Geiger in the German research institute, the Physikalisch-Technische Reichsanstalt in Charlottenburg near Berlin. He was interned for the duration of World War I and joined Rutherford at Cambridge after the war. In 1935 he was awarded the Nobel Prize in physics for his discovery of the neutron. He also built Britain's first cyclotron at Liverpool University in 1935.

CHAETA: A bristle made of chitin occurring in annelid worms such as earthworms and projecting from the skin of each segment and function on locomotion.

CHAETOGNATHA: A phylum of marine coelomate invertebrate, the arrow worms. They have slender, transparent bodies, usually under 2.5 cm long. The head bears hooks for catching prey and the trunk and tail support paired lateral and tail fins.

CHAETOTAXIS (CHAETOTACTIC): The arrangement of bristles on the exoskeleton of an insect.

CHAFF: 1. Thin dry scales or bracts. An example is the bracts on the receptacle of the heads of the Compositae (Asteraceae). 2. Short lengths of cut straw, used as feed for ruminants and as a component of manures. 3. Husks of corn which separate from the grain during threshing. Chaff is used as livestock fodder, ploughed back into the soil or burnt.

CHAFFING AND WINNOWING: Dual components of a privacy-enhancement scheme that does not require encryption. The technique consists of adding false packets to a message at the source and then removing the false packets at the destination. The false packets obscure the intended message and render the transmission unintelligible to anyone except authorized recipients.

CHAIN: 1. In mathematics, it is any of the following: (i) any set that has a linear ordering; it may be a subset of a partially ordered set on which the ordering is connected. (ii) A collection of nested sets. (iii) Markov chain, another term for Markov process. (iv) In graph theory, any path joining two vertices in a graph. A Eulerian chain uses each edge exactly once; and a Hamiltonian chain uses each vertex exactly once. 2. Atoms of carbon that bond together in forming long

molecules. In polymers, it is the whole or part of a macromolecule, an oligomer molecule or a block, comprising a linear or branched sequence of constitutional units between two boundary constitutional units, each of which may be an end-group, a branch point or an otherwise-designated characteristic feature of the macromolecule.

3. A series of associated geographic features or formations such as mountains, lakes or islands. 4. A series of interlocking links forming a flexible cable. Simple chains are used in, for example, haulage and hoisting. Roller chains are used in power transmission (as on a bicycle), with the links of an endless chain meshing with the teeth of toothed wheels called sprockets. Efficiency may be as high as 99%.

CHAIN AXIS: For a polymer, it is a straight line parallel to the direction of chain extension, connecting the centres of mass of successive blocks of chain units, each of which is contained within an identity period.

CHAIN BRANCHING: When in a chain reaction there is a net increase in the number of chain carriers there is said to be chain branching. A simple example of a chain-propagating reaction leading to chain branching is $O + H_2 \rightarrow OH + H$, in which there is one chain carrier (an oxygen atom) on the left and two chain carriers (a hydrogen atom and a hydroxyl radical) on the right.

CHAIN CARRIER: A species, such as an atom or free radical, which is involved in chain-propagating reactions.

CHAIN CONDITION: An ascending chain condition or a descending one.

CHAIN ENTANGLEMENT: Interlocking of polymer chains in a polymer material forming a transient or permanent network junction over the time-scale of the measurement.

CHAIN FISSION YIELD: The fraction of fissions giving rise to nuclei of a particular mass number.

CHAIN FOLDING: In polymer crystals, it is the conformational feature in which a loop connects two parallel stems belonging to the same crystal.

CHAIN INITIATION: The process in a chain reaction that is responsible for the formation of a chain carrier.

CHAIN ISOMERISM: A type of molecular isomerism seen in carbon compounds. As the number of carbon atoms in the molecule increases, the linkage between the atoms may be a straight chain or branched chains producing isomers that differ from each other by possessing different carbon skeletons.

CHAIN LENGTH: In a chain reaction, the average number of times the closed cycle of reactions (involving the chain propagating reactions) is repeated. It is equal to the rate of the overall reaction divided by the rate of the initiation reaction.

CHAIN POLYMERISATION: A chain reaction in which the growth of a polymer chain proceeds exclusively by reaction(s) between monomer(s) and reactive site(s) on the polymer chain with regeneration of the reactive site(s) at the end of each growth step. A chain polymerisation consists of initiation and propagation reactions and may also include termination and chain transfer reactions.

CHAIN REACTION: 1. A series of chemical reactions in which the product of each reaction causes the next one to occur. For example, the fission of uranium-235 nucleus by a slow neutron can produce chain reactions because neutrons released by the splitting of the first uranium-235 nucleus go on to split more nuclei, releasing even more neutrons. Uncontrolled chain reaction produces a nuclear explosion, for example as in an atom bomb. 2. A chemical change forming products which themselves cause more changes so that new compounds are produced or a succession of reactions usually involving free radicals, where the products of one stage are the reactants of the next. There are three separate stages involved which are initiation, propagation and termination. A typical chain reaction is the chlorination of methane in presence of sunlight.

CHAIN RULE: A theorem that may be used in the differentiation of compositions of functions. It states that $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$, where y is a differentiable function of t and t is a differentiable function of x . This enables a function $y = f(x)$ to be differentiated by finding a suitable function u , such that f is a composition of y and u , y is a differentiable function of u and u is a differentiable function of x . Similarly, for partial differentiation, $\frac{\partial f}{\partial x} = \left(\frac{\partial f}{\partial u} \times \frac{\partial u}{\partial x}\right) + \left(\frac{\partial f}{\partial v} \times \frac{\partial v}{\partial x}\right)$, where f is a function of u and v , each of which is a function of x .

CHAIN SCISSION: In a polymer, it is a chemical reaction resulting in the breaking of skeletal bonds.

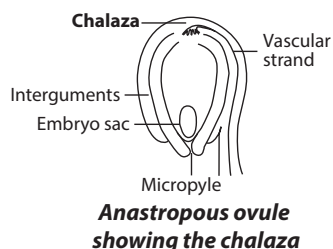
CHAIN TERMINATION STEP: Any chemical reaction leading to the destruction of the reactive intermediate in a chain propagation step. Example is the combination of two radicals, which removes the reactive species that propagate the chain reaction.

CHAIN TRANSFER: The abstraction, by the radical end of a growing chain-polymer, of an atom from another molecule. The growth of the polymer chain is thereby terminated but a new radical, capable of chain propagation and polymerisation, is simultaneously created.

CHAIN-ORIENTATIONAL DISORDER: In polymer crystals, it is a structural disorder resulting from the statistical coexistence within the crystals of identical chains with opposite orientations. A typical example is provided by the up-down statistical coexistence of antilined chains in the same crystal structure.

CHAIN-PROPAGATING REACTION (PROPAGATING REACTION): An elementary step in a chain reaction in which one chain carrier is converted into another. The conversion can be a unimolecular reaction or a bimolecular reaction with a reactant molecule.

CHALAZA: 1. A twisted strand of fibrous albumen in a bird's egg that connects the yolk to the lining membrane at either end of the shell, holding the yolk in position in the albumen. 2. The part of a plant ovule that is opposite the micropyle, where the nucellus and integuments merge. **Chalazogamy** is the pollen tube entrance through the chalaza.



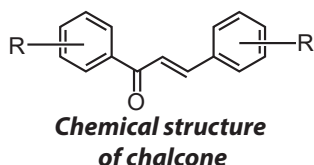
CHALAZOGAMY: A method of fertilisation in seed plants observed in certain trees and shrubs, such as beech, elm and hazel, in which the pollen tube does not reach the nucellus through the micropyle (porogamy) but grows or digests its way through the placenta and chalaza instead.

CHALCEDONY: A mineral, with the chemical formula SiO_2 , composed of a microcrystalline variety of quartz and moganite. It comes in various shades of white, grey, yellow, brown, green and blue occurring in several forms, such as onyx, agate and jasper.

CHALCONE (1,3-DIPHENYL-2-PROPEN-1-ONE; BENZYLIDENEACETOPHENONE; PHENYL STYRYL KETONE):

An aromatic ketone, 1,3-Diphenyl-2-propen-1-one, with the molecular formula $\text{C}_{15}\text{H}_{12}\text{O}$, density 1.071 g/cm^3 , melting point $55\text{--}57^\circ\text{C}$ ($328\text{--}330 \text{ K}$) and boiling point $345\text{--}348^\circ\text{C}$ ($618\text{--}621 \text{ K}$). It forms the central core for a variety of important biological compounds known collectively as **chalcones**. Chalcones are intermediates in the biosynthesis of flavonoids. They show antibacterial, antifungal, anti-tumour and anti-inflammatory properties.

CHALCONIDES: Binary compounds formed between metals and groups of elements, for example, oxide and selenides.



CHALCOPYRITE: Copper iron sulfide with composition CuFeS_2 , which is the most common ore of copper. Chalcopyrite has brass-yellow colour so it is sometimes called fool's gold, but is brittle while gold is sectile.

CHALK: A soft, fine grained whitish sedimentary rock composed of calcium carbonate (CaCO_3) extensively quarried for use in cement, lime and mortar and in the manufacture of cosmetics and toothpaste. It is made from the skeletons or shell of microscopic sea creatures.

CHALLENGE-HANDSHAKE AUTHENTICATION PROTOCOL (CHAP): A process of authenticating a user to a network entity, which may be any server, for example, the Web or Internet service provider (ISP). CHAP is primarily used for security purposes. For example, users provide authenticated plain text passwords when accessing remote servers, which also are authenticated prior to user access.

CHALLENGE-RESPONSE AUTHENTICATION MECHANISM (CRAM): The two-level scheme for authenticating users of the network, used as part of the Internet's Hypertext Transfer Protocol (HTTP). Most smart card systems use challenge-response authentication. These systems require at least two things for authentication and entry: the smart card and the user's password. Examples of CRAM are: (i) password authentication, in which a server asks the client for a password to authenticate the client's identity so that the client can be served; (ii) the use of CAPTCHA, a form of reverse-Turing test for the system to determine if the client is a human or not, used to prevent spam and auto-registration of new accounts for a website or email; (iii) biometric systems are another form of challenge-response authentication.

CHAMAEPHYTE: Low-growing shrubs, in which the overwintering or hibernating buds are borne above ground (not more than 25 cm) but near the surface to minimise exposure to the wind. They are especially important in stressful environments, such as Alpine, Arctic or dry ecosystems often in combination with herbivores and nutrient poor soils or on rocks. Typical examples are *Thymus vulgaris*, *Calluna vulgaris* and African wild olive.

CHAMBER SATURATION: In gas chromatography, it is a uniform distribution of the eluent vapour throughout the chamber prior to chromatography.

CHANCE: Likelihood or probability of an event taking place, expressed as a fraction or percentage.

CHANCE VARIABLE (RANDOM VARIABLE; STOCHASTIC VARIABLE): A variable whose value is determined by the result of a random experiment and may take any of a range of values, which are either continuous or discrete that cannot be predicted with certainty but only described probabilistically. It may also be defined as a measurable function defined on a probability space having range in the interval $[0, 1]$.

CHANDRASEKHAR LIMIT: The maximum possible mass of a star (about $1.44 M_{\text{sun}}$) for it to become a white dwarf. Above this limit a white dwarf cannot exist in a stable configuration.

CHANGE DATA CAPTURE (CDC): The process of capturing changes made at the data source and applying them throughout the enterprise. CDC minimises the resources required for ETL (extract, transform and load) processes because it only deals with data changes. The goal of CDC is to ensure data synchronicity.

CHANGE MANAGEMENT: An IT services management (ITSM) strategy in which a systematic approach ensures the efficient and seamless flow of change in an organisation's IT infrastructure. Change management helps all involved parties, including both individuals and teams, to move from a current state to the next desired state. It also helps to minimise the impact of related incidents on service. Change is prevalent in any IT organisation and it may arise reactively in response to problems or be externally imposed.

CHANGE OF PHASE (CHANGE OF STATE): A change in the physical state of matter, from one physical phase (solid, liquid or gas) to another or the movement of a molecule of a material to and from the solid, liquid and gas state to another state. The change is invariably accompanied by the evolution or absorption of energy even if it takes place at constant temperature. Examples are the gas-liquid

transition (condensation), the liquid-solid transition (freezing) and the normal-to-superconducting transition in electrical conductors.

CHANGE OF VARIABLES: A transformation in which new expressions are substituted for the variables of a given expression, especially where such substitutions are linear and the effect is to change the position of a curve relative to the coordinate axes.

CHANGE PROBLEMS: Mathematical situations involving a change in the initial amount of a quantity over time. Such problems involve three quantities: an initial amount, a change amount, and the resulting amount, which may be an increase or a decrease of the initial amount.

CHANGING THE SUBJECT OF A FORMULA: The transformation or rearrangement of a formula to find the value of a subject of formula.

For example, if $s = ut + \frac{1}{2}at^2$, then in $a = \frac{2(s-ut)}{t^2}$, a is made the subject of formula by transforming or rearranging the formula.

CHANNEL: 1. A pore formed by a protein molecule in semi permeable membranes through which certain substances such as charged ions or lipid-insoluble molecules can diffuse into and out of the cell. 2. In computing, this refers to the physical transmission medium, i.e., a set of frequencies, a wire or a logical connection in a multiplexed medium. It can also be defined as a route by which discrete pieces of information and messages are transmitted from one source (sender) to the other (receiver), defined by the input set, the output set and the probability set for the channel.

CHANNEL ACCESS METHOD (CAM): It is used in telecommunications and computer networks to allow network terminals to share media capacity through a multipoint transmission medium. CAM examples include bus, hub, wireless and ring networks.

CHANNEL BANK: A device used for multiplexing or demultiplexing a group of communication channels, including analogue or digital telephone lines combined as a channel of higher bandwidth or digital bit rate. Channel banks are located in a telephone exchange or enterprise telephone closet, where individual telephone lines are separated from high-capacity telephone trunk lines originating from a central telephone office or enterprise private branch exchange (PBX).

CHANNEL BONDING (ETHERNET BONDING): A computer networking arrangement in which two or more network interfaces on a host computer are combined for redundancy or increased throughput.

CHANNEL CONFLICT: A situation in which channel partners have to compete against one another or the vendor's internal sales department. Channel conflict can cost a company and its partners money as partners try to undercut one another. It can also lower morale within the channel and cause some partners to consider other vendors. To prevent channel conflict, partners sometimes enact agreements such as deal registration.

CHANNEL PARTNER: A third-party organisation or individual that markets and sells products, services or technologies for a manufacturer or service provider via a partnering relationship.

CHANNEL SERVICE UNIT, CSU (DIGITAL SERVICE UNIT, DSU): Two digital hardware devices that provide an interface between digital circuits and data terminal equipment (DTE). A CSU/DSU unit operates at the physical layer of the Open Systems Interconnection (OSI) model. A CSU/DSU unit is required for both ends of a T1 or T3 connection.

CHANNELLING: The direct transfer of a reaction intermediate from one enzyme to the next.

CHANNERY SOIL: Accumulation of soil consisting of thin and flat limestone, sandstone, or schist fragments with diameters of up to 15 cm.

CHAOS: Unpredictable behaviour occurring in a system that should be controlled by deterministic laws. In such systems, the equations that describe the way the system changes with time are nonlinear and involve several variables. Consequently, they are very sensitive to the initial conditions and a very small initial difference may make an enormous change to the future state of the system, for example,

turbulent flow of fluids, long time unpredictability of the weather, oscillations in electrical circuits, combustion processes and oscillating reactions in chemistry.

CHAOS THEORY (CHAOLGY): The branch of mathematics that is concerned with irregular, unpredictable systems such as weather that rely upon an underlying order, so that very simple or small systems and events can cause very complex behaviours or events.

CHAOTIC ATTRACTOR (STRANGE ATTRACTOR): A type of attractor in a complex dynamical system's phase space that shows sensitivity to initial conditions. Because of this property, once the system is on the attractor nearby states diverge from each other exponentially fast. Consequently, small amounts of noise are amplified. Once sufficiently amplified the noise determines the system's large-scale behaviour and the system is then unpredictable.

CHAOTIC REACTION: A type of chemical reaction in which the concentrations of reactants show chaotic behaviour. This may occur when the reaction involves a large number of complex interlinked steps. Under such conditions, it is possible for the reaction to display unpredictable changes with time.

CHAPARRAL: A type of stunted woodland found in temperate regions with little summer rainfall. It is dominated by drought resistant evergreen shrubs.

CHAPERON: In chemistry, it is a species that facilitates a combination reaction between atoms or radicals or the reverse process. It is a special case of a third body.

CHAPERONE (MOLECULAR CHAPERONE): In biology, it is a large family of heat shock proteins that assist in correct protein folding and assembly. Heat shock proteins are a family of proteins that are produced by cells in response to exposure to stressful conditions.

CHAPERONIN: A class of chaperones that assist in the correct folding and assembly of polypeptides in bacteria, plasmids, eukaryotic cytosol, and mitochondria.

CHAR: 1. A solid decomposition product of a natural or synthetic organic material. If the precursor has not passed through a fluid stage, char will retain the characteristic shape of the precursor but becomes smaller. For such materials the term 'pseudomorphous' has been used. Some simple organic compounds, such as sugar, melt at an early stage of decomposition and then polymerise during carbonisation to produce chars. 2. In some programming language, for example, C and CH, it is a description of a data type used in storing character strings.

CHARACTER: 1. Also known as **trait**, a distinctively inherited particular feature such as colour, height or shape of an organism that can be expressed in two or more alternative forms and distinguishes an individual, a group or a thing from another. Character may be acquired as a response to the environment; or inherited from genes transmitted from parent to offspring, with their expressions being often modified by environmental conditions. **Phenotype** is the observable properties of an organism that are produced by the interaction of the genotype and the environmental pressures the organism is subjected to. Trait is one of the alternative forms of character, for example, tall, short, red, blue and anger and may be inherited or acquired. A phenotypic trait is an obvious and observable trait, which is the expression of genes in an observable way. An example of a phenotypic trait is hair colour. 2. A multiplicative functional from a group to the complex numbers. In other words, the mapping from a given group, G , to a field, that corresponds to the representation, R , of G , under which the image of an element, x , is the trace of $R(x)$. 3. Also known as **finite character**, the property of a collection of sets or a property that a set is in the collection or has the property if and only if this is also true of every non-empty finite subset of the given set. 4. Any written or printed letter, number or other symbol. 5. A single letter, number or symbol that can be displayed on a computer screen or printer and represents one byte of data.

CHARACTER DISPLACEMENT: The effect of competition or reproductive barriers characterised by a more pronounced difference in a particular character occurring where two similar species coexist

than where their populations do not overlap. This pattern results from evolutionary change driven by competition among species for limited resources such as food, water, etc. So in a stable environment two competing species must differ in their respective ecological niche.

CHARACTER ENCODING: A mapping of a coded character set onto a character representation for manipulation in a computer. A single character set may have more than one character encoding. Many character encodings, such as the ISO 8859 series, assign a single byte to each character. Other character encodings, such as the Unicode UTF-8, UTF-16 and UTF-32 encoding forms, define a number of ways of encoding the same character.

CHARACTER RECOGNITION: A method of inputting information usually to a computer, by optically or magnetically scanning printed or hand-written characters. The process involves document analysis (extracting individual character images), recognising these images (based on shape) and contextual processing (either to correct misclassifications made by the recognition algorithm or to limit recognition choices). Therefore an Optical Character Recognition (OCR) system contains three logical components: an image scanner, OCR software and hardware and an output interface.

CHARACTER SET: 1. A term used in computing to denote the complete set of symbols that can be used in a program or recognised by a computer. It may include letters, digits, spaces, punctuation marks, etc. 2. A group of unique symbols used for display and printing. Character sets for languages that use the English alphabet generally contain 256 symbols, which is the number of combinations one byte can hold. Except for special fonts, such as Dingbats and Greek symbols, the symbols are the same for the first 128 characters. The letters may have different styling due to their typeface, but an "M," for example, is an "M" in the same sequential order in each character set.

CHARACTER THEORY: The study of the traces (sums of the diagonal elements) of the matrix representations of a group. The information gained is listed in character tables, the properties of which give insight into the group's properties.

CHARACTERISTIC: 1. A distinguishing quality, trait, feature or property of an individual or thing that helps to identify it. Characteristic may also define one's profession or job, for example, a scientist has the following characteristics: observational, objective, curious, creative, sceptical and accurate. 2. The integral part of a common logarithm or the whole number in logarithm that represents the order of magnitude but not the digits of the given number. It is equal to the exponent of the greatest power of 10 less than it. For example, the characteristic of log 450 is 3 and of log 4.5 is 1.

CHARACTERISTIC CURVE: 1. The curve in a graph determined by the characteristic equations in partial differential equations or the relation between two variables. 2. Also known as **H and D curve** or **sensitometric curve**, it is the curve indicating the variation of density in a developed photographic image that results from the increase of exposure expressed logarithmically. 3. A curve showing the relationship between frequency and recording intensity or frequency and reproducing intensity in sound recording.

CHARACTERISTIC EQUATION: 1. An equation derived from a given square matrix A in which the characteristic polynomial is set equal to 0. It is of the form $\det[A - \lambda I]$, where A is the square matrix, I is the unit matrix and λ is a scalar variable. 2. The second

order partial differential equation for which $a\left(\frac{dy}{dx}\right) - b\left(\frac{dy}{dx}\right) + c = 0$,

where $au_{xx} + bu_{xy} + cu_{yy} + du_x + eu_y + fu = h$ is the general form of a second order partial differential equation whose solutions give the characteristic curves of the partial differential equation, allowing it to be written in a simpler canonical form.

CHARACTERISTIC FUNCTION: 1. Of a set, the function taking the value 1 for arguments that are members of the given set and 0 otherwise. 2. Also known as **characteristic polynomial**, it is a function derived from the probability distribution function that in particular enables the distribution of the sums of random variables

to be analysed since two distributions have the same characteristic function only if they are identical almost everywhere. 3. Also known as **characteristic polynomial**, it is the polynomial $\det[A - \lambda I]$ derived from a given square matrix A , where I is the unit matrix and λ is a scalar variable. For example, the characteristic polynomial of

$$\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} \text{ is } \lambda^2 - 3\lambda + 1.$$

CHARACTERISTIC GROUP: A single heteroatom, for example, $-\text{Cl}$ and $=\text{O}$; a heteroatom bearing one or more hydrogen atoms or other heteroatoms, for example, $-\text{NH}_2$, $-\text{OH}$, $-\text{SO}_3\text{H}$, $-\text{PO}_3\text{H}_2$ and IO_2 ; or a heteroatomic group attached to or containing a single carbon atom, for example, $-\text{CHO}$, $-\text{C}\equiv\text{N}$, $-\text{COOH}$ and $-\text{NCO}$ attached to a parent hydride.

CHARACTERISTIC LENGTH (CHARACTERISTIC SCALE): The parameter which characterises the density profile of a given physical quantity. Characteristic length gives a measure of the average distance that the products of burned fuel must travel to escape.

CHARACTERISTIC MASS: The mass of analyte which, when atomized electrothermally, produces the uniquely defined absorption signal.

CHARACTERISTIC MASS FOR INTEGRATED ABSORBANCE: That mass of analyte which produces an integrated absorbance signal whose net area is equal to 0.0044 s. Its symbol is m_0 .

CHARACTERISTIC MASS FOR PEAK ABSORPTION: That mass of analyte which produces a defined peak absorbance of 0.0044 (or 1% absorbance).

CHARACTERISTIC OF A FIELD: The smallest positive natural number n such that 0 is the sum unit element with itself n times. If no such n exists, the field is said to be of characteristic 0.

CHARACTERISTIC POLYNOMIAL (CHARACTERISTIC FUNCTION): The polynomial $\det[A - \lambda I]$ derived from a given square matrix A , where I is the unit matrix and λ is a scalar variable. For

example, the characteristic polynomial of $\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$ is $\lambda^2 - 3\lambda + 1$.

CHARACTERISTIC POTENTIAL: An applied potential that is characteristic of a charge-transfer process and the experimental conditions such as the composition of the solvent and supporting electrolyte and the temperature under which it is investigated and whose nature depends on the technique that is employed. Some typical characteristic potentials are the half-wave potential in polarography, the quarter-transition time potential in chronopotentiometry, the peak potential and the half-peak potential in linear-sweep voltammetry and the summit potential in alternating current (a.c.) polarography.

CHARACTERISTIC ROOT (CHARACTERISTIC NUMBER/VALUE; EIGENVALUE; LATENT ROOT): In a characteristic polynomial, for a square matrix A , in which x is a nonzero vector called the eigenvector that does not change its direction under the associated linear transformation and satisfies the equation $Ax = \lambda x$, the scalar, λ is the eigenvalue. In other words, it is a root, λ , of the characteristic equation $\det(A - \lambda I)$. For a linear operator A , it is a solution, λ , of the equation $AX = \lambda X (X \neq 0)$.

CHARACTERISTIC SUBSET (CHARACTERISTIC SUBGROUP): A subset or subgroup of a group that is mapped by all automorphisms of the group onto itself. The derived subgroup and the centre of a group are characteristic and any characteristic subgroup is a normal subgroup.

CHARACTERISTIC VECTOR (EIGENVECTOR; LATENT VECTOR): For a square matrix or linear operator, A , it is a nonzero vector, x that does not change its direction under the associated linear transformation and satisfies the equation $Ax = \lambda x$, where λ is a scalar of A called an eigenvalue.

CHARACTERISTIC X-RAY EMISSION (CHARACTERISTIC X-RADIATION): X-ray emission originates from the radiative decay of electronically highly excited states of matter. Excitation by electrons

is called **primary excitation**; if by photons, it is called **secondary** or **fluorescence excitation**. Particle induced X-ray emission (PIXE) is produced by the excitation of heavier particles such as protons, deuterons or heavy atoms in varying degrees of ionisation. Emission of photons in the X-ray wavelength region also occurs from ionised gases or plasmas at high temperatures, from nuclear processes (low-energy end of the gamma-ray spectrum) and from radiative transitions between muonic states. Characteristic X-ray emission consists of a series of X-ray spectral lines with discrete frequencies, characteristic of the emitting atom. Other features are emission bands from transitions to valence levels. In a spectrum obtained with electron or photon excitation the most intense lines are called diagram lines or normal X-ray lines. They are dipole allowed transitions between normal X-ray diagram levels.

CHARACTERS (TAXONOMIC CHARACTERS): Features, such as form, physiology, structure and behaviour that are assessed in isolation from the rest of the plant by taxonomists, in order to make comparisons and interpretations. It is important to distinguish between characters and character states. For example, leaf width may be a character, while leaves 4 mm wide are an expression of that character, i.e. its character state. Characters are often referred to as 'good' or 'bad' but this is strictly relative. Thus a good diagnostic character, such as compound leaves in a group of plants that mainly have entire leaves, would be bad for separating taxa in a group in which leaf divisions were either variable or often compound.

CHARCOAL: A black porous material containing 85-98% carbon produced by burning objects such as cellulose, wood, peat or other organic materials at 500-600 °C (773-873K) in the absence of air. Charcoal has highly reactive inner surfaces and a low sulfur content. It is used as a fuel for cooking, in the smelting of metals such as copper and zinc and for making activated charcoal for use in industry.

CHARGAFF'S RULE: A rule which states that deoxyribonucleic acid (DNA) from any cell of organisms should have 1:1 ratio of the complimentary bases or the ratios of adenine (A) to thymine (T) and guanine (G) to cytosine (C) are equal. It is named after its discoverer, Erwin Chargaff (1905-2002), an Austrian American biochemist.

CHARGE (ELECTRIC CHARGE; ELECTROSTATIC CHARGE): A fundamental property of matter, which defines the extent to which matter has more or fewer electrons than protons. It occurs in nature in two forms, conventionally described as positive and negative in order to distinguish between the two kinds of interaction between particles. Two particles that have similar charges (both negative or both positive) interact by repelling each other; two particles that have dissimilar charges (one positive, one negative) interact by attracting each other. The size of the interaction is determined by Coulomb's Law. Protons and the nuclei of atoms have a positive charge; electrons have a negative charge; neutrons have no charge. Generally charged substances have an imbalance of positive and negative charges and a net charge that exerts a force on other charged substances. The unit of electrical charge quantity in the International System of Units is the coulomb. One coulomb (1 C) is equal to approximately 6.24×10^{18} elementary charges. This is a very large quantity and therefore, measured in micro-coulombs ($1 \mu\text{C} = 1 \times 10^{-6} \text{ C}$) and nanocoulombs ($1 \text{ nC} = 1 \times 10^{-9} \text{ C}$). **Elementary charge**, with the symbol e is defined as the amount of charge carried by one electron (electrons have negative charge). The value of elementary charge is approximately 1.602×10^{-19} coulombs (C), and the charge of the electron is approximately $-1.602 \times 10^{-19} \text{ C}$. **Charge quantization** is the principle that the charge of all other objects in the universe is an integer multiple of the elementary charge and has many times the charge of a single electron it has.

CHARGE CARRIER: The free particle whose movement constitutes electric current. It is an electron, hole or ion that transports the electric charge in an electric current. The nature of the carrier depends on the type of conductor. In metals, the charge-carriers are electrons, in semi-conductors they are electrons (n-type) or positive holes (p-type), in gases they are positive ions and electrons, in solutions (electrolytes), they are positive and negative ions. The magnetic force on a moving

charge is responsible for the sign of the charge carriers in a conductor. A current flowing from right to left in a conductor can be the result of positive charge carriers moving from right to left or negative charges moving from left to right or some combination of each.

CHARGE CONJUGATION: A symbol of elementary particles that tells the difference between a particle and its antiparticle.

CHARGE CONSERVATION (LAW OF CONSERVATION OF CHARGE): The law which states that the net charge of an isolated system remains constant during any physical process, such as two charge objects making contacting and separating. In other words, any system that is not exchanging mass or energy with its surroundings will never have a different total charge at any two times.

CHARGE DENSITY: The amount of electric charge per unit volume of a medium or body (volume charge density) or the electric charge per unit surface area of a body (surface charge density). It is measured in coulombs per metre (C/m), coulombs per square metre (C/m²) or coulombs per cubic metre (C/m³) respectively. In chemistry, it refers to the charge distribution over the volume of an ion, molecule or atom.

CHARGE HOPPING: Electron or hole transfer between equivalent sites.

CHARGE NUMBER: 1. Ratio of the charge of a particle to the elementary charge. Its symbol is z . 2. In inorganic nomenclature, the charge number is the magnitude of the charge on an ion, given in Arabic numerals followed by the sign of the charge, in parentheses without a space, immediately after the name of an ion. An example is iron(2+), hexacyanoferrate(4-). 3. In a cell reaction, charge number is the number of electrons transferred according to the cell reaction equation. It is a dimensionless positive or negative integer which determines the net charge of an ion. Its value is equal to the number of electrons which must be gained or lost by an ion in order to achieve a stable state according to the octet rule. The charge of an ion is the product of the specific charge number and the elementary charge.

CHARGE POPULATION: The net electric charge on a specified atom in a molecular entity, as determined by some prescribed definition.

CHARGE RECOMBINATION: The reverse of charge separation. In using this term it is important to specify the resulting electronic state of the donor and acceptor.

CHARGE SEPARATION: A process in which, under the influence of a suitable driving force such as provided by photoexcitation, electronic charge moves in a direction that increases the difference in local charges between donor and acceptor sites. Electron transfer between neutral species is one of the most important examples.

CHARGE SHIFT: A process in which under the influence of a suitable driving force such as that provided by photoexcitation, electronic charge moves without changing the difference in local charges between donor and acceptor sites. Examples are electron transfer reversing the charges in a system composed of a neutral donor and a cationic acceptor or of a neutral acceptor and an anionic donor.

CHARGE TRANSFER COMPLEX (CT COMPLEX; ELECTRON-DONOR-ACCEPTOR COMPLEX): A complex compound in which charge transfer between two molecules is possible because of weak bonds that hold them together. Charge transfers are created by an electronic transition into an excited electronic state, such that a fraction of electronic charge is transferred between the molecular entities. The source molecule from which the charge is transferred is called the electron donor and the receiving molecule is called the acceptor. An example is phenoxinone, in which the phenol and quinone molecules are not held together by formal chemical bonds but are associated by transfer of charge between the compounds' aromatic ring systems.

CHARGE TRANSFER TRANSITION TO SOLVENT (CTTS): Electronic transition which is adequately described by single electron transfer between a solute and the solvent, as opposed to excitation followed by electron transfer to solvent.

CHARGE-COUPLED DEVICE (CCD): A light-sensitive integrated circuit that uses a layer of silicon that releases electrons when struck by incoming light to record images. The electrons are in pixels and

read off into a computer at the end of the exposure. CCDs are used in digital cameras and video cameras to record still and moving images, and have now almost entirely replaced photographic film for applications such as astrophotography, where extreme sensitivity to light is paramount.

CHARGE-EXCHANGE IONISATION (CHARGE-TRANSFER IONISATION): This occurs when an ion/atom or ion/molecule reaction takes place in which the charge on the ion is transferred to the neutral species without any dissociation of either. An example is $A^+ + B \rightarrow A + B^+$.

CHARGE-PERMUTATION REACTION: A general term to describe an ion/neutral species reaction where there is a change in the magnitude and/or sign of the charges of the reactant.

CHARGE-STRIPPING REACTION: An ion/neutral species reaction wherein the charge on the reactant ion is made more positive.

CHARGE-TO-MASS RATIO (CHARGE TO MASS RATIO): The ratio of electric charge of a charged particle in coulombs to its mass in kilograms. Charge-to-mass ratio gives an indication of how much a particle will be deflected as it moves through an electric or magnetic field. The greater the charge the greater is the force on the particle; and the smaller the mass, the greater is the deflection produced by the force. The CODATA recommended value, e/m_e for an electron is $1.758820088 \times 10^{11}$ C/kg.

CHARGE-TRANSFER ADSORPTION: Oxidative or reductive chemisorption where 'reductive' and 'oxidative' refer to electron gain or loss on species in the solid. In simple cases it is non-dissociative, i.e. there is a mere transfer of charge between adsorptive and adsorbent in forming the adsorbate.

CHARGE-TRANSFER COMPLEX: An electron-donor-electron-acceptor complex, characterised by electronic transition(s) to an excited state in which there is a partial transfer of electronic charge from the donor to the acceptor moiety.

CHARGE-TRANSFER DEVICE: A charge-transfer device has a metal oxide semiconductor (MOS) structure that is composed of many independent pixels where charge is stored in such a way that the charge pattern corresponds to the irradiation pattern. These devices can be linear or two-dimensional. According to the method used to detect the charge pattern, two types of charge-transfer devices can be distinguished: charge-coupled devices (CCDs) and charge-injection devices (CIDs).

CHARGE-TRANSFER STEP: In an electrode reaction, it is an elementary step in which charge is transferred from one phase to the other.

CHARGE-TRANSFER TRANSITION: An electronic transition in which a large fraction of an electronic charge is transferred from one region of a molecular entity, called the electron donor, to another, called the electron acceptor (intramolecular CT) or from one molecular entity to another (intermolecular CT). It is typical for donor acceptor complexes or multichromophoric molecular entities. In some cases the charge-transfer absorption band may be obscured by the absorption of the partners.

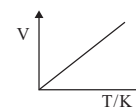
CHARGED COUPLED DEVICE (CCD): Sensors used in digital cameras and video cameras to record still and moving images. The CCD captures light and converts it to digital data that is recorded by the camera. For this reason, a CCD is often considered the digital version of film.

CHARGED PARTICLE BEAM: High energy streams of charged particles such as electrons, protons, alpha particles and positrons. Charged particles are able to interact with matter causing various effects important in many applications such as inducing x-ray (PIXE) for chemical analysis. Charged particle beams are being developed as weapons.

CHARLES, JACQUES ALEXANDRE CÉSAR (1746-1823): French chemist and physicist and aeronaut, best remembered for discovering Charles' law (1787), relating to the volume and temperature of a gas. He became Professor of Physics at the Paris Conservatoire des Arts et Métiers. In 1783 he made the first balloon using hydrogen gas and ascended to a height of nearly 3 km. He also discovered the

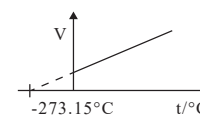
relationship between the volume of gas and temperature, known as Gay-Lussac's law or Charles's law in 1787; and was elected to the French Academy in 1785.

CHARLES' LAW (GAY-LUSSAC'S LAW): One of the three Gas Laws, which states that the volume of a given mass of gas at constant pressure is directly proportional to its absolute temperature, i.e. $V = kT$, where V = volume of gas and T = absolute temperature and k is a constant.



CHARLES' THEOREM: Any general displacement of a rigid body can be represented by a translation plus a rotation.

CHARM: A property possessed by certain elementary particles that is expressed as a quantum number and is used in the quark (very small particles found inside protons and neutrons) model. It was believed that it accounts for the abnormally long life time of the psi particle. Charm is also thought to be conserved in strong and electromagnetic interactions.



V = Volume
 T = Absolute temperature

Graphical illustration of Charles' law

CHAROPHYTA: A division of green aquatic algae containing one order, the Charales that live entirely submerged in freshwater ponds attached to the bottom by rhizoids. Charophyta includes spirogyra, stoneworts and desmids. They have a distinct main axis with whorls of branches arising from the nodes and superficially resemble certain higher plants. The thallus often becomes encrusted with calcium carbonate. Sexual reproduction is oogamous and very complex. The zygote germinates to give a filamentous protonema, a situation only found among algae in the Charales.

CHARPIT'S METHOD: A method for solving a first-order partial differential equation of the form $F\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right) = 0$, in which a

solution a solution of a specimen of Lagrange's linear equation is used to provide a second first-order partial differential equation

$f\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right) = 0$, that has the property that solving these equations

for $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ produces an integrable total differential equation

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

CHARRING: Pyrolysis of samples containing organic matter. In the presence of oxygen this is referred to as ashing.

CHART: 1. A graphical way of showing detailed information. 2. Also called **(local) coordinate system**, a neighbourhood of a point in a manifold together with its mapping into Euclidean n -space; formally, a pair $(U_\lambda, \Phi_\lambda)$ where U_λ is an element of a cover of the manifold and Φ_λ is a homeomorphism that maps it to an open subset of $\mathbb{R}^n$. A collection of charts that cover the manifold is called an **atlas**. 3. In meteorology, an outline map that shows weather patterns. 4. In astrology, a map showing the relative positions of the planets at the time of somebody's birth, on which his or her horoscope is based. 4. In navigation, maps prepared for marine or air navigation showing geographic features useful to the navigator, as well as indications of direction such as true north (the direction of the geographic North Pole), magnetic north (the direction indicated by the north-seeking end of a magnetic compass needle) and magnetic declination (the difference between these two directions). Marine charts would show details such as the outline and nature of coasts, with landmarks; currents and undercurrents (both direction and force); winds; tides; location and type of lighthouses, buoys, beacons and lightships; position of rocks, bars, reefs, shoals, wrecks or other dangers; contour and nature of bottom (mud, sand, rock or gravel); and depth.

CHARTACEOUS: Having a thin, papery texture which is usually not green, such as in a bract or sheath.

CHASMANTHERIC POLLINATION PATHWAY: Pollination in which pollen is transferred from a normally dehiscent anther by a pollinating agent with pollen grain germination on the stigma and subsequent growth of the pollen tube through stigma, style and the ovule into the embryo sac.

CHASMOGAMY: The production of flowers that open to expose the reproductive organs. Chasmogamous flowers allow cross pollination but does not preclude self pollination. On the other hand, **cleistogamous** refers to flowers which self-fertilise without opening.

CHAT (CHATTING): Text-based real-time communication between two or more users via computer using keyboard. Usually, it is the exchange of typed-in messages on a local network (LAN) or over the Internet. Videos can also be exchanged using the webcam. A group of users can chat on the Internet.

CHAT ROOM: An area on the Internet or other computer network where users can communicate, typically limiting communication to a particular topic. Forums and discussion groups, in comparison, allow users to post messages but do not have the capacity for interactive messaging. Most chat rooms do not require users to have any special software; those that do, such as Internet Relay Chat (IRC) allow users to download it from the Internet.

CHATTERBOT (CHATBOT; CHATTERBOX; ARTIFICIAL CONVERSATIONAL ENTITY, ACE): An artificial intelligence (AI) program that attempts to simulate the conversation of a human being.

CHATTY PROTOCOL: An application or routing protocol that requires a client or server to wait for an acknowledgement before it can transmit again. On a local area network (LAN), this wait time is measured in milliseconds and does not affect performance. On a wide area network (WAN), however, the wait time, which is called latency is longer because the acknowledgements have to travel farther. Extended wait times can degrade the quality of service (QoS) dramatically. Chatty protocols can also lower the productivity of remote employees.

CHEBYSHEV APPROXIMATION: The kind of approximation which when truncated the error spreads over $[-1, 1]$.

CHEBYSHEV NORM (SUPRENUM NORM; INFINITY NORM; UNIFORM NORM): The norm placed on continuous or bounded functions on a set S that assigns to each function the supremum of the moduli of the values of the function on the set: $\|f\|_{\infty} = \sup\{|f(x)| : x \in S\}$. The real or complex continuous functions on a compact set S form a Banach space that is denoted $C(S)$.

CHEBYSHEV POLYNOMIALS OF THE FIRST KIND: The orthogonal polynomials defined by $T_n(x) = \cos(n \arccos x)$. These arise as the polynomials of degree n of smallest Chebyshev norm on $[-1, 1]$, with leading coefficient unity.

CHEBYSHEV'S INEQUALITY: 1. In statistics, the fundamental theorem that the probability of a random variable differing from its mean by more than k standard deviations is less than or equal to $1/k^2$.

More generally, the result that, for all $\varepsilon > 0$, $P(|X - c| > \varepsilon) \leq \frac{1}{\varepsilon^2} E[(X - c)^2]$, where X is a random variable and c is a constant. 2. In statistics, the inequality for two non-increasing sequences of real numbers, $a_1, \dots, a_n$ and $b_1, \dots, b_n$ that $\frac{1}{n^2} \sum_{j=1}^n a_j \sum_{i=1}^n b_i \leq \frac{1}{n} \sum_{k=1}^n a_k b_k$, with identity if and only if all the a_i and all the b_i are equal.

CHECK BITS (PARITY BITS): Bits of data that are used to check for errors in the transmission or storage of a unit of information before it is accepted. In the process, a 0 or 1 is added to each group of bits so that it will have either an odd number of 1's or an even number of 1's; for example, if the parity is odd then any group of bits that arrives with an even number of 1's must contain an error.

CHECK DIGIT: A numeric digit attached to an important code number as a validation check, to ensure that account numbers are entered accurately into the computer.

CHECK SUMMATION: A technique for error detection to ensure that data or program files have been accurately copied or transferred.

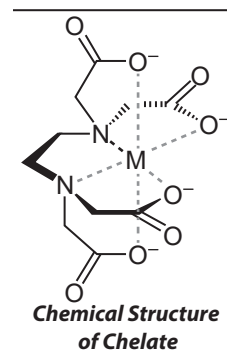
CHECKSUM: A digit that is calculated from data when data is sent from one place to another, representing the sum of the correct digits and against which later comparisons can be made to detect errors in the data. If the checksums do not match those of the original files, the data may have been altered or corrupted. A checksum can be computed in many different ways, using different algorithms.

CHEDDITE: 1. A group of high explosives such as nitrobenzene or nitrotoluene made from nitro aromatic compounds mixed with sodium or potassium chlorate and small amount of paraffin as moderant for the chlorate. Several different types were made and they were principally used in quarrying. Cheddite explosives are named after the town of Chedde in Haute-Savoie, France, where they were first manufactured. 2. The commercial name for an explosive compound used as an explosive primer for shotgun cartridges, containing 90% potassium chlorate, 7% kerosene (paraffin), 3% petroleum jelly and traces of carbon black.

CHEETAH: A slender, black-spotted feline (*Acinonyx jubatus*) belonging to the order Carnivora, inhabiting most of Africa and Asia. It has a small head and the claws are not retractile. It is recognised as the world's fastest mammal with a maximum running speed of 120 km/hr.

CHELA: The terminal segment appendage of an arthropod such as a lobster, crab, scorpion or similar animal arthropod. It is often enlarged and modified as a pincer in lobster.

CHELATE: A cyclic metalorganic complex, formed when a cation is bonded by two or more donor groups of a multidentate ligand. The multidentate ligand is called a chelating agent or chelator and the product is known as a metal chelate compound or metal chelate complex.



CHELATION: The formation or presence of bonds between two or more separate binding sites within the same ligand and a single central atom. A molecular entity in which there is chelation and the corresponding chemical species is called a **chelate**. The terms bidentate or didentate, tridentate, tetradentate, multidentate are used to indicate the number of potential binding sites of the ligand, at least two of which must be used by the ligand in forming a chelate. For example, the bidentate ethylenediamine forms a chelate with Copper(I) iodide or Cuprous iodide (CuI) in which both nitrogen atoms of ethylenediamine are bonded to copper.

CHELETROPIC REACTIONS: A type of addition reaction in which a conjugated molecule forms two single bonds from terminal atoms of the conjugated system to a single atom on another molecule to give a cyclic adduct. An example is the addition of sulfur dioxide to butadiene.

CHELICERAE: The first pair of appendages on the head of arachnids such as a spider and other arthropods of the phylum Chelicerata. It is often modified for grasping and piercing.

CHELICERATA: A phylum of invertebrate animals that includes arthropods belonging to the classes Arachnida, Merostomata and Pycnogonida. It includes horseshoe crabs, scorpions, spiders and mites.

CHELONIA (TESTUDINES): An order of reptiles, which includes the turtles and tortoises that evolved during the late Permian or Triassic Period and have changed little since. Chelonians are characterised by a protective hard shell that consists of bony plates fused to the vertebrae and ribs that covers most of the body and with no teeth. Unlike all other living reptiles, the skulls of chelonians lack temporal openings, which is characteristic of the earliest known reptiles, called anapsids.

CHEMI-IONISATION: In mass spectrometry, it refers to a process whereby gaseous molecules are ionised when they interact with other internally excited gaseous molecules or molecular moieties. Chemi-ionisation and chemical ionisation are two terms which should not be used interchangeably.

CHEMICAL: 1. A single pure substance that is either an element or a compound or a substance obtained by or used in a chemical process. 2. Produced by or involving in the process of chemistry.

CHEMICAL ACTINOMETER: A detector in which the amount of a chemical formed is proportional to the numbers of photons absorbed.

CHEMICAL ACTIVATION: It occurs when some of the energy required for a reaction is provided by a preceding exothermic chemical reaction.

CHEMICAL AMPLIFICATION: A process consisting of a chemical reaction that generates a species that catalyses another reaction and also the succeeding catalysed reaction. Chemical amplification can lead to a change in structure and by consequence to a change in the physical properties of a polymeric material. An example of chemical amplification is the transformation of [(*tert*-butoxycarbonyl)oxy]phenyl groups in polymer chains to hydroxyphenyl groups catalysed by a photo-generated acid.

CHEMICAL ANALYSIS: The determination of the compounds or elements comprising a chemical substance. Qualitative analysis deals with what a sample contains; and quantitative analysis finds the amounts. The methods available depend on the size of the sample: macro, which is more than 100mg; semi micro is between 1 and 100mg; micro is between 1µg and 1mg; or submicro which is less than 1µg.

CHEMICAL BONDS: The forces of attraction, which are ionic, covalent and metallic bonds that hold particles together in a compound or elements. The strength of bonds varies considerably; covalent or ionic bonds are generally strong bonds; weak bonds include dipole-dipole interactions, the London dispersion force and hydrogen bonding.

CHEMICAL CHANGE: A change which occurs when two or more substances known as reactants interact with each other, leading to the formation of different substances with different chemical compositions. For example, when carbon burns in oxygen, it produces carbon dioxide. In a chemical change, bonds are broken and new bonds are formed between different atoms. A chemical change may also be accompanied by energy change.

CHEMICAL COMBINATION: The reverse of decomposition, in which elements or molecules combine to form new compounds. There are three laws of chemical combination: (i) The Law of Constant Composition or the Law of Definite Proportions states that the proportions of the elements in a compound are always the same, no matter how the compound is made. (ii) The Law of Multiple Proportions states that when two elements A and B combine to form more than one compound, then the masses of B that combine with a fixed mass of A are in simple ratio to one another. For example, carbon forms two oxides. In one, 12 grams of carbon is combined with 16 grams of oxygen (CO); in the other 12 grams of carbon is combined with 32 grams of oxygen (CO₂). The oxygen masses combining with a fixed mass of carbon are in the ratio 16:32, i.e. 1:2. (iii) The Law of Equivalent Proportions states that if two elements A and B each form a compound with a third element C, then a compound of A and B will contain A and B in the relative proportions in which they react with C. For example, sulfur and carbon both form compounds with hydrogen. In methane, 12 g of carbon react with 4 g of hydrogen. In hydrogen sulfide, 32 g of sulfur react with 2 g of hydrogen (i.e. 64 g of S for 4 g of hydrogen). Sulfur and carbon form a compound in which the C:S ratio is 12:64 (i.e. CS₂). Combination reactions are always exothermic. For example, barium metal and fluorine gas will combine in a highly exothermic reaction to form the salt barium fluoride: $\text{Ba} + \text{F}_2 \rightarrow \text{BaF}_2$.

CHEMICAL COMPOSITION: The proportion by weight of each element present in a chemical compound. The **law of definite proportions** discovered by J. L. Proust, states that pure compounds have a fixed and invariable composition. A few compounds, termed non-stoichiometric, disobey this law, because they have lattice vacancies or extra atoms and the composition varies within a certain range depending on the formation conditions. The **law of multiple proportions**, discovered by John Dalton, states that if two elements A and B form more than one compound, the various weights of B which combine with a given weight of A are in small whole-number ratios.

CHEMICAL COMPOUND: A pure substance with unique properties that is composed of two or more separate elements that are chemically bonded together in fixed proportions that cannot be readily separated through simple physical means. The four different types are classified by the way the constituent atoms are bonded or held together, which include the following: molecules held together by covalent bonds; salts held together by ionic bonds; intermetallic compounds held together by metallic bonds; and certain complexes held together by coordinate covalent bonds.

Examples of chemical compounds include acetic acid (CH₃COOH), carbon dioxide (CO₂), chalcone (C₁₅H₁₂O), lactose or sugar (C₁₂H₂₂O₁₁), sodium chloride or common salt (NaCl), sodium hydroxide (NaOH) and water (H₂O).



Fig. I: Carbon dioxide, CO₂

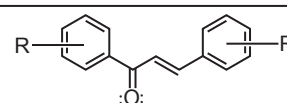


Fig. II: chalcone, (C₁₅H₁₂O)

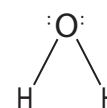


Fig. III: water, H₂O

Bonding of atoms in a chemical compound

CHEMICAL CONTROL: The use of pesticides for pest control. It includes herbicides, which kill unwanted plants or weeds, insecticides which kill insect pests, fungicides which kill fungi and rodenticides which kill rodents.

CHEMICAL DATING: The determination of the relative or absolute age of minerals and of ancient objects and materials by measurement of their chemical compositions.

CHEMICAL DECOMPOSITION: The breakdown of a single entity (normal molecule, reaction intermediate, etc.) into two or more fragments.

CHEMICAL DOSIMETER: A dosimeter in which the dose is measured by observing the extent, under specified conditions, of a chemical reaction caused by the ionising radiation to be measured.

CHEMICAL ELEMENT: A substance made up entirely of atoms that each contains the same number of protons in their nuclei (and therefore the same number of electrons). In other words, it is a substance that cannot be separated into simpler substances by chemical means. Most atoms may have different numbers of neutrons in their nuclei, leading to different isotopes of the same element. There are 92 naturally occurring elements, ranging from the lightest, hydrogen, to the heaviest, uranium.

CHEMICAL ENERGY: The energy that is within the chemical bonds of a substance, which are liberated in a chemical reaction. Such energy is changed into heat and light energy during the combustion of fuels. The rearrangement of the atoms in reacting compounds to produce new compounds causes a change in chemical energy. By the law of conservation of energy, a chemical compound must undergo a change equal and opposite to the change of heat energy in a reaction.

CHEMICAL ENGINEERING: Academic discipline and industrial activity concerned with developing processes and designing and operating plants for large scale industrial chemical processing to produce chemicals. The field includes research, design, construction,

operation, sales and management activities. Chemical engineers must master chemistry, physics, mathematics and engineering. They are usually employed in the chemical and oil industries as well as in processing industries such as foods, paper, textiles, plastics, nuclear and biotechnology.

CHEMICAL EQUATION: A way of denoting the essential features of a chemical reaction using chemical symbols for the participating particles such as atoms, molecules and ions. In other words, it is an expression representing a chemical reaction using chemical symbols and formulae. By convention, reactants are put on the left and products on the right. A single arrow between them denotes an irreversible reaction, a double arrow a reversible reaction. Sometimes the reaction conditions (temperature, pH, catalyst, etc.) are put on top or below the arrow. A chemical equation gives information on the reactants, the reacting proportions and how many units of each reactant and product are involved. For example, $xA + yB \rightarrow mC + rD$, where x , y , m and r show the relative numbers of molecules reacting and are called **stoichiometric coefficients**. An example is the dissociation of salt into sodium and chloride ions: $\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$.

CHEMICAL EQUILIBRIUM: The condition in which the rates of forward and reverse reactions are equal and there is no net change in concentration of all the species involved. In a reversible reaction, this can be denoted, for example, by $3\text{H}_2 + \text{N}_2 \rightleftharpoons 2\text{NH}_3$.

CHEMICAL FERTILISER (SYNTHETIC FERTILISER): Any inorganic synthetic material that is added to the soil to sustain plant growth. Examples include ammonium nitrate, ammonium phosphate, superphosphate, and potassium sulfate. Although they make plants grow very well, they have some disadvantages that include the following: most chemical fertilisers do not contain micronutrients; they do not support microbiological life in the soil; they do not add organic content to the soil; etc.

CHEMICAL FORMULA: A symbolic representation of the composition of a molecule. It identifies each constituent element by its chemical symbol and indicates the number of atoms of each element found in each discrete molecule of that compound. If a molecule contains more than one atom of a particular element, this quantity is indicated using a subscript after the chemical symbol. An example is the hexane molecule with 6 carbon atoms and 14 hydrogen atoms, written as C_6H_{14} .

CHEMICAL FOSSIL: Any of various organic compounds found in ancient geological strata that appear to be biological in origin and are assumed to indicate that life existed when the rocks were formed.

CHEMICAL FUEL: A substance that requires an oxidiser in order to achieve combustion and develop thrust. Chemical fuels include those used by liquid-propellant rocket engines and solid-propellant rocket motors, jets and internal-combustion engines. They contrast with nuclear fuel.

CHEMICAL FUNCTIONALITY: Ability of functional groups present within a polymer or polymer network to participate in chemical reactions. The chemical functionality of a network formed by a sol-gel process from a precursor such as $(\text{RO})_3\text{Si}-\text{CH}=\text{CH}_2$ is that of the vinyl group.

CHEMICAL FUSOGEN: Any chemical that is used in the fusing of two cells or protoplast. Examples are potassium chloride, sodium chloride, potassium nitrate, sodium nitrate and polyvinyl alcohol.

CHEMICAL HYGIENE PLAN (CHP): A written program developed and implemented by an employer designating procedures, equipment, personal protective equipment and work practices that are capable of protecting employees from the health hazards presented by hazardous chemicals used in that particular workplace.

CHEMICAL INDUCTION (COUPLING): The acceleration of one reaction by another in a chemical system such as oxidation-reduction reactions, chain reactions and biological reactions. It is due to an intermediate or product of the inducing reaction participating in the second reaction.

CHEMICAL IONISATION: The process in which new ionisation species are formed when gaseous molecules interact with ions. The process may involve transfer of an electron, proton or other charged species between the reactants. When a positive ion results from chemical ionisation, the term may be used without qualification. When a negative ion is involved, it is called **negative ion chemical ionisation**. Negative ion formation by attachment of a free electron is not the same as negative ion chemical ionisation. Chemical ionisation is not the same as chemi-ionisation. **Chemi-ionisation** is ionisation that occurs as a result of the collision of a particle with a neutral species, usually excited, such as a metastable atom.

CHEMICAL ISOTOPE EXCHANGE: Exchange of isotopes between different types of molecules or ions in the course of a chemical reaction.

CHEMICAL KINETICS (REACTION KINETICS): The area of chemistry concerned with the speed or rate at which chemical reaction occurs. It involves the change in concentration of a reactant or a product with time; and includes investigations of how different experimental conditions can influence the rate of a chemical reaction; provide information on reaction mechanisms as well as the construction of mathematical models that can describe the characteristics of a chemical reaction.

CHEMICAL LASER: A continuous wave or pulsed laser in which the excitation and population inversion of the emitting species result from a chemical reaction. Typical examples are HF and DF lasers emitting many lines in the IR region.

CHEMICAL MEASUREMENT PROCESS (CMP): Analytical method of defined structure that has been brought into a state of statistical control, such that its imprecision and bias are fixed, given the measurement conditions. This is a prerequisite for the evaluation of the performing characteristics of the method or the development of meaningful uncertainty statements concerning analytical results.

CHEMICAL MODIFICATION: A process by which at least one feature of the chemical constitution of a polymer is changed by chemical reaction(s).

CHEMICAL NOMENCLATURE: A system of naming chemical substances, devised by Antoine Lavoisier and Claude Louis Berthollet and others in 1787. A set of rules adopted by the International Union of Pure and Applied Chemistry (IUPAC) is the basis for a standardised name for any organic compound.

CHEMICAL OXYGEN DEMAND (COD): A measure of the amount of oxygen, divided by the volume of the system, required to oxidise the organic and inorganic matter in waste water using a chemically oxidising agent. In practice, it is usually expressed in milligrams O_2 per litre.

CHEMICAL PERIODICITY: The change in properties of elements with their position in the Periodic Table. The Periodic Table classifies the chemical elements by means of the periodicity of their chemical properties.

CHEMICAL POTENTIAL: The change in Gibbs free energy (thermodynamic property which determines whether a chemical reaction is spontaneous or not) with respect to a change in the amount of the component, with pressure, temperature and amounts of other components being constant. It was first described by the American engineer, chemist and mathematical physicist Josiah Willard Gibbs (1839-1903). Its symbol is μ_g .

CHEMICAL PROPERTY: Any property of a substance that becomes evident during a chemical reaction or any one of the chemical changes (chemical reaction) that can happen to a substance. For example, the chemical property of sulfuric acid is that it can neutralise bases. In determining the chemical properties of a substance, the substance's internal structure must be affected for it to be investigated.

CHEMICAL REACTION: A process in which one or more substances may be transformed into one or more new substances. Energy is given out or taken in during a chemical reaction but the masses of the reactants and products in a reaction are in a fixed ratio to each other.

CHEMICAL REACTION EQUATION: A symbolic representation of a chemical reaction, in which the reactant entities are given on the left hand side and the product entities on the right hand side. The coefficients next to the symbols and formulae of entities are the absolute values of the stoichiometric numbers. Different symbols are used to connect the reactants and products with the following meanings: $\equiv$ for a stoichiometric relation; $\rightarrow$ for a netforward reaction; $\rightleftharpoons$ for a reaction in both directions; and $\rightleftharpoons$ for equilibrium.

CHEMICAL RELAXATION: If the equilibrium mixture of a chemical reaction is disturbed by a sudden change, especially of some external parameter such as temperature, pressure or electrical field strength, the system will readjust itself to a new position of the chemical equilibrium or return to the original position, if the perturbation is temporary. The readjustment is known as chemical relaxation.

CHEMICAL ROCK: Sedimentary rock that forms when minerals that were dissolved in a solution or colloidal suspension crystallise. Chemical rocks frequently possess crystalline texture.

CHEMICAL SHIFT: The position of a signal resulting from a change in the absorption or emission wavelength in a process in which there is a nuclear charge relative to a certain standard peak. In nuclear magnetic resonance (NMR), chemical shift represents the magnetic field strength at which absorption occurs. It is highly dependent on the magnetic environment of each absorbing nucleus for example, ^1H , ^{13}C , ^{15}N . The magnetic environment depends on two factors, namely, magnetic field generated by circulation electrons of the nucleus and magnetic fields that result from nearby magnetic nuclei. The size of the chemical shift is given with respect to a reference frequency or reference sample, usually tetramethylsilane (TMS).

CHEMICAL SPECIES: Any set of chemically identical molecular entities, the members of which have the same composition and can explore the same set of molecular energy levels within the time scale of a particular experiment. Two conformational isomers may interconvert sufficiently slowly to be considered separate chemical species on an NMR time scale, but in a slow chemical reaction they may interconvert sufficiently quickly to be considered as a single chemical species.

CHEMICAL STRUCTURE: The arrangement of atoms and the chemical bonds between the atoms in a molecule (or in an ion or radical with multiple atoms), specifically which atoms are chemically bonded to what other atoms with what kind of chemical bond, whether ionic, covalent, dative; single or multiple bonds, together with any information to the spatial arrangement of the atoms in the molecule that is needed to uniquely identify the type of molecule. Its determination includes specifying the molecular geometry and where possible and necessary, the electronic structure of the target molecule or other solid.

CHEMICAL SUBSTANCES: Matter which can be prepared and shows its own distinct chemical property and composition. An example is pure water; it has the same properties and the same ratio of hydrogen to oxygen whether it is isolated from a river or made in a laboratory. Chemical substances exist as a solid, liquid, gas or plasma and may change between these phases of matter with changes in temperature or pressure. Some typical chemical substances are diamond, gold, sodium chloride (salt) and sucrose (sugar).

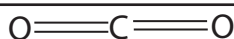


Fig. I: Carbon dioxide, CO_2

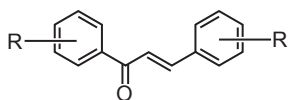


Fig. II: chalcone, $(\text{C}_{15}\text{H}_{12}\text{O})$

Examples of structure of chemical compounds

CHEMICAL SYMBOL: An abbreviation for the name of an element. They are used for writing the formulae for compounds and chemical equations, for example, sodium (Na), carbon (C), silver (Ag), tungsten (W) and potassium (K).

CHEMICAL VAPOUR GENERATION: A system in which the analyte is separated from the sample matrix by the generation of gaseous species as a result of a chemical reaction.

CHEMICAL WARFARE: The use of poisonous agents and other harmful chemicals in war. Chemical weapons are inert agents that come in four categories: pulmonary/choking agents such as chlorine, carbonyl chloride and diphosgene; blister agents such as sulfur mustard, nitrogen and mustard; blood agents such as cyanogen chloride and hydrogen cyanide; and nerve agents such as cyclosarin, sarin, soman and tabun.

CHEMICAL WEATHERING: A form of weathering brought about by a chemical change in the rocks affected. It involves the chemical breakdown of the minerals within a rock and usually produces a more stable residue. Examples of the chemical reactions involved include oxidation, hydrolysis, ion-exchange and dissolution.

CHEMICAL YIELD: The fraction of the amount of an element or chemical compound following a specified chemical reaction or separation.

CHEMICALLY BONDED HYBRID: Hybrid material in which the different components are bonded to each other by covalent or partially covalent bonds.

CHEMICALLY INDUCED DYNAMIC ELECTRON POLARIZATION (CIDEP): Non-Boltzmann electron spin- state population produced in thermal or photochemical reactions, either from a combination of radical pairs known as radical-pair mechanism or directly from the triplet state known as triplet mechanism and detectable by ESR spectroscopy.

CHEMICALLY INDUCED DYNAMIC NUCLEAR POLARIZATION (CIDNP): A mechanism enabling nuclear spin to influence the direction of a chemical reaction. This can occur in certain cases, even through the gap between energy levels of nuclear spin states in a magnetic field very much smaller than the dissociation energies of chemical bonds.

CHEMICALLY INITIATED ELECTRON EXCHANGE LUMINESCENCE (CIEEL; CATALYSED/CATALYZED CHEMILUMINESCENCE): A type of luminescence resulting from a thermal electron-transfer reaction.

CHEMIEXCITATION: Generation, by a chemical reaction, of electronically excited molecular entities from reactants in their ground electronic states.

CHEMILUMINESCENCE (CHEMOLUMINESCENCE): The emission of light (ultraviolet, visible or infrared) with limited emission of heat (luminescence), when a chemical reaction produces molecules with electrons in excited energy levels that can then radiate light. An example is yellow phosphorus oxidising in air, emitting green luminescence. Many organisms, such as bacteria and fish, make light this way by manufacturing substances called **luciferase** and **luciferin**.

CHEMILUMINESCENCE ANALYSER: An instrument consisting of a reaction chamber with separate inlets for the sample and reagent gas, an optical filter, a photomultiplier and a signal processing device. The reactive gas is introduced in excess. The quantity of light produced is proportional to the sample flow rate and the concentration of the measured substance in the sample under specified temperature and pressure conditions. The filter limits the wavelength to the region of interest and helps to eliminate interferences.

CHEMILUMINESCENT METHODS OF DETECTION: Methods which utilise gas or liquid phase reactions between two molecules which produce a third molecule in an excited state. The wavelength distribution of the light emission from the excited molecule is characteristic of the species; in some circumstances this may be

used to identify and to determine the concentration of one of the reacting species.

CHEMIOSMOTIC COUPLING: The coupling of adenosine triphosphate (ATP) synthesis to an electrochemical potential gradient in mitochondria and chloroplasts.

CHEMIOSMOTIC THEORY: A theory propounded by Peter Mitchell to explain the formation of adenosine triphosphate (ATP) in the mitochondrial electron transport chain. As electrons are transferred along the electron carrier system in the inner mitochondrial membrane, hydrogen ions (protons) are actively transported into the space between the inner and outer mitochondrial membrane which thus contains a higher concentration of protons than the matrix. This creates an electrochemical gradient across the inner membrane, down special channels associated with adenosine triphosphate (ATP) synthase, the enzyme that catalyses the conversion of adenosine diphosphate (ADP) to ATP and is coupled with the phosphorylation of ADP.

CHEMISORPTION (CHEMICAL ADSORPTION): A type of adsorption in which a chemical reaction occurs at the exposed surface. A new chemical species such as corrosion, metallic oxidation is generated at the adsorbant surface. The strong interaction between the adsorbate and the substrate surface creates new types of electronic bonds, ionic or covalent, depending on the reactive chemical species involved. Common absorbing liquids are water, aqueous amine solutions and sodium carbonate.

CHEMIST: 1. A scientist who specialises in chemistry. Chemists study the composition of matter and its properties such as density and acidity, substance proportions, reaction rates and other chemical properties including level of molecules and their component atoms including details for measuring, weighing, etc. Chemists use this knowledge to learn the composition and properties of unfamiliar substances, as well as to reproduce and synthesise large quantities of useful naturally occurring substances and create new artificial substances such as synthetic fibres. They develop drugs and cosmetics and processes, including oil refining and petrochemical processing that reduce energy use and pollution. Chemists may specialise in any number of sub disciplines of chemistry such as analytical organic, inorganic, physical and theoretical, macromolecular, medical and materials chemistry. 2. A person who prepares and sells medicines and usually also sells cosmetics, toiletries, etc.

CHEMISTRY: A science concerned with the study of matter, its composition such as gas, liquid or solid and of the changes that take place in them under certain conditions or the study of the elements and the compounds they form. The various sub disciplines include **analytical chemistry**, which is the analysis of material samples to gain an understanding of their chemical composition and structure; **biochemistry**, the study of the chemicals, chemical reactions and chemical interactions that take place in living organisms and also associated with molecular biology and genetics; **inorganic chemistry**, the study of the properties and reactions of inorganic compounds; **materials chemistry** is the preparation, characterisation and understanding of substances with a useful function. Some others include neurochemistry, nuclear chemistry organic chemistry, physical chemistry, theoretical chemistry, agrochemistry, astrochemistry and cosmochemistry, atmospheric chemistry, chemical engineering, chemical biology, etc.

CHEMOAUTOTROPH: An organism such as bacterium or protozoan, which obtains nourishment through the oxidation of an inorganic substance, rather than through photosynthesis.

CHEMOHETEROTROPH: An organism that obtains the energy required for the synthesis of organic molecules from the oxidation of organic compounds. All heterotrophs, with the exception of the photoheterotrophs, are chemoheterotrophs.

CHEMOINFORMATICS: The branch of chemistry representing molecules and reactions and with the design of databases for the storage of chemical information.

CHEMOKINE: Any of a large family of small proteins (cytokines), secreted by many types of cells, that attract and guide white blood cells to sites of infection or in other settings, such as lymphocyte development and migration in lymph nodes. They are crucial in innate immunity during initial response to injury or pathogen invasion and influence lymphocyte behaviour in adaptive immune responses. Chemokines are found in all vertebrates, some viruses and some bacteria, but none have been described for other invertebrates.

CHEMOLITHOTROPH: Any bacterium that is able to obtain its energy by the oxidation of inorganic compound, including those of iron, nitrogen and sulfur.

CHEMOMETRICS: The application of statistics to the analysis of chemical data from organic, analytical or medicinal chemistry and design of chemical experiments and simulations.

CHEMOORGANOTROPH: An organism that obtains its energy by oxidation of organic compounds such as glucose and acetate. All animals, fungi, protozoa and some bacteria are chemoorganotrophs.

CHEMORECEPTOR (CHEMOSENSOR): Any of the special cells or organs which are stimulated by chemical substances. There are two main classes: (i) direct chemoreceptors, which include taste buds in the gustatory system and carotid bodies and aortic bodies, which detect changes in the oxygen, carbon dioxide and hydrogen ion concentration in the blood; and (ii) Distance chemoreceptors include olfactory receptor neurons in the olfactory system and neurons in the vomeronasal organ that detect pheromones.

CHEMOSELECTIVITY (CHEMOSELECTIVE): The preferential reaction of a chemical reagent with one of two or more different functional groups. A reagent has a high chemoselectivity if reaction occurs with only a limited number of different functional groups. For example, sodium tetrahydroborate is a more chemoselective reducing agent than lithium tetrahydroaluminate. The concept has not been defined in more quantitative terms. The term is also applied to reacting molecules or intermediates which exhibit selectivity towards chemically different reagents.

CHEMOSTAT: Is a bioreactor to which fresh medium is continuously added, while culture liquid containing left over nutrients, metabolic end products and microorganisms are continuously removed at the same rate to keep the culture volume constant.

CHEMOSYNTHESIS: The biological conversion of one or more carbon-containing molecules and nutrients into organic matter using chemical energy from oxidation of inorganic compounds such as hydrogen gas, hydrogen sulfide and methane, rather than sunlight, as in photosynthesis.

CHEMOSTAT: A bioreactor in which constant growth conditions for microorganisms are maintained over prolonged periods of time by supplying the reactor with a continuous input of nutrients and continuous removal of medium.

CHEMOSTERILANT: Any chemical used to sterilise pests and hence reduce the number of offspring in the subsequent generation. A common technique is to regularly release large numbers of laboratory-reared sterile males into the wild population, which compete with normal males. The method has been used successfully to control various fruit flies.

CHEMOSYNTHESIS: A method of synthesising compounds, using the energy from chemical reactions, rather than the sunlight used for energy in photosynthesis. The process is used by certain bacteria which can synthesise organic compounds from carbon dioxide using the energy from oxidation of compounds. Chemosynthetic autotrophs are a necessary part of the nitrogen cycle.

CHEMOTAXIGEN: Any substance that facilitates the formation of an agent that is chemotactic for cells.

CHEMOTAXIS: A movement in response to a chemical stimulus by somatic cells, bacteria and other single-cell or multicellular organisms. For example, bacteria exhibit positive chemotaxis, when they move towards higher concentration of nutrients. Conversely, negative chemotaxis occurs if the contained in G_k .

CHEMOTAXONOMY: The biological classification of plants and animals based on similarities and differences in biochemical composition.

CHEMOTROPISM: The growth or orientation of a cell or an organism in response to chemical stimulus from outside. It is called positive chemotropism if the movement is toward the chemical; and negative chemotropism if the movement is away from the chemical. An example of chemotropism is the growth of a pollen tube down the style during fertilisation in response to the presence of sugars in the style.

CHERENKOV LIGHT: Blue light that is emitted when charged particles travel through a medium faster than light travels through that medium.

CHERNOZEM: Russian for black **soil/earth**, it is an order of dark fertile grassland soil, rich in organic matter, 7% - 15% humus with a high agricultural yield, found in the temperate plains. This order of soil has four different surface colours: brown, dark brown, black, and dark grey that reflect the soil organic matter content: the greater the organic matter, the darker the surface colour. Chernozemic soil has a large depth, often more than 1 metre. Such soils are very fertile over a long period of time, and used extensively for growing cereals or for raising livestock.

CHI-SQUARE DISTRIBUTION (χ^2 DISTRIBUTION): A probability density function that gives the distribution of the sum of the squares of a number of independent random variables, each with a normal distribution with zero mean and unit variance that has the property that the sum of two or more random variables with such a distribution also has one. It is used in testing statistical hypotheses especially about the theoretical and observed values of a quantity and about population variances and standard deviations.

CHI-SQUARE STATISTICS (χ^2 STATISTICS): A measurement of how expectations compare to results. It shows any discrepancies between the expected results and the actual results. The data used in calculating a chi square statistic must be random, raw, mutually exclusive, drawn from independent variables and drawn from a large enough sample.

CHI-SQUARED TEST (χ^2 TEST): A statistical test for determining whether data obtained by sampling correspond with those predicted hypothetically and hence to test the validity of the hypothesis.

CHIASMA (CHIASMATA, plural): The cross-shaped connection produced by the crossing over of paired chromosomes during the first metaphase of meiosis. 2. The crossing over of two parts or structures, such as the fibres of the optic nerves at the point below the brain.

CHICKEN (*Gallus gallus domesticus*): A type of domesticated fowl, a subspecies of the red jungle fowl *G. gallus*. It is one of the most common and widespread domestic animals in world, raised worldwide for its meat and eggs. Chickens averagely have a height of less than 70 cm and weigh approximately 2.6 kg. A chicken has a wattle beneath its beak and a comb on top of the head, but these are more pronounced in males than females. It has a high-arched tail. Chickens are omnivores and feed on plants, seeds, insects, and also larger prey such as small mice and lizards and are commonly seen scratching the ground looking for food. Chickens are very sociable birds and live together in flocks. They reproduce through eggs. Large commercial farms breed them in large numbers.

CHICKEN POX (VARICELLA): A highly infectious viral disease caused by the herpes zoster virus. It mainly affects children, adolescents, adults and people with weak immune system. Its symptoms include blister-like rash, itching, fatigue and fever. It is spread by direct contact with people or through the air by coughing or sneezing.

CHIEF CELL PEPTIC CELL (ZYMOGENIC CELL): A cell that lines the lumen of the gastric glands of the stomach and secretes the enzymes pepsin and rennin.

CHILDBIRTH: The expulsion of a baby from its mother's uterus following pregnancy. It is the period of time involving labour and delivery of the baby, including the effort and pain involved. The process of normal human childbirth is categorised in three stages of labour: the shortening and dilation of the cervix, descent and birth of the infant and birth of the placenta.

CHILDHOOD: The stages of life of a newly born human baby before the beginning of sexual maturity. Stages of childhood include infancy (birth to 2 years old), early childhood (3 to 8 years old), middle childhood (9 to 11 years old), and adolescence (12 to 18 years old).

CHILL HARDENING: A technique used in casting whereby the mould is chilled. This accelerates the cooling of the molten metal poured into it, thereby increasing its surface hardness.

CHILL HAZE: A term in brewing indicating precipitation that occurs when beer is stored at cold temperatures. It consists of proteins that can be broken down by using protease enzyme. Chill haze appears at 0°C (273.15 K) and disappears at 20°C (293.15K); permanent haze remains at 20°C (293.15K) but there is no fundamental difference.

CHILLED BEAM: A type of heating, ventilating and air-conditioning (HVAC) system designed to heat or cool large buildings which acts as a radiator chilled by recirculated water. It uses natural convection principles (warm air rises, cool air falls) and the superior energy capacity of water to make cooling much more economical. Warm air rises to the chilled beam, which is located in the ceiling and passes through a heat exchange coil where it cools and then falls to the floor, where the cycle starts over. There are two main types: active and passive chilled beams. In an active beam, air duct and nozzle blowers provide ventilation air, which helps draw the warm room air up through the heat exchange coils where it cools and then falls to the floor, where the cycle starts over. In passive beam, chilled beams require ventilation air to be delivered by a separate air-handling system and the air is drawn to the heat exchanger through the natural rise of warm air and does not need fans to push air through.

CHILLED CEILING: A type of heating, ventilating and air-conditioning (HVAC) system that uses radiation rather than convection as used by chilled beams for heating and cooling. It consists of a metal sheet with water pipes running above it. The pipes heat or cool the metal panel, which then radiates the energy to the building.

CHILLI (CHILLI PEPPER; CHILE; *Capsicum annuum*): Plants from the genus *Capsicum* and members of the nightshade family, *Solanaceae*. The term also applies to the small seed pod or fruit produced by various types of capsicum pepper plant, which may be green (unripe fruit), red, yellow in colour with a very hot taste. The plant is a large perennial shrub with a densely branched stem which grows up to 0.6 metre. The solitary flowers produced are purplish or off-white. The flowers develop into fruits (berries) which may be red, green or yellow when ripe. It is used as flavouring sauces and relishes, as topical (applied directly) analgesic for pain. Red chillies contain high amounts of vitamin C and carotene or provitamin A. Yellow and especially green chillies contain a considerably lower amount of both substances. They also contain vitamins B and vitamin B₆ in particular; and very high in potassium, magnesium and iron. The high vitamin C content can also substantially increase the uptake of non-haeme iron from other ingredients in a meal, such as beans and grains. The term also applies to any of the five cultivated species and twenty-six wild species in the genus *Capsicum*, native to tropical areas.

CHILLING-SENSITIVE PLANT: A plant that is badly injured or killed by low but non-freezing temperatures, up to about 20 °C. This causes injury and may even be fatal to some tropical and subtropical plants such as cotton, soybeans, cucumbers, squash and some vegetable

species. For example, when plants such as rice, velvet beans, cotton and peanut are exposed to a temperature of between 1 °C and 5 °C for 24 to 36 hours it is injurious or fatal to them.

CHILOPODA: A class of wormlike terrestrial arthropods belonging to the phylum Myriapoda comprising centipedes, found in all kinds of habitats, on all continents except Antarctica. Centipedes can have between 30 to over 300 legs, depending on the species. All centipedes are predators and feed on soft-bodied insects, spiders, other centipedes and worms. Larger species will attack small mice, frogs, toads, birds, lizards and snakes. Animals are killed by venom injected by the fangs, then grasped by the fangs by using the first several pairs of walking legs. A few species may eat plant materials if they cannot find animal prey.

CHIMERA (CHAEMERA): An organism organ or part composed of two or more tissues that are genetically different. This condition is either inherited or acquired through organ transplant, blood transfusion or genetic engineering.

CHIMERIC DNA: Recombinant DNA, the components of which originate from two or more sources.

CHIMNEY EFFECT: A vertical movement of a localised mass of air or other gases which occurs due to local temperature differences.

CHIMPANZEE: A highly social anthropoid ape, *Pan troglodytes* classified in the phylum Chordata, subphylum Vertebrata, class Mammalia order Primates, family Pongidae. It inhabits the rain forests and woodland savannas of equatorial Africa living in groups of 15-100 or more members. It also includes the much smaller bonobo, *Pan paniscus*. It is the closest living relative to humans, exhibiting human-like behaviour and a high degree of intelligence including problem solving and the use of tools. It has long dark hair and a bare face. It is about 1-1.7 metres tall when standing upright and weigh between 32-60 kg. It eats mostly fruits, berries, leaves and seeds, some termites and ants and also hunts animals such as monkeys, a young baboon or bush pig. It can live up to 45 years in the wild and more than 50 years in captivity.

CHINA CLAY (KAOLIN): A complex aluminosilicate compound composed chiefly of the mineral kaolinite and formed by the decomposition of feldspar in granite rock. It is fine white clay, used as porcelain, ceramics, medicine and as filler in, for example, paper making.

CHINESE MEDICINE (TRADITIONAL CHINESE MEDICINE): A medical system that was developed in China more than 2,000 years based on tradition and used to prevent, diagnose, and treat diseases. It includes acupuncture, diet, herbal therapy, meditation, physical exercise, and massage.

CHINESE REMAINDER THEOREM: In number theory, the basic result that if a set of integers, m_i , are pairwise relatively prime, then the congruences $x \equiv a_i \pmod{m_i}$ have a unique solution for x , modulo the product of all the m_i . In other words, the Chinese remainder theorem determines a number n that, when divided by some given divisors, leaves given remainders.

CHIP (SILICON CHIP, INTEGRATED CIRCUIT, IC; MICROCIRCUIT; MICROCHIP): A complete electronic circuit on a slice of silicon or other semiconductor crystal only a few millimetres square. Integrated circuits are used in almost all electronic equipment in use today including computers, cell phones and all digital appliances.

CHIPSET: A group of integrated circuits that work together. It may refer to the design of a single component or may describe the relationship of multiple components within a computer system. For example, the chipset of video card describes the design of the card, while a motherboard chipset describes its layout and the different components it supports.

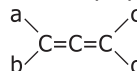
CHIRAL COMPOUND: A compound that has no internal plane of symmetry and cannot be superimposed on its mirror image. For a

molecule to be chiral it must have an asymmetric centre: at least one carbon atom should be bonded to four different substituents. A chiral molecule and its mirror image are enantiomers.

CHIRAL SELECTOR: The chiral component of the separation system capable of interacting enantioselectively with the enantiomers to be separated.

CHIRAL STATIONARY PHASE: A stationary phase which incorporates a chiral selector. If not a constituent of the stationary phase as a whole, the chiral selector can be chemically bonded to (chiral bonded stationary phase) or immobilised onto the surface of a solid support or column wall (chiral coated stationary phase) or simply dissolved in the liquid stationary phase.

CHIRALITY: 1. In chemistry, the handedness of an asymmetric molecule or the property of existing in left and right handed structural form. 2. In physics, the property of an object that cannot be superimposed upon its mirror image by rotations and translations alone. If the object is superposable on its mirror image the object is described as being achiral. Another definition is the characteristic of particles of spin $\frac{1}{2}$ that are allowed to have only one spin state with respect to an axis of quantisation parallel to the particle's momentum. If the particle's spin is always parallel to its momentum, it has positive chirality. If it is antiparallel, it has negative chirality.

CHIRALITY AXIS (AXIS OF CHIRALITY): An axis about which a set of ligands is held so that it results in a spatial arrangement which is not superposable on its mirror image. For example, with an allene  the chiral axis is defined by the C=C=C bonds; and with an *ortho*-substituted biphenyl the atoms C-1, C-1', C-4 and C-4' lie on the chiral axis.

CHIRALITY CENTRE (CENTRE OF CHIRALITY): An atom holding a set of ligands in a spatial arrangement which is not superposable on its mirror image. A chirality centre is thus a generalised extension of the concept of the asymmetric carbon atom to central atoms of any element. An example is the central carbon in serine, in which the amino group and hydrogen can rotate about the carbon.

CHIRALITY ELEMENT (ELEMENT OF CHIRALITY): The part of a molecule that makes it exist in left and right-handed forms.

CHIRALITY PLANE: A planar unit connected to an adjacent part of the structure by a bond which results in restricted torsion so that the plane cannot lie in a symmetry plane. For example, with (*E*)-cyclooctene the chirality plane includes the double bond carbon atoms with all four atoms attached to the double bond. With a monosubstituted paracyclophane the chirality plane includes the monosubstituted benzene ring with its three hydrogen atoms and the three other atoms linked to the ring (i.e. from the substituent and the two chains linking the two benzene rings).

CHIRALITY SENSE: The property that distinguishes enantiomorphs. The specification of two enantiomorphous forms by reference to an oriented space, for example, a screw, a right threaded one or a left threaded one. An enantiomorph is either of a pair of crystals, molecules or compounds that are mirror images of each other but are not identical and that rotate the plane of polarised light equally, but in opposite directions.

CHIRON: A minor planet discovered in 1977. It has an orbit of 50-68 years.

CHIROOPTICAL SPECTROSCOPY: Spectroscopy making use of the properties of chiral substances when they interact with polarised light of various wavelengths. It is used in the analysis of the structure of molecules.

CHIROPTERA: An order of flying mammals comprising nearly 1000 bat species. Bats are the only vertebrates apart from birds that are capable of prolonged powered flight. Their membranous wings are supported by very elongated forelimbs and digits and stretch along the

sides of the body to the hind limbs and tail. An important characteristic of bats is their ability to fly and hunt in the dark, using echolocation, although they are not blind. Most species have good eyesight.

CHIROPTEROPHILY: Pollination of plants by bats. Flowers of such plants tend to be large, bell-shaped and showy, white or light coloured, open at night and have strong odours. Bats use smell and echolocation initially to find the flowers and excellent spatial memory is used to visit them repeatedly.

CHIROPTIC (CHIROPTICAL): Any of the optical techniques using refraction, absorption or emission of anisotropic radiation for investigating chiral substances. Examples are measurements of optical rotation at a fixed wavelength, optical rotatory dispersion (ORD), circular dichroism (CD) and circular polarisation of luminescence (CPL).

CHIROTOPIC: The description of an atom (or point, group, face, etc. in a molecular model) that resides within a chiral environment. One that resides within an achiral environment has been called achirotopic.

CHISEL: A tool with a sharp cutting edge at the end for shaping wood, stone or metal.

CHITIN: A nitrogen-containing polysaccharide widely found in invertebrates. It forms the exoskeleton of arthropods, crustaceans and a component of fungal walls. It has widespread industrial application such as in purifying waste water, thickening and stabilising food and pharmaceuticals, as ion-exchanger, etc.

CHITINASE: An enzyme that catalyses the hydrolysis of chitin, a major constituent of fungal cell walls and of the exoskeleton of insects and other arthropods. It is synthesised by certain insect-eating animals and certain plants as parts of their defence against fungal infection.

CHITINOPHOSPHATIC: A term used to describe the shells of invertebrates that are composed of chitin impregnated with phosphate salts or alternating layers of chitin and phosphatic salts. The phosphatic layers are usually thickened where they overlie the body cavity. An example of such invertebrate is Brachiopoda.

CHLADNI FIGURES: Stationary wave pattern produced on thin metal or glass plates in which the plates covered in sand are made to vibrate. The sand collects in the nodal lines producing symmetrical patterns. It was invented by Ernst Friedrich Chladni (1756–1827), a German physicist.

CHLAMYDEOUS: Pertaining to or with a floral whorl.

CHLAMYDIA: A single celled bacterium that lives parasitically only in animal cells. The species are responsible for trachoma, the world's leading cause of blindness, pneumonia in infants and certain sexually transmitted diseases.

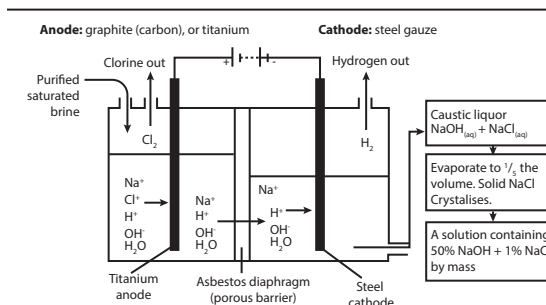
CHLAMYDOBACTERIALES (FILAMENTOUS IRON BACTERIA): An order of colourless rod-shaped gram-negative bacteria that aggregate into filaments, which sometimes become branched. The filaments are surrounded by a sheath of organic material impregnated with ferric and manganese oxides. Such bacteria derive energy from oxidising organic substrates. However they also oxidise ferrous compounds to ferric oxides and probably derive additional energy from these reactions.

CHLAMYDOMONAS: A common oval-shaped single-celled green alga characterised by the presence of two excretory contractile vacuoles and a red eyespot. It lives in fresh water bodies and moist soil.

CHLAMYDOSPORE: A thick-walled asexual spore that is produced by some fungi. It is a resting spore (spore that remains dormant for long periods before germination), capable of withstanding unfavourable growing condition.

CHLOR-ALKALI PROCESS: The production of chlorine gas by the electrolysis of aqueous sodium chloride (NaCl) solution. Depending on the method, several products beside hydrogen can be produced. If the products are separated, chlorine and sodium hydroxide are produced.

By mixing, sodium hypochlorite or sodium chlorate is produced, depending on the temperature. The chlor-Alkali Process, which forms the basis of one of the largest inorganic chemical industries, electrolyzes concentrated aqueous NaCl to produce Cl_2 and several other important chemicals. The chlor-Alkali Process uses an asbestos diaphragm to separate the anode and cathode compartments, electrolysis NaCl brines and yields Cl_2 , H_2 and industrial grade NaOH which is an important base. At the anode, chloride (Cl^-) is oxidised to chlorine (Cl_2). The ion-selective membrane allows the counterion Na^+ to freely flow across, but prevents anions such as hydroxide (OH^-) and chloride from diffusing across. At the cathode, water is reduced to hydroxide and hydrogen gas (H_2). The net process is the electrolysis of an aqueous solution of sodium chloride (NaCl) into useful products such as NaOH and Cl_2 gas.



A diaphragm of Chlor-alkali cell

CHLOR-ZINC IODIDE (CZI; SCHULTZES SOLUTION): A temporary stain containing a mixture of zinc dissolved in hydrochloric acid added to iodine dissolved in potassium iodide. It is used to detect the presence of cellulose in plant tissue, which it turns blue. Lignified walls stain blue-green, lignin and suberin yellow and starch blue-black.

CHLORATE: Salts of trioxochloric(V) acid (HClO_3), for example, calcium chlorate. The chlorate anion has the formula ClO_3^- , ClO_2^- , ClO_3^- or ClO_4^- . Chlorates are powerful oxidising agents and should be kept away from organic compounds. Mixtures of chlorate salts with virtually any combustible material such as sugar, sawdust, charcoal organic solvents and metals will deflagrate.

CHLORAZOL BLACK: A temporary acid stain that colours walls grey or black and is especially suitable for demonstrating pitting. It is also used for chromosome counts, in which case the tissue is first fixed in acetic acid. After staining, a drop of acetocarmine is added and the chromosomes stain a deep reddish-black colour. The stain gives a clear-cut picture and is therefore useful for photomicrography.

CHLORENCHYMA: Parenchyma tissue that contains chloroplasts and is photosynthetic. Chlorenchyma makes up the mesophyll tissue of plant leaves and is also found in the stems of herbaceous plants.

CHLORIC ACID: Any of the oxoacids of chlorine and the formal precursor of chlorate salts. It includes oxochloric(I) acid (HClO); dioxochloric(III) acid (HClO_2); trioxochloric(V) acid (HClO_3) and tetraoxochloric(VII) acid (HClO_4). All these acids are oxidising agents, the oxidising power increasing with the amount of oxygen atoms in the molecule. The acid strength also increases in the same order.

CHLORIC(I) ACID (HYPOCHLOROUS ACID; CHLORIC(I) ACID; CHLORANOL; HYDROXIDOCHLORINE; HYDROGEN HYPOCHLORITE; CHLORINE HYDROXIDE): A weak liquid acid, with the chemical formula HOCl , which becomes stable only when in solution and may be prepared by the reaction of chlorine with an agitated suspension of mercury(II) oxide. It is used as a bleaching agent, an oxidiser, a deodorant and a disinfectant.

CHLORIC(III) ACID (CHLOROUS ACID): A pale-yellow weak acid, with the chemical formula HClO_2 that is known only in solution and formed by the reaction of chlorine dioxide and water. It is an oxidising agent.

CHLORIC(V) ACID (CHLORIC ACID): A colourless unstable liquid, an oxoacid of chlorine and the formal precursor of chlorate salts, with the chemical formula HClO_3 , which decomposes at 40°C (313.15 K). It is both a strong acid and a powerful oxidising agent. Hot solutions of the acid or its salts have been known to detonate in contact with readily oxidised organic material.

CHLORIC(VII) ACID (PERCHLORIC ACID): An unstable liquid acid, which explodes at about 90°C (363.15 K) at atmospheric pressure, with the chemical formula HClO_4 . It is a powerful oxidising agent, dangerously corrosive and readily forms explosive mixtures. It is used for preparing perchlorate salts, especially ammonium perchlorate, an important rocket fuel.

CHLORIDE SECRETORY CELL: Any of the cells in the gills of marine bony fish that remove chloride from the blood and secrete it into the sea, to maintain the osmotic composition of the body.

CHLORIDE SHIFT (HAMBURGER SHIFT): Exchange of chloride (Cl^-) and bicarbonate (HCO_3^-) across the membrane of red blood cells, named after Hartog Jakob Hamburger (1859–1924), a Dutch physiologist.

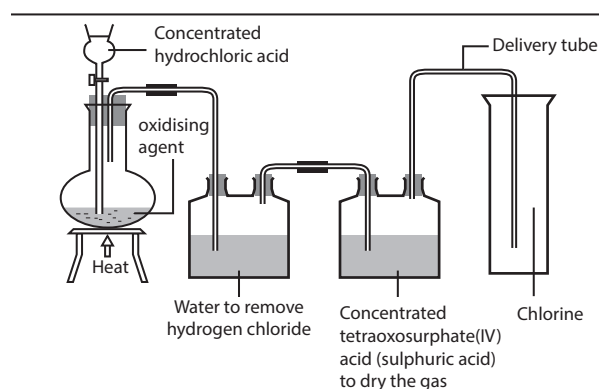
CHLORINATED HYDROCARBON: Any hydrocarbon with any or all of the hydrogen atoms replaced by chlorine atoms. Such compounds are often used as insecticides, examples being aldrin, dieldrin and DDT.

CHLORINATED SOLVENT (ORGANOCHLORIDE; ORGANOCHLORINE; CHLOROCARBON; CHLORINATED HYDROCARBON): Any liquid organic compound such as chloroform that contains two or more chlorine atoms and used as solvent. These compounds are very effective solvents for fats and greases but may be toxic.

CHLORINATING AGENT: A chemical reagent that introduces chlorine atoms into a compound or substitutes chlorine for some of the groups. Typical examples are thionyl chloride (SOCl_2), phosphorus trichloride (PCl_3) and phosphorus pentachloride (PCl_5).

CHLORINATION: 1. A common and effective method for the disinfection of drinking water, swimming pool water and wastewater; the most important process for assuring the bacteriological safety of potable water supplies. Water which has been treated with chlorine is effective in preventing the spread of waterborne diseases. Chlorine can be added directly as chlorine gas or indirectly as sodium-hypochlorite solution. The lower the pH of the water, the more effective chlorine is for disinfection. After the chlorine is added to the water, there must be sufficient contact time for the chlorine to effectively destroy the bacteria. Chlorine is also used to treat waste water. 2. The treatment of a substance with chlorine such as bleaching. 3. The reaction between chlorine or appropriate chlorinating agent and an organic compound to form the corresponding chlorinated compound. Examples are chlorination of benzene to produce chlorobenzene or production of acyl chlorides from carboxylic acid using thionyl chloride.

CHLORINE: A greenish-yellow gaseous non-metallic element with a pungent smell with the symbol Cl , atomic number 17, relative atomic mass 35.455, freezing point -100.98°C (374.13 K), boiling point -34.6°C (238.55 K) and specific gravity 1.56 at -33.6°C (239.55 K). It is a member of the halogen group and is widely distributed in combination with the alkali metals such as sodium chloride (NaCl), potassium chloride (KCl), magnesium chloride (MgCl_2), etc. It occurs in acid compounds such as hydrochloric acid (HCl), perchloric acid (HClO_4), phosphorus chloride (PCl_5), etc. Chlorine gas can be made in various ways; one such method is reacting concentrated hydrochloric acid (HCl), which contains 97% chlorine, with an oxidising agent such as potassium manganate(VII) or Manganese(IV) Oxide (MnO_2). It is an important bleaching agent and germicide, used for disinfection of drinking water, wastewater and swimming pool water.



Preparation of chlorine from concentrated hydrochloric acid with oxidising agents

CHLORINE DIOXIDE (CHLORINE(IV) OXIDE): A yellowish-red explosive gas, with the chemical formula ClO_2 , melting point -59°C (214.15 K) and boiling point 11°C (284.15 K). It is soluble in cold water but decomposes in hot water to give chlorine and oxygen. It is one of the oxides of chlorine. It is a potent and useful oxidising agent. Chlorine dioxide is widely used as bleach and in water purification.

CHLORITE: A layer silicate mineral, normally green or white in colour, that is similar to the mica structure and crystallises in the monoclinic system. Its chemical formula is $(\text{MgAlFe})_{12}(\text{SiAl})_8\text{O}_{20}(\text{OH})_{16}$. Chlorite is composed of complex silicates of aluminium, magnesium and iron.

CHLOROBENZENE: An aromatic organic compound, colourless and highly flammable, with the chemical formula $\text{C}_6\text{H}_5\text{Cl}$, melting point -45°C (228 K) and boiling point 131°C (404 K). It is prepared by the direct chlorination of benzene using a halogen carrier such as iron or FeCl_3 . It is used mainly as industrial solvent and as an intermediate in the production of some herbicides, dyestuffs and rubber.

CHLOROCARBONS: Compounds consisting wholly of chlorine and carbon. Examples of chlorocarbons are carbon tetrachloride and tetrachloroethylene.

CHLOROCOCCALES: An order of the Chlorophyceae containing nonmotile unicellular and colonial algae. Individual specimens are sometimes found in soil, but mostly in fresh and marine waters. Motility is restricted to the gametes and zoospores. Sexual reproduction is isogamous or anisogamous. Asexual reproduction is by motile zoospores or nonmotile aplanospores (autospores).

CHLOROCHUORIN: A greenish iron-containing respiratory pigment that occurs in the blood of polychaeta worms. It has a giant macromolecule structure and therefore floats freely in blood. Its affinity for oxygen is weaker than that of most haemoglobin.

CHLOROETHANE (MONOCHLOROETHANE; ETHYL CHLORIDE; ECHLORENE; MURIATIC ETHER): A colourless flammable gaseous compound, with the chemical formula $\text{C}_2\text{H}_5\text{Cl}$, melting point -139°C (134 K) and boiling point 12.3°C (285.4 K). It is made by the reaction of ethane and hydrogen chloride over an aluminium chloride catalyst at temperatures ranging from 130°C - 250°C (403.15 K - 523.15 K). It is used in making lead tetraethyl for petrol. Its use in petrol has been banned in most parts of the world due to the negative health effect of lead on man and other organisms.

CHLOROETHANOIC ACIDS (CHLOROACETIC ACIDS): Three organic compounds, comprising of monochloroethanoic acid (CH_2ClCOOH), dichloroethanoic acid (CHCl_2COOH) and trichloroethanoic acid (CCl_3COOH), in which hydrogen atoms in the methyl group of ethanoic acids have been replaced by chlorine atoms. The acid strength in these acids increases with the number of chlorine atoms in the compound.

CHLOROETHENE (VINYL CHLORIDE): A colourless, flammable, toxic, gaseous compound belonging to the family of organic compounds of halogens, with the chemical formula CHClCH_2 , melting point -153.8°C (119.4 K) and boiling point -13.4°C (259.8 K). It

is made by reacting ethylene or acetylene with hydrochloric acid. It is produced in very large quantities and used principally in making polyvinylchloride (PVC).

CHLOROFLUOROCARBON (CFC): Synthetic chemicals consisting of chlorine, fluorine, carbon and hydrogen that have been used as propellants in aerosol cans, as refrigerants in refrigerators, air conditioners and in the manufacture of foam packing. They were sold under the trade name FREON. CFCs are odourless, non-toxic, non-flammable and chemically inert. They have been implicated in the destruction of the ozone layer, so most developed countries have stopped CFC production.

CHLOROFORM (FORMYL TRICHLORIDE; METHANE TRICHLORIDE; METHYL TRICHLORIDE; TRICHLOROMETHANE; METHENYL TRICHLORIDE): A colourless volatile sweet-smelling liquid haloform with the chemical formula CHCl_3 , relative density 1.48, melting point -63.5°C (209.5 K) and boiling point 61.7°C (334.7 K). It can be made by chlorination of methane followed by separation of the mixture of products or by the haloform reaction. It is an effective anaesthetic but can cause liver damage and it has now been replaced by other halogenated hydrocarbons. It is used as a solvent and raw material for making other compounds.

CHLOROFORM (TRICHLORO METHANE): An odourless, toxic carcinogenic liquid, with the chemical formula CHCl_3 , melting point -63.5°C (210 K) and boiling point 61.2°C (334 K). It has a characteristic pungent, sickly sweet smell and taste. It is produced by reaction of chlorine with ethanol and by the reduction of carbon tetrachloride with moist iron. It dissolves freely in ethanol and ether but does not mix with water. When exposed to sunlight and air it reacts slowly to form phosgene (COCl_2), a very poisonous gas. It was once used as a general anaesthetic in surgery but has been replaced by less toxic, safer anaesthetics, such as ether. It is employed as a solvent and in the synthesis of organic chemical compounds.

CHLOROMETHANE (METHYL CHLORIDE; R-40; HCC 40): A colourless flammable gas, with the chemical formula CH_3Cl , relative density 2.22, melting point -97.7°C (176 K) and boiling point -24.2°C (249 K). It is a haloalkene made by direct chlorination of methane and used as a local anaesthetic and once used as a refrigerant. Due to concerns about its toxicity, it is no longer present in consumer products.

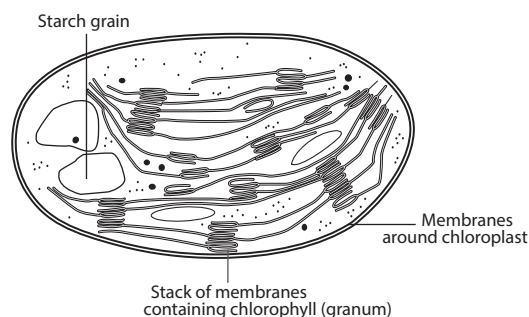
CHLOROPHENOL: One of the several substitution products of chlorination of phenol ($\text{C}_6\text{H}_5\text{Cl}$). This compound and other chlorinated phenols are acidic and are used in antiseptics, disinfectants, herbicides, insecticides, etc.

CHLOROPHYCEAE: A class of the Chlorophyta. In some classifications it contains all the orders of the Chlorophyta except the Charales. Members may be unicellular, colonial or filamentous. Cells of unicellular and colonial chlorophytes may have two or more flagella. Vegetative reproduction usually takes place by fragmentation or by formation of spores. Asexual reproduction is by flagellated zoospores. The sexual reproduction shows considerable variation in the type and formation of sex cells and it may be isogamous, anisogamous or oogamous.

CHLOROPHYLL: The green pigment located in the chloroplast and found in plants, algae and cyanobacteria. It absorbs energy from sunlight during photosynthesis. It absorbs the red and blue-violet parts of sunlight but reflects the green, thus giving green plants their characteristic colour. Chemically, chlorophyll is an organometallic compound containing magnesium. There are four groups of chlorophylls: a, b, c and d. Chlorophyll a is found in all autotrophic plants, chlorophyll b in the Chlorophyta, Charophyta and land plants and chlorophylls c and d only in certain algae. **Chlorophyllous** means of or containing chlorophyll.

CHLOROPHYTA (GREEN ALGAE): A large phylum of algae, of the kingdom Protista consisting of the photosynthetic organisms commonly known as green algae. The members possess chlorophylls a and b, store food reserves as starch and have cellulose cell walls.

CHLOROPLAST: A membrane-bound organelle unique to plant cells and other eukaryotic organisms containing the green pigment chlorophyll, which is responsible for photosynthesis. Chloroplasts capture light energy to conserve free energy in the form of adenosine triphosphate (ATP) and reduce NADP to NADPH through a complex set of processes called photosynthesis. In higher plants they are usually disc-shaped and on the order of 20 to 100 chloroplasts per leaf cell. Within the chloroplast are a number of membranes called thylakoids. A few extend all the way across the chloroplast. Most are in the form of disc-shaped stacks of membranes called grana, with 40 to 60 grana in each chloroplast. The thylakoids are embedded in a matrix called stroma, which is largely protein.



The structure of a chloroplast

CHLOROPLATINIC ACID (PLATINIC CHLORIDE): A solution of platinum chloride used in platinising glass and ceramics, with the chemical formula H_2PtCl_6 . It is obtained as red-brown deliquescent crystals. It is also a reagent for qualitative analysis of potassium.

CHLOROSIS: A plant condition in which the synthesis of the green pigment chlorophyll is inhibited. It is caused by a deficiency in essential elements such as magnesium, iron or manganese, a lack of light, viral infection or genetic factors. This causes the leaves to have pale green, yellow or pale yellow colourisation.

CHLOROTIC: Yellowing in plants due to deficiency in nitrogen or iron caused by breakdown of chlorophyll.

CHLOROXYBACTERIA (GRASS-GREEN BACTERIA; PROCHLOROPHYTES): Green pigmented cyanobacteria that resemble the chloroplasts. They are found as organelles in green plants and green algae.

CHOKER: 1. A coil of wire with low resistance and high inductance that is used to prevent electric signals and energy from being transmitted along undesired paths or into inappropriate parts of an electric circuit or system. It is also used to smoothen the output of a rectifying circuit. For example, power-supply chokes prevent alternating-current components from entering the electronic equipment; radio-frequency chokes (RFCs) prevent radio-frequency signals from entering audio-frequency circuits. 2. In electricity, it is an inductor that is used in a tube light (fluorescent tube) to ionise the gas in the tube for the tube to produce light by providing a very high voltage initially between the filaments (located at the two ends of the tube light). After the gas in the tube has been ionised the choke provides a low voltage. The operation of a choke is done in conjunction with a device called starter that switches the current through the choke on and off frequently until the gas is ionised. 3. In an internal-combustion engine, it is a valve in the carburettor of a petrol engine that is used to partially shut off the air intake of the carburettor in order to enrich the fuel mixture when the engine is started. 4. To interfere with the respiration by compression or obstruction of the larynx or trachea resulting in difficulty in breathing, swallowing or speaking.

CHOLECALCIFEROL (VITAMIN D; CALCIOL; VITAMIN D₃): A form of vitamin D with a white colour and needle-like crystals. The chemical formula is $\text{C}_{27}\text{H}_{44}\text{O}$ and melting point is 83°C - 86°C (356.15 K-359 K). It is found naturally in fish-liver oils and egg yolks. Cholecalciferol is also produced industrially from lanolin found in sheep's wool for use in vitamin supplements and to fortify foods.

Vitamin D deficiency causes rickets in children and results in bony deformities including bowed long bones. Deficiency in adults leads to the disease osteomalacia, a disease occurring mainly in women, causing softening of the bones and resulting in pain and weakness.

CHOLECYSTECTOMY: In medicine, it is the removal of the gall bladder by conventional surgery or by laparoscopy that is usually done to remove gallstones or to treat gall bladder diseases.

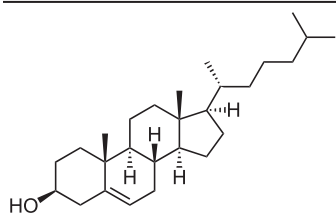
CHOLECYSTOKININ, CCK: Formerly called **pancreozymin**, it is a peptide hormone produced by the duodenal region of a small intestine, that induces the gall bladder to contract and eject bile into the intestine and stimulates the pancreas to secrete its digestive enzymes.

CHOLERA: An infectious and often fatal disease, which causes severe diarrhoea and vomiting. It is caused by the gram-negative, rod-shaped bacterium, *Vibrio cholerae*, which is transmitted by the faecal-oral route.

CHOLESKY DECOMPOSITION (CHOLESKY FACTORISATION): The factorisation of a positive definite matrix A as $LL^* = R^*R$, where L is lower-triangular, R is upper-triangular and L^* and R^* are their respective transposes. The matrix R is sometimes called a Cholesky factor or 'square root' of A and can be computed directly from element-by-element comparison, starting in the first row.

CHOLESTERIC CRYSTAL (CHOLESTERIC LIQUID CRYSTAL): One of the liquid crystal (LC) phases, which can be classified as nematic crystal because only chiral compounds can give rise to it. It was first discovered for cholesterol derivatives, giving it the name. In a nematic phase, the rod-shaped organic molecules have no positional order, but they self-align to have long-range directional order with their long axes roughly parallel. The molecules are therefore free to flow and their centres of mass positions are randomly distributed as in a liquid, but still maintain their long-range directional order. Cholesteric liquid crystals diffract light and have colours that depend on the temperature. Examples of cholesteric compounds are hydroxypropyl cellulose and cholesteryl benzoate.

CHOLESTEROL: A fatty substance, with the chemical formula $C_{27}H_{45}OH$, relative density 1.052, melting point 148°C - 150°C (421 K - 423 K) and boiling point 360°C (633.15 K) and decomposes. It is found in animal fluids and tissue. The principal component is sterol. It is involved in the manufacture of steroid hormones and fat-soluble vitamins including Vitamins A, D, E and K. It is provided in the diet by foods such as eggs, meat and butter. A high level of serum cholesterol causes hardening or blockage of the arteries, an indicator for heart disease.



CHOLINE (2-HYDROXY-N,N,N-TRIMETHYLETHANAMINIUM; (2-HYDROXYETHYL)TRIMETHYLAMMONIUM; BILINEURINE): An amino alcohol, with the chemical formula $C_5H_{15}NO_2$ and boiling point 305°C (578.15 K). It occurs widely in living organisms as a constituent of certain types of phospholipids (the lecithin and sphingomylin) and in the neurotransmitter acetylcholine. It is usually classified with the B vitamins because it resembles them in function and in its distribution in foods. Choline has various uses in medicine, nutrition and the processing of foods and feeds.

CHOLINERGIC: Describing or relating to sympathetic and parasympathetic nerve fibres that release neurotransmitter acetylcholine, when stimulated or is stimulated by acetylcholine. Cholinergic also describes agents such as drugs or medications that produce the same effects as the parasympathetic nervous system.

CHOLINESTERASE (ACETYLCHOLINESTERASE): Enzymes that hydrolyse the neurotransmitter acetylcholine to choline and acetic acid, a reaction necessary to allow a cholinergic neuron to return to its resting state after activation. They are secreted by nerve cells at synapse and by muscle cells at neuromuscular junction.

CHONDRICHTHYES: A class of vertebrates comprising the fishes with cartilaginous skeleton. They are jawed fish with paired fins, paired nares, scales, two-chambered hearts and skeletons made of cartilage rather than bone. The majority belongs to the subclass *Elasmobranchii*.

CHONDRIIN: The matrix of cartilage, which is made up of chondrocytes embedded in chondroitin sulfate. It is a bluish-white gelatin-like substance and a protein-carbohydrate complex. It can be obtained by boiling cartilage in water.

CHONDROCYTE: Any of the cells that make up the matrix of cartilage. The only cells found in cartilage, they produce and maintain the cartilaginous matrix, which consists mainly of collagen and proteoglycans.

CHORD: 1. In geometry, a straight line joining any two points on a curve. 2. In astronomy, a line crossing a foreground astronomical object during an occultation which gives an indication of the objects size and/or shape. 3. In graph theory, an edge joining two not-adjacent nodes in a cycle.

CHORDATA: A phylum of animals characterised by a hollow dorsal nerve cord and at some stage in their development, a flexible skeletal rod (the notochord) and gill slits opening from the pharynx. Urochordata, Cephalochordata and Craniata are the three subphyla.

CHORDATE: An animal belonging to the phylum Chordata, which includes vertebrates and protochordates such as sea squirts and amphioxii.

CHOREA: A neurological condition characterised by involuntary irregular, jerky muscular movements of the face, shoulders and hips. It is seen in a number of neurological diseases, such as Huntington's disease.

CHOREOGRAPHY: In a Web services context, it refers to specifications for how messages should flow among diverse, interconnected components and applications to ensure optimum interoperability.

CHORION: 1. The outermost membrane enclosing the embryo of reptiles, birds and mammals. It has a dense concentration of blood vessels and aids in the formation of the placenta in mammals. 2. The protective shell of an insect egg produced by the ovary. It is pierced by a small pore (micropyle) that allows the entry of spermatozoa for fertilisation.

CHORIONIC GONADOTROPHIN (CHORIONIC GONADOTROPHIN; HUMAN CHORIONIC GONADOTROPHIN): A glycoprotein hormone, with the chemical formula $C_{17}H_{28}$. It is synthesised in the placenta by the corpus luteum after fertilisation of the ovum during the early stages of pregnancy. It promotes the maintenance of the corpus luteum, causing it to secrete progesterone. The blood level rises to a detectable concentration 7-9 days after conception and may be detected in urine 1-2 days later, which yields a highly sensitive and specific test for pregnancy. It also stimulates Leydig cells to synthesise testosterone.

CHORIONIC VILLI SAMPLING (CVS; CHORIONIC VILLUS BIOPSY, CVB): A diagnostic method of prenatal testing in which a sample of chorionic villi from the developing placenta is extracted from the uterus of a pregnant woman and analysed for chromosomal and biochemical defects. The sex of the child can also be ascertained.

CHORIPETALOUS (POLYPETALOUS): A flower such as a rose or carnation in which the corolla consists of separate petals.

CHOROID (CHOROIDEA; CHOROID COAT): A pigmented layer of cells rich in blood vessels that lie between the retina and the sclerotic of the vertebrate eye. It provides oxygen and nourishment to the outer layers of the retina. By absorbing light that has already passed through the retina, it stops back reflecting and so aids vision.

CHOROID PLEXUS: The area on the ventricles of the brain where cerebrospinal fluid (CSF) is secreted by modified ependymal cells.

CHRISTMAS FACTOR (FACTOR IX): A protein in blood that plays an important part in the clotting mechanism. A deficiency of factor IX causes a rare genetic bleeding disorder known as Christmas disease, also called haemophilia B.

CHROMA: The relative quality of a colour's purity, intensity, or saturation. In other words, it is a measure of how weak or strong (intense) a colour is, usually rated from 0 (neutral) to 8 (strong) values. For example, grey is a neutral colour rated at 0 and has an extreme low chroma. Chroma is one of the three components of Munsell colour system that is used to visually identify and match colours. The others are **hue**, which is similarity to a colour such as red, yellow, green, and blue; and **value** which is how light or dark a colour is, ranging from 0, black to 1, white.

CHROMAFFIN TISSUE: A group of cells found in the medulla of the adrenal glands and in the paraganglia that synthesises noradrenaline. Chromaffin tissue also contains the enzyme that converts noradrenaline into adrenaline.

CHROMALVEOLATES: An assemblage of eukaryotic organisms constituting the subgroup chormalveolata, which comprises the stramenopiles and alveolates.

CHROMATE: A salt containing the ion CrO_4^{2-} . Chromate compounds are good oxidisers. Examples are sodium chromate (Na_2CrO_4) used in making pigments and other chromium chemicals and in dyeing and processing textiles; and barium chromate (BaCrO_4) used as a pigment.

CHROMATIC: Relating to colour or of bright colours. For example, **metachromatic stain** is a stain that has the ability to produce different colours in various histological or cytological structures.

CHROMATIC ABERRATION (ACHROMATISM; CHROMATIC DISTORTION): An optical defect, in which the image formed by a lens but not a mirror has a series of coloured images, even though only white light enters the system, as a result of the different extent to which light of different colours is refracted by the lens. The refractive index of glass for light at the blue end of the spectrum is marginally greater than the refractive index for light at the red end. Hence the focal length is slightly different and white images are weakly coloured at the edges. It is present when using more than one wavelength of light. Chromatic aberrations are classified as transverse or lateral and longitudinal or axial. **Transverse chromatic aberration (TCA)** occurs when the size of the image changes with wavelength. In other words, when white light is used, red, yellow and blue wavelengths focus at separate points in a vertical plane. **Longitudinal chromatic aberration** occurs when different wavelengths focus at different points along the horizontal optical axis as a result of dispersion properties of the glass. Longitudinal chromatic aberration can be either primary or secondary. **Primary chromatic aberration** is where the far ends of the spectrum, red and blue, focus differently.

Secondary chromatic aberration is the aberration that remains after primary chromatic aberration is corrected. This can be completely corrected with apochromatic lenses.

CHROMATIC NUMBER: The minimum number of colours needed to colour the vertices of a connected graph so that the edges at a common vertex have distinct colours and no two adjacent vertices have the same colour. The chromatic number of a square, cube or sphere, for example, is 4, as it is impossible to place more than four differently coloured regions on one of these figures so that any pair has a common boundary. A graph with $\chi(G) = k$ is said to be k -colourable. All bipartite graphs are two-colourable; and all planar graphs are four-colourable as a consequence of the four colour theorem.

CHROMATIC SILENCING: The inactivation or "switching off" of gene transcription as a result of condensation of chromosomal DNA into heterochromatin.

CHROMATICITY: An objective specification of the colour quality of

a visual stimulus that does not depend on its luminance but which, together with the luminance completely specifies the colour.

CHROMATID: One of the pair of strands produced when a chromosome divides longitudinally and copies itself during meiosis or mitosis.

CHROMATIN: The substance of which chromosomes are composed, found inside the nuclei of eukaryotic cells. It consists of proteins, DNA and small amounts of RNA. Chromatin contains genetic material-instructions to direct cell functions.

CHROMATIN LOOP: A region of relatively unwound DNA, about 200 nm long, extending outwards in a loop from the chromosome.

CHROMATOGRAM: The term used in chromatography to describe a picture or graphical presentation of detector response (on the ordinate) versus retention time or effluent volume. In paper chromatography or thin layer chromatography (TLC) chromatogram may refer to the paper or layer with the separated zones.

CHROMATOGRAPHIC DETECTOR: A device that measures the change of composition of the effluent. There are two general types of detectors: destructive and non-destructive. The destructive detectors perform continuous transformation of the column effluent (burning, evaporation or mixing with reagents) with subsequent measurement of some physical property of the resulting material (plasma, aerosol or reaction mixture). The non-destructive detectors are directly measuring some property of the column effluent (for example UV absorption) and thus ensuring for the further analyte recovery.

CHROMATOGRAPHY: The collective term for a set of techniques for the separation of mixtures. The different types of chromatography include column, paper, gas, gas-liquid and ion. It involves passing a mixture dissolved in a "mobile phase" through a stationary phase, which separates the components in the mixture. The separation is based on differential partitioning of the solutes between the mobile and stationary phases. Subtle differences in the compounds' partitions result in differential retention on the stationary phase and thus causing the separation. Some of the applications of chromatography include analysing foods, drugs, blood and petroleum products.

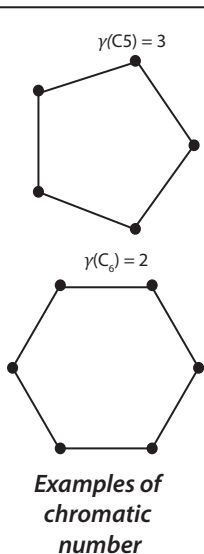
CHROMATOPHORE: 1. A pigment containing cells formed in the skin of many lower vertebrates such as the chameleon and in the integument of crustaceans that bring about changes in colour or brightness. A concentration or dispersion of the pigment granules in the cytoplasm of the cell causes the colour of the animal to alter to match its surroundings. 2. Any of the particles isolated from photosynthetic organisms that contain photosynthetic pigments, such as a chloroplast or membrane fragments from photosynthetic bacteria. The principle photosynthetic pigment in the blue-green algae is chlorophyll *a*; in purple bacteria, bacteriochlorophyll *a* or *d*; and in green bacteria, bacteriochlorophyll *b* or *c*.

CHROME: 1. Chromium, especially when used as a protective coating on metals. The chromed layer can be decorative, provide corrosion resistance, ease cleaning procedures or increase surface hardness. 2. A yellow colouring matter obtained from a compound of chromium and used in paints.

CHROME IRON ORE: A mixed iron-chromium oxide, with the chemical formula $\text{FeO} \cdot \text{Cr}_2\text{O}_3$. It is used to make ferrochromium for chromium steels.

CHROME RED: A basic lead chromate, with the chemical formula $\text{PbO} \cdot \text{PbCrO}_4$. It is used as a red pigment.

CHROME YELLOW: A natural yellow pigment of lead(II) chromate, with the chemical formula PbCrO_4 . Because the pigment tends to oxidise and darken on exposure to air over time and contains lead, a toxic, heavy metal, it has been largely replaced by another pigment, cadmium yellow mixed with enough cadmium orange to produce a colour equivalent to chrome yellow. Cadmium is also a toxic metal.



CHROMIC ACID (DIHYDROXIDODIOXIDOCROMIUM; CHROMIC(VI) ACID; TETRAOXOCHROMIC ACID): A collection of acids, known only in chromate salts, with the chemical formula H_2CrO_4 . It has a relative density of 1.201, melting point 197°C (470 K) and boiling point 250°C , 523 K (decomposes). It is produced by the acidification of aqueous solutions of chromium trioxide or sodium chromate or potassium dichromate with sulfuric acid. Chromic acid (H_2CrO_4) is a strong and corrosive oxidising agent.

CHROMIC COMPOUNDS: Compounds containing chromium in a higher (+3 or +6) oxidation state.

CHROMISTA: 1. In some classification systems, a kingdom consisting of brown algae, golden algae and diatoms. 2. Multicellular eukaryotic life forms characterised by photosynthesis, in which chemical energy is produced from water, minerals and carbon dioxide with the aid of pigments and the radiant energy of the Sun.

CHROMITE: A mineral belonging to the spinel group, with the chemical formula FeCr_2O_4 , the principal ore of chromium. It is black with a metallic lustre and mostly occurs in massive form.

CHROMIUM: A hard brittle, grey-white metallic element, with the symbol Cr, atomic number 24, relative atomic mass 51.990, melting point 1890°C (2163.15 K) and boiling point 2482°C (2755.15 K). It is used in making alloys such as stainless steel, as a shiny, protective coating on other metals and also as a catalyst. Its compounds are used for tanning leather and for making alums. It was discovered by Louis Nicolas Vauquelin in the mineral crocoite (lead chromate) in 1797.

CHROMIUM POTASSIUM SULFATE: A red crystalline solid, with the chemical formula $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$. It is used as a mordant.

CHROMIUM STEEL (CHROME STEEL; STAINLESS STEEL): Any group of stainless steel containing at least 10% of chromium content by mass. Stainless steel differs from carbon steel by the amount of chromium present. Chromium steel might contain 18% of chromium, 8% of nickel and 0.15% of carbon. Chromium steels are highly resistant to corrosion and are used for cutlery, chemical plant, ball bearings, etc.

CHROMIUM(II) OXIDE: A black water insoluble powder prepared by oxidising chromium amalgam with air, with the chemical formula CrO and melting point 300°C , 573.15K (decomposes). It is readily oxidised by the atmosphere. At high temperatures hydrogen reduces it to the metal.

CHROMIUM(III) OXIDE: A green crystalline water-insoluble salt, with the chemical formula Cr_2O_3 , melting point 2435°C (2708.15 K) and boiling point 4000°C (4273.15K). It is obtained by heating chromium in a stream of oxygen or by heating ammonium dichromate. In nature, it occurs as the rare mineral eskolaite. It is amphoteric, dissolving in acids to give chromium(III) ions and in concentrated solutions of alkalis to give chromites. It is anti ferromagnetic up to 33.9°C (307 K), the Neel temperature. It is used as a green pigment in glass, porcelain and oil paint.

CHROMIUM(IV) OXIDE (CHROMIUM DIOXIDE): A black insoluble powder prepared by the action of oxygen on chromium(VI) oxide, with the chemical formula CrO_2 and melting point 375°C (648 K), decomposes. It is highly unstable. It is a synthetic magnetic substance widely used in magnetic tape emulsion until the arrival of CDs and DVDs. However, it is still used in data tape applications for large storage systems.

CHROMIUM(VI) OXIDE (CHROMIUM TRIOXIDE; CHROMIC ANHYDRIDE): A red compound, with the chemical formula CrO_3 , melting point 197°C (470 K), boiling point 251°C (524 K) and decomposes. It can be made by the careful addition of concentrated sulfuric acid to an ice-cooled concentrated aqueous solution of sodium dichromate with stirring. Chromium(VI) oxide is an extremely powerful oxidising agent, especially to organic matter; it immediately inflames ethanol. It is an acidic oxide and dissolves in water to form chromic acid, a powerful oxidising agent and cleansing fluid for glassware, used mainly for electroplating.

CHROMOMERE: A particulate portion of a chromosome that becomes apparent as contraction and condensation of chromatin occurs in meiosis. Chromomeres vary in size and shape, homologous chromosomes having identical patterns along their lengths. An average chromomere has been found to contain sufficient DNA to code for approximately 30-35 polypeptides of average length. Genetic evidence, however, suggests that chromomeres contain only a single functional structural gene.

CHROMOPHORE: 1. Any group of atoms in a molecule that includes an unsaturated chemical group capable of absorbing ultraviolet radiation. 2. Chemical grouping that causes compounds to have colour. It is a region in the molecule, where the energy difference between two different molecular orbitals falls within the range of the visible spectrum. The colour arises when a molecule absorbs certain wavelengths of visible light and transmits or reflects others. It is the $-\text{N}=\text{N}-$ group in an azo dye.

CHROMOPLAST: Any of various pigment-containing plastids in plant cells other than chloroplast. Red orange and yellow chromoplasts contain carotenoid pigments and are responsible for the colouration of fruits, flowers and other organs such as tuber of carrots and beet root.

CHROMOSOMAL ANALYSIS (CYTOGENETICS): A branch of genetics that is concerned with the study of the whole chromosome set as photographed and laid out in a recognised order.

CHROMOSOMAL MUTATION (CHROMOSOMAL REARRANGEMENT): A genetic alteration in DNA that occurs during cell processes like mitosis and meiosis which changes whole genes on chromosomes. Types of chromosomal mutations include insertions or duplications (addition of short stretches of nucleotide on one chromosome), deletions (removal of short stretches of nucleotide sequences on one chromosome), inversions (when genes change order on the same chromosome), and translocations (the transfer of a gene from one chromosome to another).

CHROMOSOME: Any of the tiny threads in animal and plant cell nucleus that carries the genes. It is a self-replicating structure consisting of DNA complexed with various proteins and involved in the storage and transmission of genetic information. Each species has a characteristic number and type of chromosome. Humans have 23 pairs of chromosomes, which include 22 pairs of numbered chromosomes (non-sex), called **autosomes** and one pair of sex chromosomes, X and Y, making a total of 46 chromosomes. A full set or complement, of chromosomes is carried by each sperm or ovum in animals and each pollen grain or ovule in plants. This constitutes the haploid genome of that organism and contains a complete set of the genes characteristic of that organism.

CHROMOSOME ABERRATION (CHROMOSOME MUTATION): An alteration in the number order or arrangement of genes caused by structural changes to a chromosome. Sometimes chromosome mutation may involve a large number of genes and the aberration may be visible by microscopy. There are four classes of chromosome mutations, which are deletions, duplications, inversions and translocations.

CHROMOSOME BANDING: Staining of bands or chromosome segments, allowing the precise identification of individual chromosomes or parts of chromosomes. Banding patterns lend each chromosome a distinctive appearance so the 22 pairs of human numbered (non-sex) chromosomes and the X and Y sex chromosomes can be identified and distinguished without ambiguity. Banding also permits the recognition of chromosome deletions (lost segments), chromosome duplications (surplus segments) and other types of structural rearrangements of chromosomes.

CHROMOSOME DIMINUTION (CHROMOSOME ELIMINATION): The separation or elimination of certain chromosomes from cells that form somatic tissues in embryonic nematode worm during embryogenesis (early stage of embryo development).

CHROMOSOME MAP: Any plan that shows the positions of genes, genetic markers or other landmarks along the length of a chromosome. There are essentially two complementary types of map: linkage maps, which give the relative positions of genetic sites (loci) determined by studies of how frequently recombination occurs between the loci; and physical maps, which show the arrangement of the chromosomal material.

CHROMOSOME MUTATION: A change in the gross structure of a chromosome, which usually causes severely deleterious effects in the organism. Chromosome mutations often occur due to an error in pairing during the crossing over stage of meiosis. The main types of chromosome mutation include translocation, duplication, deletion and inversion.

CHROMOSOME NUMBER: The number of chromosomes possessed by a species, which may be given as either the haploid (n) or diploid ($2n$) number. In any given asexually reproducing species, the chromosome number is always the same. In sexually reproducing organisms, the number of chromosomes in the body (somatic) cells is diploid ($2n$; a pair of each chromosome), is twice the haploid ($1n$) number found in the sex cells or gametes. The haploid number is produced during meiosis.

CHROMOSOME PAINTING: A technique based on fluorescence in situ hybridisation (FISH) that uses a palette of fluorescently labelled probes to identify specific chromosomes or chromosomal regions by 'painting' them in different colours. It is used diagnostically in clinical cytogenetics to screen for translocations or other structural aberrations, such as hereditary diseases, cancer and in cytogenetics to determine the structural changes in genomes occurring during evolution.

CHROMOSOME PUFF (BALBIANI RING): A swollen region at a site along the length of a polytene chromosome (Balbiani chromosome), where the swelling reflects a transcription is actively taking place.

CHROMOSOME THEORY OF INHERITANCE (BOVERI-SUTTON CHROMOSOME THEORY): A fundamental unifying theory of genetics that chromosomes are the cellular components that physically contain genes. It also states that chromosomes which are seen in all dividing cells and pass from one generation to the next, are the basis for all genetic. This theory was proposed in 1903 by Walter Sutton and Theodore Boveri.

CHROMOSOME WALKING: A technique, used in constructing a physical map for selecting contiguous overlapping clones from a DNA library and thus reconstructing the order of genes along a segment of chromosome. For example, one can effectively 'walk' in either direction from a known 'marker' gene to identify adjacent genes; a technique that is useful in fine structure mapping of a genome.

CHROMOSPHERE: A complex structure of warm gas above the visible surface or photosphere, of the Sun and most stars. The chromosphere is normally only visible when the photosphere is totally eclipsed by the Moon.

CHROMOUS COMPOUNDS: Chromium containing compounds in their lower (+2) oxidation state. For example, chromous chloride is chromium(II) chloride (CrCl_2).

CHROMYL CHLORIDE (CHROMIUM OXYCHLORIDE; CHROMIUM(VI) DICHLORIDE DIOXIDE; DICHLORODIOXOCHROMIUM; CHROMIC ACID CHLORIDE): A dark red liquid with the chemical formula CrO_2Cl_2 , boiling point 117°C (390 K) and melting point -96.5°C (176.65 K). It is evolved as a dark red vapour on addition of concentrated sulfuric acid to a mixture of sodium potassium dichromate and sodium chloride. The compound is a powerful oxidising agent, exploding on contact with phosphorus and inflaming sulfur and many organic compounds.

CHRONIC: Lasting for a long time, continually, recurring, especially of a disease. For example, diabetes mellitus and hypertension are

chronic diseases, which cannot be cured but can only be controlled, through medication and therapy.

CHRONIC CARRIER: An individual that harbours a pathogen with duration of 6 months or more without showing any symptoms.

CHRONIC FATIGUE SYNDROME, CFS (MYALGIC ENCEPHALOMYELITIS, ME): A chronic degenerative neuro-immune disease, having serious immune and cardiovascular abnormalities. Its causes are not known but have been linked to viral infectious diseases such as herpes or hepatitis process, exposure to chemicals, bacterial infection, etc. Signs and symptoms are varied and include short-term memory loss, concentration difficulties, muscle and nerve pain, muscle weakness, noise and light sensitivity, sleep and temperature disturbance, inability to sustain upright activity such as standing, sitting or walking and sensitivity to food, alcohol, chemicals and medicines.

CHRONIC TOXICITY: The ability of a substance or mixture of substances to cause harmful effects over an extended period, usually upon repeated or continuous exposure, sometimes lasting for the entire life of the exposed organism. **Acute toxicity** is the adverse effects of a substance which result either from a single exposure or from multiple exposures in a short space of time, usually less than 24 hours.

CHRONISTICS: The study of the time taken for evolutionary events to occur. Organisms that, from fossil evidence, appear to have evolved rapidly are termed **tachytelic** while those that appear to have changed slowly are termed **bradytelic**. Organisms showing moderate rates of evolution are termed **horotelic**.

CHRONOLOGY: The science of fixing the dates of historical events or the arrangement or list of events in the order in which they occurred.

CHRONOLOGY PROTECTION CONJECTURE: A conjecture proposed by Stephen Hawking (1942-), British theoretical physicist and mathematician in 1992 asserting that the fundamental laws of physics should forbid time travel.

CHRONOMETER: An instrument that keeps very accurate time. It is used especially for navigating at sea and designed to remain accurate through all conditions of temperature and pressure. Some features are: (i) a heavy balance wheel, the axis of which is kept always vertical; (ii) a balance spring wound in cylindrical shape, instead of a nearly flat helix; (iii) a special escapement, which is a device that transfers energy to the timekeeping element and enables counting the number of oscillations of the timekeeping element.

CHRONON: A hypothetical 'particle' of time, defined as the time taken for electromagnetic radiation to cross an electron, that is the diameter of an electron divided by the speed of light. This value is approximately 2×10^{-23} second. The term was introduced by Robert Lévi (1927).

CHRONOSEQUENCE: A sequence of soils that share similar attributes such as climate, organisms, relief, and parent materials but differ from one another in certain properties as a result of time as a soil-forming factor (the soils are of different ages).

CHROOCOCCALES: An order of the Cyanophyta containing the unicellular and colonial blue-green algae. The colonial (or palmelloid) forms arise when daughter cells become bound together in a mucilaginous sheath after cell division.

CHRYALIS: A stage in the life of an insect when it changes from a grub to an adult insect, especially a butterfly changing from caterpillar or larva to the adult. The term also refers to the hard case that encloses an insect during this stage.

CHRYSONOMADA (CHRYSOPHYTA; GOLDEN-BROWN ALGAE): A large phylum of the kingdom Protista of mostly freshwater organisms consisting of the diatoms (class Bacillariophyceae), the golden or golden-brown algae (class Chrysophyceae) and the yellow-green algae (class Xanthophyceae) that possess carotenoid pigments in addition to chlorophylls.

CHRYSTOPHYTA (CHRYSTOPHYTES; GOLDEN-BROWN ALGAE):

A division made up predominantly of flagellate unicellular algae but also containing some colonial and filamentous forms. These algae usually have two golden-brown plastids in each cell and food is stored as leucosin or fat. In some classifications, the Chrystophyta also includes the yellow-green algae, the diatoms and the Haptophyceae on the basis that these three groups share a number of features with the golden-browns. These include accumulating food reserves as the polysaccharide leucosin, having siliceous or calcareous cell walls more or less conspicuously divided into two parts and having a flimmer flagellum.

CHURCH'S THEOREM: In logic, the result that no decision procedure exists for arithmetic. It was named after the US mathematician and logician Alonzo Church (1903-1995).

CHURCH'S THESIS: The hypothesis that a function is recursive if and only if it is effectively computable.

CHURN RATE (RATE OF ATTRITION): The percentage of subscribers to a service who discontinue their subscriptions to the service within a given time period.

CHYLE: A milky fluid consisting of lymph that contains absorbed food materials especially emulsified fats, which travels in the lymphatic vessels draining the small intestine. Most chyle occurs in the lymphatic ducts (lacteals) in the villi of the small intestine during the absorption of fat.

CHYLOMICRON: Any of the lipoprotein particles synthesised by intestinal epithelial cells and consisting of triglycerides (85-92%), phospholipids (6-12%), cholesterol (1-3%) and proteins (1-2%). Chylomicrons are the form in which dietary fat is transported in the circulatory system.

CHYME: A thick creamy fluid made up of partly digested food, hydrochloric acid and a range of enzymes including amylase and pepsin. Secretion of hydrochloric acid by the stomach makes the chyme strongly acidic.

CHYMOTRYPSIN: An endopeptidase enzyme in pancreatic juice that is secreted into the duodenum. The enzyme is secreted as an inactive precursor, chymotrypsinogen, which is activated by another pancreatic protease, trypsin. Chymotrypsin preferentially cleaves peptide amide bonds, where the carboxyl side of the amide bond (the P₁ position) is a tyrosine, tryptophan or phenylalanine.

CHYTRID: Any of the simple, algae-like fungi constituting the class Chytridiomycetes order Chytridiales, of aquatic and soil environments, having flagellated zoospores and little or no mycelium.

CHYTRIDIOMYCOTA: A phylum of motile single-celled microorganisms, the chytrids that live in damp soil or fresh water and have affinities with true fungi. The body is unicellular and is anchored to the substrate by threadlike hyphae or rhizoids and moves by means of a single backward-pointing whiplike appendage flagellum.

CICADA: An insect such as a grasshopper of the order Hemiptera, suborder Auchenorrhyncha, which is common in tropical to temperate regions. Cicadas are medium-sized and reach a length of 2.5-5 cm. They have short heads, with bristle-like antennae, large eyes and sucking mouthparts. The two pairs of large wings are transparent, with branching veins. The male makes a small chirping noise by vibrating membranes near the base of the abdomen. Cicadas can cause damage to several cultivated crops, shrubs and trees, when the females lay their eggs deep in branches.

CILIARY BODY: The circular band at the front of the vertebrate eye that connects the pigmented layer choroid of the eyeball with the iris diaphragm composed of the ciliary muscle and ciliary processes.

CILIARY FEEDING: A method of feeding used by lancelets and many other aquatic invertebrates. The movement of cilia causes a current of water to be drawn towards and through the animal and microorganisms in the water are filtered out by the cilia.

CILIARY MUSCLE: A tissue or ring of striated smooth muscle surrounding and controlling the lens of the vertebrate eye, which is

used in accommodation for viewing images of objects at different distances to be focused on the retina.

CILIATE: 1. A term used in botany to describe a structure, such as a leaf margin, fringed with numerous cilia or fine hairs. 2. Any of various protozoans of the phylum Ciliophora whose body are covered by cilia.

CILIATED EPITHELIUM: A region consisting of columnar or cuboidal cells bearing hair-like appendages that are capable of beating rhythmically.

CILIOPHORA: A phylum in the kingdom Protista containing ciliated protozoa that possess two types of nuclei, a micronucleus and macronucleus. An example is Paramecium.

CILUM: Short minute hair-like structures present in numbers on the surface of many animals and a few plant cells. Cilia beat rhythmically to aid the movement of a fluid past the cell or the movement of the organism through liquid. There are two types of cilia: motile cilia and non-motile or primary cilia, which typically serve as sensory organelles. **Ciliate** means covered with hair-like structures which are relatively short and dense.

Ciliolate means a marginal fringe of minute hairs.

CINCINNUS (SCORPIOID CYME): A cymose inflorescence or monochasium in which successive branches alternate on opposite sides of the stem. An example is that of *Strelitzia reginae* (the bird of paradise flower). The inflorescence is often bent to one side.

CINDER CONE (ASH CONE; SCORIA

CONE): A steep cone formed from volcanic ash and loose rock that deposits around a volcanic vent. Cinder cones are the simplest type of volcano.

CINE SUBSTITUTION: A type of substitution reaction in which the entering group takes a position on an atom adjacent to the atom to which the leaving group is attached.

CINE-CAMERA: A camera that takes rapid sequence of still photographs (24 pictures each second). When the pictures are projected one after the other at the same speed on to a screen, they appear to show movements because the human eyes hold on to the image of one picture before the next one appears.

CINNABAR: A reddish brown mineral form of Mercury(II)sulfide, with the chemical formula HgS, which crystallises in the hexagonal system. It is deposited in veins and impregnations near recent volcanic rocks and hot springs. It is the only commercially useful ore of mercury.

CINNAMIC ACID (3-PHENYLPROPENOIC ACID): A white crystalline organic compound which is insoluble in water but with a pleasant odour. Its chemical formula is C₉H₈O₂, relative density 1.2475, melting point 133 °C (406 K) and boiling point 300 °C (573 K). It is obtained from oil of cinnamon and also occurs in shea butter. It is used in flavours and certain pharmaceuticals, but its primary use is in the synthesis of the methyl, ethyl and benzyl esters for the perfume industry. Cinnamic acid is a key intermediate in shikimate and phenylpropanoid pathways of the biosynthesis of several compounds including alkaloids, aromatic amino acids and indole derivatives as well as flavonoid class of compounds.

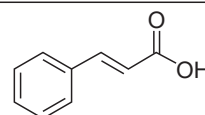


Fig. I: Structure of trans-cinnamic acid

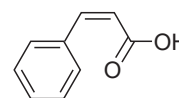
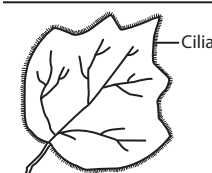
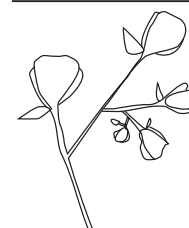


Fig. II: Cis-cinnamic acid structure

CINNAMON: Any of several tropical plant species in the genus *Cinnamomum*, family Lauraceae, including *Cinnamomum verum*, commonly known as Ceylon cinnamon; *Cinnamomum aromaticum* (Chinese cinnamon); and *Cinnamomum burmanni*, with the common



Ciliate leaf



Cincinnus inflorescence

name Indonesian cinnamon, grown for their leaves and inner bark which is used as a flavouring agent. The trees are small evergreens with aromatic bark and leaves. In the wild, the trees can grow up to 20 metres high, but the cultivated ones are just about 10 metres tall. The leaves of the tree are thick and oval or lanceolate in shape, growing on smooth grey branches. The tree produces small yellow or green flowers on panicles and berry-like fruits. Cinnamon is propagated by seed or vegetatively. The bark of cinnamon is used as a spice in cooking and as a flavouring agent in beverages. The oil is used in products such as toothpaste, perfume and soap. It is also used to treat various ailments such as nausea, vomiting, diarrhoea, indigestion, backache, menstrual problems, etc.

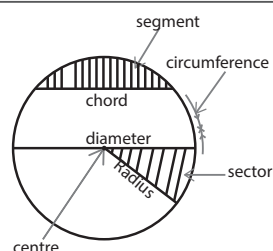
CINQUEFOIL: Various annual and perennial plants and shrubs of the rose family, native to the temperate and cold regions of the world. The leaves are composed of three leaflets and the small white, yellow or red flowers have five petals.

CIPHER (CYPHER): 1. An algorithm for performing encryption or decryption 2. Any of the Arabic numerals, 1,..., 9 used for counting.

CIRCADIAN RHYTHM (DIURNAL RHYTHM): A metabolic rhythm found in most organisms which generally coincides with daylight/darkness or any 24-hour periodicity in the behaviour or physiology of animals or plants. Its most obvious manifestation is the regular cycle of sleeping and waking up.

CIRCINATE VENATION: A form of venation in which the leaf primordia are rolled in on themselves from the apex to the base, so that the apex is in the middle of the coil. It is seen in most ferns, except the Ophioglossales and in certain of the cycads and extinct seed ferns.

CIRCLE: 1. A two-dimensional geometric figure enclosed by a curved line, with every point on it being the same distance from the centre. Every circle has a radius, a circumference and a diameter. Its equation is $(x-h)^2 + (y-k)^2 = r^2$, where r is the radius and (h, k) is the centre; its parametric equations are $x = r \cos \theta$; $y = r \sin \theta$. Its circumference is $2\pi r$. 2. The figure enclosed by such a curve, with area πr^2 . 3. The path of a point that moves so as to keep a constant distance from a fixed point (the centre). It is the locus of points equidistant from a fixed point. It is also the locus of points equidistant from a fixed point. 4. A **great circle** is a circle described by the intersection of the surface of a sphere with a plane passing through the centre of the sphere. Sections of the sphere that do not contain a diameter are called **small circles**. It is also a segment of such a circle representing the shortest distance between two terrestrial points. A great circle becomes a straight line in a gnomonic projection. 5. **Vicious circle** is a form of explanation given in terms that cannot be understood independently of what was to be explained. It consists of finite sequences $p_1, p_2, \dots, p_n$, in which the definition of p_1 involves that of p_2 and that of p_2 involves that of p_3 , etc.



A circle with its various parts

CIRCLE GRAPH (PIE CHART): A method of displaying statistical information where a circle is divided into different-sized sectors by radial lines. The angle of each sector is directly proportional to the figure it represents and it is expressed as a fraction of the of figures in the group data: $\frac{\text{Angle of sector}}{360^\circ} = \frac{\text{Figure sector represents}}{\text{Sum of figures in data}}$.

CIRCLE INVOLUTE: The simplest kind of spiral to draw and understand. It is, for example, the path that a goat, tethered to a post, would follow if it walked around and around in the same direction, keeping its tether taut until it wound its way to the centre. The radial distance between adjacent loops of the spiral is equal to the circumference of the central circle. Except for the innermost loop,

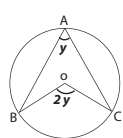
the circle involute is hard to distinguish from the Archimedean spiral, though the two curves are never identical.

CIRCLE OF CONVERGENCE: A circle on the Argand diagram such that a given power series converges at all points within the circle and diverges at all points outside it. For any power series $\sum_i c_i (z-a)^i$, there is an R such that the series converges, if $R \neq 0$, for all z such that $|z-a| < R$, and diverges for all z such that $|z-a| > R$. The radius R may also be infinite, in which case the circle of convergence is the entire plane or zero so that it is a single point and is the absolute value of the limit of the ratio of each term to the next. The series may either converge or diverge at points for which $|z-a|$ equals the radius that lies on the circumference of the circle.

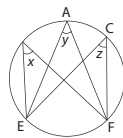
CIRCLE OF CURVATURE (OSCULATING CIRCLE): The circle that touches a curve on the concave side and has the same tangent and curvature as a given curve at a given point with its radius equal to the radius of curvature at the point. For a function $y = f(x)$, the formula for the curvature and radius of curvature (its radius), at a point $[x, f(x)]$ on the curve $y = f(x)$, the circle of curvature is tangent to the curve and has a radius $r(x)$. The radius of curvature is normal to the concave side of the curve at the point and is equal to the reciprocal of its curvature.

CIRCLE OF WILLIS: A circular arrangement of large blood vessels on the undersurface of the brain formed by linked branches of the arteries that supply the brain. Most cerebral aneurysms occur on or near the circle of Willis.

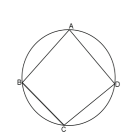
CIRCLE THEOREM: Any of a set of theorems used in solving circle problems in plane geometry as follows: (i) Theorem 1 states that the angle subtended at the centre of a circle is twice the angle subtended at the circumference: $\angle BOC = 2 \times \angle BAC$ or $\angle BOC = 2y$ and $\angle BAC = y$. (ii) Theorem 2 states that angles subtended by an arc in the same segment of a circle are equal: $\angle EBF = \angle EAF = \angle ECF$ or $x = y = z$. (iii) Theorem 3 states that opposite angles in a cyclic quadrilateral sum up to 180° : $\angle BAD + \angle BCD = 180^\circ$ or $\angle ABC + \angle ADC = 180^\circ$. (iv) Theorem 4 states that the angle in a semicircle is a right angle (90°). In other words the angle subtended by a diameter on the circumference of the circle is a right angle or 90° : $\angle ABC = 90^\circ$. (v) Theorem 5 states that the angle between a tangent and the radius drawn to the point of contact is 90° : $\angle OBC = 90^\circ$. (vi) Theorem 6 states that only two tangents can be drawn from a point outside a circle: $|AB| = |AC|$. The tangents are also equal in length. (vii) Theorem 7 states that the angle between the tangent of a chord through the point of contact is equal to the angle subtended by the chord in the alternate segment: $\angle CAD = \angle DEA$, $\angle BAE = \angle EDA$.



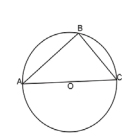
Circle Theorem 1



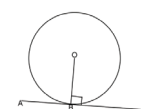
Circle Theorem 2



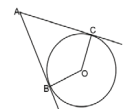
Circle Theorem 3



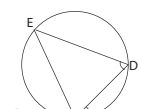
Circle Theorem 4



Circle Theorem 5



Circle Theorem 6



Circle Theorem 7

CIRCUIT: 1. An arrangement of electrical components through which a current flows. The two types are series and parallel circuits. 2. A route or path that follows a curved course and finishes at the point at which it began. 3. A single complete trip around a circular route or path. 4. An area that lies inside a circular route or path. 5. A simple closed path in a graph that ends at the vertex it begins. An

Euler circuit uses every edge of a graph exactly once; a Hamiltonian circuit uses each every vertex of a graph exactly once.

CIRCUIT BENDING: The practice of modifying existing electronics items, most often toys, to create other items, usually musical instruments.

CIRCUIT BREAKER: A safety device or switch that automatically cuts off a current to protect an electric circuit from excessive current. It has the same function as a fuse, but the difference is that circuit breakers can be reset, without replacing a fuse element. There are two main types of circuit breakers: magnetic circuit breakers and thermal magnetic circuit breakers. Magnetic circuit breakers have a solenoid and an iron plunger inserted partially inside the solenoid. When current flows through the solenoid, it produces a magnetic field, whose strength depends on the amount of current flowing through it. When the amount of current exceeds the amount that the circuit is designed to hold, the magnetic field in the solenoid is so strong that it pulls the iron plunger completely into the solenoid, breaking contact with the circuit at the end of the plunger and stopping the flow of current. Thermal magnetic circuit breakers work on the principle that when current is flowing through a strip of bimetallic sensor composed of two different metals sandwiched together, heat is produced and the two metals expand at different rates. As the current increases more heat is generated in the bi-metallic strip. Since the two metals are mechanically bonded together, they expand at different rates, causing the strip to bend. At a pre-determined temperature, the strip, which is attached to a lever and latch moves the lever far enough to unlock a spring loaded handle which then moves toward the "off" position opening the electrical contacts and interrupting the current flow. Thermal magnetic circuit breakers also have the magnetic system such as described above, which responds instantaneously to large surges in current, often produced by short circuit faults. The thermal bimetallic strip responds to longer-term over-current conditions. When a current is interrupted, an arc is generated and in order for the gap between the contacts to withstand the voltage in the circuit to prevent the arc which is created from jumping to complete the circuit, the arc must be cooled and extinguished in a controlled way. Different circuit breakers use vacuum, air, insulating gas or oil as the medium to cool and dispose of the arc.

CIRCUIT DIAGRAM: A simplified electric circuit which is drawn to indicate the positions of connecting wires by straight lines and dots, where wires join with the circuit's components. The components are represented by internationally recognised symbols.

CIRCUIT-SWITCHED: A type of network in which a physical path is obtained for and dedicated to a single connection between two end-points in the network for the duration of the connection. An example is voice phone service. The telephone company reserves a specific physical path to the number being called for the duration of the call. During that time, no one else can use the physical lines involved.

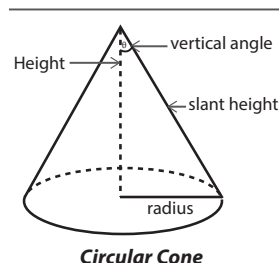
CIRCULANT: A determinant or matrix in which the elements of a given row are the elements of the preceding row that are moved cyclically one element to the right. As a result, all entries on the main diagonal are identical.

CIRCULAR: 1. Resembling a circle in shape or having the shape of a perfect circle. 2. Following a curved route or path that ends at the point where it began. 3. **Circular argument** is a discussion that makes a conclusion that has been used in the premises or based on an assumption that has already been used in the argument. 4. Of an explanation or construction, given in terms of something that is itself explained in terms of or constructed from, the very thing that is supposedly being explained or constructed.

CIRCULAR BIREFRINGENCE: A phenomenon in which there is a difference between the refractive indices of the molecules of a substance for right and left-circularly polarised light. Circular birefringence depends on the way in which the electromagnetic field interacts with the molecule and it is affected by the handedness of the molecule and its polarisation.

CIRCULAR CONE: A cone whose base is a circle. To find the volume V of a circular cone with height h and radius r , the following formula is applied: $V = \frac{1}{3}\pi r^2 h$. The total surface area is calculated as $A = \pi r(L + r)$,

where r is the radius of the circular base and L is the length of the slant edge.



CIRCULAR DICHROISM (CD): A spectroscopic method which measures the difference in absorbance of left- and right-handed circularly polarised light by a material, as a function of the wavelength. Most biological molecules, including proteins and nucleic acids, are chiral and show circular dichroism in their ultraviolet absorption bands, which may be used as an indication of secondary structure. Metal centres that are bound to such molecules, even if they have no inherent chirality, usually exhibit circular dichroism in absorption bands associated with ligand-based or ligand-metal charge-transfer transitions.

CIRCULAR MEASURE: A ratio of the length of an arc by which it is subtended at the centre of the circle to the length of the radius of the circle.

CIRCULAR MOTION: A motion of a body along a circular path. The speed of the body in circular motion remains constant, but its direction changes continuously so the velocity changes and hence the body accelerates. Since this acceleration is strictly a change in direction and is a result of a force directed toward the centre of the circle, the force is a centripetal force, causing centripetal acceleration. An object moving in a circular motion around another object is said to be orbiting the latter.

CIRCULAR NUMBER: Any number whose powers end on the same figure as it does itself. Examples are numbers ending in 0, 1, 5 or 6.

CIRCULAR PERMUTATION: The arrangement of n objects in a circular form. This is given by $(n - 1)!$.

CIRCULAR POINT: In Euclidean geometry, a type of umbilical point in which the radii of normal curvature are equal.

CIRCULAR POLARISATION: The process in which the tip of an electric vector describes a circular helix about the direction of propagation with a frequency equal to the frequency of the light. If the wave is frozen in time the electric field vectors describe a helix along the direction of propagation.

CIRCULAR TRIANGLE: A triangle in which each side is the arc of a circle. The sum of the angles between the tangents to the arcs at the point of intersection falls between 0 and 1080° (67π radians). If only minor arcs are considered, the upper bound is 4π radians; a limiting case in which these arcs are semicircles.

CIRCULAR VELOCITY: The horizontal velocity of an object required to keep it in a circular orbit at a given altitude about a planet or other gravitating mass.

CIRCULATING DECIMAL (RECURRING DECIMAL; REPEATING DECIMAL): Decimal numbers that repeat indefinitely in a pattern after the decimal point, for example, 0.3333 or 2.1818181... The repeating figure or figures of a recurring decimal fraction is called repetend(s). Recurring decimals can also be written by drawing a horizontal line called a vinculum above the repetend, placing dots above the outermost numerals of the repetend or enclosing the repetend in parentheses. For example, 0.3333 can be written as 0. $\overline{3}$, 0.3 or 0.(3); and 2.18, can be written as 2. $\overline{18}$, 2.18 or 2.(18).

CIRCULATION: 1. The movement of blood round the body from and back to the heart. 2. The free movement of something such as air, water or any fluid in an enclosed space. 3. The curvilinear integral $C(\Gamma, t)$ defined as the $\int_{\Gamma} v \cdot dx$, around a curve Γ in the configuration of a body t , where v is the velocity at the point with position vector $\mathbf{x}$.

CIRCULATION PRESERVING MOTION: A motion in which circulation is independent of time.

CIRCULATORY SYSTEM:

The system of vessels including the heart, blood vessels, lymphatic vessels, lymph and other circulatory fluids in an animal's body that transport essential substances such as nutrients, gases, hormones and blood cells to and from the different parts of the body and filter waste materials from the body.

CIRCUM: A prefix, which means round about or around.

CIRCUMCENTRE: The centre of the circle circumscribing a triangle or polygon. It is simply the point where the perpendicular bisectors of a triangle meet. A perpendicular bisector is a line or a ray which cuts another line segment into two equal parts at 90 degrees. A circumcentre lies inside an acute triangle, outside an obtuse triangle and at the centre of the hypotenuse of a right triangle.

CIRCUMCIRCLE (CIRCUMSCRIBED CIRCLE): The circle that passes through all vertices or corner points of a triangle/polygon. The centre of this circle is called the **circumcentre** and its radius is called the **circumradius**.

CIRCUMCISE: 1. To cut off the foreskin of the penis as a religious rite or for hygienic purposes. 2. To cut away the skin prepuce covering the clitoris or to remove the clitoris of a girl or woman, usually as part of a religious ritual.

CIRCUMCISION: The surgical removal of all or part of the foreskin (prepuce) of the penis, usually performed on the newborn for religious (for example, among Jews and Muslims as a sign of God's covenant with the prophet Abraham), cultural or health reasons. **Female circumcision** or **clitoridectomy** is the removal of the labia minora and/or clitoris of adolescents and babies for religious, cultural or health reasons. It is also sometimes done to curb sexual desire in females.

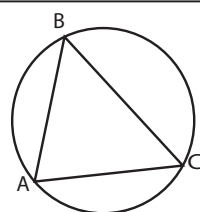
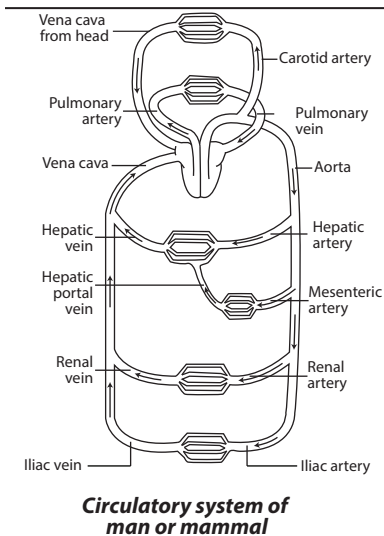
CIRCUMFERENCE: The curved line that encloses a curved plane figure, such as a circle or an ellipse or a line that marks out a circle or other curved figure or a distance round a circle. The formula for finding the circumference C of a circle is: $C = 2\pi r$, where r is the radius.

CIRCUMLUNAR RHYTHM: A biorhythm that corresponds with the lunar cycle. The reproductive cycle of many organisms, especially marine organisms are linked to changing levels of moonlight and the tidal cycle, both of which are governed by the phase of the moon.

CIRCUMNUTATION: 1. An irregular period of oscillation of the Earth's poles. 2. Also known as **nutation**, it is the irregular spiral or elliptical rotation of the apex of a growing stem, root or shoot, caused by differences in the rate of growth of the opposite sides.

CIRCUMSCISSILE: A dehiscent seed vessel that splits or opens along a circumference, with the upper part coming off as a lid.

CIRCUMSCRIBE: To draw a geometric figure around another which passes through all the vertices, in a way that both figures are in contact, but do not intersect. For example, a circumscribed polygon has its sides tangential to a given figure or pass through the endpoints of its sides.



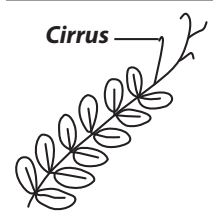
A diagram showing a circle passing through all the vertices of the triangle ABC

Circumcircle (circumscribed circle)

CIRQUE (CORRIE): A steep sided hollow in a mountain side much like a concave amphitheatre. It is formed at the head of a valley glacier by erosion.

CIRRHOSIS: A chronic, non-infectious, often fatal, degenerative disease of the liver associated with drinking excessive alcohol, consumption, malnutrition, viral hepatitis B and C, fatty liver disease, etc. The liver shows formation of scar tissue, hardening and contraction. People with cirrhosis may develop jaundice (yellowing of the skin, eyes and tongue), itching, loss of appetite, easy bruising and extreme tiredness.

CIRRUS: 1. A type of light wispy clouds composed of tiny crystals of ice which occur at above 6 km high in the sky. 2. A tendril on a leaf, as shown in the diagram. **Cirriforous** is bearing a tendril.



CIS: It is a notation which is normally used in complex numbers to represent $\cos\theta + i\sin\theta$.

CIS ARRANGEMENT (COUPLING): The situation in which an individual is heterozygous for two linked genes and one chromosome carries both the recessive or mutant alleles. The homologous chromosome thus carries both the dominant or normal alleles and will mask the effects of the recessives in the phenotype.

CIS DOMINANCE: A property of a sequence or a gene that exerts a dominant effect on a gene to which it is linked.

CIS-: A prefix used to indicate a type of isomerism that atoms or groups are located on the same side of a plane that passes through two carbon atoms linked by a double bond. Example is *cis*-2-butene.

CIS-FUSED: Steric relations at saturated bridgeheads common to two rings are denoted by *cis* or *trans*, followed by a hyphen and placed before the name of the ring system, according to the relative positions of the exocyclic atoms or groups attached to the bridgeheads. Such rings are said to be *cis*-fused or *trans*-fused.

CIS-TRANS ISOMERISM (GEOMETRICAL ISOMERISM): A type of geometrical isomerism found in alkenes in which it is possible for each of the doubly bonded carbons to carry two different atoms or groups. The *cis* isomer has two similar atoms or groups on the same side of a plane bisecting the alkene carbons and perpendicular to the plane of the alkene system. In the *trans* isomer, the atoms or groups are on opposite side.

CIS-TRANS ISOMERS: Stereoisomeric olefins or cycloalkanes or hetero-analogues which differ in the positions of atoms or groups relative to a reference plane. In the *cis*-isomer the atoms are on the same side, in the *trans*-isomer they are on opposite sides.

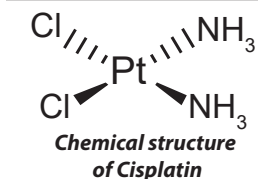
CIS-TRANS TEST (COMPLEMENTATION TEST): A test used to determine whether two independent mutations, affecting the same phenotypic function, occur within the same cistron or on different cistrons. If the mutations occur in the same cistron, only heterozygotes with the *cis* arrangement will show the normal phenotype because the gene in the homologous chromosome is dominant wild type and codes for normal protein. In the *trans* arrangement both copies of the cistron contain a mutation and thus there is no wild-type copy of the gene and hence no normal protein. If the mutations occur in different cistrons, specifying different polypeptides, all heterozygotes will be normal in appearance because the homologous chromosomes carry wild-type copies of the genes.

CISC: In computing, it is acronym for **complex instruction set computer**, a type of microprocessor design and its instruction set with an architecture that contains a large set of computer instructions that varies in size and complexity.

CISCO BORDERLESS NETWORKS: The brand name for a set of hardware and software technologies which allow "anyone, anywhere, anytime and on any device" to connect to an organisation's network.

CISPLATIN (CISPLATINUM; CIS-DIAMMINEDICHLOROPLATINUM(II), CDDP):

A platinum complex with the chemical formula $\text{H}_2\text{Cl}_2\text{N}_2\text{Pt}$ and molar mass 301.1 g/mol. It is used to treat various types of cancers, including sarcomas, some carcinomas such as small cell lung cancer and ovarian cancer, lymphomas and germ cell tumours. It acts by binding between strands of DNA.



CISSOID: A plane curve with the equation $y^2(2a - x) = x^3$. The polar equation is $r = 2a \tan \theta \sin \theta$. It has all points on a variable line passing through a fixed point and whose distance from the fixed point is equal to the distance between the intersections of the line with two given curves.

CISSUS: A genus of about 350 species of tropical and subtropical, chiefly woody vines in the grape family, Vitaceae. Most are climbing plants that clamber up objects by means of tendrils. They generally have shiny, often compound leaves that are often fleshy and somewhat succulent. The berries may be blue, black, red or purple and are generally inedible. The shrubs and vines are grown mainly for their attractive foliage. The species *Cissus quinquangularis*, commonly known as veld grape, is a large succulent climber or trailer with tendrils, which reaches up to a height of 1.5 m. It is widely distributed in the drier regions of Africa and very abundant in the Sahel region. It also grows in parts of Asia (Southeast Asia, Sri Lanka, India, etc). Medicinally, when *cissus quadrangularis* is combined with a healthy diet and exercise plan, it can help promote weight loss. It also reduces blood pressure, blood sugar, triglycerides and cholesterol, thereby reducing the risk of heart diseases. It is used for the treatment or prevention of obesity, diabetes, heart disease, high cholesterol, bone fractures, osteoporosis, scurvy, cancer, upset stomach, haemorrhoids, peptic ulcer disease, painful menstrual periods, asthma, malaria and pain. It is also used in bodybuilding supplements as an alternative to anabolic steroids. The species *Cissus incisa*, commonly known as ivy treebine, possum grape, sorrel vine, is a rapid-growing, semi-evergreen vine native to the southern and south-central United States. It grows up to 9 m long and has compound leaves with three leaflets, bearing a black fruit. *Cissus sicyoides*, known commonly as princess vine, is native of southern Florida to tropical America and has long, slender aerial roots. The plant is used in treating several diseases, including rheumatism, epilepsy, stroke and diabetes.

CISTERNA (CISTERNAE): Any of the flattened membrane-bound sacs that form the branches of the endoplasmic reticulum and Golgi apparatus. The cisternae carry Golgi enzymes to help or to modify cargo proteins travelling through them destined for other parts of the cell.

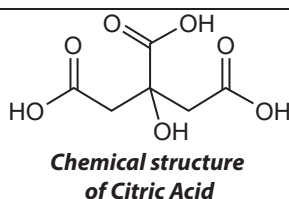
CISTRON: The segment of DNA that is required to synthesise a complete polypeptide chain. The end of one cistron and the beginning of the next can be determined by the cis-trans test.

CITES: Abbreviation for **Convention on International Trade in Endangered Species of Wild Fauna and Flora**, an international agreement under the auspices of the World Conservation Union with the aim of ensuring that international trade in endangered species of animals and plants does not threaten their survival.

CITRATE: 1. Any conjugate base of citric acid such as trisodium citrate, with the chemical formula $\text{C}_3\text{H}_5\text{O}(\text{COOH})_3^{3-}$. 2. Any ester of citric acid, for example, triethyl citrate.

CITRIC ACID (2-HYDROXYPROPANE-1,2,3-TRICARBOXYLIC ACID; 3-CARBOXY-3-HYDROXPENTANEDIOIC ACID):

A weak organic acid, with the chemical formula $\text{C}_3\text{H}_5\text{O}(\text{COOH})_3$, relative density 1.665, melting point 153 °C (426 K) and boiling point 175 °C (448 K). It has a sharp, sour taste and is widely distributed in the plant kingdom and found in high concentrations in citrus fruits, such as oranges and lemons. It is important as an intermediate in the citric acid cycle and therefore, occurs in the metabolism of virtually all living things. It is a natural preservative used in foods and soft drinks.



CITRIC ACID CYCLE (KREBS' CYCLE; TRICARBOXYLIC ACID CYCLE, TCA CYCLE):

The final part of the chain of biochemical reactions essential for the oxidative metabolism of glucose and other simple sugars to form carbon dioxide, water and energy. The reaction in the Krebs' cycle is cyclical since the end product, oxaloacetic acid is also an initial compound for the cycle of reactions.

CITRULLINE: A crystalline amino acid with the molecular formula $\text{C}_6\text{H}_{13}\text{N}_3\text{O}_3$ that is not commonly found in proteins. It was first isolated from the watermelon (*Citrullus vulgaris*) and formed especially as an intermediate in the conversion of ornithine to arginine during urea cycle in the liver.

CITRULLUS: A small dicot genus of herbaceous flowering vines, having two cotyledons (embryonic leaves) in the seed which usually appear at germination. It belongs to the family Cucurbitaceae. The most important species, *Citrullus lanatus* commonly known as watermelon because the fruit contains about 93% water is an annual vine-like flowering plant with stems up to 10 m long, lying or creeping on the ground, with curly tendrils. There are more than 1200 varieties, ranging in weight from less than 0.5 kg to more than 90 kg; with red orange, yellow or white flesh. The leaves are hairy, usually deeply palmate with 3-5 lobes. The flowers are pale green, monoecious, solitary and borne on pedicels with 5 shortly united pale green petals. The fruits vary considerably in morphology. The fruit in the wild form is subglobose, indehiscent and up to 200 mm in diameter. The fruits in the cultivated forms are globose to ellipsoid or oblong, having colours of green or yellowish, evenly coloured or variously mottled or striped. The fruit of water melon has a sweet, pulpy flesh, containing about 6% sugar and about 93% water by weight. It also contains important nutritional compounds including sugars, lycopene and cardiovascular health-promoting amino acids, such as citrulline, arginine and glutathione. The seeds are numerous, more or less ovate in outline, sometimes bordered. In wild forms they are usually black or dark brown; while in cultivated forms they are also white or mottled, mostly 6 -12 mm long. The seeds of water melon have a much higher food value than the flesh and have a nice nutty taste. Significant amounts of vitamin C, minerals, fat, starch and riboflavin have been obtained from them. The seeds can be dried, roasted and eaten as such or ground into flour to make bread. The seed contains a high percentage of oil which is similar to pumpkin seed oil and can be used in cooking. Also, the fruit pulp, fruit juice and seed have all been credited with diuretic properties. A preparation of the seed can lower blood pressure. Some of the other species of *lanatus* include *Citrullus ecirrhosus*, a desert perennial found in both Namibia and South Africa, in particular the Namib Desert; and *Citrullus colocynthis*, a wild vine growing in the desert areas of the west coast of northern Africa, eastward through the Sahara, Egypt until India and reaches also the north coast of the Mediterranean and the Caspian seas. It has analgesic and anti-inflammatory properties.

CITRUS: Any of a group of related fruit trees including lemon, lime orange, grape fruit, etc. Citrus is a common term of the genus *Citrus*, of flowering plants in the rue family, Rutaceae. They grow in tropical and subtropical regions of the world.

CLADDING: 1. The covering of one material with another. 2. In nuclear reactors, cladding is the outer layer of the fuel rods. It is made of zircaloy, a zirconium aluminium alloy, which is corrosion-resistant and has low absorption cross section for thermal neutrons. 3. In metallurgy, cladding is the bonding together of dissimilar metals. It is often achieved by extruding two metals through a die as well as pressing or rolling sheets together under high pressure. 4. In telecommunication, cladding is one or more layers of material of lower refractive index, in intimate contact with a core material of higher refractive index. The cladding causes light to be confined to the core of the fibre by total internal reflection at the boundary between the two. 5. In building construction, **siding** or cladding may refer to the application of one material over another to provide a skin or layer intended to control the infiltration of weather elements or for aesthetic purposes.

CLADE: A group of organisms such as a species that have homologous features derived from a common ancestor. A clade consists of all the descendants of an ancestral organism and the ancestor itself.

CLADISTICS: A method of classification in which the relationships between organisms are represented by a diagram or cladogram, based on selected shared characteristics. In other words, it is a method of biological classification that uses a formal step-by-step procedure for objectively assessing the extent to which organisms share particular characters and for assigning them to taxonomic groups called clades. Within a clade, species that share other characters unique to them are grouped together and so on, until a cladogram or a branching diagram that resembles a family tree is formed.

CLADODE (PHYLLOCLADE; CLADOPHYLL): A flattened stem that is leaf-like in appearance and functions as the main photosynthetic organs of the plant. It is often an adaptation to dry conditions because the stem contains fewer stomata than a leaf and water loss is thus minimised. Examples of plants having cladodes are some cacti and asparagus.

CLADOGENESIS (ADAPTIVE RADIATION; BRANCHING EVOLUTION; CLADOGENESIS; DIVERGENT EVOLUTION): A pattern of evolutionary change that produces biological diversity by budding one or more new species from a parent species.

CLADOGRAM: A type of branching diagram (dendrogram) in which the endpoints of the branches represent specific species of organisms. It is used to illustrate phylogenetic relationships and show points at which various species have diverged from common ancestral forms.

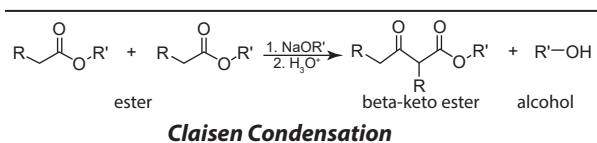
CLADOPHORALES: An order of the Bryopsidophyceae containing both freshwater and marine algae. The thallus of such algae is divided into a number of multinucleate cells.

CLADOPTOSIC: Leaves, branches and stems dropping simultaneously, as seen in, for example, *Taxodium*.

CLAIRAUT'S EQUATION: The differential equation written as $xy' - y + f(y') = 0$.

CLAIRAUT'S FORM: A first-order ordinary or partial differential equation written as $z = \sum_{i=1}^n x_i \frac{\partial z}{\partial x_i} + f\left(\frac{\partial z}{\partial x_1}, \frac{\partial z}{\partial x_2}, \dots, \frac{\partial z}{\partial x_n}\right)$, where f is a differential function. If $n > 1$, it has the complete solution $z = \sum a_i x_i + f(a_1, a_2, \dots, a_n)$, where $a_1, a_2, \dots, a_n$ are arbitrary constants; the general solution for $n = 1$ has the same form. The equation may also have a singular solution or a solution that is not obtainable from the complete solution.

CLAISEN CONDENSATION (CLAISEN REACTION): A condensation reaction between two carbonyl compounds that occurs between two esters or one ester and another carbonyl compound in the presence of a strong base, resulting in the formation of β -keto ester or β -diketone. At least one of the reacting carbonyl compounds must possess an α -proton. It is named after Rainer Ludwig Claisen (1851–1930), a German chemist, who first published his work on the reaction in 1881.



CLAM: A bivalve mollusc found mainly in marine waters of the class Pelecypoda, including members of the genera *Venus* and *Mya*, many of which are edible. It is usually partially buried in sand or mud and respire by means of siphons, which reach to the surface. A clam's shell consists of two, usually equal, halves, which are connected by a hinge joint and a ligament which can be external or internal. Two adductor muscles contract to close the shells. Clam lack head or eyes, but have kidneys, a heart, a mouth and an anus. A fleshy part called the mantle, lies next to the shells. Clams feed on plankton.

CLAMBERING (SCANDENT): Plants often without tendrils that sprawl weakly on other plants.

CLAMP: A mechanical device for holding things tightly together, usually by means of a screw, a piece of wood, metal, etc. It is also used for straightening other materials or fastening things together.

CLAMP CONNECTION: A bulgelike hyphal outgrowth in many basidiomycetous fungi that is formed at cell division over the cross wall between the two daughter cells. It provides a connection through which one of the involved nuclei may pass in order to maintain the binucleate state of the resulting cells.

CLAMPED BOUNDARY CONDITION: A type of boundary condition in spline-fitting in which the first derivatives of the function and the spline agree at the end points.

CLAMSHELL (FLIP PHONE): A cell phone design that has a top cover connected to the lower half, usually the main phone with a hinge that can be opened or closed by slightly pushing the upper half by a fingertip.

CLAN: A social group whose membership is restricted to those claiming descent from a common ancestor, usually through the male line, though sometimes through the female line.

CLAPEYRON-CLAUSIUS EQUATION (CLAUSIUS-CLAPEYRON EQUATION; CLAPEYRON EQUATION): A differential equation that governs the phase transition of a substance. In a two-phase system of the same substance, for example, solid and liquid, heat change is done very slowly so that one phase can convert reversibly into the other while the system remains at equilibrium. If the two phases are denoted by A and B , the Clapeyron-Clausius equation is given by: $\frac{dp}{dT} = \frac{\Delta H}{T(V_B - V_A)}$ where p is the pressure, T is the thermodynamic temperature, ΔH is the change in heat content per mole in the change from A to B and V_B and V_A are the volumes of B and A respectively. In the case of a transition from liquid to vapour, the volume of the liquid can be ignored.

CLAPPER: A piece of metal inside a bell that strikes its sides, making it ring.

CLARIFICATION: The process of removing impurities or solid matter during the production of, for example, fruit juice, milk, sugar, wine, and treatment of water, sewage treatment, using clarifiers. **Clarifiers** are circular settling tanks or basins used to remove solid particulates or suspended solids from liquid by holding the liquid for a period sufficient to allow the impurities and other suspended materials to settle to the bottom. Concentrated impurities, discharged from the bottom of the tank are known as **sludge**, while the particles that float to the surface of the liquid are called **scum**. Clarifiers are also used in many applications such as sewage treatment, power generation, oil and gas, petrochemical, mining and paper industries to aid product separation and recovery.

CLARK CELL: A type of voltaic cell invented by English engineer Josiah Latimer Clark in 1873, consisting of an anode made of zinc-amalgam and a cathode of mercury immersed in a saturated solution of zinc sulfate. It produces a highly stable voltage usable as a laboratory standard.

CLARKE GENERALIZED DIRECTIONAL DERIVATIVE: The function associated with a real-valued locally Lipschitz function, f on a normed space X , by the formula $f^0(a;h) = \limsup_{x \rightarrow a; t \rightarrow 0+} \frac{f(y+th) - f(y)}{t}$.

The function $f^0(x;)$ is always sublinear and coincides with the ordinary directional derivative for a convex or a continuously differentiable function. The **Clarke generalized gradient** is defined as the set of linear functionals with $\phi(h) \leq f^0(x;h)$ for all h in X . This set is nonempty, weak-star compact and convex. It coincides with the subgradient for a convex function and with the gradient for a continuously differentiable function. If the space is finite dimensional it

can be realised as the closed convex hull of all limit points of sequences of gradients of the function at arguments approaching the point.

CLARKE ORBIT (GEOSTATIONARY ORBIT): A circular orbit positioned approximately 35,900 km above Earth's equator and having a period of the same duration and direction as the rotation of the Earth. An object in this orbit will appear stationary relative to the rotating Earth. Communications and weather satellites are usually placed in a geostationary orbit.

CLASPING: A leaf which completely or partially surrounds a stem.

CLASS: 1. A taxonomic category ranking below a phylum or division and above an order, for example, all mammals belong to the class Mammalia and all birds to the class Aves. 2. A set, collection, group or configuration containing members regarded as having certain attributes or traits in common. 3. In computer science, **class** is used in object-oriented programming to describe one or more objects. It serves as a template for creating or instantiating, specific objects within a program. While each object is created from a single class, one class can be used to instantiate multiple objects. Several programming languages support classes, including Java, C++, Objective C and PHP 5 and later.

CLASS EQUATION: The equation $|G| = |Z(G)| + \sum |cl(x_i)|$, where G is a finite group, $Z(G)$ is its centre and $cl(x_1) \dots cl(x_n)$ are all the distinct non-singleton conjugacy classes of G .

CLASS (A) METAL ION: A metal ion that combines preferentially with ligands containing ligating atoms that are the lightest of their Periodic Group.

CLASS (B) METAL ION: A metal ion that combines preferentially with ligands containing ligating atoms other than the lightest of their Periodic Group.

CLASS BOUNDARIES: When data is grouped, it is the boundaries in which the classes are located. Class boundary is the number used to separate the two different classes. It is the midpoint between the upper class limit of a class and the lower class limit of the next class. Examples of class boundaries are 0.5 - 5.5, 5.5 - 10.5 and 10.5 - 15.5. In each class boundary, the lower value is the lower class boundary and the higher value is the upper class boundary.

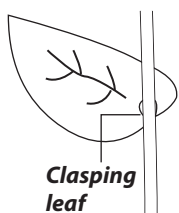
CLASS FREQUENCY: The number of observations in the various class boundaries. For example, in class boundaries of 0.5 - 5.5 and 5.5 - 10.5 with frequencies of 3 and 4, respectively, the class frequencies are 3 and 4, as shown in the table.

Class boundaries	Frequency
0.5-5.5	3
5.5-10.5	4

CLASS LIMITS: They are the smallest and largest data value in a numerical group or set of data; the end values which specify a class interval. Therefore, each class interval has two limits: **a lower and upper**. Lower class limit is the lowest data value or the least value that can go into any class of data. Upper class limit is the largest data value or the largest value that can go into any class of data.

CLASS INTERVAL: The limit of a numerical group or set of data, especially one of the divisions of the base line of a bar chart or histogram. Class intervals may be set by equal groupings at natural breaks in the data or as part of a logarithmic sequence. Examples are 1-5, 6-10, 11-15, etc.

CLASS MARK: A value within the class interval, usually the average of the upper class limit and the lower class limit. For example, in a class interval of 1 - 5 the class mark is expressed as $\frac{1+5}{2} = 3$, and



Clasp leaf

for a class interval of 6-10, it is $\frac{6+10}{2} = 8$, as shown in the table.

CLASS NUMBER: The finite number h_f of equivalence classes of equivalent ideals in the ring of integers, D , of an algebraic number field, F . It is 1 if and only if D is a principal ideal domain.

CLASSICAL: 1. A **classical theory** is one distinguished from some, usually later, variant with a more complex structure. The term is often used to distinguish a variant that the author regards as no longer interesting. 2. A **classical logical** or **mathematical system** is one having the law of excluded middle as an axiom or theorem, so that every statement is known to be either true or false even although that particular statement may not be known. 3. Well-behaved, in terms of some classical theory.

CLASSICAL AUTISM (AUTISM): A developmental disorder that appears in children by the age of three and affects the normal functioning of the brain. Autism is variable in expression but normally characterised and diagnosed by three distinctive behaviours: (i) difficulties with social interaction; (ii) problems with verbal and nonverbal communication; (iii) repetitive behaviours or narrow, obsessive interests rather than with flexible and imaginative ones.

CLASSICAL CATEGORY: In mathematics, a type of category in which the categories are small and the morphisms are functions.

CLASSICAL CONDITIONING: A type of associative learning; the association of a normally irrelevant stimulus with a fixed behavioural response.

CLASSICAL EIGENVALUE PROBLEM: It consists of finding a scalar λ and a non-zero vector x such that $Ax = \lambda Bx$, where A is a square, complex or real matrix and B is the identity matrix or operator.

CLASSICAL FIELD THEORY: A theory that describes a field in terms of classical physics rather than quantum mechanics. Examples of classical field theories include classical electrodynamics, described by Maxwell's equations and the general theory of relativity, describing classical gravitation.

CLASSICAL GENETICS (MENDELIAN GENETICS): The branch of genetics concerned with the study of inheritance through observation of the progeny obtained in breeding experiments.

CLASSICAL PROBABILITY (MATHEMATICAL PROBABILITY): The probability of an event consisting of n out of m possible equally likely occurrences, defined as n/m . 2. The study of classical probabilities.

CLASSICAL PHYSICS: Theoretical physics up to approximately the end of the 19th century, before the concept of quantum theory (1900) and special relativity (1905).

CLASSICAL THEORY: Any physical theory that shares certain features with Newtonian physics, including the assumption that the future is completely determined by the present and that the act of observation has no effect on the system studied. The term is used mainly to label any theory that is not part of quantum mechanics. Einstein's general theory of relativity is considered to be a classical theory.

CLASSIFICATION: 1. A method by which things that share properties together are placed in groups. Biologists group and categorise organisms according to similarities of structure origin, etc. or on the basis of their similarities in biochemical, anatomical or physiological characters. The basic grouping is a species, several of which may constitute a genus, which in turn are grouped into families, then orders, classes, phyla and kingdom. This shows how closely they are related to each other and also indicate existing relationships. 2. In statistics, it is the grouping together of similar objects. If each object is characterised by p variables, classification can be performed according to rational criteria. Depending on the criteria used, an object could potentially belong to several classes.

CLASSROOM ASSESSMENT TECHNIQUES, CATs (ASSESSMENT TECHNIQUES): Methods or categories educators identified in the syllabus, used to gather evidence about students' achievement (students' comprehension of key points during a lesson or a course). The techniques are meant to be a type of formative assessment that also allow teachers to make adjustments to a lesson based on students' needs.

CLASSROOM PERFORMANCE SYSTEM (CPS): A technology system that is used for educational purposes within a classroom. It includes both hardware and software that work together to create a modern, interactive learning environment for students. A typical CPS classroom setup includes a projector, a CPS Chalkboard, computer running CPS software and response pads that are given to the students.

CLASTIC ROCK: A rock made up of fragments or clasts of other pre-existing rocks and minerals. The most common types of clastic rocks are sandstone and shale or mudrock. Clastic rocks are one of the three broad categories of sedimentary rocks. The others are chemical rocks and organic rocks.

CLATHRATE (INCLUSION COMPOUNDS; CAGE COMPOUNDS): A solid compound with a physical structure in which small molecules of one compound or element are trapped in holes in the crystal lattice of another substance. **Clathrate** also describes a lattice-like appearance.

CLATHRATION: A method of removing water from an aqueous solution and, therefore, increasing the solution's concentration.

CLATHRIN: A protein that is the major constituent of the coated pits and vesicles formed during endocytosis of materials at the surface of eukaryotic cells. It was first isolated and named by Barbara Pearse (1948 -), a British biological scientist in 1975. The clathrin coat is a polyhedral lattice of subunits; each subunit or clathrin triskelion comprises three heavy chains and three light chains.

CLAUSIUS-MOSSOTI EQUATION: A relationship between polarisability of a molecule and the dielectric constant of a dielectric substance made up of molecules with this polarisability. The Clausius-

Mossoti equation is given by: $\left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \cdot \frac{M}{d} = \frac{4\pi N_A \alpha}{3}$ where ϵ is the

dielectric constant of a substance, ϵ_0 is the permittivity of a vacuum, M is the molar mass of the substance, d is its density, N_A is Avogadro's constant and α is the molecular polarisability.

CLAUSIUS, RUDOLF JULIUS EMMANUEL (1822-88): A German mathematical physicist and one of the founders of the science of thermodynamics. He was educated at the universities of Berlin and Halle and became a professor at the Polytechnic Institute in Zürich and the universities of Würzburg and Bonn. He is best known for formulating the second law of thermodynamics in 1850, independently of William Thompson (Lord Kelvin). He developed the kinetic theory of gases with Maxwell, showing that molecules are very small relative to the distance between them and that they are in constant, random motion and frequently collide with each other and with the walls of any container.

CLAVATE: Having a shape like a club, which gradually widens towards the apex, as seen in some leaves, as shown in the diagram.

CLAVICLE (COLLAR BONE): A flat bone that forms part of the shoulder by linking the acromion to the sternum (breast bone) in humans. It functions as a rigid support from which the scapula and free limb (arm) are suspended; protects the neurovascular bundle that supplies the upper limb; and transmits physical impacts from the upper limb to the axial skeleton. The clavicle does not resist compression forces very well and is easily fractured.



In some animals such as cats and primates it is the bone or structure with a function similar to that of the human clavicle.

CLAW: 1. The hard, sometimes hooked, pointed nail growth on the end of each toe in birds, some reptiles and some mammals. 2. An appendage used for grasping in crabs and other invertebrates. 3. A mechanical device such as a claw hammer used for gripping and lifting things.

CLAW HAMMER: A hammer with one end of its head bent and divided for pulling out nails.,

CLAY: 1. A naturally occurring particle-size material, with a diameter below 0.002 mm containing aluminosilicate, that becomes hard when dried or baked, making it impermeable. It is a very fine-grained sedimentary deposit that has undergone a greater or lesser degree of consolidation and has very small particles, few air spaces and little humus. It may be white, grey, red, yellow, blue or black, depending on its composition and it is used for making bricks and pottery. The various types of clay include adobe, alluvial clay, building clay, brick, cement, china clay, etc. **Clayey** means containing large amounts of clay, or having properties similar to those of clay.

CLAY FILMS (CLAY COATINGS, CLAY SKINS): Thin coatings of oriented clays on the surfaces of soil peds and mineral grains, and in soil pores.

CLAY HYBRID: Organic-inorganic composite material in which one of the components is clay, with the particles dispersed in a polymer.

CLAY LOAM: Soil material that contains 27% to 40% clay and 20% to 45% sand.

CLAY MINERALS: Very small particles, chiefly hydrous silicates of aluminium, sometimes with magnesium and/or iron substituting for all or part of the aluminium, that are the major constituents of clay materials. The particles are essentially crystalline (either platy or fibrous) with a layered structure, but may be amorphous or metalloid. The clay minerals are responsible for the plastic properties of clay; the particles have the property of being able to hold water. Some examples are kaolinite $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$, halloysite $\text{Al}_2\text{Si}_2(\text{OH})_6 \cdot 4\text{H}_2\text{O}$ and illite $\text{KAl}_2(\text{Si}_4\text{Al})\text{O}_{10} \cdot 2\text{H}_2\text{O}$.

CLAYING: The improvement of sandy and black fen soils by the addition of clay.

CLEAN CATTLE: Cattle that has not been used for breeding.

CLEAN CULTURE: Any practice that removes the breeding or overwintering sites of a pest.

CLEAN INSTALL: An operating system (OS) installation that overwrites all other content on the hard disk. Unlike a typical OS upgrade, a clean install removes the current operating system and user files during the installation process. When a clean install finishes, the hard disk only contains the new operating system, similar to a computer that is used for the first time.

CLEAN ROOM: A highly purified and regulated room for producing equipment sensitive to contamination. For example, the production of microprocessors is done in a clean room, since the smallest amounts of dust could create flaws in their production. Therefore, the air in a clean room is constantly filtered. Other aspects such as temperature, humidity and pressure may be regulated as well.

CLEAN SURFACE: An experimental surface having no surface contamination observable by means of the used method.

CLEAN UP: A process in spectrochemical analysis caused by sputtering, trapping or adsorbing metallic atoms, gas atoms or molecules on the walls of the vessel or anode. This has the effect of reducing the gas pressure in sealed lamps and may be useful for cleaning up or gettering unwanted gases.

CLEANLINESS UNIT: A unit for measuring air pollution.

CLEAR-CUT: Removal of all marketable trees from an area.

CLEARANCE: 1. Product of the concentration of a component in an output system and the volume flow rate of that output system divided by the concentration of this component in the input system. 2. In toxicology, it is the volume of blood or plasma or mass of an organ effectively cleared of a substance by elimination (metabolism and excretion) divided by time of elimination. **Total clearance** is the sum of the clearances of each eliminating organ or tissue for a given substance.

CLEARING: A stage in the preparation of permanent slides for microscopical examination in which the stained dehydrated cells or tissue sections are immersed in a clearing agent to remove the alcohol that was used to dehydrate the preparation. The process is essential if the dehydrating agent and the mounting or embedding medium are not miscible. Examples of clearing agents include xylene (xylol), toluene and 1,1,1-trichloro-ethane and various essential oils, such as clove oil and thyme oil.

CLEARING POINT: The temperature at which the transition between the mesophase with the highest temperature range and the isotropic phase occurs.

CLEARTEXT: In cryptography, it is the unencrypted transmitted or stored original text that does not require decryption. Any text that can be read on a computer with the appropriate program but without the need for decryption is considered cleartext. Plaintext is not a synonym of cleartext. **Plaintext** is the input to an encryption algorithm or meant to be encrypted prior to transmission or storage or the result of decryption of received or retrieved data.

CLEARTYPE: Technology developed by Microsoft in 1998 for displaying computer fonts so that they appear clear and smooth and, therefore, easier to read for long periods of time without experiencing eyestrain and mental fatigue. It is used in electronic book readers and personal digital assistants and first used on Pocket PC devices in 2000. By addressing an area smaller than a pixel, ClearType makes screen type easier to read, especially on LCD devices, including flat-panel monitors, mobile PCs and smaller hand-held devices.

CLEAVAGE: 1. Splitting, division or separation of an object. 2. The tendency of a mineral or a crystallised substance to split along defined parallel planes related to its internal structure, yielding smooth surfaces. It occurs where bonding between atoms is weakest. 3. The series of cell divisions by which a single fertilised egg cell is transformed into a multicellular body, the blastula.

CLEAVAGE FURROW: A groove formed from the cell membrane in a dividing cell as the contractile ring tightens.

CLEAVAGE POLYEMBRYONY: The splitting of an embryo into several identical parts each of which is capable of developing into a mature embryo. It is seen in some gymnosperms and is particularly common in Pinus, where the zygote divides to form four nuclei, each of which divides to give a chain of cells or suspensor. At the tip of each suspensor is a group of embryo initials. Usually, however, only one embryo develops to maturity.

CLEAVAGE SITE (RECOGNITION SITE; RESTRICTION RECOGNITION SITE): The site in a polynucleotide chain as to where the restriction enzyme cleaves nucleotides by hydrolysing the phosphodiester bond between them. The restriction enzyme recognises the site as to where it cuts via a specific sequence of nucleotides along the polynucleotide chain. For example, when the restriction enzyme, Eco RI, recognises the palindromic sequence GAATTC, it cuts between G and A on the top and bottom strands.

CLEFT: 1. An opening or fissure which extends about halfway to the midrib or base of certain leaves, as shown in the diagram. 2. An abnormal fissure or indentation which results from failure of parts to fuse during embryonic development. 3. A hollow between ridges or protuberances such as the anal cleft in humans.



Cleft leaf

CLEFT PALATE (PALATOSCHISIS; URANOSCHISIS; URANOSTAPHYLOSCHISIS): A common developmental deformity of the palate often

associated with harelip, in which the two halves do not meet in the midline. It can be familial or follow disease in early pregnancy, but may appear spontaneously. It causes a characteristic nasal quality in the voice. Plastic surgery can close the defect and allow more normal development of the voice and teeth.

CLEIDOIC EGG: An egg enclosed by a shell for protection against the environment and containing all the food the embryo needs to develop until hatching. An example is a reptile egg.

CLEISTANTHERIC POLLINATION PATHWAY: Pollination in which pollen is not transferred from a normally dehiscent (split at maturity) anther by a pollinating agent. The pollen grain germinates within the anther with subsequent growth of the pollen tube through the anther wall and ovary wall into the ovule and embryo sac.

CLEISTOCARP (CLEISTOTHECIUM): The type of ascocarp characteristic of fungi in the Plectomycetes. It is a globose structure with no specialised opening to the exterior.

CLEISTOGAMY: Flowers that never fully open and that are automatically self-fertilised and contain numerous seeds. Cleistogamous fertilisation takes place within a flower that does not open and enables seed production in the absence of normal cross-pollination. Examples of cleistogamous flowers are dayflower and spiderwort and other Commelina species.

CLEISTOGENE: A plant bearing cleistogamous flowers. **Cleistogenes** is a perennial grass genus of the family Gramineae that is endemic to the steppes of northern China, the Caucasus, Siberia and Middle Asia. They have bisexual spikelets with one or several flowers clustered in panicles. Cattle feed on some species such as *C. squarrosa*.

CLEMMENSEN REDUCTION: A method of reducing a carbonyl group (C=O) to CH₂ using zinc amalgam and concentrated hydrochloric acid. The Clemmensen reduction is particularly effective at reducing aryl-alkyl ketones. With aliphatic or cyclic ketones, zinc metal reduction is much more effective.

CLEOMACEAE: A small family of flowering plants in the order Brassicales, comprising about 300 species in 10 genera, or about 150 species in 17 genera. Their leaves are palmately compound, and they have six stamens. Some species have medicinal properties, for example, *Cleome viscosa*, commonly called the Asian spiderflower or tick weed is used to treat (mainly the leaves and seeds) various diseases such as fever, rheumatism, headache, gonorrhoea, inflammation of the middle ear, wounds and ulcers. In certain cultures, the leaf is eaten as vegetable.

CLEPSYDRA (WATER CLOCK): An instrument in which the discharge of water from a storage tank is monitored in order to measure the passing of time.

CLEVIS: An U-shaped iron attachment, used to couple an implement to a tractor towbar.

CLICK: In computing, it describes the action of pressing and releasing a mouse button one or more times. This generates an action. For example, using a mouse, to point to an icon on screen and pressing the left hand mouse button twice in quick succession (called **double clicking**) to select a file, program, or function for various operations such as copy, cut, delete, drag, etc. **Click and drag** involves moving a selected icon to some other location by holding down the mouse

button and moving or dragging the icon until the location is reached and the mouse button is released. **Left-clicking** is pressing the left hand side button used for many common computer tasks, such as selecting objects, opening hyperlinks, and closing windows. **Right-click** is pressing the right hand side button, which is less used but opens pop-up menus, etc. **Middle-click** (pressing the middle button on a three-button mouse or using the mouse wheel as a button) will perform any pre-programmed option setup through the mouse settings or a function setup through the program. For example, in most Internet browsers, using the mouse wheel as a button on a link opens that link in a new. **Shift click** (click, holding down the shift key, and clicking again) can highlight text, icons, or other objects. **Ctrl click** is to click, holding down the Ctrl key, and clicking again to select multiple files or folders.

CLICK-BEETLE (SKIPJACK; SNAPPING BEETLE): Any of a group of beetles that turn themselves over by snapping their bodies and throwing themselves into the air. They make an audible click in the process. Their long, cylindrical larvae are called **wireworms**.

CLICK-THROUGH RATE (CTR): A term used in online advertising to refer to revenue Web publishers generate from advertisers each time a visitor clicks on one of the advertisements (the PPC model). The higher the click-through rate, the more revenue a publisher will generate.

CLIENT-SERVER ARCHITECTURE (CLIENT/SERVER): A system in which the programs the supply data, such as files, devices and even processing power, called servers are separated from the programs that use the data, called clients. The server might be a central database, typically located on powerful computers or processes. The client would be an ordinary program located in users PCs that requests data from the server as needed.

CLIFF: A high steep ice or rock face, especially a rock face extending along a coastline, mountainous areas, escarpments and along rivers. Cliffs are formed as erosion landforms due to the processes of erosion and weathering that produce them; and by rock that is resistant to erosion and weathering.

CLIMACTERIC: 1. A stage of fruit ripening associated with ethylene production and cell respiration rise. Fruits such as apples, bananas, melons, apricots and tomatoes are climacteric fruits. Citrus, grapes, strawberries are non-climacteric (they ripen without ethylene and respiration bursts). 2. The period of life when physical powers begin to decline. For women in menopause, this may be evident. A corresponding period sometimes occur in men that may be marked by a reduction in sexual activity, although fertility is retained.

CLIMATE: The long term or regular pattern of weather conditions of a particular region. These conditions include temperature, humidity, atmospheric pressure, wind, rainfall, atmospheric particle count and other meteorological elements. The primary factors that determine the variations of climate over the surface of the Earth include latitude, movement of wind belts, location of area in relation to ocean currents, etc.

CLIMATE CHANGE: A change in the climate patterns of global or regional dimensions over an appreciable period of time that can be identified, especially a change due to an increase in the average atmospheric temperature. Climate fluctuations are natural phenomena, but there is increasing evidence that human activity, such as deforestation and the increased levels of atmospheric carbon dioxide produced by the use of fossil fuels affect the global climate, particularly in terms of global warming. Some of the effects of climate change are global widespread temperature increase; increase in average Arctic temperatures of almost twice the global average rate in the past 100 years; land regions have warmed faster than the oceans; global increased or declining precipitations; etc.

CLIMATE CLASSIFICATION: Description of different types of climate, taking into account the averages, extremes and frequencies of all meteorological elements such as temperature, atmospheric pressure,

precipitation, wind, humidity and sunshine, together with the factors that influence them. The different types of climate can be described as tropical (hot), warm temperate (or subtropical), cool temperate, cold (or polar) and arctic. There are many different methods of classifying climate, which can be put under two main methods: empiric or genetic.

Genetic classifications group climates by their causes: (i) those based on the geographic determinants of climate, (ii) those based on the surface energy budget and (iii) those derived from air mass analysis; and **empirical classifications** such as the Köppen climate classification, also known as Köppen–Geiger climate classification system that group climates based on one or more aspects of the climate system such as local natural vegetation, which is considered as the best expression of climate. Thus, climate zone boundaries have been selected with vegetation distribution in mind. It combines average annual and monthly temperatures and precipitation and the seasonality of precipitation. The climate zone boundaries are grouped as Group A comprising tropical/megathermal climates, Group B comprising dry (arid and semiarid) climates, Group C comprising temperate/mesothermal climates, Group D comprising continental/microthermal climates and Group E comprising polar and alpine climates.

CLIMATE FEEDBACK: An interaction mechanism between processes in the climate system, when the result of an initial process triggers changes in a second process that in turn influences the initial one. A positive feedback intensifies the original process, and a negative feedback reduces it. For example, a positive feedback accelerates a temperature rise, whereas a negative feedback decelerates it. It is important in the understanding of global warming because feedback processes may amplify or diminish the effect of each climate forcing, and so play an important part in determining the climate sensitivity and future climate state.

CLIMATE FORCING: Any influence on climate that originates from outside the climate system itself. In other words, climate forcings are different factors that affect the Earth's climate. It has to do with the amount of energy that is radiated from the Sun, and the amount of energy that is radiated back into space. Variances in climate forcing are determined by physical influences on the atmosphere such as orbital and axial changes as well as the amount of greenhouse gas in the atmosphere. Climate forcings are a major cause of climate change. Examples of external forcings include: surface reflectivity (albedo), human induced changes in greenhouse gases, atmospheric aerosols (volcanic sulfates, industrial output). These examples all influence the balance of energy entering and leaving the Earth system. These types of forcings are often referred to as **radiative forcing** and can be quantified in units of the extra energy in watts per metre squared ($W m^{-2}$) entering the Earth near the top of the atmosphere (TOA).

CLIMATE IMPACT ASSESSMENT (IMPACT ASSESSMENT): The practice of identifying and evaluating the detrimental and beneficial consequences of climate change on human systems, infrastructure, ecosystems, or valued natural resources.

CLIMATE IMPACTS: Consequences of climate change on natural and human systems. Depending on the consideration of adaptation, one can distinguish between potential impacts and residual impacts. Potential impacts are all impacts that may occur given a projected change in climate, without considering adaptation. Residual impacts are the impacts of climate change that would occur after adaptation.

CLIMATE MODEL/HIERARCHY: A numerical representation of the climate system based on the physical, chemical, and biological properties of its components, their interactions and feedback processes, and accounting for all or some of its known properties. It is essentially a representation of the many interactions within the climate which includes the atmosphere, oceans, land surface, and ice. Climate models can be used to study and simulate the climate, as well as to predict monthly, seasonal, and interannual global temperature, and increases in temperature due to greenhouse gases.

CLIMATE PREDICTION/FORECAST: The result of an attempt to produce a most likely description or estimate of the actual evolution of the climate in the future, for example, at seasonal, interannual, or long-term time-scales, using deterministic models.

CLIMATE PROJECTION: The simulated response of the climate system to emission or concentration scenarios of greenhouse gases and aerosols, often using climate models. Climate projection is not the same as climate prediction/forecast. A prediction/forecast is a projection that is "most likely" to occur in the future.

CLIMATE SCENARIO: A plausible and often simplified representation of the future global climate, based on an internally consistent set of climatological relationships that has been constructed for explicit use in investigating the potential consequences of anthropogenic climate change, often serving as input to impact models.

CLIMATE SENSITIVITY: In Intergovernmental Panel on Climate Change (IPCC) assessments, "equilibrium climate sensitivity" refers to the equilibrium change in global mean surface temperature following a doubling of the atmospheric (equivalent) carbon dioxide (CO₂) concentration.

CLIMATE SYSTEM: The highly complex interactive system consisting of five major components: the atmosphere, the hydrosphere, the cryosphere, the land surface and the biosphere. The climate system is forced or influenced by various external forcing mechanisms, the most important of which is the Sun. It also includes volcanic eruptions, solar variations, and human-induced forcings such as the changing composition of the atmosphere and land-use change.

CLIMATE VARIABILITY: Short term changes in the pattern of the climate. This contrasts with climate change, which is long-term. An example of climate variability is the fluctuation of temperature around the average without causing the long-term average itself to change. Variability may be due to natural internal processes within the climate system, called internal variability, or to variations in natural or anthropogenic external forcing, called external variability.

CLIMATIC FACTORS: Aspects of the weather, such as temperature, rainfall, light, humidity and air movement that influence the life of organisms.

CLIMATIC INDEX: A simple, calculated numerical value that expresses climatic relationships or used to describe the state and the changes in the climate system.

CLIMATIC MODEL (CLIMATE MODEL): A computer simulation, based on physical and mathematical data of the entire climatic system of the earth. It is used for a variety of purposes including projections of future climate.

CLIMATIC REGION: A geographical area which experiences distinct fairly uniform weather conditions. No two localities on Earth may be said to have exactly the same climate, but widely separated areas of the world can possess similar climates. These climatic regions have some comparable physical and environmental features as well as having similar weather patterns. Climatic regions are differentiated by weather conditions (including temperature, humidity, precipitation type and amount, wind speed and direction, atmospheric pressure, sunshine, cloud types and cloud coverage) and by weather phenomena such as thunderstorms, fog and frost that have prevailed there over a long period of time, usually 30 years.

CLIMATOLOGY: The scientific study of climate, including the causes and long-term effects of variation in regional and global climates over a defined period of time, past, present or future. Climatology is also the study of how climate changes over time and is affected by human actions. Climatologists are scientists who study long-term trends in climate. In contrast, meteorologists study current weather conditions and make short-term forecasts.

CLIMAX COMMUNITY: A relatively stable ecological community that is achieved at the end of a succession. This equilibrium occurs because the climax community is composed of species best adapted to average conditions in that area. However, there are often factors

that halt or slow down succession and prevent the establishment of a climax community. These factors include changes in climate, natural hazards, disease, animal interference, fire and human activity. Often intensive grazing prevents the growth of shrubs and trees. The equilibrium achieved in this way is termed a biotic climax, disclimax, plagioclimax or subclimax vegetation. **Climax vegetation** are the plants that inhabit a climax community.

CLIMBERS: Opportunistic plants with long, flexible stems, often not self-supporting but using neighbours or some type of structure for support. Climbing plants grow by twining or supporting themselves on another structure.

CLIMBING ROOT: Any of the short adventitious roots that develop from the stems of certain climbing plants, such as ivy (*Hedera helix*) and serve to attach the plant to its support. Climbing roots are negatively phototropic and thus grow into darkened tissues in the bark or crevices in walls. Their function is enhanced by the secretion of mucilaginous substances from the root tip.

CLIMOGRAPH: A graphical representation of annual cycle of temperature and rainfall for a particular geographical area.

CLIMOSEQUENCE: A sequence of related soils, usually on the same parent material, that differ from one another in certain properties primarily as a result of the effect of climatic conditions as a soil-forming factor. It is usually found along mountain slopes in certain highland areas.

CLINANDRIUM: A hollow in the upper part of the column of an orchid flower that contains the anthers.

CLINE: A gradual variation in the characteristics of a species or population over its geographical range of the population to the other. It occurs in response to varying environmental factors, such as soil type or climate. In species of birds and mammals, there is usually a cline in body size, with smaller individuals in warm climates and larger individuals tending to be found in colder climates.

CLINIC: 1. A private or specialised hospital or a place or session at which specialised medical treatment or advice is given to visiting patients. 2. An occasion in a hospital when students learn by watching how a specialist examines and treats his patients.

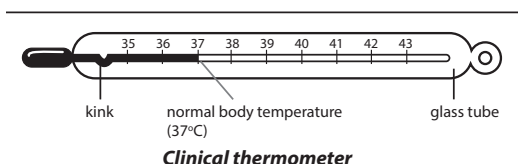
CLINICAL: Relating to the examination and treatment of patients and their illnesses.

CLINICAL DEATH: The medical term for cessation of blood circulation and breathing, the two necessary criteria to sustain life. It occurs when the heart stops beating in a regular rhythm, a condition called **cardiac arrest**. At the onset of clinical death, consciousness is lost within several seconds. Measurable brain activity stops within 20 to 40 seconds. The term is also sometimes used in resuscitation research. During clinical death, all tissues and organs in the body steadily accumulate a type of injury called **ischaemic injury**.

CLINICAL ECOLOGY: A psychological field that studies environmental factors involved in human health, especially illnesses such as fatigue, depression, allergic reactions and immune-system malfunctions that do not manifest specific symptoms and prescribing means of avoiding or minimising these effects.

CLINICAL THERMOMETER: An instrument for measuring the temperature of the human body. It has the temperature range of 30 °C to 40 °C. Normal body temperature varies by person, age, activity and time of day. The average normal body temperature is generally 37 °C. The stem of the thermometer is thickened on the side far from the graduations so that it acts as a lens, to magnify the mercury thread. At room temperature, the mercury retreats right into the bulb. Between the bulb and the graduation is a constriction or kink. When the bulb is warmed the mercury is forced through the constriction into the stem. When the bulb is cooled, the mercury does not flow back to the kink but stays in the bulb so that the temperature can be read at leisure. After the temperature is recorded, the thermometer is shaken to

force the mercury back into the bulb for another temperature reading.



CLINICAL TRIAL: Any research study that evaluates the effectiveness of medical treatment in a systematic fashion and prospectively assigns human participants or groups of humans to one or more health-related interventions to evaluate the effects on health outcomes. It may involve drugs, cells and other biological products, surgical procedures, radiological procedures, devices, behavioural treatments, process-of-care changes, preventive care, etc. Ethical standards are maintained by ethics committees and they also ensure that the clinical trial procedure is explained to patients.

CLINKER: 1. A rough, partially fused product left in a furnace after firing, which is ground for use in cement. 2. A vitrified or partially vitrified residue of coal which has been burnt in a furnace.

CLINOGRAPH: 1. A diagram that shows both the average monthly temperature and precipitation of a place. 2. An instrument that records the deviation of boreholes, well or shaft from the vertical. 3. A drawing instrument having a pair of straightedges hinged together so as to be adjustable to any angle.

CLINOSEQUENCE: A sequence of related soils that differ from one another as a result of the effect of the angle of slope of the soil surface on which they were formed.

CLINOSTAT: A mechanical device that rotates whole plants, usually with seedlings attached to it and by so doing, removes the effect of gravity on the plants. It is used to study the growth of plants when the effect of gravity has been removed.

CLINOTAXIS (KLINOTAXIS; CLINOTACTIC; KLINOTACTIC): 1. A directed reaction or orientation response of a motile organism to a gradient of stimulation. 2. A taxis in which the movement is one of turning.

CLIP ART: A collection of pictures or images that can be imported into a document or another program. The images may be either raster graphics or vector graphics. Clip art galleries may contain from a few images to hundreds of thousands of images. It is typically organised into categories, such as people, objects, nature, etc., which is especially helpful when browsing through thousands of images.

CLIPBOARD: The location in a computer operating system such as Microsoft Windows that stores information that has been cut or copied from a document or other location. The clipboard will hold this information until it has been overwritten by new information.

CLIPPER CHIP: An encryption microchip that contains an identifying code inserted into encrypted transmissions that allow authorized third parties access to the encrypted data. It was developed by the US National Security Agency as part of its Capstone project to insert the chip in all devices such as computers, modems, telephones and televisions that might use encryption.

CLIQUE: A connected subgraph of another graph such that its induced subgraph is complete with every two distinct vertices in the clique adjacent.

CLISERE: A succession of different climax communities in a particular area brought about by changes in the climate.

CLITELLUM: A thickened glandular section, which is part of the reproductive system in the body wall of some worms such as earthworms and leeches and some other annelids, similar in shape to a saddle, that secretes a viscid sac in which the eggs are deposited. It slides off the body once the eggs have been deposited.

CLITORIDECTOMIES (FEMALE CIRCUMCISION): The excision of the labia minora and/or clitoris, carried out in some societies

in Africa and the Middle East that is usually performed as part of female initiation rites. It is also sometimes done to curb sexual desire in females.

CLITORIS: An erectile rod tissue in female mammals and also some reptiles and birds that is the equivalent of the male penis, but the clitoris does not contain the distal portion of the urethra as in penis, except in the spotted hyena. In some women it may be quite visible, in others it may be small and barely visible and may be so concealed that it can be viewed only when the labia majora are separated. In humans, the upper portion is located near the anterior junction of the labia minora, above the opening of the urethra and vagina. In human females, when stimulated, it may produce sexual excitement, clitoral erection, sexual pleasure and orgasm.

CLOACA: The enlarged posterior single chamber of the alimentary canal in most vertebrates into which the terminal part of the digestive, urinary and the reproductive ducts open. Cloacae are found in most reptiles, birds, amphibians and many fishes. Placental mammals have a separate anus and urogenital opening.

CLOACAL SPUR (PELVIC SPUR; VESTIGAL LIMBS): A claw in boas and pythons, located on each side of the cloaca, used by the male snake, while mating.

CLOCHE: A portable glass or plastic cover used to protect outdoor plants.

CLOCK: Any device that measures time in seconds, minutes and hours. It is usually shown by means of pointers moving over a dial, called analogue clocks or by a digital display. Mechanical clock's recurring action depends on the swing of a pendulum or the oscillation of a balance wheel and balance spring or hairspring or the vibration of a tuning fork, mechanisms capable of repeating their cyclic movements with great regularity.

CLOCK ARITHMETIC: Arithmetic to a given finite modulus, analogous to the numbers on a clock face, for which $12 + 1 \equiv 1 \pmod{12}$.

CLOCK CYCLE: A single electronic pulse of a CPU. During each cycle, a CPU can perform a basic operation such as fetching an instruction, accessing memory or writing data. Since only simple commands can be performed during each cycle, most CPU processes require multiple clock cycles.

CLOCK PULSE: A continuous, precisely spaced changes in voltage, which can be used to measure the speed of a computer processor by counting each pulse. The speed of the computer's processor, measured in MHz or GHz, refers to its number of clock pulse cycles per second. Clock pulse is also used to synchronise the timing of hardware components or any other electronic system.

CLOCK REACTION (OSCILLATING REACTION): A complex mixture of reacting chemical in which the concentrations of the products and reactants change periodically, either with time or with position in the reacting medium. Thus, the concentration of a component may increase with time to maximum, decrease to a minimum, then increase again and so on, continuing the oscillation over a period of time. They are a class of reactions that serve as an example of non-equilibrium thermodynamics, resulting in the establishment of a nonlinear oscillator.

CLOCK SPEED: The rate at which a processor can complete a processing cycle. It is typically measured in megahertz or gigahertz. One megahertz is equal to one million cycles per second, while one gigahertz equals one billion cycles per second. This means a 1.8 GHz processor has twice the clock speed of a 900 MHz processor.

CLOCKING: 1. In binary synchronous communication, the use of clock pulses to control synchronisation of data and control characters. 2. A method of controlling the number of data bits sent on a telecommunication line in a given time.

CLOCKWISE: Rotation in the direction normally followed by the hands of an analogue clock. **Anticlockwise**, also referred to as **counterclockwise** or **conrtraclockwise** is the opposite direction to the rotation of the hands of a clock.

CLOD: A large compact, coherent lump of earth or clay produced by digging or ploughing.

CLONAL FRAGMENTATION (FRAGMENTATION): A common type of vegetative reproduction in plants by which many trees, shrubs, perennials and ferns form clonal (cell, group of cells or organisms descended from and genetically identical to a single common ancestor) colonies by reproducing new rooted shoots by rhizomes or stolon which increases the diameter of the colony.

CLONAL SELECTION THEORY: A theory explaining the specificity of the adaptive immune response and how the cells of the immune system produce large quantities of the right antibody at the right time, each having a genetically predetermined specificity for only one of a vast array of possible antigens. Upon encountering an antigen, the lymphocytes sensitive to it reproduce much more rapidly than the others, thus leading to a build-up of antigen-specific cells large enough to mount the response.

CLONE: 1. A population of genetically identical cells produced from a common ancestor; or a plant, animal or other organism that is genetically identical to its parents, having developed by vegetative reproduction from a bud, cutting or other part or, in experimental conditions, from a single cell. 2. A computer, hardware device or software program that is made to be similar to an existing device or technology. A cloned product is almost always made by a third-party company and not the company that originally designed it.

CLONING: The process of making identical copies of an organism, cell or DNA sequence. **Molecular cloning** is a process of amplifying a desired DNA sequence. The target sequence is isolated, inserted into another DNA molecule (known as a vector) and introduced into a suitable host cell. Then, each time the host cell divides, it replicates the foreign DNA sequence along with its own DNA. Cloning also can refer to asexual reproduction.

CLONING VECTOR: A self-replicating piece of DNA to which a foreign DNA can be covalently attached for purposes of amplification in host cells.

CLONOTYPE: 1. An attribute or specificity that is unique to a particular clone of cells, especially the structure of the surface receptors of a T-cell or of the secreted antibodies of a plasma cell. 2. A plant or line of plants that has been propagated by the cloning of a type of specimen.

CLOSE: 1. In computing, it is an action used to exit a window or a program thereby removing it from the screen. If a file is opened and is being worked on and the program is closed, any changes made to the opened file may be lost if the changes are not saved before closing. 2. Close can also be used to refer to a button or a feature in a graphic user interface (GUI) operating system, (the close box or close button) that allows a user to terminate a window from a program or the complete program.

CLOSE BREEDING: Mating of a male parent to a female offspring, female parent to male offspring or between offspring of the same parents in certain farm animals.

CLOSE JUNCTION (TIGHT JUNCTIONS; OCCLUDING JUNCTIONS): Cell-cell junctions that seal adjacent epithelial cells together, preventing the passage of most dissolved molecules from one side of the epithelial sheet to the other.

CLOSE PACKING: The packing of spheres so as to occupy the minimum amount of space.

CLOSED: 1. Of a set under an operation, containing all members of the produced by that operation acting on members of the set. For example, the positive integers are closed under addition but not under subtraction, since $n + m$ is also a positive integer for any positive integers n and m , but $n - m$ may be negative or zero and so not in the set. 2. Of a curve or surface, completely enclosing an area or volume. 3. Of a set in a topology, containing all its limit points, being the complement of an open set. 4. Of a set, being the algebraic closure

of some set. 5. Of a function or multivalued function, possessing a graph that is topologically closed. 6. Of a function between two topological spaces, sending closed sets to closed sets. 7. Of a path in a graph, having the same vertex at each end. 8. Of a differential form, having exterior differential equal to zero. 9. Of a branch in a semantic tableau, containing inconsistent proportions. If every branch is closed, then the tableau is said to be closed; this shows the given set of propositions to be inconsistent.

CLOSED BALL: A closed set in a metric space consisting of all the points, whose distance from a given point is less than or equal to a given constant. It is denoted by $B(a, \quad)$ or $B(a)$.

CLOSED CIRCUIT: 1. An electrical circuit that provides a continuous path through which current can flow without interruption. 2. **Closed-circuit television (CCTV)** is a localised television system in which programs are sent over relatively short distances, with the camera, receiver and controls being linked by cable. It is mostly used as a security device in watching different parts or rooms of a building.

CLOSED CIRCULATORY SYSTEM: A type of internal transport in which blood is confined to blood vessels (arteries, veins and capillaries) as found in vertebrates.

CLOSED CURVE: A continuous curve such as a loop with no end points, completely enclosing an area. It is the continuous image of a closed set and has coordinate functions such that each coordinate is a continuous function $f_i(t)$ on the interval $[0,1]$ of the real line and $f_i(0) = f_i(1)$. A **simple closed curve** is a closed curve that does not intersect with itself.

CLOSED FORM: An expression for a given function or quantity, especially an integral, in terms of known and well-understood quantities such as the evaluation of $\int_{-\infty}^{\infty} \exp(-x^2) dx$ as $\sqrt{\pi}$.

CLOSED GRAPH THEOREM: A theorem characterising continuous functions in terms of their graphs that a given function or multi-valued function with a closed graph is continuous or semi-continuous. The classical closed graph theorem proves that a linear operator between Banach or Frechet spaces is continuous if and only if it has a closed graph.

CLOSED INTERVAL: A set of numbers that lie between and including its endpoints. It is denoted by square brackets written as $[x, y]$, where x is both the minimum and infimum of the interval and y both its maximum and supremum. For example, the closed interval of 4 and 10 is $[4 \leq x \leq 10]$ and also $[x, \infty]$ and $[-\infty, x]$ are regarded as closed.

CLOSED LOOP CONTROL SYSTEM: A set of mechanical or electronic devices that automatically regulates an operation, process, or mechanism by feedback variable to a desired state or set point without human interaction. This enables automation and are widely used in industry applications including agriculture, chemical plants, quality control, nuclear power plants, water treatment plants and environmental control. On the other hand, open loop system does not monitor or measure the condition of its output signal as there is no feedback.

CLOSED MAPPING: A function or multivalued function between topological spaces that maps closed sets to closed sets.

CLOSED NEIGHBOURHOOD: In mathematics (Euclidean or metric space), a neighbourhood defined by $\{x: d(x, a) \leq \varepsilon\}$.

CLOSED PATH: A graph whose ending point is identical with its initial point; it begins and ends at the same vertex without traversing any edge more than once.

CLOSED SENTENCE: In logic, an expression that contains no free occurrence of any variable, so that all its variables are bound by quantifiers. In other words, a closed sentence is always true or always false. Examples of closed sentences are: 20 is an even number is always true, therefore, is closed; and 19 is an even number is always false, therefore, is closed. A sentence is open when it is not known if it is true or false.

CLOSED SET: A concept used in mathematics to denote a set of data for which an operation such as addition or multiplication done on any member of the set gives a result that is also a member of the set. A set is closed under a given operation if and only if the operation on two elements of the set produces an element of the same set. For example in the set $X = \{2, 4, 6, 8, 10\}$ is closed under subtraction.

CLOSED SHELL MOLECULAR SYSTEMS: Even-electron atomic or molecular systems whose electron configurations consist of doubly occupied orbitals.

CLOSED SYSTEM: 1. In thermodynamics, a system that can exchange heat and work (energy) but not matter with its surroundings. 2. In computing, a software for which the specifications and detail of implementation are kept secret, as opposed to open source systems.

CLOSEST PACKING: The most efficient arrangements for packing atoms, molecules or ions in a crystal.

CLOSO-: An affix used in names to denote a cage or closed structure, especially a boron skeleton that is a polyhedron having all triangular faces.

CLOSURE: 1. The smallest closed set containing a given set. It is equal to the intersection of all closed sets containing that set. For example, the closure of the positive integers under subtraction is the set of all integers. 2. The set of points in a space of every neighbourhood, which has a non-empty intersection with a given set. The closure of A is written $\bar{A}$ or $\text{Cl } A$. For example, the closure of the open interval, $(0, 1)$ is the closed interval $[0, 1]$; the closure of the rationals is the reals. 3. In logic, the closed sentence formed by prefixing quantifiers to a given open sentence to bind all its free variables; especially the universal closure of the given sentence, formed by binding its free variables with universal quantifiers. Mathematical identities written without quantifiers are abbreviations for their universal closures; thus one writes $a + b = b + a$ as the commutative law for addition, indicating that the result of adding any two elements is independent of the order. The operation forming such a set or sentence is also called closure.

CLOSTRIDIUM: A genus of rod-shaped, usually flagellated, Gram-positive endospore-forming bacteria belonging to the Firmicutes. Many cause serious infectious diseases such as gas gangrene, tetanus and botulism, while some species fix atmospheric nitrogen and are of agricultural importance.

CLOT (THROMBUS): The final product of the blood coagulation step in haemostasis, in which a mass of semisolid material at the site of an injury closes the wound, helping to prevent further blood loss and bacterial invasion. Damage to tissue triggers a series of reactions involving thromboplastin, which is a glycoprotein; calcium ions; phospholipids from platelets; and clotting factors, which result in the conversion of prothrombin in the blood to form the enzyme thrombin, which catalyses the formation of insoluble fibrin from soluble fibrinogen. The fibrin forms a fibrous network in which blood cells become enmeshed, producing a clot.

CLOTTING-FACTORS (COAGULATION FACTORS): Chemical and cellular constituent substances that cause a blood clot to form at the site of an injury. Calcium ions and platelets trigger the release of the enzyme thromboplastin from injured tissues. This converts prothrombin in the blood to form the enzyme thrombin, which catalyses the formation of insoluble fibrin from soluble fibrinogen. The fibrin forms a fibrous network in which blood cells become enmeshed, producing a clot.

CLOUD: 1. An aerosol of the atmosphere which is dense enough to be perceptible to the eye. In other words, it is an assembly of water droplet aerosols suspended in the atmosphere, although dust clouds are also recognised. A similar definition is the separate mass of visible water droplets, ice crystals or mixture of the two floating in the sky. They are created by upward-moving air mass. Clouds are classified according to the heights at which they occur and their shape. All weather producing clouds form in the troposphere, the lowest major

layer of the atmosphere. 2. A term used for early network diagrams, in which the image of a cloud was used to indicate a large network, such as a WAN. The term is now associated with the entire Internet and the two terms are now used synonymously. Cloud may also be used to describe specific online services such as web applications, SaaS, online backup and other types of online storage, which are collectively called cloud computing.

CLOUD ARCHIVE: A storage procurement model in which data that no longer needs to be accessed on a regular basis is maintained and backed up remotely by a cloud storage service provider.

CLOUD BACKUP (ONLINE BACKUP): A method for backing up data that involves sending a copy of the data over a proprietary or public network to an off-site server. The server is usually hosted by a third-party service provider, who charges the backup customer a fee based on capacity, bandwidth or number of users.

CLOUD CHAMBER (WILSON CHAMBER): A device for detecting ionising radiation by making visible the paths of the particles in a cloud, invented by Charles T. R. Wilson (1869-1959), Scottish physicist and Nobel laureate. It consists of a dust-free sealed environment containing a super cooled, supersaturated water or alcohol vapour. When an alpha particle or beta particle interacts with the mixture, it ionises it. As an ion moves through the cloud chamber, drops of water or mist form around it. Because the ion moves very quickly, the mist looks like a continuous path marking the movement of the ion. When a strong light is directed at the cloud chamber the path of the ions become visible. When any uniform magnetic field is applied across the cloud chamber, positively and negatively charged particles will curve in opposite directions, which can be photographed for analysis.

CLOUD COMPUTING: A term used for applications and services offered over the Internet by a collection of servers from data centres all over the world. They are broadly divided into three categories: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS). Examples of popular cloud-based services include web applications, SaaS, online backup and other types of online storage. A cloud service has three distinct characteristics that differentiate it from traditional hosting. It is sold on demand, typically by the minute or the hour; it is elastic, a user can have as much or as little of a service as they want at any given time; and the service is fully managed by the provider (the consumer needs nothing but a personal computer and Internet access).

CLOVER (TRIFOLIUM): A genus of about 300 species of plants in the leguminous pea family, Fabaceae, consisting of small annual, biennial or short-lived perennial herbaceous plants. The plants have smooth stems and are commonly trifoliate. The flowers are small with yellow, white, red or purple colours. The pollen-bearing structures (stamens) are concealed in the fold of the lower petal and are not exposed except by the action of insects such as bees, which pollinate them. The plants are distributed worldwide and fix nitrogen in the soil, reducing the application of synthetic fertilisers. Several species are cultivated for fodder. The most commonly cultivated is white clover, *Trifolium repens*.

CLUB DRUG: A drug, especially one derived from opium that may produce effects of pain relief and sleep to stupor, coma and convulsions. Examples are ecstasy and methamphetamines used typically by some young people for recreational reasons.

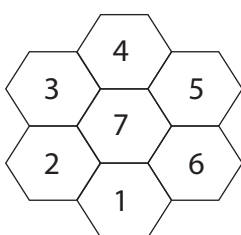
CLUB MOSS: Plants belonging to the family Lycopodiaceae. It is a non-flowering plant that typically has creeping stems with small overlapping leaves and reproduces by spores, often borne in club-shaped organs strobili. They occur world-wide.

CLUB ROOT (FINGER-AND-TOE DISEASE): A disease caused by *Plasmodiophora brassicae* affecting cabbages, turnips and allied plants of the Cruciferae family. The roots become swollen and malformed. Above ground, the symptoms, which may not appear for a while, are wilting and later stunting and yellowing of infected plants. Infection occurs via the root hairs and once the cambium is infected club roots develop rapidly. Spores can remain viable in the

soil for several years making control difficult. Liming of the soil is a traditional control method.

CLUSIUS COLUMN: A long vertical tube with an electrically heated wire along its axis that produces a temperature gradient, causing the lighter isotope to concentrate around the wire and the heavier isotope to concentrate near the walls. It is used for separating isotopes by thermal diffusion.

CLUSTER: 1. A group of associated stars that travel together through space and the largest known gravitationally bound objects to have arisen thus far in the process of cosmic structure formation. There are two basic morphological types: open clusters and globular clusters. The stars in a galaxy usually orbit in a plane around a common centre and almost all open clusters lie close to the plane of the galaxy in which they are located; forming the densest part of the large scale structure of the universe. In models for the gravitational formation of structure with cold dark matter, the smallest structures collapse first and eventually build the largest structures, clusters of galaxies. Clusters are formed relatively recently, between 10 billion years ago and now. Groups and clusters may contain from ten to thousands of galaxies. They range from about 5 light-years to about 70 light-years in diameter. 2. In computer science, cluster refers to a group of sectors on a disk. While a sector is the smallest unit that can be accessed on a hard disk, a cluster is a slightly larger unit that is used to organise and identify files on the disk. Most files take up several clusters of disk space. The term is also used for several machines grouped together, all performing a similar function. For example, a cluster may consist of eight PCs, all connected via high-speed Ethernet, processing scientific data. This type of setup is often referred to as **parallel computing**, since all the computers in the cluster are acting as one machine. Clusters are typically used for high-end processing, such as performing scientific calculations or decrypting algorithms. 3. In wireless communication, it refers to a group of cells within which no channels are reused. In multicell systems, each cell will have a base station with a low-power transmitter. Adjacent cells cannot use the same frequencies.



Cluster (a group of cells in wireless communication)

CLUSTER ANALYSIS: 1. In statistics, it is the partitioning of a data set into subsets or clusters, so that the degree of association is strong between members of the same cluster and weak between members of different clusters according to some defined distance measure. Several methods of performing cluster analysis exist; examples are partitional clustering and hierarchical clustering. 2. The clustering or grouping, of large data sets such as chemical and/or pharmacological data sets, on the basis of similarity criteria for appropriately scaled variables that represent the data of interest.

CLUSTER COMPOUND: A compound in which groups of metal atoms are joined together by metal-metal bonds. The formation of such compounds is a feature of the chemistry of certain transition metals, particularly molybdenum and tungsten, but also vanadium, tantalum, niobium and uranium. **Isopoly compounds** are ones in which the cluster contains atoms of the same element; **heteropoly compounds** contain a mixture of different elements.

CLUSTER ION: An ion formed by the combination of more ions or atoms or molecules of a chemical species often in association with a second species. For example, $[(H_2O)_nH]^+$ is a cluster ion.

CLUSTER OF DIFFERENTIATION (CD): Any group of antigens that is associated with a specific sub-population of human T cells. The differentiation and antigens expressed by a T cell vary with its stage of development and thus with its role in the immune response.

CLUSTER SAMPLING: A sampling technique in which the entire population is divided into groups or clusters and a random sample of

these clusters are selected. All observations in the selected clusters are included in the sample. Each cluster consists of individuals that are supposed to be representative of the population. Cluster sampling is typically used when the researcher cannot get a complete list of the members of a population they wish to study but can get a complete list of groups or 'clusters' of the population. It is also used when a random sample would produce a list of subjects so widely scattered that surveying them would prove to be far too expensive, for example, people who live in different districts in Ghana.

CLUTCH: 1. Any device for connecting or disconnecting rotating shafts, used especially in a car's transmission system. Mechanical clutches provide either a positive or a friction-dependent drive; centrifugal clutches provide automatic engagement. 2. A set of eggs that a hen sits on and that hatch together or a group of young chickens that hatch together from these eggs.

CMYK COLOUR (FULL COLOUR; 4 COLOUR PROCESS): The four basic colours representing **C**yan, **M**agenta, **Y**ellow and **K** used for printing colour images. Unlike RGB (red, green, blue), which is used for creating images on a computer screen, CMYK colours get darker as they are blended. Since RGB colours are used for light, not pigments, the colours grow brighter as they blend or increase their intensity.

CNESTIS: A genus of woody liane, climbing shrubs or small trees belonging to the *Connaraceae* family. The leaves are imparipinnate and the leaflets are opposite or subopposite. Inflorescences are terminal or axillary racemes, often from old leaf bases. Flowers are bisexual or unisexual; and the fruit is a dense velutinous follicle, having an apex with a curved or twisted beak. Seeds are solitary, attached to the base of the follicle. One of the species, *Cnestis ferruginea* is a short tropical shrub or sometimes a climber with cylindrical twigs, that grows commonly in bushes in the savannah, often in secondary regrowths on farm land, in fences, fringing lagoons and near the beach. It occurs in throughout Senegal to West Cameroons and in other parts of tropical Africa. The young twigs are densely covered with rusty brown velvety hairs. It has imparipinnate leaves; and small white fragrant star-like flowers in auxiliary panicle or groups of racemes and conspicuous red fruit like pears. It is an important medicinal plant widely used throughout West African region with the roots and leaves used mainly for the treatment of infections, dysentery, venereal diseases, bronchitis and snakebite. The fruit is reputed to be poisonous. Some of the other species include *Cnestis racemosa* Don, a shrub, with woody, glabrous, terete branches found in Upper Guinea Sierra Leone; and *Cnestis palala*, a woody vine with short-pubescent branches found in Malay Peninsula, Sumatra, Borneo and the Philippines in dry woods, in forests or thickets.

CNIDARIA (COELENTERATES): A group of aquatic invertebrates, whose bodies consist of mesoglea, a non-living jelly-like substance sandwiched between two layers of epithelium that are mostly one cell thick. They include corals, hydras, jellyfish, Portuguese man-of-war and sea anemones. They may reproduce by means of asexual reproduction, sexual reproduction or both.

CNIDOBLAST: In cnidarians such as jellyfishes, sea anemones, corals and hydras, it is a cell that develops into mature cnidocytes or nematocytes.

CNIDOCYST (NEMATOCYST; CNIDA): A specialised capsule-like structure found within the nematocyte (cnidocyte) of cnidarians containing a thread-like tube. It contains a poisonous or paralysing substance, used for capturing prey and as a defence mechanism. It is characteristic of cnidarians such as jellyfishes, sea anemones, corals and hydras.

CNIDOCYTE (NEMATOCYTE): A cell found in cnidarians such as jellyfishes, sea anemones, corals and hydras that contains the cnidocysts (nematocysts or cnidae).

CO-EXISTENCE: In agriculture, it refers to the presence, within the same area of land, of crops grown in different ways, for example, crops grown in a conventional manner, crops grown according to the principles of organic farming and crops grown from seeds that have been genetically modified.

CO-IONS: 1. Ions of low relative molecular mass with a charge of the same sign as that of a given colloidal ion. 2. In an ion exchanger, mobile ionic species with a charge of the same sign as the fixed ions.

CO-LOGARITHM: The logarithm of the reciprocal of a number.

CO-OLIGOMER: An oligomer derived from more than one species of monomer.

CO-REPRESSOR: A small molecule that activates a particular genetic repressor by combining with it.

CO-SOLVENCY: The dissolution of a polymer in a solvent comprising more than one component, each component of which by itself is a non-solvent for the polymer.

COACERVATION: The separation into two liquid phases in colloidal systems. The phase more concentrated in colloid component is the **coacervate** and the other phase is the **equilibrium solution**.

COADAPTATION: The mutual adaptation of two species that occurs during co-evolution.

COAGULATION (FLOCCULATION): 1. The process in which colloidal particles come together irreversibly to form larger masses. Coagulation can be brought about by adding ions to change the ionic strength of the solution and thus stabilise the colloid. 2. The process of change from a liquid to a thick and semi-solid state or clot by the action of an enzyme, heat, acid or other means.

COAGULATION FACTORS (BLOOD CLOTTING FACTOR; CLOTTING FACTOR; BLOOD COAGULATION FACTORS): Endogenous substances, usually proteins that are involved in the blood coagulation process. The absence or mutation of these factors can lead to haemophilia and blood clotting disorders.

COAL: A readily combustible brown or black fossil fuel derived from the accumulation and alternation of ancient vegetation which originated largely in swamps and other moist environments. It is composed primarily of carbon along with variable quantities of other elements such as sulfur, hydrogen, oxygen and nitrogen. It is burnt to supply heat and used to make coal gas and coal tar, as well as for generation of electricity. Coal is extracted from the ground by mining, either underground or in open pits.

COAL GAS: A gas produced when coal is destructively distilled or heated out of contact with the air. It consists mainly of hydrogen, carbon monoxide and methane.

COAL TAR: A black highly viscous oily material resulting from the destructive distillation of bituminous coal. It is made up of complex and variable mixtures of phenols, polycyclic aromatic hydrocarbons (PAHs) and heterocyclic compounds. Coal tar is sometimes used for heating or to fire boilers and also as a component of road pavement material, especially when mixed with ironwork slag. It is also used in medicated shampoo, soap and ointment, as a treatment for dandruff and psoriasis.

COAL TAR PITCH: A residue produced by distillation or heat treatment of coal tar. It is a solid at room temperature, consisting of a complex mixture of numerous predominantly aromatic hydrocarbons and heterocyclics and exhibits a broad softening range instead of a defined melting temperature.

COAL-DERIVED PITCH COKE: The primary industrial solid carbonisation product obtained from coal tar pitch mainly produced in a chamber or delayed coking processes. Coal-derived pitch coke, although exhibits a pregraphitic microstructure, has often a lower graphitizability than petroleum coke. Fractions of coal tar pitches obtained by extraction or filtration may form cokes with needle-like structures and have an improved graphitizability. The usually lower graphitizability compared to petroleum coke is due to an inhibition of mesophase growth because of chemical and physical differences of the cokes.

COALESCENCE: The disappearance of the boundary between two particles (usually droplets or bubbles) in contact or between one of these and a bulk phase followed by changes of shape leading to a

reduction of the total surface area. The flocculation of an emulsion or the formation of aggregates, may be followed by coalescence.

COALIFICATION: A geological process of formation of materials with increasing content of the element carbon from organic materials that occurs in a first, biological stage into peats, followed by a gradual transformation into coal by action of moderate temperature (about 500 K) and high pressure in a geochemical stage. Coalification is a dehydrogenation process with a reaction rate slower by many orders of magnitude than that of carbonisation. Some specific reactions approach completion before others have started. The dehydrogenation remains incomplete. The degree of coalification reached by an organic material in the process of coalification increases progressively and can be defined by means of the measured C/H ratio and of the residual contents of oxygen, sulfur and nitrogen.

COARSE: Consisting of large particles, not fine, rough or loose in texture, low quality, inferior, not refined.

COARSE FRAGMENTS: Rock or mineral particles with a diameter greater than 2.0 mm.

COARSE-TEXTURE SOIL (SANDY SOIL): Loose and single grained soil. Coarse textured soils have mainly large particles with large pores in between. In coarse soils, the rain or irrigation water enters and moves more easily into larger pores. Therefore, the infiltration rate is higher for coarse textured soils than for fine textured soils. Fine textured soils have mainly small particles with small pores in between.

COARSER: Of or pertaining to a topology, that is strictly contained in another topology. For example, the topology of open intervals on the reals is coarser than the discrete topology. The coarsest topology is the indiscrete topology.

COAST: Land bordering the sea. The line of contact between the land and water surfaces is called the **shoreline**. It fluctuates with the waves and tides.

COASTAL DEPOSITION: The laying down of materials such as sand, rock particles and pebbles on the coast by the sea when it loses energy or when large inputs of sediment are made into the coastal system, perhaps due to the arrival of fluvial sediment at a river estuary.

COASTAL EROSION: The erosion of land bordering the sea by the constant battering by sea waves, with the result that they are worn away. It may take the form of long-term losses of sediment and rocks or redistribution of coastal sediments.

COASTAL PROTECTION: Measures taken to prevent coastal erosion. Coastal erosion is the wearing away of land and the removal of beach or dune sediments by the action waves, tidal currents, wave currents or drainage. Measures to protect the coast against erosion include: (i) Building seawalls usually made of concrete to absorb the energy of the wave and protect the coast against strong waves especially during storms. (ii) Creating granite breakwaters. This creates a zone of shallow water between itself and the coast so that waves will break against it before reaching the coast. (iii) Building low walls called groynes at right angles to prevent materials from being transported away into the sea.

COAT: 1. The fur, hair or wool covering an animal's body. 2. The outer covering of an organ or structure. For example, **seed coat** is the hard covering of a seed.

COAT PROTEIN (CAPSID): The coating of a protein that encloses the nucleic acid core of a virus.

COATING: The application of a thin film of heavy metal, commonly gold, to the surface of a specimen during preparation for scanning electron microscopy. Coating is particularly useful in preparation of biological specimens because it results in the production of many more secondary electrons and the improved conductivity helps prevent a charge building up in the specimen and causing image

deterioration. Coating units are of several types; some use fine filaments as the source of coating metal while others, known as sputter coaters, have a large solid block of metal that is ionised by a beam of ions.

COAXIAL CABLE: A cable consisting of a metal conductor surrounded by an insulator which is in turn contained in an earthed sheath of another conductor. In other words, it is a cable consisting of one conductor, usually a small copper tube or wire, within and insulated from another conductor of larger diameter, usually copper tubing or copper braid. The inner conductor and the outer conductor have a common axis (coaxial). In an ideal coaxial cable the electromagnetic field carrying the signal exists only in the space between the inner and outer conductors and so is protected from external electromagnetic interference. This allows coaxial cable runs to be installed next to metal objects such as gutters without the power losses. Coaxial cable is used as a transmission line for radio frequency signals, in applications such as connecting radio transmitters and receivers with their antennas, computer network connections and distributing cable television signals.

COBALT: A hard, lustrous grey, transition metallic element, with the chemical formula Co, atomic number 27, relative atomic mass 58.933, melting point 1,495 °C (1768 K) and boiling point 2,900 °C (3173 K). It is found in association with nickel, silver, lead, copper and iron ores. It is used in the preparation of magnetic wear resistant and highly strong alloys. Its compounds are used in inks, paints and varnishes and it is a major element which is also used with iron in the manufacture of magnets.

COBALT STEEL: Any of a group of alloy steels containing 5-12% of cobalt, 14-20% of tungsten, usually with 4% of chromium and 1-2% uranium.

COBALT(II) OXIDE: A pink solid, with the chemical formula CoO, relative density 6.45 and melting point 1935 °C (2208 K). It is formed by adding potassium hydroxide to a solution of cobalt(II) nitrate giving a bluish-violet precipitate, which on boiling is converted to pink impure cobalt(II) hydroxide. Heating this in the absence of air forms cobalt(II) oxide.

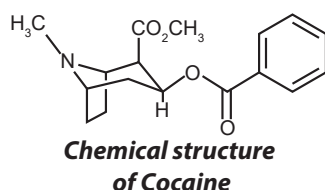
COBALT(III) OXIDE (COBALT SESQUIOXIDE): A hexagonal or rhombic black grey insoluble solid, with the chemical formula Co₂O₃, relative density 6.45 and decomposes at 895 °C (1168 K). It is produced by the ignition of cobalt nitrate. On heating it readily forms Co₃O₄ which contains both CO(II) and CO(III). Cobalt(III) oxide dissolves in strong acid to give unstable brown solutions of trivalent cobalt salts. With dilute acids cobalt(II) salts are formed.

COBBS-DOUGLAS FUNCTION: A commonly used production function of the form $AL^a K^b M^c$ where L , K and M measure the quantity of labour, capital and raw material consumed for a given output rate, the constants A , a , b , c , are parameters. Increasing return occurs when $a + b + c > 1$.

COBOL: Acronym for **Common Business Oriented Language**, which is a high level computer programming language designed for use in common commercial data processing problems. It has become the major language in the field and it features powerful facilities for file handling and business arithmetic.

COCA: A South American shrub of the family Erythroxylaceae from whose dried leaves cocaine is obtained. Its alkaloids include cocaine, a powerful stimulant.

COCAINE: An alkaloid, with the chemical formula C₁₇H₂₁NO₄, melting point 98 °C (371.15 K) and boiling point 187 °C (460.15 K). It is extracted from the leaves of the coca tree. It has limited medical application, being used mainly as a local anaesthetic agent that is relatively absorbed by mucus membranes of the nose



and throat. Cocaine increases heart rate and blood pressure and causes feelings of alertness and euphoria. It is both toxic and addictive and its use as a stimulant is illegal. Long term use may cause mental and physical deterioration.

COCCIDIOIDOMYCOSIS: A lung disease of humans and animals caused by inhaling spores of the fungus *Coccidioides immitis*.

COCCIDIOSIS: A parasitic disease of domestic animals and birds and occasionally, humans, leading to diarrhoea caused by infestation of the alimentary canal by protozoans of the order Coccidia.

COCCIDIUM: Parasitic obligate, intracellular sporozoan belonging to the apicomplexan class Conoidasida of the *Coccidi* order that can cause disease in the gut of humans and animals.

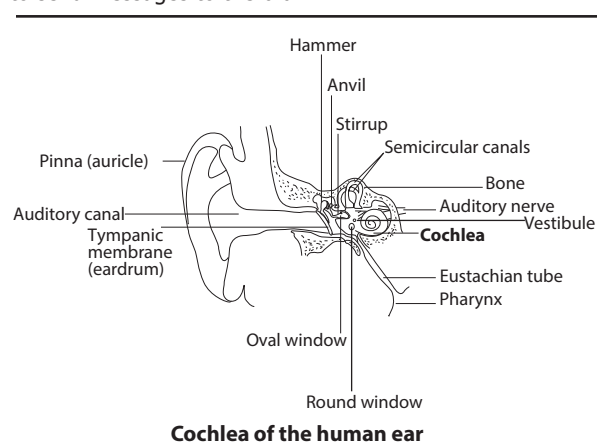
COCCOLITH: A round, chalky platelet, only one or two thousandths of a millimetre across, formed by a single-celled flagellate of the class Coccolithophorida, a class of algae of the phylum Chrysophyta. Many limestone and chalk cliffs are made up entirely of the remains of such platelets.

COCCUS (COCCI): 1. Spherical or nearly spherical micro-organisms, especially bacteria, some of which are harmful to humans. Subgroups include streptococci and staphylococci. 2. A subdivision of a fruit that contains a single seed and resembles a berry.

COCCYX: The last bone in the vertebral column in apes and man. It is formed by the fusion of three to five separate or fused vertebrae below the sacrum.

COCHINEAL: A crimson red colouring agent and dye obtained from the pulverized, dried bodies of female scale insects of the species *Dactylopius coccus*, with a length of about 3 mm. These soft-bodied cactus-eating insects are found in Mexico and other parts of Central America. Synthetic aniline dyes have largely replaced cochineal commercially.

COCHLEA: The auditory portion of the inner ear of mammals, birds and some reptiles that transforms sound waves into nerve impulses and detects sound. It is equipped with approximately 10,000 hair cells which move in response to sound waves and thus stimulate nerve cells to send messages to the brain.



COCHLEAR IMPLANT: A small, complex electronic device that can help provide a sense of hearing to a person who is profoundly deaf or severely hard-of-hearing. The implant consists of an external portion that sits behind the ear and a second portion that is surgically placed under the skin.

COCHLEATE: Shaped like the coiled shell of a snail.

COCHLOSPERMUM: A genus of tropical trees belonging to the family Cochlospermaceae with about 15 known species. One of the species, *Cochlospermum tinctorium* is a yellow flowering low bushy shrub. It occurs widely in savannah and scrub land, especially in devastated, rocky and annually burnt areas throughout the drier parts of West

Africa up to Cameroun, as well as in Sudan and Uganda. The root stock (rhizome) is yellow, long, irregular and woody. It is one of the first plants which flower after the annual bush fires. In the rainy season it produces annual leafy shoots of 30-80 cm height. The leaves are very variable, mostly more or less deeply 5-7 lobed. The large bright yellow flowers are very conspicuous against the burnt ground and normally appear near the ground while the plant is leafless. The fruit is a subglobose, obovoid to spindle-shaped capsule, often hairy, dehiscing by 5 valves. Seeds are kidney-shaped, usually covered with long woolly, yellowish hair. The yellow root is used traditionally for the treatment of jaundice and also for the treatment and prevention of diseases due to liver damage. *Cochlospermum vitifolium* commonly known as buttercup tree is found in Central America and the West Indies and has bright-yellow, cup-shaped flowers. In some areas, its bark is used to make rope. The seeds of *Cochlospermum angolense*, an African species, yield a red dye; the roots of *Cochlospermum planchonii*, yield a yellow dye; and *Cochlospermum gossypium* with the common name cotton tree from India and Burma produces a gum from the bark, which is made into a powder used as thickener, binder, emulsifier and stabiliser in food and pharmaceutical products.

COCK: 1. A tap or valve controlling the flow of a liquid or gas in a pipe. 2. The hammer of a gun. 3. The adult male bird of the domestic fowl or the male of any other bird especially of a game bird.

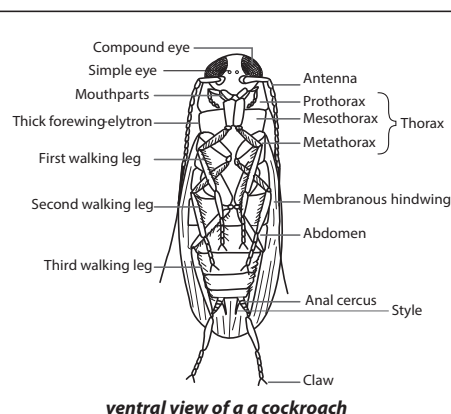
COCKCHAFER: Any of various European large, scarabeid beetles whose grubs live underground and feed on the roots of plants. Some species are particularly harmful to trees.

COCKCROFT-WALTON GENERATOR: The first proton accelerator which is a simple linear accelerator producing a potential difference of some 800kV from a circuit of rectifiers and capacitors fed by a lower voltage. It is named after John Douglas Cockcroft and Ernest Thomas Sinton Walton, who in 1932 used this circuit design to power their particle accelerator.

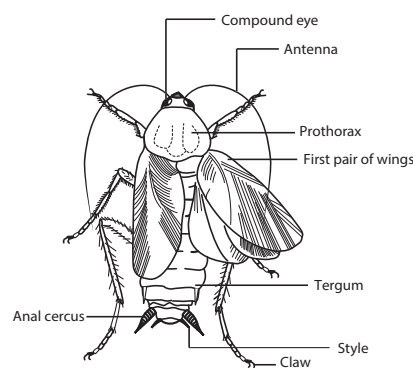
COCKEREL: A young male chicken or pheasant.

COCKROACH: An insect, of the order Blattaria or Blattodea, with about 4,500 known species. Some of the species include German cockroach (*Blattella germanica*), a common household pest; the American cockroach (*Periplaneta americana*) is 30 to 50 mm long, reddish brown and lives outdoors or in dark, heated indoor areas; the Oriental cockroach (*Blatta orientalis*), which is about 25 mm long; and the brown-banded cockroach (*Supella supellectilium*). They are nocturnal insects and spend most of the day hiding. Cockroaches live in a wide range of environments around the world, but prefer a warm, humid, dark environment and is usually found in tropical or other mild climates. Some cockroaches live in sewers and wet, decaying areas, such as basements and piles of leaves; some live in warm and damp places, like kitchens, bathrooms and places where people eat and drink. They have broad, flattened bodies and relatively small heads, long whip-like antennae and chewing mouthparts located on the underside of the head. They have large compound eyes, two ocelli and long, flexible, antennae. The first pair of wings (the tegmina) are tough and protective, lying as a shield on top of the membranous hind wings. All four wings have branching longitudinal veins and multiple cross-veins. The legs are sturdy, with large coxae and five claws each. The abdomen has ten segments and several cerci. Cockroaches use pheromones to attract mates. The female usually attaches the egg case to a substrate, inserts it into a suitably protective crevice or carries it about until just before the eggs hatch. Some species, however, are ovoviviparous, keeping the eggs inside their bodies, with or without an egg case, until they hatch. Cockroaches are among the hardest insects. Some species are capable of remaining active for a month without food and are able to survive on limited resources and can go without air for 45 minutes. Cockroaches are one of the most common

household pest insects. They feed on almost anything, including soiled dressings, faecal matter and food for human consumption. They are capable of carrying various food poisoning germs on their bodies and can be responsible for the spread of gastro-enteritis, dysentery and typhoid. They are also linked with allergic reactions in humans.



ventral view of a cockroach



Dorsal view of a cockroach

COCKROFT, SIR JOHN DOUGLAS (1897-1967): British physicist, who was educated at the University of Manchester and St. John's College, Cambridge and a fellow of St. John's College (1928-46) and professor of natural philosophy at Cambridge (1939-46). He joined Rutherford at the Cavendish Laboratory in Cambridge and together with Ernest Walton (1903-95) built a linear accelerator. In 1923, using the apparatus to bombard lithium nuclei with protons, they produced the first artificial nuclear transformation, for which they were awarded the 1951 Nobel Prize. From 1941 to 1944 he was chief superintendent of the Air Defence Research and Development Establishment; from 1944 to 1946 he directed the atomic energy division of the National Research Council of Canada. In 1946, he became the director of the British Atomic Energy Research Establishment.

COCKSCOMB: 1. A tropical plant, *Celosia crista* often grown as a houseplant. It has orange or red flower heads, appearing as a broad crest or plume resembling a cockscomb on flowers. 2. The red fleshy crest that grows on the top of a domestic cock's head.

COCKSFOOT: A high-yielding, deep-rooting perennial grass, *Dactylis glomerata*, native to Europe, Asia and North Africa, which is resistant to drought and sometimes used for hay and as a forage grass.

COCKTAIL EFFECT: The effect of two toxic or potentially toxic chemicals when taken together rather than separately. This sometimes occurs because both chemicals require the same enzyme to break them down. To avoid this, chemicals such as pesticides and food additives are only tested singly.

COCOA (CACAO; THEOBROMA CACAO): Also called **cacao bean** or **cocoa bean**, it is the dried partly fermented fatty seeds of the evergreen tree, *Theobroma cacao* of the family Malvaceae, native to tropical America, but now grown mainly in West Africa. The term

is also used for the spindly, evergreen tree, which is 5 to 8 metres tall and grows in the shade of giant trees occupying the top layer of the rain forest. It has a taproot, which penetrates far below the soil surface. Its dark green leaves are shiny, leathery, egg-shaped or elliptic in shape and 20 to 35 cm long and 7 to 8 cm wide. The leaf surfaces are hairless or covered in scattered star-shaped hairs. The base of the leaf is rounded or heart-shaped and the tip is long and drawn out allowing water to drip from it. The flowers are small, yellowish white to pale pink and grouped together in clusters arising directly from the trunk (cauliflory). Flowers are produced throughout the year. The fruit is an egg-shaped ripening yellow to orange and weighs about 500 g, 15 to 25 cm long, with a more or less knobbly surface and lines from top to bottom. The pod contains 30 to 40 seeds, each of which is surrounded by a bitter-sweet white pulp. When the seeds are dried and fermented in the sun they are brownish red and known as cocoa beans. Cocoa has a high food value, containing as much as 20 percent protein, 40 percent carbohydrate and 40 percent fat. The dried and processed cocoa beans are used in making the beverage cocoa drink, chocolate and cocoa butter. The beans also contain up to 10% of phenols and flavonoids which are antioxidants and can inhibit cancer or cardiovascular diseases. They also contain potassium, magnesium, calcium, iron; and theobromine and caffeine, which are alkaloids that stimulate the central nervous system. Cocoa is also used for medicine and cosmetics. Unfermented cocoa seeds and the seed coat are used to treat a variety of ailments, including diabetes, digestive and chest complaints. Cocoa powder, prepared from fermented cocoa beans, is used to prevent heart disease. Cocoa butter is also taken to lower cholesterol levels.

COCONUT PALM (COCONUT TREE): A tall, tropical palm, *Cocos nucifera* of the family Arecaceae with an erect or slightly curved stem, which grows from a swollen base reaching up to 30 metres tall. It has a crown of 60-70 spirally arranged leaves. The leaves are pinnate and 4-6 m long. It has a fibrous root system. The inflorescence is a spike produced at the leaf axil with 20-60 branches, each with a female flower at the base and many male flowers; and bears large, hard-shelled fruits called coconuts enclosed in a thick, fibrous husk. The fruit is a drupe, ovoid in shape with three sides divided by ridges. The exocarp and the mesocarp make up the husk of the coconut. The drupe contains a hard shell that encloses the embryo with its endosperm composed of both a white edible flesh and a sweet milky liquid. The flesh of the matured fruit, called copra can be made into coconut oil. The oil is a semi-solid fat, consisting principally of the glyceride of lauric acid, $\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$. The oil is used to manufacture soaps, vegetable fats, candles and cosmetics. The husk yields coir, a fibre that is used in the manufacture of ropes, mats, baskets, brushes and brooms.

COCOON: 1. A protective covering for the pupal stage of many insects; or a protective covering for eggs or larvae produced by many invertebrates. For example, the larvae of many insects spin a cocoon in which the pupa develops and earth worms secrete a cocoon for the developing eggs. Insect cocoons are largely formed of silk, a quick-hardening material with high tensile strength, secreted from special glands. 2. A cover or protective spray used to seal machinery to make it waterproof. An example is military equipment when being transported or in storage.

COCOYAM (*Xanthosoma sagittifolium*): Herbaceous tropical perennial plant belonging to the genus *Xanthosoma*, family Araceae, with the common English names of tannia, new cocoyam or elephant ear. It thrives very well in fertile, well-draining, sandy loam soil. The corm or main underground stem is like a rhizome from which swollen secondary shoots, called cormels and suckers or stolons sprout. Its leaves sprout from the main stem and are sagittate and erect with long, ribbed petioles. The blades are large and heart-shaped and can reach up 50 cm in length. The inflorescences sprout between the leaves in a spadix. The spadix is cylindrical, slightly longer than the spathe, with female flowers on the lower portion, male flowers on the upper portion and sterile flowers in the middle portion. It is grown

primarily for its starchy edible corms, although the young fresh leaves are cooked and eaten as vegetable. Some other cultivated species include *Xanthosoma atrovirens*, *Xanthosoma caracu* and *Xanthosoma nigrum* (*X. violaceum*). A closely related genus is *Colocasia*. The most common cultivated species both as an ornamental plant and as a food source is *Colocasia esculenta* also known as *Colocasia antiquorum* with common names taro, swamp taro, old cocoyam and arrowroot.

COCURRENT FLOW: Flow of two fluids in the same direction with transfer of matter or heat between them.

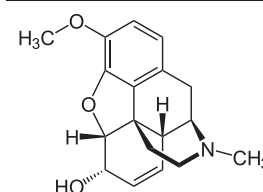
COD LIVER OIL: Oil rich in vitamins A and D that is extracted from the liver of cod, a fish belonging to the *Gadus* family. It is principally a mixture of the glycerides of many fatty acids, but its minor constituents, the fat-soluble vitamin A and vitamin D, give it its importance. It is often used as a food supplement and also as a remedy for joint pain caused by arthritis.

CODE: 1. In computer science, data or instructions in a computer program written in a programming language or the language itself. 2. In coding theory, a set of n -tuples of elements drawn from an alphabet, S ; where each such string is a word, and n is the word-length of the code, the code is thus a subset of S^n .

CODE-DIVISION MULTIPLE ACCESS (CDMA): Any of several channel access protocols used in second-generation (2G) and third-generation (3G) wireless communications. It is a form of multiplexing, which allows numerous signals to occupy a single transmission channel, optimising the use of available bandwidth. CDMA is based on Spread Spectrum Technology and uses orthogonal codes to avoid interference or the transmitted signal is spread over a wide frequency band to avoid interference. The technology is used in ultra-high-frequency (UHF) cellular telephone systems in the 800-MHz and 1.9-GHz bands.

CODEC: In computing, it is abbreviation for **coder-decoder**, an electronic device or component combining the circuits needed to convert digital signals to and from analogue (Pulse Code Modulation) form.

CODEINE: A naturally occurring alkaloid found in opium, with the chemical formula $\text{C}_{18}\text{H}_{21}\text{NO}_3$. It is a synthetic derivative of morphine. It is used as analgesic for relieving pain, to treat diarrhoea and also as cough suppressant.



Chemical structure of Codeine

CODER-DECODER (CODEC): An electronic device or component combining the circuits needed to convert digital signals to and from analogue (Pulse Code Modulation) form. Audio and video files are compressed with a certain codec when they are saved and then decompressed by the codec when they are played back. Common codecs include MPEG and AVI for video files and WAV and AIFF for audio files. Codecs can also be used to compress streaming media (live audio and video) which makes it possible to broadcast a live audio or video clip over a broadband Internet connection.

CODIMENSION: In mathematics, the term is used in a number of algebraic and geometric contexts to indicate the difference between the dimension of certain objects and the dimension of a smaller object contained in it. If an object sits inside or is associated to another object of dimension n , then it is said to have codimension k if it has dimension $n - k$. Alternatively, the difference between the dimension of a space and the dimension of a given subspace of the first one.

CODING STRAND (PLUS STRAND; SENSE STRAND): The strand in duplex DNA that by convention contains the same sequence of bases as that in the messenger RNA transcribed from the DNA.

CODING THEORY: The branch of mathematics concerned with transmitting data across noisy channels and recovering the message. Coding theory is about making messages easy to read and is thus not to be confused with cryptography, which is the art of making messages hard to read.

CODOMAIN: For a given function or mapping, a set within which the values of the function lie. This is different from the set of values, known as the range that the function actually takes. For example, the

function $y = \frac{-1}{x}$ for integral values of x can be said to have the interval $[-1, 1]$ as codomain, although 0 is not the value of the function for any

argument; its range is in fact only the set of rationals of the form $\frac{1}{x}$.

CODOMINANCE: The condition that arises when both alleles in a heterozygous organism are dominant and are fully expressed. For example, the human blood group AB is the result of two alleles, A and B, both being expressed.

CODON: A triplet of bases in a molecule of DNA or RNA that directs the placement of a particular amino acid during the process of protein synthesis or the sequence of three bases that represent a particular amino acid in a messenger RNA molecule.

COEFFICIENT: 1. Proportionality constant between two quantities of different dimension. It can also be defined as the quantity placed in front of an algebraic term, signifying multiplication. For example, in $4x^2 + 2xy - x$, the coefficient of x^2 is 4, that of xy is 2 and that of x is -1. 2. The multiplier of a given power of a variable in an expression. For example, the coefficient of x in $3xyz + zx^2$ is $3yz$. 3. In physics, a measure of a specified property of a substance under specified conditions, for example, coefficient of expansion.

COEFFICIENT OF DETERMINATION: The percentage of the variance of the dependent variable that is explained by the independent variables used to fit the data. Normally the symbol r^2 is used to represent the coefficient of determination. This is given by $r^2 = (\text{explained variation})/(\text{total variance})$.

COEFFICIENT OF EXPANSION (EXPANSIVITY): The ratio of the change in length or volume of a body to the original length or volume for a unit change in temperature.

COEFFICIENT OF FRICTION (FRICTION COEFFICIENT; FRICTION FACTOR): The ratio of frictional force to the normal reaction. It is a number characterising the force necessary to slide or roll one material along the surface of another. The formula is $\mu = \frac{F}{R}$,

where μ is the coefficient of friction, F is the amount of force that resists motion and R is the normal reaction force. The normal reaction force is the force the surface exerts on the body that is resting on it. If a rock that weighs 50 newtons is lying on the ground, then the normal force is 50 newtons. The higher μ is, the more force resists motion if two objects are sliding past each other. The force required to move an object starting from rest is usually greater than the force required to keep it moving at constant velocity once it is started. Therefore, two coefficients of friction are sometimes quoted for a given pair of surfaces: a coefficient of static friction and a coefficient of kinetic friction.

COEFFICIENT OF HAZE (COH): The measurement of visibility interference in the atmosphere. In other words, it is the measurement of the amount of filterable particulate matter suspended in air. A measurement is made of the intensity of light transmitted through the dust spot formed relative to that transmitted through an identical clean filter. The dirtiness of the air is reported in terms of the COH unit. This relates to the quantity of particulate material which produces an optical density, $\log_{10}(I_0/I)$ of 0.01, when measured by light transmission at 400 nm and relative to the transmission of an identical dust-free filter taken as 100%. Thus a filter which transmitted 50% relative to the clean filter has an absorbance of 0.301 or 30.1 (COH) units. This is not a recommended measure of filterable particulate matter since the size, colour and other properties of the aerosol and the air in which it is suspended affect the results.

COEFFICIENT OF KINETIC FRICTION: The ratio of the frictional force, parallel to the surface of contact that opposes the motion of a body which is sliding or rolling over another, to the force, normal to the surface of contact, with which the bodies press against each other. It is expressed mathematically as $\mu_k = F_k/N$, where μ_k is the coefficient

of kinetic friction, F_k is the force of kinetic friction and N is the normal force or the force perpendicular to the contacting surfaces. It is the friction that applies when an object is moving along a surface and the force of friction points in a direction parallel to the surface and directly opposing the direction of motion of the object along the surface.

COEFFICIENT OF LINEAR EXPANSION: It is the fractional change in length per degree rise in temperature. It is represented by the symbol α . Different materials expand by different amounts. How much a particular material expands depends on the forces of cohesion between neighbouring atoms or molecules. If these forces are strong, the material can only expand slightly for a given rise in temperature. A metal such as lead which has a low melting point and is relatively easily compressed, expands a great deal for a given rise in temperature. Coefficient of linear expansion is expressed as

$\alpha = \frac{1}{l_1} \frac{l_2 - l_1}{\Delta\theta}$, where l_1 = original length, l_2 = final length and $\Delta\theta$ = change in temperature.

COEFFICIENT OF PROBABILITY: The single-valued real function, P that determines the probability that a member of a system lies within an elemental volume, dv , of phase space. This probability may be represented by an expression of the form $dp = Pdv$. P also fulfils the condition $\int Pdv = 1$, where the integral is over all phase space; otherwise P is arbitrary.

COEFFICIENT OF RELATIONSHIP (COEFFICIENT OF RELATEDNESS): The probability that any two individuals share a given gene by virtue of being descended from a common ancestor.

COEFFICIENT OF RESTITUTION: The ratio of the velocity of an object after impact to that before impact. The coefficient of restitution is given by $C_R = \frac{V_2 - V_1}{u_1 - u_2}$ for two colliding objects, where V_1 is the final velocity of the first object after impact, V_2 the final velocity of the second object after impact, u_1 the initial velocity of the first object before impact and u_2 the initial velocity of the second object before impact.

COEFFICIENT OF STATIC FRICTION: The ratio of the maximum possible frictional force, parallel to the surface of contact that acts to prevent two bodies in contact at rest from sliding, to the force that is normal to the surface of contact. It is expressed mathematically as

$\mu_s = \frac{F_s}{F_n}$, where F_s is static friction, μ_s is the coefficient of static friction and F_n is the normal force.

COEFFICIENT OF VARIATION: The coefficient of variation for a list of numbers is equal to the standard deviation for those numbers divided by the mean. The coefficient can be used to compare the dispersions of quantitative variables that are not expressed in the same units (for example, when comparing the salaries in different countries, given different currencies) or the dispersions of variables that have very different means. It is expressed as: Coefficient of Variation (Cv) = Standard Deviation/Mean.

COEFFICIENT OF VOLUME EXPANSION: Fractional change in volume per degree change in temperature. Its symbol is γ .

COELACANTH: A bony fish of the genus *Latimeria*, which was believed to be extinct. However, in 1938, the first modern specimen was found around the Comoros Islands.

COELENTERATE: Any fresh water or marine organism of the phylum Coelenterata or Cnidaria. They have very simple tissue organisation, with only two external and internal layers of cells. Examples are the jellyfish, hydra and coral.

COELIAC DISEASE (COELIAC SPRUE; NON-TROPICAL SPRUE; ENDEMIC SPRUE; GLUTEN ENTEROPATHY; GLUTEN-SENSITIVE ENTEROPATHY; GLUTEN INTOLERANCE): An autoimmune nutritional disorder of the small intestine that occurs in genetically predisposed people of all ages. It is caused by a reaction to gluten, a protein constituent of wheat, barley, malt and rye flours. Symptoms may include chronic diarrhoea having a high fat content, malnutrition, a distended abdomen, muscle wasting and fatigue.

COELOM: A fluid filled cavity that forms the main body cavity of vertebrates and most invertebrate animals. It separates the body wall from the gut and associated organs and allows the gut muscles to contract independently of the rest of the body. It also absorbs shock or provides a hydrostatic skeleton.

COELOMYCETES: A class of the Deuteromycotina containing mycelial imperfect fungi in which the conidia are borne in pycnidia, as in the Sphaeropsidales or acervuli, as in the Melanconiales. It contains over 7000 species in some 870 genera.

COELOSTAT: A device that enables light from the same area of the sky to be continuously reflected into the field of view of an astronomical telescope. It consists of a plane mirror driven by a clockwork or electrical mechanism so that it rotates from East to West to compensate for the apparent West to East rotation of the celestial sphere.

COENOBIUM (COENOBIA): A loose association of unicellular organisms that live in a colony and may be surrounded by a membrane. A coenobium often gives rise to internal daughter colonies. An example is Volvox.

COENOCARPIUM: A multiple fruit with a fleshy axis such as pineapple (*Ananas cotnosus*) that incorporates the ovaries, floral parts and receptacles of many coalesced flowers.

COENOCYTE: A mass of protoplasm containing many nuclei and enclosed by a cell wall, found in certain algae, slime mould and fungi.

COENOSORUS: A continuous line of sporangia seen in certain ferns, such as Pteridium, where discrete sori are not produced.

COENZYME: An organic non-protein molecule that associates with the enzyme molecule in catalysing biochemical reactions or an organic molecule that associates with enzymes and affects their activity. Examples of important coenzymes are **nicotinamide adenine dinucleotide (NAD)**, also known as **coenzyme I** and nicotinamide adenine dinucleotide phosphate (NADP), **coenzyme A** and biotin.

COENZYME A (CoA; CoASH; HSCoA): A sulfur-containing molecule that occurs in all living cells and is essential to the metabolism of carbohydrates, fats and some amino acids. It is formed from adenosine triphosphate (ATP), pantothenic acid and cysteine and acts in conjunction with enzymes involved in various energy-producing biochemical reactions, notably the oxidation of pyruvate via the Krebs cycle and fatty-acid oxidation and synthesis. An example is the reaction of acetyl CoA and oxaloacetate to form citrate.

COENZYME I (NICOTINAMIDE ADENINE DINUCLEOTIDE, NAD): A pyridine-based nucleotide found in all living cells that functions as a coenzyme in many oxidation-reduction reactions, according to the general reaction:

$\text{NAD}^+ + \text{reduced substrate} \leftrightarrow \text{NADH} + \text{H}^+ + \text{oxidised substrate}$. An example of such a reaction is the oxidation of citrate to α -ketoglutarate in the TCA cycle. Reduced NAD can act as a reducing agent or it may be reoxidised in the respiratory chain with coupled adenosine triphosphate (ATP) formation. NAD is not usually tightly bound to the enzyme with which it is acting but instead acts as a second substrate, binding only during the reaction and being released into the medium with the reaction products.

COENZYME II (NICOTINAMIDE ADENINE DINUCLEOTIDE PHOSPHATE; NADP): A pyridine-based nucleotide that functions as a coenzyme to several different oxidoreductases. NADP usually acts as an electron donor in enzymatic reduction reactions. In this it differs from its counterpart NAD, which acts principally as an electron acceptor in biological oxidations. NADP is sometimes called TPN, an abbreviation for its trivial name triphosphopyridine nucleotide.

COENZYME Q (UBIQUINONE; COENZYME Q10; VITAMIN Q10): A lipid-soluble electron-carrying coenzyme that participates in the transport of electrons from NAD to oxygen in the respiratory chain. Its respiratory chain is a reversibly reducible quinone. Plastoquinone, a closely related compound, performs a similar function but in photosynthetic electron transport.

COERCIVE FORCE (COERCIVITY): The magnetising force required to reduce the flux density in a magnetic material to zero after the magnetisation of the sample has been driven to saturation. It is measured in oersted or ampere/metre and its symbol is H_c .

COETANEOUS: 1. Occurring at the same duration or time. 2. Plants with its flowers and leaves developing at the same time or flowering as the leaves expand; synanthrous.

COEVOLUTION: The evolution of certain structures and behaviours within a species that can best be understood in relation to another species. An example is how insects and flowering plants have evolved.

COEXTRACTION: Formation of mixed-species aggregates in a low-polarity organic phase.

COFACTOR: 1. A non-protein chemical compound bound to a protein essential for the normal catalytic activity of an enzyme. They are usually heat stable, unlike most enzyme proteins, which are denatured by heat. Many enzymes require metal ions as cofactors. In some enzymes the metal ion is the primary catalytic centre, while in others it may act to bind enzyme and substrate together or to stabilise the enzyme in its catalytically active conformation. 2. The determinant obtained by deleting the row and column of a given element of a matrix or determinant. Consider the square matrix $A [a_{ij}]$. The cofactor, A_{ij} of the entry a_{ij} which is given by $(-1)^{i+j}|A|$, where $|A|$ is the determinant of the matrix A , obtained by deleting the i^{th} row and j^{th} column of the matrix A .

COFFEE: Tropical evergreen shrub of the genus *Coffea*, of the family Rubiaceae or its seeds, called beans. The term also applies to the beverage made by brewing the roasted and ground beans with water, which contains caffeine and has a mildly stimulating effect. They are shrubs or small trees native to tropical and southern Africa and tropical Asia. It usually grows to a height of 3-3.5 metres. Most commonly cultivated coffee species grow best at high elevations, but are intolerant of freezing temperatures. The trees produce red or purple fruits called cherries that look like drupes, but are epigynous berries. The cherries contain two seeds. Several species including *Coffea arabica* and *Coffea canephora* are grown for the beans. The beans constitute one of the world's most valuable and widely traded commodity crops and an important export product of several countries. Dried seeds or beans are roasted, ground and brewed to make one of the most important beverages in the world. Coffee is also widely used as flavouring, for example, in ice cream, pastries, candies and liqueurs. Because it contains caffeine, dried ripe seeds are used as a stimulant, nervine and diuretic, acting on central nervous system, kidneys, heart and muscles.

COFFEE BERRY BORER: A small beetle, *Hypothenemus hampei*, which lays its eggs inside green coffee berries. It is the most harmful pest to coffee crops worldwide.

COFFEE CHERRY: The ripe red fruit of the coffee plant, which contains the coffee beans.

COFFEE RUST: A fungus disease of coffee plants caused by *Hemileia vastatrix* of the order Uredinales. Infection appears as yellow-orange powdery lesions or spots on lower surface of leaves.

COFFEEScript: A programming language that predictably transcompiles to JavaScript. Programs can be written with less code and in more human-readable terms, without sacrificing runtime performance, compared with JavaScript.

COFFERDAM: A temporary dam used to divert a stream in the early stages of dam-building. The term is also used for a structure similar to an open caisson. Cofferdams are usually constructed of pile-driven steel sheeting and posts.

COFINAL: Of a subset, having the property that if A is a set and $\leq$ a binary relation on A , then a subset B of A is said to be cofinal if it satisfies the following condition: For every $a \in A$, there exists some $b \in B$, such that $a \leq b$. In a partially ordered set, all countable directed sets without a greatest element are of one cofinal type.

COFINITE SUBSET: Any set whose complement is finite. A cofinite subset of a set X is a subset A whose complement in X is a finite set. A contains all but finitely many elements of X .

COFUNCTION (COMPLEMENTARY FUNCTION): A trigonometric function whose values for any argument are equal to the given function of the complementary angle. Each trigonometric function has a cofunction. Cosine is the cofunction for sine,

since $\sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$, cotangent is the cofunction for tangent and cosecant is the cofunction for secant. The cofunction of the trigonometric function $f(\theta)$ is equal to $f\left(\frac{\pi}{2} - \theta\right)$. It is a cofunction

because $\frac{\pi}{2} - \theta$ is a complement of θ . For example, $\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$.

COGENERATION (COMBINED HEAT AND POWER, CHP):

An energy production process in which the waste heat from the generation of electricity, such as exhaust from gas turbines, is used for other purposes. It involves producing power from a specific fuel source, such as natural gas, biomass, coal, or oil. During fuel combustion, cogeneration captures the excess heat which would have otherwise been wasted. The captured heat can be used to boil water, create steam, heat buildings, for industrial purposes, etc. When the heat captured is used to produce electricity, the process is referred to as **combined cycle**.

COGNITION: In psychology, it is conscious mental activities involved in synthesising information. Examples are perception such as seeing and hearing; attention; memory; and reasoning.

COGNITIVE DISSONANCE: A state of psychological tension occurring when two or more contradictory beliefs, ideas or values are presented at the same time or if one is confronted by new information that conflicts with existing beliefs, ideas or values. An example is if a choice has to be made between two equally attractive or equally unpleasant alternatives. The dissonance is greater the closer the alternatives are in attractiveness or unpleasantness.

COGNITIVE ENHANCERS (Nootropics; Smart Drug):

Any drug or combination of nutrients such as vitamins, amino acids, minerals and sometimes herbs that is said to enhance the functioning of the brain such as improvement of memory, learning, attention, concentration, problem solving, reasoning, social skills, decision making and planning, etc. The drugs have been used to treat people with neurological or mental disorders, but there is no scientific evidence to suggest that these drugs have significant effect on healthy people.

COGNITIVE PSYCHOLOGY: The branch of psychology that studies information-processing functions in humans and animals, covering their role in learning, memory, reasoning and language development. In other words, it evaluates the relationship between what happens within the mind that links stimulus (input) and response (output). Cognitive psychologists use a number of experimental techniques that include laboratory-based research with normal and brain-damaged subjects, as well as computer and mathematical models to test and validate theories.

COHERENCE: In physics, it is a property of two or more waves in which there is a fixed relationship between the phase of waves in a beam of radiation of a single frequency. Two beams of light are coherent when the phase difference between their waves is constant. They do not have to be in phase to be coherent, but must have the same frequency and wavelength and are normally produced from one source. The waves will interfere constructively if in phase, thereby enhancing one another. If they interfere destructively, they are out of phase, hence, cancelling one another. Light from two lamps will brighten a room but there will be no interference pattern, because the two light sources are not coherent. Coherent light is usually monochromatic and the most common source of such light for practical uses is from a laser.

COHERENCE LENGTH: A length-scale giving the range within which electrons in a superconductor can interact to form Cooper pairs, a concept put forward by Sir Brian Pippard. The significance is that interference will be strong within a coherence length of the source, but not beyond it; a concept commonly used in telecommunication engineering. In long-distance transmission systems, the coherence length may be reduced by propagation factors such as dispersion, scattering and diffraction.

COHERENCE TIME: The average time required for the relative phase of two waves or the phase of a single wave, to fluctuate appreciably.

COHERENT: 1. In botany, it is the joining together of like parts, but only superficially. The attachment is not as firm as a connate. 2. In physics, two wave sources are perfectly coherent if they have a constant phase difference and the same frequency, and the same waveform; they are noncoherent if there is a random or changing phase relationship.

COHERENT ANTI-STOKES RAMAN SPECTROSCOPY (CARS):

A form of Raman spectroscopy enabling the intensity of Raman transitions to be increased. In this technique, two laser beams with different frequencies pass through a sample producing electromagnetic radiation of several frequencies. A practical application is to estimate temperatures in different parts of a flame from the intensity of the transitions.

COHERENT BACKSCATTERING (CBS): A phenomenon in which partial waves traversing time-reversed (momentum-reversed) scattering paths interfere constructively in the backscattering direction leading to the appearance of an intensity cone.

COHERENT LIGHT: Two sources of light waves that are in phase with each other (the waves emitted from them have the same frequency). For example, two waves are coherent if the crests of one wave are aligned with the crests of the other and the troughs of one wave are aligned with the troughs of the other. Otherwise, these light waves are considered incoherent. Light produced by lasers are coherent light. Light from light bulbs or the Sun, however, are incoherent light.

COHERENT RADIATION: An electromagnetic radiation in which two or more sets of waves have a constant phase relationship, with peaks and troughs always similarly spaced.

COHERENT SCATTERING: A scattering in which there is a definite phase relationship between incoming and scattered particles or photons.

COHERENT SOURCE: Those sources such as lasers, where the radiation has a constant phase relationship between waves spatially as well as temporally. Non-coherent optical sources emit radiation which is randomly distributed in phase, spatially as well as temporally. Most sources which are used in spectroscopy and for spectrochemical analysis conform to this latter group.

COHERENT STATE: The quantum-mechanical state of a system that most closely resembles the corresponding classical state of the system. An example of a coherent state is the quantum state of a harmonic oscillator or of an electromagnetic field. The concept of a coherent state is very important in quantum optics. It was the first example of quantum dynamics when Erwin Schrödinger derived it in 1926 while searching for solutions of the Schrödinger equation that satisfy the correspondence principle.

COHERENT UNITS (COHERENT DERIVED UNITS): A system of units of measurement such as the International System, in which derived units are obtained by multiplying or dividing base units without the use of a numerical factor other than 1. For example, for a derived quantity of acceleration with the symbol a , the SI coherent derived unit is metre per second squared (m/s^2), the metre, and second are base units of the system.

COHESIN: A multiprotein complex that regulates the separation of sister chromatids during cell division, either mitosis or meiosis. It includes members of the structural maintenance of chromosomes (SMC) family of proteins.

COHESION (COHESIVE ATTRACTION; COHESIVE FORCE):

The attraction between particles of the same type or a phenomenon in which interaction between two surfaces of the same material in contact makes them stick together. An example is water, which is strongly cohesive as each molecule may make four hydrogen bonds to other water molecules in a tetrahedral configuration. When it happens between two different materials with similar phenomenon, it is called **adhesion**.

COHESION THEORY: In botany, it is the currently accepted theory explaining the ascent of sap in the xylem. It postulates that the reduction in water potential in the leaf, due to an increase in the rate of evaporation, causes water to flow through the plant to replace the lost water. The column of water in the xylem can withstand considerable tension before breaking because of the very strong cohesive forces between the water molecules.

COHESIVE FORCES: All the forces of attraction between similar particles. For example, there is a strong cohesive force between molecules of mercury. That is why mercury detaches itself from the wall of the vessel in which it is contained and forms globules, when poured on the ground without clinging to it.

COIL: 1. A device that supplies a high voltage to the spark plugs in an internal-combustion engine. 2. A spiral of wire through which an electric current is passed to function as an inductor or to create a magnetic field.

COINAGE METALS: A group of three malleable ductile transition metals, forming Group 11 (IB) of the Periodic Table. The metals include copper (Cu), silver (Ag) and gold (Au). They are durable and do not readily react with many other materials. They generally do not oxidise in air and are more resistant to corrosion. They are malleable, ductile and are also good conductors. These metals use their d orbital in chemical bonding and so exist in several oxidation states. They can also form complex ions and demonstrate paramagnetism.

COINCIDENCE CIRCUIT: An electronic circuit that gives an output only if two input signals are fed to it simultaneously or within a specified time of each other. Coincidence circuits are widely used in particle physics experiments and in other areas of science and technology.

COINCIDENCE RESOLVING TIME: The greatest time interval that can elapse between the occurrence of two or more consecutive signal pulses, in order that the measuring device processes them as a coincidence.

COINCIDENT: 1. In mathematics, it refers to configurations having all points in common. 2. Two lines or shapes that lie exactly on top of each other. Any two lines can be coincident, parallel or intersecting lines.

COKE: A hard black solid residue containing 90% carbon. It is obtained after coal has been heated to a high temperature in an air-tight oven. The heating drives off volatile components and renders the coke porous and cleaner as fuel. It is a useful domestic and industrial fuel used by the iron and steel industries.

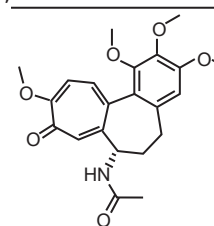
COKE BREEZE: A by-product of coke manufacture, which is the residue from the screening of heat-treated coke. The particle size is less than 10mm. Generally, coke breeze has a volatile matter content.

COKING (FOULING): The phenomenon in which many reactions involving hydrocarbons and particularly those running at higher temperatures lead to the deposition on the catalyst of high molecular weight compounds of carbon and hydrogen which deactivate the catalyst. Catalysts so deactivated can often be regenerated.

COL: 1. Also known as **saddle point**, it is a pass between two peaks or ridges. Landscape cols may have been formed by the headward erosion of a cirque, river capture, the beheading of dip-slope valleys by scarp retreat or the localised differential erosion of a ridge. 2. In meteorology, it is a narrow belt of relatively low pressure, but not a

depression, between two anticyclones. Here, isobars are few and therefore winds are slack.

COLCHICINE: An alkaloid derived from the autumn crocus, *Colchicum autumnale*, with the chemical formula $C_{22}H_{25}NO_6$. It inhibits spindle formation in cells during mitosis so that chromosomes cannot separate during anaphase, thus inducing multiple sets of chromosomes, a condition called **polyploidy**. It is used to inhibit cell division and cause chromosome doubling in plants, used in cancer therapy and as an immunosuppressive drug and also to treat gout.



Chemical structure of Colchicine

COLD BACKUP (OFFLINE BACKUP): A database backup when the database is offline and thus not accessible for updating. This is the safest way to back up because it avoids the risk of copying data that may be in the process of being updated. However, a cold backup involves downtime because users cannot use the database while it is being backed up.

COLD BLOODED: Organisms having a blood temperature which varies with the temperature of the surroundings. The temperature of cold-blooded animals is always maintained slightly below the outside temperature to prevent the loss of body moisture through evaporation. Cold-blooded organisms tend to favour cool, dark places during the day to avoid excessive body temperatures. Examples are insects, reptiles, frogs and fish.

COLD BOOT (HARD BOOT): The process of turning on the computer after it has been powered off, in contrast to a **warm boot**, which refers to restarting a computer once it has been turned on. A cold boot is typically performed by pressing the power button on the computer.

COLD BUFFER: In data processing, it is a segment of computer memory reserved for temporary data storage (buffer) that has not been used or accessed recently.

COLD EMISSION: The emission of electrons from a metal or semiconductor into vacuum under the influence of a strong electric field. This occurs without the use of high temperature but either as a result of field emission or secondary emission.

COLD FRONT: A zone of transition between warm and cold air masses where cold air moves in under a warm air mass. It is the leading portion of a cold atmospheric air mass moving against and eventually replacing a warm air mass at ground level. Cold fronts typically move from northwest to southeast and bring cooler temperatures to an area and can cause changes in weather patterns including thunderstorms or other types of extreme weather. On a weather map cold fronts are shown as blue lines with triangles.

COLD FUSION: The fusion of atomic nuclei at room temperature, which could provide a limitless, cheap and pollution free source of energy. It occurs at lower temperatures than those necessary to overcome the electrostatic repulsion between nuclei.

COLD NEUTRONS: Neutrons with a neutron temperature considerably lower than normal room temperature.

COLD SERVER: A recovery backup server that operates only if the main server is interrupted, lost or fails. A cold server holds all main server files and programs and remains in an unpowered state until a backup is required. It is turned on periodically to have software installed and configured and then is turned off until needed. A **warm server** is a backup server that is turned on periodically to receive updates from the server being backed up. Warm servers are often used for replication and mirroring. A **hot server** is a backup server that receives regular updates and is on standby to take over immediately in case of a failure of the system.

COLD WORKING: A method of shaping metal at or near atmospheric temperature by plastic deformation. The processes include rolling, drawing, pressing, spinning, extruding and heading. It is carried out below the recrystallisation point usually at room temperature. Hardness and tensile strength are increased with the degree of cold work whilst ductility and impact values are lowered. The cold rolling and cold drawing of steel significantly improves surface finish.

COLDFUSION: A product from Macromedia, for building Web sites and serving pages to users. It can be used to build a content database using input templates and combining these with application programs to create a Web site in which pages are developed dynamically as they are served.

COLDFUSION MARKUP LANGUAGE (CFML): A scripting language for web development that runs on the JVM, the .NET framework and Google App Engine.

COLDSET WEB OFFSET PRINTING: A printing technique in which the ink is allowed to dry naturally through evaporation and absorption. **Heatset Web offset printing** dries the ink rapidly by forced-air heating. **Web offset printing** is a printing technique in which a continuous roll of paper is fed through a printing press.

COLEMAN-GROSS THEOREM: A theory that non-Abelian gauge theories with unbroken gauge theories are the only renormalisable quantum field theories able to have asymptotic freedom.

COLEOPTERA: An order of insects comprising the beetles and weevils and containing 330,000 known species, the largest order in the animal kingdom. Beetles are found worldwide, in virtually all terrestrial and aquatic habitats. Most adult beetles have a hard, dense exoskeleton that covers and protects most of their body surface. They have chewing mouthparts, sometimes located at the tip of a beak or snout. The front wings, known as elytra are hard and serve as covers for the hind wings and meet in a line down the middle of the back. The hind wings are large, membranous and folded beneath the elytra. In flight, a beetle holds the elytra out for balance and uses its membranous hindwings for movement. Many beetles are regarded as major pests of agricultural plants and stored products. They attack all parts of living plants as well as processed fibres, grains and wood products. Scavengers and wood boring beetles are useful as decomposers and recyclers of organic nutrients. Predatory species, such as lady beetles, are important biological control agents of aphids and scale insects.

COLEOPTILE: The protective sheath that surrounds the young shoot tip of the embryo in plants of the grass family during its passage through the soil to the surface. Coleoptiles have two vascular bundles, one on either side.

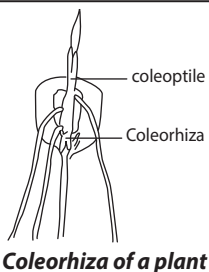
COLEORHIZA: A protective sheath that covers the embryonic root of grasses.

COLIC: Pain or discomfort in any part of the intestinal tract or abdomen, which starts and stops abruptly.

COLIFORM BACTERIA: A group of Gram-negative rod-shaped bacteria that are found in the vertebrate gastrointestinal tract, their presence in water is an indicator of faecal pollution. Examples are *Escherichia coli* (*E. coli*) and *Salmonella* species.

COLIPHAGE: Any DNA bacteriophage that infects the bacterium *Escherichia coli* (*E. coli*). Examples are the intensively studied T series of phages (T1 through to T7), which have been developed on a particular strain of *E. coli*.

COLLABORATION DIAGRAM: A diagram that shows object interactions organised around the objects and their links to each other. Sequence and collaboration diagrams express similar information, but show it in different ways. A collaboration diagram shows the relationships between the objects, shown as rectangles with labels. The labels are preceded by colons and may be underlined. The relationships between the objects are shown as lines connecting the rectangles. The messages between objects are shown as arrows connecting the relevant rectangles along with labels that define the message sequencing. In a sequence diagram, object interaction occurs through messages on the vertical and horizontal dimensions and are designated by horizontal arrows and message names. The initial sequence diagram message begins at the top and is located on the diagram's left side. Subsequent messages are added just below previous messages.



COLLABORATION PLATFORM: A type of business software that adds broad social networking capabilities to work processes. The goal of a collaboration platform is to foster innovation by incorporating knowledge management into business processes so employees can share information and solve business problems more efficiently.

COLLABORATIVE BROWSING (CO-BROWSING): Synchronising browser access to the same sites. As one user browses the Web, the other users trail along automatically and link to and view the same pages from their own Web browsers. For example, call centre agents might use collaborative browsing to help customers navigate the company Web site.

COLLAGEN: A naturally occurring insoluble fibrous protein found extensively in the connective tissue of skin tendons and bones of mammals. It is the chief component of connective tissue.

COLLAPSTAR: A neutron star, white dwarf or any other star composed of degenerate matter. Black holes are also regarded as collapsars.

COLLAR: Light green or white banded area on the outside of a leaf blade.

COLLATERAL: 1. Running side by side or parallel. 2. A living organism such as a plant or an animal which descends from a common ancestor but through different lines.

COLLATERAL BUNDLE: A vascular bundle in which the phloem occurs on only one side of the xylem. In the stem the xylem is usually internal to the phloem. This type of bundle is common in both angiosperms and gymnosperms.

COLLECTING DUCT: Any of the tubules and ducts in the mammalian kidney that drains into the renal pelvis, which leads to the ureter. It participates in electrolyte and fluid balance through reabsorption and excretion, processes regulated by the hormones aldosterone and antidiuretic hormone.

COLLECTION: In chemistry, it is the planned removal from solution of a micro- or macro-component by the intentional formation of a contaminated host precipitate or by the adsorption or entrapment of the component by an added solid.

COLLECTION EFFICIENCY: A term which characterises an entire sampling and sample pre-treatment procedure, usually represented as a percentage of the original amount of the analyte which is left for measurement (signal formation) after having passed through this procedure. This term also applies to the efficiency of collection of an air pollutant by an arrestment plant. For particulates, the collection efficiency is size related.

COLLECTION GAS FLOW SYSTEM: A system in which the analyte is carried to a collecting device called the collector which may be an amalgamator for mercury, a cold trap for some of the more stable hydrides, an absorbing solution in which the analyte is absorbed by some chemical reaction, an electrothermal atomizer or a reservoir.

COLLECTIVE EXCITATION: A quantized mode in a many-body system, occurring because of cooperative motion of the whole system as a result of interaction between particles.

COLLECTIVE OSCILLATION: A mode of oscillation in a many-body system, in which there is cooperative motion of the system as a result of interaction between the particles in the system.

COLLECTOR: 1. The terminal of a transistor towards or through which charge carriers flow. 2. A term which is often applied to cleaning devices in exhaust systems for a device for removing and retaining contaminants from air or other gases. 3. A device for removing and retaining samples from media in different environmental compartments to be investigated. 4. Also known as **scavenger**, it is a solid substance added to or formed within a solution to collect a micro- or macro-component.

COLLEMBOLA: An order of small wingless insects, the springtails, belonging to the subclass Apterygota. They are less than 10 mm long, undergo a metamorphosis and leap by means of specialised forked organ that is attached on the underside of the abdomen by a special catch, which acts as a spring. When released, the spring flies backward, catapulting the insect.

COLLENCYMA: Plant tissue composed of relatively elongated living cells with irregularly thickened cell walls, in particular at the corners. They provide structural support to growing shoots and leaves and herbaceous plants. The thickened cell walls of collenchyma cells are sometimes used as a supply of cellulose for the other tissues in times of shortage. Examples are young herbaceous stems and leaves.

COLLIDING-BEAM EXPERIMENTS: In a colliding-beam experiment two beams of high-energy particles are made to cross each other. Since the collision occurs at relativistic speeds, the relevant measure of available energy is the centre-of-mass frame. The advantage of this arrangement is that both beams have significant kinetic energy, so a collision between them is more likely to produce a higher mass particle than would a fixed-target collision (with the one beam) at the same energy.

COLLIGATION: The combination of two radicals to form a covalent bond. An example is $\text{H}_3\text{C} \cdot + \text{Cl} \cdot \rightarrow \text{CH}_3\text{Cl}$.

COLLIGATIVE PROPERTIES: Properties of a solution that depend on the concentration or number of solute particles present and not on the nature of the particles. Colligative properties of a solution include freezing point, boiling point, osmotic pressure and solvent vapour pressure.

COLLIMATION: The limiting of a beam of radiation to the required dimensions and angular spread.

COLLIMATOR: 1. Any device for producing a parallel beam of radiation by causing the directions of motion to become more aligned in a specific direction or by causing the spatial cross section of the beam to become smaller. 2. A small telescope attached to a large optical instrument to fix its line of sight.

COLLINEAR: 1. Of or pertaining to two points, lying on the same straight line. 2. Of or pertaining to the state of congruence between the order of amino acids along a polypeptide chain and the order of the corresponding codons along the polynucleotide chain encoding the polypeptide.

COLLINEAR REACTION: A reaction, which is assumed to occur via an activated complex in which all of the atoms directly involved in the process lie along a straight line.

COLLINEARITY: Variables are mathematically collinear if one of them is a linear combination of the other variables. They are known as statistically collinear if one of them is approximately a linear combination of other variables. In the case of a regression model where the explanatory variables are strongly correlated to each other, there is a collinearity or multicollinearity between the explanatory variables. In the first case, it is simply impossible to define least squares estimators and in the second case, these estimators can exhibit considerable variance.

COLLINEATION: A mathematical transformation that maps points or lines as corresponding elements to another plane or space or onto itself.

COLLISION: A phenomenon that occurs when two or more bodies or particles intersect and physically exert relatively strong forces against each other or when the path of one body is changed by the influence of another body. A **perfectly elastic collision** occurs when both the kinetic energy and momentum are conserved; and an **inelastic collision** is when only the momentum is conserved but the kinetic energy is not conserved and is changed into some other form of energy. Forces acting in a system of colliding bodies are action and reaction forces.

COLLISION COMPLEX: An ensemble formed by two reaction partners for which the distance is the sum of their van der Waals radii. As such it constitutes a subclass of the species indicated as encounter complex.

COLLISION CROSS-SECTION: In simple hard sphere collision theory the area of the circle with radius is equal to the collision diameter.

COLLISION DENSITY (COLLISION NUMBER): The total number of collisions that occur in unit volume per unit time when a given neutron flux passes through matter. Its symbol is Z_{AA} .

COLLISION DIAMETER: In simple hard sphere collision theory the sum of radii of two colliding spherical particles. Its symbol is d .

COLLISION FREQUENCY: The average number of collisions experienced by a molecule A with other molecules A (or B) in a time interval divided by that time interval.

COLLISION THEORY: The theory that explains how chemical reactions take place and why rates of reactions differ for different reactions. It states that for a reaction to occur, the reactant particles must collide, but only a fraction of the total collisions have the energy to form bonds and cause the reactants to transform into the products. The minimal amount of energy needed to break any existing bonds and form new ones is known as the **activation energy**. As temperature increases, the average kinetic energy and speed of the molecules increase but this only slightly increases the number of collisions. The rate of the reaction increases with temperature increase because a higher fraction of the collisions overcome the activation energy.

COLLISION-INDUCED DISSOCIATION: An ion/neutral species process wherein the projectile ion is dissociated as a result of interaction with a target neutral species. This is brought about by conversion of part of the translational energy of the ion to internal energy in the ion during the collision.

COLLISIONAL ACTIVATION (COLLISIONALLY ACTIVATED DISSOCIATION): An ion/neutral species process wherein excitation of a projectile ion of high translational energy is brought about by the same mechanism as in collision-induced dissociation. The ion may decompose subsequently.

COLLISIONAL BROADENING: Collisions of the emitting or absorbing particle with other particles cause collisional broadening as well as collisional shift of the spectral line. When collisions occur between unlike, neutral particles the term **foreign-gas broadening** is used or **van der Waals' broadening** when both collision partners are neutral. When the colliding particles are of the same species, the term **resonance broadening** is employed. **Stark broadening** refers to collisions with charged particles or particles with a strong permanent electrical dipole moment. Whereas a strong chaotic electrical field causes Stark broadening, an applied static electrical field induces a Stark shift.

COLLISIONAL EXCITATION: An ion/neutral species process wherein there is an increase in the reactant ion's internal energy at the expense of the translational energy of either (or both) of the reacting species. The scattering angle may be large.

COLLODION: A thin film of cellulose nitrate made by dissolving the cellulose nitrate in ethanol or ethoxyethane, coating the surface and evaporating the solvent. When collodion is spread on a surface and exposed to air, the solvents evaporate, leaving a thin, colourless, elastic film on the surface.

COLLOID: A substance composed of extremely small particles of one substance, called the **dispersed phase**, which is evenly distributed in another material called the **dispersion medium**. The particles are so small that they are unaffected by gravity and remain in suspension permanently. An example is emulsion, which is a colloidal dispersion of liquid particles in another liquid. For example, mayonnaise is a suspension of tiny globules of oil in water.

COLLOID MILLS: Machines used to grind aggregates into very fine particles in suspension in a liquid or to reduce the droplet size of a liquid suspended in another liquid by applying very high shearing forces within a fluid to produce colloid suspensions or emulsions in which the particle sizes are less than 1 micrometre. There are three types of colloid mills: disc mill, valve mill and orifice mill.

COLLOID OSMOTIC PRESSURE (DONNAN PRESSURE): The pressure difference which has to be established between a colloidal system and its equilibrium liquid to prevent material transfer between the two phases when they are separated by a membrane, permeable to all components of the system, except the colloidal ones.

COLLOIDAL: The term refers to a state of subdivision, implying that the molecules or polymolecular particles dispersed in a medium have at least in one direction a dimension roughly between 1 nm and 1 μ m or that in a system discontinuities are found at distances of that order.

COLLOIDAL CARBON: A particulate carbon with particle sizes below 1000 nm in at least one dimension. Colloidal carbon exists in several morphologically distinct forms.

COLLOIDAL DISPERSION: A system in which particles of colloidal size of any nature, for example, solid, liquid or gas are dispersed in a continuous phase of a different composition or state. The name dispersed phase for the particles should be used only if they have essentially the properties of a bulk phase of the same composition.

COLLOIDAL ELECTROLYTE: An electrolyte which gives ions of which at least one is of colloidal size. It includes hydrophobic sols, ionic association colloids and polyelectrolytes.

COLLOIDAL GEL: Gel in which the network component comprises particles of colloidal dimensions.

COLLOIDAL NETWORK: Network comprising particles of colloidal dimensions.

COLLOIDAL PROCESSING: Sol-gel processing in which a network of precipitated colloidal particles is treated by a conventional processing technique, such as cold pressing, hot pressing or sintering, in order to produce a ceramic article.

COLLOIDAL SUSPENSION: A suspension in which the size of the particles lies in the colloidal range.

COLLOIDALLY STABLE: Particles which do not aggregate at a significant rate: the precise connotation depends on the type of aggregation under consideration. For example, a concentrated paint is called stable by some people because oil and pigment do not separate out at a measurable rate and unstable by others because the pigment particles aggregate into a continuous network.

COLLUVIAL (COLLUVIUM): Sediment or soil deposited by gravity at the bottom of a low-grade slope or against a barrier on that slope.

COLOCASIA: A genus of perennial flowering plants in the family Araceae, native to tropical regions with a tuberous rootstock called a corm growing just below the ground surface. It is found throughout the tropics and much of the subtropics, where water is available most of the year, sometimes growing wild in swampy areas with a pH between 4.2-7.5. They can be grown in the ground in moist soil or in large containers. The leaves are large to very large and are arrowhead-shaped that point earthward. The new leaves and stems push out of the innermost stalk, unrolling as they emerge with a stalk which is attached almost in the middle and not at the edge. The species, *Colocasia esculenta* also known as *Colocasia antiquorum* with common names taro, swamp taro, old cocoyam and arrowroot, is the most common form cultivated both as an ornamental plant and as a food source. When the plant reaches maturity, it produces a flower stalk in some leaf axils. Near the apex of the flower stalk appears yellow-white, tubular spathe or modified leaf, which covers and protects the flower cluster within. Inside the flower cluster grows an erect spike called the spadix, which bears male and female flowers. Its corm is shaped like a top with rough ridges, lumps and spiny roots. Its primary use is the consumption of the corm, which provides easily digested starch. In its raw form, the plant is toxic due to the presence of calcium oxalate. Some of the numerous colocasia species include *Colocasia affinis*, *Colocasia latifolia*, *Colocasia formosan*, *Colocasia humilis*, etc. A closely related genus is *Xanthosoma*. The most common cultivated species as a food source is *Xanthosoma sagittifolium* with common names of tannia, new cocoyam or elephant ear. It is a herbaceous tropical perennial plant grown primarily for its starchy edible corms, although the fresh leaves are cooked and eaten as vegetable. Some other cultivated species in the genus include *Xanthosoma atrovirens*, *Xanthosoma caracu* and *Xanthosoma nigrum* (*X. violaceum*).

COLOGARITHM: Abbreviated colog, it is the logarithm of the reciprocal of a number, which is equal to the negative base or the additive inverse of its logarithm. For example, $\text{colog}100 = \log 0.01 = \log 10^{-2} = -2 = -\log 100$.

COLUMBIFORMES: An order of birds consisting of two families, the pigeons and doves referred to as Columbidae, and dodos referred to as Raphidae. The order has about 42 genera and 310 species. The dodos are now extinct. Pigeons occur in all terrestrial habitats from deserts to forest all over the world except the Arctic, the Antarctic, and very high mountains. They have stout bodies, birds, small- to medium-sized heads, short necks, and short slender bills, and some species have fleshy ceres. The wings are usually broad with rounded tips. They primarily feed on seeds, fruits, and plants.

COLON: A region of the large intestine in mammals between the caecum and rectum, which receives undigested food from the ileum. Water and mineral salts are absorbed from digested food and the residue is formed into faeces in the colon.

COLONISATION (COLONIZATION): In ecology, it is the integration of a species into a new habitat, such as a freshly cleared field or a recently flooded valley. The first species to move in are called pioneers and may establish conditions that allow other animals and plants to move in; for example, by improving the condition of the soil. Over time a range of species arrives and becomes integrated into a community and the habitat matures. Early colonisers will probably be replaced, so that the variety of animal and plant life changes, known as succession.

COLONY: 1. A group of organisms such as animals or plants of the same species living or growing in the same place and dependent upon each other. An example is a colony of social insects such as ants and bees consisting of usually castes with different responsibilities.

Colonisation occurs when one or more species such as birds, bacteria or plants populate an area such as cleared field. 2. A group of microorganisms usually bacteria or yeasts that are considered to have developed from a single parent cell.

COLONY-STIMULATION FACTOR (CSF; COLONY STIMULATING FACTOR 1 RECEPTOR, CSF1R; MACROPHAGE COLONY-STIMULATING FACTOR RECEPTOR, M-CSFR; CLUSTER OF DIFFERENTIATION): Any one of several cytokines that stimulates development of certain type of blood cells from haemopoietic stem (progenitor) cells in the bone marrow and in other tissues.

COLORIMETER: 1. Any instrument, used for comparing or reproducing colours. 2. An instrument that determines the absorbance of a solution of a coloured substance. From the measured absorbance the concentration of the solution can be obtained by applying the Beer-Lambert law, which states that the concentration of a solute is proportional to the absorbance.

COLORIMETRIC ANALYSIS: Quantitative analysis of solutions based on the absorption of visible light at a definite wavelength.

COLOSSAL MAGNETORESISTANCE (CMR): A property of certain materials, especially manganese-based perovskite oxides, in which electrical resistance can change by several orders of magnitude in the presence of a magnetic field. The magnetoresistance of conventional materials enables changes in resistance of up to 5%, but materials featuring CMR may demonstrate resistance changes by orders of magnitude.

COLOSTRUM: A liquid with a high content of nitrogen, antibodies and vitamins that is secreted from the mammary glands in late pregnancy and just after giving birth. It is the first milk produced by mammals.

COLOUR: An aspect of any object that may be described in terms of hue, lightness and saturation. It is light that is visible, ranging from 400 (red) to 700 nanometers (violet) or the frequency of visible light, ranging from 430 THz (terahertz, 10^{12} Hz), seen as red, to 750 THz, seen as violet. They are different wavelengths of visible light or visible quality that objects have, produced by rays of light of different wavelengths being reflected by them. The frequency is the number of waves that pass a point in space during any time interval, usually one second. When all the waves are seen together, they make white light. When white light shines through a prism, the white light is broken apart into the colours of the visible light spectrum. Water vapour in the atmosphere can also break apart wavelengths creating a rainbow.

The colours red, yellow and blue, known as primary colours, can be combined in varying proportions to produce all other colours.

COLOUR BLINDNESS: Any disorder of vision in which colours are confused or inability to distinguish between certain colours, especially red, blue and green. This is caused by the absence of cone cells in the retina of the eye that absorb light in different parts of the spectrum.

COLOUR BREED: Breed of horses based on colour.

COLOUR CENTRE: 1. A defect in a crystal that changes the way in which it absorbs light or other electromagnetic radiation. Impurities in the crystal affect the bond structure and allow transitions in different regions of the spectrum. Impurity colour centres are responsible for the characteristic colours of many gemstones. 2. A region in the human brain responsible for processing colour.

COLOUR CONFINEMENT (CONFINEMENT): The phenomenon that colour charged particles such as quarks cannot be isolated singularly and therefore cannot be directly observed or the force between quarks does not diminish as they are separated. Because it would take an infinite amount of energy to separate two quarks; they are forever bound into hadrons such as the proton and the neutron.

COLOUR FASTNESS: A resistance of colour to fading or having colour that retains its original hue without fading or running after washing. The term is usually used in the context of clothes.

COLOUR FORCE: A manifestation of the strong force for which gluons are the carriers. Quarks (except for the top quark, whose short 10^{-24} seconds lifetime means that it decays before it can form a meson) are permanently confined in particles by the colour force.

COLOUR INDEX (CI): 1. A list regarded as definitive of dyes and pigments, which includes information on their commercial names, method of application, colour fastness, etc. 2. In astronomy, a method of indicating a star's colour in a quantitative manner that yields a close approximation of the star's actual surface temperature.

COLOUR INDICATOR: An indicator which is classified as one- or two-colour, depending on whether it is colourless on one side of the transition interval or possesses a different colour on each side of this range.

COLOUR PHOTOGRAPHY: Photography that uses media capable of representing colours which are produced chemically during the photographic processing phase.

COLOUR SCALE: A series of ordered numbers that represents observable gradations of a given attribute or gradations of a combination of attributes of colour perception.

COLOUR TELEVISION: A television system in which the camera filters the light from the scene into three primary colour components: red, blue and green which are detected by separate camera tubes.

COLOUR TEMPERATURE: A way of describing the colour of an incandescent light source. The power distribution in its spectrum is related to its absolute temperature in kelvins ($1\text{ K} = 273.15^\circ\text{C}$). For example, the colour temperature of unobscured midday sunlight is around 5,000 K - 5,400, and that of a fluorescent lights is 3,200 K - 7,500 K.

COLOUR VISION: The ability of the eye to recognise different frequencies in the visible spectrum of colour. The brain derives colour by comparing the responses to light from the three types of cone photoreceptors in the eye. These cone photoreceptors are sensitive to different portions of the visible spectrum, from 380 - 740 nm.

COLOURABILITY: 1. Ability of a colourless or slightly coloured photochromic material to develop colour. 2. Ability of a graph or map to be coloured with finitely many colours; or the ability to have a finite chromatic number.

COLOURING: Food additive used to alter or improve the colour of processed food.

COLOURLESS SULFUR BACTERIA: A group of non-photosynthetic bacteria found in marine and freshwater sediments, soils, and wastewater treatment systems that oxidise sulfur compounds, thus deriving their energy by this process.

COLPITTS OSCILLATOR: 1. A type of linear oscillator circuit that uses a combination of inductors (L) and capacitors (C), (called LC oscillator) to produce an oscillation at a certain frequency. Its active device is taken from a voltage divider having two centre-tapped capacitors in series connected in parallel to an inductor to produce sinusoidal oscillations.

COLPUS: An oblong to elliptic germinal aperture in a pollen grain, at least twice as long as it is broad. Pollen with such apertures is termed **colpate**.

COLT: A young male horse or pony of less than four years of age.

COLUMELLA: A small column, especially: (i) the central portion of the root cap of certain roots in which the cells are arranged in longitudinal lines; (ii) the sterile tissue in the centre of the sporangium in mosses, liverworts and fungi; (iii) a rodlike radial element in a pollen or spore wall.

COLUMN: 1. In botany, it is also known as **gynostemium** and refers to a structure formed by the fusion of a stamen and a filament as in the orchid or the union of staminal filaments. **Column foot** is a basal platform between the column and the lip. 2. A vertical linear arrangement of elements, such as those in a determinant or matrix. It could be a vertical linear arrangement in an $n \times 1$ matrix or as part of a bigger matrix with many elements. 3. As a modifier, operating upon or relating to, the columns of a matrix and contrasted with a corresponding row operation.

COLUMN ADDRESS STROBE: In computing, it refers to a signal sent from a processor or memory controller to a dynamic random access memory (DRAM) to indicate that an associated address is a column address.

COLUMN CHROMATOGRAPHY: A separation technique in which the stationary bed is within a tube. The particles of the solid stationary phase or support coated with a liquid stationary phase may fill the whole inside volume of the tube (packed column) or be concentrated on or along the inside tube wall leaving an open, unrestricted path for the mobile phase in the middle part of the tube (open-tubular column).

COLUMN DENSITY: The total amount per unit area of a material suspended in a fluid or in a cloud in space, measured along the length of a column.

COLUMN EQUIVALENCE: The relation that holds between a pair of matrices when one is obtainable from the other by a finite sequence of elementary matrix operations on its columns. Two matrices are column equivalent if and only if their transpose matrices are row equivalent.

COLUMN MATRIX: A matrix with only one column, which is an $n \times 1$. The sum of two column matrices of the same dimension is also a column matrix of the same dimension. If a row matrix is multiplied with a column matrix having the same number of elements, then the resultant has one element only.

COLUMN RANK: The rank of the column space of a matrix, which is the dimension of the vector space generated or spanned by its columns. It is the same as the dimension of the space spanned by its rows.

COLUMN SPACE: The vector space generated by the columns of a matrix, which is the set of all possible linear combinations of its column vectors. The dimension of this space is called the column rank and coincides with both the row-rank and the rank of the matrix. For

example, for a matrix, $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$,

the row vectors are $(1, 0)$, $(0, 1)$ and $(0, 0)$. The row space is the subspace of $\mathbb{R}^2$ spanned by these vectors. Since the first two vectors are the standard basis vectors for $\mathbb{R}^2$, the row space is $\mathbb{R}^2$. The column vectors are $(1, 0, 0)$ and $(0, 1, 0)$. The column space is the subspace of $\mathbb{R}^3$ spanned by these vectors. Thus, the column space consists of all vectors of the form $a \cdot (1, 0, 0) + b \cdot (0, 1, 0) = (a, b, 0)$.

COLUMN VECTOR: The single column of a matrix with its elements, written as $n \times 1$, where n is the number of elements in the column.

For example, for a matrix, $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$, the column vectors are (1,0,0) and (0,1,0).

COLUMN-REDUCED ECHELON FORM: A reduced echelon form with the additional rules attached to rows and columns in row-reduced echelon form reversed.

COLUMN-STOCHASTIC: Of or pertaining to a matrix that is non-negative or has all of its entries greater or equal to zero and the sum of the entries in each column is equal to 1.

COM PORT (RS-232 PORT; SERIAL PORT): A type of connection on PCs that is used for peripherals such as mouse, gaming controllers, modems and older printers. A serial port can only transmit one bit of data at a time, whereas a parallel port can transmit many bits at once.

COMA: 1. A prolonged period of deep unconsciousness. 2. A term used in astronomy to denote a hazy or nebulous cloud of gas and dust that surrounds the nucleus of a comet. 3. In optics, it is an aberration of a lens or mirror in which the image of a point lying off the axis has a comet-shaped appearance or one of the geometrical aberrations of a lens, whereby skewed rays from a point make a comet-shaped spot on the image plane instead of meeting at a point. 4. Tuft of hairs mostly on the tip of seeds. Comose means bearing tuft of hairs.

COMB: 1. Wax material built by honeybees for storing honey. 2. The fleshy red growth on the head of a rooster or other bird.

COMB MACROMOLECULE: A macromolecule comprising a main chain with multiple trifunctional branch points from each of which a linear side-chain emanates. If the subchains between the branch points of the main chain and the terminal subchains of the main chain are identical with respect to constitution and degree of polymerisation and the side chains are identical with respect to constitution and degree of polymerisation the macromolecule is termed a **regular comb macromolecule**. If at least some of the branch points are of functionality greater than three, the macromolecule may be termed a **brush macromolecule**.

COMB POLYMER: A polymer composed of comb macromolecules.

COMBINATION: 1. In mathematics, it is a selection of a number of objects from some large numbers of objects when no account is taken of the order within any one arrangement. For example, 6,7,8; 7,6,8; and 8,6,7 are regarded as the same combination of three digits from 6,7,8. Combinational analysis is used in the study of probability. The number of combinations of n objects, taken r of them at a time is given by $\binom{n}{r} = \frac{n!}{(n-r)!r!}$, also written as ${}_nC_r$ or nC_r . 2. In a structure such as a vector space, any finite sum of appropriate multiples of given elements such as a linear combination, affine combination or convex combination.

COMBINATION ELECTRODE: An ion-selective electrode and an external reference electrode combined in a single unit, thereby avoiding the need for a separate external reference electrode.

COMBINATION REACTION: A chemical reaction in which two substances, elements or compounds, combine to form one compound. A combination reaction can be of three types: between two elements; between two compounds; and between an element and a compound.

COMBINATIONAL LOGIC (COMBINATORIAL LOGIC): A concept in which two or more input states define one or more output states, where the resulting state or states are related by defined rules that are independent of previous states. Each of the inputs and output(s) can attain either of two states: logic 0 (low) or logic 1 (high). An example is a simple logic gate.

COMBINATIONAL OPTIMISATION: A technique of optimisation in which the values of the variables are restricted to integers. Examples include, knapsack and travelling salesman problems.

COMBINATORIAL ANALYSIS (COMBINATORICS): The branch of mathematics dealing with subdivisions of sets, which is primarily

concerned with combinations and permutations and partitions. In other words, it is the study of the ways of choosing and arranging objects from given collections.

COMBINATORIAL BIOLOGY: The technique that transfers genetic material from one microorganism such as bacteria and fungi to another that produces complex natural products. It is mostly used to synthesise products such as antibiotics; and also used in genetic engineering.

COMBINATORIAL CHEMISTRY: A technique in which very large numbers of chemical compounds are synthesised by condensing a small number of reagents together in all combinations defined by a small set of reactions. The main aim of combinatorial chemistry is to synthesise an array of chemicals or biological compounds called libraries. These libraries are then screened to identify useful components, such as drug candidates.

COMBINATORICS: The branch of mathematics that studies the number of ways of selecting or arranging objects from a finite set, with its main applications in the theory of probability for calculating the probability of an event such as r of the objects chosen have a particular property or the probability of a sequence of events with a particular property.

COMBINE DRILL: A drill which sows grain and fertiliser at the same time.

COMBINE HARVESTER: An agricultural machine that both reaps and threshes grain.

COMBINE PROBLEMS: Mathematical situations involving the combination of two discrete sets or splitting of one set into two discrete sets.

COMBINED CYCLE GAS TURBINES (CCGT): A process in which natural gas is used to power a jet engine and the exhaust fumes from the jet engine used to produce steam which turns a generator to produce electricity.

COMBINED HEAT AND POWER GENERATION (CHP GENERATION): Simultaneous production of electricity and useful thermal energy in the form of heat or steam or useful mechanical work, such as shaft power, from the same fuel source in a power station.

COMBINED LAW OF THERMODYNAMICS (GIBBS FUNDAMENTAL EQUATION): The mathematical summation of the first law of thermodynamics ($dU = dQ + PdV$) and the second law of thermodynamics ($dQ \leq TdS$) subsumed into a single mathematical expression. It is expressed as $dU - TdS + PdV \leq 0$, where dU is a variation in internal energy, T is temperature, dS is variation entropy, P is pressure and dV is variation in volume of a simple working body in which there are no flows of particles or out of the body nor external forces, other than gravity, acting on the body.

COMBINED SAMPLE: A sample obtained by removing specific fractions by separation or selection techniques such as heavy liquid, magnetic and sieving, analysing the fractions separately and combining the results mathematically. When not combined, the sample is a modified sample.

COMBINED-CYCLE GENERATION: A system of electricity generation that makes use of both a gas turbine and a steam turbine. The cycle consists of a combustion turbine-generator, with the turbine exhaust gases directed into a waste-heat-recovery boiler that converts the exhaust gas heat into steam which drives a steam turbine, extracting steam to the process while driving an electric generator.

COMBUSTION: The chemical process in which substances combine with oxygen in air producing heat and light. A slow burning candle flame and the explosion of a mixture of petrol vapour and air are extreme examples of combustion. In a complete combustion reaction, a compound reacts with an oxidising element, such as oxygen or fluorine and the products are compounds of each element in the fuel with the oxidising element. **Combustible** is capable of igniting and burning. **Combustibility** is a measure of how easily a substance will ignite, through fire or combustion. It is a necessary factor, when

materials are used for construction or is being stored to prevent fire outbreak.

COMBUSTION EFFICIENCY: The ratio of the energy actually released by the fuel during combustion to the energy contained in the fuel. A perfect combustion would release all the energy a fuel contains, in which case combustion efficiency would be 1 or 100%.

COMBUSTION GAS: A gas or vapour produced in furnaces, combustion chambers or in open burning.

COMET: A small, icy object that moves in closed orbits around the Sun, usually on a highly elliptical path. It looks like a bright star with a long luminous bright tail.

COMETABOLISM: This occurs when an enzyme produced by an organism to catalyse the degradation of its growth-substrate to derive energy and carbon from it is also capable of degrading additional compounds.

COMFORT INDEX: An index which gives a numerical value, in the general range of 70-80, of how tolerable conditions are for humans during the warm seasons of the year. It is calculated as the temperature in Degree Fahrenheit ($32^{\circ}\text{F} = 0^{\circ}\text{C}$) plus a quarter of the relative humidity expressed as a percentage. It is expressed

as; $\text{Comfort Index} = T + \frac{RH}{4}$, where T is temperature in degrees

Fahrenheit and RH is the relative humidity. The presence of wind allows higher values to be tolerated; therefore, when the comfort index is above 95° and the wind speed is low, it may be necessary to acclimatise to the conditions before attempting very strenuous activity.

Discomfort Index is the impact of heat stress on an individual taking into account the combined effect of temperature and humidity.

COMMA-SEPARATED VALUES (CSV): A file format in which individual data is organised so that each column value is separated by a comma from the next column's value and each row starts a new line.

COMMAND: In computer science, it refers to a specific directive from a user to the computer's operating system or to an application to perform a specific task.

COMMAND INJECTION: A technique used by hackers to take advantage of a vulnerability which results in the execution of operating-system commands. A malevolent hacker can exploit that vulnerability to gain unauthorised access to data or network resources. When users visit an affected Web page, their browsers interpret the code, which may cause malicious commands to execute in the users' computers and across their networks.

COMMAND INTERPRETER: The part of a computer operating system that understands and executes commands that are entered interactively by a human being or from a program. In some operating systems, the command interpreter is called the shell.

COMMAND LINE INTERFACE (CLI): Interface software that provides a blank line and cursor on screen, allowing the user to type in instructions to obtain the corresponding response. For example, a user may type in `C:\IEsetup.exe` and the computer will execute the setup files for Internet Explorer stored on the system. All major operating systems such as Windows, Mac, Unix and Linux, support command lines that programmers and power users can employ to perform file management operations directly and often more efficiently than by using a graphical user interface (GUI). After typing a command, it is executed by pressing the enter key.

COMMAND PROMPT: The command line interpreter program `cmd.exe` that allows users to interact with OS by typing in their commands at a prompt rather than using an icon based system. The command prompt looks like `c:\>` and users type in their command after the prompt.

COMMAND-LANGUAGE (CONTROL LANGUAGE): The language of an operating system; a set of commands and the rules governing their use, by which users of a computer control a program. For example, the operating systems may have commands such as `SAVE`, `DELETE`, etc.

COMMELINA: A genus of straggling herbaceous monocotyledonous plants called dayflowers, which are common in open fields because of the short lives of their flowers. Some are fodder plants.

COMMENSALISM: A relationship between two organisms whereby one benefits from the association such as obtaining food, protection, home and transportation, whereas the other neither benefits nor suffers. For example, barnacles that attach themselves to the back of whales and the shells of crabs are provided with both home and transportation. There are three other types of association: **mutualism**, where both organisms benefit; **competition**, where both organisms are harmed; and **parasitism**, where one organism benefits and the other one is harmed. **Commensal** is an organism that benefits from a symbiotic relationship with another organism in which neither gains nor loses from the partnership.

COMMENSURABLE: Two lines or distances are commensurable if the ratio of their lengths is a rational number. In other words, two quantities are in rational proportion, if both are integral multiples of the same quantity; expressible in common units. For example, minutes and seconds are commensurable, but days and light years are not. If the ratio is an irrational number, it is called **incommensurable**, for example, $\log_3$ and $\log_2$ are incommensurable because of the irrationality of $\log_3(2)$.

COMMENSURATE LATTICE: A lattice that can be divided into two or more sublattices, with the basis vectors of the lattice being a rational multiple of the basis vectors of the sublattice.

COMMENSURATE-INCOMMENSURATE TRANSITION: A transition between two states having magnetic or crystallographic structures with a basic lattice and an imposed super-periodicity. In the commensurate (lock-in) state this super-periodicity is a simple rational multiple of the basic unit cell. In the incommensurate state the ratio of the super-periodicity repeat distance to the basic lattice repeat distance is irrational and may show continuous variation with variation in some degree of freedom, for example, temperature, pressure, composition of the system. An example is the transition of Rb_2ZnCl_4 from an incommensurate structure to a commensurate structure at the lock-in temperature, T_L , which is dependent on the crystal growth method and varies in the range 128 K to 190 K. A commensurate-incommensurate transition also occurs in liquid-crystal systems where there is an incommensurability in the packing of dimers in relation to monomers.

COMMERCE SERVICE PROVIDER (CSP): An Internet service provider that hosts merchant shopping sites and processes payments for the merchants.

COMMERCIAL FARMING: Production of crops for sale and profit, although the farmers and their families may use a small amount of what they produce. Profits may be reinvested to improve the farm. Large-scale commercial farming is called **agribusiness**. The opposite of commercial farming is **subsistence farming**, in which no food is produced for sale.

COMMINUTION (PULVERISATION/PULVERIZATION): The reduction of rock debris to fine powder, usually as a result of abrasion and attrition. Comminution has caused a reduction of the shingle on beaches, which may increase problems of coastal erosion. To **comminute** is to grind or break into very small pieces or to reduce to powder.

COMMISSURE: 1. A place where two things are joined. 2. In anatomy, it is used to denote a bundle of nerve fibres that cross the midline at their level of origin or entry. 3. In botany, it is the surface or place along which two structures such as carpels join one another.

COMMON: A common cycloid, epicycloid or hypocycloid is one described by a point lying on, rather than outside or within, the circumference of a circle as it rolls without slipping round some other figure.

COMMON ANCESTOR: In biology, an ancestral species with two or more descendants, based on scientific evidence.

COMMON BUSINESS ORIENTED LANGUAGE (COBOL):

A high level computer programming language designed for use in common commercial data processing problems. It has become the major language in the field and it features powerful facilities for file handling and business arithmetic.

COMMON DENOMINATOR: A denominator, which is a common multiple of the denominators of fractions with different values. It allows arithmetic operations involving fractions to be carried out easily.

COMMON DIFFERENCE: The difference between successive terms in an arithmetic series. For example, 3 is the common difference in the series 2, 5, 8, 11.

COMMON FACTOR (COMMON DIVISOR): A number, algebraic expression or quantity that is a factor of a set of numbers, algebraic expressions or quantities. For example, 6 is a common factor for the set of numbers 24, 30 and 48; for a set of algebraic expressions, $4a^2x2y$, $6ax^3z$, and $8atx$, the common factor is $2ax$. If the common factor is the highest factor of the numbers or expressions or quantities then it is called the greatest common divisor (G.C. D.) or highest common factor (H.C. F.).

COMMON FRACTION (SIMPLE FRACTION; VULGAR FRACTION): A fraction having an integer both as a numerator and a denominator, for example, $\frac{4}{5}$ and $\frac{7}{4}$.

COMMON GATEWAY INTERFACE (CGI): A set of rules for running scripts and programs on a Web server. It is a standard way for a Web server to pass a Web user's request to an application program and to receive data back to forward to the user.

COMMON INFORMATION MODEL (CIM): A computer industry standard for defining device and application characteristics so that system administrators and management programs will be able to control devices and applications from different manufacturers or sources in the same way.

COMMON INTERNET FILE SYSTEM (CIFS): A protocol that lets programs make requests for files and services on remote computers on the Internet. With CIFS, users with different platforms and computers can share files without having to install new software. CIFS runs over TCP/IP but uses the SMB (Server Message Block) protocol found in Microsoft Windows for file and printer access; therefore, CIFS will allow all applications, not just Web browsers, to open and share files across the Internet.

COMMON ION EFFECT: The shift in chemical equilibrium caused by the addition of a compound having an ion in common with dissolved substances. For example, in the water treatment process, highly soluble sodium carbonate salt is added to precipitate out sparingly soluble calcium carbonate. The very pure and finely divided precipitate of calcium carbonate that is generated is a valuable by-product used in the manufacture of toothpaste. Another example is the addition of sodium chloride in the manufacture of soap to reduce the solubility of the soap salts. The soaps precipitate due to a combination of common ion effect and increased ionic strength.

COMMON LANGUAGE INFRASTRUCTURE (CLI): Part of Microsoft's .NET strategy, which runs high-level language program applications in different computer systems without changing the application code. Common Language Infrastructure provides a virtual execution environment comparable to the one provided by Sun Microsystems for Java programs.

COMMON LANGUAGE RUNTIME (CLR): A part of Microsoft's .NET Framework, which compiles and executes programs written in any of several supported languages, allowing them to share common object-oriented classes written in any of the languages.

COMMON LOGARITHM: The logarithm to the base 10 or the power to which 10 must be raised to equal the given number. It is indicated for a number x as $\log_{10}(x)$ or $\log x$. It consists of two parts, the integer part called the **characteristics** and the decimal part

called **mantissa**. The characteristic is one less than the number of digits to the left of the decimal point. For example, the common logarithm of 30 ($\log_{10} 30$) is 1.4771. The characteristic is 1, and the mantissa is 0.4771.

COMMON MULTIPLE: An algebraic expression or a number that is a multiple of the number or the expression. For example, 2 and 4 are common multiples of 8 and 16. Also, $(3x - 1)(x + 4)$ is a common multiple of $(3x - 1)$ and $(x + 4)$.

COMMON SHORT CODE (CSC; MOBILE SHORT CODES; SHORT NUMBERS): Special telephone numbers, usually consisting of five or six digits, that is used to address SMS and MMS messages from a cellular telephone. Each common short code is designed to be unique to each operator.

COMMON TANGENT: A line, which is a tangent to a multiple of curves or circles.

COMMON-COLLECTOR CONNECTION: A transistor, in which the collector connection is common to both input and output. The input terminal is the base and the output terminal is the collector.

COMMUNICABLE DISEASE (CONTAGIOUS/INFECTIOUS DISEASE): Any of the diseases that can be transmitted from one person to another by physical contact, a carrier or by inhalation from the air. Examples are ebola, flu, measles, rabies, chlamydia, gonorrhoea, HIV/AIDS and tuberculosis.

COMMUNICATION: 1. The exchange of thoughts, messages or information, as by speech, signals, writing or behaviour or signalling of information by visual, auditory, electrical or tactile means from one organism to another usually with the intention of altering the recipient's behaviour. 2. In biology, **communications** is the transfer of information from one molecule, cell or organism to another, as by chemical or electrical signals or by behaviours. 3. In anatomy, **communications** is an opening or connecting passage between two structures; or connection of solid fibrous structures, such as tendons and nerves. 4. **Communications** is the field of study concerned with the transmission of information by various means, such as print or broadcasting; or any of various professions such as advertising, broadcasting or journalism involved with the transmission of information.

COMMUNICATION ADAPTER: 1. A circuit card with associated software that enables a processor, controller or other device to be connected to a network. 2. A mechanism that enables communication facilities to be attached to host processors.

COMMUNICATION PORTAL: A service that allows consumers to access personal information from diverse sources using unified communications (UC) media. The most common modes available through a communication portal are e-mail, texting, Voice over IP (VoIP) and conventional telephone services.

COMMUNICATION SATELLITE: An artificial satellite that acts as a relay station in space for sending telephone, television, telex and other messages around the world. Communication satellites move in geosynchronous orbits about 35,900 km above the Earth and operate at frequencies near 4 GHz for downlinking and 6 GHz for uplinking. A number of different orbits may be used including geosynchronous elliptical and low earth orbits.

COMMUNICATION SERVER (TERMINAL SERVER): A hardware device that provides computers, printers, terminals or other devices with a common connection point to a local or wide area network. Examples are hubs, switches, terminal servers, device servers and transaction servers.

COMMUNICATIONS OVER INTERNET PROTOCOL (CoIP): A set of standards, which define the transmission of multimedia over the Internet. It is an expansion of voice over Internet Protocol (VoIP) to include text, images, video and other forms of digital data as well as voice information. Voice over Internet Protocol (VoIP) allows voice to be carried over internet connections or

other networks (which are not designed specifically to carry voice) using Internet Protocols (IP).

COMMUNICATIONS PORT (COM PORT; SERIAL PORT): A serial communication used to connect devices in earlier personal computers and usually designated as COM1, COM2, COM3 and COM4 on IBM compatible computers running Microsoft Windows. For many computer peripheral devices, the USB interface has replaced the Communication Port. Modern computers are connected to devices through a USB connection and often do not even have a Communication Port.

COMMUNITY (CONSORTIA; SYNTROPHIC; SYNERGISTIC ASSOCIATIONS): A naturally occurring population of different species living within a particular habitat and interacting with each other or an assembly of plants, animals and other organisms living within a defined area or habitat. Communities may be of varying sizes and larger communities may contain smaller ones.

COMMUNITY AGE: One of the factors that help cause the latitudinal diversity gradient. Tropical communities have had more time to evolve because they have been less disrupted by advancing ice sheets and other relatively recent climatic changes.

COMMUNITY SIMPLIFICATION: The reduction of overall species diversity in a community. It is generally caused by human activity.

COMMUNITY SUCCESSION: Predictable and orderly changes in the composition or structure of an ecological community. This may come about as a result of formation of new habitat, migration of new species, by local extinction of old ones, etc.

COMMUTATIVE DIAGRAM: A diagram of objects and arrows or edges such that all directed paths between any two given vertices in the diagram with the same start and endpoints lead to the same composition arrow.

COMMUTATIVE LAW (COMMUTATIVE OPERATION; COMMUTATIVE PROPERTY; COMMUTATIVITY): Two closely related laws of number operations relating to addition and multiplication that is independent of the order of the numbers or symbols concerned, expressed as: $a + b = b + a$ and $ab = ba$. Multiplication and addition of numbers have commutative properties, but multiplication does not necessarily hold for multiplication of conditionally convergent series. Quantities to be added or multiplied can be combined in any order. For example, addition is commutative; that is, the result of adding $4 + 2$, for example, is the same as that of adding $2 + 4$. Division and subtraction are not commutative; for example, for subtraction, $8 - 2 \neq 2 - 8$ or $8 \div 2$ is not the same as $2 \div 8$. Scalar multiplication of two vectors to give the dot product is commutative: $a \cdot b = b \cdot a$, but vector multiplication to give the cross product is not: $a \times b = -b \times a$. Multiplication of matrices is not commutative. Set operations for intersection and union are commutative, i.e., $A \cap B = B \cap A$ and $A \cup B = B \cup A$.

COMMUTATIVE RING: A ring, R with two binary operations, addition (+) and multiplication ($\cdot$), satisfying the axioms that R is an abelian group under addition (+) and multiplication ($\cdot$) is associative and the two operations are connected by the distributive laws.

COMMUTATOR: 1. Of two elements of a group, of the form $[x, y] = x^{-1}y^{-1}xy$, for x and y in the group. It is equal to the group's identity if and only if x and y commute (i.e., if and only if $xy = yx$). The subgroup of a group generated by all commutators is the derived subgroup. 2. The operator $PQ - QP$, where P and Q are given operators. 3. A rotary device for altering the direction of an electric current in direct current types of electric motors or electrical generators. It reverses the current direction in the moving coil of a motor's armature to provide a steady torque or rotating force. In a generator, it reverses the coil's connection to the external circuit to provide direct current to the external circuit.

COMMUTATOR SUBGROUP (DERIVED SUBGROUP): The subgroup generated by the set of commutators of a given group. It is denoted by G' .

COMMUTE: In mathematics, of a group, semi-group or ring operation that is commutative or obey a commutative law.

COMMUTING INDETERMINATE: An element that commutes with every element of a given ring is not the root of any polynomial equation over the ring.

COMPACT: 1. In a topological space, a set is compact if in every collection of open sets, the union of the whole space has a finite subcollection with the same property. This is equivalent to the finite intersection property. In particular, in n -dimensional Euclidean space, a set is compact if and only if it is closed (containing all its limit points) and bounded (having all its points lying within some fixed distance of each other). For example, the closed interval $[0, 1]$ is compact,

but the open interval $(0, 1)$ is not, as $\left\{\left(\frac{1}{n}, 1\right)\right\}_{n \in \mathbb{N}}$ is a cover of $(0, 1)$

that has no finite subcover. A subspace is compact if every cover in the induced topology is a finite subcover. 2. A relation is compact if it has the property that for any pair of elements in which a is related to b , there is some element c such that a is related to c and c is related to b . For example, less than is compact, on the rational numbers, since for any rationals, a and b , $1/2(a + b)$ is rational between a and b . 3. A mapping between topological vector spaces, especially Banach spaces, having the property that the image of any bounded set has a compact closure.

COMPACT BONE (CORTICAL BONE): One of the two types of osseous tissues that form bones. It is very dense and constitutes the outer layer that forms the shaft of the long bones. It is made up of concentric layers of mineral deposits surrounding a central opening. Its main functions include support of the body, protection of organs, provision of levers for movement, etc.

COMPACT DISC (CD): A small plastic disc on which digital data such as music, audio or graphic images is digitally stored and from which the information can be reproduced using reflected laser light. Digital information for computers (CD-ROM) and entertainment systems for video disc and DVD can be stored and reproduced.

COMPACT DISC INTERACTIVE (CD-i): An interactive multimedia CD player that can combine audio, video and text on high-capacity compact discs and a variant of CD-ROM. CD-i also refers to the multimedia Compact Disc standard used by the CD-i console, also known as Green Book, which was developed by Philips and Sony.

COMPACT DISC RE-WRITABLE (CD-RW): A compact disc (CD) format that enables repeated audio or other digital data to be written, read, erased and rewritten. The CD-RW drive is the most versatile CD drive and most new CD drives are CD-RWs. CD-RW uses a phase change to record data.

COMPACT DISC READ-ONLY MEMORY (CD-ROM): A device that is based on the audio compact disc and provides read-only access (but not writable) to a large amount of data, up to 640 megabytes for use on computer systems. The data may be in any form (text, sound, images or binary data or a mixture) and various CD-ROM format standards exist to handle these.

COMPACT DISC-RECORDABLE (CD-R): A type of write once, read many (WORM) compact disc (CD) format that allows one-time recording on a disc or a compact disc on which can be written only once and thereafter is read-only memory. It is used to back up and transfer data to master CD-ROMs, CD-R discs can be written and read by most CD/DVD drives.

COMPACT FLASHCARDS (CF): A form of memory card which uses flash memory to store data. They are relatively small and suitable for storing data for transfer between devices. Compact flash cards are most commonly used for storing pictures in digital cameras, but are also used in devices such as PDAs and portable music players. There are two types of Compact Flash cards, creatively named "Type I" and "Type II." Type I cards are 3.3 mm thick, while Type II are 5 mm thick.

COMPACT HTML (CHTML; CHTML): A subset of standard Hypertext Markup Language (HTML) adapted for use with small computing devices such as personal digital assistants (PDAs), cellular phones and smartphones.

COMPACTIONS: The process in which the volume and porosity of a sediment is decreased by the weight of overlying sediments as a result of burial beneath other sediments.

COMPACTIFICATION: A compact topological space containing a given topological space. A one-point compactification adds a single point, designated ∞ , to a Hausdorff space. The extended interval $[0, \infty]$ is a one-point compactification of $[0, \infty)$ in which all complements of bounded intervals are neighbourhoods of ∞ .

COMPANDING (COMPANSION): The non-linear process of reducing the distortion of an analogue signal, which takes place in the same way at all points in time during transmission, in order to preserve the signal-to-noise ratio of the original audio. This is done by a device called **compandor**. This allows signals with a large dynamic range to be transmitted over facilities that have a smaller dynamic range capability. Companding is used as a complement to the process of modulation and demodulation. In this process, the voice signal is compressed, changed from analogue to digital, transmitted and converted back from digital to analogue before it is expanded again. For audio analogue signals, the amplitude of weak signals is raised and the amplitude of strong signals is decreased, thereby compressing and expanding the dynamic range of the signals. Companding is used in AM, FM and single-side band modulation radio and it is effective in reducing buzz, hiss and low-level audio tones caused by mild interference, thus improving the quality of amplified voice and musical instrument sound. It is also used in most wireless transmission systems, audio systems and in some noise reduction systems such as Dolby NR and helpful in improving the quality of amplified voice and musical instruments.

COMPANION CELL: One of the two conducting cells located in the phloem of flowering plants, the other one is the sieve-tube elements. Companion cells are smaller and are found adjacent to the sieve-tube elements. They are believed to be closely associated with a sieve-tube element to regulate the flow of nutrients.

COMPANION MATRIX: A matrix that is obtained from a given polynomial, such as $p(x) = x^n - b_{n-1}x^{n-1} - \dots - b_1x - b_0$ by placing the negatives of the coefficients of the polynomial in ascending order of degree in the last row of a unit super-diagonal matrix.

COMPARABLE: Two elements a and b of a partially ordered set are comparable, if they are related by the ordering so that either $a \leq b$ or $a \geq b$.

COMPARATIVE ANATOMY: The scientific study of the differences and similarities in the structure of organisms. Comparative anatomy supports Darwin's theory of descent with modification, known as evolution as it indicates that various organisms share a common ancestor. Also, it assists scientists in classifying organisms based on similar characteristics of their anatomical structures. A common example of comparative anatomy is the similar bone structures in forelimbs of cats, whales, bats and humans.

COMPARATIVE BIOCHEMISTRY: The recognition of discontinuities in biochemical variation and their subsequent use in the construction of classifications or correlation with existing classifications.

COMPARATIVE INVESTIGATION: A type of investigation that involves collecting data on different populations/organisms, or collecting under different conditions such as times of year, temperatures, and locations to make comparisons. It is one of the three main types of investigations used by scientists; the others are descriptive investigation and experimental investigation.

COMPARATIVE MOLECULAR FIELD ANALYSIS (CoMFA): A 3D-QSAR method that uses statistical correlation techniques for

the analysis of the quantitative relationship between the biological activity of a set of compounds with a special alignment and their three-dimensional electronic and steric properties. Other properties, such as hydrophobicity and H-bonding, can also be incorporated into the analysis.

COMPARATOR: 1. In chemistry, it is defined as a known amount of element that is simultaneously irradiated with the test portion in the context of activation analysis. 2. An electronic circuit that produces an output voltage or current whenever two input levels simultaneously satisfy predetermined amplitude requirements. A comparator circuit may be designed to respond to analogue or digital signals and its output may be in the form of signalling pulses which occur at the comparison point or in the form of digital direct-current (dc) levels.

COMPARISON TEST: The test that gives the absolute convergence of a series by verifying that the terms are dominated by the terms of a convergent series with positive terms. The first series is said to be subordinate to the second series. For example, since

$\sum_{k=1}^{\infty} \frac{1}{k(k-1)}$ is convergent and $\frac{3}{k(k+1)} \geq \frac{1}{k^2}$, it follows by the comparison test that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges.

COMPARE: To display recognition of similarities and differences between multiple things and recognise the significance of these similarities and differences. In the scientific field, comparison is an important tool, used to understand the relationship between two or more variables by documenting observed differences and similarities.

COMPARE PROBLEMS: Mathematical situations involving comparisons of two discrete sets of objects, for example, to identify smaller or larger quantities, differences between quantities, or ratio relationships.

COMPASS: 1. A navigational instrument used for finding direction relative to the Earth's magnetic poles. It is usually a small magnet, pivoted at its central point to revolve in a horizontal plane with a face highlighting the cardinal points of north, south, east and west. Its Earth's magnet called compass needle aligns itself so that the north seeking end points to the Earth's magnetic north pole. The compass has been supplanted by modern satellite navigation systems such as the Global Positioning System (GPS). 2. A small mathematical instrument for sketching arcs. It consists of two hinged arms with one ending in a sharp point and the other in a drawing point.

COMPASS BEARINGS: Angular measurements in degrees of an object as measured from the observer.

COMPASS PLANT (ROSINWEED): A plant, *Silphium laciniatum* of the family Asteraceae that has its basal leaves pointing in a north-south direction. It is found mainly in open grasslands.

COMPASS-AND-STRAIGHTEDGE CONSTRUCTION (EUCLIDEAN CONSTRUCTION): The geometric constructions of figures and lengths using only a compass and a straight edge or ruler, for example, bisecting a line using only a compass. The straight edge is used only to draw lines and not for measurement.

COMPATIBLE POLYMER BLEND: Immiscible polymer blend that exhibits macroscopically uniform physical properties. The macroscopically uniform properties are usually caused by sufficiently strong interactions between the component polymers.

COMPATIBLE SOLUTE (OSMOPROTECTANTS): Highly water-soluble substances that accumulate within the cell of an organism and help it survive extreme osmotic stress or enables it to maintain its water balance with the external environment. For example, halophilic bacteria are adapted to living in salty water and contain intracellular solutes like glycine, betaine, trehalose, etc, that lower the water potential of their cell contents so that the organism stays in an osmotic balance with its surroundings. Some plants employ similar strategy to adjust the water potential of their cells as a means of protecting against damage due to water shortage. Compatible solutes are generally sugar alcohols, sugars, alcohols, amino acids or their derivatives.

COMPENSATION POINT: The amount of light intensity at which the rate of carbon dioxide uptake called photosynthesis is exactly equal to the rate of carbon dioxide (CO_2) production called respiration; or the point at which there is just enough light for a plant to survive. At that point all food produced by photosynthesis is used by respiration.

COMPENSATOR: An optical device capable of imposing a controllable retardance (phase shift) on a wave. Varieties include the Babinet compensator and Soleil compensator.

COMPETITOR: A species characterised by a relatively longer life span, with relatively few offspring, when compared with an opportunist species.

COMPETITION: The interaction between two or more organisms or species that use a common resource, which is in short supply. For example, plants compete with each other for sunlight or nutrients from the soil or animals competing for food, water or refuge. Competition is not always straightforward and can occur in both a direct and indirect fashion. When members of different species compete for most or all of the same resources, one species flourishes and the other declines. This is known as the **competitive exclusion principle** and means that two different species rarely occupy the same ecological niche.

COMPETITIVE BINDING ASSAY: An assay based on the competition between a labelled and an unlabelled ligand in the reaction with a receptor binding agent (e.g. antibody, receptor, transport protein).

COMPETITIVE EXCLUSION PRINCIPLE (GAUSE'S LAW; GAUSE'S LAW OF COMPETITIVE): A rule, derived by G.F. Gause in 1934, stating that two species that occupy the same habitat cannot also occupy the same ecological niche. When one species has even the slightest advantage or edge over another, then the one with the advantage will dominate in the long term. One of the two competitors will always overcome the other, leading to either the extinction of this competitor or an evolutionary or behavioural shift towards a different ecological niche. Some examples are: (i) The Red Squirrel (*Sciurus vulgaris*), native to Britain but whose population has declined due to competitive exclusion, disease and the disappearance of hazel coppices and mature conifer forests in lowland Britain. (ii) The grey squirrel (*Sciurus carolinensis*), which was introduced to Britain in about 30 sites between 1876 and 1929, has easily adapted to parks and gardens replacing the red squirrel.

COMPETITIVE INHIBITION: An inhibitor that competes with the substrate for binding to the enzyme. The inhibitor can bind to the active site but does not react to form products. An example is the inhibition of the enzyme succinate dehydrogenase by certain compounds resembling succinate, for example, malonate and oxalate. Studies of competitive inhibitors have provided valuable information about the way in which substrates are bound to the active site.

COMPILE: The process of creating an executable program from code written in a programming language that allows the computer to run and understand the program without the need of the programming software used to create it. **Compiler** is the computer program that translates programs written in a high level language (the source language) into machine code.

COMPLEMENT: 1. The set of the elements within the universal set that are not contained in the designated set. For example, given the set $U = \{1, 2, \dots, 5\}$ and $A = \{3, 4, 5\}$, then the complement of set A is given by $A' = \{1, 2\}$. 2. The difference between some given value and a fixed total value, especially the complementary angle of a given angle. 3. In vector space, a subspace disjoint from a given subspace that when added to it gives the whole space. 4. Any element of a structure that is complementary to a given element, such as orthogonal vectors or elements in a lattice of which the meeting point is zero. 5. A set of proteins in the bloodstream and body fluids that, together with antibodies, recognise and attack foreign cells such as bacteria. When activated by antigen-antibody complexes or by other agents such as proteolytic enzymes such as

plasmin, it kills bacteria and other microorganisms. It also releases peptides that enhance vascular permeability, release histamine and attract white blood cells (chemotaxis).

COMPLEMENT RECEPTOR: A protein found on the surface of certain cells of the immune system that binds specifically to complement components coating the surface of a bacterium or other pathogen. It is one of the most important complement receptors found on macrophages, monocytes, polymorphonuclear leucocytes and erythrocytes.

COMPLEMENTAL MALES: The small males of certain animals that live parasitically in or on the females of certain barnacles and are usually more or less degenerate apart from the reproductive organs.

COMPLEMENTARITY: 1. The concept that a single model may not be adequate to explain all the observations made of atomic or subatomic systems in different experiments. 2. Combining well to form a balanced whole.

COMPLEMENTARY: 1. Constituting a complement to one another. 2. Mutually exclusive and exhaustive so that each is the complement of the other; related in a way definable in such terms. For example, sine and cosine are called complementary functions (cofunctions) since, for any value of x , $\arcsin x$ and $\arccos x$ are complementary angles. 3. Of two vectors, in Euclidean space each having coordinates of one sign and being orthogonal to one another. For example, $(1, 2, 0)$ and $(0, 0, -3)$ are complementary. More generally, one allows one vector to lie in a given convex cone, the other to lie in the polar cone and the inner product of the two to be zero.

COMPLEMENTARY ANGLES: Two angles that add up to 90° . For example, 60° and 30° are complementary angles. In a right-angled triangle, the two non-right angles are complementary, because in a triangle the three angles add to 180° and 90° has already been taken by the right angle.

COMPLEMENTARY COLOURS: A pair of coloured lights of specific hue that produces the sensation of white when mixed in appropriate intensities; or two colours which mixed in the proper proportion, produce a neutral colour (grey, white or black).

COMPLEMENTARY DNA (CDNA): A form of DNA prepared in the laboratory using a mature mRNA template in a reaction catalysed by the enzyme reverse transcriptase and the enzyme DNA polymerase.

COMPLEMENTARY FUNCTION: For a linear differential equation, an arbitrary linear combination of n linearly independent solutions of the corresponding homogeneous equation, where n is the order of the given linear equation.

COMPLEMENTARY GENES: Two genes that are inter-dependent, such that the dominant allele from either gene can only produce an effect on the phenotype of an organism if the dominant allele from the other gene is also present.

COMPLEMENTARY MEDICINE: A type of medical practice that uses a combination of alternative medicine and conventional medical practice to treat patients. An example is using acupuncture to help with side effects of cancer treatment.

COMPLEMENTARY METAL-OXIDE SEMICONDUCTOR (CMOS): A memory device that stores information such as the system time and date and the system hardware settings of the computer. It is used in static RAM, digital logic circuits, microprocessors, microcontrollers, image sensors and the conversion of computer data from one file format to another. It uses reduced electrical power and less heat.

COMPLEMENTARY MINOR: The minor of an entry of a given square matrix is the determinant of the matrix obtained by deleting the elements of a given minor from it.

COMPLEMENTARY MODULUS: The quantity $\sqrt{1-k^2}$, where k is the modulus of a complete elliptic integral or elliptic function.

COMPLEMENTARY NUCLEOTIDE BASES: Members of the pairs adenine-thymine, adenine-uracil and guanine cytosine that have the ability to hydrogen bond to one another.

COMPLEMENTARY NUMBER: The number obtained by subtracting a number from its base, for example, the complementary of 7 in base 10 is 3; or two numbers that add up to a power of 10. For example, 2 and 8 are complementary, because they add up to 10.

COMPLEMENTARY PROBLEM: An optimisation model including linear and integer programming and having applications in fixed-point theory. Given a function f on Euclidean space, a vector x is sought such that x and $f(x)$ are complementary in the lattice of non-negative vectors.

COMPLEMENTARY SLACKNESS: In linear programming or linear optimisation, the condition that arises in the Kuhn-Tucker theorem and other similar results, in which the multiplier and the constraint vector are complementally vectors to one another.

COMPLEMENTATION: 1. In mathematics, it is the process of taking complements, especially in set theory. 2. In genetics, it is the interaction that occurs between two closely related mutant alleles of a gene in a diploid organism with one on each homologous chromosome and at a slightly different position to make up for each other's deficiencies and allow the organism to function normally.

COMPLEMENTATION TEST (CIS-TRANS TEST): A test used to determine whether two independent mutations, affecting the same phenotypic function, occur within the same cistron or on different cistrons. If the mutations occur in the same cistron, only heterozygotes with the cis arrangement will show the normal phenotype because the gene in the homologous chromosome is dominant wild type and codes for normal protein. In the trans arrangement both copies of the cistron contain a mutation and thus there is no wild-type copy of the gene and hence no normal protein. If the mutations occur in different cistrons, specifying different polypeptides, all heterozygotes will be normal in appearance because the homologous chromosomes carry wild-type copies of the genes.

COMPLETE: 1. A **complete metric space** is one that has every Cauchy sequence converging; for example, the real numbers are complete but the rationals are not, where the metric is the absolute difference between the numbers. 2. Also called **order complete** or **Dedekind complete**, of a partially ordered set, such that every subset has a supremum and infimum, for example, the reals are not complete but the interval $[0,1]$ is. A partially ordered set is complete for every countable non-empty set S of F that has an upper bound in F , a supremum exists in F and for countable non-empty set S of F , which has a lower bound in F , the supremum exists in F ; an infimum exists in F . 3. A **complete graph** is one having each pair of vertices connected by an edge. 4. In logical theory, completeness property means that every validity (truth) is provable, i.e., every semantically valid formula can be proved syntactically from the axioms. It can be defined mathematically, for example, as if given $\Sigma \Phi$ then $\Sigma \Phi$: for every truth placement Z in Σ , there is T , then Φ also would get T . If the previous indeed exists, then Φ can be proved using the rules in Σ . 5. (of a sufficient statistic for a parameter θ , having the property that if the expected value of a function of the statistic is zero for all values of the parameter, then the function is identically zero. 6. Of a group, having a trivial centre and isomorphic to the group of its own automorphisms. In botany, it describes a structure with all the parts typically belonging to it. An example is a flower which contains all the floral parts. Such a flower has the sepals, petals, stamens and pistils present.

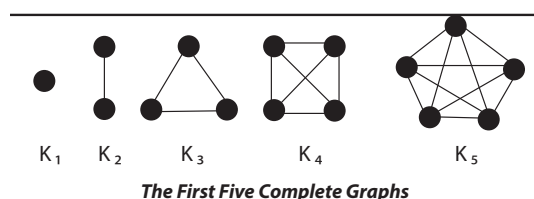
COMPLETE ACTIVE SPACE SELF-CONSISTENT FIELD (CASSCF): Computational scheme employed in multi configurational SCF theory especially suitable for studies of reactivity of excited states. The wave function is defined by selecting the set of active orbitals involved in the excitation or chemical reaction under investigation and is constructed as a linear expansion in the set of configuration functions that can be generated by occupying the active orbitals in all ways consistent with an overall spin and space symmetry (full configuration interaction, CI).

COMPLETE ELLIPTIC INTEGRAL: Any elliptic integral expressed in terms of the functions, K , which denotes the complete elliptic integral

of the first kind and E , for the second kind. The first kind and second kind are related by Legendre's identity that for any $0 < k < 1$, $K(k)E[\sqrt{1-k^2}] + E(k)K[\sqrt{1-k^2}] - K(k)K[\sqrt{1-k^2}] = \pi^2$. They are useful in finding the arc length of an ellipse and the period of a pendulum and also arise in a natural way from the theory of theta functions.

COMPLETE FERTILISER: A fertiliser having all the essential nutrients: nitrogen, phosphorus and potassium.

COMPLETE GRAPH: A connected graph in which exactly one edge connects each pair of vertices. A complete graph with n vertices, denoted K_n , has $\frac{n(n-1)}{2}$ edges (i.e., the n th triangular number), $(n-1)!$ Hamilton circuits and a chromatic number of n . Every vertex in K_n has degree $n-1$; therefore K_n has an Euler circuit if and only if n is odd. In a **weighted complete graph**, each edge has a number with a weight attached to it. Each path then has a total weight, which is the sum of the weights of the edges in the path.



COMPLETE INDUCTION (SECOND-KIND INDUCTION; GENERAL INDUCTION): A mathematical induction in which the step is from all integers less than k to all less than $k+1$, rather than from the single integer k to $k+1$.

COMPLETE MATCHING (1-FACTOR; PERFECT MATCHING): A matching of a graph in which all vertices are matched by an edge. In other words, every vertex of the graph is incident to exactly one edge of the matching.

COMPLETE MEASURE: A measure in which all subsets of a null set or measure zero are measurable. For example, Lebesgue measure is complete, but Borel measure is not.

COMPLETE METAMORPHOSIS (HOLOMETABOLISM): A metamorphosis in insects such as butterflies, beetles, flies and bees that involves the four stages of egg, larva, pupa and adult.

COMPLETE QUADRANGLE: The configuration of four points, with no three colinear and the six intersecting lines; the lines joining the vertices of a complete quadrangle that have segments that are the sides and diagonals of the quadrilateral defined by their vertices.

COMPLETE QUADRILATERAL: A configuration consisting of four distinct lines of which no three are coincident and their six points of pair-wise intersection. The dual configuration replacing lines by points is a complete quadrangle.

COMPLETE RESIDUE SYSTEM: Any set of representatives chosen one from each residue class. The most frequently used systems are the system of smallest positive residues $0, 1, 2, \dots, m^{-1}$ and the system of absolutely smallest residues: $\frac{-(m^{-1})}{2}, \dots, -1, 0, 1, \frac{(m^{-1})}{2}$ for odd m and $\frac{-m}{2}, -1, 0, 1, \dots, \frac{m}{2}$ for even m . A complete residue system modulo m is a set $v = 0, 1, \dots, m^{-1}$, of integers containing one and only one representant from every residue class modulo m . Thus the numbers av may be ordered such that for all $v = 0, 1, \dots, m^{-1}$, they satisfy $av \equiv v \pmod{m}$.

COMPLETELY MONOTONE FUNCTION: An infinitely differentiable function, f on the interval $[0, \infty)$ is completely monotonic if and only if there exists a bounded and non-decreasing function $\alpha(t)$ on $[0, \infty)$ such that $f: [0, \infty) \rightarrow \mathbb{R}$ for which the sequence of derivatives satisfies $(-1)^n f^{(n)}(x) \geq 0$.

COMPLETING THE SQUARE: A way of solving a quadratic equation by simplifying the quadratic equation by, adding an expression to the equation making it a perfect square. For example, the quadratic equation, $2x^2 - 4x - 3 = 0$ can be expressed as $2\left[x^2 - 2x - \frac{3}{2}\right] = 0$.

The linear term in this equation is divided by 2, the result squared and added to the expression on the right hand side in such a way that the equation is not altered.

Thus $2\left[x^2 - 2x - \frac{3}{2}\right] = 0$, simplifying the equation becomes

$2\left[(x-1)^2 - \frac{5}{2}\right] = 0$. Thus a pure quadratic equation without a linear term is obtained and the solution is then obtained as

$$2(x-1)^2 - 5 = 0$$

$$2(x-1)^2 = 5$$

$$(x-1)^2 = \frac{5}{2}$$

$$x-1 = \pm\sqrt{\frac{5}{2}}$$

$$x = 1 \pm \sqrt{\frac{5}{2}}$$

COMPLETENESS POSTULATE: An axiom, which states that the real line is order-complete, which is equivalent to the Archimedean property.

COMPLETENESS THEOREM (GÖDEL'S COMPLETENESS THEOREM): A theorem in mathematical logic that indicates a correspondence between semantic truth and syntactic provability in first-order logic. It is the result that a theory is consistent if and only if it has a model. It was first proved by Kurt Gödel in 1929.

COMPLETION: 1. The completion of a metric space is the minimal complete metric space that contains the given space. 2. A measure that extends a given measure to a complete measure.

COMPLEX: 1. A molecular entity formed by loose association involving two or more component molecular entities (ionic or uncharged) or the corresponding chemical species. The bonding between the components is normally weaker than in a covalent bond. 2. Of or pertaining to the form of a complex number, consisting a real and an imaginary part, either of which may be zero.

COMPLEX ADAPTIVE SYSTEM (CAS): A nonlinear, interactive, complex system with the ability to adapt to a changing environment. CAS evolves by random mutation, self-organisation, the transformation of their internal models of the environment and natural selection. Examples include living organisms, the nervous system, the immune system, the economy, corporations and societies.

COMPLEX ANALYSIS: The study of functions of a complex variable. Often, the most natural proofs for statements in real analysis or even number theory use techniques from complex analysis. Unlike real functions, which are commonly represented as two-dimensional graphs, complex functions have four-dimensional graphs and may usefully be illustrated by colour-coding a three-dimensional graph to suggest four dimensions.

COMPLEX COACERVATION: Coacervation caused by the interaction of two oppositely charged colloids.

COMPLEX COMPOUND (COORDINATION COMPLEX): A compound in which non-metal molecules or ions form coordinate bonds with a central metal atom.

COMPLEX CONJUGATE: A pair of complex numbers, both having the same real part, but with imaginary parts of equal magnitude and opposite signs. For example, $3 + 4i$ and $3 - 4i$ are complex conjugates. Its symbol is $\bar{z}$.

COMPLEX CURVE (RIEMANN SURFACE): A complex manifold with one complex dimension. The Riemann surface is a conformal structure.

COMPLEX FRACTION (COMPOUND FRACTION): A fraction in which either the numerator or denominator or both contains fractions.

For example, $\frac{\frac{1}{3}}{\frac{2}{7}}$ is a complex fraction. To simplify the complex fraction,

multiply both the numerator and the denominator by the reciprocal of

$$\text{the denominator: } \frac{\frac{1}{3}}{\frac{2}{7}} = \frac{1}{3} \times \frac{7}{2} = \frac{7}{6}.$$

COMPLEX GAIN: A term used in electronics to describe the effect circuitry has on the amplitude and phase of a sine wave signal.

COMPLEX INSTRUCTION SET COMPUTER: Abbreviated **CISC**, it is a type of microprocessor design and its instruction set with an architecture that contains a large set of computer instructions and varies in size and complexity: from very simple to very complex and specialised. In contrast, Reduced Instruction Set Computer (RISC) microprocessors usually have few instructions in the instruction set, with smaller and simpler instructions. Examples of CISC are Intel Pentium processors, with some RISC facilities built into them. The PowerPC processors and Macintosh computers are completely RISC.

COMPLEX ION: A structure consisting of a central atom or ion usually metallic, covalently bonded to a surrounding array of molecules or anions (ligands). The atom within a ligand that is directly bonded to the central atom or ion is called the donor atom; the donor atom donates at least one pair of electrons to the central atom/ion.

COMPLEX NUMBERS: A number consisting of a real and imaginary part, written in the form $a + ib$, where a and b are real numbers and i is the square root of -1 or $(\sqrt{-1})$.

COMPLEX PLANE (ARGAND PLANE): The geometric representation of the complex numbers as identified with the infinite two-dimensional space defined by the real axis and the orthogonal imaginary axis of the Argand diagram.

COMPLEX SYSTEMS: Systems that are difficult to model and predict, consisting of a collection of many simple nonlinear units that operate in parallel and interact locally with each other so as to produce emergent behaviour. The theory behind the complex system is that all the parts of this system are interconnected with each other as if they have been interwoven. Thus, to understand about the nature of these complex systems, it is necessary to know about the behaviour of its every part and how they work together to complete the entire system. Components of a complex system may appear to act spontaneously such that predicting complex systems with certainty is believed to be impossible. Complex systems include the universe, climate, ecosystem, organisms, the human brain, social and economic organisations, cities, etc.

COMPLEX VELOCITY POTENTIAL: In continuum mechanics, a holomorphic function that describes an incompressible and irrotational two-dimensional steady motion, the complex conjugate of the derivative of which is the velocity expressed in terms of complex numbers of which the real part is the velocity potential and the imaginary part is the stream function.

COMPLEX VIRUSES: Viruses that have a complicated symmetry, with capsids (proteins, that encloses the nucleic acid of a virus) that are neither icosahedral nor helical.

COMPLEX-MODE REACTION (INDIRECT REACTION): An elementary reaction which proceeds via an intermediate species having a lifetime longer than a few rotational periods.

COMPLEXIFICATION: The treatment of the direct product of a given real vector space with itself as a complex vector space. This is analogous to the identification of the complex numbers with the real plane. All complex vector spaces can be so reconstructed from the underlying real vector space, with multiplication restricted to real scalars.

COMPLEXION: The natural colour and appearance of the skin especially that of the face. A number of internal and external factors may influence a person's complexion. They include genetics and hormones, as well as environmental factors, medications and use of skin care products. Genetic factor is determined by a protein molecule known as melanin. The more melanin a person has, the darker the skin.

COMPLEXITY: The levels of self-organisation of a system. It is used to characterise something with many parts in intricate arrangement. In physical systems, complexity is associated with broken symmetry and the ability of a system to have different states between which it can make phase transmission.

COMPLEXITY THEORY: A set of concepts that attempts to explain complex phenomenon not explainable by traditional (mechanistic) theories. It integrates ideas derived from chaos theory, cognitive psychology, computer science, evolutionary biology, general systems theory, fuzzy logic, information theory and other related fields to deal with the natural and artificial systems as they are and not by simplifying them or breaking them down into their constituent parts.

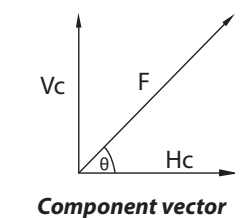
COMPLEXOMETRIC ANALYSIS: A type of volumetric analysis in which the reaction involves the formation of an inorganic complex.

COMPLEXOMETRIC TITRATION: A form of volumetric analysis in which the formation or disappearance of a coloured complex is used to indicate the end point of a titration. They are particularly useful for the determination of a mixture of different metal ions in solution.

COMPLIMENTARITY DETERMINING REGION (CDR): A region on the surface of an antibody molecule that contributes to the antigen-binding site. Each of the two arms of the Y-shaped antibody comprises a heavy chain and light chain, both of which carry hyper variable loops.

COMPONENT: 1. Also known as **component vector** or **vector component**, it is any of the vectors produced when a single vector is resolved into two or more parts in a given direction. In other words, it is the effect in a specified direction of a physical quantity such as a force. If a vector represents the magnitude and direction of the physical quantity, its component in any direction is represented by the projection of the vector or by the process of resolution of vectors. The vector sum of the components gives the original vector. In the diagram, the horizontal component of the vector, F is $F\cos\theta$ and the vertical component of F is $F\sin\theta$. For example, if a boy pulls a trolley with a force of 10 Newtons with a rope which makes an angle 60° with the horizontal, then only 5 N will pull the trolley forward, then the horizontal component, $H_c = 10\cos 60 = 5$ newtons. 2. One of the elements of the ordered set that represents a vector, for example, 2 in (1, 2, 3). 3. One of a set of two or more vectors of which the resultant is a given vector, especially one of such a set of orthogonal vectors lying in a specified direction such as parallel to a coordinate axis. In this case, the second component of (1, 2, 3) is (0, 2, 0). 4. Of a topological space, a maximal connected subset of the given space. For example, the set of positive x and y with $xy = 1$ is a component of the corresponding hyperbola. 5. In statistics, any one of the variables of a multivariate distribution. 6. Any of the set of the minimum number of chemical constituents by which every phase of a given system can be described. For example, a mixture of water and ethanol has two components but is a single phase. A mixture of ice and water has two phases but one component (H_2O). If an equilibrium reaction occurs, the number of components is taken to be the number of chemical species minus the number of reactions. 7. Also known as element, it is any electrical device, such as a resistor, that has distinct electrical characteristics and that may be connected to other electrical devices to form a circuit.

COMPONENT OBJECT MODEL (COM): In computing, it is Microsoft's framework for creating object-oriented software that was first released in 1993. The objects are discrete components, each



with a unique identity, which expose interfaces that allow applications and other components to access their features. ActiveX also is based on COM.

COMPONENT TRANSFORMATION LAW: The identity that relates tensors defined over vector spaces with different bases of the same dimension.

COMPOSITAE (ASTERACEAE; SUNFLOWER FAMILY): The largest family of flowering plants in the major group Angiosperms (Flowering plants) of about 12,000 species worldwide, consisting of numerous economic plants. Compositae is now known as Asteraceae. Most members are herbaceous, but a significant number are also shrubs, vines and trees. The stem may be solid or hollow and it may be prickly. The roots may be short and near the surface, as in the Daisy or there may be a long taproot, as in the Dandelion or Thistle. The leaves are alternate, opposite or less commonly whorled and range from simple to pinnately or palmately compound; with no stipules. They are characterised by composite flower type in the form of capitula surrounded by involucre bracts. The inflorescence, called a capitulum or head consists of several small flowers, called florets, that are crowded together, sessile, on a receptacle. This inflorescence is often mistaken for a single flower. The flowers are of two basic types: those with tubular actinomorphic corollas and those with strap-shaped or radiate zygomorphic corollas, often within the same head. Either type may be bisexual or unisexual. The fruit often possesses a ring of hairs, the pappus, to aid dispersal. Compositae is important primarily for its many garden ornamentals, such as ageratum, asters, chrysanthemums, cosmos, dahlias, marigolds (*Tagetes*) and zinnias. Some genera include noxious weeds, such as dandelion (*Taraxacum*), ragweed (*Ambrosia*) and thistle (*Carduus*, *Cirsium* and others). Several other members have economic importance as food crops. Artichokes (*Cynara*), lettuce (*Lactuca*), endive (*Cichorium*) and salsify (*Tragopogon*) are commonly eaten as vegetables and the edible seeds of safflower (*Carthamus*) and sunflower (*Helianthus*) are used in the production of cooking oils. One species, *Ageratum conyzoides*, commonly known as billygoat-weed, chick weed, goatweed, or whiteweed is effective in treating diseases such as syphilis, gonorrhoea, menstrual pains, hypertension, stroke, rheumatism and gout. This is done by chopping and drying the plant and preparing as tea or can be infused or boiled fresh. Another species, *Aspilula africana* has medicinal properties and mostly preparations of the leaves are used to treat dysentery and diarrhoea, gonorrhoea, coughing; stop bleeding.

COMPOSITE: 1. Of a number, polynomial, function, etc., having proper factors. For example, $y = x(x + 2)$ is a composite function. 2. Any purposely designed engineering material created by combining single materials such as wood or fibreglass and polymers, to produce structural or functional properties not present in any individual component. 3. In chemistry, it is a multicomponent material comprising multiple, different (non-gaseous) phase domains in which at least one type of phase domain is a continuous phase. A foamed substance, which is a multiphase material that consists of a gas dispersed in a liquid or solid, is not normally considered to be a composite.

COMPOSITE BEAM: A structural member composed of two or more dissimilar materials joined together to act as a unit in which the resulting system is stronger than the sum of its parts.

COMPOSITE FRUIT (MULTIPLE FRUIT): A type of fruit such as pineapple, mulberry and breadfruit that develops from an inflorescence or from a cluster or group of flowers into an entire fruit rather than from a single flower. Each of the flowers produces a fruit, but these all mature into a single mass to form the composite fruit. The two types of composite fruits are: (i) **Syconus**, which is a collective fleshy fruit, in which the ovaries are hidden within a hollow receptacle. An example is fig; and (ii) **Sorosis**, which develops from spike or spadix or catkin, where the fleshy and juicy part is formed from the axis, the bract and the perianth. Examples are jackfruit, pineapple, mulberry and breadfruit.

COMPOSITE FUNCTION: A function that consists of two functions arranged in such a way that the output of one function becomes the input of the other function. For example, if $f(x) = 2x + 5$ and $g(x) = 6x$, then the composite function $f(g(x))$ is the function $2(6x) + 5 = 12x + 5$. Composite of functions is not a commutative operation; that is, $f(g(x))$ is not equal to $g(f(x))$; $g(f(x))$ is also denoted by $f \circ g(x)$, i.e., the function g followed by the function f .

COMPOSITE INDEX NUMBER: Composite index numbers allow the measurement, with single number, the relative variations within a group of variables upon moving from one situation to another. Examples are the consumer price index, the wholesale price index, the employment index and the Dow Jones index. It summarises all of the simple index numbers contained in a complex number (a value formed from a set of simple values) in just one index. The most commonly used composite index numbers are the Laspeyres Index, the Paasche Index and the Fisher Index.

COMPOSITE LAMINATES: Assemblies of layers of fibrous composite materials, which can be joined to provide a wide range of engineering properties, including in-plane stiffness, bending stiffness, strength and coefficients of thermal expansion.

COMPOSITE MATERIALS: Two or more materials that have better properties when combined than the single components have by themselves. The term is usually used to describe any of various modern industrially manufactured composites, such as carbon fibre-reinforced plastics. The fibres by themselves are easily damaged, but incorporating them within a matrix provides protection from damage and solvents, holds them in place and transfers loads to the fibres resulting in high tensile and compression strength. Other related, reinforced composite materials employ a matrix of metals or ceramics when temperature, strength or some other requirement precludes the use of polymers.

COMPOSITE MECHANISM: A reaction that involves more than one elementary reaction. The terms complex mechanism, indirect mechanism and step-wise mechanism are also commonly used. There are two main kinds of evidence for a composite mechanism: (i) The kinetic equation for the reaction does not correspond to its stoichiometry. (ii) There is experimental evidence, direct or indirect, for intermediates of such a nature that it is necessary to conclude that more than one elementary reaction is involved. There are many types of composite mechanisms. Examples are: (i) Reactions occurring in parallel, are called **parallel reactions** or **simultaneous reactions**. When there are simultaneous reactions there is sometimes competition. (ii) Reactions occurring in forward and reverse directions are called **opposing reactions**. (iii) Reactions occurring in sequence are known as **consecutive reactions**. (iv) Reactions are said to exhibit feedback if a substance formed in one step affects the rate of a previous step. (v) **Chain reactions** are also composite mechanisms.

COMPOSITE MEMBRANE: A membrane having chemically or structurally distinct layers.

COMPOSITE NUMBER: A number with two or more factors. Every composite number can be written as the product of two or more prime numbers. Examples are 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, etc. The number $445 = 5 \times 89$ is a composite number.

COMPOSITE REACTION: A chemical reaction for which the expression for the rate of disappearance of a reactant (or rate of appearance of a product) involves rate constants of more than a single elementary reaction. Examples are opposing reactions, in which the rate constants of two opposed chemical reactions are involved; parallel reactions, in which the rate of disappearance of any reactant is governed by the rate constants relating to several simultaneous reactions to form different respective products from a single set of reactants; and stepwise reactions.

COMPOSITE SAMPLE: A sample comprising two or more increments selected to represent the material being analysed. It is often prepared as a representative mixture of several different (usually

bulk) samples and from which the laboratory sample is taken. For example, composite samples of effluent collected, stored, analysed, tabulated and averaged over an extended period of time provide the only verifiable indication of treatment plant performance.

COMPOSITE THEORETICAL PERFORMANCE (CTP): It is used to measure the performance of computer processors. The values returned by CTP calculations are used for benchmarking purposes, which compare the performance of different processors. For example, Intel and AMD use CTP calculations to measure how many millions of theoretical operations per second (MTOPS) their processors can perform. An Intel Pentium M 770, which runs at 2.13 GHz, has a CTP of 7100 MTOPs, while an AMD Opteron 146, which runs at 2.0 GHz, has a CTP of 7168 MTOPs.

COMPOSITE VIDEO (BASEBAND VIDEO; RCA VIDEO): The analogue waveform that conveys the image data or video in a single cable. Examples are conventional National Television Standards Committee (NTSC) and Phase Alternating Line (PAL) television signals. Composite video contains chrominance (hue and saturation) and luminance (brightness) information, along with synchronisation and blanking pulses, all transmitted together in a single signal. Composite video offers a lower quality picture than component video.

COMPOSITE VOLCANO (STRATOVOLCANO): A steep sided conical volcano formed by many layers (strata) of hardened alternate layers of lava, tephra, pumice and volcanic ash at a destructive margin.

COMPOSITION: The operation that forms a single function from two given functions by applying the second function to the value of the first for any argument; it is only defined when the range of the first is contained in the domain of the second. The composition of functions f and g is written $f \circ g$ and pronounced as "f composed with g, 'f circle g' or 'f of g'". The formula for $f \circ g$ is written $(f \circ g)(x)$ and pronounced as "f composed with g of x." The composition of, for example, $x + 3$ with x^2 is $x^2 + 3$; and that of x^2 with $x + 3$ is $(x + 3)^2$. Repeated composition is denoted by a superscript numeral f^n , for example, $f \circ f \circ f \circ f = f^{(4)}$.

COMPOSITION SERIES: In mathematics, a normal series $M_1, M_2, \dots$, of a group, where each M_i is a proper normal subgroup of M_{i-1} and no further normal subgroups both contain M_i and are contained in M_{i-1} . A composition series of M is a sequence of submodules $0 = M_0 \subset M_1 \subset M_2 \subset \dots \subset M_n = M$ such that M_{i+1}/M_i is simple for each $i=0, \dots, n-1$. The length of the composition series is then n . The modules $(M_{i+1}/M_i)_{i=0}^{n-1}$ are called the **composition factors** of the series. Every finite group has a composition series, for example, $(S_n, A_n, 1)$ is a composition series for the symmetric groups $S_n \geq 5$, where A_n is the alternating group.

COMPOSITION STOICHIOMETRY: The quantitative mass relationships between elements in compounds. For example, composition stoichiometry describes the nitrogen to hydrogen (mass) relationship in the compound ammonia.

COMPOSITIONAL HETEROGENEITY: The variation in elemental composition from molecule to molecule, usually found in copolymers.

COMPOST: Organic matter, or a mixture of organic matter such as decaying leaves, manure and other nutritive matter that have been decomposed by bacteria with soil to make a nutrient rich natural fertiliser for use in gardening. The decomposition of compost can be hastened by adding a compost activator, such as nitrochalk. Composting is the process of mixing decaying leaves, manure and other nutritive matter to improve and fertilize soil.

COMPOUND: 1. A pure chemical substance, made up of two or more elements bonded together. Chemical compounds can be divided into molecular compounds (which are held together by covalent bonds), ionic compounds (for example, salts are held together by ionic bonds) and metallic compounds hold metals together). They cannot be separated by physical means, but can be separated into simpler substances by chemical reactions. Each compound has a chemical formula, which describes the ratio in which the atoms of each element

in the compound occur. For example, the chemical formula of water is (H₂O), which means it is a compound that is made up of two hydrogen atoms for every one oxygen atom. 2. Composed of two or more anatomically or morphologically equivalent units, whether subdivided into them or an aggregate of them.

COMPOUND ANGLE FORMULAE: It is any of the trigonometric sum or difference functions, which are given as follows:

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$, $\cos(A \pm B) = \cos A \cos B \pm \sin A \sin B$, and $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$.

COMPOUND DOCUMENT: In computer science, it refers to an organised collection of user interfaces that form a single integrated perceptual environment. It includes a data structure that contains different data types, such as text, audio files and motion video files. A compound document is also an application environment containing program objects that can be interlinked and interacted with by a user.

COMPOUND EVENT: An event comprising of two or more simple events, for example, getting heads in a coin toss and an even number in rolling a die.

COMPOUND EYE: A description of the eye of most insects and some crustaceans composed of many light-sensitive elements, each having its own refractive system and each forming a portion of an image.

COMPOUND FRACTION (COMPLEX FRACTION): A fraction in which either the numerator or denominator or both contains fractions.

For example, $\frac{1}{\frac{3}{2}}$ is a complex fraction. To simplify the complex fraction,

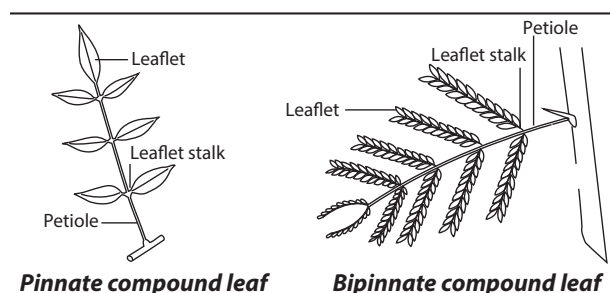
multiply both the numerator and the denominator by the reciprocal of the denominator.

COMPOUND INTEREST: The interest calculated on an amount composed of the principal and the previous interest earned together forming the new principal for another term. Compound interest on principal (p), rate (r) and time (t) is given by $C.I. = p \left[\left(1 + \frac{r}{100} \right)^n - 1 \right]$

The formula for calculating the amount A to be paid on a compound

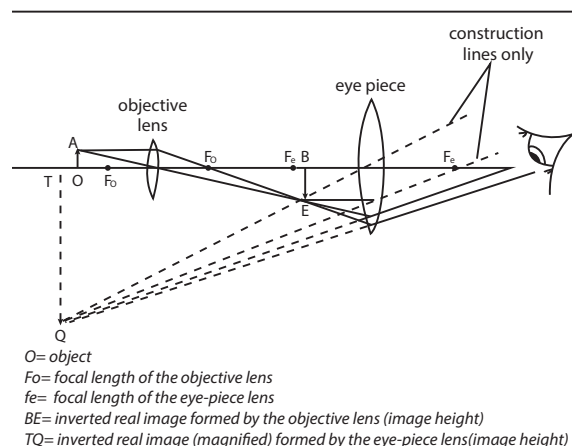
interest is $A = p \left(1 + \frac{r}{100} \right)^n$, where A is the amount of money after a certain period of time, P is the principal, r represents the interest rate, always represented as a decimal, t represents the period of time in which repayment is done and n is the number of times interest is compounded in a year.

COMPOUND LEAF: A leaf in plants such as cowpea, groundnut and neem with the blade divided into two or more separate parts, going down the midrib or petiole such that the leaf is broken into leaflets. In pinnately compound leaves, a row of leaflets forms on either side of an extension of the petiole called the rachis. A bipinnately compound leaf is divided twice, each leaflet is subdivided into smaller leaflets.



COMPOUND MICROSCOPE: A microscope that uses two lenses or systems of lenses, with the second magnifying the real image formed by the first. It consists essentially of two lenses (objective and eyepiece lenses) mounted in a tube. The objective lens has a short focal length with the object placed at a distance greater than the focal length. This forms a real image view by the eye piece which

has a long focal length. The final image produced by the eye piece is virtual, inverted and magnified. The total magnification produced by the microscope is the product of the magnification produced by the objective and the eyepiece lenses. The angular magnification of the compound microscope is given by $M = \left(\frac{V}{f_o} - 1 \right) \left(\frac{D}{f_e} + 1 \right)$, where V is the image distance of the objective lens, f_o is the focal length of the objective lens, f_e is the focal length of the eyepiece lens, D is the least distance of distinct vision.



A compound microscope

COMPOUND NUMBER: A number that is expressed with mixed units, for example, 35 kilometres 6 metres.

COMPOUND OVARY: An ovary having two or more carpels. For example, tomato is a simple fruit derived from a compound ovary.

COMPOUND SENTENCE: In logic, it is formed by joining two or more simple sentences together with one or more connectives, such as AND or, NOT or IF/THEN.

COMPOUND STATEMENT: A statement that is formed by combining two or more simple statements by the use of words or their symbols, for example, "and" ($\cdot$ or $\wedge$), "or" ($\vee$), "not" ($\neg$), "implies" ($\Rightarrow$).

COMPOUND UNIT: A quotient or product of units, or units that require two different types of measurement. Examples are kilometres per hour (km/hr), and newton per square metre (N/m²), as they require two units: distance (kilometre, km) and time (hour, hr), and force (newtons) and area (square metres), respectively to express them.

COMPRESSED GAS: 1. A gas or mixture of gases in a container with an absolute pressure exceeding 40 psi at 21.1°C (294.25 K). 2. A gas or mixture in a container, with an absolute pressure exceeding 104 psi at 54.4°C (327.55 K) regardless of the pressure at 21.1°C (294.25 K).

COMPREHENSION AXIOM: The axiom of set theory stating that for any property there exists a set consisting of all elements with the given property.

COMPRESSIBILITY: 1. A measure of the fractional change in volume of a fluid or solid as a response to a pressure or mean stress change, expressed as (increase in volume/increase in pressure) multiplied by the original volume. An increase in pressure compresses the specimen. A decrease in volume is a negative change, so the negative sign in the equation keeps the compressibility positive. Gases have much larger compressibility than solid and liquids. Its SI unit is pascal⁻¹. 2. In soil, it is the ability of a soil to decrease easily in volume when subjected to a mechanical load. **Compressible** is the ability of a body to change its volume when pressure is applied to it.

COMPRESSIBILITY FACTOR (DEVIATION FACTOR; GAS-DEVIATION FACTOR; SUPERCOMPRESSIBILITY FACTOR): A measure of how much the thermodynamic properties of a real gas deviate from those expected of an ideal gas. It is a useful thermodynamic property for modifying the ideal gas law to account for behaviour of real gases. Its symbol is Z . For an ideal

gas, Z always has a value of 1. For real gases, the value may deviate positively or negatively, depending on the effect of the intermolecular forces of the gas.

COMPRESSIBLE FLOW: A flow in which the fluid density varies significantly in response to a change in pressure. Pressure changes normally occur throughout a fluid flow and these pressure changes, in general, induce a change in the fluid density. All fluids are compressible. However, compressibility effects are more frequently encountered in gas flows than in liquid flows. It is also encountered in supersonic flight.

COMPRESSION: 1. The decrease in volume of an object by applying pressure. Gases can be compressed easily, solids and liquids to a very small degree if at all. When gases are compressed, under certain conditions of temperature and pressure, they may undergo liquefaction. An example of compression is by reducing the volume of the fuel mixture of an internal combustion engine to increase its pressure when it is ignited. Water, for example, is practically incompressible, thus making it especially useful for hydraulic machines. 2. In embryology, the shortening or omission of certain developmental stages. 3. In geology, a set of stresses directed toward the centre of a rock mass. Compressive strength refers to the maximum compressive stress that can be applied to a material before failure occurs. When the maximum compressive stress is in a horizontal orientation, thrust faulting can occur, resulting in the shortening and thickening of that portion of the crust. When the maximum compressive stress is vertical, a section of rock will often fail in normal faults, horizontally extending and vertically thinning a given layer of rock. Compressive stresses can also result in folding of rocks. 4. In computer science, **data compression** is used to reduce the size of one or more files. When a file is compressed, it takes up less disk space than an uncompressed version and can be transferred to other systems more quickly. Compression is often used to save disc space and reduce the time needed to transfer files over the Internet.

COMPRESSION FACTOR: Product of pressure and molar volume divided by the gas constant and thermodynamic temperature. Its symbol is Z .

COMPRESSION FORCE: A force that applies power, pressure, or exertion to an object to compact it. Compression force is used widely such as in compression brakes and hand tools.

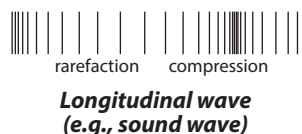
COMPRESSION RATIO: The ratio of the total volume in the cylinder of an internal combustion chamber at the beginning of the compression stroke to the volume enclosed at the end of the compression stroke. A high compression ratio is desirable because it allows an engine to extract more mechanical energy from a given mass of air-fuel mixture due to its higher thermal efficiency.

COMPRESSION WOOD: A type of reaction wood or structurally abnormal wood formed in response to various stresses, forming on the lower side of branches in gymnosperms. Compression wood is dense in structure due to the heavy lignification of tracheid walls. Tension wood forms in angiosperms on the upper side of the branches and the fibres tend to be gelatinous rather than lignified.

COMPRESSIONAL WAVE (PRIMARY WAVE; P-WAVE): A wave in which the displacement of the medium's particles is in line with or parallel to the direction of travel of the wave motion. It shakes the ground back and forth in the same direction and the opposite direction as the direction the wave is moving. It is the first wave to be recorded by a seismograph. It is the fastest of all seismic waves, travelling at a velocity of 5-7 kilometres per second in the crust and 8-9 kilometres per second in the upper mantle.

COMPRESSIVE NORMAL STRESS: Normal stress that is opposite in direction to the outward normal at a point on a surface.

COMPRESSIVE STRENGTH: A measure of the maximum compression force that a material can withstand before changing shape or breaking apart.



COMPRESSOR: A machine that increases the pressure of a gas or vapour such as air or mixture of gases and vapours. Examples are air compressor, which is used to supply high-pressure air to paint spraying, tire inflating and pneumatic tools; and refrigeration compressor used to compress the gas formed in the evaporator.

COMPROPORATION (SYMPROPORATION): An oxidation-reduction reaction in which an element in a higher oxidation state reacts with the same element in lower oxidation state to give an intermediate oxidation state. For example, potassium iodate reacts with potassium iodide under acidic conditions to form elemental iodine: $\text{IO}_3^- + 5\text{I}^- + 6\text{H}^+ \rightarrow 3\text{I}_2 + 3\text{H}_2\text{O}$. In this reaction, iodine in oxidation state +5 (iodate) reacts with iodine in oxidation state -1 (iodide) to form elemental iodine (± 0).

COMPTON EFFECT: A reduction in the energy and increase in wavelength of x-rays or gamma-ray photons, when they are scattered by free electrons, which absorb energy from the x-rays or gamma-rays. It was discovered by Arthur Holly Compton (1892-1962), US physicist in 1923, for which he shared the 1927 Nobel Prize in Physics with C.T.R. Wilson, British physicist. The discovery of the Compton Effect confirmed that electromagnetic radiation has wave and particle properties, a central principle of quantum theory.

COMPTON ELECTRON (COMPTON RECOIL ELECTRON): An electron set in motion by its interaction with a photon in Compton scattering.

COMPTON GAMMA-RAY OBSERVATORY (CGRO): A NASA (National Aeronautics and Space Administration) gamma-ray space observatory that was operated from April 1991 to June 2000 with four main telescopes to detect high energy gamma-rays.

COMPTON SCATTERING: The inelastic scattering of a photon by a free charged particle, usually an electron when a collision between a photon and a particle occurs. In inelastic scattering or incoherent scattering, part of the kinetic energy of the incident particle is lost inside the target giving rise to some internal processes and only a fraction of it goes into moving the whole target. When a photon collides with a free charged particle, the electron is ejected from its orbital position and the x-ray photon loses energy because of the interaction. Since the scattered x-ray photon has less energy, it has a lower frequency and thus a longer wavelength and is less penetrating than the incident photon. The electron that has been set into motion by the collision, called scattered or recoil electron continues to travel through the material along an altered path. The energy shift depends on the angle of scattering and not on Information Technology and the effects of computers on society.

COMPUTED TOMOGRAPHY (CT) SCAN: An X-ray imaging technique in which X-rays and detectors are rotated around a patient and a computer combines the information to produce a 3-D interior image. It can reveal anatomic details of internal organs that cannot be seen in conventional X-rays.

COMPUTER: A programmable digital or electronic device that receives information called data to perform a sequence of operations such as mathematical calculations and logical operations at high speed, based on a set of instructions called program. It basically consists of two major parts: hardware and software. The hardware is the physical machine consisting of parts such as motherboard that connects and controls all of the essential elements such as the central processing unit (CPU) and random access memory (RAM) and distributing power to the various components of the computer; central processing unit (CPU) that processes the commands or instructions; random access memory (RAM) that stores data for short-term use; and hard drive disk (HDD) that is the permanent storage device. Some other hardware include the monitor and input devices such as keyboard and mouse, which are needed to send instructions to the computer and to interact with it. The software is the set of instructions and data that a computer needs to perform specific operations. The four basic types of computers are: (i) **Supercomputers**, very powerful computers with a high level of computing performance that are used in specialised areas such as scientific and engineering applications; (ii) **Mainframe computers** are high-performance computers that can

support many users simultaneously and used in large organisations to process very large amounts of data very quickly. (ii) **Minicomputers** are small machines that are not as large as mainframes but larger than microcomputers, now replaced by personal computers (pc), they were often used in manufacturing for process control; (iii) A **microcomputer** is a small computer with a microprocessor (the central processing unit (CPU) built on a microchip), a memory system and random access memory, a bus system and input/output ports contained in a motherboard. It is widely used for entertainment, education and work purposes. Examples include desktop computers, gaming consoles, notebooks, tablet pc's, smartphones and calculators. The main manufacturers are Dell, Apple, Samsung, Sony & Toshiba.

COMPUTER ANIMATION (ANIMATION): It is a visual digital display technology that simulates inanimate objects to life on a screen or monitor. An example of animation is a cartoon shown on a television. There are two types: two-dimensional (2-D) animation, which is drawn by hand and slightly altered; and three-dimensional (3-D) animation which is always computer generated.

COMPUTER PROGRAM (SOFTWARE PROGRAM): A set of instructions written by computer programmers in a special language called programming language that directs a computer to perform specified task or combination of tasks.

COMPUTER PROGRAMMER (PROGRAMMER): A person who writes computer software needed for any computer system or application.

COMPUTER SIMULATION: The use of a computer-generated system to represent the dynamic responses and behaviour of a real or proposed system. It is used to study the behaviour of objects or systems that cannot be easily or safely tested in real life, such as weather patterns or a nuclear blast.

COMPUTER SYSTEM: A functional unit, consisting of one or more computers and associated peripheral input and output devices and associated software, that uses common storage for all or part of a program and also for all or part of the data necessary for the execution of the program. It executes user-written or user-designated programs; performs user-designated data manipulation, including arithmetic operations and logic operations; and can execute programs that modify themselves during their execution. A computer system may be a stand-alone unit or may consist of several interconnected units.

COMPUTER TERMINAL: An electronic device such as video adapter, a monitor and a keyboard which enables an operator to communicate with a computer.

COMPUTER-AIDED DESIGN (CAD; COMPUTER-AIDED DESIGN AND DRAFTING, CADD): The application of computer software in the design of precise and detailed drawings or technical illustrations of two- or three-dimensional models of physical objects, such as electronic circuit boards, mechanical parts and in construction. It is used by graphic and product designers, engineers, interior designers and architects.

COMPUTER-AIDED DESIGN AND MANUFACTURING (CADAM): The use of computers in the design and manufacture of products such as cars, airplanes, ships and computers.

COMPUTER-AIDED ENGINEERING (CAE): Software that analyses designs which have been created in the computer or elsewhere and entered into the computer. It helps engineers to use computers to design, analyse and manufacture products and processes. CAE includes computer-aided design (CAD), which is the use of a computer for drafting and modelling designs and computer-aided manufacturing (CAM), which is the use of computers for managing manufacturing processes.

COMPUTER-AIDED MANUFACTURING (CAM): The use of computers to convert engineering designs into finished products or the control of production by automating many production processes such as the control of machine tools and robots in factories. For example, CAM software helps to develop process plans order and track materials and monitor production schedules the best cutting tools for the material, sets the most effective cutting speed.

COMPUTER-AIDED SOFTWARE ENGINEERING (CASE): The use of a computer-assisted method to organise and control the development of software, especially on large, complex projects involving many software components and people. This allows designers, code writers, testers, planners and managers to share a common view of where a project stands at each stage of development.

COMPUTER-ASSISTED LEARNING, CAL (E-LEARNING): The use of all forms of electronically supported learning and teaching. The computer displays instructional materials to a student and asks questions about the information given and the student's answers determine the sequence of the lessons.

COMPUTER-ASSISTED MOLECULAR DESIGN (CAMD): The computer-assisted techniques used to discover, design and optimise compounds with desired structure and properties.

COMPUTER-ASSISTED MOLECULAR MODELLING (Camm): The investigation of molecular structures and properties using computational chemistry and graphical visualisation techniques.

COMPUTER-GENERATED IMAGERY (CGI): In computer graphics, it refers to 3D graphics used in film, TV and other types of visual media. Most modern action films include at least some CGI for special effects, while other movies, such as a Pixar animated films, are built completely from computer generated graphics.

COMPUTER-SUPPORTED COLLABORATIVE WORK (CSCW): Software tools and technology that support a group of individuals working on projects at different sites. It is based on the principle of group coordination and collaborative activities supported through computer systems.

COMPUTERIZED AXIAL TOMOGRAPHY SCANNER (CAT SCANNER; CT SCANNER): A medical technique that uses X rays, scintillation detectors and computers to generate a three-dimensional image of cross-sections of the body, including detailed views of the body's soft tissues such as blood vessels, muscle tissue and organs, such as the brain.

COMPUTERIZED BRANCH EXCHANGE (CBX): An exchange in which a central node acts as a high-speed switch to establish direct connections between pairs of attached nodes.

COMPUTING DEVICE: Any device built to perform or help computations. Examples are the abacus, slide rule or computer.

CONCATEMER: A DNA segment composed of repeated sequences linked end to end.

CONCATENATION: 1. The joining of one symbol or string a to the end of another one b to form the string ab . This forms the symbol for a new mathematical object from the given symbols. Concatenation is an associative operation and three or more strings, can be concatenated. 2. The joining of one quoted expression to another in such a way, as to form a single quoted expression. For example, a quoted conjunction, 'P&Q', is the concatenation of the first conjunct, with the symbol '&' and the second conjunct, which is written $P \wedge Q$ and can be abbreviated to the quasi-quotation $P\&Q$.

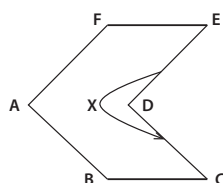
CONCAVE: 1. Of a surface, curving inwards so that at least one line segment can be drawn between two vertices that lies outside the figure; curved like a segment of the interior of a circle or hollow sphere. 2. Describing a polygon with straight sides with indentation, with at least one diagonal passing outside the figure and one or more interior angle greater than 180° . 3. Of a real-valued function or surface, having the property that the chord joining any two points on its graph lies below the graph, such that for arguments x and y in the appropriate abstract space and t in the interval $[0, 1]$, $tf(x) + (1-t)f(y) \leq f(tx + (1-t)y)$.

CONCAVE DOWN: Of or pertaining to a curve on an interval, having a decreasing derivative as the independent variable increases between two points. In other words, having a negative second derivative and curvature and being concave or having a positive second derivative and curvature and being concave. A differentiable function on an interval I , with $f'(x)$ also differentiable on I , is concave up on I , if and only if $f''(x) \leq 0$ on I .

CONCAVE LENS (DIVERGING LENS): A type of lens having at least one surface curved like the inner surface of a sphere. A parallel beam of light incident on it emerges wider after refraction through the lens.

CONCAVE MIRROR (CONVERGING MIRROR): A type of spherical mirror that is curved with a reflecting surface formed from the interior surface of a hollow sphere or paraboloid. The reflecting surface consists of silver metal deposited on the outer curved side or on the inside of the sphere. The image that is reflected from a concave mirror is a real image. It can be projected onto a screen. Common examples of concave mirrors are shaving mirrors and makeup mirrors that magnify objects placed close to them.

CONCAVE POLYGON (NON-CONVEX POLYGON): A polygon that has one or more interior angles of more than 180° . It is normally characterised by interior points, which cannot be joined by a straight line without going outside of the polygon.



Concave Polygon

CONCAVE UP: Of or pertaining to a curve on an interval, having an increasing derivative as the independent variable increases between two points. In other words, having a positive second derivative and curvature and being convex. A differentiable function on an interval I , with $f'(x)$ also differentiable on I , is concave down on I , if and only if $f''(x) \leq 0$ and if and only if $f'''(x) \geq 0$ on I .

CONCAVITY: The direction of a curve on a graph. If a curve is concave up or convex, the graph of the curve is bent upwards. If a curve is concave down or concave, then the graph of the curve is bent downwards.

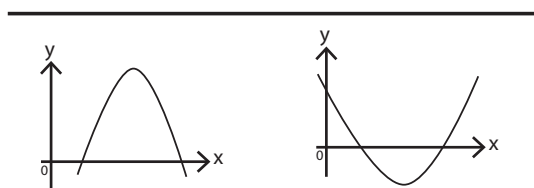


Fig. 1: Graph of concave downward, called concave

Fig. 2: Graph of concave upward, called convex

Concavity

CONCAVO-CONCAVE: Concave on both sides.

CONCAVO-CONVEX: Having one side concave and the other convex.

CONCENTRATE: 1. Feed high in total digestible nutrients such as vitamins and minerals and low in fibre. 2. Increase concentration by adding more solute or reducing the amount of solvent. 3. A drug or other preparation that has been strengthened by evaporation of its nonactive parts.

CONCENTRATED: Of a substance or solution, present in a high proportion relative to other substances; having a large amount of solute relative to the amount that could dissolve, for example, having a large amount of salt relative to that amount of water in a saltwater solution.

CONCENTRATED FLOW: In agriculture, it is the flow of a body of water over a piece of land. It often creates serious gullies and erosion thereby removing surface soil and organic matter.

CONCENTRATED SOLUTION: A solution in which a great amount of compound is dissolved in a little amount of the solvent.

CONCENTRATION (MOLARITY): The amount of a substance (solute) present in a specified amount of solution or the quantity of dissolved substance per unit quantity of solvent in a solution. In other words, it is the amount of substance of a solute dissolved in one cubic decimetre of the solution. Its unit is mol/dm^3 .

CONCENTRATION CELL: An electrochemical cell for potentiometric measurement of ionic concentrations that has two equivalent half-cells of the same material differing only in concentrations. One can calculate the potential developed by such a cell using the Nernst

Equation. A concentration cell produces a voltage as it attempts to reach equilibrium, which will occur when the concentration in both cells are equal.

CONCENTRATION DEPOLARISATION: Loss of emission anisotropy due to transfer of electronic excitation from photo-selected molecules to other molecules with different orientations.

CONCENTRATION DISTRIBUTION RATIO: The ratio of the analytical concentration of a component in the stationary phase to its analytical concentration in the mobile phase.

CONCENTRATION GRADIENT: The difference in concentration between a region of higher concentration and one of lower concentration. It is a vector quantity and may be expressed as amount-of-substance, mass, number, volume, concentration gradient.

CONCENTRATION-CELL CORROSION: The local variation of the corrosion rate due to the action of a corrosion cell resulting from inhomogeneous composition of the environment.

CONCENTRATION-SENSITIVE DETECTOR: A device, in which the response is proportional to the concentration of a sample component in the eluent.

CONCENTRIC BUNDLE (CENTRIC BUNDLE): A vascular bundle in which one vascular tissue surrounds the other so they appear to be arranged concentrically when viewed in transverse section.

CONCENTRIC CIRCLES: Two or more circles that share the same centre but different radii lengths.

CONCEPT MAP: A type of graphical visualization tool used to organise and represent knowledge of a subject. It represents ideas and information as boxes or circles, which it connects with labelled arrows in a downward-branching hierarchical structure. It is typically used by designers, engineers, technical writers, etc., to organise and structure knowledge.

CONCEPTACLE: A flask-shaped cavity with a small opening containing reproductive structures found in the swollen tip of certain algae and fungi. A female conceptacle is lined with unbranched sterile hairs or paraphysis and the oogonia develop from short stalks arising from the chamber wall. The male conceptacle is filled with branched hairs that bear antheridia. The conceptacle opens via an ostiole.

CONCEPTION: The fertilisation of a mammalian egg cell by a sperm cell; or the union of the sperm and ovum to form a viable zygote.

CONCEPTUAL SUBITIZING: To count the number of objects in a set by consciously partitioning that set into smaller subsets that can be easily recognised without counting.

CONCERTED PROCESS: Two or more primitive changes are said to be concerted (or to constitute a concerted process) if they occur within the same elementary reaction. Such changes will normally (though perhaps not inevitably) be energetically coupled. The term 'energetically coupled' means that the simultaneous progress of the primitive changes involves a transition state of lower energy than that for their successive occurrence. In a concerted process the primitive changes may be synchronous or asynchronous.

CONCERTED REACTION: A chemical reaction in which all bond breaking and bond making occur in a single step rather than a series of steps.

CONCHOID: A plane curve derived from a fixed point (the pole), with two branches situated about a line to which they are asymptotic, so that a line from the fixed point intersecting both branches is of constant length between asymptote and either branch. Mathematically, its equation is $(x-a)^2(x^2+y^2)=b^2x^2$, or in polar coordinates, it is $r \cos \theta = a \pm b \cos \theta$, where a is the distance between the pole and a vertical asymptote and b is the length of the constant segment.

CONCHOIDAL FRACTURE: The fracture of a solid in which the surface of the material is curved and marked by concentric rings. This occurs in certain brittle materials which during breakage do not follow any natural planes of separation. Fine-grained minerals such as flint and amorphous solids, for example, obsidian and some glass types show conchoidal fracture.

CONCLUSION: 1. The reasoned deduction made from results of an experiment. 2. The part of a conditional statement that comes after "then". 3. In logic, a statement that purports to follow from another or the premises by means of an argument or proof. It is also a statement that does in fact validly follow from given premises.

CONCOLOUR: Having the same or of uniform colour.

CONCOMITANT MATRIX: The matrix of coefficients in Lagrange's identity that involves an n th-order linear differential equation and its adjoint.

CONCORDE: A supersonic aircraft powered by turbojet, which travelled twice as fast as the speed of sound (2,170 km/h). It was developed by the French Aérospatiale and the British Aircraft Corporation for British Airways and Air France to carry passengers. The Concorde ended its regular passenger service in October 2003 due to unprofitability.

CONCRETE: Material mixed with cement, stone, sand and water then allowing the mixture to dry and harden and used in building construction. The cement acts as the binding material.

CONCRETE NUMBERS: Numbers that are used in counting particular items or objects, for example, six exercise books. It is an elementary stage in the identification of numbers, especially by children.

CONCRETE PATTERN: A set of concrete manipulatives, such as pattern blocks, coins, or coloured counters that are arranged following a rule.

CONCRETION: A hard, compact, irregularly-shaped nodule of hard solid mass formed by the local accumulation of matter such as calcium carbonate or iron oxide that is embedded in sedimentary rock or soil. It is of a varying size, shape, hardness, and colour. An example includes flint in chalk.

CONCURRENT LINES: Two or more lines that coincide or pass through a single point. Concurrence is the fact that three or more lines coincide or pass through a single point.

CONCURRENT PROCESSING: The simultaneous execution of more than one sequential program on a computer or network of computers. A concurrent program consists of several sequential programs to be executed in parallel. Each of the concurrently executing sequential programs is called a process.

CONCURRENT SERVER: A server that can handle many connections at the same time. It can accept new connection requests while still processing the transactions started by previous requests.

CONCYCLIC: When a circle passes through a number of points exactly without leaving any point behind, then the points are concyclic.

CONDENSATION: 1. The physical process of converting a material from a gaseous or vapour phase to a liquid or solid phase. This commonly results when the temperature is lowered and/or the vapour pressure of the material is increased. The tendency exists for condensation to occur when the partial pressure of a given component of a gaseous mixture at a given temperature exceeds the vapour pressure of the liquid or solid form of that component at the given temperature. 2. A reaction in which two organic compounds combine to form a larger molecule, accompanied by the removal of a smaller molecule, usually water. It is also known as an **addition-elimination reaction**.

CONDENSATION NUCLEI (CN): Particles, either liquid or solid or ions upon which condensation of water vapour (or other substances) begins in the atmosphere. Condensation nuclei are usually very small hygroscopic aerosols (0.001 μm to 0.1 μm in diameter), but these are not as abundant as the smaller particles. The number of CN which are active (initiate condensation) in a given air mass may be a function of the relative humidity. Usually CN are counted as the active nuclei at about 300% relative humidity, while cloud condensation nuclei (CCN) are counted as the number of active nuclei at relative humidity less than or equal to 102%.

CONDENSATION NUMBER: The ratio of the number of molecules condensing on a surface of a solid or liquid in equilibrium with its vapour phase to the total number of molecules touching the surface.

CONDENSATION POINT: 1. In topology, a point of a given a subset such that every neighbourhood of p contains an uncountable number of points of S . In Eucudean space, the set of condensation points of any set is perfect and omits only countably many points of the set. 2. Also known as **boiling point**, it is the temperature at which a material changes from a gas to a liquid.

CONDENSATION POLYMERISATION: Polymerisation reaction in which one or more monomers, with more than one reactive functional group, combine to form a polymer with the elimination of water or other small molecule. Examples of condensation polymers are polyester, polyamides and polyacetals.

CONDENSATION REACTION: A chemical reaction in which two or more small molecules combine to form a larger one, normally with the elimination of a simpler substance, usually water. For example, acetaldehyde (ethanal), CH_3CHO , condenses with hydroxylamine, H_2NOH , to form an oxime, $\text{CH}_3\text{CH}=\text{NOH}$ with the elimination of one mole of water.

CONDENSATION-PUMP (DIFFUSION PUMP): A vacuum pump in which vapourised oil or mercury vapour is diffused through a stream, which removes the gas molecules from the container in which the pressure is to be reduced.

CONDENSED PHASES: The liquid and solid phases in which particles interact strongly. Condensed phases include the superconducting phase exhibited by certain materials at low temperature, the ferromagnetic and antiferromagnetic phases of spins on atomic lattices and the Bose-Einstein condensate found in certain ultracold atomic systems.

CONDENSED-MATTER PHYSICS: The branch of physics concerned with the study of the physical properties of condensed phases of matter including the study of the solid and liquid states, dense plasmas, liquid crystals, glasses, polymers and gels. Some other condensed phases include the superconducting phase exhibited by certain materials at low temperature, the ferromagnetic and antiferromagnetic phases of spins on atomic lattices and the Bose-Einstein condensate found in certain ultracold atomic systems. These properties appear when a number of atoms, typically larger than the molecular scale, interact strongly and adhere to each other or are otherwise highly concentrated in a system.

CONDENSER MICROPHONE (ELECTROSTATIC MICROPHONE): A microphone consisting essentially of a flexible metal diaphragm and a rigid metal plate that together form a two-plate air capacitor. Sound waves set the diaphragm in vibration, producing capacitance variations that are converted into audio-frequency signals by a suitable amplifier circuit.

CONDITION: 1. A presupposition, especially a restriction on the truth of the stated result that is indispensable to the proof of a theorem, so that the truth of the result only holds if the stated precondition is satisfied. 2. In logic, a proposition on which another proposition depends; the antecedent of a conditional proposition. It is a **necessary condition**, if its truth is required for the truth of the other; if its truth guarantees that of the other, it is a **sufficient condition**.

CONDITIONAL: 1. Of an equation or inequality, true only for certain of the variables for which it can be solved. An example is $x - 1 = x + 1$ is a conditional equation, as it is true only for $x = 2$ and $x = -1$. 2. In statistics, with respect to some random variable of which the value is taken to be fixed.

CONDITIONAL DISTRIBUTION: The probability distribution of a subset of components of a random vector, conditional upon the values taken by another subset of components.

CONDITIONAL EQUATION: An equation that is true only for some values of the unknowns contained in the equation.

CONDITIONAL MUTATIONS: Mutations that are expressed only under certain conditions. For example, for some mutations to be expressed, the individual needs to be placed in a specific environment, called **restrictive condition**; if the individual grows in any other environment, called **permissive condition**, the wild type phenotype is expressed. Mutations that are only expressed at a specific temperature (usually elevated), are called **temperature sensitive mutants**.

CONDITIONAL PROBABILITY: The conditional probability that event A will occur, given that event B has occurred is written as $P_r(A|B)$ (read as "A given B). It can be found from the formula:

$$P_r(A|B) = \frac{P_r(A \cap B)}{P_r(B)}$$

CONDITIONAL RELATIVE FREQUENCY: The frequency of one event occurring given that another event occurred.

CONDITIONAL RESPONSE: A behavioural modification by which a subject learns to respond in a desired manner to a previously neutral stimulus that has been repeatedly presented along with an unconditioned stimulus that elicits the desired response.

CONDITIONAL STATEMENT: A statement of the form: "If p is true, then q is true" In other words, a statement that is true unless a true hypothesis leads to a false conclusion. Symbolically, it is written as $p \rightarrow q$ (" p implies q "), where p is a hypothesis and q is a conclusion. For example, "If a triangle has two equal sides, then it has two equal angles" is true, but the statement "If a quadrilateral has four equal sides, then it has four equal angles" is false. In this statement, the second part, called the consequent cannot be true if the first part, called the antecedent is false.

CONDITIONALISATION/CONDITIONALIZATION: 1. In logic, the process of deriving a conditional statement from an argument by taking the conjunction of the premises of the given argument as its Antecedent and its conclusion as its consequent. 2. In logic the statement so derived that is true if and only if the argument is valid.

CONDITIONALLY COMPLETE (RELATIVELY COMPLETE): Of a partially ordered set, such that every set with an upper or lower bound has a supremum or infimum, respectively.

CONDITIONALLY CONVERGENT: The series $\sum_{n=1}^{\infty} a_n$ converges conditionally if and only if $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} |a_n|$ diverges.

Convergent but not absolutely, so that although the given series converges, the series of its absolute values does not. For example, the

convergent series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is only conditionally

convergent, since $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is divergent.

CONDITIONED REFLEX (CONDITIONED RESPONSE): Acquired response in which the subject learns to associate a previously unrelated neutral stimulus with a different stimulus elicits some kind of reaction. The subject can be human or another animal. Elicitation of conditional response by conditional stimulus is known as conditional reflex.

CONDITIONING: A process by which animals learn about a relation between them or principles of behaviour modification. The two major principles are classical conditioning and operant conditioning.

CONDOM (SHEATH OR PROPHYLACTIC): A barrier contraceptive made of latex rubber which fits over an erect penis and receives the semen during ejaculation. It is an effective means of preventing pregnancy as well as protection against sexually transmitted infections such as human immunodeficiency virus (HIV). If used carefully, preferably with a spermicide it is 97% effective, alone it is 85% effective.

CONDUCTANCE: The ability of a material to carry an electrical current, usually represented by the symbol G . For a direct current

with resistance R , conductance G is the reciprocal of R : $G = \frac{1}{R}$. In an alternating current circuit, conductance is the resistance R divided by the impedance Z : $G = \frac{R}{Z}$. The SI unit of conductance is siemens. One siemens equals one ampere per volt. Siemens was formerly called the mho or reciprocal of ohm.

CONDUCTIMETRIC TITRATION: A type of titration in which changes in the electrical conductivity of a solution is monitored as the ionic composition of the solution is changed by adding another substance. A significant change in the electrical conductivity of the system occurs at the equivalent point.

CONDUCTING POLYMER (INTRINSICALLY CONDUCTING POLYMERS, ICPS): Organic polymers that conduct electricity. They may be true metallic conductors or semiconductors or organic polymers with crystalline structure in which chains of conjugated unsaturated carbon-carbon bonds are aligned. Examples of organic polymers are polyacetylene and polypyrrole.

CONDUCTING POLYMER COMPOSITE: Electrically-conducting composite comprising a non-conducting polymer matrix and an electrically conducting material. Examples of the electrically-conducting materials are carbon black and metal particles.

CONDUCTION: 1. The flow of charged particles through a material that gives rise to electric current. 2. The transfer of energy such as heat or electricity through a medium such as a metal without any apparent change in the medium. 3. The transfer of energy from particle to particle in matter.

CONDUCTION BAND: A partially filled band or a band of vacant energy levels just higher in energy than a filled band. It is located above the valence band and separated by band gap or energy gap in a solid material. It is the range of permissible energy values required to free an electron bound to an atom in solids to become mobile charge carriers to conduct electric current in metallic conductors and semiconductors. In metallic conductors there is no band gap and the valence and conduction bands overlap.

CONDUCTIVE LOSS (CONDUCTIVE HEARING LOSS): A hearing loss in which sounds are processed correctly in the inner ear but the vibrations are not reaching it. Air conduction thresholds show a hearing loss but the bone conduction thresholds are normal. Possible causes include the following: fluid accumulation in the middle ear, acute ear infections, allergies, a perforated eardrum, benign tumours, too much earwax, etc.

CONDUCTIVITY: A measure of the ability of a substance to conduct heat or electricity. Electrical conductivity, with the symbol σ , is the reciprocal of the resistivity (ρ) of a material. It is measured in siemens per metre (or $(\Omega m)^{-1}$): $\sigma = \frac{1}{\rho}$.

CONDUCTOMETRIC TITRATION: A type of titration in which the end point is determined by detecting a sudden change in the conductivity of the solution. For example, when an HCl solution is titrated with an NaOH solution, the conductivity decreases as the faster moving H_3O^+ ions are replaced by slower moving Na^+ ions. After the equivalence point the conductivity increases as the concentration of the fast-moving OH^- ions increases.

CONDUCTOR: 1. A **thermal conductor** is a material that transfers heat along it. The heat flows from the part with a higher temperature through the material to the part with a lower temperature. It is the exchange of energy between adjacent subatomic particles, atoms, or molecules when they gain kinetic energy and collide with each other. Most metals are very good thermal conductors. 2. An **electrical conductor** is a material which allows the free flow of electrons or charged particles from atom to atom when voltage is applied to it. This is because some electrons are very loosely bound to atoms, permitting a free flow. These are called 'free electrons'. For example, copper has only one electron in its fourth shell that is very loosely bound to an atom, and can move freely. The best conductor of electricity is silver; other good conductors are copper, steel, gold, aluminium and

brass. Materials which are poor conductors of electricity are called **insulators**. This is because the electrons are tightly bound to the atoms and restrict the free flow of electrons. Examples of insulators are glass, air, plastic, rubber, and wood.

CONDUPLICATE: Folded, as certain grass leaves are. The term also refers to a form of vernation in which each leaf is folded in a U-shape around the next youngest leaf.

CONDY'S FLUID: A mixture of calcium and potassium permanganates used as an antiseptic.

CONDYLE: A rounded part at the end of a bone that forms a moving joint with a cup-shaped cavity in another bone. The ball part of a ball-and-socket joint such as the shoulder or hip joint is a condyle.

CONE: 1. Also known as **cone cell**, it is a type of photo-sensitive receptor cell, found in the retinas of all diurnal vertebrate eyes. Cones are concerned with the discrimination of colour and of fine detail. In humans, they contain three types of photosensitive pigment, compared to just one in rod cells. They also have a much faster response to light than do rods, so that they can perceive more rapid changes in stimuli. 2. A solid or surface with a plane base consisting of the set of all straight lines passing through a fixed point called the vertex or the points of a circle or ellipse whose plane does not contain the vertex. If the base is a circle, the shape is a **circular cone**; if the line or **axis**, from the centre of the base to the vertex is perpendicular to the circular base, then it is a **right cone**, for example, an ice-cream cone is a right circular cone; otherwise it is an **oblique cone**. The volume of a circular cone is $\frac{1}{3}\pi r^2 h$, where r

is the radius of the base and h is the perpendicular height of the cone. The curved lateral surface of the cone is called a nappe. If the cone is extended in both directions from the vertex, the result is a double cone or bicone. A section through a double cone that has been extended indefinitely in both directions to form a conic surface is known as a conic section. 3. In a vector space, a cone is a subset of any set that is closed under positive multiplication and must be convex and include the original. 4. It is a reproductive structure occurring in gymnosperms such as cycads.

CONFIDENCE INTERVAL (INTERVAL ESTIMATE): In statistics, it is the probability that the true value of a population parameter will fall within an upper and lower bound of a probability distribution. For example, if $[F, G]$ is a 95% confidence interval of some parameter and F and G are both functions of a random sample from the given distribution, then 95% of such samples yield intervals that include the true value of the parameter. In other words, if the same sampling method is used to select different samples and an interval estimate for each sample is computed, one would expect the true population parameter to fall within the interval estimates 95% of the time. It is used to express the degree of uncertainty associated with a sample statistic.

CONFIDENCE LEVEL: In statistics, it is a measure of the reliability of a result and is the percentage of all possible samples that can be expected to include the true population parameter. It is the probability that the confidence interval constructed around an estimator contains the true value of the corresponding parameter of the population. A 95% confidence level implies that 95% of the confidence intervals would include the true population parameter. In other words, a confidence level of 95% means that there is a probability of 5% that the result is unreliable; it is often taken to mean that the probability of error is less than 5%. It is designated by $(1 - \alpha)$, where α corresponds to the risk of error; that is, to the probability that the confidence interval does not contain the true value of the parameter.

CONFIDENCE LIMIT: The lower and upper limits or the numbers at the upper and lower ends of a confidence interval, between which the true value of a population parameter lies in at least the specified percentage of estimates of the limits. Mathematically, it can be defined as: $\bar{Y} \pm t_{1-\frac{\alpha}{2}, N-1} \frac{s}{\sqrt{N}}$, where $\bar{Y}$ is the sample mean, s is the sample

standard deviation, N the sample size, α is the desired significance level and $t_{1-\frac{\alpha}{2}}$ is the 100(1- α /2) percentile of the t distribution with $N - 1$ degrees of freedom. Note that the confidence coefficient is $1 - \alpha$.

CONFIG.SYS FILE: In the OS/2 operating system, a file used by the base operating system that describes the devices, system parameters and resource options of a workstation.

CONFIGURATION: 1. Generally, it is an arrangement of parts or elements in a particular form, figure or combination. 2. In geometry, a finite set of points and lines such that each point lies on the same number of lines and each line passes through the same number of points. Any configuration has a dual in which points and lines are interchanged. For example, a complete quadrilateral and a complete quadrangle are dual configurations. A configuration in the plane is denoted by (p, ℓ, γ, π) , where p is the number of points, ℓ the number of lines, γ the number of lines per point and π the number of points per line; satisfy the equation $p\gamma = \ell\pi$. 3. Another term for **block design**, especially one for which one specifies the sizes of the components. 4. In mechanics, it is an abstraction from the notion of a body having a particular position and orientation at a time; a representation of a body by a three-dimensional geometric figure. For example, the natural configuration for a solid would be the unit cube. 5. A distribution of energy among the particles of a system with more than two particles which are not considered distinguishable; for example, if the system consists of three particles, A, B, C and the total energy of the system is one unit, then the three states $A = 1, B = 0, C = 0$; $A = 0, B = 1, C = 1$; $A = 0, B = 0, C = 1$ correspond to same configuration (1, 0, 0). 6. **Electron configuration** is a distribution of the electrons of an atom or a molecular entity over a set of one-electron wavefunctions called orbitals, according to the Pauli principle. From one configuration several states with different multiplicities may result. 7. **Stereochemical configuration** is the arrangements of atoms of a molecular entity in space that distinguishes stereoisomers, the isomerism between which is not due to conformation differences. 8. In computer science, it is the way in which software and hardware components are arranged and interconnected in a computer system. It includes processor speed, the amount of RAM, hard drive space, the type of video card in the machine and many other advanced specifications.

CONFIGURATION INTERACTION (CI): The mixing of many-electron wavefunctions constructed from different electronic configurations to obtain an improved many-electron state.

CONFIGURATION SPACE (CONFIGURATION MANIFOLD): The space of possible positions that a physical system may attain, possibly subject to external constraints. The configuration space of a typical system has the structure of a manifold. For example, in a gas of N atoms each atom has three positional coordinates, so the configuration space is $3N$ -dimensional. If the particles also have internal degrees of freedom, such as those caused by vibration and rotation in a molecule, then these must be included in the configuration space, which is consequently of a higher dimension.

CONFIGURATIONAL BASE UNIT: A constitutional repeating unit in a regular macromolecule, a regular oligomer molecule, regular block or a regular chain, the configuration of which is defined at least at one site of stereoisomerism in the main chain.

CONFIGURATIONAL DISORDER: Structural disorder resulting from the statistical co-crystallisation of different configurational repeating units.

CONFIGURATIONAL REPEATING UNIT: The smallest set of successive configurational base units that prescribes configurational repetition to one or more sites of stereoisomerism in the main chain of a regular macromolecule, a regular oligomermolecule, a regular block or a regular chain.

CONFINEMENT (COLOUR CONFINEMENT): The phenomenon that colour charged particles such as quarks cannot be isolated

singularly and therefore cannot be directly observed or the force between quarks does not diminish as they are separated. Because it would take an infinite amount of energy to separate two quarks; they are forever bound into hadrons such as the proton and the neutron.

CONFIRM: Of an experiment, to make (an hypothesis) more likely to be true. By Bayes' theorem, if an experimental result is more likely under one hypothesis than another, the occurrence of that increases the probability of the former hypothesis. For example, every single sighting of a white swan confirms the hypothesis that all swans are white, since it tends to increase the probability of its truth; however, no sequence of such sightings, however many, are ever sufficient to prove its truth.

CONFLUENT: 1. Merging or running together so as to form a mass. 2. Two water bodies meeting at a point.

CONFOCAL: Having a common focus or foci.

CONFOCAL FLUORESCENCE MICROSCOPY: A light microscopic technique that produces high-resolution images of fluorescently stained specimens without requiring elaborate preparation of the sample. The fluorescent markers, generally fluorescently labelled antibodies, are excited by light from a laser focused by the objective lens of the microscope so that it scans a single plane in the specimen, creating an optical section, under computer control.

CONFORMABLE: 1. A term used to describe matrices that are related so that the number of elements in each row of the first matrix is equal to the number of elements in each column of the second matrix. A matrix is conformable if its dimensions are suitable for defining some operation such as addition and multiplication. Multiplication of two matrices is defined if and only if the number of columns of the left matrix is the same as the number of rows of the right matrix. Two matrices are conformable for addition if their sum is defined or if they have the same order. Squaring a matrix involves multiplying it by itself; therefore, it must be a square matrix to be conformable for squaring. 2. In geology, of or relating to unbroken rock or sediment strata, that are deposited in a continuous sequence, either adjacent to each other or in vertically without disturbance by a process such as erosion or folding.

CONFORMAL: In mathematics, designating or specifying a mapping of a surface or region upon another surface so that all angles between intersecting curves are preserved so that if two arcs meet at a given angle, their images are preserved at the same angle. An analytic function is conformal at any point where its derivative is existing. Also, any conformal mapping of a complex variable that possesses continuous partial derivatives is analytic.

CONFORMAL FIELD THEORY (CFT): A quantum field theory that is invariant under conformal transformations and is concerned about angles, but not distances. In two dimensions, there is an infinite-dimensional algebra of local conformal transformations and conformal field theories can sometimes be exactly solved or classified. It has important applications to string theory, statistical mechanics and condensed matter physics.

CONFORMAL INVARIANCE: Invariance under conformal transformations and under transformations of scales. Conformal invariance is important in theories with massless particles and in the theory of phase transition.

CONFORMAL TRANSFORMATION (CONFORMAL MAPPING; CONFORMAL MAP; ANGLE-PRESERVING TRANSFORMATION; BIHOLOMORPHIC MAP; ORTHOMORPHIC MAP): A transformation that preserves the angles between curves. An analytic function is conformal at any point where it has a nonzero derivative. An example is the Mercator map, a two-dimensional representation of the surface of the Earth that preserves compass directions. Conformal mapping is extremely important in complex analysis, as well as in many areas of physics and engineering.

CONFORMATION: 1. The spatial arrangement of groups in a molecule that can be changed without breaking covalent bonds.

Often, molecules with the same configuration can have more than one conformation. 2. A particular shape of a molecule that arises through the normal rotation of its atoms or groups about single chemical bonds.

CONFORMATIONAL ANALYSIS: The determination of the relative energies associated with various geometric arrangements of a molecule and the impact it has on the properties of the molecule such as stability.

CONFORMATIONAL DISORDER: Structural disorder resulting from the statistical coexistence within the crystals of identical configurational units with different conformations.

CONFORMATIONAL ISOMER (CONFORMER): Stereoisomers that can be converted into one another by rotation around a single bond. For example, eclipsed, gauche and anti butane are all conformational isomers of one another. These molecules can be interconverted by rotating around the central carbon single bond. For example, eclipsed butane can be made into gauche butane by rotating 60° and into anti butane by rotating 180°. Similarly, gauche butane can be made into anti butane by rotating 120°.

CONFORMER: 1. Any organism such as certain marine invertebrates, whose internal environment is highly influenced by external factors. They have no need to control their internal environment since the external environment is fairly constant in terms of temperature, oxygen tension and nutrients. 2. Also known as **conformational isomer**, it is any of a set of stereoisomers, that can be converted into one another by rotation around a single bond.

CONGENERS: 1. Elements that belong to the same group in the periodic table; or as a derivative. 2. Any member of a specified genus. 3. A secondary product retained in an alcoholic beverage that contributes significantly to the final characteristics of the beverage or the extract.

CONGENITAL: A disorder of the body that is present at birth due to genetic factors or acquired while the foetus is in utero. An example of a congenital disease is Down syndrome.

CONGESTIVE HEART FAILURE (HEART FAILURE): A condition arising when the heart's pumping mechanism is defective and cannot supply adequate blood to the tissues and other organs such as the brain, liver and kidneys. Its causes include hypertension (high blood pressure); failure of the right or left or both ventricles; defective heart valves; congenital heart defects (heart problems that develop before birth) such as narrowed valves; or various types of heart disease. Some of the signs and symptoms include shortness of breath; asthma due to the heart; fatigue and weakness; oedema or swelling in the legs, ankles and feet; rapid or irregular heartbeat; and enlargement of the heart.

CONGLOMERATE: 1. Gathered into a ball or round body; to collect into a round mass; concentrated; for example, as conglomerate rays of light. 2. In geology, composed of stones, pebbles or fragments of rocks, cemented together. An example is sedimentary rock containing small pebbles or sedimentary rock containing fragments of other rock larger than 2 mm in diameter, held together with other materials such as clay and formed along beaches, as glacial drift and in river deposits. 3. In botany, closely crowded together; forming a mass or dense cluster as occurs in conglomerate flowers. Conglomerate flowers grow on a branching peduncle or foot stalk, on short pedicles, closely compacted together without order; and are opposed to diffused. 4. A conglomerate gland is composed of many smaller glands, whose excretory ducts unite in a common one, as found in the liver, kidney, pancreas and parotids. Each little granulated portion furnishes a small tube, which unites with other similar ducts, to form the common excretory duct of the gland.

CONGO RED: A scarlet benzidine dye often used to stain fungal hyphae.

CONGRUENCE: 1. The relationship between two integers relative to another (the modulus of the congruence) when the difference between the given numbers is an exact multiple of the modulus, usually written $x \equiv y \pmod{m}$. For example, $8 \equiv 2 \pmod{3}$, since $8 - 2 = 6 = 2 \times 3$. It

is an equivalence relation fundamental to number theory. In general, a congruence equation seeks to solve $f(x) \equiv 0 \pmod{m}$, where f is an integral polynomial. 2. In geometry, the fact or relation of being congruent, an isometry. 3. In logic, any equivalence relation preserved by every operation in the given structure, so that if $x \equiv y$, then $f(x) \equiv f(y)$ for every operation f .

CONGRUENCE CLASS: A set of elements each of which is congruent to every other in the class; an equivalence class under a congruence relation.

CONGRUENT: 1. Having the same shape and size, as applied to two dimensional or solid figures. **Congruent figures** are two geometric figures that are identical in size and shape. In diagrams, the corresponding congruent sides may be marked with the same number of hash marks. The symbol of "congruent to" is $\cong$. 2. Of two matrices A and B , having the property that $B = PAP^t$, where P^t is the transpose of a non-singular matrix P . Every real symmetric matrix of rank, r , is congruent (over the reals) to a diagonal matrix with entries equal to ± 1 and all others 0.

CONGRUENT MODULO m : Of a pair of integers, differing by an exact multiple of m , related by congruence to the given modulus. If two numbers b and c have the property that their difference $b - c$ is integrally divisible by a number $\frac{m(b-c)}{c}$ is an integer, then b and c are said to be "congruent modulo m ." The number m is called the modulus and the statement " b is congruent to c (modulo m)" is written mathematically as $b \equiv c \pmod{m}$.

CONGRUENT MODULO H : Of a pair of two elements, x and y , of a group, G , relative to a subgroup, H , of G being related by a left or right congruence relation. If the elements are left congruent modulo H , it is written $x \equiv_l y \pmod{H}$, if x^{-1} is in the subgroup. If they are right congruent modulo H , it is written as $x \equiv_r y \pmod{H}$, if yx^{-1} is in the subgroup.

CONGRUENT TRANSITION: A transition in which the two-phase equilibrium of melting, vaporisation or allotropism of a compound involves phases of the same composition.

CONGRUENTIAL METHOD: One of a number of methods used by a number generator to produce pseudo-random numbers by taking congruences modulo a large number.

CONIC: 1. Relating to a cone; having the form of, resembling or pertaining to a cone. 2. Any expression that represents a conic section; any second degree equation, for example, $x^2 + bxy + cy^2 + dx + ey + f = 0$. The general conic in Cartesian coordinates can be represented by $(p^2 + q^2)[(x - \alpha)^2 + (y - \beta)^2] = e^2(px + qy + r)^2$, where e is the eccentricity, (α, β) the focus and $px + qy + r$ is the equation of the directrix of the conic. In the vertex form, it is represented by $y^2 = 2px - (1 - e^2)x^2$, where $2p$ is the parameter of the conic, i.e., the length of the Latus rectum, which in the ellipse and hyperbola equals b^2/a (b and a are the lengths of the semi-axes of the conic) and e is the numerical eccentricity, e/a . 3. Conic section is a two dimensional curve such as an ellipse and a hyperbola obtained by the intersection of a plane and a right circular cone. For a plane perpendicular to the axis of the cone, a circle is produced. If the plane is at any other angle at which the intersection is a closed curve, it produces an ellipse; if it is parallel to any line joining the vertex of the cone to a point on its base and intersects only a single nappe, a parabola is produced; if the plane is at any other angle and intersects both nappes, it produces a hyperbola. Lines and points are degenerate conics obtained when the intersecting plane includes the vertex of the cone. Conic sections can be conceived geometrically as the loci of points satisfying certain distance relations from a given point, the focus and a given line, the directrix; the eccentricity is the ratio of these distance, which is constant for a particular family of similar curves; these are described algebraically by the conic equations.

CONIC SECTION: One of a group of curves formed by the intersection of a plane and a right circular cone. For a plane perpendicular to the

axis of the cone, a circle is produced. If the plane is at any other angle at which the intersection is a closed curve, it produces an ellipse; if it is parallel to any line joining the vertex of the cone to a point on its base and intersects only a single nappe, a parabola is produced; if the plane is at any other angle and intersects both nappes, it produces a hyperbola. Lines and points are degenerate conics obtained when the intersecting plane includes the vertex of the cone. Conic sections can be conceived geometrically as the loci of points satisfying certain distance relations from a given point, the focus and a given line, the directrix; the eccentricity is the ratio of these distance, which is constant for a particular family of similar curves; these are described algebraically by the conic equations.

CONICS: The branch of mathematics (geometry and coordinate geometry) that is concerned with the properties of conic sections.

CONIDIUM (CONIDIOSPORE): Asexual non-motile spores of many fungi, produced at the tip of a specialised hypha called the conidiopore.

CONIFER: Evergreen gymnosperm trees with needle-like leaves belonging to the phylum Coniferophyta. The two orders of conifers are Pinales and Taxales. **Coniferous** refers to bearing a reproductive structure occurring in gymnosperms known as **cones**.

CONIFEROPHYTA: A phylum of cone-bearing gymnosperms trees and shrubs dating from the Carboniferous period. Most are substantial trees comprising the classes Pinopsida (subdivision Pinophytina), which includes most conifers; Ginkgopsida (subdivision Ginkgophytina) and Taxopsida (subdivision Taxophytina) which in turn include the surviving orders Coniferales and Taxales (yews) and sometimes Ginkgoales as well as extinct orders such as Cordaitales (of the Carboniferous and Permian. Some examples of surviving Coniferophyta are conifers, the pines, firs and spruces.

CONJECTURE: A statement that seems to be true, but has not yet been proved.

CONJOINED TWINS (SIAMESE TWINS): Identical twins born with some part of their bodies joined together as a result of incomplete division of one fertilised ovum.

CONJUGACY CLASS: The set $cl(a)$ of all elements in a group that are conjugate to a given element, a , in the group. Each element in a group belongs to exactly one class and the identity element ($I=1$) is always in its own class. The conjugacy classes partition the group and all elements conjugate to one another, have the same order. The order of the conjugacy class in a finite group is the index of the centraliser of one such element in the group.

CONJUGACY PROBLEM (TRANSFORMATION PROBLEM): The problem of finding an algorithm that decides whether the elements represented by two words x and y in G are conjugate elements of G . There exists a finitely presented semigroup for which this problem is insoluble. The problem is to determine whether there exists an element z of G such that $y = zxz^{-1}$.

CONJUGATE: 1. With respect to angles, two angles that sum up to four right angles or 360° . 2. Any set of numbers that satisfy the same irreducible polynomial. For example, the complex conjugate of $2 + 3i$ is $2 - 3i$; the complex conjugate of $4 - 5i$ is $4 + 5i$; and the complex conjugate of the complex number 2 is just 2. 3. Of two complex numbers, differing only in the sign of their imaginary parts; for example, $2 + 3i$ and $2 - 3i$ are conjugate. 4. Of two algebraic numbers, being roots of the same irreducible algebraic equation with rational coefficients. For example, $3 + \sqrt[3]{2}$ and $3 - \sqrt[3]{2}$ are conjugate roots of the equation $x^3 - 6x + 1 = 0$. 4. Related by an automorphism that leaves the base field fixed, so that one algebraic number is the image of the other under the automorphism. 5. Of two elements of a matrix, interchanged when the rows and columns are interchanged; for example, if a_{ij} is the element at the intersection of the i^{th} row and the j^{th} column, then a_{ji} is its conjugate entry. 6. Of two lines, such that one passes through the pole of the other. 7. Of two points, such that one lies on the polar of the other. 8. Of two potential functions, being

the real and imaginary parts of a given analytic function. The pair of functions is then called **conjugate harmonics**. 9. In chemistry, conjugate refers to (i) a conjugate compound formed by the joining of two or more chemical compounds; (ii) an acid and base that differ from each other by a proton. 10. To combine two substances in such a way that they can easily be separated again, especially in chemical reactions.

CONJUGATE ACID: A species formed when a Bronsted base gains a proton from an acid in an acid-base reaction. For example, in the reaction $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$ the base (H_2O) accepts a proton from the acid (HCl) to become the conjugate acid (H_3O^+).

CONJUGATE ACID-BASE PAIR (CONJUGATED ACID-BASE PAIR): The Brønsted acid BH^+ formed on protonation of a base B is called the conjugate acid of B and B is the conjugate base of BH^+ . In other words, it is a pair of chemical species that differ in structure by a proton. Examples are HCl/Cl^- and $\text{NH}_3/\text{NH}_4^+$. The conjugate acid always carries one unit of positive charge more than the base, but the absolute charges of the species are immaterial to the definition. For example, the Brønsted acid HCl and its conjugate base Cl^- constitute a conjugate acid-base pair. In the equilibrium system $\text{HX} + \text{H}_2\text{O} \rightleftharpoons \text{X}^- + \text{XH}^+$, the conjugate acid-base pairs are HX/X^- and $\text{H}_3\text{O}^+/\text{H}_2\text{O}$.

CONJUGATE AXIS: The axis of a hyperbola that does not pass through the foci.

CONJUGATE BASE: The species that remains after a Bronsted acid has given up a proton to a base in an acid-base reaction. For example, in the reaction $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$ the acid HCl gives up a proton to H_2O . The chloride anion (Cl^-) is the conjugate base of the acid.

CONJUGATE VARIABLE: 1. An adjoint variable of a differential equation. 2. In thermodynamics, conjugate variables are sets of intensive X and extensive x variables whose product Xdx has the dimensions of energy. The internal energy of a system is expressed in terms of pairs of conjugate variables such as temperature and entropy or pressure and volume.

CONJUGATE EXPONENTS: Any two positive numbers whose reciprocals sum up to unity. 1 and ∞ are a conjugate pair. If p and q meet the following conditions: $\frac{1}{p} + \frac{1}{q} = 1$, they are conjugate exponents.

CONJUGATE POINTS (FOCI): 1. Two points in the vicinity of a lens or mirror such that an object placed at one end will form an image at the other. 2. For a conic section, two points either of which lies on the line that passes through the points of contact of the two tangents drawn to the conic from the other.

CONJUGATE GRADIENT METHODS: Any of a class of quasi-Newton methods for minimising a differentiable function of n variables by sequentially generating conjugate directions.

CONJUGATE RULED SURFACE: A ruled surface whose generators are tangent to those of a given ruled surface.

CONJUGATE SOLUTIONS: A solution formed between two liquids in which each liquid is only partially soluble in the other. Such solutions are in equilibrium at a particular temperature. Examples are phenol in water and water in phenol.

CONJUGATE SURDS: A surd that can be used to express a fraction, which has a compound surd as its denominator with a rational denominator. The compound surds $(a + \sqrt{b})$ and $(a - \sqrt{b})$, are conjugate surds.

CONJUGATE-LINEAR FUNCTIONAL: A complex functional on a complex vector space that is additive and conjugate homogeneous; equivalently, f is the conjugate of a complex linear functional g : $f(x) = \overline{g(x)}$.

CONJUGATED BONDS: Describing multiple bonds in a compound in which single bonds and multiple bonds alternate. For example, in the benzene rings and other aromatic hydrocarbons, the double bonds are conjugated.

CONJUGATED DOUBLE BONDS: Double bonds that are separated from each other by one single bond $-\text{C}=\text{C}-\text{C}=\text{C}-$.

CONJUGATED PROTEIN: A protein that contains other components besides polypeptide chains. Conjugated proteins are classified according to the nature of the nonpeptide portion of the molecule. Glycoproteins are found in plant and bacterial cell walls, while cell and organelle membranes contain lipoproteins. Many enzymes contain metal ions and some dehydrogenases are flavoproteins. Ribosomes and viruses are nucleoproteins.

CONJUGATED SYSTEM: A system of bonding p-orbitals with delocalised electrons in compounds with alternating single and multiple bonds. Lone pairs, radicals or carbenium ions can be part of the system which can be cyclic, linear or both. The compound may be cyclic, acyclic, linear or mixed. It is the overlap of one p-orbital with another across an intervening sigma bond; in larger atoms d-orbitals can be involved.

CONJUGATION: 1. In biology, conjugation refers to any one of the following: (i) the reproductive process in ciliate protozoans in which two organisms of different mating types exchange nuclear material through a temporary area of fusion. (ii) The temporary union of two bacteria, such as in *Escherichia* and related groups, in which genetic material is transferred by migration of a plasmid, either solitary or as part of a chromosome, from one bacterium, the donor, to the other, the recipient. It sometimes also includes the transfer of resistance to antibiotics. In *spirogyra*, for example, two filaments come close and a conjugation tube is formed in between them, through which the cytoplasm is transferred, forming a new *spirogyra* cell. (iii) The temporary union or permanent fusion as a form of sexual reproduction in certain green algae and fungi, with the male gametes of one organism uniting with female gametes of the other. *Spirogyra* can reproduce both sexually and asexually. Within a given population of organisms, the various forms that may engage in conjugation are known as mating types. The forms may or may not resemble each other in size, shape or motility. They differ in some physiological or genetic characteristic and are distinguished by biologists through the use of symbols such as + and - or by letters of the alphabet if more than two forms occur. **Conjugants** are partners that participate in conjugation. 2. In chemistry, it is a system of delocalised orbitals consisting of alternating single bonds and multiple bonds.

CONJUGATIVE PLASMID: Plasmid that can be mobilised or transferred from one bacterial cell to another through conjugation, or a plasmid that can encode all functions required to bring about its conjugation.

CONJUNCTION: 1. In astronomy, it is the apparent meeting or passing of two or more celestial bodies, as observed from Earth. The Moon is in conjunction with the Sun at the phase of New Moon, when it moves between the Earth and Sun and the side turned towards the Earth is dark. Inferior planets (Venus and Mercury) with orbits smaller than the Earth's have two kinds of conjunctions with the Sun. An **inferior conjunction** occurs when the planet passes approximately between Earth and Sun. If it passes exactly between them, moving across the Sun's face as seen from Earth, it is said to be in **transit**. A **superior conjunction** occurs when the Earth and the other planet are on opposite sides of the Sun, but all three bodies are again nearly in a straight line. Superior planets, with orbits larger than the Earth's, can have only superior conjunctions with the Sun. Conjunctions of planets with one another are considered of great importance in astrology. 2. An "AND" statement of the form " p and q ". It is true only if both p and q are true. The symbol for conjunction is " $\wedge$ ", but can usually be written $P \& Q$, $P \cdot Q$, $P.Q$, where P and Q are the component sentences, called the **conjuncts**. **Conjunct** defines either of the component statements or propositions of a conjunction. When two simple sentences, " p and q ", are joined in a conjunction statement, the conjunction is expressed symbolically as " p and q ". The truth table for conjunction is shown in the diagram.

CONJUNCTION ELIMINATION: In propositional logic, it is the elimination rule of sentential calculus or any of its extensions that

permits the valid immediate inference of either conjunct from a given conjunction. It infers that if the conjunction A and B is true, then both A and B must be true.

CONJUNCTION INTRODUCTION: In propositional logic, it is the introduction rule of sentential calculus or any of its extensions, that permits a valid rule of inference, of two statements as premises, of their conjunction as conclusion. In other words, it allows the inference of the conjunction of any two propositions that already appear elsewhere in a proof. It infers that if the proposition P is true and proposition Q is true, then the logical conjunction of the two propositions P and Q is true.

CONJUNCTIVA: The mucous membrane consisting of cells and underlying basement membrane covering the front of the eyeball of a vertebrate and the inner surface of the eyelid.

CONJUNCTIVE: Of two matrices A and B , such that $B = PAP^*$ with P^* the Hermitian transpose or adjoint of a non-singular matrix P . Every matrix of rank r is conjunctive (over the complex field) to a diagonal matrix with r entries equal to $\pm I$ and all others zero.

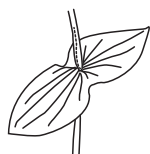
CONJUNCTIVE NAME: A name for assemblies of functionalised acyclic parent hydrides and cyclic systems implying the loss of an appropriate number of hydrogen atoms from each.

CONJUNCTIVE NORMAL FORM (CNF): The form to which every statement of sentential calculus can be reduced, consisting of a conjunction of disjunctions, each of the disjuncts of which is either an atomic formula or the negation of one. In other words, a statement is in conjunctive normal form if it is a conjunction consisting of one or more conjuncts, each of which is a disjunction (OR) of one or more literals (statement letters and negations of statement letters). Because of the associativity of both disjunction and conjunction, no brackets are necessary, as it is understood conjunction has wider scope. An example, $((P \rightarrow Q) \& P) \rightarrow Q$ reduces to the cnf.

CONJUNCTIVITIS: Inflammation of the conjunctiva, which is continually exposed to microorganisms and environmental agents and susceptible to infections or allergic reactions. It may be caused by a viral infection, such as a cold, acute respiratory infection or diseases such as measles, herpes simplex or herpes zoster. Symptoms include redness, swelling and a watery or pus-filled discharge. It may be caused by infection, allergy or other irritant.

CONNARACEAE: A small family of dicotyledonous flowering plants in the order Oxalidales. They are widely distributed in tropical regions of the world. These plants are popular in traditional medicine in Africa. One species, *Cnestis ferruginea*, for example, is used for the treatment of gonorrhoea, diabetes mellitus, constipation, migraine, miscarriages or ovarian troubles, and for teeth cleaning. Its scarlet fruits are ornamental.

CONNATE: Describing similar organs that are joined together, such as petals fused to form a tube. The term is often used of paired leaves at a node, the bases of which have become fused to completely encircle the stem. **Connate-perfoliate** is having the bases of opposite leaves fused around the stem, as shown in the diagram.



Connate-perfoliate leaf

CONNECTED: 1. Of a relation, such that either it or its converse holds between any two members of the domain. 2. Of an ordering, constituting a total ordering. 3. Of a set of real numbers, having the property that if a and b are in the set and c lies between them, then c is also in the set, constituting a continuum. 4. Of a topological set, not able to be partitioned into two non-empty open subsets, such that each subset has no points in common with the set closure of the other. For example, the rationals are not connected, although the reals are. If X is a topological space, a connected set in X is a set A subset $= X$ that cannot be partitioned into two non-empty subsets which are open in the relative topology induced on the set A . The set is called path-, arc-, pathwise or arcwise connected if each is joined by a path in the set; a connected set is

not necessarily connected. 5. Of a graph, having the property that there is a path between every pair of vertices along a sequence of edges in the graph.

CONNECTED GRAPH: A graph in which a path exists between all pairs of vertices. If the graph is also a directed graph and there exists a path from each vertex to every other vertex, then it is a strongly connected graph. If a connected graph is such that exactly one edge connects each pair of vertices, then it is said to be a complete graph.

CONNECTION MATRIX (ADJACENCY MATRIX): A matrix with rows and columns corresponding to graph vertices, with a 1 or 0 in position of which the i,j^{th} entry is 1 if vertex i is adjacent to vertex j and 0 if not. For a simple graph with no self-loops, the adjacency matrix must have 0s on the diagonal. For an undirected graph, the adjacency matrix is symmetric.

CONNECTION TABLE: A way of representing a molecule as a table showing atoms, their coordinates and the links between them. They are useful ways of storing molecular data.

CONNECTIONISM: A computational approach to modelling the brain that relies on the interconnection of many simple units to produce complex behaviour.

CONNECTIVE: 1. In botany, the parenchymatous tissue that joins the pollen sacs in the anther. It is a continuation of the filament and contains the vascular strand supplying each lobe. 2. In mathematics, (logic), a function or the symbol for it in a formal language, forming compound sentences from simple ones, corresponding to English conjunctions such as "or" and "not" that takes one or more truth values as input and returns a single truth value as output.

CONNECTIVE TISSUE: Supporting and packaging tissue in vertebrates, made up of different kinds of cells and extracellular fibres and ground substance, which varies in consistency from thin gel to rigid structured. Examples are skin, bones, tendons, cartilage and adipose tissue (fat). There are also small amounts of connective tissue in organs such as the brain and liver.

CONNECTIVITY: 1. In a chemical context, the information content of a line formula, but omitting any indication of bond multiplicity. 2. Number of covalent bonds emanating from a constitutional unit of an oligomer molecule or a macromolecule.

CONNEXON: A hexagonal array of protein subunits, typically 10nm in diameter and arranged as a short hollow cylinder that penetrates the plasma membrane of certain cells forming part of a gap junction.

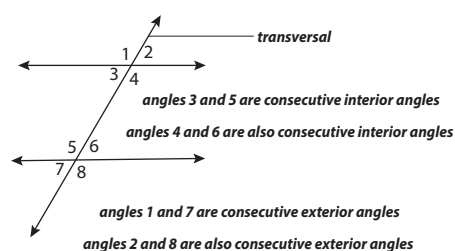
CONNIVENT: Converging and touching without being fused as, in stamens or an insect's wing.

CONOID: A geometrical surface or solid formed by rotating a conic section such as a parabola, ellipse or hyperbola about one axis.

CONOSCOPE: A polarising microscope designed to examine the transmission property of a sample plate at various directions of incidence simultaneously.

CONSECUTIVE: Following one after another in an uninterrupted order. For example, A, B, C, and D are four consecutive letters of the alphabet; 1, 2, 3, 4, and 5 are five consecutive whole numbers.

CONSECUTIVE ANGLES: In geometry, they are angles that are formed when two or more lines are crossed by another line, called the **transversal** or transverse line in the same plane at different points. The pairs of angles on one side of the transversal but inside the two lines are called **consecutive interior angles**. **Exterior consecutive angles** are pairs of angles on one side of the transversal but on the outside of both lines. In a right-angled triangle, the **consecutive sides**, also called **adjacent sides** are any two sides that have a common end point, or meet at a polygon vertex. In



Consecutive Angles

a polygon, they are two sides having a common vertex; and in a polyhedron, two sides with a common edge.

CONSENSUAL: Relating to movement that takes place through the action of the nervous system independently of the will.

CONSENSUS SEQUENCE: A sequence of nucleotides found in comparable regions of DNA or RNA. For example, the promoter region of different genes, in which certain bases occur with a frequency significantly greater than that expected by chance.

CONSEQUENCE: In logic, it is a valid deducibility, which is the relation between the conclusion and the premises of a valid argument.

CONSEQUENT (SUCCEDENT; CONCLUSION OF A CONDITIONAL STATEMENT): Part of a conditional statement that is true if the other part, called the antecedent is true. The consequent is the "then part of a conditional statement. For example, in the statement "If he likes kenkey, then he likes fried fish", the consequent is "then he likes fried fish".

CONSERVATION: 1. Action taken to protect and preserve the natural world usually from pollution, over exploitation and other harmful features of human activity or the sensible use of the Earth's natural reserves in order to avoid excessive degradation and impoverishment of the environment. Conservation methods may include the setting aside of large parcels of existing habitat, the elimination of foreign species and the restoration of areas previously altered by human activity. 2. **Soil conservation** is the protection of the soil against physical loss by erosion or against chemical deterioration such as excessive loss of fertility by either natural or artificial means so as to maintain soil fertility and productivity.

CONSERVATION LAW: A law stating that the total magnitude of a certain physical property of a system such as mass, energy or charge, remain unchanged even though there may be exchanges of that property between components of the system. Conservation laws make it possible to predict the macroscopic behaviour of a system without having to consider the microscopic details of a physical process or chemical reaction.

CONSERVATION OF ANGULAR MOMENTUM: The law of conservation of angular momentum states that when no external torque acts on an object or a closed system of objects, no change of angular momentum can occur. Hence, the angular momentum before an event involving only internal torques or no torques is equal to the angular momentum after the event.

CONSERVATION OF BARYON NUMBER: The principle that the total baryon number remains constant in all processes involving the interaction of elementary particles. A fundamental conservation law of particle physics that states that the total baryon number of a closed system must remain constant. Baryon number conservation would be violated if a proton could decay.

CONSERVATION OF CHARGE: The principle that the total charge of any isolated system is constant and independent of changes that take place within the system. In all known processes in nature, the total amount of a charge in a system has been observed to remain constant.

CONSERVATION OF ENERGY: The principle which states that in a chemical reaction, the total amount of energy in the system remains unchanged or energy cannot be created or destroyed, although it can be changed from one form to another.

CONSERVATION OF LINEAR MOMENTUM (LAW OF CONSERVATION OF LINEAR MOMENTUM): It states that if no external force such as friction acts on a system of colliding objects, the total momentum of the object in a given direction before collision equals total momentum after collision in the same direction.

CONSERVATION OF MASS/MATTER (LAW OF CONSERVATION OF MASS; PRINCIPLE OF MASS/MATTER CONSERVATION):

The principle which states that in a chemical reaction the sum of all the masses of the substances involved in the reaction or reactants is equal to the sum of all the masses of the substances produced by the reaction or products and that matter cannot be created nor destroyed in a chemical reaction.

CONSERVATION OF MASS-ENERGY: It states that mass and energy are interchangeable and in any interaction mass-energy is conserved.

CONSERVATION OF MECHANICAL ENERGY (LAW OF CONSERVATION OF MECHANICAL ENERGY): The total amount of mechanical energy (i.e., the sum of the potential plus kinetic energies), in a closed system in the absence of dissipative forces, for example, friction, air resistance, etc., remains constant.

CONSERVATION OF MOMENTUM: A fundamental law of motion, which is equivalent to Newton's laws of motion. It states that the vector sum of the momenta (mv) of all the objects of a system cannot be changed by interactions within the system; momentum is neither created nor destroyed, but only changed through the action of forces as described by Newton's laws of motion.

CONSERVATION OF NUMBER: The number of objects in a set remains the same no matter how they are arranged or counted. A person is able to understand that the number of objects remains the same even when rearranged. It is the recognition by a young child that quantity does not change with physical rearrangement.

CONSERVATION OF ORBITAL SYMMETRY: The orbital symmetry control of concerted reactions, which requires transformation of the molecular orbitals of reactants into those of products to proceed continuously by following a reaction path along which the symmetry of these orbitals remains unchanged. Reactions which adhere to this requirement are classified as symmetry-allowed reactions and those which do not as symmetry forbidden reactions.

CONSERVATION OF PARITY: The principle that the parity of the total wave function describing a system of elementary particles is conserved. It is not conserved in weak interactions.

CONSERVATION OF SHAPE: The shape of an object remains the same no matter how an object is transformed, whether translated, reflected, or rotated.

CONSERVATION OF SYMMETRY: A theorem of physics, stating that for every case of invariance, there exists a unique conservation law and for every conservation law, there exists a unique invariance.

CONSERVATION TILLAGE: Techniques of soil preparation, planting and cultivation with limited mechanical operations without disturbing the soil through tillage that leave the maximum amount of plant residue on the surface. This increases the amount of water in the soil, decreases erosion as well as increase the amount and variety of life in and on the soil but may require increased herbicide usage.

CONSERVATIVE: In mathematics, of a force, able to be described by a conservative vector field, such that the work done when the point of action moves from A to B does not depend on the path taken.

CONSERVATIVE EXTENSION (EXTENSION): A formal theory whose primitive terms, formation rules and axioms include those of a given theory and that contains the given theory in the sense that everything true in the given theory is also true in the extended theory. For example, Zermelo-Frankel set theory is a conservative extension of Peano arithmetic and first-order predicate calculus of sentential calculus.

CONSERVATIVE FIELD: A field of force in which the work done in moving a body from one point to another is independent of the path taken. An example of such field is gravitational force field.

CONSERVATIVE FORCE: Any force whose work done is independent of the path taken. Examples are gravitational force, elastic/spring force and electrostatic force.

CONSERVATIVE PLATE MARGIN: In plate tectonics, it is the boundary between two plates that slide past each other in opposite directions or at different speeds and become attached building up pressure. When the pressure is released it creates a massive earthquake. No volcanic eruptions occur along conservative plate margins because the crust is neither being created nor destroyed. The San Andreas Fault in California lies above the North American and Pacific plates and is an example of a conservative plate margin.

CONSERVATIVE REPLICATION: A hypothesis suggesting that DNA replication occurs by one DNA molecule initiating the synthesis of a new molecule while remaining intact.

CONSERVATIVE SUMMABILITY METHOD: A method of summation that assigns a finite limit to every convergent sequence, but that may assign a number different from the limit.

CONSERVATIVE SYSTEM: A system in which work done by a force is independent of path, equal to the difference between the final and initial values of an energy function and is completely reversible. The two most notable conservative systems are gravitational and electric fields. With gravity for example, the gravitational potential energy acquired or lost by a mass depends only on the difference between heights or between distances from the origin of the force and not on the path taken to get from one state to the other.

CONSERVATIVE VECTOR FIELD: A vector field whose curl is zero. Its coordinates arise as the gradient of a potential. An example is gravity.

CONSERVED SEQUENCE: Any sequence of bases in comparable segment of nucleotides that tend to show similarity greater than that due to chance alone.

CONSIGNMENT: A quantity of material transferred on one occasion and covered by a single set of shipping documents. It may consist of one or more lots or portions of lots. The presence of different lots in a consignment is important from the point of view of the sampling plan and the interpretation of the results of analysis. The term 'population' is used as the general term for the quantity of parent material being sampled when it is immaterial if the parent body is a consignment, lot, batch, entity, etc.

CONSILAR: Being entirely similar to one another.

CONSISTENCY THEOREM: In logic, the result that open theory is consistent exactly when no disjunction of negations of instances of its non-logical axioms is a quasi-tautology.

CONSISTENT: 1. In mathematics, consistent equations are two or more equations that can be solved simultaneously by using one set of values for the variables or infinite number of solutions. Examples are $x + y + z = 3$, $x + y + 2z = 4$ as well as $y = -3x - 1$, $3x + y = -1$. If the graphs of equations intersect, then there is one solution that is true for both equations, or If the graphs of equations are the same, then there are an infinite number of solutions that are true for both equations, and are therefore, consistent. 2. In logic, it is also known as sound and defines a set of statements, capable of all being true at the same time in the same circumstances or under the same interpretation. 3. In statistics, of a sequence such that as the sample size increases the probability of accepting a fixed alternative hypothesis when it is true tends to unity. 4. Of a congruence or an equation, possessing a solution. Alternatively, able to be simultaneously satisfied, possessing a solution, as in a system of equations. 5. Of a formal system, not enabling the deduction of a contradiction from the axioms or generally, not having an atomic sentence as a theorem.

CONSISTENT ESTIMATOR: An estimator that tends to converge toward the true value of the parameter it is trying to estimate as the sample size becomes larger.

CONSISTENT HISTORIES: An interpretation of quantum mechanics that makes use of the concept of decoherence to explain how the classical world emerges from quantum mechanics. The consistent histories interpretation avoids the problem of observers and has greatly clarified our understanding of the problem of measurement in quantum mechanics.

CONSOCIATION: A climax plant community that is dominated by one particular species and therefore, named after one dominant species, for example, pine forest, if the community is dominated by pine. Consociations can be grouped together to form an association.

CONSOLE: The combination of a monitor and keyboard. It is a rudimentary interface in which the monitor provides the output and the keyboard is used for input. While any computer with a monitor and keyboard may be considered a console, the term most often refers to a system used to control one or more servers. Modern consoles often provide a graphical interface that can be used to control machines on a local network (LAN) or provide remote access to remote systems. These consoles typically include a mouse for navigating graphical interfaces.

CONSOLUTE TEMPERATURE (UPPER CONSOLUTE TEMPERATURE; UPPER CRITICAL SOLUTION TEMPERATURE): The upper temperature at which two partially miscible liquids become fully miscible.

CONSORTIUM (MICROBIAL CONSORTIUM): Two or more organisms symbiotically living symbiotically together, where each organism benefits from the other, for example, by performing functions that may not be possible to carry out individually.

CONSPECIFIC: Describes animals and plants that belong to the same species.

CONSTANT: 1. A fixed quantity or one that does not change in value in an expression. For example, in the expression $y^2 = 5x - 3$, the numbers -3 and 5 are constants. 2. A quantity that is regarded as fixed and unchanging for the purposes of a particular calculation. For example, $y = mx + c$, the general equation of a line, m and c are constants, i.e., the terms independent of x are termed as constants; while x and y are variables. 3. A specific invariant quantity whose value is determined a priori, such as π or e , the base of the natural logarithms. 4. An unspecified numerical value, for example, if a is proportional to b then a/b is a constant. 5. In physics, it is a number that appears in equations and formulae that does not vary or change. Examples are Planck's constant and the speed of light in a vacuum.

CONSTANT DEVIATION PRISM: A prism which has the property that the minimum deviation is always at the same angle. In other words, it is a prism whose deviation is constant and does not depend on the index of refraction or wavelength. Examples include the Pellin-Broca prism and Abbe prism.

CONSTANT MATRIX: A matrix whose entries are constant.

CONSTANT OF INTEGRATION: If $g(x)$ is an antiderivative of the function $h(x)$, which is continuous, then the antiderivative of $h(x)$ differs from $g(x)$ by a constant k ; which can be written as $\int h(x)dx = g(x) + k$, where k is an arbitrary constant, which is the constant of integration.

CONSTANT-BOILING MIXTURE: A solution of two or more substances that retains the same composition in the vapour state as in the liquid state when distilled or partially evaporated under a certain pressure. The composition of the liquid phase at the boiling point is identical to that of the vapour in equilibrium with it and such mixtures or azeotropes form constant-boiling solutions.

CONSTANTAN: An alloy of 55% copper and 45% nickel having electrical resistance which is constant over a wide range of temperatures. Another alloy that shows similar low temperature coefficient is manganin, containing copper, manganese and nickel.

CONSTELLATION: A group of stars that seem to make a pattern as seen in the night sky. It is one of the 88 internationally defined areas into which the sky is divided for the purposes of identifying and naming celestial objects. Constellations are usually named after mythological characters, people, animals and objects. Examples are Andromeda, Antlia, Aries, Auriga, Bootes, Cancer, Indus, Libra and Pisces.

CONSTIPATION: A condition in which a person or animal has difficulty in eliminating solid waste from the body; usually accompanied by the passage of hard, dry stools that may result in a tear in the

lining of the lower rectum or painfully swollen and itchy veins in the lower rectum or anus, known as haemorrhoids.

CONSTITUENT: Chemical species present in a system; often called a component, although the term component has a more restricted meaning in physical chemistry.

CONSTITUTION: The description of the identity and connectivity (and corresponding bond multiplicities) of the atoms in a molecular entity (omitting any distinction arising from their spatial arrangement).

CONSTITUTIONAL HETEROGENEITY: The variation in constitution from molecule to molecule in polymers with molecules uniform with respect to elemental composition. An example is a polymer composed of linear and branched molecules; another example is a statistical copolymer comprising two isomeric constitutional units.

CONSTITUTIONAL ISOMERISM (STRUCTURAL ISOMERISM):

Isomerism of chemical compounds that have the same molecular formula but different structural formulae, physical and chemical properties. For example, propanal (C_3H_5CHO) and propanone (acetone), CH_3COCH_3 , have the same molecular formula, C_3H_6O , but different structures and are, therefore, structural isomers.

CONSTITUTIONAL UNIT: An atom or group of atoms (with pendant atoms or groups, if any), which comprises a part of the essential structure of a macromolecule, an oligomer molecule, a block or a chain.

CONSTITUTIVE EQUATION: An equation describing the form of the stress tensor and other quantities for a given body. For example, the constitutive equation for the stress in an incompressible Newtonian

viscous fluid is $\sigma = -p(x,t)I + 2\eta \left[\Sigma - \frac{1}{3}(tr\Sigma)I \right]$, where p is a scalar

function of the density (pressure), η , is a constant (the viscosity) and Σ is the Eulerian strain rate.

CONSTITUTIVE ENZYME: An enzyme whose concentration in a cell is constant and is not influenced by substrate concentration. A constitutive enzyme is part of the permanent and basic machinery of a cell. The enzymes of glycolysis are examples of constitutive enzymes in nearly all organisms.

CONSTITUTIVE EQUATIONS: A relation between two physical quantities often described by tensors that is specific to a material or substance and approximates the response of that material to external forces. An example is the equation $D = \epsilon E$ and $B = \mu H$, where D is the electric displacement, ϵ is the permittivity of the medium, E is the electric field intensity, B is the magnetic flux density, μ is the permeability of the medium and H is the magnetic field strength.

CONSTITUTIVE PROMOTER: An unregulated promoter that allows for continual transcription of its associated gene.

CONSTRAINED OPTIMISATION/OPTIMIZATION: An optimisation that is subject to constraints, as, for example, in linear programming. It is a condition of an optimisation problem that the solution must satisfy.

CONSTRAINT: In mathematics and programming, a Boolean relation, often expressed as an equality, inequality relation and integer constraints between the values of one or more mathematical variables. For example, $x > 3$ is a constraint on x . Constraint satisfaction attempts to assign values to variables so that all constraints are true.

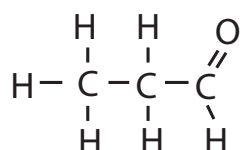


Fig I: Structure of propanal

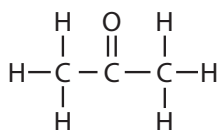


Fig II: Structure of propanone (acetone)

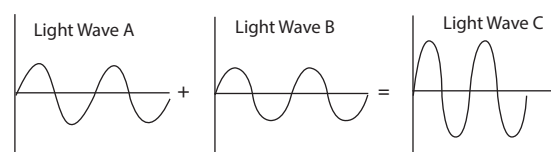
CONSTRAINT QUALIFICATION: A regularity condition imposed in order to guarantee that some necessary condition obtains upon as with Lagrange multipliers. Two of the most common of such conditions are to suppose linear independence of the derivatives of equality constraints and to suppose Slater's conditions for convex inequality constraints.

CONSTRINGENCE (ABBE CONSTANT; ABBE NUMBER; ABBE-NUMBER; NU-VALUE; RELATIVE DISPERSION; V-VALUE): The reciprocal of the dispersive power of a substance, which inversely quantifies the amount of dispersion in an optical medium. It is given by a function of the refractive index of a material at the Fraunhofer spectral lines f (486.1nm), d (587.6nm) and c (656.3nm) wavelengths. Its symbol is V and expressed as $V = n_d - 1/n_f - n_c$, where n_d , n_f and n_c are the refractive indices for the Fraunhofer spectral lines d (587.6 nm), f (486.1 nm) and c (656.3 nm). A material with a high Abbe number produces less chromatic aberration than one with a low number, which means that different wavelengths will have nearly the same index of refraction in that medium, yielding less separation between wavelengths of light.

CONSTRUCTIBLE: 1. Of a regular polygon, able to be constructed in finitely many steps with a straightedge and compass. This corresponds to determining the constructible numbers or quantities whose solution by radicals involves only quadratic surds. 2. **Constructible number** is a number that can be represented by a finite number of additions, subtractions, multiplications, divisions and finite square root extractions of integers. Such numbers correspond to line segments which can be constructed using only a straightedge and compass.

CONSTRUCTION: 1. In mathematics, drawing of a geometric shape, line or angle accurately using only a ruler or straight edge and compass. Certain instruments such as a protractor are permitted and no measurement of lengths or angles is allowed. It is used in solving a geometrical problem or to prove a theorem. 2. In architecture and civil engineering, it is a process that consists of the building or assembling of infrastructure. It includes all the on-site work such as land clearance, excavation, erection and the assembly and installation of components and equipment.

CONSTRUCTIVE INTERFERENCE (ADDITIVE INTERFERENCE): A phenomenon, which occurs when two or more waves, which are in phase overlap and their individual displacements add up to give a displacement that is greater than any of the individual displacements.



Constructive interference waves

CONSTRUCTION TECHNOLOGY: The study of the methods and equipment used to build structures. The field includes the study of engineering and construction, as well as business and construction management.

CONSTRUCTIVE PROCESSES: Processes that build up landforms on Earth's surface such as crust formation, volcanoes, and sediment deposition. Some of these processes take millions of years to occur, called **slow constructive process**. Examples are the formation of rocks such as sedimentary and metamorphic rock; creation of mountains by collision of tectonic plates, etc. Some others occur quite quickly, called **quick constructive processes**. These occur

through volcano and take seconds instead of millions of years to create some landmarks. For example, a volcano can erupt violently sending lava and ash shooting out within seconds. When that lava cools, it hardens into a new rock.

CONSTRUCTIVISM: The philosophical doctrine that mathematical entities do not exist independently of our construction of them, implying that they have no independent existence.

CONSUMABLE: Able or intended to be discarded after use, rather than being reused.

CONSUMERS: Living things that cannot make their own food and have to feed on other animals or organisms below them in the food chain. Primary consumers are herbivores that eat plants, for example sheep, cattle and sheep. Secondary consumers are carnivores that eat the flesh of other animals, for example, lions and dogs.

CONSUMPTION: 1. Using up of food, energy, resources, etc. 2. Any condition that causes the progressive wasting of the tissues, especially tuberculosis of the lungs.

CONTACT: 1. In medical science, an animal or person seen as a possible carrier of an infectious disease. 2. Describing an insecticide or herbicide that kills an insect or plant on contact rather than relying on ingestion or absorption. Examples of contact insecticides are DDT and pyrethrin. Contact herbicides include paraquat and diquat.

CONTACT CORROSION: The enhancement of the corrosion rate of the material with the more negative corrosion cell resulting from the contact between different electron-conducting phases.

CONTACT FORCE: In continuum mechanics, it is the force experienced by the points of a body due to their contact with other points of the body or with external boundaries, such as by the application of an external pressure to its boundary or if elastic, by its need to return to its original position. Formally, the effect of the contact forces on a sub-body is given by the integral deformation: $\int t(x, \partial\mathbb{R}_i) da$ over the surface, $\partial\mathbb{R}_i$, of its current configuration, where $t(x, \partial\mathbb{R}_i)$ is the contact force density (load, traction or stress vector on the surface).

CONTACT HERBICIDE: A herbicide that will not move or translocate within the plant, and need to come in direct contact with a plant. Examples of such herbicides are paraquat, oxyfluorfen, diquat and bromoxynil. Its effectiveness depends on many factors such as weather conditions and sunlight.

CONTACT INSECTICIDE: Any insecticide that kills its target insects by being absorbed through the cuticle or by blocking the spiracles rather than by being ingested. Examples are horticultural oil and insecticidal soap. They must be sprayed directly on the insects to be effective; any insects that evade the spray survive.

CONTACT METAMORPHISM: Recrystallisation of rocks surrounding an igneous intrusion in response to the heat supplied by the intrusion. There is no significant increase in pressure and the affected rocks do not melt in the process.

CONTACT POTENTIAL DIFFERENCE (CONTACT ELECTROMOTIVE FORCE; CONTACT POTENTIAL; VOLTA EFFECT): The potential difference that occurs between two electrically connected metals or between the base regions of two semiconductors.

CONTACT PROCESS: A process for making sulfuric acid from sulfur dioxide (SO_2) in the high concentrations needed for industrial processes. It is made by burning sulfur or by roasting sulfide ores. The SO_2 is oxidised to SO_3 by air using vanadium pentoxide (V_2O_5) as catalyst. The SO_3 is then dissolved in dilute sulfuric acid to form oleum $\text{H}_2\text{S}_2\text{O}_7$, which is then diluted to the required concentrated acid.

CONTACT TORQUE: In continuum mechanics, it is the torque experienced by the points of a body due to their contact with other points of the body or with external boundaries, for example, due to the internal magnetic interactions of its material points. Formally, the effect of the contact torque on a sub-body, $\mathbb{R}$ is given by the

sum of the integrals: $\int x \cdot t(x, \partial\mathbb{R}_i) da + \int c(x, \partial\mathbb{R}_i) da$ over the surface, $\partial\mathbb{R}_i$, where $t(x, \partial\mathbb{R}_i)$ and $c(x, \partial\mathbb{R}_i)$ are respectively the contact force density and the contact torque density on the surface and x is the position vector of a point of $\mathbb{R}$.

CONTACT TRANSFORMATION: A transformation of curves in the plane under which tangent curves are taken into tangent curves. A contact transformation of surfaces in space is defined similarly.

CONTAGIOUS (INFECTIOUS): Having a disease that can be spread to others by contact or spreading easily from one person to another.

CONTAGIOUS ABORTION: An infectious or contagious disease of farm animals such as brucellosis that is characterised by spontaneous abortion.

CONTAIN: 1. In mathematics, capable of being divided by a number exactly. 2. In geometry, to form the boundary of something. 3. To have as subset; a set strictly contains its proper subsets. 4. To keep something under control, within limit or hold back.

CONTAINER GROWN: Plants that are grown in pots or other types of container and are shipped in the container.

CONTAINMENT: 1. The prevention of the escape of radioactive materials from a nuclear reactor or the process of preventing the plasma in a thermonuclear reactor from touching the walls of the vessel by means of magnetic fields. 2. A structure or system designed to prevent the accidental release of radioactive materials from a reactor.

CONTAMINATE: 1. To add material that will change the purity or usefulness of a substance. 2. To make an item or object impure by adding dangerous disease-carrying substances.

CONTAMINATED LAND: Any land in such a condition, by reason of substances in, on or under the land, that is causing significant harm or there is significant possibility of such harm being caused to humans and animals. When dealing with land contamination, there are three essential elements which link together to result in a risk: a contaminant, a receptor and a pathway. These three elements can exist independently but may result in a risk when they link together. The link between a source, pathway and receptor is described as a pollutant linkage.

CONTAMINATION: The presence of a minor and unwanted constituent in another material, metal, chemical or mixture, often at the trace level.

CONTEXTUAL MENU: A pop-up menu that appears upon right-clicking on a certain area of the screen. Most modern operating systems and applications include contextual menus as part of the interface design. Contextual menus provide a choice of options that depend on where the cursor is when the right mouse button clicked. For example, by right-clicking on the computer's desktop, the menu that appears will have desktop-specific commands. These may include options to change the desktop background, create a new folder, clean up the icons or view the desktop properties. If the desktop window is right-clicked, options to create a new folder, change the view options or view the folder properties may appear.

CONTIG MAP (CONTIGUOUS MAP): A chromosome map showing the locations of those regions of a chromosome where contiguous DNA segments overlap. Contig maps are important because they provide the ability to study a complete and often large segment of the genome by examining a series of overlapping clones which then provide an unbroken succession of information about that region.

CONTINGENCY: 1. A statement or proposition that may be either false or true, its truth or falsity depending on the actual state of the world. A contingent statement or proposition is neither necessarily true nor necessarily false; it is true under certain conditions and false under others. It can also be defined as a statement or proposition, whose truth-table contains both true and false. 2. The idea that evolution consists of many branching points and that an organism's future evolution depends on previous branching points. This theory

would predict that evolution is unlikely to produce similar species on different planets. It stands in contradistinction to convergent evolution.

CONTINGENCY TABLE: In statistics, a table consisting of at least two rows and two columns showing the frequency of occurrence of events in each of a number of samples.

CONTIGUOUS: In botany, it refers to touching but not adnate, connate, adherent or coherent.

CONTINENT: Any of the seven large continuous land masses of the Earth as distinct from the oceans. They are Asia, Africa, North America, South America, Europe, Australia and Antarctica. More than two-thirds of the world's continental land area lies north of the equator.

CONTINENTAL CLIMATE: Climate that is situated inland or in the interior of a landmass in the interior of a continent. It is found mostly in middle latitudes, 40° to 55° in the Northern Hemisphere. It has large temperature differences between seasons: hot summers, cold winters, and little rainfall.

CONTINENTAL CRUST: A section of the Earth's crust composed of a layer of granitic, sedimentary and metamorphic rocks that makes up the Earth's landmasses and the continental shelves. It is between 25 to 70 km thick.

CONTINENTAL DRIFT: The theory that the Earth's continents once formed a single mass and have since moved relative to each other over the course of geologic time. The idea was first proposed by A. Wegener in 1912 but was discarded by the geophysicists of the time. However, evidence from biogeographic and oceanographic studies and the emergence of plate tectonics theory has established its validity.

CONTINENTAL GLACIERS: Continuous masses of thick ice sheets covering more than 50,000 km², i.e., all or most of a continent. Examples are those that cover most of Antarctica (covering 1,710,000 square kilometres) and Greenland (14 million square kilometres) that are up to 3500 m thick. They contain vast quantities of fresh water, enough that if both melted, global sea levels would rise by over 70 m.

CONTINENTAL POLAR: Cold, dry air mass that forms over northern Canada.

CONTINENTAL RISE: The portion of the ocean floor rising gently from the abyssal plain toward the steeper continental slope.

CONTINENTAL SHELF: The submerged edge of a continent, a gently sloping plain that extends from the coast to depths of 100-200 metres into an ocean. Continental shelves average about 65 km in width. According to the UN Convention on the Law of the Sea (1982), the continental shelf has a legal limit of up to 200 nautical miles (about 370.4 km) from the coastline and countries have exclusive rights to all natural resources within that limit.

CONTINENTAL SLOPE: The sloping, submarine portion of a continent. It extends from the edge of the continental shelf. The world's combined continental slope is about 300,000 km long and descends at an average angle of about 4° from the edge of the continental shelf to the beginning of the ocean basins at depths of 100 – 3,200 metres.

CONTINENTAL TROPICAL: Dry air mass that forms over land in areas close to the equator. An example is the dry, hot air mass that forms over Mexico in the summer.

CONTINUED FRACTION: A number that is an integer plus a fraction whose denominator is itself an integer plus a fraction and so on. For example, the continued fraction representation of a real number x is one of the form:

$$x = a_0 + \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{a_3 + \dots}}}$$

where $a_0, a_1, \dots, a_3$ and b_1, b_2, b_3 , are integers. Every irrational number can be represented as a continued fraction.

CONTINUED PRODUCT: A sequence of terms $x_k, x_{k+1}, \dots, x_n$ written as $x_i, x_{i+1}, \dots, x_k = \prod_{i=1}^k x_n$.

CONTINUITY: 1. In mathematics, a property of function of a real variable that has no break. 2. Logical connection between parts of a sequence.

CONTINUITY EQUATION (SPATIAL EQUATION OF CONTINUITY): In physics, it is a mathematical statement about the conservation of mass in a continuum material body defined as $\rho + \rho \operatorname{div} v = 0$, where v is the velocity of the particles of a body with density ρ and the divergence is taken with respect to its current configuration.

CONTINUITY INVERSION: A change in the mutual dispersion of two phases in contact.

CONTINUOUS: 1. In botany, it refers to not jointed or not separating at maturity along well-defined line of dehiscence. 2. A function is said to be continuous if it has a value that changes gradually as the independent variable or variables change, so that at every a of the independent variable, the difference between $f(x)$ and $f(a)$ approaches zero as x approaches a . A real function $y = f(x)$ is continuous at a point a if and only if at $x = a$, $f(x) = f(a)$ precisely, if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ for all x such that $|f(x) - f(a)| < \delta$. A function is continuous on the left at the point if the above condition holds only for values x less than a and continuous on the right if it holds for values greater than a . It is continuous at a point if and only if it is continuous both on the left and on the right at that point. The function itself is said to be continuous if it is continuous at all points. The function is said to be **uniformly continuous** on a set if the value of δ depends only on ε and not on the point a in the set. 3. Of a curve, representing a continuous function. 4. Defining any mathematical entity in terms of simpler entities and operations, such as those of set theory.

CONTINUOUS BED PACKING: A column packing, which is a single entity, rather than being composed of individual particles.

CONTINUOUS CULTURE: A technique used to grow microorganisms or cells continually in a particular phase of growth.

CONTINUOUS DATA: Data that can take any of an infinite number of values between whole numbers and therefore may not be measured completely accurately. The opposite or contrast is discrete data in which the variable can only take one of a definite set of values. For example, the sizes of mangoes on a tree form continuous data whereas the number of mangoes forms discrete data.

CONTINUOUS DATA PROTECTION (CDP; CONTINUOUS BACKUP): The backup of computer data by automatically saving a copy of every change made to that data, essentially capturing every version of the data that the user saves. It allows the user or administrator to restore data to any point in time.

CONTINUOUS DIFFERENTIABLE: Of a function, possessing a continuous derivative; a mapping from m to n is said to be $C^{(r)}$ if it is r -times continuously differentiable.

CONTINUOUS DISTRIBUTION FUNCTION: The distribution function of a continuous random variable is defined to be the probability that the random variable takes a value less than or equal to a real number. The function is defined by $F(b) = P(X \leq b) = \int_{-\infty}^b f(x) dx$ is called the distribution function of a continuous random variable.

CONTINUOUS FLOW ENTHALPIMETRY (CONTINUOUS FLOW ENTHALPIMETRIC ANALYSIS): A term used to describe methods in which a reagent is continuously fed into a flowing analyte stream and the temperature difference is measured before and after a reactor

(mixing chamber). Alternatively, the temperature may be measured in a differential manner between a reference and a reactor chamber. The analyte concentration is directly proportional to the measured temperature difference.

CONTINUOUS FUNCTION: A function $f(x)$ is continuous at $x = a$, if the limit of $f(x)$ as x approaches 'a' is $f(a)$.

CONTINUOUS INDICATION ANALYSER: An analysis with a permanent indication related to the sample concentration. To obtain a continuous indication, the sampling and measuring cell need to be continuous. A time lag may exist between sampling and indication of measured concentration.

CONTINUOUS OPERATION: The operation of an analytical instrument in such a way that no analytical procedure needs to be completed on any sample before the next procedure can be started.

CONTINUOUS PHASE (DISPERSION MEDIUM; EXTERNAL PHASE): The liquid in a disperse system in which solids are suspended or droplets of another liquid are dispersed.

CONTINUOUS PRECIPITATION (NUCLEATION AND GROWTH): A diffusional reaction in a multi-component system in which atoms are transported to the growing nuclei by diffusion over relatively large distances in the parent phase and during which the mean composition of the parent phase changes continuously towards its equilibrium value.

CONTINUOUS SPECTRUM: A spectrum of radiation that contains an unbroken sequence of frequencies over a relatively wide range. It is produced by incandescent solids, liquids and compressed gases.

CONTINUOUS TRANSFORMATION (HOMEOMORPHISM; TOPOLOGICAL ISOMORPHISM): A one-to-one transformation between points in two geometric figures or topological spaces that is continuous in both directions. A homeomorphism which also preserves distances is called an **isometry**. Two objects are homeomorphic if they can be deformed into each other by a continuous, invertible mapping.

CONTINUOUS TRANSITION (SECOND-ORDER TRANSITION): A transition in which a crystal structure undergoes a continuous change and in which the first derivatives of the Gibbs energies (or chemical potentials) are continuous but the second derivatives with respect to temperature and pressure (i.e. heat capacity, thermal expansion, compressibility) are discontinuous. Example is the order-disorder transition in metal alloys, for example, CuZn.

CONTINUOUS VARIATION (QUANTITATIVE VARIATION): The range of differences that can be observed in many characteristics in a population. For example, the wide range of foot sizes and height observed in adult human population. Continuous variation is common in characteristics that are strongly affected by environmental influences and in characteristics determined by many genes, each with a small effect.

CONTINUOUS WAVE (CONTINUOUS WAVEFORM): A wave of constant amplitude and frequency that is transmitted continuously rather than in pulses.

CONTINUOUS WAVE LASER: Continuous wave operation of a laser means that the laser is continuously pumped and continuously emits electromagnetic radiation. The emission can occur in a single cavity mode or on multiple modes. Some lasers are called CW mode-locked. This indicates that the pulse average power is constant, i.e., there is no Q-switched mode locking.

CONTINUOUS-FLOW CELL: A cell which allows liquid (or gaseous) samples to pass through the cell continuously while absorption measurements are made.

CONTINUUM: 1. In mathematics, a set that is infinite and everywhere continuous such as the set of points on a line. 2. A system of axes that form a frame of reference.

CONTINUUM HYPOTHESIS: The conjecture that there is no set whose cardinality is greater than that of the integers and smaller than that of the reals. The generalised continuum hypothesis states that for any infinite cardinal the next greater cardinal is that of its power set.

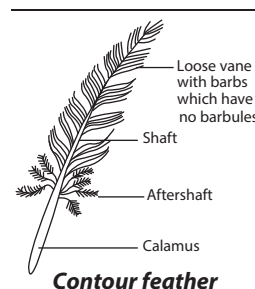
CONTINUUM MECHANICS: A branch of mechanics that deals with the analysis of the kinematics and the mechanical behaviour of materials modelled as a continuous mass rather than as discrete particles.

CONTORTED: Twisted or bent around a central axis.

CONTOUR: 1. The outward line of a body, for example, a coast, mountain range or body thought of as defining its shape, for example, a level line around a hill. 2. The particular path in the complex plane used to compute the integral in contour integration. 3. Also known as **contour line**, it is a line drawn on a map that joins points with equal elevation above or below a level, usually mean sea level.

CONTOUR FEATHERS: Feathers that are arranged in regular rows on a bird's body, giving the body a streamline shape.

CONTOUR INTERVAL: The vertical distance between the elevations represented by adjacent contour lines on a map.



CONTOUR LINE: 1. Also known as **equipotential curve**, it is a curve in two dimensions on which the value of a function $f(x,y)$ is a constant. 2. Also known as **contour**, it is a line drawn on a map that joins points with equal elevation above or below a level, usually mean sea level.

CONTOUR MAP: A map that uses contour lines to indicate the physical features of land, such as elevation or temperature. An example is a topographic map, which shows valleys and hills, and the steepness or gentleness of slopes.

CONTOUR PLOUGHING: A method of ploughing parallel to the contours rather than up or down a slope or ploughing the soil at right angles to the slope, creating channels which trap water and prevent run off. Contour ploughing helps to prevent or limit soil erosion.

CONTRACEPTIVE: Any drug, device or technique that prevents conception. Examples are female/male condoms and certain drugs.

CONTRACT: 1. Catch, acquire or develop an illness. 2. Constrict or make to become smaller or shorter, tighter or narrower. For example, most metals, solids, gases and liquids on heating expand but contract, when cooled.

CONTRACT CURVE: The locus of points in an Edgeworth box at which the respective indifference curves of the two consumers are tangential to one another. This curve is optimal in the sense that if the consumers are at a point off the contract curve, neither loses and at least one benefits from moving to a point on the curve.

CONTRACTED TROCHOID (CURTATE TROCHOID): A trochoid in which the point lies inside the rolling circle. A curtate trochoid is traced by a pedal when a bicycle is pedalled along a straight line.

CONTRACTILE ROOT: A long, straight, thick and fleshy root formed by many perennial herbaceous plants such as bulb and corn. They grow straight downwards, without branches or root hairs and pull the plant to an appropriate level in the ground. They also last for two or three months and they contain at first abundant stores of glucose, which are rapidly consumed. As the reserves disappear, the cortex collapses in a series of transverse zones which shorten the root by 30 to 40%, thereby pulling the organ down.

CONTRACTILE VACUOLE: A membrane-bound organelle found in certain protists that pumps fluid in a cyclical manner from within the cell to the outside by alternately filling and then contracting to release its contents at various points on the surface of the cell. It functions in maintaining osmotic equilibrium.

CONTRACTION: 1. Spontaneous tightening of muscular tissues. For example, contractions of the womb occur at regular intervals preceding childbirth. 2. Constrict or make to become smaller or shorter, tighter or narrower. For example, most metals, solids, gases and liquids on heating expand but contract, when cooled. 3. Also known as **contraction mapping**, on a metric space, a function that uniformly decreases distances by a common factor, which is less than 1 by

contracting all points toward some fixed point, *P*. When the common factor is greater than 1 the transformation is called a **dilation**.

CONTRADICTION: A statement that is necessarily false because of its logical structure, regardless of the facts. For example, the statement “*p* AND (NOT *p*)” is false, regardless of what *p* represents. The negation of a contradiction is a tautology.

CONTRAGREDIENT MATRIX: The inverse of the transpose of a matrix.

CONTRAIL: A condensation trail that is left behind by a passing plane, typically a jet plane. Contrails form when hot humid air from jet exhaust mixes with environmental air of low vapour pressure (pressure due to water vapour alone) and low temperature. The mixing occurs directly behind the plane due to the turbulence generated by the engine. If condensation occurs, then a contrail becomes visible. Since air temperatures at these high atmospheric levels are very cold (generally colder than -40°C), only a small amount of liquid is necessary for condensation to occur.

CONTRALATERAL: Of, relating to or affecting the opposite side of the body.

CONTRAPOSITION: In logic, the logical principle that if *p* implies *q* then *q* implies not *p*. Also, the derivation of its contrapositive from a given statement.

CONTRAPOSITIVE: In logic, a statement derived from a given statement by changing its subject and predicate terms and altering each from positive to negative or vice versa. The statement *A B* is the statement (NOT *B*) (NOT *A*). The contrapositive is equivalent to the original statement. If the original statement is true, the contrapositive is true; If the original statement is false, the contrapositive is false. For example, the statement “if *x* is a rational number, then *x* is a real number” has the contrapositive “If *x* is not a real number, then it is not a rational number”.

CONTRARY: 1. In logic, of a pair of statements such that they cannot both be true at the same time, under the same circumstances or in the same interpretation, although they may both be false. 2. Of a single statement, that cannot be true when a given statement is true. 3. A statement that is contrary to a given statement.

CONTRAST: 1. The difference in luminance and/or colour that makes an object (or its representation in an image or display) distinguishable. 2. The difference between the colour or shading of the printed material on a document and the background on which it is printed, for example, in optical character recognition.

CONTRAST MEDIUM (CONTRAST AGENT; CONTRAST MATERIAL): A substance introduced into the body in order to create a difference in the apparent density of various organs and tissues, thus making it easier to delineate adjacent body tissues and organs. Contrast media are used in conjunction with various medical imaging techniques, including X-rays, computed tomography (CT) scanning and magnetic resonance imaging (MRI).

CONTROL: 1. A parallel experiment carried out at the same time as a main experiment so that the result can be compared and the extent of any change quantified. 2. Also known as **control variable**, it is a variable that is held constant or whose impact is removed in order to analyse the relationship between other variables without interference. 3. The field of automating and managing processes using machines including the use of robots. 4. The process by which a tissue or an organism, a population or an ecosystem maintains itself in a balanced stable state. 5. The field of automating and managing processes using machines including the use of robots. 6. The process by which a tissue or an organism, a population or an ecosystem maintains itself in a balanced stable state.

CONTROL BUS: The electrical pathway or bus used by CPUs for communicating with other devices within the computer.

CONTROL CHART: In statistics, it is a chart on which observed values of a variable are plotted, usually relative to its expected value

under some condition and its allowable deviation, so that excessive variations can be detected.

CONTROL CONDITION: The condition in which the subjects of an experiment are not subjected to the treatment whose effectiveness is under the condition before experimental intervention, such as the course of a disease without therapy, with which the effect of a treatment is to be compared.

CONTROL GRID: A wire mesh electrode placed between the cathode and anode in a thermionic valve or a cathode ray tube used to control the flow of electrons from one end to the other.

CONTROL GROUP: In a scientific experiment, it is a standard for checking or verifying the results of an experiment where the effect of all variables must be held constant to ensure that the effects of the variable are being studied except the independent variable. The control group is used as a baseline in the experiment to compare groups and assess effects produced by the variable are being measured to ensure that the results obtained are valid.

CONTROL KEY: A key on a keyboard labelled Ctrl that is kept pressed down while pressing another character to perform a function. For example, CTRL + C will copy a highlighted text or object.

CONTROL MATERIAL: A material to be used for the assessment of the performance of an analytical procedure or part thereof.

CONTROL MECHANISM: Any mechanism that regulates a biological process such as a metabolic pathway or enzyme-controlled reaction or helps to maintain the internal environment.

CONTROL PANEL: A feature of the Windows operating system that allows the user to modify system settings and controls. It includes several small applications or control panels, that can be used to view and change hardware or software settings. Some examples of hardware control panels are Display, Keyboard and Mouse settings. Software control panels include Date and Time, Power Options, Fonts and Administrative Tools. Many control panels are included as part of the Windows operating system, but others can be installed by third-party applications or utilities.

CONTROL PROGRAM: 1. A computer program designed to schedule and to supervise the execution of programs of a computer system. 2. The part of the AIX operating system that determines the order in which basic functions should be performed. 3. In VM/ESA, a component that manages the resources of a single computer so that multiple computing systems appear to exist. Each of these apparent systems or virtual machines, is the functional equivalent of an IBM System/370, 370-XA or ESA computer.

CONTROL ROD (ABSORBING ROD): One of the rods made of chemical elements such as boron, silver, indium and cadmium that is capable of absorbing many neutrons to control chain reaction to ensure that nuclear fission reaction continues safely.

CONTROL SAMPLE: The test portion or test solution used for assessment of the performance of an analytical procedure.

CONTROL THEORY (OPTIMAL CONTROL): The branch of mathematics and engineering that is relevant to the control of certain physical processes and systems and finding solutions to them, as well as cost effectiveness. In other words, it deals with the design, identification and analysis of systems with a view towards controlling them, in order to perform specific tasks or make them behave in a desired way.

CONTROL TOTAL: A term used in computing to refer to a validation check in which an arithmetic total of a specific group of records is calculated.

CONTROL UNIT: The component of the central processing unit that directs and synchronizes execution program. It controls the operation of arithmetic logic unit (ALU) memory and I/O devices.

CONTROL VARIABLE (CONTROL): A variable that is held constant or whose impact is removed in order to analyse the relationship between other variables without interference.

CONTROLLED ATMOSPHERE: A synthetic gaseous sample of pure air which may contain carefully determined amounts of certain contaminants. This may be used as a standard for the calibration of analytical techniques, as a simulated environment for the study of biological responses or for other purposes.

CONTROLLED CHAIN REACTION: The process by which a nuclear reactor maintained so that chain reaction does not get out of control by releasing energy very quickly. To maintain a sustained controlled nuclear reaction, for every two or three neutrons released, only one must be allowed to strike another uranium nucleus. If this ratio is less than one then the reaction will die out; if it is greater than one it will grow uncontrolled and result in an atomic explosion. Most reactors use **control rods** made of chemical elements such as boron, silver, indium and cadmium that are capable of absorbing many neutrons to control chain reaction to ensure that nuclear fission reaction continues safely. Controlled nuclear fission is used for power production, whilst uncontrolled nuclear fission is used for nuclear weapons.

CONTROLLED HYPERTHERMIA (FEVER; PYREXIA): A condition characterised by an elevation of temperature above the normal range of 36.5 to 37.5 °C as measured in the mouth. It is a symptom of many disorders, such as infection by a virus or a bacterium and it is not itself a disease. The first signs may be chilly sensations, with associated periods of flushed or warm feelings. The temperature may rise slowly or rapidly and may fluctuate. A rise in temperature may be accompanied by shaking chills. A falling temperature may bring on heavy sweating.

CONTROLLED LIGHTING: In poultry, it is the artificial light provided in a poultry housing to increase or decrease the number of hours of light during the day. Large commercial farmers rely more on artificial light than natural light in order to control processes such as sexual maturity, fertility, and shedding of feathers (moulting). Day length influences many processes in poultry, such as welfare, laying eggs, growth rate and skeletal development.

CONTROLLER CARD (CONTROLLER): A piece of hardware that acts as the interface between the motherboard and the other components of the computer. For example, hard drives, optical drives, printers, keyboards and mice all require controllers to work. Most computers have all the necessary controllers built in the motherboard as chips, not full-sized cards. However, by adding additional components such as a SCSI hard drive, one may need to add a controller card as well. Controller cards are typically installed in one of the computer's PCI slots.

CONUS ARTERIOSUS: A small thick-walled chamber of the heart of vertebrate embryos that receive blood from the single ventricle and leads to the ventral aorta.

CONVECTION: 1. Transfer of heat from one part of a liquid or gas to another by means of a heated substance or the transfer of energy through a medium which involves movement against the gravity of higher temperature particles. The molecules of fluid in contact with the source of heat expand and cooler ones tend to take their place, setting up convection current. 2. Vertical motion of the air induced by the expansion of the air heated by the Earth's surface and its resulting buoyancy.

CONVECTION OVEN: An oven which uses a fan to circulate hot air to augment the traditional oven.

CONVECTIONAL CURRENT: 1. Any current of air involved in convection in the atmosphere; usually, the upward-moving portion of a convection circulation, such as a thermal or the updraft in cumulus clouds. 2. The current that occurs within the geosphere as the result of temperature variations; this causes mass movement of subcrustal or mantle material.

CONVECTIONAL RAINFALL: Rainfall associated with hot climates, resulting from the uprising of convection currents of warm air, in which air warmed by heat rises to great heights, cooled abruptly, with its water vapour content condensing and falling as rain.

CONVECTIONAL TILLAGE: A practice where land is ploughed, turning over all crop residues.

CONVENIENCE SAMPLE: A sample chosen on the basis of accessibility, expediency, cost, efficiency or other reason not directly concerned with sampling parameters.

CONVENTION ON INTERNATIONAL TRADE IN ENDANGERED SPECIES OF WILD FAUNA AND FLORA: Abbreviated CITES, it is an international agreement under the auspices of the World Conservation Union with the aim of ensuring that international trade in endangered species of animals and plants does not threaten their survival. The agreement came into force in 1975 and has now been signed by 160 states. It prohibits any trade in a category of 8,000 highly endangered species and controls trade in a further 30,000 species. There are three categories of appendices with Appendix 1 classifying that all animal and plant species are endangered and all trade in them is banned. Appendix 2 is classified as vulnerable and trade in the species is controlled. Appendix 3 is subject to domestic controls while national governments request help in controlling international trade.

CONVENTIONAL CURRENT: The direction of positive charge flow, therefore, it is the flow of positive charge from the positive terminal to the negative.

CONVENTIONAL TILLAGE SYSTEM: A system that uses a combination of mechanical operations with implements to manipulate the soil to get it mostly free of weeds and plant residue cover. It typically includes methods, such as ploughing and harrowing, to produce a fine seedbed. Conventional tillage systems typically leave less than 15 percent residue cover after planting.

CONVENTIONAL TRUE VALUE: A value attributed to a particular quantity and accepted, sometimes by convention, as having an uncertainty appropriate for a given purpose. Examples are the Committee on Data for Science and Technology (CODATA) recommended values of fundamental physical constants.

CONVERGE: 1. In mathematics, to approach a finite limit as the number of terms in an infinite series increases. 2. In biology, to develop independently of other species, superficially similar characteristics in response to a set of environmental conditions, for example, the development of wings in birds and insects.

CONVERGENCE: 1. The property of a series of numbers in which the difference between consecutive terms gradually decreases. 2. The coordinated turning of the eyes inward to focus on an object at close range. 3. The development of superficially similar structures, such as the wings of birds and insects, in unrelated species subjected to similar environments.

CONVERGENT EVOLUTION (CONVERGENCE): The development of superficially similar structures in organisms that evolved from unrelated ancestors, usually because the organisms live in the same kind of environment. An example is the wings of insects and birds for flying, although the wings did not evolve from the same origin.

CONVERGENT IN MEAN: Of a sequence of integrable functions on a set, having the property that integral of the absolute value of the differences of the functions from the limit function tends to zero. Thus, the sequence $\{f_n\}$ converges to f in mean on an interval

$[a, b]$ if $\int_a^b |f_n(x) - f(x)| dx \rightarrow 0$, as n tends to ∞ . Convergence in mean

requires that $\int_a^b |f_n(x) - f(x)|^2 dx \rightarrow 0$.

CONVERGENT IN MEASURE: Of a sequence, $\{f_n\}$, of measurable functions, convergent with respect to some measure, P , for every $\varepsilon > 0$ $P(\{x : |f_n(x) - f(x)| > \varepsilon\})$ tends to zero as n tends to infinity; f is then the limit of the sequence.

CONVERGENT PLATE BOUNDARY (DESTRUCTIVE PLATE BOUNDARY; TENSIONAL PLATE MARGIN): A type of plate boundary where plate material is subducted. This occurs when two

plates converge or move towards each other. The denser and heavier plate is forced under the lighter plate. Friction causes melting with the molten crust becoming magma which may be forced to the surface of the earth causing a volcanic eruption. One of the converging plates must be an oceanic plate for this to occur. If two plates of equal density collide, mountain ranges, such as the Himalayas are formed.

CONVERGENT SERIES: A series in which the difference between consecutive terms decreases as the number of terms increases.

CONVERGING LENS (MIRROR): A lens or mirror that can refract or reflect a parallel beam of light so that it converges at a point, the principal focus. Such a mirror is concave and it is thicker at the centre than at its edges.

CONVERSE: A term used in mathematics to refer to the reversed order of a conditional statement. For example, the converse of the statement 'if a then b ' is 'if b , then a '. Also, a set of points is convex if, for any two points in the set, all the points on the line segment joining them are also in the set.

CONVERSION: 1. Changing from one number system to another or a change from one measuring or calculating system to another. 2. The transformation of materials by chemical process. 3. A term used to define the percentage of reactants converted to products inside a chemical reactor or unit process.

CONVERSION ELECTRON: Electron ejected from the atom in the process of internal conversion.

CONVERSION FACT: A fixed relationship such as 1 inch = 2.54 centimetres that can be used to convert measurements within or between systems of measurement.

CONVERSION FACTOR: A numerical ratio or fraction used to express a measurement given in one unit as another unit without changing the values between the units. It comprises of a part in which the denominator is equivalent to the numerator. It usually changes the units by multiplying or dividing. A conversion factor is always equal to 1.

CONVERSION OF DECIMAL TO FRACTION: Expressing the relevant decimal as a number divided by 10, 100, etc.

CONVERSION SPECTRUM: A plot of a quantity related to the absorption (absorbance, cross section, etc.) multiplied by the quantum yield for the considered process against a suitable measure of photon energy, such as frequency wave number or wavelength.

CONVERT: 1. To change the form of a measurement, units, etc., from one system of units to another or to obtain an equivalent value without a change in the size or amount. For example, to change kilometres into miles, one divides the kilometres by 1.61. 2. To obtain the converse of a given proposition or relation. 3. To transform, for example, a substance, state or product, from one form into another.

CONVERTER: 1. An electrical device for converting alternating current into direct current or less frequently, the vice versa. 2. The reaction vessel in the Bessemer process or some similar steel-making process. 3. A computer device for converting information coded in one form into some other form. 4. A room heater that warms the air by passing it over a hot surface and then circulating it.

CONVERTER REACTOR: A nuclear reactor that converts fertile material such as thorium -232 into fissile material such as uranium -233.

CONVERTIPLANE: A vertical takeoff and landing craft (UTOL) with rotors on its wings that spin horizontally for takeoff, but tilt to spin in a vertical plane for forward flight.

CONVEX: 1. Of a geometric surface, having all edges curving outwards or towards the eye, with no part of its curves or faces pointing inwards. For example, the outer surface of a ball appears convex. 2. A **convex polygon** is one that has no interior angle greater than 180° and a line segment drawn between any two points on the shape lies entirely inside the shape. 3. A **convex function** is a continuous function whose value at the midpoint of every interval

in its domain does not exceed the arithmetic mean of its values at the ends of the interval. Alternatively, it is a real-valued function $f(x)$ defined on an interval if the chord joining any two points on its graph lies above it, in a Euclidean space or a vector space of at least two dimensions such that for arguments x and y in the appropriate abstract space and t in the interval $[0, 1]$, $tf(x) + (1-t)f(y) \geq f(tx + (1-t)y)$. Equivalently, a function is convex if the set of points in a real vector space are in the set then so are all the points on the line segment joining them so that if x and y are any two points in the set, then so is $tx + (1-t)y$, for all t between 0 and 1. Examples of convex functions are the quadratic function $f(x) = x^2$ and the exponential function $f(x) = e^x$ for any real number x .



Fig. I: Convex



Fig. II: NOT Convex

CONVEX COMBINATION: A linear combination, $\sum t_i a_i$, of a finite set of elements, a_i , such that all coefficients, t_i , of the elements are non-negative and $\sum t_i$ sums to 1.

CONVEX HULL: For a set of points S in n dimensions, it is the intersection of all convex sets containing S .

CONVEX LENS (CONVERGING LENS): A lens with at least one face formed from the exterior surface of a sphere.

CONVEX MIRROR (DIVERGING MIRROR): A type of spherical mirror whose outer bulging surface faces the source of light and reflects light outwards and therefore not used to focus light. A convex mirror is curved outwards and therefore, provides a wider field of view than flat or concave mirrors, but objects which appear in them generally look smaller and, therefore, farther away than they actually are. It also produces an upright (erect), diminished and virtual images. It is used as a rear-view mirror (driving mirror) in vehicles; in shops, offices, etc., as a security monitoring device; and to eliminate blind spots at corners, concealed entrances and exits, car parks, etc.

CONVEX POLYHEDRON: A polyhedron that has no interior angle greater than 180° and a line segment drawn between any two points on the shape lies entirely inside the shape. Algebraically, it is defined as the set of solutions to a system of linear inequalities $mx \leq b$, where m is a real $s \times 3$ matrix and b is a real s -vector.

CONVEX POLYTOPE: A type of polytope with a convex set of points in the n -dimensional space R^n .

CONVEX QUADRANGLE: A quadrangle with both diagonals lying inside.

CONVOLUTE: Coiling longitudinally upon itself with whorls overlapping as in certain leaves and petals.

CONVOLUTION: 1. A function or series derived from two given functions or series by integration. The integral convolution of $f(x)$ and $g(x)$ is given by $\int_0^x f(t)g(x-t)dt$, while the convolution of two

sets produces the Cauchy product. 2. In statistics, a method of determination of the sum of two random variables by integration or summation. 3. Also known as **gyrus**, any of the irregular convex folds or ridges of the surface of the brain especially of the cerebrum.

CONVULSION: A sudden violent involuntary movement, caused by the contraction of muscles of the body, often accompanied by loss of consciousness. It may occur in conditions such as epilepsy, poisoning, disturbances of calcium or phosphorus metabolism, alkalosis, diabetes, oxygen insufficiency and low blood sugar content and also injury of the brain.

COOKIE (WEB COOKIE; BROWSER COOKIE; HTTP COOKIE): In computer science, it refers to a small amount of data generated by a website and stored in a user's web browser while a user is browsing a website. This data commonly contains the domain, path, lifetime and value of a variable that is set. If the lifetime of this variable is longer than the time the user spends at a site, then this string is saved to

file for future reference, so that if the user browses the same website in the future, the data stored in the cookie can be retrieved by the website to notify the website of the user's previous activity.

COOLANT: 1. An agent that produces cooling, especially a fluid that draws off heat by circulating through an engine or by bathing a mechanical part; or fluid used to remove heat from a system by convection (usually forced) either to control the temperature or to extract energy. 2. A fluid that circulates around the control rods in the nuclear power plant and which transfer the energy released by the fission processes to the steam boilers of the turbines.

COOLING CURVE: A line graph that represents the change of phase of matter, typically from gas to solid or liquid to solid. The independent variable (x -axis) is time and the dependent variable (y -axis) is temperature.

COOLING DEGREE DAYS (CDD): A measure of how hot the temperature is on a given day or over a period of days, calculated against a standard reference temperature of 18° Celsius. It is the number of degrees that the average temperature over a day is above 18 °C, for example, a day with an average temperature of 20°C counts as 2 cooling degree days (CDD). It is used to assess the climate and the heating needs for different locations, or demand for energy needed to heat a building.

COOLING FORCE: The time average force on an atom, which is equal in magnitude and opposite in sign to the time-averaged rate of change of the momentum of the photons the atom is scattering.

COOMASSIE BLUE (COOMASSIE BRILLIANT BLUE; BRILLIANT INDOCYANINE G; BRILLANTINDOCYANIN G; XYLENE BRILLIANT CYANINE G; SERVA BLUE G): A biological dye used for staining protein, with the chemical formula $C_{47}H_{50}N_3NaO_7S_2$. It is slightly soluble in cold water, soluble in hot water and ethanol. The name "Coomassie" is a registered trademark of Imperial Chemical Industries.

COOPER PAIRS: The name given to two electrons or other fermions that situated a certain way among positive ions, no longer repel each other but instead develop an attraction for each other. It was named after American physicist, Leon Cooper (1930-), who discovered it. Cooper showed that an arbitrarily small attraction between electrons in a metal can cause a paired state of electrons to have a lower energy than the Fermi energy, which implies that the pair is bound. In conventional superconductors, this attraction is due to the electron-phonon interaction. The Cooper pair state is responsible for superconductivity, as described in the BCS theory developed by John Bardeen, John Schrieffer and Leon Cooper for which they shared the 1972 Nobel Prize.

COOPER, LEON NEIL (1930-): An American physicist and 1972 Nobel Prize laureate, who taught at Ohio State University (1954-58) and Brown University (from 1958). Together with John Bardeen and John Robert Schrieffer, they developed the BCS theory of superconductivity for which they shared the 1972 Nobel Prize for Physics. His principal contribution to the theory was his discovery of Cooper electron pairs (1956), electrons that repel each other under normal conditions but are attracted to each other in superconductors.

COOPERATIVE BINDING: A situation in which the binding of one substituent to a macromolecule favours the binding of one another. If the binding of a ligand at one site increases the affinity for other ligands, it results in positive cooperativity. The converse causes negative cooperativity.

COOPERATIVE PHENOMENON: A phenomenon in which the constituents of a system cannot be regarded as acting independently from each other.

COOPERATIVE TRANSITION: A transition that involves a simultaneous, collective displacement or change of state of the atoms and/or electrons in the entire system. Examples are order-disorder transition of atoms or electrons, as in an alloy, a ferromagnet or

superconductor; a Jahn–Teller or ferroic transition; and a martensitic transition.

COOPERATIVITY: Interaction between the substrate binding sites of an allosteric enzyme. Binding of a substrate molecule to one binding site changes the affinity of the binding sites on the other subunits to the substrate by induction of a conformation change at the other binding sites. Cooperative enzymes typically display a sigmoid (S-shaped) plot of the reaction rate against substrate concentration.

COORDINATE: 1. In geometry, **coordinates** are a set of numbers that describes the exact position of something such as a place on a map with reference to a set of axes (x, y, z). If the point is known to be on a given plane, two numbers are needed; if located in space, three numbers are needed. Alternatively, in the x - y plane it is an ordered pair (x, y) in the coordinate plane indicating the position of a point, the first variable x indicates the distance of the point from the vertical axis and the x -coordinate. The second variable indicates the distance of the point from the horizontal axis, which is the y -coordinate. 2. A variable used with others to describe the state of a chemical or physical system.

COORDINATE CHANGE: A mapping $\phi\psi^{-1}:\psi(U\cap V)$, where (U, Φ) and (V, ψ) are charts.

COORDINATE COVALENT BOND (COORDINATE BOND; DATIVE BOND; DIPOLAR BOND; SEMIPOLAR BOND; COORDINATE LINK): A covalent bond formed between two atoms in which one atom alone donates the pair of electrons to be shared between the two.

COORDINATE FUNCTION: A function that defines one coordinate of a curve in terms of a parameter; if $y = f(x)$ is satisfied by the set of points($u(t), v(t)$), then $x = u(t)$ and $y = v(t)$ are the coordinate functions.

COORDINATE GEOMETRY (ANALYTIC GEOMETRY; ANALYTICAL GEOMETRY; CARTESIAN GEOMETRY): The branch of mathematics that studies geometrical relations using algebra with coordinates as references. An example of coordinate geometry is plotting points, lines and curves on an x - and y -axis.

COORDINATE GRID (RECTANGULAR COORDINATE GRID): A reference frame for locating points in a plane by means of ordered pairs of numbers. A rectangular coordinate grid is formed by two perpendicular lines, or axes which are number lines that intersect at right angles at their zero points. The horizontal axis is called x -axis and the vertical axis is called the y -axis.

COORDINATE INDUCTION: The simultaneous expression of two or more genes.

COORDINATE PLANE: A two-dimensional system of coordinates.

COORDINATE SPACE: A space of s dimensions representing a system with s degrees of freedom in which the rectangular coordinates specify the positions of the points within the system.

COORDINATE SYSTEM: A system for locating points on a plain or in space by using their coordinates. An example is the Cartesian coordinate system.

COORDINATION: 1. The process involved in the reception of sensory information and the subsequent response of the organism. Coordination is controlled by regions of the brain that deal with specific functions, such as locomotion and breathing and is carried out by the nervous system. 2. The formation of a covalent bond, the two shared electrons of which have come from only one of the two parts of the molecular entity linked by it, as in the reaction of a Lewis acid and a Lewis base to form a Lewis adduct; or the bonding formed in this way. In the former sense, it is the reverse of unimolecular heterolysis.

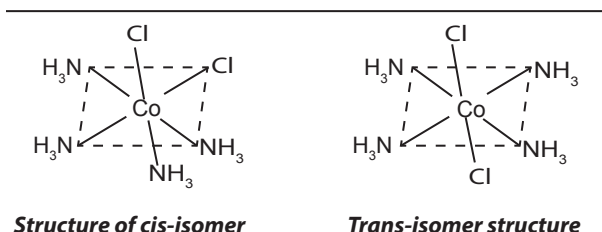
COORDINATION BOND (DATIVE BOND; DIPOLAR BOND; SEMIPOLAR BOND): A covalent bond formed between two atoms, in which one of the atoms provides both electrons that form the bond.

COORDINATION COMPOUND: A metal complex in which a central atom, usually a metal is connected by surrounding ions or molecules called ligands, bound to the metal by coordinate covalent bonds, by

donating a lone pair of electrons into empty metal orbitals. It is also possible for certain ligands to accept electrons from the metal atom into their empty antibonding orbitals.

COORDINATION ENTITY: An assembly consisting of a central atom, usually metallic, to which is attached a surrounding array of other groups of atoms (ligands).

COORDINATION ISOMERISM: A type of structural isomerism in which the composition of the complex ion varies. The example below shows the geometric isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ coordination complex:



Structure of cis-isomer

Trans-isomer structure

COORDINATION ISOMERS: Isomers involving exchanges of ligands between complex cation and complex anion of the same compound.

COORDINATION NUMBER: The number of groups, molecules, atoms or ions immediately surrounding a given atom or ion in a complex or crystal. For example, the coordination number of carbon in methane is four, it is five in protonated methane, CH_5^+ .

COORDINATE PLANE: A two-dimensional system of coordinates.

COORDINATION POLYHEDRON: In a coordination entity, the solid figure defined by the positions of the ligand atoms directly attached to the central atom.

COORDINATION SPHERE: The central metal ion and its coordinating ligands but not any uncoordinated ions.

COORDINATIVELY SATURATED COMPLEX: A transition metal complex that has formally 18 outer shell electrons at the central metal atom.

COORDINATIVELY UNSATURATED COMPLEX: A transition metal complex that possesses fewer ligands than exist in the coordinatively saturated complex. These complexes usually have fewer than 18 outer shell electrons at the central metal atom.

COPENHAGEN INTERPRETATION: The standard interpretation of quantum mechanics associated with the ideas of Neil Bohr and the most widely accepted by physicists above the various interpretations of quantum mechanics.

COPEPODA: A subclass of crustaceans occurring in marine and freshwater habitats. Copepods are usually 0.5-2mm long and lack both a carapace and a compound eye.

COPERNICAN ASTRONOMY (HELIOCENTRISM; HELIOCENTRICISM): The system of astronomy proposed by Nicholas Copernicus, (1473-1543), Polish astronomer. It is the astronomical model in which the Earth and planets revolve around a stationary Sun at the centre of the universe.

COPERNICIUM: A synthetic radioactive element of the transactinide series, with the symbol, Cn, atomic number 112, relative atomic mass of most stable isotope 277. It has only been produced in trace amounts and has nine observed isotopes with mass numbers ranging from 277 to 285. The most stable isotope has a mass number of 285 and a half-life of about 34 seconds.

COPERNICUS, NICOLAUS (1473-1543): Polish astronomer, who studied medicine and law and canon law at the University of Bologna in Italy. He also studied mathematics and physics. He is best known for his astronomical theory that the planets, including the Earth orbit the Sun in circular paths, although it was not formally published until the year he died. This refutation of an Earth-centred universe raised hostile opposition from the church as well as from other astronomers.

COPING RANGE: The variation in climatic stimuli that a system such as communities, enterprises, or ecosystems can absorb without producing significant adverse impacts or the crossing of critical thresholds. The coping range is used to link the understanding of current adaptation to climate with adaptation needs under climate change.

COPLANAR: A term used in geometry to describe lines or points that lie on the same plane.

COPLANAR FORCES: Three or more forces whose lines of action all lie in the same plane and their points of application are in the plane.

COPOLYMER: A polymer derived from more than one species of monomer. Copolymers that are obtained by copolymerisation of two monomer species are sometimes termed **bipolymers**, those obtained from three monomers are called **terpolymers** and those obtained from four monomers are called **quaterpolymers**, etc.

COPOLYMER MICELLE: A micelle formed by one or more block or graft copolymer molecules in a selective solvent.

COPOLYMERISATION: The formation of a polymer that contains two or more different monomers. Examples of copolymers are styrene-divinylbenzene, styrene-acrylonitrile and ethylene-vinyl acetate.

COPPER: 1. An orange pink, very malleable and ductile metallic element, with the symbol Cu, atomic number 29, atomic weight 63.54, melting point 1083 °C (1356 K), boiling point 2595 °C (2868 K), specific gravity 8.96, valence 1, 2 and relative atomic mass of 63.546. Pure copper is rarely found in nature. It is usually combined with other chemicals in the form of copper ores. There are about 15 copper ores commercially mined in 40 countries around the world. The most common ores are known as sulfide ores in which the copper is chemically bonded with sulfur. An example is chalcopyrite, also known as copper pyrites and similar sulfide ores. The other ores are oxide ores, carbonate ores or mixed ores depending on the chemicals present. Many copper ores also contain significant quantities of gold, silver, nickel and other valuable metals. The ores typically contain low percentages of copper and have to be concentrated by, for example, froth flotation before refining. It is widely used for electrical wiring, water piping and corrosion-resistant parts, either pure or in alloys such as brass and bronze because of its durability, pliability, high thermal and electrical conductivity and resistance to corrosion. 2. A dietary essential trace metal, involved in many processes including red blood cell formation, respiration and bone formation. The Reference Nutrient Intake is 1.2 mg/day. In the UK, the adult Reference Nutrient Intake for copper is 1.2 mg each day. In the USA, the estimated safe and adequate daily intake of copper is 1.5-3.0 mg. Rich sources include meat, poultry, game, fish and shellfish, avocado, nuts, pulses, bread, chocolate, beer, cider, coconut, mushrooms. Low intakes of copper can lead to anaemia.

COPPER FUNGICIDE: An inorganic fungicide containing copper. Some of the earliest known fungicides were copper compounds including the Bordeaux and Burgundy mixtures. Other copper-containing fungicides are often based on copper oxychloride or copper oxide and may be formulated with other types of fungicide.

COPPER ORE: Any mineral from which copper is extracted. This includes native copper (Cu), chalcocite (Cu_2Si), chalcopyrite (CuFeS_2), cuprite, malachite and tennantite.

COPPER VAPOUR LASER: A pulsed source of coherent radiation emitting at 578.2 nm and 510 nm from excited copper atoms.

COPPER(I) CHLORIDE (COPPER MONOCHLORIDE; CUPROUS CHLORIDE): A white solid cubic compound, with the chemical formula CuCl , relative density 4.14, melting point 430 °C (703 K) and boiling point 1490 °C (1763 K). It is obtained by boiling a solution containing copper(II) chloride, excess copper turnings and

hydrochloric acid. Copper(I) is present as the $[\text{CuCl}_2]^-$ complex ion. On pouring the solution into air-free distilled water copper(I) chloride precipitates. It must be kept free of air and moisture since it oxidises to copper(II) chloride under those conditions.

COPPER(I) OXIDE (CUPROUS OXIDE; DICOPPER OXIDE; CUPRITE; RED COPPER OXIDE): A brick red insoluble solid, with the chemical formula Cu_2O , relative density 6.0 and melting point 1235°C (1508 K). It is found in nature as the mineral cuprite and produced in the laboratory by reduction of an alkaline solution of copper(II) sulfate with, for example, an aldehyde. This reaction is the basis for the Fehlings reaction. Since the addition of alkalis to a solution of copper(II) salt results in the precipitation of copper(II) hydroxide the copper(II) ions are complexed with tartrate ions. Under such conditions the concentration of copper(II) ions is so low that the solubility product of copper(II) hydroxide is not exceeded. Cu_2O is a component of some antifouling paints.

COPPER(II) CHLORIDE (CUPRIC CHLORIDE): A brown-yellow powdery compound, with the chemical formula CuCl_2 , relative density 3.386 and melting point 620°C (893 K). It exists as a blue-green dihydrate $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$. The anhydrous solid is obtained by passing chlorine over heated copper. It is predominantly covalent and adopts a layer structure in which each copper atom is surrounded by four chlorine atoms at a distance of 0.23nm and two more at a distance of 0.295nm.

COPPER(II) NITRATE (CUPRIC NITRATE): A blue deliquescent solid, with the chemical formula $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, relative density 2.32 and melting point 114.5°C (387.65 K). It is obtained by reacting either copper(II) oxide or copper(II) carbonate with dilute nitric acid and crystallising the resulting solution. Other hydrates containing 6 or 9 molecules of water are known. On heating it readily decomposes to give copper(II) oxide, nitrogen dioxide and oxygen. The anhydrous form can be obtained by reacting copper with a solution of nitrogen dioxide in ethyl ethanoate.

COPPER(II) OXIDE (CUPRIC OXIDE): A black insoluble solid, with the chemical formula CuO , relative density 6.3 and melting point 1326°C (1599 K) monoclinic. It is obtained by heating either copper(II) carbonate or copper(II) nitrate. It decomposes on heating above 800°C (1073 K) to copper(I) oxide and oxygen. Copper(II) oxide reacts readily with mineral acids on warming, with the formation of copper(II) salts. It is also readily reduced to copper on heating in a stream of hydrogen. Copper(II) oxide is soluble in dilute acids forming blue solutions of cupric salts.

COPPER(II) SULFATE (CUPRIC SULFATE; COPPER SULFATE): A blue crystalline solid, with the chemical formula $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. The pentahydrate loses four molecules of water at 110°C (383 K) and the fifth H_2O at 150°C (423 K) to form the white anhydrous copper sulfate. It is prepared either by reacting copper(II) oxide or copper(II) carbonate with dilute sulfuric acid. The solution is heated to saturation and the blue pentahydrate crystallises out on cooling, with a few drops of dilute sulfuric acid being generally added to prevent hydrolysis. It is obtained on an industrial scale by forcing air through a hot mixture of copper and dilute sulfuric acid. Industrial uses of Copper(II) sulfate include electroplating, textile dyeing, wood preservative, pesticides and preparation of copper compounds.

COPPICE (COPSE): An area of densely growing small trees, especially one in which the trees are regularly cut back to encourage more growth. **Coppicing** is the management of woodland by cutting down trees to near ground level at regular intervals, typically every 3-20 years to promote the growth of numerous shoots from the base.

COPRA: The dried flesh or endosperm of a coconut, from which coconut oil is obtained. It is also shredded and used in candies and baked goods.

COPRECIPITATION (CPT; CO-PRECIPITATION): 1. In chemistry, the removal of a substance from solution by its association with a precipitate of some other substance. Coprecipitation is an important issue in chemical analysis, where it is often undesirable, but in some cases it can be exploited. 2. In medicine, the precipitation of an

unbound antigen along with an antigen-antibody complex.

COPRIME (RELATIVELY PRIME): A set of numbers which do not have any other common factor other than 1 or numbers whose highest common factor is 1. For example, 8 and 9 have no other common factor other than 1 so they are co-prime numbers.

COPROGENOUS EARTH: A type of organic soils composed of, at least 50% by volume of faecal pellets less than 0.5 mm in diameter. The faecal material is derived from aquatic animals.

COPROLITE: Fossilised faecal matter, which helps to determine the animal's behaviour through diet rather than morphology.

COPROPHAGY: The ingestion of faeces by some organisms such as beetles as a means of obtaining nutrients. The term **coprophilous** describes organisms such as certain fungi that live on dung or excrement.

COPSE (COPPICE): An area of densely growing small trees, especially one in which the trees are regularly cut back to encourage more growth.

COPULATION: The act of mating in animals with internal fertilisation or the sexual union of two individuals, resulting in insemination or deposition of the male gametes in proximity to the female gametes.

COPY NUMBER: The number of copies of a plasmid or a gene within a cell.

COPYRIGHT: A legal means of protecting an author's work. It is a type of intellectual property that provides exclusive publication, distribution and usage rights for the author. This means whatever content the author created cannot be used or published by anyone else without the consent of the author. The length of copyright protection may vary from country to country, but it usually lasts for the life of the author plus 50 to 100 years.

CORAL: Any of a group of sedentary colonial marine invertebrates belonging to the class Anthozoa of the phylum coelenterate. It is found in warm seas worldwide.

CORAL BLEACHING: The paling in colour of corals resulting from the loss of the symbiotic algae living in their tissues. The corals turn completely white. Bleaching occurs in response to physiological shock in response to abrupt changes in temperature, salinity, and turbidity. If this is sustained, the corals will disappear which will have many consequences, such as the loss of some fish populations that inhabit the corals; and the erosion of the local coastline as it is no longer protected by the reefs from the sea.

CORAL REEFS: Large underwater structures composed of the deposits of skeletons of corals found at or near the surface of warm, clear, shallow tropical waters. Coral reefs are economically and environmentally important ecosystems providing shelter and nutrients to diverse marine organisms such as fish, plants, and others. There are three types of coral reefs: (i) a barrier reef is one that surrounds an island and creates a lagoon between the reef and the beach; (ii) atolls are circular reefs with a central lagoon, formed on top of submerged volcano cones; (iii) fringing reefs are found close to newer volcanic islands.

CORAL TREES (*Erythrina*): A genus of over 100 species in the pea family, Fabaceae, distributed in the tropics and subtropics. They are evergreen trees and shrubs plants that come in a large variety of appearances. The trunk, young branches, petioles and petiolules are often armed with blunt, conical thorns or recurved prickles. Leaves are pinnately trifoliate, often clustered at the ends of branches; leaflets are broad-ovate, elliptic, often deltoid or rhomboid, entire, lateral leaflets often asymmetric. *Erythrina* species exhibit great diversity in floral structure, inflorescence orientation, fruit morphology, seed coat colouration and vestiture and epidermal ornamentation of foliage and calyces. Flowers appear before or with the first leaves, very showy, mostly red, some salmon, pink orange or yellow, solitary, paired or fasciated in erect, terminal racemes leafy at the base or in axillary

racemes. The species are used as a shade and support tree, as fences and as green manure and animal feeding. Several species are planted as ornamentals. Medicinally, it has been used traditionally to treat various ailments such as insomnia malaria fever, venereal disease, asthma and toothache.

CORALLUM: The skeleton of a zoophyte, which can be calcareous or in the formation of horns. For example, the set of parallel vertical grooves which are present on the sides of salamanders and newts.

CORCHORUS: A genus of shrubby plants belonging to the Tiliaceae family. Some members are bast fibres, source of jute fibres.

CORD: A unit for measuring the volume of wood cut for fuel.

CORDATE: A term used to describe a heart-shaped structure such as the leaves of the sweet violet (*Viola odorata*) or the lemmas of quaking grasses (*Briza*).

CORDIERITE: A magnesium iron aluminium cyclosilicate mineral, $(\text{MgFe})_2\text{Al}_4\text{Si}_5\text{O}_{18}$, which is blue to purplish in colour. It is obtained from metamorphic rocks. The natural mineral has little commercial use. When clear, cordierite is sometimes cut as a gem. Synthetic magnesium cordierite has a low thermal expansion and is used as a semirefractory material because of its resistance to thermal shock.

CORDILLERA: A group of mountain ranges and their valleys all running in a specific direction, formed by the continued convergence of two tectonic plates along a line.

CORDITE: A smokeless powder made from nitrocellulose, nitroglycerine and petroleum used as a propellant for guns. It was developed and produced in the United Kingdom from 1889 to replace gunpowder as a military propellant.

CORDON: A trained fruit tree, whose growth is restricted to the main stem by pruning. It is grown as a single stem at an angle against a support.

CORE: 1. A rod or frame of magnetic material that increases the inductance of a coil through which it passes. 2. The inner part of a nuclear reactor in which the nuclear reaction takes place. 3. The devices that make up the memory in certain types of computers. 4. The central region of a star or planet. 5. The innermost part of the Earth. 6. Of a set in a vectorspace, the points of the set such that an open line segment may be constructed in the set containing the point. 7. A solution concept that seeks Pareto optimal outcomes in games in which coalitions are allowed between players. 8. An intersection of all the conjugates of a subgroup, H, in group G. It is the largest normal subgroup of G contained in H and is denoted core H.

CORE ATOM: The primary reference atom in a mechanistic change. It is usually either of the two atoms of a transformed multiple bond or the single atom at which substitution occurs.

CORE ELECTRONS (INNER ELECTRONS): Electrons that fill all the energy levels of an atom except the valence level.

CORE MEMORY (MAGNETIC CORE MEMORY): In computer science, it is a form of random-access computer consisting of an array of magnetic cores. It uses tiny magnetic toroids (rings), the cores, through which wires are threaded to write and read information. Each core represents one bit of information. It is now replaced by semiconductor memories.

CORE ORBITAL: An atomic orbital that is part of the inner closed shell of an atom. Unlike the valence orbitals, which are the atomic orbitals of the valence shell of an atom, core orbitals on one atom have a small overlap with core orbitals on another atom. They are therefore usually ignored in simple calculations of chemical bonding.

COREACEOUS (CORIACEOUS): Having a thick and leathery texture.

CORECEPTOR: A receptor molecule on the surface of a cell that enhances the activity of another receptor following binding of its extracellular ligand. For example, optimal activation of T cells in an

immune response depends on an involvement of relevant coreceptors in a cluster with T cell receptors. The coreceptors are CD4 or CD8 proteins, which lie alongside the T cell receptor membrane.

CORIOLIS ACCELERATION: An acceleration which, when added to the acceleration of an object relative to a rotating coordinate system and to its centripetal acceleration, gives the acceleration of the object relative to a fixed coordinate system.

CORIOLIS EFFECT: The curving of the path of a moving object caused by coriolis force. The effect tends to deflect moving objects clockwise in the Southern hemisphere and counterclockwise in the Northern Hemisphere and is important in the formation of cyclonic weather systems. Coriolis force is an apparent deflective force acting on a moving object due to the rotation of the earth.

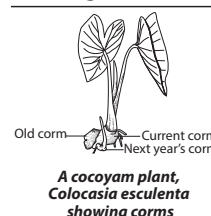
CORIOLIS FORCE: An apparent force caused by the Earth's rotation, which causes winds and currents to follow a curved path across the Earth's surface. In the Northern Hemisphere, air is deflected to the right (clockwise) of its path, while in the Southern Hemisphere, it is deflected to the left (anti-clockwise). It is dependent on the latitude and the speed of the moving object and is responsible for the direction of the winds in hurricanes. It is equal to $-2\Omega \times V$, where Ω is the angular velocity of the Earth and V the relative velocity of the air stream.

CORIOLIS PARAMETER: A measure of planetary rotation as a function of latitude. It is twice the vertical component of the Earth's angular velocity.

CORK (PHELLEM): The light, water proof outer layer of the bark of the stem and roots of almost all trees and shrubs. Commercially, only the bark of the evergreen cork oak, *Quercus suber*, is called cork. The importance of cork lies in the fact that it is made of air-filled, watertight cells that provide very effective insulation. Cork is used as stopper for bottles of wine and other beverages.

CORK CAMBIUM (PHELLOGEN): A type of cambium arising within the outer layers of the stems and roots of woody plants. The cells of the cork cambium divide to produce an outer corky tissue, cork or phellem, of non-living cells and an inner secondary cortex of living cells, called phelloderm.

CORM (BULBO-TUBER; BULBOTUBER): An underground stem surrounded by scaly leaves or a short swollen, underground plant stem, surrounded by protective scale leaves. It stores food and is an organ for perennation and vegetative reproduction. The protective brown scales surrounding the corm are the remains of the previous season's leaf bases.



CORN (MAIZE): A tall annual cereal plant, *Zea mays* that originated from Central America, but now cultivated in most tropical countries as well as some parts of temperate regions as a summer crop. It varies greatly in height but is more commonly 2-3 m high. It has a single main culm with several internodes, often with prop roots from the lower internodes. The apex of the culm ends in the tassel, an inflorescence of male flowers. The sword-shaped leaf blades are broad and alternate, usually drooping. Its upper surface is hairy, with a hairless lower surface. The leaf colour is usually green, but may be variegated white, yellow or purple-red. Inflorescence is monoecious, born in separate parts of the plant. The pistillate (female) inflorescence called ears arise from axillary bud apices; while the staminate (male) inflorescence called tassels arise from the apex of the culm. The ovary develops a long style or silk, which looks like tufts of hair and extends from the cob receiving the pollen from the tassel. The ears consist of a robust rachis (cob), with 4-30 rows of almost sessile spikelets bearing the kernels (caryopses). Kernels are of various colours such as white, yellow, blackish, purple, green and red. Corn is used in many forms as a staple food in many regions of the world as vegetable; used as livestock feed; it also produces cooking oil known as corn oil. It is

also used to produce alcohol such as Bourbon whiskey and also used to brew beer. It is used in many industrialised food products; and as a feedstock for the production of ethanol fuel.

CORNEA: The transparent, outer covering at the front section of the vertebrate eyeball that covers the iris and pupil. It shields the eye from germs, dust and other harmful matter. It also acts as the eye's outer lens; controls light entering the eye; and contributes to the eye's focusing ability.

CORNEA: Cooked and then preserved in salt or brine. An example is corned beef, which is beef preserved in brine, chopped and pressed and sold in tins.

CORNICULATE: 1. In botany, it defines structures having horns or horn-like protuberances, as shown in the diagram. 2. **Corniculate cartilage** is either of two small nodules of yellow elastic cartilage articulating with the apex of the arytenoid.

CORNIFIED LAYER: The upper layer of the skin where the cells have died, with their cytoplasm being replaced by keratin. Cornification gives the skin its protective waterproof quality.

COROLLA: The collective name for the petals of a flower and usually of a colour other than green. A corolla tube is formed when the margins of the individual petals are completely or partially fused.

COROLLARY: In mathematics, it is a statement that can be proved easily once a major theorem has been proved; or the outcome that follows from a theorem almost instantly, usually without further proof.

CORONA: 1. A very hot layer of the solar atmosphere above the chromospheres with a temperature of 2,000,000 °C (2,000,273 K). Gas flows away from the corona to form the solar wind. The corona emits most of its radiation at very short ultraviolet and X-ray wavelengths. Much of the ultraviolet and X-ray radiation that hits Earth's atmosphere is absorbed by atoms and molecules in the atmosphere. 2. A crown-like outgrowth from a corolla tube. It is especially prominent in daffodils (Narcissus). 3. A crown-like structure at the tip of the oogonium in certain algae of the Charophyta.

CORONARY THROMBOSIS (HEART ATTACK; MYOCARDIAL INFARCTION): A sudden onset of gripping central chest pain, often accompanied by sweating and vomiting, when the flow of blood to the heart is blocked due to death of a portion of the heart muscle following obstruction of a coronary artery by thrombosis (formation of a blood clot). This is most often caused by a build-up of fat, cholesterol and other substances, which form a plaque in the coronary arteries, obstructing the circulation of blood. Some other symptoms include nausea, indigestion, heartburn or abdominal pain, shortness of breath, fatigue, light-headedness or sudden dizziness. A heart attack is a major cause of death and disability worldwide and death can occur within the first two hours, if immediate medical help is not sought.

CORONARY VESSELS: Two pairs of blood vessels, the coronary arteries and coronary veins that supply blood to the muscles of the heart itself.

CORONIFORM: In botany, it refers to parts or structures such as flowers, which look like a crown, as shown in the diagram.

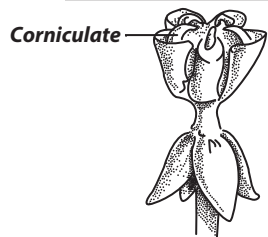
CORPSE (CADAVER): Dead body especially of a human being.

CORPUS CALLOSUM: A wide, flat bundle of axons beneath the cortex in the eutherian brain at the longitudinal fissure that connects the right and left hemispheres of the cerebrum.

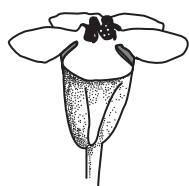
CORPUS CARDIACUM (CORPORA CARDIAC): Any of a pair of separate or fused bodies of nervous tissue that lies immediately behind the brain in insects.

CORPUS LUTEUM: A yellow mass of temporary endocrine structure in mammals, a part of the ovary Graafian follicle after ovulation that secretes the hormone progesterone

CORPUSCLE: 1. A small free floating biological cell, such as a blood



Corniculate



Coroniform

or lymph cell. 2. A rounded globular mass of cells, such as the pressure receptor on certain nerve endings. 3. A discrete particle, such as a photon or an electron.

CORPUSCULAR THEORY (CORPUSCULAR THEORY OF LIGHT): The hypothesis about the nature of light, proposed by Isaac Newton (1642-1727), who postulated that light consists of a stream of particles or corpuscles which travel in straight line with a finite velocity and possess kinetic energy.

CORRASION: The erosion of solid rock surfaces by particles carried by water or ice. It involves wearing away surfaces that the water or ice flows over through the impact or grinding action of particles moving with the water.

CORRELATION: 1. Also called **association**, it is the degree of relationship between two sets of information. If one set of data increases at the same time as the other, the relationship is said to be positive or direct and if one set of data increases as the other decreases the relationship is said to be negative or inverse. Mathematically, a correlation is expressed by a correlation coefficient that ranges from -1 (never occur together), through 0 (absolutely independent), to 1 (always occur together). 2. A bijection of the set of points of projective geometry onto the set of highest-dimension subgeometries of another or itself.

CORRELATION ANALYSIS: The use of empirical correlations relating one body of experimental data to another, with the objective of finding quantitative estimates of the factors underlying the phenomena involved. Correlation analysis in organic chemistry often uses linear free-energy relations for rates or equilibria of reactions, but the term also embraces similar analysis of physical (most commonly spectroscopic) properties and of biological activity.

correlation analysis

CORRELATION COEFFICIENT: A standardised statistical measure of the linear relationship between two variables. Its range is from -1.0 (perfect negative correlation), through 0 (no correlation), to +1.0 (perfect positive correlation). The correlation coefficient between two random variables X and Y , written as r or ρ , is defined as

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{[E(X^2) - (E(X))^2][E(Y^2) - (E(Y))^2]}}$$

$\text{Cov}(X, Y)$ is the covariance between X and Y ; σ_X and σ_Y are the standard deviations of X and Y , respectively and E stands for expectation. The correlation coefficient is always between -1 and 1. It tells whether or not there is a linear relationship between X and Y . If $Y = ax + b$, where a and b are constants and $a > 0$, then r will be close to 1. If X and Y are completely independently, then $r = 0$.

CORRELATION COEFFICIENT SAMPLE (PRODUCT-MOMENT CORRELATION COEFFICIENT; PEARSON'S CORRELATION COEFFICIENT): Where n pairs of values of random variables X and Y in a random sample are denoted by $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$,

the correlation coefficient sample is given by $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$,

where $S_{xy} = \sum_{j=1}^n x_j y_j - \frac{1}{n} \left(\sum_{j=1}^n x_j \sum_{j=1}^n y_j \right)$, $S_{xx} = \sum_{j=1}^n x_j^2 - \frac{1}{n} \left(\sum_{j=1}^n x_j \right)^2$ and S

is analogously defined to S_{xx} . Alternatively, $S_{xy} = \sum_{j=1}^n x_j y_j - n\bar{x}\bar{y}$ and

$$S_{xx} = \sum_{j=1}^n x_j^2 - n(\bar{x})^2.$$

CORRELATION DIAGRAM (WALSH DIAGRAM; ANGULAR

COORDINATE DIAGRAM): A diagram that relates the energy levels of separate atoms to the energy levels of diatomic molecules and united atoms, the two limiting states being correlate with each other. It explains why molecules are more stable in certain spatial

configurations. It is used for making predictions about the geometries of small molecules.

CORRELATION ENERGY: The difference between the Hartree-Fock energy calculated for a system and the exact non-relativistic energy of that system. The correlation energy arises from the approximate representation of the electron-electron repulsions in the Hartree-Fock method.

CORRELATION MATRIX: A square $n \times n$ matrix of which the entries are the pairwise correlations of the variables of a random vector of length n , the $(i, j)^{\text{th}}$ entry is the correlation between the i^{th} and the j^{th} variables. The diagonal elements are each equal to 1.

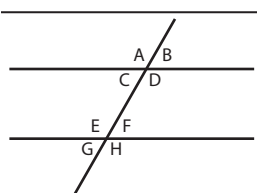
CORRELATION SPECTROSCOPY (COSY): A type of two-dimensional nuclear magnetic resonance (NMR) spectroscopy in which the pulse sequence is $90^\circ x - t_1 - \text{acquire } t_2$. The delay t_1 is variable and a series of acquisitions is made. Fourier transforms are then performed on both the delay t_1 and the real time t_2 . The information thus gained can be shown on a diagram that plots contours of signal intensity. This representation of the information enables NMR spectra to be interpreted more readily than in one-dimensional NMR, particularly for complex spectra.

CORRELATIVE INHIBITION: The suppression of the growth of certain plant parts by a compound, such as a food substance or growth substance, produced in another area of the plant. An example is the production of a substance by the root system that suppresses the development of roots on the stem. If the influence of the root is removed a detached petiole or stem may subsequently produce a profusion of adventitious roots.

CORRESPONDENCE: In mathematics, it is the relation between two sets, where an operation on the members of any one set maps onto some more members of the other.

CORRESPONDENCE THEOREM: In algebra, the result that there is a bijection from the set of subgroups of the image of a group under a homomorphism onto the set of subgroups that contain the kernel of the homomorphism; similar results hold for rings, modules, etc.

CORRESPONDING ANGLES: 1. When a transversal cuts two lines, it forms four pairs of corresponding angles, non-adjacent angles on the same side of the transversal, one internal and one external. In the diagram shown, the following angles A and E , C and G , B and F and D and H are corresponding angles. In Euclidean geometry, if a transversal cuts two parallel lines, then the pairs of corresponding angles that are formed will be equal. 2. When two polygons are congruent or similar, each angle on one polygon is equal to a corresponding angle on the other polygon.



Corresponding Angles
 $A = E, C = G, B = F, D = H$

CORRESPONDING SIDES: Sides in the same relative position in similar or congruent figures. Pairs of corresponding sides are marked with the same number of hash marks. **Corresponding vertices** are vertices in the same relative position in similar or congruent figures. Pairs of corresponding vertices can be identified by their corresponding angles. Sometimes corresponding vertices have the same letter name, but one has a "prime" symbol as in A and A' .

CORRIE (CIRQUE): A steep sided hollow in a mountain side resembling a concave amphitheatre. It is formed at the head of a valley glacier by erosion.

CORROSION: 1. An oxidation-reduction reaction in which the corroding material, especially a metal becomes oxidised. In other words, it is an electrochemical oxidation of a metal into one of its ions in the presence of oxygen and water. It is an irreversible interfacial reaction of a material with its environment which results in consumption of the material or in dissolution into the material of a

component of the environment. Often, but not necessarily, corrosion results in effects detrimental to the usage of the material considered. Exclusively physical or mechanical processes such as melting or evaporation, abrasion or mechanical fracture are not included in the term corrosion. To **corrode** is to destroy slowly by a chemical or electrochemical attack. **Corrosive** is having the ability to corrode. 2. In earth science, it is an alternative name for solution, the process by which water dissolves rocks such as limestone.

CORROSION CELL: A galvanic cell resulting from inhomogeneities in the material or in its environment.

CORROSION POTENTIAL: The electrode potential spontaneously acquired by a corroding material in a particular environment.

CORROSION-RESISTANT STEEL, CRES (INOX STEEL; STAINLESS STEEL): A widely used alloy of iron, chromium and nickel that resists rusting. It contains a minimum of 10.5 or 11% chromium content by mass. Stainless steel does not stain, corrode or rust as easily as ordinary steel, but it is not stain-proof. There are different grades and surface finishes of stainless steel to suit the environment to which the material will be subjected in its lifetime. It is used where both the properties of steel and resistance to corrosion are required. Stainless steels are used in petroleum refineries and chemical plants for the pipes and tanks, for jet planes and for space capsules, for surgical instruments and equipment, for replacing broken bones because the steels can withstand the action of body fluids. It is also commonly used for cutlery and kitchen fittings because it does not taint food and can be easily cleaned.

CORRUGATED: In botany, it refers to parts shaping into folds or alternating furrows and ridges.

CORRUPTION OF DATA: Introduction or presence of errors in data.

CORTEX: 1. Also known as **bark** or **rind**, it is the outer portion of the stem or root of a plant. 2. Unspecialised tissue in plant stems and roots between the vascular bundle and epidermis. 3. The outermost layer of any organ or part of the body. An example is the outer layer of grey matter that covers the surface of the cerebral hemisphere.

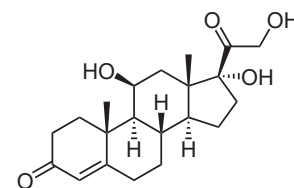
CORTI CELL: Any of the hair cells found in the organ of corti in the inner ear that plays a role in the detection of sound. They receive sound vibrations from the middle ear and send them on to the brain via the auditory nerve, where they are recognised and interpreted as specific sounds.

CORTICAL NEPHRON: One of the major types of the kidney involved in urine production, the other one is the juxtamedullary nephron. Cortical nephrons have their Loop of Henle situated in the renal medulla near its junction with the renal cortex. Its Loop of Henle, is shorter than that of juxtamedullary nephron. A nephron is the functional unit of the kidney, that regulates the concentration of water and soluble substances like sodium salts by filtering the blood, reabsorbing what is needed and excreting the rest as urine. There are about 800,000 to 1.5 million in each human kidney.

CORTICOSTEROID: Any of several steroid hormones such as cortisol and cortisone produced by the cortex of the adrenal glands. Glucocorticoids such as cortisol regulate the use of carbohydrates, proteins and fats metabolism in the body. Mineralocorticoids such as aldosterone control electrolyte and water levels, mainly by promoting sodium retention in the kidney.

CORTISOL (HYDROCORTISONE; 17-HYDROXY-CORTICOSTERONE): A hormone produced by the outer layer or

cortex of the adrenal glands, with the chemical formula $C_{21}H_{30}O_5$. It promotes the synthesis and storage of glucose. Like cortisone, cortisol is classed as a glucocorticoid, which stimulates liver glycogen formation while it decreases the rate of glucose utilisation in body cells. Cortisol affects the metabolism of carbohydrate, protein and fat; the maturation of white blood cells; the

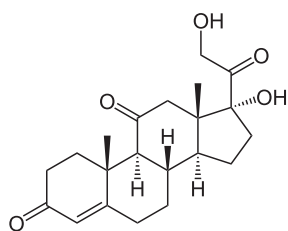


Chemical structure of Cortisol

retention of salt and water in the body; the activity of the nervous system; and the regulation of blood pressure. Secretion of cortisol from the adrenal cortex is stimulated by the pituitary hormone ACTH. It is important in the normal response to stress; and also prevents or suppresses inflammation. A preparation of this hormone obtained from natural sources or produced synthetically is used medically in the treatment of rheumatoid arthritis, adrenal insufficiency, certain allergies and gout.

CORTICOTROPIN-RELEASING HORMONE (CRH): A peptide hormone produced by the hypothalamus that stimulates the adrenal glands to produce adrenocorticotrophic hormone (ACTH) into the bloodstream, which is essential for normal metabolism.

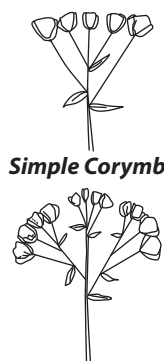
CORTISONE: A naturally occurring corticosteroid, with the chemical formula $C_{21}H_{28}O_5$. It is itself biologically inactive and is formed naturally in the adrenal gland from the active hormone cortisol, which is structurally very similar to it. Cortisone and cortisol are necessary in regulating fat, carbohydrate and protein metabolism. Medically, it is used in the treatment of rheumatoid arthritis, adrenal insufficiency, certain allergies and gout.



Chemical structure of Cortisone

CORUNDUM: An aluminium oxide mineral with traces of iron, titanium and chromium, with the chemical formula Al_2O_3 , melting point $2044^\circ C$ ($2317 K$) and specific gravity 3.95-4.10. It is the hardest naturally occurring mineral with a hardness of 9 on the Mohs scale, second only to diamond. Corundum occurs as a rock-forming mineral in both metamorphic and igneous rocks, which are relatively poor in silica. Igneous rocks which most commonly contain corundum include syenites, nepheline syenites and syenite pegmatites. It can also be synthesised. Its gem varieties are sapphire and ruby. It is commonly used as an abrasive.

CORYMB: A clustered inflorescence with a convex or flat top, which has its lower pedicels longer than the upper. Stalks grow upward from various points to the main stem such that they reach the same height. **Corymbiform** is having the appearance of, but not necessarily meeting the strict definition of a corymb, as seen in the capitulescence of *Doellingeria umbellata*. **Corymbose** refers to having flowers in corymbs.



Simple Corymb

Compound Corymb

CORYZA: 1. Severe nasal congestion caused by the inflammation of the mucous membranes lining the nasal cavity. 2. A respiratory disease of birds, especially chicken and turkey caused by the bacterium *Haemophilus paragallinarum* or *Haemophilus gallinarum*. Symptoms include nasal discharge, facial swelling, sneezing and laboured breathing.

COSECANT: The reciprocal of the sine of an angle, expressed as cosec . $A = \frac{1}{\sin A}$. In a right-angled triangle, it is the ratio of the hypotenuse

to the side opposite the angle A . Its abbreviation and symbol is **cosec** (**csc**). The symbol for its hyperbolic function, hyperbolic cosecant, which is the reciprocal of the hyperbolic sine function, $\sinh$ is **cosech**⁻¹. If θ is the angle measured in radians from the x -axis of a coordinate system towards an anticlockwise direction by a radius of length r centred on the original point, then $\text{cosec} \theta = \frac{r}{y}$, where y is the ordinate of the end of the radius.

COSET: A subset of a given group, written aH or Ha , whose members are the products of a given element, a , of the group with a given subgroup, H . The cosets of H in G are disjoint and form a partition of G . The left and right cosets, aH or Ha , may be distinct in a non-communicative group and if $aH = Ha$ for all a in G , then H is said to be normal in G .

COSH: The symbol for the hyperbolic cosine, which is related to the cosine function by the identity $\cosh z = \cos iz$, for z a complex number and $i = \sqrt{-1}$. It can be defined in terms of the exponential function as $\cosh z = \frac{1}{2}(e^z + e^{-z})$. It is an even function of which an

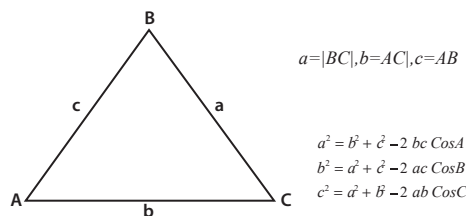
antiderivative or indefinite integral is the sine function, $\sinh$ and the hyperbolic sine function. The functions, $\cos z$ and $\sin z$ both satisfy $\cosh^2 z - \sinh^2 z = 1$

$$\cosh(2z) = \cosh^2 z + \sinh^2 z$$

COSH⁻¹: The symbol for the inverse hyperbolic cosine function, arc-cosh.

COSINE: A trigonometric function of an angle in a right-angled triangle, found by dividing the length of the side adjacent to the angle by the length of the hypotenuse, the longest side. Its symbol and abbreviation is **cos**. Its symbol for the inverse function, arc-cosine is **cos⁻¹**. **Cosh** is the symbol for the hyperbolic cosine, which is related to the cosine function by the identity $\cosh z = \cos iz$, for z a complex number and $i = \sqrt{-1}$. In other words, if θ is the angle measured in radians from the x -axis of a coordinate system towards an anticlockwise direction by a radius r centred on the origin, then $\cos \theta = \frac{x}{r}$, where r is the abscissa of the end radius. It is an even function of which an antiderivative or indefinite integral is the sine function, $\sin$ and the derivative, is $-\sin$. The functions, $\cos z$ and $\sin z$ both satisfy $\cos^2 z + \sin^2 z = 1$ and $\cos(2z) = \cos^2 z - \sin^2 z$. The **cosine law** or **rule** in any triangle, with sides of length a , b and c , is

$$c^2 = a^2 + b^2 - 2ab \cos C, \quad b^2 = a^2 + c^2 - 2ac \cos B \quad \text{or} \quad a^2 = b^2 + c^2 - 2bc \cos A.$$



Cosine law/rule

COSMIC: Of the whole universe; describing outer space or a part of the universe other than the Earth.

COSMIC BACKGROUND EXPLORER, COBE (EXPLORER 66): An orbiting satellite launched in 1989 for cosmological research. Its goals were to investigate the cosmic microwave background radiation (CMB) of the universe and provide measurements that would help shape the understanding of the cosmos.

COSMIC BACKGROUND RADIATION (BACKGROUND RADIATION): Long wavelength electromagnetic radiation that hits Earth uniformly from all directions. Background radiation represents energy left over from the "big bang," the explosion at the beginning of the universe.

COSMIC CENSORSHIP (COSMIC CENSORSHIP HYPOTHESES): A hypothesis concerning singularities and black holes in the general theory of relativity.

COSMIC DUST: Fine particles of matter that gather into clouds in outer space. It is composed of particles in space which are a few molecules to $0.1 \mu m$ in size.

COSMIC RADIATION: Streams of very high energy particles from outer space consisting of protons, alpha particles and light nuclei, which collide with atomic nuclei in the Earth's atmosphere to produce secondary nuclear particles that fall to the Earth.

COSMIC RAY SHOWER (AIR SHOWER; SHOWER): The simultaneous appearance of a number of downward-directed ionising

particles, with or without accompanying photons, caused by a single cosmic ray.

COSMIC RAYS: Elementary particles or atomic nuclei from outer space with high energy as a result of moving at nearly the speed of light reaching the Earth. Primary cosmic rays consist mostly of protons, some alpha particles and lesser amounts of nuclei of carbon, nitrogen, oxygen and heavier atoms. These nuclei collide with nuclei in the upper atmosphere, producing secondary cosmic rays of protons, neutrons, mesons, electrons and gamma rays of high energy, which in turn hit nuclei lower in the atmosphere to produce more particles. These cascade processes continue until all the energy of the primary particle is dissipated. The secondary particles shower down through the atmosphere in diminishing intensity to the Earth's surface and even penetrating it.

COSMIC STRING: Hypothetical 1-dimensional (spatially) topological defects which may have formed during a symmetry breaking phase transition in the early universe, when the topology of the vacuum manifold associated to this symmetry breaking is not simply connected.

COSMID: Fragment of DNA from the human genome inserted into a bacterial cell. It is useful for cloning large fragments of DNA.

COSMOGONY: The study of the origin and evolution of cosmic objects especially the solar system.

COSMOLOGICAL CONSTANT: A term proposed by Albert Einstein to be added to the Einstein field equation for general relativity theory. It is independent of space and time and should allow for the possibility of a static universe. It is believed that the constant is small and has a non zero value, which has not been yet determined.

COSMOLOGICAL DECADE (CD): A division of the lifetime of the cosmos. The divisions are logarithmic in size, on base 10. Each successive cosmological decade represents a ten-fold increase in the total age of the universe.

COSMOLOGICAL DISTANCE: A distance far beyond the boundaries of our Galaxy. When viewing objects at cosmological distances, the curved nature of spacetime could become apparent. Possible cosmological effects include time dilation and red shift.

COSMOLOGICAL PRINCIPLE: A hypothesis that any observer anywhere in the universe has the same view that we have, that is, the universe is not expanding from any centre but all galaxies are moving away from one another. It is the claim that on extremely large scales, the universe is homogeneous and isotropic, having physical properties that do not vary with direction. The cosmological principle is useful because it allows cosmologist to rule out many different possible cosmologies (models of the universe). However, the principle itself is an assertion rather than a statement of proven fact. There is some evidence that, contrary to the principle, distant galaxies may be rather different from our own. All these are not to say that our view of the universe remains the same with the progress of time; the universe is not static and an expanding universe implies a falling density.

COSMOLOGY: The study of the nature origin, evolution and the structure of the universe.

COSMONAUT: A crew member of a space mission launched by Russia or the former Soviet Union. In the U.S. they are called astronauts.

COSMOPOLITAN: In ecology, it defines species that have a worldwide distribution and are not restricted to specific areas.

COST PER ACTION (CPA): A term used in online advertising for how much revenue a publisher receives when a user clicks an advertisement on his website and then completes certain actions such as clicking a link, being directed to the advertiser's website, completing a form or survey. If the action has been completed, the advertiser pays the publisher a certain amount based on the CPA.

COST PER CLICK (CPC): The amount of revenue a publisher receives each time a user clicks an advertisement link on his website.

COST PER LEAD (CPL): The amount of revenue a publisher receives when he creates a lead for an advertiser.

COSTA: The central strand of a bryophyte leaf or a prominent mid-rib.

COSTAL: Pertaining to a rib or the region of the ribs. Thus, costal cartilage is the cartilaginous portion of the ribs which joins them to the sternum. The term intercostal refers to the space between the ribs.

COSTAL CARTILAGE: A cartilage that connects a rib to the sternum. The first seven ribs (true ribs) are directly connected to the sternum by individual costal cartilages. The next three ribs are indirectly connected to the sternum by three costal cartilages, each of which is connected to the one immediately above it.

COSTATE: 1. Having a costa or costae; ribbed. In botany, it defines leaves having veins or ridges, especially parallel one. 2. In anatomy, it defines having ribs.

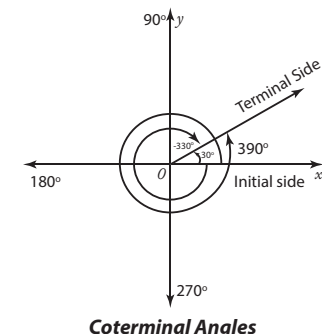
COSTULAR: Pertaining to the ribs or veins.

COTANGENT: A trigonometric function, which is the reciprocal of the tangent of an angle in a right-angled triangle and is equal to the ratio of cosine to sine or to the length of the side adjacent to the angle divided by that of the side opposite the angle: $\cot \theta = \text{adjacent/opposite}$. It is one of the six fundamental ratios of trigonometry, together with tangent, sine, cosine, secant and cosecant. The abbreviation and symbol for cotangent is **cot** or **cotan**. The symbol for the inverse cotangent function, arc-cotangent is **cot⁻¹** or **cotan⁻¹**. The symbol for the hyperbolic cotangent is **cotanh** or **coth**. The hyperbolic cotangent is the reciprocal of hyperbolic tangent function tanh and equal to the ratio of cosh to sinh. Its derivative is $-\text{cosech}^2 x$, with an antiderivative or indefinite integral of $\text{logsinh} x$. **Cotanh⁻¹** or **cot** is the inverse hyperbolic cotangent function or **arc-cotanh**.

COTECTIC: The conditions of pressure, temperature and composition under which two or more solid phases crystallise at the same time from a single liquid over a finite range of decreasing temperature.

COTERMINAL ANGLES:

Angles that have the same terminal side when placed in standard position (angles with the initial side on the positive x-axis). For example, 30° , -330° and 390° are coterminal, as shown in the diagram. To find a positive and a negative angle coterminal with a given angle, add and subtract 360° if the angle is measured in degrees or 2π if the angle is measured in radians.



COTRANSMITTER: A substance, usually a peptide that is released from neurons along with a primary neurotransmitter in order to modify the action of the neurotransmitter. For example, vasoactive intestinal peptide functions as a cotransmitter with acetylcholine at cholinergic synapses and neuropeptide Y performs a similar role with norepinephrine.

COTRANSPORTER: A type of transport protein that transports two or more substances simultaneously across a cell membrane.

COTTON: The soft white wool-like fibre for making fabrics obtained from the fruit or capsule of *Gossypium* species. The material is the epidermal hair of the seeds and is made of pure cellulose. The plant is mostly grown in the sub-tropical and tropical regions. Its seed provides edible vegetable oil and protein. Cotton also provides important raw materials for the chemical industry.

COTTON EFFECT: The displacement of the path of a plane polarised monochromatic light from its zero position or the change in direction of a circularly polarised monochromatic light wave into an elliptical path within the neighbourhood of an absorption band of an optically active substance through which it passes.

COTTON GIN: A machine that separates cotton fibres from the seed. It was invented by American, Eli Whitney in 1793. It used a combination of a wire screen and small wire hooks to pull the cotton through, while brushes continuously removed the loose cotton lint to prevent jams.

COTTON STAINER: A leaping insect, belonging to the Genus *Dysdercus* and family *Gryllidae* that has piercing and sucking mouthparts, long legs and antennae. It pierces cotton bolls and stains the fibres. The male produces a chirping sound by rubbing its forewings together.

COTTON WOOL: A mass of cotton that is used especially for applying liquids or creams to the skin or wound.

COTTRELL PRECIPITATOR (ELECTROSTATIC PRECIPITATOR; ELECTROSTATIC AIR CLEANER): A highly efficient filtration device used to remove fine particulate matter such as dust and smoke from industrial waste gases by attracting them to charged grids or wires. The particles are charged before passing through the field and migrate to an oppositely charged electrode. They are used in removing particles from power plant plumes and in other industrial applications.

COTYLEDON (SEED LEAF): An embryonic leaf in the embryo of a seed, which supplies food during germination. In most plants the cotyledons emerge above the soil with the seedling as it grows to carry out photosynthesis for a time before withering. The seed of angiosperms (flowering plants) has either one in monocotyledons or two in dicotyledons.

COTYLIFORM: 1. In physiology, having the form of a cotyle; shaped like a cup, with a tube at the base. 2. In botany, dish-shaped, with the border erect, as the leaves of *Victoria Amazonica* (Giant Water Lily).

COTYLOID JOINT (BALL-AND-SOCKET JOINT; ENARTHRODIAL JOINT; ENARTHROSIS; SPHEROID JOINT): Asynovial joint in which a spherical knob or knoblike part of one bone fits into a cavity or socket of another, allowing considerable free movement in three dimensions. An example is the joint between the pelvis and the femur.

COUGH: The sudden explosive release of air from the lungs, which clears respiratory passages of obstruction or irritants such as smoke and mucus. It occurs both on reflex and on volition, when the nerve endings in the airways become irritated. Certain irritants and allergens, medical conditions and medicines can irritate these nerve endings. Air flow may reach high velocity and potentially infectious particles spread a great distance if the mouth is uncovered. A cough is a natural reflex that protects the lungs. By clearing the airways, it also helps prevent infection. However, persistent cough points to an underlying medical condition.

COULOMB: The SI unit of electrical charge, which is a derived unit defined in terms of two base units, the ampere and the second. One coulomb is the quantity of electricity conveyed by a constant current of one ampere in one second, which is equal to the charge of approximately 6.241×10^{18} electrons. Its symbol is *C*.

COULOMB BARRIER: An energy barrier resulting from electrostatic interaction that two nuclei must overcome in order that they can approach closely enough to undergo nuclear fusion. It is produced by electrostatic potential energy.

COULOMB COLLISION: A binary elastic collision between two charged particles interacting through their own electric field. As with any inverse-square law, the resulting trajectories of the colliding particles is a hyperbolic Keplerian orbit. This type of collision is common in plasmas where the typical kinetic energy of the particles is too large to produce a significant deviation from the initial trajectories of the colliding particles and the cumulative effect of many collisions is considered instead.

COULOMB COUNTING: A method of determining the state of charge of a battery by integrating the ingoing and outgoing discharge currents of a battery over time.

COULOMB DAMPING: A type of constant mechanical damping in which energy is absorbed via sliding friction. The friction generated by the relative motion of the two surfaces that press against each other is a source of energy dissipation. Its direction is opposite to the direction of the velocity of the particle.

COULOMB EXCITATION: A technique in experimental nuclear physics to probe the electromagnetic aspect of nuclear structure. In coulomb excitation, a nucleus is excited by an inelastic collision with another nucleus through the electromagnetic interaction.

COULOMB EXPLOSION: The sudden disruption of a molecule moving at high velocity after striking a solid. This results in the electrons that bond the molecule being torn off rapidly to leave only the nuclei, which repel each other because of their electric charge. It is a mechanism for coupling electronic excitation energy from intense electromagnetic fields into atomic motion.

COULOMB FIELD: The electric field created by a stationary charged particle or object. The interaction between two such fields produces Coulomb's Force.

COULOMB FORCE: The electrostatic force of attraction or repulsion between two charged particles or bodies. The magnitude of the force *F* is proportional to the product of the charges q_1 and q_2 and inversely proportional to the square of the distance between them r^2 : $F = \frac{Kq_1q_2}{r^2}$

where *k* is a constant that depends on the measurement system being used; q_1 and q_2 are expressed in coulombs, *r* is expressed in metres and *F* is given in newtons. The Coulomb force can be one of repulsion, such as the force between two objects having like charges or it can be attractive, such as the force between two objects having opposite charges.

COULOMB IONISATION: Ionisation produced by Coulomb's forces between a moving particle (projectile) and another particle it interacts/ collides with (target).

COULOMB, CHARLES AUGUSTIN DE (1736-1806): French physicist, best remembered for his work with electric charges. In 1789 he retired from his posts as military engineer and as superintendent of waters and fountains and devoted himself to continuing his scientific research. He was known for his work on electricity, magnetism and friction and he invented a magnetoscope, a magnetometer and a torsion balance that he employed in determining torsional elasticity and in establishing Coulomb's law. The unit of quantity of electric charge, the coulomb, was named in his honour.

COULOMB'S LAW (COULOMB'S LAW OF FORCE): The law that the force exerted between two electric or magnetic charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them. Mathematically, it is given by $F = \frac{Kq_1q_2}{r^2}$, where *k* is a constant that depends on the measurement

system being used, q_1 and q_2 are the charges expressed in coulombs, *r* is the distance between the charges expressed in metres and *F* is given in newtons. If the bodies are oppositely charged, one positive and one negative, they are attracted toward one another; if the bodies are similarly charged, both positive or both negative, the force between them is repulsive. Coulomb's law applies exactly only when the charged bodies are much smaller than the distance separating them and therefore can be treated approximately as point charges. When combined with principles of quantum physics, Coulomb's law helps describe the forces that bind electrons to an atomic nucleus, that bind atoms together into molecules and that hold together solids and liquids. The law was deduced in 1785 by C. A. de Coulomb from experimental measures of the forces between charged bodies. The experiments were made using his torsion balance.

COULOMBIC ATTRACTION: The positive-negative attraction which takes place when two charged particles are in close proximity.

COULOMBMETER: An instrument which measures electric charge in coulombs. The charge *Q* to be measures is allowed to flow on to a capacitor of known capacitance *C*. The potential difference *V* across the capacitor is then measured and the value of the charge calculated using the capacitor equation, $Q = CV$.

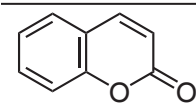
COULOMETRIC DETECTION METHOD: A detection method in which the current is directly proportional to the flow rate of the substance involved in the electrochemical reaction. The amount of charge which flows is proportional to the amount of substances taking part in the reaction.

COULOMETRY: The quantitative application of Faraday's Law to chemical analysis based on the measurement of a quantity of

electricity, measured in coulombs, in an electrode reaction. There are two basic categories of coulometric techniques: **Potentiostatic coulometry** involves holding the electric potential constant during the reaction using a potentiostat; **Coulometric titration** or amperostatic coulometry, keeps the current, measured in amperes, constant using an amperostat.

COULTER: A vertical blade attached to a plough in front of a ploughshare that cuts into the soil.

COUMARIN: Any of a group of fragrant organic compounds, with the chemical formula $C_9H_6O_2$, melting point 71°C (344 K) and boiling point 301°C (574 K). It is found widely in plants and derived from the amino acid phenylalanine. It is used in making perfumes, to scent tobacco and as an anticoagulant in medicine and a warfarin derivative.



Chemical structure of Coumarin

COUNT: 1. To enumerate, to check a collection of items in order to determine the cardinal number. This is to put them in one-to-one correspondence with an initial segment of the natural numbers. 2. To recite numbers in ascending order (up to a stated maximum); for example, to count up to a thousand. Also, to recite in ascending order the multiples of a given number. For example, to count in threes is to count '3, 6, 9, 12, ...'.

COUNT RATE: The number of count per second recorded by a detector near a radioactive source. It depends on the strength, the distance and the direction of the source as well as on the size and geometry of the detector. Count rate is proportional to activity provided the source and the detector are maintained in their positions. The unit of count rate is count per second. The measured count rate should be corrected for the effect of background radiation.

COUNTABLE: A countable set is a set that can be put into one-to-one correspondence with a subset of the set of natural numbers; it may be either finite or countably infinite set (denumerable or enumerable).

COUNTABLY ADDITIVE: Of a set function, S , defined on a class of sets, having the property that, for any countable pairwise disjoint family of sets $\{A_n\}_{n \in N}$ in the class, $S\left(\bigcup_{n \in N} A_n\right) = \sum_{n \in N} S(A_n)$, whenever the union lies in the given class. For example, by definition, a measure is countably additive on its sigma-algebra.

COUNTABLY INFINITE (ENUMERABLE; DENUMERABLE): A set is countably infinite if its elements can be put in one-to-one correspondence with the set of natural numbers. For example, the set of integers $\{0, 1, -1, 2, -2, 3, -3, \dots\}$ is infinite. In other words, whether finite or infinite, the elements of a countable set can always be counted one at a time and, although the counting may never finish, every element of the set will eventually be associated with a natural number.

COUNTER (RADIATION COUNTER): An electronic circuit such as Geiger counters and scintillation counters, which is able to detect or measure radioactivity by counting the resultant ionising events even at high rates.

COUNTERCURRENT MULTIPLIER SYSTEM (COUNTERCURRENT MULTIPLICATION): A biological mechanism that expends energy to create a concentration gradient. It occurs especially in mammalian organs. For example, the loop of Henle in the kidney is responsible for the production of concentrated urine in the collecting duct of the nephrons.

COUNTER EXAMPLE: An example that refutes or disproves a general statement, a proposed theorem or that shows an argument to be invalid by satisfying its antecedent or premises while rendering its consequent or conclusion obviously false. For example, the proposed theorem "All integers have rational square roots" can be disproved by finding a counterexample – in this case, by showing that $\sqrt{2}$ is not rational.

COUNTER ION: An ion of opposite charge accompanying a given ion. For example, in a crystal of sodium chloride, the chloride ion is the counter ion to the sodium ion and vice versa.

COUNTER TUBE: Radiation detector consisting of a gas-filled tube or valve whose gas amplification is much greater than one, in which the individual ionising events give rise to discrete electrical pulses. Often an expression is added indicating the geometry, composition of the gas, for example, helium; or the physical process for its operation for example, proton recoil or fission.

COUNTER-CURRENT HEAT EXCHANGE (COUNTER CURRENT FLOW): A counter flow mechanism that enables fluids such as liquids and gases at different temperatures flowing in channels in opposite directions to each other to exchange their heat content without mixing. For example, in a distillation column, the vapours bubble up through the downward flowing liquid while exchanging both heat and mass.

COUNTER-DOMAIN: Of a relation, the set of objects for which there is something that has the given relation. If the relation is regarded as a function, the counter-domain is the range rather than the codomain of the function.

COUNTER-HARMONIC MEAN: The fourth mean of ten defined by the later Pythagorean school in terms of proportions. It is given as: $\frac{a^2 + b^2}{a + b}$.

COUNTERFACTUAL (COUNTERFACTUAL CONDITIONAL): A subjunctive conditional statement in the form "If A had not occurred, C would not have occurred", whose antecedent is in fact false and that is usually interpreted modally in terms of truth of the consequent in the nearest possible world in which antecedent is true. In other words, it is a conditional statement that contains a clause, which is contrary to the fact. In the above example, A is the antecedent and B is the consequent.

COUNTERIMAGE (INVERSE IMAGE): The set of elements whose image under a mapping lies in a given set. It is denoted by $f^{-1}(S)$ and is well defined even if the inverse mapping is not; for example, the counterimage of the square root function on the non-negative reals is all the reals, even though this is a set-valued function.

COUNTERPART: In logic, the object in one possible world that most closely resembles a given object in another and that therefore figures in certain accounts of the semantics of counterfactual statements about the latter. On this account, counterparts of the same need not be identical and indeed if a counterfactual statement is true they will differ in exactly that respect. For example, Nelson might have been celibate is true if and only if there is some possible world (which may be the actual world) in which the predicate celibate is satisfied by that world's counterpart of Nelson.

COUNTERSHADING: A type of camouflage used in animals such as whales and dolphins, great white sharks and some fish and birds in which they develop dark colours on the areas exposed to the sun and light colours on the undercarriage.

COUNTING EFFICIENCY (EFFICIENCY): The ratio between the number of particles or photons counted with a radiation counter and the number of particles or photons of the same type and energy emitted by the radiation source.

COUNTING LOSS: A reduction of the counting rate resulting from phenomena such as the resolving time or the dead time.

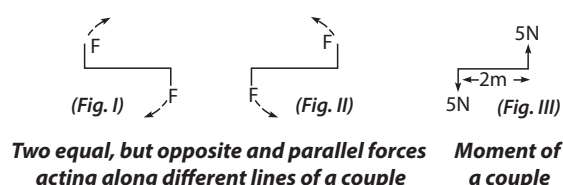
COUNTING MEASURE: The measure function whose value for each finite subset of a given set is its cardinality. The counting measure can be defined on any measurable set, but is mostly used on the sigma-algebra of all countable subsets or on the entire powerset.

COUNTING NUMBERS (NATURAL NUMBERS): Any number that is used for counting, which does not include zero, decimals, negative numbers and fractions. Examples are 1, 2, 3, 4, 5, ...

COUNTING-UP SUBTRACTING: Subtracting two numbers by counting or adding up from the smaller number to the larger number. For example, to calculate $87 - 49$, start at 49, add 30 to reach 79, and then add 8 more to reach 87. The difference is $30 + 8 = 38$.

COUPE: An area of a forest in which trees have been cut down.

COUPLE: In mechanics, it is a pair of equal and parallel forces that are acting on a body in opposite directions. The turning effect or moment, of a couple is measured by the product of the magnitude of either force and the perpendicular distance between the action lines of the forces. A body acted upon by a couple will rotate in a clockwise direction or anticlockwise direction. The turning effect or moment of a couple is the product of the magnitude of either force and the perpendicular distance between the action lines of the forces. If the two forces acting on the body have the same line of action, then the moment becomes zero and there is no turning effect. Some examples of the turning effect of two equal and opposite forces not acting in the same straight line are the steering wheel and the pedals of a bicycle. For example, in a steering wheel, when making a turn to the left, the left hand pulls with force F on the steering wheel towards anticlockwise direction, while the right hand pushes with the same force F , but in a clockwise direction. In riding a bicycle, one pedal is being pushed forward while the other is being pushed back. This rotates the sprocket wheel and the attached chain anticlockwise. Some other examples are the forces exerted by the tip of a screw-driver on the head of a screw; and drag forces acting on a spinning propeller. In the diagram (Fig I), the turning effect or moment, is $\text{Moment} = \text{Force (N)} \times \text{Perp. Distance (m)}$ between the forces $= 5 \text{ N} \times 2 \text{ m} = 10 \text{ Nm}$.



COUPLED: Of a pair of equations, interdependent or interrelated in some fashion.

COUPLED AXLE DRIVE: A transmission arrangement in which one or more motors are coupled to a group of axles.

COUPLED CLUSTER METHOD: A method employed in *ab initio* quantum mechanical calculations providing for a higher-level treatment of electron correlation beyond MP4.

COUPLED SIMULTANEOUS TECHNIQUES: The application of two or more techniques to the same sample when the two instruments involved are connected through an interface, for example, simultaneous thermal analysis and mass spectrometry.

COUPLING: 1. An interaction between two different parts of a system or between two or more systems. 2. An interaction between two electrical or electronic circuits so that energy is transferred from one to the other. 3. The linking of electron transport and oxidative phosphorylation in isolated mitochondria. 4. The covalent linking of one chemical entity to another. An example is the formation of an azo compound by coupling of a diazonium ion with a benzene ring. 5. The mechanical device that connects shafts together for power transmission or connect the shaft of a machine to an external drive. An example is the connection between a motor that drives a pump. 6. In computer science, it is also called **dependency** and it is the degree to which each program module relies on each one of the other modules.

COUPLING CONSTANT: 1. A physical constant that is a measure of the strength of interaction between particles. Its symbol is g . 2. Also known as **coefficient of coupling** or **coupling coefficient**, it is a measure of the strength of the coupling between two systems, especially electric circuits. The maximum coupling is 1 and no coupling is 0.

COURTSHIP BEHAVIOUR: A type of animal behaviour which establishes contact between the sexes, resulting in copulation and reproduction. It plays a part in the initial attraction of a mature male to a female, a prelude to copulation. An example is a bird song and

display. It may be chemical, visual or auditory stimuli or it may be a highly complex series of acts by two or more individuals using several modes of communication.

COVALENCY OF AN ELEMENT: The number of covalent bonds formed by the atoms of a given element. It is sometimes equal to the number of unpaired electrons in the atom.

COVALENT BOND: A chemical bond produced when two atoms share one or more pairs of valence electrons. The outermost orbitals of the atoms overlap so that unpaired electrons in each of the bonding atoms can be shared. By overlapping orbitals, the outer energy shells of all the bonding atoms are filled. The shared electrons move in the orbitals around both atoms. As they move, there is an attraction between these negatively charged electrons and the positively charged nuclei. This attractive force holds the atoms together in a covalent bond. A single covalent bond is formed when two electrons are shared between the same two atoms, one electron from each atom. A double covalent bond is formed when four electrons are shared between the same two atoms, two electrons from each atom. A triple covalent bond is formed when six electrons are shared between the same two atoms, three electrons from each atom. **Covalent compounds** are compounds which contain covalent bonds and tend to be nonionic. An example is methane (CH_4). Some properties of covalent bonds include the following: (i) they have lower melting and boiling points than ionic compounds; (ii) they have more flexibility than ionic compounds; (iii) they are generally not very soluble in water, for example, plastics are covalent compounds and many plastics are water resistant; (iv) they generally do not conduct electricity when dissolved in water, for example, iodine dissolved in pure water does not conduct electricity.

COVALENT CRYSTAL (VALENCE CRYSTAL): A crystalline compound such as diamond in which the dominant forces of interaction are covalent bonds; or a crystalline compound held together by covalent bonds.

COVALENT RADIUS (ATOMIC RADIUS): The effective radius of an atom involved in a covalent bond. It is measured as half the distance between two nuclei of identical covalently bonded atoms.

COVARIANCE: In probability theory, covariance is a measure of how much two random variables change together. Covariance of two random variables is the difference between the expected value of their product and the product of their separate expected values. Mathematically, it is given by $\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$, where X and Y are random variables. If X and Y are independent then $\text{Cov}(X, Y) = 0$ not necessarily independent.

COVARIANCE MATRIX (VARIANCE-COVARIANCE MATRIX): A square matrix, usually denoted Σ in which variance $[\text{Var}(X_i)]$ and covariance $\text{Cov}(X_i, X_j)$ are displayed with the variances appearing along the main diagonal and the off-diagonal elements (those in the i, j position) contain the covariances between all possible pairs of variables. The matrix is then symmetrical and non-negative definite. If $X_1, X_2, \dots, X_n$ are random variables then the variance-covariance matrix is given by $\Sigma_{ij} = \text{cov}(x_i, x_j) = [(x_i - \mu_i)(x_j - \mu_j)]$.

COVARIATE (CONTROL VARIABLE): A variable that has no effect that is of no direct interest and analysis of the variable of interest. It can affect the outcome of an experiment or study and is made more accurate by controlling for variation in the covariance.

COVELLITE: A mineral form of copper sulfide (CuS). It occurs with other copper minerals and is mined as an ore. It forms hexagonal, platy crystals, which have a deep indigo blue colour, often tinged with purple. It cleaves into thin, flexible plates. Its Moh scale hardness is 1.5-2.0, with a relative density of 4.7.

COVER (COVERING): 1. In mathematics, a collection of sets whose union contains a given subset. 2. In a graph, a vertex cover is a set of vertices with at least one endpoint of each edge; a vertex cover is minimum if no other vertex cover has fewer vertices.

COVER CROP: Close-growing plant grown for specific reasons rather than the crop yield. The reasons include suppressing weeds, managing soil erosion, helping to build and improve soil fertility and

quality, and controlling diseases and pests. Cover crops are typically grasses or legumes but other green plants can be used. Examples include rye that looses compact soil and suppress weeds; buckwheat that helps prevent erosion and suppress weeds; clover for fixing nitrogen in the soil, adding fertility; and sorghum to add biomass, and suppress weeds.

COVER SLIP: A piece of thin glass used to cover a specimen on a microscope slide.

COVER-MANAGEMENT FACTOR (C-FACTOR): The effect of cropping and management practices on erosion rates that policy makers and farmers can readily apply in order to help reduce soil loss rates. The C-factor is the ratio of the soil loss from land with a specific type of land use/land cover to the soil loss from continuously fallow and tilled land under the same environmental conditions. The C-factor is unitless, with a value between 0 and 1. Some of the land use and management factors that affect soil loss are vegetation, type of crop, and tillage practice.

COVERSED SINE: A trigonometric function that is equal to $1 - \sin x$.

COW: A fully grown female of usually large, herbivorous animals that constitute the genus *Bos*, of the family Bovidae, especially the domestic kind kept by farmers to produce milk and beef. The name is also applied to the mature female of various, usually large, animals such as elephant, rhinoceros and whale.

COW BELL: A bell worn round the neck of a cow to make it easier for the farmer to locate the animal.

COWPEA (SOUTHERN PEA; BLACK EYE PEA; BLACK EYE BEAN):

A legume, *Vigna unguiculata* belonging to the subfamily Papilionoideae, family Fabaceae formerly Leguminosae. It is grown all over the tropics as a green vegetable and also for fodder and as green manure. Its symbiotic nitrogen-fixing abilities help maintain soil fertility.

COWPER'S GLAND (BULBOURETHRAL GLAND): Either of two small exocrine glands, about the size of a pea present in the reproductive system in male mammals, and connected to the urethra. It produces a thick and clear liquid that mixes with semen as it passes down the urethra, to protect the spermatozoa against excess vaginal acidity.

COWPOX (VACCINIA): An infectious viral disease of cattle, characterised by pustular eruptions, especially on the udders and teats caused by virus in the genus *Orthopoxvirus*. It can be transmitted to humans by direct contact. Persons infected with cowpox become immune to smallpox, a similar but more serious disease. The virus is used as component of the vaccine against smallpox.

COXA: The first segment attached to the thorax (the side of the lower surface) and found on each thoracic segment of the leg of an insect such as cockroach, spider or certain other arthropods. The coxa is typically short and rather stout, although it varies in shape among taxa. It is set in a coxal cavity and articulates with the thorax at the coxal process of the pleural sulcus (groove).

COXAL GLANDS: Paired ducts in arthropods that lead from the coelom to the exterior and are normally involved in excretion.

CP INVARIANCE (CHARGE CONJUGATION AND PARITY): The symmetry generated by the combined operation of changing charge conjugation and parity.

CpG ISLAND: A stretch of DNA, generally several hundred to several thousand bases long, relatively rich in dinucleotides containing the bases of cytosine and guanine with the cytosines (C) located 'upstream' of a guanine (G) residue. The 'p' denoted a phosphodiester bond between the cytosine and the guanine.

CPMG SEQUENCE (CARR-PURCELL-MEIBOOM-GILL SEQUENCE):

A sequence of pulses used for spin-echo experiments in nuclear magnetic resonance (NMR) in which the initial pulse is 90° followed by a series of 180° pulses.

CPRU: Acronym for **Computer Professionals for Social Responsibility**, it is a non-profit global organisation that was founded in 1981 to promote the responsible use of computer technology and

educates the public and policymakers with realistic assessments of the power, promise and problems of Information Technology and the effects of computers on society.

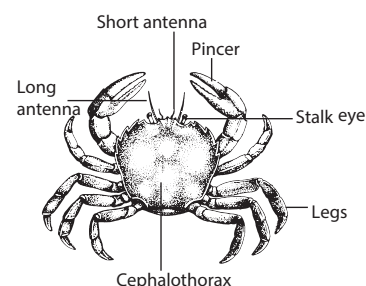
CPT THEOREM: The theorem that the combined operation of changing charge conjugation *C*, parity *P* and time reversal *T*, denoted by CPT, is a fundamental symmetry of relativistic quantum field theory. No violation of the CPT theorem is known experimentally. When *C*, *P* and *T* or any two of them are violated, the principles of relativistic quantum field theory are not affected. However, violation of CPT invariance would drastically alter the fundamentals of relativistic quantum field theory. It is not known whether superstrings obey versions of the CPT theorem.

CPU (CENTRAL PROCESSING UNIT; PROCESSOR): The main unit of a computer, where most of the processing takes place. It includes the control unit (CU) and the arithmetic/logic unit. Its function is to fetch instructions from memory, decode them and execute the program. It also provides timing signals. An integrated circuit that has a complete CPU on a single silicon chip is called a **microprocessor**.

CRAB: The common name for a number of crustaceans in the order Decapoda, which live in the sea and fresh water and also on marshy land all over the world. There are about 5,000 different species of crabs; about 4,500 are true crabs and about 500 are hermit crabs, which do not have a very hard shell and use other animals' old shells for protection. They vary in size from a few millimetres wide, to the Japanese spider crab (*Macrocheira kaempferi*), with a leg span of up to 4 metres. They have a flattened body, two feelers (antennae) and two eyes located at the ends of stalks. They have a very hard shell called **carapace** that both protects them from predators and provides support. The carapace moults during growth. They have five pairs of walking legs, with the first pair modified as pincers or claws called **chelipeds**. Crabs typically walk sideways because of the articulation of the legs which makes a sidelong gait more efficient; although some crabs are known to walk forwards or backwards. The crab's abdomen, known as the **apron** is a moveable flap. In the male it is shaped like a narrow triangle or an inverted T, but in the female it is broader. Marine crabs breathe underwater using gills, which are located in two cavities under the carapace. True land crabs have enlarged, modified cavities that act like lungs so that they can breathe air. Many crabs are omnivores (plant- and meat-eaters), others are carnivores (meat-eaters) and some are herbivores (plant-eaters).

CRACK: 1. The common name for a chemical derivative, bicarbonate of cocaine in hard, crystalline lumps, with the chemical formula $C_{17}H_{21}NO_4$. It is heated and inhaled as a stimulant. 2. In chemistry, to break into simpler molecules by means of heat. 3. In computing, it is to break into a computer system or to break or decipher codes or software modification to remove the unwanted features by the person cracking.

CRACKING: A reaction in which molecular bonds of large alkanes are broken down by thermal, catalytic or hydrocracking means into smaller alkanes and alkenes. For example, the higher boiling hydrocarbon fractions, the large molecules, such as crude oil can be processed to lower boiling fractions; the smaller molecules, such as liquefied petroleum gas (LPG) or gasoline. Catalytic cracking involves the use of a mixture of silica and alumina as catalyst and at lower temperature and pressure to convert higher boiling petroleum fraction into smaller fraction.



External features of a crab

CRADLE KNOLL: A small knoll formed by earth that was raised and left by a large tree uprooted tree.

CRAG: A large lump of rock often found in previously glaciated areas that a glacier has been unable to wear away; or a high steep or rugged mass of rock projecting upward or outward.

CRAMER'S RULE: A method for solving a set of simultaneous linear equations using determinants. It was named after Gabriel Cramer (1704-1752), who published the rule for an arbitrary number of unknowns in 1750. For the 2×2 system:

$a_1x + b_1y = k_1$, $a_2x + b_2y = k_2$. The rule states that:

$$x = \frac{\begin{vmatrix} k_1 & b_1 \\ k_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} \text{ and } y = \frac{\begin{vmatrix} a_1 & k_1 \\ a_2 & k_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

The vertical lines symbolise determinant of the whole matrix of coefficients. This determinant appears in the denominator of the solution for each variable. To calculate the numerator of the solution for x , set up the same matrix but make one substitution: cross out the column that contains the coefficients of x and replace the column with the column of constants from the other side of the equal sign. To use the rule to solve a system of n equations in n unknowns, you will have to calculate $n+1$ determinants of dimension $n \times n$. For an example of the method, we can find the solution of this two equation system: $3x - 2y = 5$ and $2x + y = 7$. The determinant in the denominator is $\begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} = 3 \times 1 - 2 \times (-2) = 7$. The following are the two determinants in the numerators:

$$\begin{vmatrix} 5 & -2 \\ 7 & 1 \end{vmatrix} = 5 \times 1 - 7 \times (-2) = 19 \text{ and}$$

$$\begin{vmatrix} 3 & 5 \\ 2 & 7 \end{vmatrix} = 3 \times 7 - 5 \times 2 = 11; \text{ then } x = \frac{19}{7} \text{ and } y = \frac{11}{7}.$$

CRAMP: A painful sensation caused by muscle contraction. The common cause of cramp may be due to sodium imbalance and tiredness of the muscles. It can also be caused by menstruation or gastroenteritis, if it is felt in the smooth muscles.

CRANE: A machine for lifting, lowering or placing in position heavy objects. It is often powered by electricity, diesel or hydraulic. It uses devices such as a wire wound on a drum, wire ropes or chains and sheaves to create mechanical advantage and thus move loads beyond the normal capability of humans.

CRANIAL: A term describing activities and parts of the body related to the skull and cranium.

CRANIAL NERVE: Either of a pair of nerves that originate in the brain stem and pass out of the skull to the surface of the body. There are 12 pairs of cranial nerves in mammals, birds and reptiles and usually 10 in fish and amphibians.

CRANIATA (VERTEBRATA): A subphylum of the phylum Chordata that includes the Agnatha (jawless vertebrates) and Gnathostomata (jawed vertebrates). Examples of Agnatha are lampreys and hagfish.

CRANIUM: 1. The skull of a vertebrate. 2. The part of the skull of vertebrates that contains the brain.

CRANK: A handle bent at right angles and connected to the shaft of a machine. It is used to transmit motion or convert reciprocating (back and forwards or up and down) movement into rotary movement or vice versa.

CRANKSHAFT: A part of some machines that converts reciprocating motion or up and down motion into rotating motion. For example, the pedals on a bicycle are joined to a simple crankshaft, which converts the pedalling action to the rotation of the wheels to propel the bicycle.

CRASH: 1. To break into pieces violently and noisily. 2. To collide, especially violently and noisily, as occurs, for example, when a motor vehicle collides with something. 3. To cause, for example, an aircraft to make a landing in an abnormal manner, usually damaging or wrecking the aircraft. 4. Also known as **system crash**, it is a software program error or hardware problem that is encountered without warning. A

crash is often caused by either a software error where either the programming has errors or with conflicts encountered with other programs. A crash may also occur because of a hardware problem, such as a failing component. For example, when a hard drive dies it is often referred to as a hard drive crash.

CRASPEDROMOUS (PENNI-PARALLEL): A form of leaf venation in which there is a single primary vein running down the centre of the leaf with secondary veins branching off this in an essentially parallel fashion. An example is the leaf of the sweet chestnut (*Castanea sativa*).

CRASSULA (BAR OF SANIO): One of a pair of bar-like thickenings made up of primary wall and intercellular material found adjacent to the bordered pits of most gymnosperm tracheids.

CRASSULACEAN ACID METABOLISM (CAM PHOTOSYNTHESIS): A carbon fixation pathway, which is characteristic of plants such as cacti and bryophyllum that are adapted to arid regions, in which carbon dioxide (CO_2) is taken up only at night and stored to use the following day, to carry out photosynthesis. During the night, when the weather is relatively cooler and more humid, the stomata open for the uptake of carbon dioxide (CO_2) that is stored as a four-carbon acid, malate, which is released during the day-time to provide the enzyme, rubisco with high concentration of CO_2 to carry out photosynthesis. During the hot day-time, the stomata close to reduce water loss through evapotranspiration. Plants that undergo this pathway are called CAM plants.

CRATER: A circular bowl shaped depression, usually with steep sides. Craters are formed by explosive events such as the eruption of a volcano or by the impact of a meteorite.

CRATERIFORM: Shaped in the form of a bowl or like a bowl.

CRATON: A core of a continental crust or lithosphere that has remained stable without any deformation for, perhaps hundreds of million years. Its interior layer, called a **platform** is composed of igneous or metamorphic rocks covered by a thin layer of sedimentary rock.

CRAYFISH: These are freshwater crustaceans (aquatic) which have similar features as lobsters. They are also known as crawfish, crawdads, freshwater lobsters, mountain lobsters and mud bugs. Taxonomically, they are members of the super families, Astacoldea and Parastacoldea. They breathe through feather-like gills. Its flesh is eaten as food.

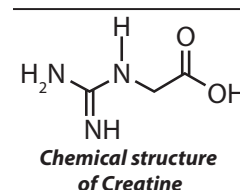
CRAZING: Formation of cavities, when a polymer is stressed, which contain load-bearing fibrils spanning the gap between the surfaces of each cavity.

CREAM: 1. The fatty part of milk, which rises to the surface when the liquid is allowed to stand when not homogenized. 2. A soft solid or thick liquid containing medicaments or other specific ingredients, applied externally for a prophylactic, therapeutic or cosmetic purpose.

CREAMER: Chemically bonded hybrid material which is a crosslinked organic-inorganic polymer. Ceramers are usually prepared by sol-gel processing of oligomers or polymers with reactive silyloxy substituents.

CREAMING: The macroscopic separation of a dilute emulsion into a highly concentrated emulsion, in which interglobular contact is important and a continuous phase under the action of gravity or a centrifugal field. This separation usually occurs upward, but the term may still be applied if the relative densities of the dispersed and continuous phases are such that the concentrated emulsion settles downward.

CREATINE: A nitrogenous organic acid, with the chemical formula $\text{C}_4\text{H}_9\text{N}_3\text{O}_2$ that occurs naturally in the muscle tissue of vertebrates. It is synthesised from the amino acid arginine, glycine and methionine. In the form of creatine phosphate, it is an important reserve of energy for muscle contraction.



CREATIONISM: The theory concerned with the origins of matter and life according to the bible in Genesis that the world and humanity were created by a supernatural creator not more than 6,000 years ago, rather than by natural processes such as evolution. A proponent of the theory of special creation is called a **creationist**.

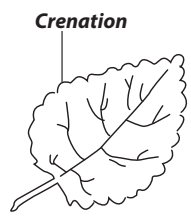
CRECHE: In biology, it is a flock of birds, not necessarily belonging to the same species that gather together for protection.

CREEP: 1. In material science, it is the property of a solid material, typically metal, under continuous stress to slowly deform below its yield point, the point at which any elastic solid normally stretches without any increase in load or stress. 2. **Creeping** is the growth of some plants such as vines along the ground, up walls, and on trees, often winding around other plants to support themselves. 3. Also called **soil creep** or **downhill creep**, it is the gradual movement of rock materials and soil down a slope under the influence of water and gravitation.

CREEP FEEDING: In animal breeding, it is a means of providing nutrition to young animals such as piglets, calves, and lambs to supplement their mother's milk. It is accomplished by giving them access to extra feed or better pasture in partitioned part of their environment to exclude the parents or older animals. Creep feeding teaches young animals to feed by themselves. It reduces the stress of early weaning. The rumens of creep-fed young animals will develop faster.

CREMOCARP: A dry dehiscent fruit such as fennel, carrot and parsnip that is a type of schizocarp. It is formed from two one-seeded carpels, which separate at maturity and remain pendant from the carpophore or from a supporting axis.

CRENATION: 1. A medical condition in which the red blood cells lose water and shrink after exposure to a hypertonic solution. Crenation occurs because in a hypertonic environment, the cell has a lower concentration of solutes than the surrounding extracellular fluid, resulting in a net movement of water out of the cell by osmosis, causing the cytoplasm to decrease in its volume. 2. A rounded projection or tooth along the margin of a leaf, as shown in the diagram.

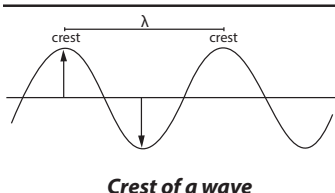


CREOSOTE (CREOSOTE OIL; PITCH OIL): A thick, black or brown oily liquid derived from coal tar and used to preserve wood. Medicinal creosote is transparent and oily and derived from wood tar and used as an antiseptic and expectorant.

CRESCENT: The curved shape of the Moon when it appears less than half illuminated. It also refers to any object or symbol resembling the crescent Moon.

CRESOL (METHYLPHENOL): One of the three poisonous, colourless isomeric methylphenols, comprising o-cresol, m-cresol and p-cresol with the chemical formula $\text{CH}_3\text{C}_6\text{H}_4\text{OH}$. It is used in the production of phenolic resins, disinfectants and also as solvents.

CREST: A point on the wave profile where the particles of the medium have the maximum upward displacement. A **trough** is a point on a wave at which the displacement is at a minimum.



CRETACEOUS: A period of geological time spanning 146-65 million years ago. It is the last period of the Mesozoic era, during which angiosperm or seed bearing plants evolved and dinosaurs reached a peak before their complete extinction at the end of the period.

CRETINISM: An iodine deficiency congenital condition characterised by stunted physical and mental development. It is due to deficiency of thyroxine, a hormone secreted by the thyroid gland, which comes as a result of iodine deficiency in the diet.

CREUTZFELDT-JAKOB DISEASE (CJD): A degenerative neurological disease of humans characterised by intellectual and cognitive deterioration, dementia and destruction of brain tissue. It is a type of mad cow disease, MCD (bovine spongiform encephalopathy, BSE). It is believed to be caused by an abnormal prion protein and is transmissible, although there is also an inherited familiar form.

CREVASSE: A large deep crack in the upper surface of a glacier, several metres in depth. Crevasses often occur as a result of the movement and resulting stress associated with the shear stress generated when two semi-rigid pieces above a plastic substrate have different rates of movement. The resulting intensity of the shear stress causes a breakage along the faces. **Crevasse fillings** are ridges or hummocks formed from glacial sediments that were deposited by water in the cracks and crevasses of the ice.

CREVICE: A narrow opening or crack, especially in rock.

CRG PROCESS (CACTALYTIC RICH GAS PROCESS): An industrial process for producing fuel gas from naphta and other hydrocarbon sources.

CRIBIFORM: In the form of a sieve.

CRINITE: Having soft hair or tuft.

CRINKLE: Forming wrinkles or twists.

CRISTA: A shelf-like structure in a mitochondrion formed by in-folding of the inner membrane. The cristae greatly increase the surface area of the inner mitochondrial membrane, providing a large area for the reactions of the electron transfer system to occur. Without the cristae, the inner membrane of the mitochondrion would be reduced to a single spherical shape, with less reaction surface available and reduced reaction efficiency.

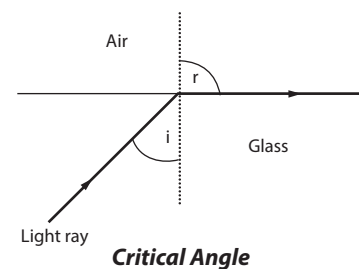
CRISTOBALITE: A mineral form of silicon(IV) oxide, with the chemical formula SiO_2 . Cristobalite is stable only above 1470 °C (1743 K). It occurs in small cavities as lithophysae in volcanic rocks. It crystallises in the tetragonal system at low temperatures and the isometric system at high temperatures.

CRITERIA: A property, dimension or characteristic by which something is judged or evaluated. For example, the six criteria for scientific theories are consistent, observable, natural, predictable, testable, and tentative (CONPTT). Another example of criteria are the various Scholastic Aptitude Test (SAT) scores which evaluate a student's potential for a successful educational experience at college.

CRITH: A unit of mass used in weighing gases. One crith is the mass of one litre of hydrogen gas (H_2) at standard temperature and pressure. It is found experimentally to equal 8.9885×10^{-5} kilograms.

CRITICAL ANGLE: 1. The angle of incidence in a dense medium at which the angle of refraction is 90°.

When a light ray emerges from a denser medium into a less dense medium, it is refracted and bends away from the normal. As the angle of incidence (i) increases so is the angle of refraction (r). When the angle of incidence (i) equals the critical angle, the refracted ray is just touching the glass surface making an angle of 90° with the normal. Critical angle is a function of frequency; the higher the frequency the larger the critical angle. The relationship between the refractive index (μ)



of a medium and the critical angle (c) is given as $\mu = \frac{1}{\sin i_c}$.

2. The minimum angle of incidence at which a radio wave of a particular frequency can strike an ionosphere and still be reflected to the earth. 3. Also known as critical angle of attack, stalling angle or critical angle, it is the angle between the chord line of an aerofoil and the undisturbed relative airflow at which airflow abruptly changes, causing changes in the lift and drag of an aircraft. At this angle, stalling occurs.

CRITICAL ANODE VOLTAGE: The minimum value of the anode voltage at which anode ignition occurs for a stated grid voltage.

CRITICAL ANODIC CURRENT DENSITY: The maximum anodic current density observed in the active region for a metal or alloy electrode that exhibits active-passive behaviour in an environment.

CRITICAL HABITAT: An area essential for the survival and conservation of a critically endangered or endangered species, though the area need not actually be occupied by the species at the time it is designated.

CRITICAL CURRENT DENSITY: The electric current density at the critical current using either the cross-section of the whole conductor (overall) or of the non-stabiliser part of the conductor if there is a stabiliser.

CRITICAL DAMPING: Any effect that is just enough to reduce the amplitude of oscillations in an oscillatory system. A critically damped system returns to equilibrium as quickly as possible without oscillating, preferred for the damping of systems such as doors.

CRITICAL DAY LENGTH: The period of daylight, specific in length for any given species, that appears to initiate flowering in long-day plants or inhibit flowering in short-day plants. Long-day plants will not flower if the dark period exceeds a certain maximum and conversely short-day plants will not flower unless the dark period exceeds a certain minimum. These periods are termed **critical dark periods** and must be continuous to have effect.

CRITICAL DENSITY: 1. The density of a substance at its critical temperature and under its critical pressure. Below this density, saturated granular material will lose strength and above it will gain strength when subjected to rapid deformation. 2. The minimum density of matter that stops the universe from expansion. Its unit is ρ_c .

Its value which depends on the Hubble Constant, H can be calculated as: $\rho_c = \frac{3H^2}{8\pi G}$, where G is Newton's gravitational constant. Its value is approximately 10^{-26} kg/m^3 (or 10 hydrogen atoms per cubic metre).

CRITICAL ENERGY (THRESHOLD ENERGY): The total energy available, which must be greater than the rest energy of a body for a chemical reaction to occur. If critical energy is not available, a chemical reaction cannot occur, except by quantum-mechanical tunnelling.

CRITICAL EXPONENTS: Numbers that quantify how thermodynamic quantities, such as the specific heat capacity, diverge at the critical temperature of a second-order phase transition. It is an exponent n that characterises the temperature dependence of a thermodynamic property of a substance near its critical point. The temperature dependence has the form $|T - T_c|^n$, where T is the temperature and T_c is the critical temperature.

CRITICAL FIELD: 1. The maximum critical value of a magnetic flux density of a magnetic field required to destroy the superconductivity of a superconductor. 2. Also known as **cutoff field**, it is the smallest theoretical value of steady magnetic flux density that would prevent an electron emitted from the cathode of a magnetron at zero velocity from reaching the anode.

CRITICAL GROUP: A large group of related organisms that cannot be divided into smaller groups of equivalent taxonomic rank to the parent group, although variations exist between them.

CRITICAL MASS: 1. The minimum mass of fissionable material that will sustain a nuclear chain reaction. 2. A condition in software development in which a piece of software acquires so many features that it ceases to be useful.

CRITICAL PATH ANALYSIS (CRITICAL PATH METHOD, CPM): A mathematically based procedure for scheduling a set of project activities to minimise the amount of time taken. The analysis shows which subprojects can run in parallel with each other and which have to be completed before other subprojects can follow.

CRITICAL PHENOMENA: Physical properties of liquids and gases at the critical point (conditions at which two phases are just about to become one). For example, critical pressure is needed to condense a gas at the critical temperature and above the critical temperature the gas cannot be liquefied at any pressure.

CRITICAL POINT: In physics, also called **critical state**, it is the point at which a substance in one phase, as the liquid, has the same density, pressure, and temperature as in another phase, as the gaseous. 2. In mathematics, of a function of a single variable, a point at which the derivative of the function is zero.

CRITICAL PRESSURE: The pressure of fluid at its critical temperature and volume or the pressure of a substance when at the critical point between liquid and vapour phase. It is the minimum applied pressure required at the critical temperature to liquefy a gas. Each substance has its critical pressure.

CRITICAL REACTION: A nuclear chain reaction in which a reactive product or by-product causes additional reactions to take place so that the chain reaction is self-sustaining.

CRITICAL REGION: If the calculated value of a test statistic falls within the critical region, then the null hypothesis is rejected.

CRITICAL STATE: The unique conditions of pressure, temperature and composition of a fluid at which the liquid and gas phases both have the same density.

CRITICAL TEMPERATURE: 1. The temperature above which it is impossible to compress a gas into a liquid. 2. The temperature at which a material becomes a superconductor.

CRITICAL VALUE: 1. In hypothesis testing, the critical value is the limit value at which the decision to reject the null hypothesis H_0 for a given significance level is taken. 2. In calculus, it is the stationary value.

CRITICAL VOLUME: 1. The volume of a fixed mass of a fluid in its critical state, that is, when it is at its critical temperature and pressure. 2. The volume of 1 mole of a substance when at the critical point between liquid and vapour phases.

CRITICAL-POINT DRYING: A technique for removing the water from biological specimens without causing shrinkage or collapse. The specimen is dehydrated by suspending it in acetone or alcohol and driving off the solvent under controlled conditions of temperature and pressure.

CRITICALLY ENDANGERED (CR) SPECIES: A categorisation of threatened species by the International Union for Conservation of Nature (IUCN) and defined as a species that is facing a very high risk of extinction in the wild. It is the highest risk category assigned by the IUCN Red List for wild species. Conservation biologists use the IUCN Red List, to categorise endangered and critically endangered species. There are currently 2129 animals and 1821 plants on this list.

CROCODILE: A large semi-aquatic reptile belonging to the order crocodylia, which inhabits tropical regions. It has a symmetrical and streamlined body, making it an excellent swimmer. Its skin is mostly covered by tough scales with lots of pores in them. The size of some of the various species are as follows: saltwater crocodile between 4.3 – 5.2 m long and a mass of between 400 – 1,000 kg; the length of the Nile crocodile is about 4.2 m and weighs 220 – 550 kg, American crocodile weighs 400 – 500 kg with a length of between 4.1 – 4.8 m. Crocodiles have short limbs, and on land, their means of movement include crawling on their belly, walking with their body raised off the ground, or galloping. A crocodile has a very long powerful tail used for swimming without involving the webbed limbs at all. Crocodiles' jaws are very powerful, filled with between 60–72 teeth; the jaws are often wide open to cool themselves because they do not have sweat glands. A crocodile looks like an alligator but usually has a longer and narrower snout, described as V-shaped, while the alligator's snout is wider and U-shaped. It catches its preys, which consists of fish and any animal they are capable of attacking, by ambushing them and dragging

what they catch on land into the water. Crocodiles have been raised or hunted for food; its meat is a low-fat, low cholesterol and high protein meat, believed to be good for the heart, lungs, blood circulation and known to cure asthma and respiratory problems. Its skin is used in making leather goods such as bags, shoes, belts, wallets, etc.

CRON (CRON JOB): A process or task that runs periodically on a UNIX system. Examples include syncing the time and date via the Internet every ten minutes, sending an e-mail notice once a week or backing up certain directories every month.

CROOK: In agriculture, it is a long-handled staff with a hooked end, used by shepherds to catch sheep.

CROOKES, SIR WILLIAM (1832-1919): British chemist and physicist, who in 1861 used spectroscopy to discover thallium. In 1875 he invented the radiometer, an evacuated transparent bulb that has an arrangement of vanes, black on one side and silvered on the other, that spins when placed in a light and also developed an improved vacuum tube named Crookes' tube for studying gas discharges.

CROP: 1. Cultivated plants or agricultural produce, such as grain, vegetables or fruit, grown in one season or very short harvest that can be grown and harvested extensively for profit or subsistence. 2. The bag-like part of a bird's throat, where food is prepared for digestion before passing into the stomach. 3. In computer science, it refers to editing part of the image by cutting some of the image from any side to one's desire.

CROP BREEDING: The development of new varieties of crops by genetic principles to improve cultivated plants.

CROP ROTATION: The systematic practice of growing two or more annual crops on a given piece of land. By using crops which exploit nutrients at different depths of soil, it minimises depletion of soil nutrients and the inclusion of legumes which fix atmospheric nitrogen enriches the soil.

CROP SCIENCE: The use of modern scientific principles and technology in growing and managing crops.

CROPPING SYSTEMS: Crops, sequences and management techniques used on a piece of land over a period of years. Examples of cropping system are mono cropping, crop rotation, sequential cropping, inter cropping and relay cropping.

CROSS: 1. A mating between two selected individuals. 2. Also known as **hybrid**, it is an organism resulting from a mating between two selected individuals.

CROSS MULTIPLICATION: 1. The process of rewriting a proportion by calculating its cross products. It can be used in solving open proportions. 2. The process of solving a fractional equation by multiplying the numerator of each fraction by the other's denominator and equating the two products obtained. For example,

$$\frac{a}{b} = \frac{c}{d}$$

$$\Rightarrow a \times d = b \times c = ad = bc.$$

CROSS OVER VALUE (COV): The percentage of genetic material exchanged between homologous chromosomes during the process of crossing over during the first prophase of meiosis.

CROSS PRODUCT (VECTOR PRODUCT): The product of two vectors **A** and **B**, with components a_1, a_2, a_3 and b_1, b_2, b_3 , respectively. This can be expressed as

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \text{ or}$$

$\mathbf{A} \times \mathbf{B} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$. The vector product is itself a vector, perpendicular to both **A** and **B** and of length $AB\sin\theta$, where θ is the angle between **A** and **B**. The cross product of two vectors is a vector rather than a number, unlike the dot product. The determinant notation is a useful scheme for remembering the cross product. If

$$\mathbf{A} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \text{ then the determinant notation is}$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}. \text{ The symbols } \mathbf{i}, \mathbf{j} \text{ and } \mathbf{k} \text{ are vectors while the other elements are numbers.}$$

CROSS PROTECTION: The inoculation of a young plant with a mild strain of a virus to protect it from other virulent strains. This procedure has been used in the control of tobacco mosaic virus and citrus tristeza virus.

CROSS RATIO (CROSS-RATIO): If P, Q, R and S are four points lying on a straight line, the ratio of PQ to QR divided by the ratio of PS to SR is called the anharmonic or cross-ratio of the four points and is denoted by (PQRS). Thus

$$\begin{aligned} (PQRS) &= \frac{PQ}{QR} \div \frac{PS}{SR} \\ &= \frac{PQ}{QR} \cdot \frac{SR}{PS} \\ &= \frac{PQ \cdot SR}{QR \cdot PS}. \end{aligned}$$

CROSS REACTION: The ability of substances other than the analyte to bind to the binding reagent and the ability of substances other than the binding reagent to bind the analyte in competitive binding assays.

CROSS SECTION: 1. A surface formed by the intersection of a figure in 2-dimensional space with a line or of a body in 3-dimensional space with a plane; or cutting through something especially at right angles, resulting in many parallel cross-sections or a plane shape that results from cutting through a solid. 2. Also called **collision cross section**, it is a measure of the probability that an encounter between particles will result in the occurrence of a particular atomic or nuclear reaction. 3. Typical or representative sample.

CROSS-BREED: An organism such as an animal or plant produced by the breeding of different species or varieties.

CROSS-BREEDING: Mating or artificial insemination of animals of different breeds in order to combine the best characteristics of the two breeds. A cross-breed is the product of such a process.

CROSS-BROWSER: A website that is developed for multiple browsers.

CROSS-CAP: A two-dimensional surface in 3-space that is one-sided and the continuous image of a Möbius strip that intersects itself in an interval.

CROSS-CORRELATION: A statistical measure between two different information sets of a series of information.

CROSS-CUT: A simple arc between two distinct points on a surface.

CROSS-FERTILISATION (ALLOGAMY): The fertilisation of an ovum from one individual with the spermatozoa of another. In plants, it is the use of pollen from one plant to fertilise the flower of another plant.

CROSS-FLOW FILTRATION: A method of operating a filtration device where retained fluid is circulated over the membrane (filter) surface thus preventing undue build-up of filtered material on membrane (filter).

CROSS-LINKING: Covalent or ionic bonds linking two or more long chain molecules in a polymer such as synthetic polymers or natural polymers such as proteins. Cross linking in synthetic polymer gives the polymer a higher melting point and makes it harder.

CROSS-MULTIPLICATION: A method of simplifying an equation in which one or both terms are fractions. An equation between two fractions or rational expressions can be cross-multiplied to simplify the equation or determine the value of a variable.

CROSS-OVER CONCENTRATION: The concentration range at which the sum of the volumes of the domains occupied by the solute molecules or particles in solution is approximately equal to the total volume of that solution. The term 'domain' refers to the smallest convex body that contains the molecule or particle in its average shape.

CROSS-POLLINATION (CROSS-FERTILISATION): Transfer of pollen grains from the anther of a flower to the stigma of same flower or to the stigma of another plant of the same species. Some species require cross-pollination in order to set seed.

CROSS-SITE SCRIPTING (XSS): A method by which an attacker can use a website to insert malicious coding into a link that appears to be from a trustworthy source. When someone clicks on the link, the embedded programming is submitted as part of the client's Web request and can execute on the user's computer, typically allowing the attacker to steal information.

CROSSED ELECTRIC AND MAGNETIC FIELDS: Electric and magnetic fields, in which the direction of the electric field is at right angles to the magnetic field direction.

CROSSED-BEAM REACTION: A chemical reaction in which two molecular beams are crossed; one beam is regarded as the incident beam of gas and the other as the target gas. By controlling the target and projectile molecules, information can be gained about the chemical reaction.

CROSSING-OVER (RECOMBINATION): An exchange of portions of chromatids between homologous chromosomes during meiosis, which creates new combinations of genes in the gametes that are not found in either parent, contributing to genetic diversity.

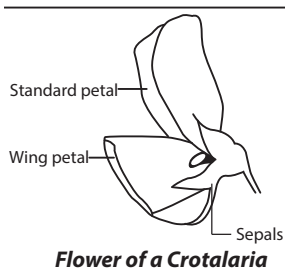
CROSSING LINES (SKEW LINES): Lines that lie in different planes and do not intersect one another.

CROSSLINK: A small region in a macromolecule from which at least four chains emanate and formed by reactions involving sites or groups on existing macromolecules or by interactions between existing macromolecules. The small region may be an atom, a group of atoms or a number of branch points connected by bonds, groups of atoms or oligomeric chains. In the majority of cases, a crosslink is a covalent structure but the term is also used to describe sites of weaker chemical interactions, portions of crystallites and even physical entanglements.

CROSSLINK DENSITY: Number of crosslinks per unit volume in a polymer network.

CROSTALK: Interference caused by the electric or magnetic fields of one telecommunication signal affecting a signal in an adjacent circuit. Crosstalk can cause errors, noise or prevent cables from transmitting data. Twisted-pair cable, shielded cable and keeping cables further apart help prevent crosstalk.

CROTALARIA: Commonly known as **rattlepods**, it is a genus of herbaceous plants and woody shrubs in the Family Fabaceae or Leguminosae, Subfamily Faboideae. There are about 600 species distributed worldwide, mostly in the tropics; with at least 500 species in Africa. *Crotalaria* are erect, herbaceous, variably hairy plants and may be annual or perennial. The leaves are simple, alternate, lanceolate to obovate, with a finely haired undersurface. The flowers are yellow, with the leguminous calyx longer than the corolla. The fruit is a leguminous pod, inflated, hairless, becoming black with maturity and contains 10 to 20 glossy black, heart-shaped seeds, which often detach and rattle with the pod. Some species of *Crotalaria* including *Crotalaria sagittalis*, *Crotalaria spectabilis* and *Crotalaria retusa* are toxic and can poison all livestock, including domestic fowl. Some of the species include *Crotalaria juncea*, commonly known as brown hemp, indian-hemp, madras-hemp,



sann-hemp, sunn crotalaria and sunn-hemp; *Crotalaria capensis*; *Crotalaria sagittalis*, *Crotalaria spectabilis* and *Crotalaria retusa*. *Crotalaria juncea* is an annual shrub, with a deep strong tap root and well-developed lateral roots bearing numerous multi-branched and lobed nodules. It is used for fishnets, paper and fabrics and also grown for hay and as green manure crop.

CROTON: A genus of about 400 species of annual herbaceous herbs, shrubs and trees in the Euphorbiaceae family. Some of the known species include *Croton lobatus*, occurring in West Africa and widespread across tropical Africa and Arabia; *Croton Eluteria*, also known as Cascarilla occurring in the West Indian; and *Croton tiglium* with the common name Croton oil plant, which occurs in Asia and the W African region. The plants have flowers without petals and contain bitter milky juice. *Croton lobatus* grows up to 1 metre high with erect stem woody at base, branched and hairy. The plant has a strong taproot and is not easy to root out in cultivated land. The leaves and roots are commonly used in African traditional medicine to treat eye diseases, unconsciousness, female sterility, menstrual cycle, pain-killers, paralysis, epilepsy, convulsions, spasm etc. It is also used as fodder and for land conservation. *Croton eluteria* is a shrub or small tree, which grows up to 4 metres in height. The young stems are covered with brown lepidote scales. The leaves are arranged alternately and have an ovate shape, with silvery brown lepidote scales on the lower side. *Croton eluteria* is used medicinally to treat coughs, diarrhoea, flu, indigestion, stomach pain and to prevent vomiting. The bark known as Cascarilla yields an aromatic spicy oil used as a fragrance and a flavouring, for example, for the liquor Campari. The bark and leaves can be used as an inhalant to clear sinuses and the leaves are used in bath water to refresh the body. *Croton tiglium* is an erect or more or less spreading shrub or small tree with alternate, ovate leaves. All parts of the plant contain toxic substances. *Croton tiglium* is used as laxatives; arrow-poisons; as antidotes for venomous stings and bites; etc. The seed produces oil, used for soap-making and illumination.

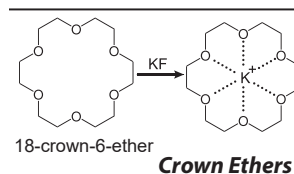
CROTOVINA (KROTOVINA): A disused animal burrow in one soil horizon that has become filled with material from another horizon.

CROWDING: The displacement of an impurity from an extract phase by contact with a solution containing the main extractable solute. The main solute need not be present in a pure solution but should have a higher distribution ratio than the impurities present.

CROWN: 1. Part of the tooth that is visible outside the gum or artificial replacement of this. 2. An unelongated stem of major meristematic tissue of turf grass. 3. A molecular entity comprising a monocyclic ligand assembly that contains three or more binding sites held together by covalent bonds and capable of binding a guest in a central (or nearly central) position. The adducts formed are sometimes known as **coronates**. The best known members of this group are macrocyclic polyethers, such as '18-crown-6', containing several repeating units $-\text{CR}_2-\text{CR}_2\text{O}-$, where R is most commonly H and known as **crown ethers**.

CROWN CONFORMATION: A conformation of a saturated cyclic molecular entity, containing an even number (≥ 8) of atoms in the ring, in which these atoms lie alternately in each of two parallel planes and are symmetrically equivalent (for cyclooctane, for cyclodecane etc). It is analogous to the chair conformation of cyclohexane.

CROWN ETHERS: Cyclic organic compounds containing several ether groups. Common compounds have repeating units of ethylene oxide $-\text{CH}_2\text{CH}_2\text{O}-$ arranged into a cyclic molecule shaped like a crown. They have a hollow middle cavity within which metallic atoms, ions and even neutral organic molecules can be chelated. The oxygens, which can be replaced by other heteroatoms, serve as coordinating points to the metals. They can be used to selectively remove certain metals



from a solution since the cavity in the middle can only accommodate metals of a certain size.

CROWN GRAFT: A type of graft, in which a branch of a tree is cut across at right angles, slits are made in the bark around the edge of the stump and shoots are inserted into the slits.

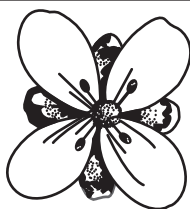
CROWN RUST: A fungal disease affecting oats and certain other grasses caused by varieties of *Puccinia coronata* and marked by light-orange masses of fungi on the leaves.

CRUCIBLE: 1. A dish or other vessel in which substances can be heated to a high temperature and still maintain its mechanical strength and rigidity and does not react in an undesirable way with its contents. Some types are used for melting ores or metals. Porcelain, iron and platinum crucibles are used in the laboratory; graphite crucible is usually used in industry. 2. The hollow part at the bottom of a furnace where molten metal collects. 3. A place or set of circumstances where people or things are subjected to forces that test them and often make them change.

CRUCIFER: A plant in the mustard family Cruciferae or Brassicaceae whose flowers have four petals arranged in the shape of a cross. Examples are cabbage, radish, broccoli and alyssum.

CRUCIFORM: 1. Having the shape of a cross. In architecture, it is usually used to describe the plan of a church. 2. In botany, it refers to structures or organs such as flowers, which are shaped like a cross, as shown in the diagram. 3. In mathematics, it defines a geometric curve, shaped like a cross and has four similar branches asymptotic to two mutually perpendicular pairs of lines. For example, in the equation

$x^2y^2 - a^2x^2 - a^2y^2 = 0$, where $x = y = \pm a$ are the four lines.



Cruciform

CRUD: A deposit or emulsion at the interface between two partially settled phases. The phenomenon of crud formation arises from many causes and this definition does not imply any single one.

CRUDE: 1. A substance such as ore or petroleum in its natural or unrefined state. Unrefined petroleum is called crude oil. 2. Of a statistical data, not analysed, consisting of the collected values of variables.

CRUDE OIL (PETROLEUM): A thick, flammable, yellow-to-black mixture of gaseous, liquid and solid hydrocarbons that occurs naturally beneath the Earth's surface, made up of a mixture of over 300 different hydrocarbons. It can be separated into fractions including natural gas, gasoline, naphtha, kerosene, fuel and lubricating oils, paraffin wax and asphalt. It is used as raw material for a wide variety of derivative products.

CRUISE MISSILE: An unpiloted aircraft with an air-breathing engine which uses an onboard automatic navigation system to guide it to its target.

CRUM BROWN'S RULE: A rule that predicts how substituents will enter the benzene ring. It states that substitution takes place in the meta position to group on a benzene ring if the compound can be converted by direct oxidation. If not, a mixture of 1,2 (ortho) and 1,4 (para) compounds will be formed. It was proposed in 1892 by the British chemist, Alexander Crum Brown (1838–1922).

CRUMB STRUCTURE: A type of soil structure in which the structural units or peds have a spheroidal or crumb shape occurring in a good loam soil or very porous soil horizons. It provides optimal pore space for soil fertility and good plant growth. Soil crumbs are rich in minerals and they improve the drainage of the soil. Soil crumbs also allow more air to circulate through the soil.

CRUMPLE ZONE: A structural feature at the front and rear of a motor vehicle that is designed to crumple gradually during a collision to absorb deformation energy from the impact, so reducing the risk of serious injury to passengers.

CRUNODE: A point where a curve intersects itself so that two branches of the curve have distinct tangent lines.

CRUPPER: A strap fixed to the back of a saddle and looped under the horse's tail to keep a saddle, harness or other equipment from sliding forward.

CRUSH: 1. A steel or wooden appliance such as a strong stall, used to restrain livestock when administering injection or when the animal is being inspected by a veterinary surgeon. 2. To press something with a heavy weight, such as crushing seeds to extract oil. 3. The application of pressure so as to destroy the natural condition, shape or integrity of the parts.

CRUSHING STRENGTH: 1. The force required to crush a mass of dry soil, or the resistance of a mass of dry soil to crushing. It is expressed in units of force per unit area (pressure). 2. The greatest compressive stress that a brittle solid such as stone or concrete can sustain without fracture.

CRUST: 1. The outermost structure of the Earth consisting of two distinct parts, the oceanic crust and the continental crust. 2. In soil, it is a thin layer on the surface that is from a few millimetres to 2.5 cm thick. When dry, it is much more compact, hard, and brittle than the underlying material. Its main cause is the impact of rain on exposed soil. If it is too dry, it can prevent seedlings from germinating and plants from growing well.

CRUSTACEAN: A very large group of arthropods, which are aquatic organisms with exoskeleton that moults during growth or a class of arthropods with a body made up of a head joined to a segmented thorax and abdomen. It includes crabs, shrimps, lobsters and barnacles and they are found in fresh water and marine habitats. They are distinguished from other groups of arthropods, such as insects, myriapods and chelicerates by the possession of biramous (two branches) and by the nauplius form of the larvae.

CRUSTOSE (CRUSTACEOUS): In botany, it refers to lichens that have a crust-like thallus closely appressed to and virtually inseparable from the substratum. An example is Lecanora.

CRUTCH: 1. A forked pole, used when dipping sheep, to push the animal's shoulders down into the liquid. 2. An object that sustains or supports somebody or something that is otherwise liable to collapse, fail or falter. 3. A staff with a handgrip and a rest for the forearm or armpit used to assist a person to walk unaided.

CRYOBIOLOGY: The study of the effects of very low temperatures on organisms, including proteins, cells, tissues and organs.

CRYOGENIC: Low temperature processes, apparatus, etc; usually applied to systems operated at the temperature of liquid nitrogen, helium or other condensed gas which boils at a at atmospheric pressure.

CRYOGENIC PROPELLANT: A form of liquid propellant for rocket engines that must be kept at very low temperatures to remain liquid. The commonest examples are liquid hydrogen and liquid oxygen. Cryogenic propellants require special insulated containers and vents to allow gas from the evaporating liquids to escape. The liquid fuel and oxidiser are pumped from the storage tanks to an expansion chamber and injected into the combustion chamber where they are mixed and ignited by a flame or spark.

CRYOGENIC PUMP: A high-speed vacuum pump that can produce an extremely low vacuum in which pressure is reduced by converting gases on surfaces maintained at about -253.15 °C (20 K) by means of liquid hydrogen or at -269.15 °C (4 K) by means of liquid helium.

CRYOGENIC SAMPLING: The collection of trace compounds from gaseous media by condensation with a major constituent, for example, water vapour, CO₂, N₂, Ar) of the matrix.

CRYOGENICS: The science of very low temperatures defined as below -123.15 °C (150 K). Some of the phenomena that occur at cryogenic temperatures include liquefaction and solidification of ambient gases; loss of ductility and embrittlement of some structural materials such as carbon steel; increase in the thermal conductivity

of relatively pure metals, ionic compounds and crystalline dielectrics such as diamond, sapphire, solidified gases and so forth to a maximum value, followed by a decrease as the temperature is lowered further; decrease in the thermal conductivity of metal alloys and plastics; decrease in the electrical resistance of relatively pure metals; decrease in the heat capacity of solids; decrease in thermal noise and disorder of matter; and appearance of quantum effects such as superconductivity and superfluidity.

CRYOHYDRATE: 1. Eutectic mixture of ice and some other substances such as an ionic salt, obtained by freezing a solution. 2. A salt that contains water of crystallisation at low temperatures, also known as **cryocel**.

CRYOLITE: A mineral compound of sodium hexafluoroaluminate(III), with the chemical formula Na_3AlF_6 , which crystallises in the monoclinic system. It is used as flux to lower the melting point of aluminium oxide during the production of aluminium by electrolysis.

CRYOLOGY: The study of the properties of snow, ice, and frozen ground.

CRYOMETER: A thermometer usually filled with alcohol or other low freezing point substances and used to measure low temperatures. Examples include magnetic thermometers, which can be used to measure temperatures below -272.15°C (1 K) and carbon resistors used at ultra low temperatures [-173°C (100.15 K) to -273°C (0.15 K)].

CRYOPHILE (PSYCHROPHILE): An organism that reproduces and grows best at low temperatures, typically in the range of -10 to 20°C . Chrophiles are a type of extremophile. Some have been found to thrive in the Arctic and Antarctic oceans, which remain frozen for much of the year. Psychrotolerant organisms, by contrast, are mesophiles that can survive at low temperatures but grow more slowly.

CRYOPHILOUS: Having an affinity for or thriving at low temperatures. A cryophilous plant is a plant that needs a period of cold weather to grow properly.

CRYOPHYTE: 1. A plant that forms winter buds below the soil surface. 2. Any plant, such as certain algae, mosses, fungi and bacteria, that grows on ice or snow.

CRYOPROTECTANT: An antifreeze compound such as sugars and glycerol that protect an organism's tissues from the effects of freezing. Arctic and Antarctic insects, fish, amphibians and reptiles create cryoprotectants in their bodies to minimise freezing damage during cold winter periods.

CRYOSCOPIC CONSTANT: In thermodynamics, the cryoscopic constant, with the symbol K_f , relates molality to freezing point depression and used to calculate the freezing-point-depression effects of a solute. In the expression, $\Delta t = K_f C_m$, C_m is the molar concentration of dissolved solute and K_f is the cryoscopic constant for the solvent used. K_f is expressed in degrees per mole of pure solvent. For example, the K_f for water is $1.853^\circ\text{C}\cdot\text{kg}/\text{mol}$.

CRYOSCOPY: The use of depression of freezing point to determine relative molecular masses of a solute by dissolving a known quantity of it in a solvent and recording the amount by which the freezing point of the solvent drops.

CRYOSPHERE: The frozen water part of the climate system consisting of all snow, ice, and permafrost on and beneath the surface of the earth and ocean. Examples of such places are waters surrounding Antarctica and the Arctic.

CRYOSTAT: A device used to maintain cryogenic temperatures or very low temperatures of samples or devices. The Dewar flask is the most satisfactory vessel for controlling heat leakage into the vessel by radiation, conduction or convection. Cryostats usually consist of two or more flasks nesting in each other to prevent heat leaking into the flask by radiation, conduction or convection.

CRYOTRON: A switch that functions by superconductivity. It consists of a coil of wire of one super conducting material such as niobium surrounding a straight wire of another super conducting material such as tantalum, with both immersed in liquid-helium bath. When current is passed through the niobium coil wrapped around tantalum wire it

produces a magnetic field, which in turn reduces the superconductivity of tantalum and hence reduces the amount of the current that can flow through it. It is thus possible to control the amount of the current that can flow in the tantalum wire with the help of a small current in the niobium wire.

CRYOTURBATION: Frost action on water in the soil, which results in the mixing of materials from various horizons of the soil caused by freezing and thawing.

CRYPTANDS: A group of organic compounds with three dimensional ether linked carbon units joined in a manner similar to crown ethers with all the ether units joined together by three-coordinate nitrogen atoms into a macropolycyclic compound. They also contain cavities into which ions can be chelated into a complex.

CRYPTARITHM (VERBAL ARITHMETIC; ALPHAMETICS; CRYPTARITHMETIC; CRYPT-ARITHMETIC; WORD ADDITION):

A number puzzle in which an indicated arithmetical operation has some or all of its digits replaced by letters or symbols and where the restoration of the original digits is required. Each letter represents a unique digit.

CRYPTIC COLOURATION: The type of colouring or marking of an animal that helps to camouflage it in its natural environment.

CRYPTOBIOSIS (ANHYDROBIOSIS): A state of ametabolic state of life or apparent suspended animation entered by certain invertebrate animals in order to survive desiccation or extreme stresses such as desiccation, freezing and oxygen deficiency. In such a state, all metabolic procedures stop, preventing reproduction, development and repair. Such animals are called **cryptobionts**.

CRYPTOCURRENCY (CRYPTO CURRENCY): A digital or virtual currency that uses cryptography for security and is used as a medium of exchange. A cryptocurrency is difficult to counterfeit because of this security feature. It is not issued by any central authority, rendering it theoretically immune to government interference or manipulation. An example of cryptocurrency is bitcoin and sometimes used synonymously.

CRYPTOGAM: A name applied to the lower plants producing spores but no seeds, thus the reproductive structures are not borne in conspicuous flowers or cones. It includes algae, liverworts, mosses and ferns.

CRYPTOGRAPHIC ALGORITHM: A set of rules that specify the mathematical steps required to encipher and decipher data.

CRYPTOGRAPHIC KEY: A parameter that determines cryptographic transformations between plaintext and ciphertext.

CRYPTOGRAPHY: The science of enciphering and deciphering of messages in secret code in order to render them unintelligible to all but the intended receiver. Computers encrypt data by applying an algorithm to a block of data. Modern cryptography intersects the disciplines of mathematics, computer science and electrical engineering. Applications of cryptography include ATM cards, computer passwords and electronic commerce.

CRYPTOMONADA (CRYPTOPHYTA): A phylum of unicellular eukaryotic organisms occurring in both fresh and salt water. They have flattened ovoid cells and two undulipodia (flagella) arising from an obliquely situated gullet (crypt). Cryptomonads include freshwater and marine algae; and heterotrophic protozoa, which may be carnivorous or parasitic in animal intestines.

CRYPTOZOIC: 1. Animals that live mainly in dark or concealed places such as under stones or in caves or holes, within soil, litter or detritus (organic matter) and being rarely seen in the open air. Most are invertebrates such as earthworms, although burrowing or hole-dwelling vertebrates such as lizards, snakes and rodents could also be said to be cryptozoic. 2. Precambrian; or of belonging to the geologic time period between Hadean Time and the Cambrian Period, geologic time between 4,650 to 700 million years ago; often subdivided into the Archean and Proterozoic eras, comprising most of the Earth's history and marked by the appearance of primitive forms of life.

CRYPTS OF LIEBERKUHN (INTESTINAL GLAND; INTESTINAL CRYPT): Tubular glands found in the epithelial lining of the small intestine and colon. The cells of these glands called Paneth cells secrete various enzymes, including sucrase and maltase, along with end peptidases and exopeptidases.

CRYSTAL: A solid inorganic substance of definite chemical composition that has its atoms (molecules or ions) arranged in a geometrical three dimensional pattern or the regular many sided shape which the molecules of a substance forms when it is solid. Within a crystal the atoms or molecules are arranged periodically and many identical unit cells, each containing a group of atoms, are packed together to fill all space. The term crystal in scientific nomenclature usually refers to a single crystal, i.e. a single periodic arrangement of atoms.

CRYSTAL COUNTER: A type of particle detector in which a potential difference is applied across a crystal. When the crystal is struck by an elementary particle or photon, the electron ion pairs created cause a transient increase in conductivity. The resulting current pulses are counted electronically.

CRYSTAL DEFECT: An imperfection in the regular lattice of a crystal. It can result from impurities, vacancies, dislocations or disorder causing a departure from the crystal symmetry.

CRYSTAL DIFFRACTION SPECTROMETER: A spectrometer, in which diffraction by a crystal is used to obtain the energy spectra of electromagnetic radiation as well as of slow neutrons.

CRYSTAL FIELD: The average static electric field experienced by an ion, molecule or atom in a crystal generated by all the other surrounding atoms, molecules or ions. The concept introduced by Becquerel and developed by Bethe has been extended to refer to the field generated by the ligands (anions or neutral species) surrounding a transition metal ion in a coordination compound. With the understatement that some covalency may exist in the metal-ligand bond, it is often referred to as ligand field.

CRYSTAL FIELD SPLITTING: The energy difference between two sets of d-orbitals in a metal atom when ligands are present.

CRYSTAL FIELD STABILISATION ENERGY (CFSE): A measure of the net energy of stabilisation gained by an ion's nonbonding electrons as a result of complex formation.

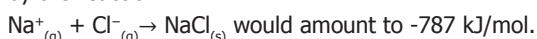
CRYSTAL FIELD THEORY: A theory which assumes that coordinated compounds are made up of a central atom or ion surrounded by ligands. These ligands are thought to be the source of negative charges which perturb the energy levels of the central atom or ion and thus split the energy level of the d-orbital of the central metal atom or ion and thus subject it to an electric field analogous to that within an ionic crystalline lattice.

CRYSTAL HABIT: In mineralogy, it is an overall description of the visible external shape and size of a mineral. This description can apply to an individual crystal or an assembly of crystals or aggregates. A crystal can be described by how well it is formed, using terms such as euhedral (perfect to near-perfect), subhedral (moderately formed) and anhedral (poorly formed to no discernable habit seen).

CRYSTAL LASER: A solid state laser in which the active medium is composed of ionic species supported in a crystalline host. The first laser used a ruby crystal in which chromium ions as the active medium were supported in an aluminium oxide (sapphire) crystal. One of the most common crystal lasers involves neodymium atoms in a yttrium aluminium garnet host crystal.

CRYSTAL LATTICE: The regular pattern of atoms, ions or molecules in a crystalline substance composed of a pattern, a set of atoms arranged in a particular way and a lattice exhibiting long-range order and symmetry.

CRYSTAL LATTICE ENERGY: The amount of energy that holds a crystal together; the energy change when a mole of solid is formed from its constituent molecules or ions (for ionic compounds) in their gaseous state. It is always negative. For example, the energy released by the reaction:



CRYSTAL MICROPHONE: A microphone in which the sound waves fall on a plate of Rochelle salt or similar material with piezoelectric properties, which uses light modulation to produce an electrical voltage signal from mechanical vibration.

CRYSTAL OPTICS: The optical properties of crystals. The transmission of light by crystals differs from that by glass since the refractive index may depend on the direction of incidence of the light (crystals are not in general isotropic). Uniaxial crystals belonging to the tetragonal, hexagonal and trigonal systems have two principal refractive indices and display double refraction. Biaxial crystals (orthorhombic, monoclinic, triclinic) have three principal refractive indices.

CRYSTAL OSCILLATOR (PIEZOELECTRIC OSCILLATOR): An electronic circuit in which the mechanical resonance of a vibrating piezoelectric crystal is used to determine the frequency.

CRYSTAL PHOTOCHEMISTRY: A photochemical process in the crystal lattice of a photoactive compound or of a chromophore in the crystal lattice of host molecules. Often asymmetric photochemistry takes place.

CRYSTAL PICK-UP: A pick-up in a record player in which the mechanical vibrations produced by undulations in the record groove are transmitted to a piezoelectric crystal which produces a varying electric field of the same frequency as the sound.

CRYSTAL STRUCTURE: The arrangement in a particular way of atoms, ions or molecules in a crystalline liquid or solid. Crystals are solids having, in all three dimensions of space, a regular repeating internal unit of structure. The points can be thought of as forming identical tiny boxes, called **unit cells**, that fill the space of the lattice. The lengths of the edges of a unit cell and the angles between them are called the **lattice parameters**. A crystal's structure and symmetry play a role in determining many of its physical properties, such as cleavage, electronic band structure and optical transparency.

CRYSTAL SYSTEM: A method of classifying crystalline substances on the basis of their unit cell. There are seven crystal systems into which a crystal may be classified according to the shape of the unit cell of its Bravais lattice. They are cubic, hexagonal, tetragonal, trigonal orthorhombic, monoclinic and triclinic.

CRYSTAL TEST: A type of presumptive test done usually by microscope, in which a substance is identified by the formation of characteristic crystals when a certain reagent is added. An example is the acetone-chlor-haemin test for blood.

CRYSTALLINE: Having the regular internal arrangement of atoms, ion or molecules highly characteristic of crystals. Crystalline materials need not necessarily exist as crystals but are packed in a regular array.

CRYSTALLINE ELECTRODES: Electrodes which contain mobile ions of one sign and fixed sites of opposite sign. They may be homogeneous or heterogeneous. Homogeneous membrane electrodes are ion-selective electrodes in which the membrane is a crystalline material prepared from either a single compound or a homogeneous mixture of compounds (i.e. Ag_2S , $\text{AgI}/\text{Ag}_2\text{S}$). Heterogeneous membrane electrodes are formed when an active substance or mixture of active substances, is mixed with an inert matrix, such as silicone rubber or PVC or placed on hydrophobized graphite or conducting epoxy, to form the heterogeneous sensing membrane.

CRYSTALLINE POLYMER: A polymer showing crystallinity. One or two-dimensional order leads to mesophase structure. The range of order may be as small as about 2 nm in one (or more) crystallographic direction(s) and is usually below 50 nm in at least one direction.

CRYSTALLINE SOLID: A solid that possesses rigid and long-range order. Its atoms, molecules or ions occupy specific position and exist in a regular, well-defined arrangement. The smallest repeating pattern of crystalline solids is known as the **unit cell**. There are four types of crystalline solids: (i) **Ionic solids**, such as table salt (NaCl) which is made up of positive and negative ions and held together by electrostatic attractions. They have very high melting points, brittle and are poor conductors in the solid state. (ii) **Molecular solids** such

as sucrose made up of atoms or molecules held together by London dispersion forces, dipole-dipole forces or hydrogen bonds. They have low melting points and flexibility and are poor conductors. (iii)

Covalent-network or atomic solids such as diamond and graphite and the fullerenes, made up of atoms connected by covalent bonds; the intermolecular forces are covalent bonds as well. They are very hard with very high melting points and being poor conductors. (iv)

Metallic solids, made up of metal atoms that are held together by metallic bonds. They have high melting points, can range from soft and malleable to very hard and are good conductors of electricity.

CRYSTALLINITY: The presence of three-dimensional order on the level of atomic dimensions. Crystallinity may be detected by diffraction techniques, heat-of-fusion measurements, etc. The amount of disorder within the crystalline region is not incompatible with this concept.

CRYSTALLISATION (CRYSTALLIZATION): The natural or artificial process by which crystals are formed from a solution, melt or more rarely deposited directly from a gas. Crystallisation is the phase transition of matter from the state of a supercooled or supersaturated mother medium to a crystalline state with lower energy. Excess energy is evolved during crystallisation in the form of the latent heat of crystallisation. Some of this heat may be converted into mechanical energy; for example, a growing crystal may lift a weight placed upon it by exerting a crystallisation pressure on the order of several dozen kilograms-force per sq cm. The crystals of salts that form in the pores of concrete piers immersed in seawater may lead to the destruction of the concrete. The crystallisation of water plays an important role in atmospheric and soil phenomena. Crystallisation leads to the formation of minerals and is the basis of metallurgy and the production of semiconducting, optical and piezoelectric materials, films for microelectronics and metallic coatings. It is used widely in the chemical, food and medical industries (purification and the production of fertilisers, salt, sugar, chemicals and pharmaceuticals).

CRYSTALLITE: A small often microscopic crystal, such as one of the small crystals forming part of a microcrystalline substance. It may be formed from incomplete crystallisation process.

CRYSTALLOGRAPHY: The scientific study of crystal structures. It helps chemists to determine the internal structures and bonding arrangements of minerals and molecules, including the structures of large complex molecules such as proteins and DNA.

CRYSTALLOID: 1. A non-colloid substance, that forms a true solution and hence able to pass through a semipermeable membrane. 2. A mass of protein resembling a crystal that often occurs in seeds and other storage organs.

CRYPTOCHROME: A plant pigment that absorbs mainly blue and ultraviolet light and is thought to be involved in various plant responses to light, including phototropism and hypocotyls elongation. They are found in the mitochondrial inner membrane and endoplasmic reticulum of eukaryotes, in the chloroplasts of plants, in photosynthetic microorganisms and in bacteria.

CRYTONEPHRIDIAL SYSTEM: A specialised excretory system found in certain beetles and lepidopteran larvae in which the distal blind ending of the malpighian tubules are held against the wall of the rectum by a perinephric membrane.

CS GAS (2-CHLOROBENZYLIDENE-MALONONITRILE; O-CHLOROBENZYLIDENE MALONONITRILE): A white irritant organic compound, with the chemical formula $C_{10}H_5ClN_2$, relative density 1.04, melting point 93 °C (366.15 K) and boiling point 310 °C (583.15 K). It is used as a tear gas for riot control.

CSC: Abbreviation and symbol for the cosecant function.

CSC⁻¹: Symbol for the inverse cosecant function.

CSCH: Abbreviation and symbol for the hyperbolic cosecant function.

CSCH⁻¹: Symbol for the inverse hyperbolic cosecant function.

CSCW (COMPUTER-SUPPORTED COLLABORATIVE WORK): Software tools and technology that support a group of individuals working on projects at different sites. It is based on the principle of

group coordination and collaborative activities supported through computer systems.

CSR STRATEGIES (C-S-R TRIANGLE THEORY): A concept in plant ecology developed by John Philip Grime, British ecologist and emeritus professor at the University of Sheffield, which classifies plants based on their life history as Competitive (C), Stress-tolerant (S) and Ruderal (R) or some combination of these. It is based on the proposition that habitats are affected by two principal external factors: the level of disturbance such as grazing and disease and the degree of stress like drought, nutrient deficiencies, fire and shading.

CT SCANNER (CAT SCANNER): A ring shaped x-ray machine that rotates through 180° around a patient or an object, making numerous x-ray measurements every few degrees. Data produced is used in generating three dimensional image of the inside of the object for medical diagnosis or nondestructive material testing.

CTN: Abbreviation and symbol for the cotangent function.

CTN⁻¹: Symbol for the inverse cotangent function.

CTNH: Abbreviation and symbol for the hyperbolic cotangent function.

CTNH⁻¹: Symbol for the inverse hyperbolic cotangent function.

CTENIDIA: The gills of aquatic molluscs, which are present on both sides of the mantle cavity and are held in place by specialised membranes.

CTENOPHORA: A phylum of marine invertebrates that contains the comb jellies. It has the following characteristics: radially or biradial symmetrical; multicellular with body few tissues, an internal cavity and a mouth and anal pores; swims by means of plates of cilia; it reproduces mostly sexually as hermaphrodites, occasionally asexually; and has a distinct larval stage which is planktonic and carnivorous.

CTRL: Short form of **control**. This is a modifier key on modern computer keyboards that are IBM compatible. It is used in combination with other keys on the keyboard to execute a command, for example, CTRL+V is used to paste copied items.

CUBAGE: The volume or content of a solid body or figure.

CUBANE: A crystalline hydrocarbon, with the chemical formula C_8H_8 , molar mass 104.15 g/mol, relative density 1.29 and melting point 131 °C (404 K). It consists of eight carbon atoms arranged at the corners of a cube, with one hydrogen atom attached to each carbon atom. The compound is unusual because it was thought that due to the sharp bond angle of 90° it should be highly strained; but it is kinetically stable.

CUBATURE: The determination of volume of a solid or its calculation.

CUBE: 1. A regular solid figure whose faces are all squares. It has 6 equal faces, 12 equal length edges and 8 vertices.

2. The result of multiplying a number, quantity or an expression by itself three times, for example, 27 is the cube of $3 \times 3 \times 3$. 3. The third power of a number, quantity or an expression. For example, the cube of 3 is $3 \times 3 \times 3$ (3^3) = 27.

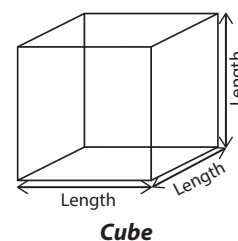
CUBE ROOT: A number raised to the number one third. For example, $x^{1/3} = \sqrt[3]{x}$.

CUBE ROOT FUNCTION: A function where the outputs are the cube roots of

the inputs. For example, $f(x) = x^{\frac{1}{3}}$.

CUBED NUMBER: The number obtained by multiplying a number three times.

CUBIC: 1. Having the shape of a cube; or a cubic equation, function, expression or term. 2. Of or to the third power. 3. Of an algebraic expression, equation or term, containing a variable raised to the third power and none raised to a higher power; of the third degree. 4. Denoting a measure of three-dimensional space that is derived from a linear measure by raising it to the third power. For example, a cubic metre is the volume enclosed by a cube with each side one metre long. **Cubical** is of or related to a cube.



CUBIC CRYSTAL (ISOMETRIC CRYSTAL): A crystal in which the unit cell is a cube. This is one of the most common and simplest shapes found in crystals and minerals. There are three main varieties of these crystals: **Primitive cubic**, also called **simple cubic**; **Body-centred cubic**; and **Face-entered cubic** also called **cubic close-packed**. Gemstones occurring in cubic crystal forms include diamond, the garnets, pyrite and spinel.

CUBIC DECIMETRE: Metric system measure of volume corresponding to the volume of a cube whose edges are all 1dm (10cm) long. Its symbol is dm³. One cubic decimetre (dm³) is equal to one litre.

CUBIC EQUATION: An equation in which the highest power of the variable is three. For example, $ax^3 + bx^2 + cx + d = 0$, where a is nonzero. In other words, it is a polynomial of degree three. The derivative of a cubic function is a quadratic function. The integral of a cubic function is a quartic function. Setting $f(x) = 0$ and assuming $a \neq 0$ it produces a cubic equation of the form:

$ax^3 + bx^2 + cx + d = 0$. Usually, the coefficients a, b, c, d are real numbers. However, most of the theory is also valid if they belong to a field of characteristic other than 2 or 3. Solving a cubic equation amounts to finding the roots of a cubic function.

CUBIC EXPANSION (COEFFICIENT OF VOLUMETRIC EXPANSION; VOLUMETRIC EXPANSION; VOLUME EXPANSION): The fractional change in volume of a substance per kelvin rise in temperature. It expressed mathematically as

$$\gamma = \frac{V_2 - V_1}{V_1(\theta_2 - \theta_1)}, \text{ where } \gamma \text{ is the coefficient of volumetric expansion, } V_1$$

is the volume at θ_1 and V_2 is the volume at θ_2 .

CUBIC FUNCTION: A polynomial of the third degree. Its general formula is $f(x) = ax^3 + bx^2 + cx + d$, where a is not equal to zero.

CUBIC METRE: An S.I. unit of measure for large volumes or capacities of objects, with the symbol m³. It is the volume of a cube with sides of one metre in length. 1 m³ is equal to 1000 litres. For volumes of smaller objects, cubic centimetre (cc or cm³) is often used.

CUBIC POLYNOMIAL: A polynomial of degree 3. Examples are $x^3 - 1$, $4a^3 - 100a^2 + a^6$ and $m^2n + mn^2$.

CUBIC RESOLVENT EQUATION: An intermediate equation that can be used to solve for the roots of quartic in terms of the root of the cubic which can in turn be solved for using the cubic equation.

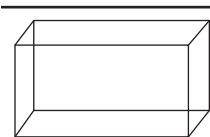
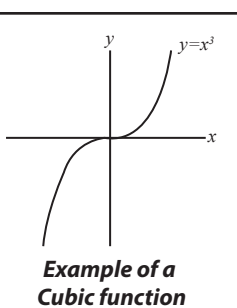
CUBIC UNIT: A unit such as cubic centimetres, and cubic metres used to measure volume or capacity.

CUBIC ZIRCONIA (CZ): A crystal form of zirconium(IV) oxide (zirconium dioxide, ZrO₂) made by fusing ZrO₂ and allowing it to cool under controlled conditions. It is used as a diamond substitute in jewellery.

CUBIT: An ancient unit of length, measured from the point of the elbow to the end of the middle finger. Its value is between 46 and 52 centimetres.

CUBOCTAHEDRON: A polygon obtained by cutting the corners off a cube or an octahedron. It has 8 faces that are equilateral triangles and 6 faces that are squares.

CUBOID: A six sided three-dimensional prism whose faces are all rectangles, forming a convex polyhedron. If the lengths of the sides are denoted by a, b and c , a cuboid with all sides equal ($a = b = c$) is called a **cube** and a cuboid with integer edge lengths $a > b > c$ and face diagonals is called an **Euler brick**. If the space diagonal is also an integer, the cuboid is called a **perfect cuboid**. The volume of



Cuboid

a cuboid is given by $V = abc$; its total surface area is given by $S = 2(ab + bc + ca)$.

CUCULATE: In botany, it refers to structures that are hooded or shaped like a hood.

CUCULLUS: In botany, it refers to the outer layer of a seed coat.

CUCUMBER: The trailing vine (*Cucumis sativus*), of the gourd (Cucurbitaceae) family. It bears long warty green fruits with white crisp flesh used as raw in salads or pickled with spices and eaten as a relish. The leaves are large, about 15-20 cm long, with three lobes and the flowers are bell-shaped and yellow.

CUCURBIACEAE (CUCURBITS; THE GOURD FAMILY): A family of plants belonging to the order Campanulales consisting of about 965 species in about 95 genera consisting of mostly herbaceous vines, some woody lianas, thorny shrubs, or trees. They are found mainly in tropical and subtropical regions. The large, yellow or white flowers in many species are unisexual, with male and female flowers on different plants (dioecious) or on the same plant (monoecious). Examples include squash, pumpkin, calabash, cucumber, some gourds, watermelon, etc. Some species have some medicinal properties. For example, *Momordica charantia*, commonly called bitter melon can do the following: lower blood sugar and high cholesterol levels; has antimicrobial, anticancer and antiviral properties; fight obesity; used to treat ulcers; to manage pain; inhibit pregnancy, and can cause infertility in men.

CUCUMIFORM: Having a cylindrical shape as that of the cucumber.

CUD: A bolus of semi-digested food regurgitated from the reticulorumen of a ruminant for chewing.

CULEX: A genus of mosquito (*Culex pipiens*) consisting of about 2,000 species found on every continent except Antarctica. Several species transmit diseases such as West Nile virus, filariasis, Japanese encephalitis, St. Louis encephalitis and avian malaria to humans and domestic animals. Males do not bite. The developmental cycle takes two weeks and is by metamorphosis. It lays eggs in water singularly or in batches, depending on the species. During the larval stage the mosquito lives in water and feeds on organic matter and plants, then develops into a pupa. The adult mosquito can measure from 4-10 millimetres and morphologically has three body parts: head, thorax and abdomen.

CULL: To remove an animal, especially a weak or sick one, from a flock or herd or a reduction of the numbers of an animal population achieved by killing some of its members.

CULM: The jointed stem of members of the Gramineae and Cyperaceae. In grasses it is usually hollow or more rarely filled with pith. In sedges it is usually solid and triangular in cross section.

CULTIVAR: A variety of a plant developed by agricultural or horticultural techniques and distinguished by one or more characteristics such as crop yield and disease resistance. Plants of a particular cultivar retain their special characteristics. The term is derived from cultivated variety.

CULTIVATION: The tilling or manipulation of the soil to prepare land for the purpose of planting and breeding crop plants. It involves manipulation of the soil primarily to remove weeds that compete with crops for water and nutrients and also to increase water infiltration. It also involves the investigation of new means of increasing crop yield and quality. To **cultivate** is to prepare land and grow crops.

CULTIVATOR: 1. Agricultural implement or tool for breaking up the soil to remove weeds or to loosen the soil to increase water infiltration. 2. A person who cultivates.

CULTURAL CONTROL: Pest control methods such as using tillage, sanitation, harvesting, and other techniques to modify the pest's growing environment to reduce its prevalence.

CULTURE: A particular kind of organism growing in a laboratory medium or the cultivation of cells in the laboratory for a wide spectrum

of scientific research. Cultivated cells can be microorganisms or of animal or plant origin, which are grown under specific conditions of nutrient levels, temperature, pH, oxygen levels, osmotic factors, light, pressure and water content. A culture medium provides the appropriate conditions for growth.

CULTURE MEDIUM (GROWTH MEDIUM; NUTRIENT MEDIUM):

A solid or liquid nutrient material, which includes simple sugars and complex substances such as meat broth, used to support the growth and reproduction of microorganisms or to maintain tissue or organ cultures. The most common growth media for microorganisms are nutrient broths and agar plates.

CUMENE PROCESS: An industrial process for making phenol from benzene. A mixture of benzene vapour and propene is passed over a phosphoric acid catalyst at 250 °C (523 K) and high pressure to form the intermediate product, $C_6H_6 + CH_3CH=CH_2 \rightarrow C_6H_5CH(CH_3)_2$. The product is called cumene and it can be oxidised in air to peroxide, $C_6H_5C(CH_3)_2O_2H$. This reacts with dilute acid to give phenol (C_6H_5OH) and propanone (acetone, CH_3COCH_3), which is a valuable by-product.

CUMULANT: In statistics, any of the coefficients, K_n , of $t^n/n!$ in the Taylor series of the natural logarithm of the moment generating function, $M_x(t)$, random variable. K_1 is the mean and K_2 is the variance. If the moment-generating function of a distribution exists, then its natural logarithm exists and is called the cumulant generating function (cgf). The coefficient of $\frac{t^n}{n!}$ in the Taylor expansion of the cgf is called the n th cumulant and is denoted by K (where K is kappa). The cumulant can be expressed in terms of the central moments and vice versa in particular, denoting the mean by μ and the central moment by $\mu_1, \mu_2, K_1 = \mu, K_2 = \mu_2, K_3 = \mu_3, K_4 = \mu_4^3 - 3\mu_2^2, K_5 = \mu_5 - 10\mu_3\mu_2$.

CUMULATIVE DISTRIBUTION FUNCTION (C.D.F.): In statistics, a function that gives the probability that a random variable at each point will be less than or equal to a specific value. The function F , for a random variable X , is defined for all real values x by $F(x) = P(X \leq x)$. From the definition, $F(-\infty) = 0$ and $F(\infty) = 1$ are the limits of $F(x)$ as x tends to $-\infty$ and ∞ respectively. The function is a non-decreasing function such that if $x_2 > x_1$ then $F(x_2) \geq F(x_1)$. If X is a continuous random variable then F is a continuous function and conversely, if X

has probability density function f , the $F(x) = \int_{-\infty}^x f(t)dt$ and

$f(x) = F'(x)$, where $F'(x)$ is the derivative of $F(x)$.

CUMULATIVE DOUBLE BONDS: Double bonds joining at least three contiguous carbon atoms in a single structure, for example, $H_2C=C=CH_2$ (allene). Non-cumulative double bonds comprise every other arrangement of two or more double bonds in a single structure. The generic name **cumulene** is given to compounds containing three or more cumulative double bonds.

CUMULATIVE FISSION YIELD: The fraction of fissions which have resulted in the production of a nuclide either directly or indirectly, up to a specified time. If no time is specified, the yield is considered to be the asymptotic value.

CUMULATIVE FREQUENCY: The total frequency of a given value up to and including a certain point in a set of data.

CUMULATIVE ODDS RATIO: A function used in models for ordinal variables. Let j denote a possible value of the ordinal variable, Y and let E_1 and E_2 be two possible events, the ratio R_j given by

$$R_j = \frac{P(Y \leq j | E_1)P(Y > j | E_1)}{P(Y \leq j | E_2)P(Y > j | E_2)}$$

is called cumulative odds ratio. If E_1 and

E_2 represent different values for a vector of explanatory variables, the model $R_j = R$, where R is a constant, is a proportional odds model.

CUMULATIVE PROBABILITY: The probability that the value of a random variable falls within a specified range. For discrete random variable X that can take only the ordered values $x_1 < x_2 < \dots$ the cumulative probability of the value less than or equal to x_k is $\sum_{j=1}^k P(X = x_{(j)})$.

CUMULATIVE SAMPLE: A sample collected over an extended period of time.

CUMULENES: Hydrocarbons (and by extension, derivatives formed by substitution) having three or more cumulative double bonds, for example, $R_2C=C=CR_2$.

CUMULO LAYER: A layer of sandy, silty, or clayey material in an organic soil.

CUMULONIMBUS CLOUD: Heavy masses of dense cloud whose cumuliform summits rise in the forms of towers, the upper parts having a fibrous texture and often spreading out into the shape of an anvil. These clouds generally produce showers of rain and sometimes of snow, hail or soft hail and often develop into thunderstorms.

CUMULUS CLOUD: Detached, dense clouds with a dome-shaped upper surface with sharp-edged, rounded protuberances and a nearly horizontal base. The height at which the cloud forms (the cloud base) depends on the amount of moisture in the thermal that forms the cloud. Humid air will generally result in a lower cloud base. In temperate areas, the base of the cumulus clouds is usually below 1,800 metres above ground level, but it can range up to 2,400 metres in altitude. In arid and mountainous areas, the cloud base can be in excess of 6,100 metres.

CUNEATE: Describing leaves that are wedge shaped with the point of the wedge forming the base of the lamina.

CUP: In mathematics, the symbol for set union, denoted, $\cup$, written as $S \cup T$ or iSi .

CUP FUNGI (PEZIZALES): An order of the Discomycetes in which the asci are operculate and are contained within a hymenium. It contains about 700 species in some 90 genera. The hymenium lines the surface of the cuplike apothecium and may be brightly coloured, as in *Sarcoscypha coccinea*, the scarlet elf cup. The group also contains the edible morels (*Morchella*).

CUPELLATION: A method of separating noble metals such as gold or silver from base metals such as lead, zinc and arsenic by melting the mixture with a blast of hot air in a shallow porous dish (the cupel). The base metals are oxidised, the oxide being carried away by the blast of air or react forming slags.

CUPOLA: A vertical shaft furnace used for melting metals. The melting of ores is done in a blast furnace.

CUPRIC COMPOUNDS: Compounds containing copper in its higher (+2) oxidation state, for example, cupric chloride is copper(II) chloride ($CuCl_2$).

CUPRIC SULFATE (COPPER(II) SULFATE): A white rhombohedral crystals or amorphous powdery compound, with the chemical formula $CuSO_4$ and decomposes at 650 °C (923 K) to cupric oxide (CuO). It is insoluble in alkali solutions, a property used in the preparation of Bordeaux mixture. It is fairly soluble in water and when dissolved forms the pentahydrate, $CuSO_4 \cdot 5H_2O$, which can be collected as blue triclinic crystals. Cupric sulfate is used in copperplating, in dyeing as a mordant, in wet-cell batteries, in pigments and in insecticides, fungicides and algicides.

CUPRITE: Copper(I) oxide ore found in crystalline form or in earthy masses, with the chemical formula Cu_2O . It occurs where deposits of copper have been subjected to oxidation. It is red to black in colour and is often called ruby copper. It is an ore of copper.

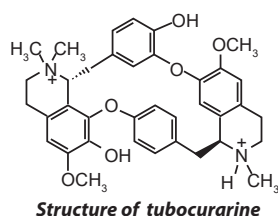
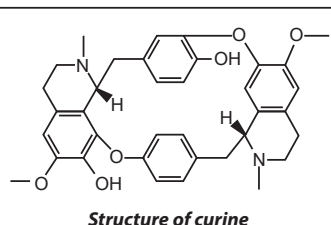
CUPRONICKEL (COPPER-NICKEL): An alloy of copper alloy containing 75% copper and 25% nickel, which is highly resistant to corrosion in seawater. It is used in hardware products, for coinage and especially in piping, heat exchangers and condensers in seawater systems.

CUPROUS COMPOUND: Compounds containing copper in its lower (+1) oxidation state, for example, cuprous chloride is copper(I) chloride ($CuCl$).

CUPSID (EYE TOOTH; EYETOOTH): In humans, any pointed tooth located behind the incisors and in front of the premolar teeth.

CUPULE: 1. A hard or membranous cup-shaped structure formed from bracts and enclosing various fruits such as the hazelnut and acorn. 2. Gemmae-bearing cub of the genus *Marchantia* of liverworts. 3. The bright red tissue around the seed of yew (*taxus*) forming the yew 'berry.'

CURARE: An alkaloid obtained from the bark of South American trees of the genera *Strychnos* and *Chondrodendron* that causes paralysis of voluntary muscle. Curare contains two alkaloids: curine, $C_{18}H_{19}NO_3$, which paralyses the muscle fibres of the heart; and curarine, $C_{19}H_{26}N_2O_2$, which paralyses the motor nerve. Curare is used as an arrow poison by South American Indians and was formerly used as a muscle relaxant in surgery. The structures of curine and tubocurarine are depicted in their respective diagram.



CURATIVE: A treatment or substance that can restore health.

CURD: Thick solid material formed by the proteins in milk by the action of bacteria or the enzyme rennin, extracted from the stomach of calves. It is acted on by bacteria and fungi after the liquid has been drained to form cheese.

CURDLE: Of an emulsion or colloid, to breakdown into large parts of different composition through the processes of flocculation, creaming, and coalescence. There are two categories: intentional and desirable, used in making cheese and tofu; unintentional and undesirable in making sauces and custards. Curdling occurs naturally in milk if the milk is not used by the expiration date, or if the milk stays out in warm temperature.

CURIE (CI): The former unit of radioactivity, named after Pierre Curie (1859-1906), French physicist and chemist; now replaced by the becquerel. One curie equals 3.7×10^{10} disintegrations per second or approximately the amount of radioactivity given off by 1 gram of radium.

CURIE POINT (CURIE TEMPERATURE): The temperature above which a magnetic material cannot be magnetised or the temperature at which a ferromagnetic substance loses its ferromagnetism and becomes paramagnetic. For nickel it is 356°C (629 K) and for iron it is 760°C (1033 K).

CURIE, MARIE SKŁODOWSKA (1867-1934): Polish-born French chemist and physicist, a pioneer in the field of radioactivity with her husband Pierre Curie (1859-1906), French physicist and the first woman to win two Nobel Prizes. She was also the first female professor at the University of Paris. Her works include founding the Curie Institutes in Paris and Warsaw; the creation of a theory of radioactivity, a term she coined and techniques for isolating radioactive isotopes; and the discovery of two new elements, polonium and radium. She also directed researches into the treatment of neoplasms (cancers) using radioactive isotopes. Her daughter Irène Joliot-Curie (1897-1956), French physicist and son-in-law, Frédéric Joliot-Curie (1900-1958), French nuclear physicist, also shared the 1935 Nobel Prize for chemistry for their work in the synthesis of radioactive substances.

CURIE'S LAW: A law which states that the susceptibility (χ) of a paramagnetic substance is inversely proportional to the thermodynamic temperature (T). This can be expressed as $M = C.B/T$, where M is the resulting magnetisation, B is the magnetic field, measured in teslas, T is absolute temperature, measured in kelvins and C is

a material-specific Curie constant. The law is named after Pierre Curie (1859-1906), a French physicist, a pioneer in crystallography, magnetism, piezoelectricity and radioactivity.

CURING: 1. A word used to describe the hardening of resins (polymer material). 2. Chemical process of converting a prepolymer or a polymer into a polymer of higher molar mass and then into a network. Curing is achieved by the induction of chemical reactions which might or might not require mixing with a chemical curing agent. 3. A method of processing and flavouring meat such as bacon, ham, beef. The curing solution usually consists of a mixture of sodium salts and is either rubbed onto or injected into the raw meat.

CURIUM: A radioactive metallic transuranic element belonging to the actinoids, with the chemical symbol Cm, atomic number 96, melting point 1340°C (1613 K) and boiling point 3110°C (3383 K). It is produced by bombarding plutonium with alpha particles (helium ions). It has several isotopes, its longest lived isotope is Cm 247 with a half-life of 16.4 million years.

CURRENT: 1. **Electric current** is the flow of electric charges, usually electrons or electron-deficient atoms through a conductor. It can also be defined as the rate at which electric charges move past a specific point in a circuit. Its symbol is I and its unit is the ampere with the symbol A. One ampere of current is defined as one coulomb of electrical charge (6.24×10^{18} charge carriers) that passes through a specific point in one second. The current in a circuit can be determined if the quantity of charge, Q passing through a cross section of a conductor in a time, t (second) can be measured. It is the ratio of the quantity of charge and time, which can be expressed as $I = Q/t$. Conventionally, current flows from positive points to negative points; electrons, however are negatively charged and flow from negative points to positive points. There are two types of electric current: direct current, such as produced by a battery or electrochemical cell and abbreviated DC; and alternating current abbreviated AC, for example, as produced by electricity grid to industries, commercial entities and homes. Direct current (DC) flows in the same direction at all points in time, although the instantaneous magnitude of the current might vary. In alternating current (AC), the flow of charge carriers reverses direction periodically. The number of complete cycles per second is the frequency, which is measured in hertz. When current flows through a conductor, it produces the following effects: heat, magnetic and motor. Heat effect results when current flows in a component possessing resistance and electrical energy is converted into heat energy, W with the unit joule (J) expressed as $W = I^2 R t$, where R is the resistance in ohms (Ω) of the resistor (for example, a heater), I is the current flowing through the resistor in time, t in seconds (s). The magnitude of the resistance, R depends on the material from which it is made so that the current in the circuit is related to voltage and resistance by Ohm's law: $I = V/R$, where V , is the voltage or electromotive force (EMF) that causes current to flow through a circuit and R is the resistance of the conductor measured in ohms with the symbol, Ω . When current flows through a conductor, magnetic field is generated around it, but ceases when no current flows through it. The strength of the magnetic field is directly proportional to the current in the conductor. The direction of the field created around a conducting wire may be determined by Maxwell's screw rule. In the motor effect, a conductor carrying current in a magnetic field experiences a force, which moves in a direction perpendicular to both the direction of the current and the direction of the magnetic field. The direction of motion may be determined by Fleming's left-hand rule. The motor effect occurs, when a current-carrying conductor is placed in a magnetic field, resulting in an interaction between the magnetic field produced by the current and the permanent field, which leads to a force being experienced by the conductor. When the conductor is wound into a coil that can rotate between the poles of a magnet, it forms the basis of an electric motor and generator. 2. Any current of air involved in convection in the atmosphere; usually, the upward-moving portion of

a convection circulation, such as a thermal or the updraft in cumulus clouds. 3. The current that occurs within the geosphere as a result of temperature variations; this causes mass movement of subcrustal or mantle material. 4. **Ocean current** is a continuous, directed movement of seawater from one location to another driven by wind, water density differences and tides. It is caused or generated mainly by forces acting upon this mean flow, such as breaking waves, wind, the Coriolis effect, cabbeling and temperature and salinity differences, while tides are caused by the gravitational pull of the Sun and Moon. Currents affect the Earth's climate by driving warm water from the Equator and cold water from the poles around the Earth. Currents also occur in a river making the water flow from high points to lower ones and eventually down to a larger body of water. The force of gravity, which makes the water flow downward, creates river currents. It is also influenced by the volume; the river's steepness (stream gradient); and the riverbed's topography, including sand bars, basins, rocks and dams. 5. A **rip current** is a powerful, narrow channel of fast-moving water flowing from the beach back to the water, when excess water that has accumulated along a shore due to wind and waves rushes back suddenly. It is also influenced by the beach's topography such as dunes or marshes, sandbars, piers and reefs. Topography is the surface features of an area.

CURRENT BALANCE (AMPERE BALANCE; KELVIN BALANCE): An electromechanical device used to measure current absolutely, on the basis of definition of the ampere.

CURRENT CONFIGURATION: The representation of the arrangement of a body under motion at a given time.

CURRENT DENSITY: 1. A vector quantity whose magnitude is the ratio of the magnitude of current flowing in a conductor to the cross-sectional area perpendicular to the current flow and whose direction points in the direction of the current. Its symbol is j or J . 2. The number of subatomic particles per unit time crossing a unit area in a designated plane perpendicular to the direction of movement of the particles.

CURRENT SHARING: A condition in a composite superconducting wire in which the electric current is shared between the superconducting filaments and the resistive matrix material.

CURRY TREE (*Murraya koenigii*): A small tree belonging to the family, Rutaceae growing to the height of 4-6 metres with pinnately compound leaves, found in tropical to sub-tropical regions. The leaves are often used in Indian cooking in curries as well as in many dishes to add flavour. The leave, when fresh has a short shelf life and loses most of its flavour when dried. The plant produces flowers, which can self-pollinate to small shiny-black berries containing a single, large viable seeds.

CUSOR (POINTER): The movable symbol that indicates the current entry position on a computer screen or the movable symbol used to point to some element on screen.

CUSOR CONTROL KEY: A group of keys on the keyboard that are used to direct the movement of the cursor on the display screen.

CURTATE TROCHOID (CONTRACTED TROCHOID): A trochoid in which the point lies inside the rolling circle. A curtate trochoid is traced by a pedal when a bicycle is pedalled along a straight line.

CURVATURE: The reciprocal of the radius of the circle that best matches a curve at a given point.

CURVATURE ABERRATION: A lack of spatial correspondence that causes the image of a straight extended object to appear curved.

CURVATURE OF FIELD (FIELD CURVATURE): A lens aberration causing the plane of sharp focus to be curved. The image plane formed by a single lens is naturally curved. While one part of the field will be in good focus, the rest will need refocusing to be sharp. While the eye may partially correct for this, a camera lens will not and the final image as photographed will not be in perfect focus over the entire image

plane. In the presence of astigmatism, this problem is compounded because there are two separate astigmatic focal surfaces.

CURVATURE TENSOR: The basic mathematical object in Einstein's general theory of relativity. It determines how the tipping of light cones changes from time to time and place to place in the history of the universe.

CURVATURE VECTOR: Of a space curve, the derivative of a unit tangent vector with respect to arc length; the second derivative of the position vector of the curve when parametrized by arc length; the product of the curvature and the unit vector in the normal direction.

CURVE: 1. The locus of a point moving according to specified

conditions or the circle in the locus of all points equidistant from a given point, the centre. A curve can be classified into two categories, an open curve or a closed curve. In an open curve the starting and end points are different; and in a closed curve the starting point is always joined to the endpoint. An example of an open curve is a parabola; that of a closed curve is a circle. 2. A system of points whose coordinates satisfy a given equation; a locus of points. 3. A line plotted on a graph from statistical data or a line whose points are defined by an equation and whose coordinates are functions of an independent variable.

CURVE FITTING: A curve which is obtained by plotting a set of data points. Examples are circular, elliptical, parabolic, hyperbolic, sine and cosine functions.

CURVE OF CONSTANT WIDTH: A curve that, when rotated in a square, makes continuous contact with all four sides.

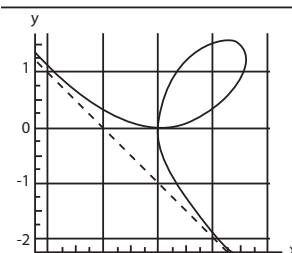
CURVE-CROSSING MODEL: Model of organic reactivity that generates a reaction profile from curves that describe the energies of reactant, product and intermediate electronic configurations (or, alternatively, reactant, product and intermediate electronic states) as a function of the reaction coordinate.

The crossing reflects the electronic reorganisation that accompanies the transformation of reactants and products.

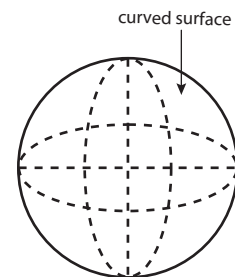
CURVED SURFACE: A 2-dimensional surface that does not lie in a plane. Spheres, cylinders, and cones each have one curved surface.

CURVILINEAR: 1. Consisting of or characterised by bounded by curved lines or lines. 2. Of a set of coordinates, determined by or determining a system of three usually orthogonal surfaces, which are the coordinate planes in a Cartesian coordinates system.

CURVILINEAR INTEGRAL (LINE INTEGRAL): An integral in which the function to be integrated is evaluated along a curve, which may be of the first kind or second kind. The line integral of the second kind is a linear combination of line integrals of the first kind. An ordinary definite integral is defined over a line segment, whereas a line integral may use a more general path, such as a parabola or a circle. Line integrals are used extensively in the theory of functions of a complex variable. A **linear integral of the first kind** is denoted by $\int_C \phi(x) dx$, of a scalar field, $\phi(x)$, along a curve C in



An example of a curve on the Cartesian plane



Curved surface of a sphere

n -dimensional Euclidean space, with respect to i^{th} position variable; if C can be represented as $x(t)$, in terms of a parameter, t in the interval $[a, b]$, then it is equal to $\int_a^b \phi(x(t)x_i) dt$.

CURVILINEAR RELATION: A relation between two variables that is not linear but appears as a curve when the relation is graphed.

CUSP: 1. A sharp raised protuberance on the surface of a molar tooth that helps in grinding food during chewing. 2. A flap forming part of a valve. 3. In geometry, it is also called spinode and it is a point at which two arcs of a curve meet and the tangents to each coincide. 4. In botany, a point, ending abruptly or unexpectedly in a triangular, stiff or sharp apex, seen in some leaves, as shown in the diagram. **Cuspidate** means suddenly narrowing to a point. The term is especially used of leaves.

CUSTOMER INFORMATION CONTROL SYSTEM (CICS): An IBM licensed program that provides online transaction processing services and management for critical business applications. CICS runs on many IBM and non-IBM platforms (from the desktop to the mainframe) and is used in various types of networks that range in size from a few terminals to many thousands of terminals. The CICS application programming interface (API) enables programmers to port applications among the hardware and software platforms on which CICS is available. Each product in the CICS family can interface with the other products in the CICS family, thus enabling interproduct communication.

CUT: 1. To remove a portion of the complex plane, leaving a cut plane, so that some function has an analytic branch in the remaining region; thus the principal value of logarithm lies in the cut plane excluding $]-\infty, 0]$. 2. In topology, to remove a point, a cut point, from a set so that the remainder of the set is disconnected. For example, the interval $(-1, 1)$ is disconnected by cutting zero. 3. In logic, a rule of elimination in sequent calculus, according to which $A, C \vdash D$ can be derived from $A \vdash B$ and $B, C \vdash D$. 4. In optimisation, to remove a portion of the feasible set by adding a linear inequality, called a cutting plane, as in integer programming so as to remove the current estimate of the solution while maintaining all possible solutions. 5. In computer science, it is a command located under the edit menu, allowing users to cut desired text, graph or picture and move text or files to a new location on the computer.

CUT FLOWER: A flower that is harvested for its brightly coloured and showy blossom.

CUT OFF: The size of particles at which the retention efficiency of an instrument device drops below a specified value under defined conditions.

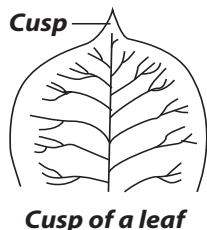
CUT-OFF FILTER: An optical device which only permits the transmission of radiation of wavelengths that are longer than or shorter than a specified wavelength. Usually, the term refers to devices which transmit radiation of wavelengths longer than the specified wavelength.

CUT-ON FILTER: An optical device that only permits the transmission of radiation of wavelengths shorter than a specified wavelength.

CUTAN: 1. A modification of the texture, structure, or fabric at natural surfaces in soil materials composed of any of the component substances of the soil material. 2. A hydrocarbon polymer, of uncertain structure but thought to be similar to kerogen, found in some plants.

CUTANEOUS RESPIRATION: Breathing through the skin in some vertebrates such as members of the class Amphibia that have soft highly vacularised skin for gaseous exchange.

CUTANEOUS STREPTOTHRICHOSIS (DERMATOPHILUS INFECTION; LUMPY WOOL; STRAWBERRY FOOTROT): A bacterial disease, *Dermatophilus congolensis*, which affects sheep,



cattle and horses causing ulcers. It also affects people, who attend to infected animals if elementary hygiene measures are not observed. It causes severe skin infections.

CUTE NUMBER: A number n such that a square can be cut into n squares of, at most, two different sizes. For example, 4 and 10 are cute numbers.

CUTICLE: 1. Waxy protective covering of the epidermis of aerial plant organs. 2. The horny non-cellular surface layer of many invertebrates such as insects.

CUTICULARISATION: The secretion by the epidermis of plants and many invertebrates of substances that then harden to form a cuticle.

CUTIN: A mixture of waxy substances that impregnates the cell wall and covers the outer surface of mature epidermal plant cells, making the walls more or less impervious to water. **Cutinisation** is the deposition of cutin in plant cell walls, principally in the epidermis.

CUTLASS (MACHETE): A large heavy broad blade, used as a weapon and an implement for cutting vegetation, small trees and harvesting crops such as cereals.

CUTOFF: In chemistry, it is a valve that controls the flow of gas or fluid through a pipe.

CUTOFF FREQUENCY: The upper and lower frequency beyond which no appreciable energy is transmitted.

CUTTING: In botany, it is a part of a living plant such as a stem, bud or leaf which has been cut so that it develops into a new daughter plant if grown in soil or in a suitable culture medium. Often the cut end of a stem cutting is dipped into a rooting powder, which contains a natural or synthetic auxin that stimulates the rapid development of adventitious roots and the establishment of a new daughter plant.

CUTTING AGENT (ADULTERANT): A substance such as flour, starch, sugar or caffeine used to dilute illegal drugs such as heroine and cocaine.

CUTWORM: Any of the various types of larvae of many moths of the family Noctuidae that eat the stems of young plants near the ground. The cutworm is a serious insect pest of tobacco.

CYANAMIDE: 1. A colourless crystalline solid, with the chemical formula H_2N-CN , relative density 1.28, melting point $44^\circ C$ (317 K) and boiling point $260^\circ C$ (533 K). It is made by the action of carbon dioxide on hot sodamide. It is a weakly acidic compound (the parent acid of cyanamide salts) that is soluble in water and ethanol. It is hydrolysed to urea in acidic solutions. Calcium cyanamide is used in agriculture as a fertiliser, some is used as a weed killer, as a pesticide and as a cotton defoliant. 2. An inorganic salt containing the ion CN_2^{2-} .

CYANAMIDE PROCESS: A process used in the manufacture of calcium cyanamide ($CaCN_2$), a colourless crystalline powder, in which nitrogen gas is passed through finely divided calcium carbide at temperatures of between $1000^\circ C$ - $1100^\circ C$ (1273 K-1373 K).

CYANATES: Salts and esters of cyanic acid, $HOC\equiv N$. Examples are KOCN potassium cyanate and $PhOCN$ phenyl cyanate.

CYANIC ACID (ISOCYANIC ACID; FULMINIC ACID): An unstable explosive acid, with the chemical formula $HOCN$, relative density 1.14, melting point $-86^\circ C$ (187K) and boiling point $23.5^\circ C$ (296.65K). The compound has the structure $H-O-C\equiv N$. Its salts and esters are cyanates or fulminates. It is a volatile liquid, which readily polymerises. In water, it hydrolyses to ammonia and carbon dioxide. Its salts and esters are isocyanates.

CYANIDE: An inorganic salt containing the CN^- cyanide ion. Examples are potassium cyanide (KCN), calcium cyanide ($Ca(CN)_2$) and hydrocyanic or prussic acid (HCN). Cyanides are extremely poisonous because of the ability to coordinate the iron in haemoglobin, thereby blocking the uptake of oxygen by blood. The free acid is hydrocyanic (or prussic) acid (HCN). Chemically, the simple inorganic cyanides resemble halides in many ways. Organic compounds containing this group are called nitriles.

CYANIDE PROCESS (GOLD CYANIDATION; MACARTHUR-FORREST PROCESS): A method of extracting gold by dissolving it in sodium cyanide to form the complex ion $[\text{Au}(\text{CN})_2]$. The ion can be reduced back to gold with zinc, which displaces the gold in the complex.

CYANOACRYLATE: A compound formed by the condensation of an alkyl cyanoethanoate and methanol, $\text{CH}_2\text{C}(\text{CN})\text{COOR}$. The methyl and ethyl esters form the basis of superglues, because of their rapid polymerisation in air.

CYANOBACTERIA (BLUE-GREEN ALGAE; BLUE-GREEN BACTERIA; CYANOPHYTA): A phylum of single-celled organisms that resemble bacteria in their internal cell structure and lack an enclosed nucleus and other specialised cell structures. They are prokaryotic and are the earliest known form of life on the earth. They are widely distributed in aquatic habitats, on the damp surfaces of rocks and trees and in the soil; where they are joined together in colonies or filaments; and obtain energy through photosynthesis and reproduce by simple cell division or by fragmentation of the filaments. Some fix nitrogen and thus are necessary to the nitrogen cycle, while others live a symbiotic existence; for example, living in association with fungi to form lichens. Sometimes they form algae blooms on fresh water bodies, where the water is polluted by nitrates and phosphates from fertilisers and detergents and cover the water's surface. Overtime, they give off toxins as they decay, which kill fish and other wildlife and can be harmful to domestic animals, cattle and people.

CYANOCOBALAMIN: The chemical name for vitamin B_{12} , which is normally produced by microorganisms in the gut. Cyanocobalamin is not synthesised by plants or animals and commercial production is from bacterial cultures such as *Streptomyces olivaceus*. It is not needed by plants but animals need it for the normal development of erythrocytes.

CYANOGEN: A colourless, highly poisonous gas, with the chemical formula C_2N_2 , relative density 0.95, melting point -28°C (245 K) and boiling point -21°C (252 K). It is made by the action of acids on cyanides. It is very toxic, as it readily undergoes reduction to cyanide and can poison the cytochrome c oxidase complex, thus interrupting the mitochondrial electron transfer chain. It is used as a rocket propellant, an insecticide and a chemical weapon.

CYANOGENESIS: The natural release and generation of hydrogen cyanide glycosides that occurs in some plants including cherry laurel (*Prunus laurocerasus*), birdsfoot trefoil (*Lotus corniculatus*) and white clover (*Trifolium repens*). The glycosides, found largely in water solution in the vacuole, yield a sugar and hydrogen cyanide on hydrolysis.

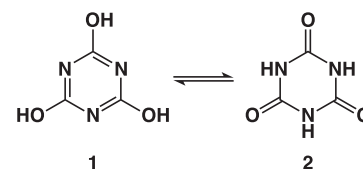
CYANOGENIC: Compounds able to produce cyanide. Examples are the cyanogenic glycosides such as amygdalin in peach and apricot stones.

CYANOHYDRINS: Organic compounds formed by the addition of hydrogen cyanide to aldehydes or ketones in the presence of a base. Cyanohydrins have the formula $\text{R}_2\text{C}(\text{OH})\text{CN}$, where R is H, alkyl or aryl. The first step is attack by a CN^- ion on the carbonyl carbon atom. The final product is a compound in which a $-\text{CN}$ and $-\text{OH}$ group are attached to the same carbon atom.

CYANOPHYTA (CYANOBACTERIUM; BLUE-GREEN ALGAE; BLUE-GREEN BACTERIA; CYANOBACTERIA): A phylum of single-celled organisms that resemble bacteria in their internal cell structure and lack an enclosed nucleus and other specialised cell structures. They are prokaryotic and are the earliest known form of life on the earth. They are widely distributed in aquatic habitats, on the damp surfaces of rocks and trees and in the soil; where they are joined together in colonies or filaments; and obtain energy through photosynthesis and reproduce by simple cell division or by fragmentation of the filaments. Some fix nitrogen and thus are necessary to the nitrogen cycle, while others live a symbiotic existence; for example, living in association with fungi to form lichens. Sometimes they form algae blooms on fresh water bodies, where the water is

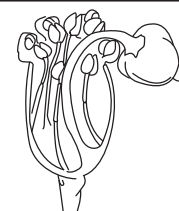
polluted by nitrates and phosphates from fertilisers and detergents and cover the water's surface. Overtime, they give off toxins as they decay, which kill fish and other wildlife and can be harmful to domestic animals, cattle and people.

CYANURIC ACID (1,3,5-TRIAZINE-2,4,6-TRIOL): A white crystalline water soluble trimer of cyanic acid, with the chemical formula $(\text{HNCO})_3$. It is a cyclic compound having a six cyanamide membered ring made of alternating imide (NH) and carbonyl (CO) groups. It is used as a precursor or a component of bleaches, disinfectants and herbicides.



Chemical structure of Cyanuric Acid

CYATHIUM: A specialised cup-shaped type of inflorescence found in members of the Euphorbiaceae. It resembles a single flower and consists of a female flower in the centre surrounded by many simple male flowers enclosed within an involucre. The involucre also bears a number of petaloid glands along its rim.



Cyathium inflorescence

CYBER: A prefix often used to describe computers, electronics, the Internet or anything in the virtual world.

CYBERBULLYING: The use of electronic communication to bully a person, typically by sending messages of an intimidating or threatening nature.

CYBERCAFÉ: A cafe that has a number of personal computers connected to the Internet and available for use by customers.

CYBERCRIME: The use of electronic communication to do criminal activities such as downloading illegal music files, stealing money from online bank accounts, creating and distributing viruses on other computers or posting confidential business information on the Internet.

CYBERJOCKEY: A volunteer who works online, in online chat rooms and Internet discussion forums, answering questions, primarily to new users.

CYBERNETICS: The science concerned with how systems organise, regulate and reproduce and also how they evolve and learn. It is a relatively new interdisciplinary science that concerns the way in which all types of biological, mechanical and social systems function and operate. Cybernetics research is devoted to: (i) the internal autonomous regulatory processes within systems; (ii) the control and regulation of and within systems; and (iii) the communication or transmission of information between different parts of the system and its environment.

CYBERSPACE: A notional, non-physical domain characterised by the use of electronics and the electromagnetic spectrum to store, modify and exchange data via networked systems and associated physical infrastructures. It is the interconnection of human beings through computers and telecommunication, without regard to physical geography.

CYBERSQUATTING: A term used to describe an individual or company who intentionally purchases a domain and holds that domain with the hope of reselling it at a profit. It also includes stealing or misspelling a domain name in order to profit from an increase in website visits.

CYBERSTALKING: Repeated threats or harassing behaviour of an individual, group, or organisation conducted over any electronic communication medium, such as e-mail, instant messaging, messages posted to a Web site, or other Internet communications method, using anonymity afforded by the Internet. Such behaviours, which

constitute criminal offences include false accusations, defamation, slander and libel.

CYBERTERRORISM: U.S. Federal Bureau of Investigation defines cyberterrorism as any "premeditated, politically motivated attack against information, computer systems, computer programs, and data which results in violence against non-combatant targets by sub-national groups or clandestine agents."

CYBERWARFARE: Politically motivated attacks on computer and information systems conducted through the Internet. An example is the attack on Estonia's Internet system in 2007 by the Russian state or its proxies.

CYBOTACTIC REGION: That part of a solution in the vicinity of a solute molecule in which the ordering of the solvent molecules is modified by the presence of the solute molecule.

CYBOTAXIS: In physics, it is a three dimensional arrangement of molecules in a liquid crystal evidenced by x-ray diffraction effects, in which there is combined mobility of the constituent molecules, as well as a recognisable spatial arrangement.

CYCADALES: An order of tropical and subtropical organisms in the Gymnospermae that contains many extinct species and a few of the present gymnosperms. Cycads have a thick, generally unbranched, stem bearing a terminal rosette of large palmlike leaves. Scale leaves are also present on the upper part of the stem and apex. They are very slow-growing and vary considerably in height between species

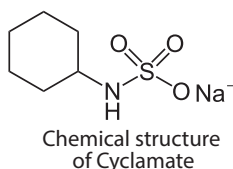
CYCADOFILICALES (PTERIDOSPERMALES; SEED FERNS): An extinct order of gymnosperms that flourished in the Carboniferous period. They possessed characteristics of both the fern and the seed plants in reproducing by means of seed and yet retaining fernlike leaves.

CYCADOPHYTA (CYCAD): A phylum of seed plants that contains many extinct species; the few modern representatives of the group include *Cycas* and *Zamia*. Cycads are characterised by a large crown of compound leaves and a stout trunk and inhabit tropical and subtropical regions, sometimes growing to a height of 20 metres. The stem bears a crown of fernlike leaves. These species are among the most primitive of living seed plants.

CYCAS: A single genus in the family Cycadaceae of about 95 species consisting of widely distributed evergreen tropical trees having pinnate leaves and columnar stems covered with persistent bases of old leaves, native to equatorial regions. They are dioecious plants which take many years to grow, with the female plants producing megasporophylls (a group of leaf-like structures) each with naked seeds on the lower margins. The male plants bear strobila (singular: strobilus) on them. The leaves are pinnately compound, arranged spirally with thick and hard keratinose. Sexual reproduction takes place during an exclusive vegetative growth which occurs by bulbils arising at the base of the trunk. The best-known species is *Cycas revoluta*, commonly known as sago palm, king sago, sago cycad or Japanese sago palm, native to southern Japan. Sago, an edible starch is obtained from the plant which is used in cooking as well as stiffening textiles. It is also cultivated as an ornamental plant.

CYCLAMATE (SODIUM CYCLAMATE; SODIUM N-CYCLOHEXYLSULFAMATE):

Salts of cyclohexylsulfamic acid, with the chemical formula $C_6H_{11}.NH.SO_3H$ and formerly used as an artificial sweetener. It is 30–50 times sweeter than sugar.



CYCLE: 1. A single recurring series of changes in, for instance, a wave, a vibration or rotation. In other words, it is a process or series of events that ends at the point at which it begins or that repeats itself continuously at regular intervals. 2. A permutation in which one subset of elements are moved cyclically and

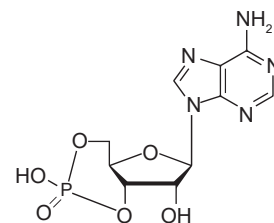
the remainder are not moved, for example, (1,2,5,4,5) (1,4,5,2,5). If γ is a cycle that permutes exactly l elements, then l is the length of the cycle and γ has order l , that is $\gamma^l = e$, where e is the identity permutation. Every permutation has a unique factorisation as a product of disjoint cycles. 3. A simple closed graph.

CYCLE PATTERN: A representation of the unique decomposition of a permutation as a product of disjoint cycles. If γ is a permutation of n symbols and the decomposition consists of λ cycles of length j for each j between 1 and n , then the cycle pattern is written $1^{\lambda_1} 2^{\lambda_2} \dots n^{\lambda_n}$. Two elements of the permutation group S are conjugate if and only if they have the same cycle pattern and the number of permutations with a given cycle pattern is $\frac{n!}{(1^{\lambda_1})(2^{\lambda_2}) \dots (n^{\lambda_n})(\lambda_1)!(\lambda_2)! \dots (\lambda_n)!}$.

CYCLIC: 1. Occurring in cycles or repeated regularly such as pattern of weather changes. 2. In botany, it refers to having parts arranged in a whorl; or forming a whorl. 3. In chemistry, it refers to the molecule of any compound whose atoms form a ring. For example, benzene and the cycloalkanes are simple cyclic compounds. 4. In mathematics, the term is used in several contexts, for example, **Cyclic graph** in algebra is a diagram representing the cycles determined by taking powers of group elements; **Cyclic notation** is a way of writing permutations; **Cyclic permutation** is a permutation with one nontrivial orbit; **Cyclic polygon**, is a polygon which can be given a circumscribed circle; **Cyclic quadrilateral** is a four-sided figure with all four of its vertices lying on the circumference of a circle; etc.

CYCLIC ADENOSINE MONOPHOSPHATE (cAMP, CYCLIC AMP; 3'-5'-CYCLIC ADENOSINE MONOPHOSPHATE):

A second messenger, which plays a key role in metabolic regulation. It is derived from adenosine triphosphate (ATP) and used for intracellular signal transduction in many different organisms, conveying the cAMP-dependent pathway.



Chemical structure of Cyclic Adenosine Monophosphate

CYCLIC ADP-RIBOSE (cADPR; ADENOSINE; CYCLIC DIPHOSPHORIBOSE): A second messenger synthesised by the multifunctional transmembrane ectoenzyme CD38 in various systems particularly platelets, microsomes and sea urchin eggs. It is an endogenous regulator of intracellular calcium.

CYCLIC AMP (CYCLIC ADENOSINE MONOPHOSPHATE): A derivative of adenosine triphosphate (ATP) that is widespread in animal cells as an intermediate messenger in many biological reactions induced by hormones. It is also an important component of regulatory processes in bacteria and other prokaryotes and slime moulds.

CYCLIC BEHAVIOURS: Innate behaviours, such as migration, circadian rhythms or hibernation that occur in a repeated pattern, often in response to changes in the environment.

CYCLIC COMPOUND: Any group of organic chemicals that have rings of atoms in their molecules, giving them a closed chain structure. Examples are benzene, cycloalkanes and heterocyclics.

CYCLIC COORDINATES (IGNORABLE COORDINATE): A generalised coordinate on which the Lagrangian of a system does not depend explicitly.

CYCLIC FLOWER: A flower arranged in a series of whorls and sets of identical organs attached around the axis at the same point. A typical flower has a cyclic arrangement, consisting of a single whorl of sepals known as a **calyx**; a single whorl of petals called a **corolla**; one or more whorls of stamens called the **androecium**; and a single whorl of carpels termed the **gynoecium**.

CYCLIC GUANOSINE MONOPHOSPHATE (CGMP; CYCLIC GMP): A derivative of the nucleotide guanosine triphosphate (GTP)

that, like cyclic AMP, acts as a second messenger in signaling pathways within cells. It activates protein kinase G, which in turn activates specific intracellular proteins by phosphorylation. It also regulates light-dependent opening and closing of ion channels in the rod-cell plasma membrane of the retina of most mammals.

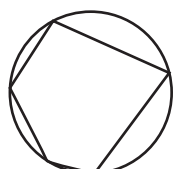
CYCLIC GROUP: A group in which all the members are powers of a given element, called the generator. Examples are the integers modulo n under addition, for which $n - 1$ is a generator and any subgroup of a cyclic group.

CYCLIC HYDROCARBONS: One of the two major groups into which hydrocarbons are classified. Cyclic hydrocarbons form a closed ring and are classified into alicyclic or aromatic hydrocarbons. Alicyclic hydrocarbons contain only carbon and hydrogen atoms joined to form one or more rings and have the properties of both aliphatic and cyclic substances. Aromatic hydrocarbons contain one or more benzene rings. Examples of cyclic hydrocarbons are benzene, which is aromatic (C_6H_6) and cyclopentane, which is alicyclic (C_5H_{10}). The other major group of hydrocarbons is aliphatic or open chain hydrocarbons.

CYCLIC NUMBER: A number with n digits, which, when multiplied by 1, 2, 3, ..., n produces the same digits in a different order. For example, 142857 is a cyclic number: $142857 \times 2 = 285714$; $142857 \times 3 = 428571$; $142857 \times 4 = 571428$; $142857 \times 5 = 714285$; $142857 \times 6 = 857142$; etc. It has been conjectured, but not yet proven, that an infinite number of cyclic numbers exist.

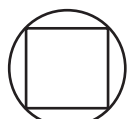
CYCLIC PERMUTATION (CIRCULAR PERMUTATION): A set that results from mapping each element of an ordered set to its successor. This is given by $(n - 1)!$. The number is $(n - 1)!$, instead of the usual factorial $n!$ since all cyclic permutations of objects are equivalent because the circle can be rotated.

CYCLIC POLYGON: A polygon in which each vertex (corner) lies on the circumference of a circle.



Cyclic polygon

CYCLIC QUADRILATERAL: A four-sided figure with all four of its vertices lying on the circumference of a circle. In a cyclic quadrilateral, opposite angles are supplementary; exterior angles are equal to the opposite interior angles; and the product of the two diagonals is equal to the sum of the products of opposite sides.



Examples of cyclic quadrilateral

CYCLIC REDUNDANCY CHECK, CRC: A method of checking for errors in data that has been transmitted on a communications link, in which data is sent with a block of data that helps verify all data is received correctly. It is a hash function that detects accidental changes to raw computer data commonly used in digital telecommunications networks and storage devices such as hard disk drives. Mathematically, data receivers check on the check value attached by finding the remainder of the polynomial division of the contents transmitted. If it seems that an error has occurred, a negative acknowledgement is transmitted asking for data retransmission.

CYCLIN: A family of proteins that controls the progression of cells through the cell cycle by activating cyclin-dependent kinase (Cdk) enzymes.

CYCLING: In mathematics, it is the behaviour of an algorithm when the same approximate solution recurs repeatedly. It can happen in Newton's method for finding a zero of a real function that every second value produced is the same.

CYCLISATION (CYCLIZATION): The formation of a cyclic compound from an open-chain compound.

CYCLITOLS: Hydroxylated cycloalkanes containing at least three hydroxy groups, each attached to a different ring carbon atom.

CYCLO-: A prefix designating a cyclic compound, for example, a cycloalkane or a cyclosilicate.

CYCLO-OCTATETRAENE (COT): A flammable yellow liquid cyclic compound, with the chemical formula C_8H_8 , melting point -5 to -3 °C (268-270 K) and boiling point $142-143$ °C (415-416 K). It has eight carbon atoms and four double bonds in its ring. It is not aromatic and made by polymerising ethyne (acetylene) in the presence of nickel salts. Its reactivity is characteristic of an ordinary polyene.

CYCLO-OXYGENASE (COX; PROSTAGLANDIN-ENDOPEROXIDE SYNTHETASE): An enzyme that catalyses the key reactions in the formation of prostaglandins from arachidonic acid, linolenic acid and eicosa-5, 8, 11, 14, 17-pentaenoic acid. Non steroidal anti-inflammatory drugs such as aspirin and ibuprofen, inhibit the cyclo-oxygenase enzymes COX1 and COX2, thus blocking the synthesis of prostaglandins and related compounds, such as prostacyclins and thromboxane A_2 and relieving the symptoms of fever, inflammation and pain associated with them.

CYCLOADDITION: A reaction in which two or more unsaturated compounds form a cyclic adductor in which a cyclic compound is formed by addition between unsaturated parts of same molecule.

CYCLOALKANES: Saturated hydrocarbons containing one or more rings of carbon atoms joined by single bonds. They have the general formula C_nH_{2n} , for example cyclohexane, C_6H_{12} . They are named with the prefix "cyclo". In general, they behave like the alkanes but are rather less reactive.

CYCLOALKYL GROUPS: Univalent groups derived from cycloalkanes by removal of a hydrogen atom from a ring carbon atom, for example, 2-methylcyclopropyl.

CYCLOBUTADIENE: A short-lived cyclic diene hydrocarbon, with the chemical formula C_4H_4 , with its four carbon atoms in a square ring. It is made by degradation of its metal complexes and has a short lifetime of a few seconds.

CYCLODEXTRINS: A group of structurally related natural products formed during bacterial digestion of cellulose. These cyclic oligosaccharides consist of (α -1,4)-linked α -D-glucopyranose units and contain a somewhat lipophilic central cavity and a hydrophilic outer surface.

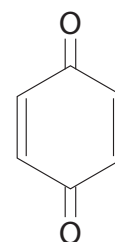
CYCLOELIMINATION (CYCLOREVERSION): The reverse of cycloaddition. The term is preferred to the synonyms 'cycloreversion', 'retro-addition' and 'retrocycloaddition'.

CYCLOGENESIS: The process of forming storm systems or cyclone in the atmosphere in a low-pressure area.

CYCLOGRAPH (ARCOGRAPH): An instrument for drawing arcs of large diameter circles, whose centres are inconveniently or inaccessibly located, therefore does not use a central point.

CYCLOHEXADIENE-1,4-DIONE (BENZOQUINONE; QUINONE):

A yellow solid, with the chemical formula $C_6H_4O_2$, relative density 1.3 and melting point 116 °C (389 K). It has a six-membered ring of carbon atoms with two opposite carbon atoms linked to oxygen atoms ($C=O$) and the other two pairs of carbon atoms linked by double bonds ($HC=CH$). The compound is used in making dyes.

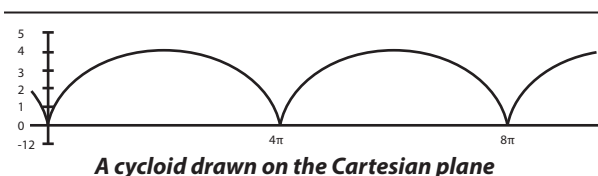


Structure of cyclohexadiene-1,4-dione (benzoquinone or quinone)

CYCLOHEXANE: A colourless liquid cycloalkane, with the chemical formula C_6H_{12} , relative density 0.78, melting point 6.5 °C (279.65K) and boiling point 81 °C (354K), which occurs in petroleum. It is made by passing benzene and hydrogen under pressure over a heated Raney nickel catalyst at 150 °C (423K) or by the reduction of cyclohexanone. It can also be

oxidised using hot concentrated nitric acid to hexanedioic acid (adipic acid). It is used as a solvent and paint remover.

CYCLOID: A curve traced by a point on the circumference of a circle that rolls along straight lines. It can be defined by the parametric equations $x = x_0 + a(\theta - \sin \theta)$, $y = y_0 + a(1 - \cos \theta)$. Its applications include the study of the motion of wheeled vehicles along roads and tracks.



CYCLOMETRIC FUNCTIONS (TRIGONOMETRIC FUNCTIONS): Functions defined in terms of angle and sides of a right-angled triangle and widely used in the solution of many mathematical problems; or any of the six functions: sine, cosine, tangent, cotangent, secant and cosecant that represent ratios of sides of right angle triangle. For example, for acute angle x in a right triangle ABC, the trigonometric functions are as follows:

$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}}, \cos x = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan x = \frac{\text{opposite}}{\text{adjacent}}, \cot x = \frac{\text{adjacent}}{\text{opposite}}$$

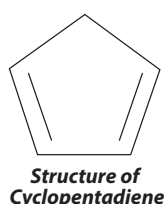
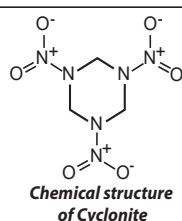
$$\sec x = \frac{\text{hypotenuse}}{\text{adjacent}}, \operatorname{cosec} x = \frac{\text{hypotenuse}}{\text{opposite}}.$$

CYCLOMORPHOSIS (SEASONAL POLYPHENISM): The occurrence of seasonal changes in the phenotype of an organism through successive generations. It occurs in small aquatic invertebrates that reproduce by parthenogenesis and give rise to several generations annually, such as rotifers and cladoceran crustaceans. An example is the cyclomorphic species of *Daphnia* that undergoes changes in the shape of the head.

CYCLONE: 1. Also known as **depression** or **area of low atmospheric pressure**, it is a system of winds rotating inwards around a calm area of low pressure; or a violent, rotating, destructive wind storm or a large-scale storm system or tornado with heavy rain and winds. It rotates anticlockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere towards a low-pressure centre. 2. In atmospheric chemistry, it is a dust, grit or droplet separator utilising essentially the centrifugal force derived from the motion of the gas. The flow of gases containing suspended particles into the device is transformed into a confined vortex from which centrifugal forces tend to drive the suspended particles to the wall of the cyclone body. The agglomerated particles are subsequently removed from the cyclone by gravitational action.

CYCLONITE (HEXAGEN; RDX): A powerful explosive derived from hexahydro-trinitro-triazine, with the chemical formula $(\text{CH}_2)_3\text{N}_3(\text{NO}_2)_3$.

CYCLOPENTADIENE: A colourless liquid cyclic hydrocarbon, with the chemical formula C_5H_6 having a ring of five carbons, containing two carbon-carbon double bonds. It forms the cyclopentadienyl ion C_5H_5^- present in compounds like ferrocene.



CYCLOPHANE: A compound consisting of one or more aromatic rings forming part of a larger ring system in which aliphatic chains of the CH_2 groups link the aromatic rings.

CYCLOPOLYMERISATION: A polymerisation in which the number of cyclic structures in the constitutional units of the resulting macromolecules is larger than in the monomer molecules.

CYCLOPROPANE: A colourless gas, with the chemical formula C_3H_6 , whose molecules contain a triangular ring of carbon atoms. It is made by treating 1,3-dibromopropane with zinc metal and is used as a general anaesthetic.

CYCLOSILOXANES: Compounds having rings of alternating silicon and oxygen atoms, for example, 2,2-dimethylcyclotrisiloxane.

CYCLOSIS (CYTOPLASMIC STREAMING):

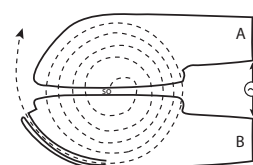
The movement of cytoplasm within cells, which is not seen in prokaryotic cells. The viscosity of cytoplasm can vary as a result of changes in the nature of protein molecules, particularly actin. In the fluid form it flows freely and can be observed carrying organelles and other inclusions in the cell. Cyclosis is very variable in plant cells. For example, in leaves of *Elodea* (waterweeds) it is more active in cells in the midrib region than in cells near the margin. The rate of flow increases with rise in temperature but is normally less than 0.1 mm per second.

CYCLOSTOMATA: A class of marine and fresh water vertebrates in the order of bryozoans in the class Stenolaemata that lack jaws.

CYCLOSYMMETRIC: A symmetric function whose value is unchanged by cyclic permutation of its variables.

CYCLOTOMIC: Of or pertaining to n th roots of unity. The cyclotomic equation is $z^n - 1 = 0$. The n th cyclotomic polynomial is the polynomial, whose roots are all the primitive roots of unity and thus, for prime n , is the cyclotomic equation of degree n . A cyclotomic field is an extension field of the rationals obtained by adjoining a root of unity.

CYCLOTRON: A cyclic particle accelerator in which charged particles fed into the centre of the device are accelerated in an outward spiral path inside two hollow D-shaped conductors placed to form a split circle. A cyclotron is capable of generating particle energies between a few million and several tens of millions of electron volts. The ion to be projected is placed at S in between the D-shaped conductors A and B placed between the poles of a powerful electromagnetic. It the poles are connected to an a.c. source, the ion will be accelerated each time it crosses the gap between the two conductors.



Cyclotron

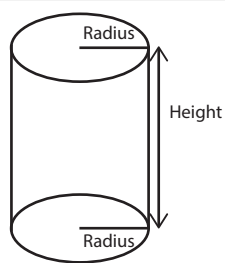
CYCLOTRON FREQUENCY (GYROFREQUENCY): The number of times per second that a particle orbits a magnetic field line. The frequency is completely determined by the strength of the field and the particle's charge-to-mass ratio. Cyclotron frequency is twice the Larmor frequency of precession.

CYCLOTRON RADIATION: Electromagnetic radiation given off by charged particles that are spiraling along magnetic field lines. The cyclotron frequency, also known as the gyrofrequency, which is the number of times per second that a particle orbits a magnetic field line, is completely determined by the strength of the field and the particle's charge-to-mass ratio. The cyclotron frequency is twice the Larmor frequency of precession.

CYCLOTRON RADIUS (GYRORADIUS; LARMOR RADIUS): Radius of orbit of charged particle about a magnetic field line.

CYCLOTRON RESONANCE: Charged particles in a magnetic field at the same frequency with (and absorb energy from) an electric field (perpendicular to the magnetic field) which oscillates at the particles' cyclotron frequency or at a harmonic (multiple) of that frequency.

CYLINDER: 1. A tube-like solid figure with a circular base or a solid or hollow curved body with circular ends and straight sides. In its everyday use, the term applies to a right cylinder, the curved surface of which is at right angles to the base. 2. A three-dimensional geometric solid bounded by two equal parallel circles and a curved surface formed by moving a straight line so that its ends lie on the circles. For a closed curve, the total surface area (T.S.A.) = $2\pi rh + 2\pi r^2$. The curved surface area = $2\pi rh$. 3. A three-dimensional geometric surface formed by a straight line moving in a circle around and parallel to a fixed straight line, forming a hollow tube shape. 4. In a Cartesian product, it is a set that is the direct product of a set and non-trivial vector space. 5. A rotating part of a revolver containing chambers into which cartridges are loaded. 6. A revolving drum of a printing press that produces or receives the impression.

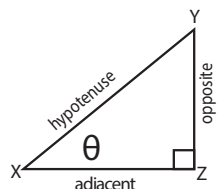


Cylinder

CYLINDER BLOCK: An integrated structure comprising the cylinders of a reciprocating engine and often some or all of their associated surrounding structures such as coolant passages, intake and exhaust passages and ports and crankcase.

CYLINDER HEAD: A detachable plate that covers the closed end of a cylinder chamber in a reciprocating engine or pump.

CYLINDRICAL COORDINATES: A set of coordinates that locate a point in space by the spherical coordinates of the foot of a perpendicular from it to a coordinate plane and its height above that plane; for example, the point (x, y, z) has cylindrical coordinates (r, θ, z) , where (r, θ) are the polar coordinates of (x, y) .



Cylindrical coordinates of a right-angled triangle

CYLINDRICAL POLAR COORDINATES:

A three-dimensional coordinate system that specifies point positions by the distance from a chosen reference axis, the direction from the axis relative to a chosen reference direction and the distance from a chosen reference plane perpendicular to the axis (r, θ, z) . The radius and the azimuth are together called the polar coordinates, as they correspond to a two-dimensional polar coordinate system in the plane through the point, parallel to the reference plane and z is the same as cartesian coordinates.

CYLINDROID: A cylinder with an elliptical cross section.

CYMBIFORM: Shaped like a boat.

CYMOSE INFLORESCENCE (DEFINITE INFLORESCENCE CYME): A flat-topped determinate inflorescence or panniculate in which the first formed flower develops from the growing region at the top of the flower stalk and the other flowers form later on lateral shoots or on branches arising below. **Cymule** refers to a small cyme or a section of a compound cyme.

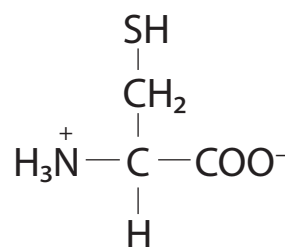
CYPHER (CIPHER): 1. An algorithm for performing encryption or decryption. 2. Any of the Arabic numerals, 1, ..., 9 used for counting.

CYSELA: A dry non-dehiscent single seeded fruit formed from a double ovary in which only one ovule develops into a seed. It is typical of the Compositae, where the fruit is surrounded by hairs derived from the calyx (pappus). This type of fruit is traditionally considered a pseudocarp.

CYST: 1. A hollow organ, bladder or vesicle in an animal containing liquid matter. 2. A sac or capsule that encloses and protects some organisms, such as certain bacteria and protozoans in a dormant or larval stage or a protective covering around a parasite produced by the parasite itself or the host. 3. A spore that is not undergoing cell division as in some algae and fungi.

CYSTEINE (β -MERCAPTOALANINE; -AMINO- β -THIOLPROPIONIC ACID; 2-AMINO-3-MERCAPTOPROPANOIC ACID):

A natural sulfur-containing nonessential amino acid, with the chemical formula $C_3H_7NO_2S$. It is converted to cystine during metabolism. Cysteine is involved in the synthesis of the cofactor biotin. Sulfur-substituted derivatives of cysteine are found in the storage organs of many plants and the sulfide thus stored can be utilised in periods of rapid growth. Its isoelectric point is 5.05.



Chemical structure of cysteine

CYSTINE: A naturally occurring amino acid which is derived from cysteine, with the chemical formula $(-\text{SCH}_2\text{CH}(\text{NH}_2)\text{COOH})_2$. Cystine is necessary for the synthesis of thiamin pyrophosphate, a coenzyme in many decarboxylation reactions. Its isoelectric point is 4.8.

CYSTITIS: Inflammation of the urinary bladder, occurring primarily in young women and frequently resulting from bacterial invasion of the urethra from the adjacent rectum. Symptoms include burning pain during and right after urination, unusually urgent or frequent urination and lower back pain. Its isoelectric point is 4.8.

CYSTOCARP: A swollen urn-shaped fruiting body found in some red algae such as Polysiphonia. It forms following fertilisation of a carpogonium and contains the developing carpospores. The surrounding pseudoparenchymatous tissues develop from cells of the filament below the carpogonium.

CYSTOLITH: A large intracellular structure formed by the deposition of lime on an ingrowth of the cell wall. An enlarged epidermal cell containing such a structure is termed a **lithocyst**. Species in which cystoliths are formed include *Fiats elastica* (rubber plant) and *Urtica dioica* (stinging nettle).

CYTIDINE: A component of RNA, it is a nucleoside comprising one cytosine molecule linked to a D-ribose sugar molecule.

CYTOBIOTAXIS (CYTOCLEISIS): The influence exerted by one cell on the action of other cells. It is the vital principle in all living tissues. It is a form of energy, totally unrelated to electricity, light, heat or sound, which is generated by living tissues.

CYTOCHROME: Any of the respiratory pigments found widely in aerobic organisms which transfer energy within cells. They are proteins containing a chelate complex of an iron prosthetic group and a porphyrin system and so include haemoglobin. Cytochromes undergo a series of oxidation and reduction reactions, transferring electrons from a substrate to each other and finally to an electron-acceptor. In cellular respiration the substrates are the hydrogen acceptors in the citric acid cycle. The electron acceptor is oxygen and the energy produced is stored in adenosine triphosphate (ATP).

CYTOCHROME OXIDASE: An enzyme complex comprising the terminal two cytochromes (a and a_3) of the respiratory chain in the mitochondria. It is responsible for the reduction of oxygen, receiving two electrons from cytochrome c and combining them with two hydrogen ions and oxygen atoms to form water.

CYTOCHROME P450: A member of a superfamily of haeme-containing monooxygenases that catalyses the oxidation of organic compounds through radical intermediates.

CYTOGENETIC MAP: The visual appearance of a chromosome when stained and examined under a microscope. Particularly important are visually distinct regions, called light and dark bands, which give each of the chromosomes a unique appearance. This feature allows a person's chromosomes to be studied in a clinical test known as a karyotype, which allows scientists to look for chromosomal alterations.

CYTOGENETICS: The study of inheritance in relation to the structure and functions of chromosomes. This includes examination of chromosome structure, learning and describing the relationships between chromosome structure and phenotype and seeking out the causes of chromosomal abnormalities. Generally, chromosomes are examined and characterised by obtaining an individual's karyotype, which is a description of the number and structure of the chromosomes. Cytogenetics studies can be performed on fresh blood, bone marrow, prenatal specimens and solid tissue specimens and on fixed specimens. Together with genomics, they are considered an essential component of the diagnostic studies for individuals with congenital malformations, mental retardation, multiple spontaneous miscarriages or infertility and for individuals with haematologic neoplasms and solid tumours.

CYTOKINE: Any soluble factor secreted by cells of the lymphoid system that acts as a signal to other lymphoid cells. There are two types: lymphokines, which are secreted by lymphocytes; and monokines secreted by macrophages. Cytokines are involved in reproduction, growth and development, normal homeostatic regulation, response to injury and repair, blood clotting and host resistance (immunity and tolerance).

CYTOKINESIS: The process of cytoplasmic division as distinct from nuclear division or karyokinesis. As telophase proceeds, a cell plate is organised in the cytoplasm in the equatorial region of the spindle. The cell is thus divided into two daughter cells.

CYTOKININ (KININ): Plant hormones that stimulate cell division in plant roots and shoots. They are primarily involved in cell growth and differentiation, but also affect apical dominance, axillary bud growth and leaf senescence.

CYTOLOGY: A branch of biology that studies cells and their functions. The discipline includes cytogenetics, cytochemistry and microscopic anatomy, which involve investigations employing various microscopes, such as light, phase, interference and electron microscopes.

CYTOLYSIS: The breakdown of cells usually as a result of destruction or dissolution of their outer membrane. It is an important immune function.

CYTOMEGALOVIRUS: A virus belonging to the herpes group, which remains latent within the body over long periods. It causes enlargement of epithelial cells, usually resulting in mild infections but causing more serious disorders in newborn babies, people with AIDS or weakened immune system.

CYTOMIXIS: The apparent migration of cytoplasm and nuclear material from one pollen mother cell to another through pores (cytomictic channels) in the cell walls. It occurs prior to and during the prophase stage of meiosis. The reason for this is unknown though it may be a response to the nutritional problems of developing reproductive cells.

CYTOPHARYNX: A channel connecting the surface with the protoplasm in certain protozoans. It functions as a gullet in ciliates.

CYTOPLASM: The fluid part of the cell in which the nucleus is embedded or the part of the protoplasm of a cell bounded by the cell

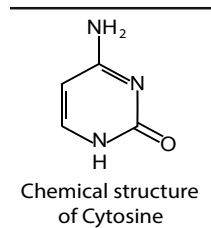
membrane but excluding the nucleus. Often it refers to the jelly-like matter in which the organelles are embedded or the jelly-like material contained in living cells.

CYTOPLASMIC INHERITANCE (EXTRACHROMOSOMAL INHERITANCE): The inheritance of genes contained in the cytoplasm of a cell, rather than the nucleus. The expression of a cytoplasmic determined character is therefore not related to the behaviour or movement of chromosomes and consequently such characters fail to segregate in Mendelian ratios. They are normally transmitted through the female gamete, which contributes most of the cytoplasm to the zygote. The male gametes only contribute a nucleus. Genes inherited in this manner are found on DNA present in small quantities in the chloroplasts, mitochondria and sometimes the cytosol itself. An example of a characteristic inherited in this fashion is a form of male sterility in maize (*Zea mays*).

CYTOPLASMIC MEMBRANE: A selectively permeable membrane which is present around the cytoplasm of the cell and separates the interior of all cells from the outside environment. It consists of a lipid bilayer with embedded proteins; and functions to regulate the passage of substances into and out of the cell.

CYTOPLASMIC STREAMING (CYCLOSIS): The directional movement of cytoplasm in both plant and animal cells, which allows transport of substances such as nutrients and metabolites and also genetic information through the cell, especially around the cell's periphery. Cytoplasmic streaming is also responsible for the locomotion of some protists, such as amoeba.

CYTOSINE: One of the four principal component bases of nucleotides and the nucleic acids DNA and RNA. It is a pyrimidine derivative, with a heterocyclic aromatic ring and two substituents attached: an amine group at position 4 and a keto group at position 2.



CYTOSKELETON: A matrix of protein filaments and tubules that occurs within the liquid part of the cytoplasm present in both plant and animal cells. It gives the cell a definite shape, transports vital substances around the cell and may also be involved in cell movements.

CYTOSOL (HYALOPLASM): The liquid portion of the cytoplasm, including the macromolecules but not including the larger structures, such as subcellular organelles or cytoskeleton.

CYTOSTOME: A mouth-like structure of certain protozoists such as the ciliates and excavates through which particulate food is ingested. The cytostome is typically at the base of an indentation in the cell.

CYTOTAXONOMY: The classification of organisms based on cellular structure and function, especially on the structure and number of chromosomes.

CYTOTOXIC: Destructive or toxic to living cells, inhibiting cell division, cell-killing. Examples are cytotoxic T cells, which are a group of cells in the immune system that attack and destroy virus-infected cells (they also destroy cells of transplanted organs if they are recognised as foreign); or any class of drugs or therapy such as chemotherapy and radiotherapy that inhibit cell division and used to treat cancer. Cytotoxic T cells are also known as **T cells**, **T-killer cells**, **cytolytic T cells** and **killer T cells**.

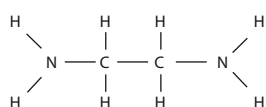
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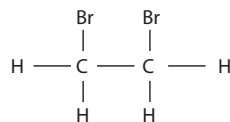
1,2-DIAMINOETHANE (ETHYLENEDIAMINE): A colourless fuming liquid that smells of ammonia, with the molecular formula $C_2H_8N_2$, density 0.899 g cm^{-3} , melting point 9°C (282 K) and boiling point $116 - 118^\circ\text{C}$ (389 – 391 K). It has the structural formula $H_2NCH_2CH_2NH_2$ and therefore a primary amine. It is made from ammonia 1,2-dichloroethane, which is heated under pressure with copper(I) chloride catalyst. With fatty acids it produces soaps, used as emulsifying agents and detergents, solvent, paper and textiles. It is a widely used building block in chemical synthesis.



Chemical structure of
1,2-Diaminoethane
(Ethylenediamine)

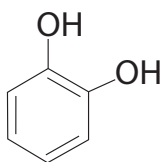
1,2-DIBROMOETHANE (ETHYLENE DIBROMIDE; GLYCOL BROMIDE; ETHYLENE BROMIDE): A volatile haloalkane organic liquid, colourless with a sweet odour that exists in two isomeric forms. Its molecular formula is $C_2H_4Br_2$, density 2.17 g/cm^3 , liquid; melting point $9 - 10^\circ\text{C}$ (282 – 283 K) and a boiling point of $131-132^\circ\text{C}$ (404 – 405 K). It is detectable at 10 ppm.

Although trace amounts occur naturally in the ocean, where it is formed probably by algae and kelp, it is mainly synthetic, made by the addition of bromine to ethene (ethylene). The reaction is $CH_2=CH_2 + Br_2 \rightarrow BrCH_2CH_2Br$ in a classic halogen addition reaction. It is added to petrol to remove lead during combustion and it reacts with lead residues to generate volatile lead bromides. It has also been used as a pesticide in soil and various crops; used as a fumigant for the treatment of logs to control termites and beetles and also for the control of moths in beehives. It is used in the course of preparing dyes and waxes. It is also used in organic synthesis as a source of bromine, for example, to brominate carbanions and to activate magnesium for certain Grignard reagents.



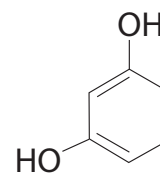
Chemical structure of
1,2-dibromoethane

1,2-DIHYDROXYBENZENE (1,2-BENZENEDIOL; 2-HYDROXYPHENOL; CATECHOL; PYROCATECHOL): A white, feathery crystalline phenol, with the molecular formula $C_6H_4(OH)_2$, density 1.344 g/cm^3 (solid), melting point 105°C (378 K) and boiling point of 245.5°C (519 K). It is the ortho isomer of the three isomeric benzenediols, which occurs naturally in trace amounts and very soluble in water. It is used as a photographic developer.



Chemical structure of
1,2-dihydroxybenzene

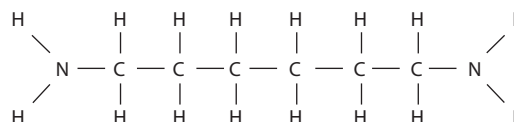
1,3-DIHYDROXYBENZENE (RESORCINOL; RESORCIN; DIHYDROXY PHENOL): A colourless crystalline aromatic compound, with the molecular formula $C_6H_4(OH)_2$, density 1.28 g/cm^3 (solid) melting point, 110°C (383 K) and boiling point 277°C (531 K). It is a 1,3-isomer of benzenediol. It is made by the fusion of benzene-disulfonic acid with sodium hydroxide and used mainly in dyestuffs industry, lasticizers, resins, etc.



Chemical structure of
1,3-dihydroxybenzene

1,6-DIAMINOHEXANE

(HEXAMETHYLENEDIAMINE): A solid colourless organic compound, which is an amine, with the molecular formula $H_2N(CH_2)_6NH_2$, density 0.84 g/cm^3 (solid), melting point, 42°C (315 K) and boiling point 205°C (478 K). It is a comonomer of nylon. It is yellowish for some commercial samples with a strong amine odour, similar to piperidine. It is made by the hydrogenation of adiponitrile: $NC(CH_2)_4CN + 4 H_2 \rightarrow H_2N(CH_2)_6NH_2$. The compound is used, with hexanedioic acid, for producing nylon 66.



Chemical structure of 1,6-diaminohexane
(Hexamethylenediamine)

2-D (2D): Abbreviation for **two dimensional**, which is any virtual object that has no appearance of depth or any figure having width and height but no thickness. Examples are squares, circles and triangles. If a graphic or image of a computer is 2D, it can only be viewed at one angle; whereas, a 3D object may be seen from multiple angles.

2,4-DICHLOROPHENOXYACETIC ACID (2,4-D; HEDONAL; TRINOXOL): A synthetic auxin with the molecular formula $C_8H_6Cl_2O_3$, melting point 140.5°C (414 K) and boiling point 160°C (433 K). It is frequently used as a weed killer of broad-leaved weeds and in laboratories for plant research and as a supplement in plant cell culture media such as MS medium (Murashige and Skoog medium). At very low concentrations, it is used as a growth regulator. It is currently the most widely used herbicide in the world.

3-D (3D): 1. Abbreviation for three dimensional, which defines an object that has height, width and depth. Examples include cones, cubes, cylinders, prisms, pyramids and spheres. 2. A three-dimensional medium, display or performance, especially a cinematic or graphic medium in three dimensions.

3-D ACCELERATOR: A graphics card with additional processing power and memory designated to improve performance and enhance the appearance of 3-D images.

d: 1. Denoting a small increment in a given variable. 2. The notation for differential or derivative. For example, if y is a function of x : $y = f(x)$, then $\frac{dy}{dx}$ is the differential of y with respect to x . 3. Abbreviation for deci-, which represents one tenth is used in symbols for fractions of the physical units in the metric system (S.I).

D: A notation for the number 500 in the Roman numerals.

D GALAXY (DIFFUSE GALAXY): A luminous giant elliptical galaxy found near the centre of a cluster of galaxies. D galaxies are similar